

1	1	/	P?S/8 TC01/TC08 SYSTEM HANDLER	1
2	2			2
3	3	/	COPYRIGHT (C) 1975-1989, CHARLES LASNER ASSOCIATES; 1990-2014, CLA SYSTEMS.	3
4	4			4
5	5	/	LAST EDIT: 18-AUG-2014 CJL.	5
6	6			6
7	7	/	TAKEN FROM A 1976 P?S/8 SYSTEM TAPE IMAGE [KNOWN TO NOT BE A FINAL RELEASE].	7
8	8			8
9	9	/	CONTAINED WITHIN THIS FILE IS THE COMPLETE DEFINITION OF THE SYSTEM DEVICE	9
10	10	/	HANDLER AND ALL DEVICE-RELATED COMPONENTS THAT DISTINGUISH ONE P?S/8 BOOTABLE	10
11	11	/	SYSTEM FROM ANOTHER. THIS IS THE TC01/TC08 DECTAPE-SPECIFIC SET OF SYSTEM	11
12	12	/	COMPONENTS WHICH INCLUDES THE FOLLOWING:	12
13	13			13
14	14	/	1) THE P?S/8 TC01/TC08 SYSTEM DEVICE HANDLER. THIS IS THE BASIC READ/WRITE	14
15	15	/	DEVICE HANDLER FOR ALL OPERATIONS THROUGHOUT THE OPERATING SYSTEM AND ALL	15
16	16	/	SYSTEM COMPONENTS.	16
17	17			17
18	18	/	THE SYSTEM DEVICE HANDLER INCLUDES THE ABILITY TO RELOAD THE HANDLER AND	18
19	19	/	BASE OPERATING SYSTEM BY RESTART AT 07600. THIS IS THE CODE NORMALLY	19
20	20	/	RESIDENT IN 07600-07777 WITH VARIOUS INTERNAL WORDS RESERVED FOR SPECIFIC	20
21	21	/	PURPOSES. [THIS IS EXPLAINED IN DETAIL BELOW.]	21
22	22			22
23	23	/	WHEN THE SYSTEM IS STARTED AT 07600, THE SYSTEM DEVICE HANDLER CODE AND THE	23
24	24	/	SYSTEM KEYBOARD MONITOR ROUTINES ARE RELOADED USING A STRICTLY READ-ONLY	24
25	25	/	PROCESS. IF DESIRED, USER PROGRAMS CAN BE WRITTEN TO THE SYSTEM DEVICE BY	25
26	26	/	INVOKING A VARIETY OF COMMANDS RELATED TO LOADING BINARY FILES AND MEMORY	26
27	27	/	SAVE OPERATIONS; HOWEVER, THIS SELDOM-NEEDED FUNCTION IS NOT PERFORMED	27
28	28	/	DURING ROUTINE SYSTEM RESTARTS. THIS DESIGN WAS CHOSEN TO ALLOW SAFER	28
29	29	/	OPERATION AS WELL AS HIGHER PERFORMANCE; GENERALLY, TRANSIENT SYSTEM DEVICE	29
30	30	/	PROBLEMS CANNOT CORRUPT THE DEVICE-RESIDENT OPERATING SYSTEM IF THE DRIVE	30
31	31	/	IS WRITE-LOCKED. BY DESIGN, SYSTEM DEVICE FAILURES CAN BE SAFELY RETRIED.	31
32	32	/	THE USER IS ENCOURAGED TO WRITE-ENABLE THE SYSTEM DRIVE ONLY WHEN	32
33	33	/	PERFORMING DELIBERATE WRITE OPERATIONS.	33
34	34			34
35	35	/	VARIOUS STAND-ALONE TC01/TC08 STANDARD BOOTSTRAP ROUTINES EXIST, EACH	35
36	36	/	GENERALLY DESIGNED TO MINIMIZE THE EFFORT TO START THE SYSTEM FROM	36
37	37	/	UNINITIALIZED MEMORY. [NOTE: VARIOUS EFFORTS HAVE BEEN MADE OVER THE YEARS	37
38	38	/	LEADING TO SUCCESSIVELY SMALLER ROUTINES AND/OR LESS MANUAL ENTRY EFFORT.]	38
39	39	/	THE P?S/8 SYSTEM DEVICE HANDLER FOR TC01/TC08 IS COMPATIBLE WITH ALL KNOWN	39
40	40	/	BOOTSTRAPS [INCLUDING THOSE KNOWN TO ONLY SUPPORT P?S/8 AND A FEW OTHER	40
41	41	/	OPERATING SYSTEMS WHILE MINIMIZING THE MANUAL EFFORT]. [NOTE: ONCE P?S/8	41
42	42	/	IS STARTED UP, SECONDARY PROGRAMS COULD BE DEPLOYED TO BOOT ANY PDP-8	42
43	43	/	OPERATING SYSTEM INCOMPATIBLE WITH THESE SHORTER BOOTSTRAP ROUTINES].	43
44	44			44
45	45	/	APPENDIX "A" CONTAINS DOCUMENTATION OF WHAT IS PURPORTED TO BE THE SHORTEST	45
46	46	/	KNOWN MANUAL BOOTSTRAP FOR TC01/TC08 DECTAPE SYSTEMS. THIS METHOD CAN BE	46
47	47	/	USED TO BOOTSTRAP P?S/8 AS WELL AS SEVERAL OTHER PDP-8 OPERATING SYSTEMS.	47
48	48			48
49	49	/	UNLIKE THE OS/8 OPERATING SYSTEM, BOTH THE P?S/8 SYSTEM DEVICE HANDLER AND	49
50	50	/	INITIALIZATION CODE RESIDE IN BLOCK 0000 OF THE DECTAPE; THERE IS NO BOOTUP	50
51	51	/	"HELPER" BLOCK TO ASSIST LOADING A SEPARATE HANDLER. OVER THE LONG TERM,	51
52	52	/	THIS HAS PROVEN TO BE A SUPERIOR METHOD FOR ALL SUPPORTED DEVICES. [NOTE:	52
53	53	/	CERTAIN LOCATIONS WITHIN THE BLOCK ARE DEFINED AS LIST ELEMENTS DURING	53
54	54	/	PROGRAM EXECUTION ONCE THE SYSTEM IS FULLY LOADED. DURING THE BOOTSTRAP	54
55	55	/	OPERATION THESE WORDS ARE USED AS ONCE-ONLY INITIALIZING INSTRUCTIONS.]	55
56				56
57				57
58				58

1	57	/	2) A STAND-ALONE BOOTSTRAP FOR THE TC01/TC08 SYSTEM. THIS IS A ROUTINE	1
2	58	/	DESIGNED TO BE LOADED AND STARTED AT 07600 WHEN THE SYSTEM DEVICE HANDLER	2
3	59	/	IS DELIBERATELY DISPLACED FROM MEMORY; THIS OCCURS WHEN ONE OF TWO BINARY	3
4	60	/	LOADING ALTERNATIVES IS INVOKED BY RUNNING THE P?S/8 "BIN" UTILITY PROGRAM	4
5	61	/	WITH VARIOUS OPTION SWITCHES SET. THE REST OF THE LAST PAGE OF FIELD 0 IS	5
6	62	/	FILLED WITH SPECIFIC LOADER CODE AND PARAMETERS APPROPRIATE TO THE	6
7	63	/	PARTICULAR BINARY LOADING FUNCTION CHOSEN [A) P?S/8 BINARY FILES, B) BINARY	7
8	64	/	PAPER-TAPES IN BIN OR RIM FORMAT].	8
9	65			9
10	66	/	THE MAXIMUM LENGTH PERMITTED FOR THE STAND-ALONE BOOTSTRAP ALLOWS	10
11	67	/	COMPATIBILITY WITH THE PAPER-TAPE BIN AND RIM LOADERS; HOWEVER, THE ACTUAL	11
12	68	/	TC01/TC08 BOOTSTRAP CODE IS SHORTER THAN THIS MAXIMUM AMOUNT [WHICH MAY BE	12
13	69	/	IN EFFECT FOR OTHER P?S/8 SYSTEM CONFIGURATIONS]. AS CURRENTLY DEFINED,	13
14	70	/	THE ENTIRETY OF THE STAND-ALONE BOOTSTRAP MUST LOAD INTO 07600-07642.	14
15	71	/	[NOTE: THIS MEMORY RANGE MAY BE SLIGHTLY MODIFIED IN FUTURE RELEASES TO	15
16	72	/	IMPLEMENT ADDITIONAL FEATURES WITHIN THE VARIOUS BINARY LOADERS.]	16
17	73			17
18	74	/	3) SLIGHTLY MODIFIED VERSIONS OF THE DEC STANDARD BIN AND RIM LOADERS. THE	18
19	75	/	BIN AND RIM FORMAT PAPER-TAPE LOADERS CAN BE PLACED IN MEMORY WHEN LOADING	19
20	76	/	DIAGNOSTIC OR OTHER PROGRAMS NOT RELATED TO ANY STORAGE-BASED OPERATING	20
21	77	/	SYSTEM. THIS PROVIDES BACKWARD COMPATIBILITY WITH LEGACY SYSTEMS HAVING NO	21
22	78	/	OPERATING CAPABILITY BEYOND BASIC BINARY PAPER-TAPE LOADING.	22
23	79			23
24	80	/	CERTAIN KNOWN PROGRAMS MUST BE RUN FROM THIS MORE FUNDAMENTAL ENVIRONMENT	24
25	81	/	DUE TO VIOLATION OF VARIOUS PROGRAMMING CONVENTIONS ASSOCIATED WITH ALL	25
26	82	/	STORAGE-BASED OPERATING SYSTEMS. [NOTE: IN SOME INSTANCES, P?S/8 MAY BE	26
27	83	/	MORE ACCOMMODATING THAN OTHER OPERATING SYSTEMS WITH REGARD TO EXPLICIT	27
28	84	/	LOADING BOUNDARIES; HOWEVER, CERTAIN PROGRAMS ARE KNOWN TO CLEARLY VIOLATE	28
29	85	/	STORAGE-BASED LOADING LIMITS. A BRIEF DISCUSSION IS GIVEN BELOW REGARDING	29
30	86	/	POTENTIAL WORK-AROUNDS PARTICULAR TO P?S/8 FOR THESE "ROGUE" PROGRAMS.]	30
31	87			31
32	88	/	A SLIGHTLY MODIFIED BIN LOADER [ALONG WITH THE STANDARD RIM LOADER] LOADS	32
33	89	/	INTO 07643-07777 ALLOWING SEEMLESS COEXISTENCE WITH THE STAND-ALONE	33
34	90	/	BOOTSTRAP LOADING INTO 07600-07642. AS LONG AS ALL PROGRAMS LOADED BY THE	34
35	91	/	BIN OR RIM BINARY PAPER-TAPE LOADERS [OR MANUAL FRONT-PANEL ENTRY] DO NOT	35
36	92	/	VIOLATE THE DEFINED BOOTSTRAP MEMORY RANGE, P?S/8 CAN BE RELOADED BY	36
37	93	/	RESTARTING AT 07600.	37
38	94			38
39	95	/	WHILE THE BIN AND RIM PAPER-TAPE LOADERS ARE NOT SPECIFIC TO THE TC01/TC08	39
40	96	/	P?S/8 CONFIGURATION, THEY ARE INCLUDED IN THIS DOCUMENT TO ILLUSTRATE THE	40
41	97	/	INTERACTION BETWEEN THE STAND-ALONE P?S/8 BOOTSTRAP [AND THE SLURP LOADER]	41
42	98	/	AND THE BIN AND RIM LOADERS. THE BIN LOADER HAS BEEN SLIGHTLY MODIFIED TO	42
43	99	/	FACILITATE COEXISTENCE WITH THE BOOTSTRAP; DUE TO ITS SIMPLISTIC NATURE,	43
44	100	/	THE RIM LOADER IS UNCHANGED. [NOTE: FURTHER MODIFICATIONS MAY BE MADE TO	44
45	101	/	THE BIN LOADER IN FUTURE RELEASES OF P?S/8 BIN FOR THE PURPOSE OF	45
46	102	/	ELIMINATION OF KNOWN TRADE-OFFS THAT AFFECT ALL RELEASES OF THE BIN LOADER.	46
47	103	/	ASSUMING NO FURTHER OPTIMIZATION OF THE CODE IS POSSIBLE, IT MAY BE	47
48	104	/	FEASIBLE TO SHORTEN THE MAXIMUM LENGTH OF THE STAND-ALONE BOOTSTRAP TO	48
49	105	/	ACCOMPLISH THIS. A DISCUSSION OF THIS ISSUE IS INCLUDED BELOW.]	49
50				50
51				51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	107	/	4) THE P?S/8 IMPLEMENTATION OF RICHARD LARY'S UNIQUE "SLURP" LOADER FOR	1
2	108	/	TC01/TC08 SYSTEMS. THE SLURP LOADER WAS FIRST IMPLEMENTED WITHIN THE	2
3	109	/	ORIGINAL R-L MONITOR SYSTEM [SOMETIMES KNOWN AS "MS/8"]. BINARY FILES ON	3
4	110	/	THE P?S/8 SYSTEM DEVICE [AND/OR RELATED LOGICAL UNITS] CAN BE LOADED FASTER	4
5	111	/	BY THE SLURP LOADER THAN ANY OTHER METHOD; THE SYSTEM DEVICE CAN REMAIN	5
6	112	/	WRITE-LOCKED DURING THE LOADING PROCESS. [NOTE: MAJOR DETAILS OF THE BASIC	6
7	113	/	SLURP LOADER ARE DISCUSSED BELOW. FOR A COMPLETE LIST OF AVAILABLE	7
8	114	/	LOADING OPTIONS, INCLUDING THE ALTERNATIVE "VIRTUAL" LOADER, CONSULT	8
9	115	/	DOCUMENTATION OF THE P?S/8 "BIN" UTILITY PROGRAM.]	9
10	116			10
11	117	/	[NOTE: THE P?S/8 BIN UTILITY PROGRAM SHOULD NOT BE CONFUSED WITH THE	11
12	118	/	PAPER-TAPE BIN LOADER. TO MAINTAIN CLARITY, THE PAPER-TAPE BIN LOADER IS	12
13	119	/	ALWAYS CLEARLY REFERENCED AS NEEDED. ALL OTHER REFERENCES TO "BIN" REFER	13
14	120	/	TO THE P?S/8 BIN UTILITY PROGRAM. P?S/8 BIN CAN PERFORM VARIOUS FUNCTIONS	14
15	121	/	RELATED TO BINARY FILES AND BINARY PAPER-TAPES INCLUDING LAUNCHING THE	15
16	122	/	PAPER-TAPE BIN AND RIM LOADERS; THE PRIMARY FUNCTION OF P?S/8 BIN IS TO	16
17	123	/	LOAD BINARY FILES ON THE SYSTEM DEVICE (AND RELATED LOGICAL DRIVE UNITS)	17
18	124	/	INTO MEMORY. FOR MORE INFORMATION ON THE VARIOUS OPERATING MODES OF P?S/8	18
19	125	/	BIN, CONSULT ADVANCED DOCUMENTATION OF THE P?S/8 BIN UTILITY PROGRAM.]	19
20	126			20
21	127	/	THE STAND-ALONE BOOTSTRAP DESCRIBED IN 2) ABOVE IS ALSO INTEGRATED WITH THE	21
22	128	/	SLURP LOADER; THE TWO COMPONENTS WORK TOGETHER TO LOAD BINARY FILES LOCATED	22
23	129	/	ON THE SYSTEM DEVICE [AND RELATED LOGICAL DEVICE UNITS] INTO USER MEMORY	23
24	130	/	WITHOUT USING A SEPARATE BUFFER.	24
25	131			25
26	132	/	[NOTE: MOST SYSTEMS DO NOT CONCEPTUALLY SUPPORT THIS FORM OF ADVANCED	26
27	133	/	BINARY FILE LOADER; CONVENTIONAL WISDOM DICTATES IT CANNOT EXIST. DESPITE	27
28	134	/	WIDESPREAD IGNORANCE OF THIS NOVEL DESIGN, SOME FORM OF SLURP LOADER HAS	28
29	135	/	BEEN PRESENT IN SEVERAL SUPPORTED DEC PRODUCTS. DEC MANAGEMENT PERSONNEL	29
30	136	/	OFTEN ASSUMED THE PROGRAMMING STAFF KNEW WHAT THEY WERE DOING BASED ONLY ON	30
31	137	/	ERRATIC HUNCHES; FEW IF ANY OF THEM WERE COMPETENT TO UNDERSTAND THIS AND	31
32	138	/	OTHER PDP-8 SOFTWARE ISSUES; HOWEVER, WHEN PUT IN A POSITION TO PASS	32
33	139	/	JUDGEMENT ON CERTAIN SOFTWARE PROJECTS, THEY MADE ARBITRARY DECISIONS NOT	33
34	140	/	BASED ON TECHNICAL MERIT; THIS SOMETIMES RESULTED IN NEGATIVE CONSEQUENCES,	34
35	141	/	SOME OF WHICH ARE DISCUSSED ELSEWHERE IN THIS DOCUMENT.]	35
36	142			36
37	143	/	5) A SPECIAL "FILE" IMAGE USED TO RELOAD THE SYSTEM DEVICE HANDLER DIRECTLY	37
38	144	/	FROM THE SLURP LOADER WITHOUT WRITING ON THE SYSTEM DEVICE. THE RELEVANT	38
39	145	/	OPTIONS OF THE P?S/8 BIN UTILITY ARE DESCRIBED IN SUFFICIENT DETAIL TO	39
40	146	/	EXPLAIN HOW THIS TRULY NOVEL SYSTEM COMPONENT FUNCTIONS. [NOTE: THIS	40
41	147	/	MODULE WAS NEVER PART OF THE R-L MONITOR SYSTEM OR THE FEW OTHER SYSTEMS	41
42	148	/	THAT SUPPORT SLURP BINARY LOADING. AS BEYOND CONVENTIONAL WISDOM THE SLURP	42
43	149	/	LOADER IS, THE CONCEPT OF A FILE THAT CAN RELOAD THE SYSTEM DEVICE HANDLER	43
44	150	/	OVER THE SLURP LOADER (WHICH DOESN'T USE A BUFFER) IS BEYOND THE	44
45	151	/	UNDERSTANDING OF MANY PEOPLE. THIS DOCUMENT WILL ATTEMPT TO EXPLAIN	45
46	152	/	EXACTLY WHAT IS ACCOMPLISHED WHEN USING THIS OPTION. SUFFICE TO SAY, AT	46
47	153	/	LEAST THE AUTHORS OF P?S/8 UNDERSTAND HOW THIS WORKS.)	47
48	154			48
49	155	/	6) P?S/8 BOOTUP STARTS A COMBINATION KEYBOARD MONITOR AND IMBEDDED EDITOR FOR	49
50	156	/	TEXT FILES; SOME OF THE AVAILABLE COMMANDS ARE DISCUSSED IN THIS DOCUMENT.	50
51	157	/	WHILE ANY COMMAND IS BEING ENTERED, TWO KEYBOARD CHARACTERS ARE RESERVED TO	51
52	158	/	INITIATE REWIND AND UNLOAD OPERATIONS ON THE SYSTEM DEVICE. THE TC01/TC08	52
53	159	/	VERSION OF THESE OPTIONAL DEVICE-DEPENDENT COMMANDS IS DOCUMENTED BELOW.	53
54	160	/	AS APPLICABLE, ALL P?S/8 SYSTEM CONFIGURATIONS BASED ON TAPES OR OTHER	54
55	161	/	DISMOUNTABLE MEDIA SHOULD IMPLEMENT EQUIVALENT ROUTINES.	55
56				56
57				57
58				58

1	163	/	DOCUMENT LAYOUT DETAILS.	1
2	164			2
3	165	/	THIS DOCUMENT CONTAINS SECTIONS REGARDING ISSUES FAR REMOVED FROM THE PRIMARY	3
4	166	/	PURPOSE OF MERELY DEFINING THE TC01/TC08 SYSTEM DEVICE HANDLER AND RELATED	4
5	167	/	DEVICE-SPECIFIC COMPONENTS AS OUTLINED ABOVE.	5
6	168			6
7	169	/	TOPICS OF DISCUSSION INCLUDE CERTAIN ASPECTS OF OVERALL SYSTEM DESIGN,	7
8	170	/	DIRECTORY STRUCTURE TRADE-OFFS AND RELATED ISSUES. A GLOSSARY OF TERMS IS	8
9	171	/	PRESENT AT THE END OF THE DOCUMENT FOR THE BENEFIT OF ALL READERS NOT FULLY	9
10	172	/	VERSED IN THE SUBJECT OF PDP-8 SYSTEM DEVELOPMENT. BY INCLUDING THIS	10
11	173	/	ADDITIONAL MATERIAL, A BETTER UNDERSTANDING OF HOW P?S/8 OPERATES, BOOTS UP	11
12	174	/	AND OTHER CONSIDERATIONS OF ITS HISTORICAL DEVELOPMENT CAN BE OBTAINED.	12
13	175			13
14	176	/	SOME OF THE MATERIAL CONCERNS THE RELEVANT HISTORY AND DEVELOPMENT OF THE R-L	14
15	177	/	MONITOR SYSTEM, AS WELL AS SOME DETAILS ABOUT KEY PEOPLE WHO DESIGNED AND	15
16	178	/	IMPLEMENTED THIS PREDECESSOR OPERATING SYSTEM. ONE OF THE MISSIONS OF THE	16
17	179	/	P?S/8 OPERATING SYSTEM IS TAKING THE R-L MONITOR SYSTEM CONCEPTS TO THEIR	17
18	180	/	LOGICAL CONCLUSION, A PROCESS STILL UNFOLDING AS OF THE TIME OF THIS WRITING.	18
19	181			19
20	182	/	INCLUDED IS A DISCUSSION OF FUTURE DIRECTIONS FOR P?S/8 INCLUDING RECENT WORK	20
21	183	/	EXPRESSED IN THE CONTEXT OF THE SYSTEM AS IT PRESENTLY EXISTS. [NOTE: MANY	21
22	184	/	ASPECTS OF THIS REPRESENT A LEVEL OF DEVELOPMENT SOMEWHAT LATER THAN THE	22
23	185	/	SYSTEM DEVICE HANDLER AND RELATED CODING AS DESCRIBED IN THIS DOCUMENT; FUTURE	23
24	186	/	VERSIONS WILL UPDATE THE SYSTEM DEVICE HANDLER CODE TO INCLUDE SUPPORT FOR	24
25	187	/	ADDITIONAL WORK NOT YET FULLY DEFINED; CERTAIN CHANGES ARE CURRENTLY UNDER	25
26	188	/	CONSIDERATION FOR FUTURE RELEASES.]	26
27	189			27
28	190	/	WHERE APPLICABLE, NEW WORK IS INDICATED IN TERMS OF MINOR MODIFICATIONS THAT	28
29	191	/	ARE REQUIRED THAT DO NOT MATERIALLY AFFECT CURRENT PERFORMANCE OF WHAT IS NOW	29
30	192	/	DEFINED AS THE "BASIC" P?S/8 SYSTEM. SOME OF THESE CHANGES ARE REQUIRED TO	30
31	193	/	ELIMINATE KNOWN ISSUES WITHIN THE BASIC P?S/8 SYSTEM OR ADD CERTAIN	31
32	194	/	ENHANCEMENTS THAT APPLY TO POTENTIAL SYSTEM EXTENSIONS.	32
33	195			33
34	196	/	SOME OF THESE MODIFICATIONS ARE KNOWN TO BE PART OF MORE RECENT RELEASES OF	34
35	197	/	P?S/8; IT IS HOPED THAT THE CONTENT OF THESE NEWER PROGRAMS CAN BE INTEGRATED	35
36	198	/	INTO A FUTURE RELEASE OF THIS DOCUMENT.	36
37	199			37
38	200	/	ADDITIONAL MODIFICATIONS ARE CONCERNED WITH THE IMPLEMENTATION OF THE P?S/8	38
39	201	/	"SHELL" OVERLAY. THIS IS AN OPTIONAL COMPONENT IN THE EARLY STAGES OF	39
40	202	/	DEVELOPMENT; THE OVERALL GOAL IS TO PROVIDE A POWERFUL SOFTWARE BASE WITH	40
41	203	/	LEVELS OF IMPLEMENTATION SUITABLE FOR ALL USERS FROM MODEST CONFIGURATIONS	41
42	204	/	THROUGH ELABORATE SYSTEMS LARGER THAN WERE AVAILABLE DURING THE ORIGINAL	42
43	205	/	ECONOMIC LIFETIME OF THE PDP-8. [NOTE: IN THEORY, STORAGE DEVICES THAT	43
44	206	/	AGGREGATE AS MUCH AS 1.0 TB ARE SUPPORTABLE.]	44
45	207			45
46	208	/	DEVELOPMENT WILL BE OPEN-ENDED TO ALLOW ADDITIONAL CHANGES IN THE FUTURE;	46
47	209	/	EXPERIENCE SHOWS THAT ONLY AFTER CERTAIN FEATURES ARE AVAILABLE WILL THE	47
48	210	/	NEED FOR ADDITIONAL FUNCTIONALITY BE REALIZED; AS SUCH, THE IMPLEMENTATION	48
49	211	/	WILL ALLOW EASY EXPANSION AT ANY LEVEL OF DEVELOPMENT. UNLIKE CERTAIN OTHER	49
50	212	/	OPERATING SYSTEMS, NO NOVEL FEATURES WILL BE CONCEPTUALLY LOCKED OUT DUE TO	50
51	213	/	SHORT-SIGHTED DESIGN.	51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	215	/	P?S/8 SYSTEM DEFINITIONS.	1
2	216			2
3	217	0000	BOOTBLK=0000 /BOOTSTRAP BLOCK	3
4	218	7643	BINLOAD=7643 /LOWEST ADDRESS OF MODIFIED BIN LOADER	4
5	219	0020	PERFILE=0020 /HARD-WIRED BLOCK FOR % FILE	5
6	220	7600	SBOOT= 7600 /THE SYSTEM [RE]BOOTS AT 07600 AS DO ALL OTHERS	6
7	221	7611	SCORE= 7611 /SYSTEM CORE SIZE STATUS WORD [ALSO OTHER STUFF]	7
8	222	7610	SDATE= 7610 /LOW-ORDER 12-BITS OF THE CURRENT DATE	8
9	223	7756	SEQUALS=7756 /=XXXX PARAMETER PASSED HERE	9
10	224	7757	SFILES= 7757 /FILE LIST PASSED HERE	10
11	225	7634	SFUN= 7634 /SYSTEM LOADER FUNCTION WORD	11
12	226	7633	SLDADR= 7633 /SYSTEM LOADER LOADING ADDRESS	12
13	227	7635	SLDBLK= 7635 /SYSTEM LOADER STARTING BLOCK	13
14	228	7632	SLOAD= 7632 /SYSTEM LOADER ENTRY POINT	14
15	229	7607	SOUTCT= 7607 /OUTPUT FILE COUNT WORD [AND A FEW OTHER THINGS]	15
16	230	7637	SSTART= 7637 /SYSTEM LOADER STARTUP POINTER	16
17	231	7604	SWAL= 7604 /OPTION SWITCHES /A THROUGH /L PASSED HERE	17
18	232	7605	SWMX= 7605 /OPTION SWITCHES /M THROUGH /X PASSED HERE	18
19	233	7606	SWY9= 7606 /OPTION SWITCHES /Y THROUGH /9 PASSED HERE	19
20	234	0000	SYSBLK= 0000 /SYSTEM DEVICE HANDLER LOADS FROM BLOCK 0000	20
21	235	7640	SYSIO= 7640 /SYSTEM DEVICE HANDLER ENTRY POINT	21
22	236	4000	WRITE= 4000 /FUNCTION WORD WRITE BIT	22
23	237			23
24	238	/	P?S/8 KEYBOARD MONITOR AND INTEGRATED EDITOR DEFINITIONS.	24
25	239			25
26	240	0000	CURUNIT=0000 /LATEST LOGICAL DRIVE UNIT IN BITS[9-11]	26
27	241	0100	KBDCHR= 0100 /LATEST KEYBOARD INPUT CHARACTER	27
28	242	1216	KBDLUP= 1216 /KEYBOARD MONITOR INPUT ROUTINE	28
29	243	2530	KBDMON= 2530 /START THE KEYBOARD MONITOR HERE	29
30	244	0104	L200= 0104 /CONSTANT 0200	30
31	245	0064	L7756= 0064 /CONSTANT 7756	31
32	246	0040	L7770= 0040 /CONSTANT 7770	32
33	247	1342	REWUNLD=1342 /REWIND/UNLOAD ROUTINE LOADS HERE	33
34	248	1305	RWTEMP= 1305 /AVAILABLE TEMPORARY FOR TIMING LOOP	34
35	249			35
36	250	/	SLURP FORMAT CONTROL DEFINITIONS.	36
37	251			37
38	252	0000	DA= 0 /DATA WORD	38
39	253	0001	EF= 1 /END-OF-FILE	39
40	254	0003	FL= 3 /FIELD SETTING	40
41	255	0002	OR= 2 /ORIGIN SETTING	41
42				42
43				43
44				44
45				45
46				46
47				47
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51				51
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58				58

1	257	/	IN-MEMORY SYSTEM GENERATION DEFINITIONS.		1
2	258				2
3	259	/	GENERATION ADDRESSES ARE WHERE IN MEMORY THE STATED COMPONENT LOADS DURING THE		3
4	260	/	INITIAL GENERATION OF P?S/8. THIS IS UNRELATED TO THE ACTUAL MECHANISM USED		4
5	261	/	TO PERFORM THE WRITING OUT OF THE COMPONENT WHICH IS NOT DEFINED HERE. IN		5
6	262	/	ACTUAL SYSTEM GENERATION SITUATIONS, A PRE-EXISTING P?S/8 SYSTEM DEVICE		6
7	263	/	HANDLER IS RESIDENT IN 07600 AND AVAILABLE FOR THE WRITE OPERATIONS. [NOTE:		7
8	264	/	IT IS BEYOND THE SCOPE OF THIS DOCUMENT TO EXPLAIN ANY ADDITIONAL DETAILS OF		8
9	265	/	SYSTEM GENERATION, CERTAIN OF WHICH ARE STILL EVOLVING AS OF THIS WRITING.]		9
10	266				10
11	267	/	THE GENERATION FIELD IS EXPRESSED IN BITS[6-8] FOR THE PURPOSES OF GENERATION		11
12	268	/	MEMORY EXTENSION INSTRUCTIONS SUCH AS "CDF SDHFLD". FIELD SETTINGS SHOULD USE		12
13	269	/	THIS PARAMETER DIVIDED BY 10 SINCE THE FIELD DIRECTIVE IS EXPRESSED IN THE		13
14	270	/	RANGE OF 0 THROUGH 7.		14
15	271				15
16	272	/	ALL COMPONENTS WILL BE SUITABLY RELOCATED TO ENSURE THE INTENDED INSTRUCTIONS		16
17	273	/	ARE CREATED. THE NOPUNCH AND ENPUNCH DIRECTIVES WILL BE USED TO RELOCATE THE		17
18	274	/	GENERATION FIELD; THE RELOC DIRECTIVE WILL BE USED TO RELOCATE THE 12-BIT		18
19	275	/	ADDRESS.		19
20	276				20
21	277	6600	BRGADR= 6600	/GENERATION ADDRESS FOR WRITING OUT	21
22	278			/THE BIN AND RIM LOADER [WITHIN "BIN"]	22
23	279	0010	BRGFLD= 1^10	/GENERATION FIELD FOR WRITING OUT	23
24	280			/THE BIN AND RIM LOADER [WITHIN "BIN"]	24
25	281	0000	KMNADR= 0000	/GENERATION ADDRESS FOR WRITING OUT	25
26	282			/THE KEYBOARD MONITOR AND INTEGRATED EDITOR	26
27	283	0000	KMNFLD= 0^10	/GENERATION FIELD FOR WRITING OUT	27
28	284			/THE KEYBOARD MONITOR AND INTEGRATED EDITOR	28
29	285	7400	SDHADR= 7400	/GENERATION ADDRESS FOR WRITING OUT	29
30	286			/IMAGE OF THE SYSTEM DEVICE HANDLER	30
31	287	0010	SDHFLD= 1^10	/GENERATION FIELD FOR WRITING OUT	31
32	288			/IMAGE OF THE SYSTEM DEVICE HANDLER	32
33	289	7000	SLUADR= 7000	/GENERATION ADDRESS FOR WRITING OUT	33
34	290			/IMAGE OF THE BOOTSTRAP AND SLURP LOADER	34
35	291	0010	SLUFLD= 1^10	/GENERATION FIELD FOR WRITING OUT	35
36	292			/IMAGE OF THE BOOTSTRAP AND SLURP LOADER	36
37	293	7200	SLIADR= 7200	/GENERATION ADDRESS FOR WRITING OUT	37
38	294			/IMAGE OF THE /I "FILE" HANDLER RELOADER CODE	38
39	295	0010	SLIFLD= 1^10	/GENERATION FIELD FOR WRITING OUT	39
40	296			/IMAGE OF THE /I "FILE" HANDLER RELOADER CODE	40
41	297				41
42	298	/	TC01/TC08 DECTAPE DEFINITIONS.		42
43	299				43
44	300	7755	CA= 7755	/CURRENT ADDRESS REGISTER FOR 3-CYCLE DATA BREAK	44
45	301	6762	DTCA= 6762	/CLEAR STATUS REGISTER "A"	45
46	302	6766	DTLA= 6766	/LOAD STATUS REGISTER "A" FROM AC, CLEAR AC	46
47	303	6774	DTLB= 6774	/AC[6-8] TO BREAK FIELD, CLEAR AC	47
48	304	6761	DTRA= 6761	/STATUS REGISTER "A" .OR. INTO AC	48
49	305	6772	DTRB= 6772	/STATUS REGISTER "B" .OR. INTO AC	49
50	306	6771	DTSF= 6771	/SKIP ON DECTAPE FLAG OR ERROR FLAG SET	50
51	307	6764	DTXA= 6764	/AC .XOR. INTO STATUS REGISTER "A", CLEAR AC	51
52	308	7754	WC= 7754	/WORD COUNT REGISTER FOR 3-CYCLE DATA BREAK	52
53					53
54					54
55					55
56					56
57					57
58					58

1	310	/	BIN LOADER AND RIM LOADER DEFINITIONS.	1
2	311			2
3	312	7756	BEG= 7756 /STARTING ADDRESS OF THE RIM LOADER	3
4	313	7777	BINSTRT=7777 /OFFICIAL STARTING ADDRESS OF THE BIN LOADER	4
5	314			5
6	315	/	NUMERIC LOAD DEFINITIONS.	6
7	316			7
8	317	/	NOTE: THESE COMBINED MICRO-OPERATIONS MUST WORK ON ALL PDP-8 MODELS [WITH THE	8
9	318	/	LONE EXCEPTION OF THE PDP-8/S].	9
10	319			10
11	320	7330	NL4000= CLA CLL CML RAR /LOAD AC WITH 4000	11
12	321	7240	NL7777= CLA CMA /LOAD AC WITH 7777	12
13	322			13
14	323	/	MAINTENANCE DEFINITIONS.	14
15	324			15
16	325	/	THE 1976 RELEASE OF THE SYSTEM DEVICE HANDLER HAS AN EASILY FIXED MINOR BUG.	16
17	326	/	A SIMPLE REMEDY CONSISTS MERELY OF REARRANGING EXISTING INSTRUCTIONS. THE	17
18	327	/	REPAIRED VERSION OCCUPIES THE EXACT SAME MEMORY RANGE AS THE ORIGINAL VERSION;	18
19	328	/	HOWEVER, CERTAIN LOCATIONS WITHIN THE SYSTEM DEVICE HANDLER MUST BE MODIFIED	19
20	329	/	CONSISTENT WITH THE MODIFIED ADDRESSING OF KEY LOCATIONS WITHIN THE HANDLER.	20
21	330			21
22	331	/	THE SOURCE CODE CAN BE ASSEMBLED FOR THE HISTORIC ORIGINAL VERSION OR THE	22
23	332	/	CORRECTED VERSION. LEAVE UNCOMMENTED THE APPROPRIATE CHOICE OF EITHER	23
24	333	/	AUTHENTIC ORIGINAL CODE OR THE VERSION REMEDYING THE PROBLEM. [NOTE: FUTURE	24
25	334	/	RELEASES OF THE SYSTEM DEVICE HANDLER WILL ELIMINATE THE BUG AS WELL AS	25
26	335	/	IMPLEMENT ADDITIONAL FEATURES.]	26
27	336			27
28	337		IFNDEF ASMBUG <ASMBUG=1> /INHIBIT BUG FIX FOR 1976 RELEASE	28
29	339	/	IFNDEF ASMBUG <ASMBUG=0> /FIX BUG FOR FUTURE RELEASE	29
30	340			30
31	341	/	ASSEMBLY OPTIONS.	31
32	342			32
33	343	/	THIS SOURCE FILE IS DESIGNED TO BE ASSEMBLED EITHER WITH P?S/8 PAL OR OS/8	33
34	344	/	PAL8 WITHOUT ANY MODIFICATIONS. [NOTE: PAL10 CANNOT BE USED DUE TO LACK OF	34
35	345	/	CERTAIN KEY ASSEMBLY DIRECTIVES.]	35
36	346			36
37	347	/	SINCE THERE ARE NO TEXT STRINGS IN THE SOURCE CODE, THE P?S/8 PAL "/J" SWITCH	37
38	348	/	CAN BE SET IF DESIRED. IF ASSEMBLING WITH PAL8, THE CORRESPONDING PAL8 "/F"	38
39	349	/	SWITCH CAN BE SET IF DESIRED.	39
40	350			40
41	351	/	NOTE: THIS SOURCE CODE DOES NOT USE LITERALS. WHILE NOT A CONSIDERATION WHEN	41
42	352	/	USING PAL8, P?S/8 PAL HAS SEVERAL MODULAR ASSEMBLY OPTIONS. AS SUCH, IT IS	42
43	353	/	RECOMMENDED THE "/Q" SWITCH BE CLEARED TO IMPROVE ASSEMBLER EFFICIENCY [IF	43
44	354	/	DESIRED].	44
45				45
46				46
47				47
48				48
49				49
50				50
51				51
52				52
53				53
54				54
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56				56
57				57
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1	356	/	BACKGROUND MATERIAL.	1
2	357			2
3	358	/	THE P?S/8 TC01/TC08 SYSTEM DEVICE HANDLER AND RELATED DEVICE-SPECIFIC	3
4	359	/	COMPONENTS REPRESENT THE CULMINATION OF A SURPRISINGLY LARGE AMOUNT OF	4
5	360	/	PROGRAMMING EFFORT ON THE PART OF A SERIES OF INDIVIDUALS. SOMETIMES ACTING	5
6	361	/	ALONE OR IN SMALL GROUPS, THE CODE PRESENTED BELOW WAS DEVELOPED VIA A LONG	6
7	362	/	AND CIRCUITOUS PATH OVER THE COURSE OF SEVERAL DECADES.	7
8	363			8
9	364	/	VARIOUS RELATED TOPICS WILL BE INTRODUCED TO ATTEMPT TO EXPLAIN THE UNUSUAL	9
10	365	/	CIRCUMSTANCES THAT APPLY TO THIS DEVELOPMENT; THE APPLICATION OF VARIOUS	10
11	366	/	INSPIRATIONAL NOTIONS LED TO THE PRESENT DESIGN AS PRESENTED IN THIS DOCUMENT.	11
12	367			12
13	368	/	SECTION 0: THE HISTORY OF THE R-L MONITOR SYSTEM.	13
14	369			14
15	370	/	THE R-L MONITOR SYSTEM IS THE FIRST VIABLE PDP-8 OPERATING SYSTEM FOR THE	15
16	371	/	PDP-8 WITH TC01/TC08 DECTAPE. IT WAS CREATED MANUALLY OVER THE COURSE OF	16
17	372	/	SEVERAL MONTHS WITH LITTLE, IF ANY, PRIOR PLANNING; THE HARDWARE BECAME	17
18	373	/	AVAILABLE AND A GROUP OF FIVE INDIVIDUALS [LATER KNOWN AS THE "WILLIAM CLAUDE	18
19	374	/	FIELDS MEMORIAL PROGRAMMING GROUP" (WCFMPG) ESSENTIALLY JUST DECIDED TO WRITE	19
20	375	/	IT EVEN THOUGH NO DEVELOPMENT RESOURCES WHATSOEVER WERE AVAILABLE.	20
21	376			21
22	377	/	THE SYSTEM WAS INCREMENTALLY DESIGNED AND INDIVIDUAL SECTIONS OF CODE WERE	22
23	378	/	HAND ASSEMBLED FROM NOTES WRITTEN ON PAPER [INCLUDING NAPKINS FROM LOCAL	23
24	379	/	CHINESE RESTAURANTS]. THE CODE WAS WRITTEN DOWN IN OCTAL NOTATION AS WELL AS	24
25	380	/	PRIMITIVE [AND GENERALLY UNCOMMENTED] SOURCE CODE, THEN MANUALLY ENTERED INTO	25
26	381	/	MEMORY BY EXTRAORDINARY USE OF THE PDP-8 FRONT PANEL ON THE FIRST PDP-8 SYSTEM	26
27	382	/	AVAILABLE [AT BROOKLYN POLYTECHNIC INSTITUTE WHERE THEY WERE ALL STUDENTS AT	27
28	383	/	THE TIME].	28
29	384			29
30	385	/	DESPITE THE INCREDIBLE NATURE OF THE DEVELOPMENT PROCESS, THE R-L MONITOR WAS	30
31	386	/	QUITE SUCCESSFUL. A SIGNIFICANT PERCENTAGE OF THE EVENTUAL PDP-8 PROGRAMMING	31
32	387	/	GROUP BOTH WITHIN DEC AND IN THE REST OF THE WORLD STARTED THEIR PDP-8 CAREERS	32
33	388	/	ON THIS SYSTEM. IN MANY WAYS IT IS THE STARTING POINT OF THE DEVELOPMENT OF	33
34	389	/	P?S/8, WHICH IS IN PART DESIGNED TO TAKE THE PRINCIPLES OF THE R-L MONITOR TO	34
35	390	/	THEIR LOGICAL CONCLUSION.	35
36	391			36
37	392	/	SINCE THE R-L MONITOR WAS AVAILABLE TO THE "NEXT GENERATION" OF PDP-8	37
38	393	/	PROGRAMMERS AT BROOKLYN POLY, ATTENTION COULD FOCUS MORE ON IMPROVING THE	38
39	394	/	SYSTEM THAN MERELY CREATING IT FROM LITERALLY NOTHING.	39
40	395			40
41	396	/	MANY OF THE ORIGINAL CONCEPTS WERE RETAINED IN P?S/8; OTHERS WERE REFINED.	41
42	397	/	MANY ADDITIONAL CONCEPTS WERE EVENTUALLY ADDED AS WELL AS A LARGE NUMBER OF	42
43	398	/	NEW PROGRAMS OVER THE COURSE OF SEVERAL DECADES. P?S/8 IS STILL EVOLVING AS	43
44	399	/	OF THE TIME OF THIS WRITING AND WILL EVENTUALLY INCLUDE ADDITIONAL FEATURES.	44
45				45
46				46
47				47
48				48
49				49
50				50
51				51
52				52
53				53
54				54
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57				57
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1	401	/	R-L SYSTEM DEVICE HANDLER AND LIMITATIONS.	1
2	402			2
3	403	/	THE SENSE OF EXPEDIENCY INHERENT IN THE CREATION OF THE R-L MONITOR IS	3
4	404	/	UNDERSTANDABLE; HOWEVER, P?S/8 IS BASED ON MANY OF THE SAME CONCEPTS ONLY	4
5	405	/	TAKEN TO A "BETTER PLACE" OVER TIME. P?S/8 WAS NOT WORKED ON WITH A	5
6	406	/	SCHEDULE IN MIND. SPORADICALLY, THE SYSTEM WAS GRADUALLY IMPROVED; AT ANY	6
7	407	/	GIVEN TIME, OVERALL VIABILITY WAS ALWAYS MAINTAINED; A CURRENTLY WORKING	7
8	408	/	SYSTEM IS REQUIRED TO EVOLVE INTO A NEWER WORKING SYSTEM. [IN SOME CASES,	8
9	409	/	THIS MEANT PUTTING ON HOLD CERTAIN NEW DESIGN COMPONENTS BECAUSE OF THE LACK	9
10	410	/	OF A PREREQUISITE FOUNDATION. NO OTHER SIGNIFICANT SOFTWARE PROJECT HAS EVER	10
11	411	/	IMPLEMENTED IN THIS FASHION.]	11
12	412			12
13	413	/	THE "WEAKEST LINK" OF THE R-L MONITOR IS THE COMPLEX OF COMPONENTS NEEDED TO	13
14	414	/	CARRY OUT THE VARIOUS LOW-LEVEL FUNCTIONS NEEDED TO SUPPORT PDP-8 PROGRAMS	14
15	415	/	THAT CAN OCCUPY NEARLY THE ENTIRE MEMORY OF THE SYSTEM. THIS LEAVES ONLY A	15
16	416	/	SMALL RESIDENT AREA CAPABLE OF RELOADING THE OPERATING SYSTEM AND SUPPLYING	16
17	417	/	CERTAIN SYSTEM SERVICES [IF POSSIBLE].	17
18	418			18
19	419	/	THE ORIGINAL R-L SYSTEM DEVICE HANDLER WAS NOT A GOOD EXAMPLE OF ADVANCED	19
20	420	/	PDP-8 PROGRAMMING OF THE TC01/TC08 DECTAPE CONTROLLER; HOWEVER, IT WAS	20
21	421	/	ADEQUATE TO THE TASK AS DEFINED IN THE ORIGINAL SYSTEM.	21
22	422			22
23	423	/	THE R-L SYSTEM DEVICE HANDLER IS ENTERED VIA ONE OF TWO ENTRY POINTS [ONE FOR	23
24	424	/	READING, THE OTHER FOR WRITING]. THE HANDLER IS CALLED AS FOLLOWS:	24
25	425			25
26	426	/	CIF CDF 00 /THE R-L SYSTEM DEVICE HANDLER	26
27	427	/	/ONLY WORKS WITH FIELD 0.	27
28	428			28
29	429	/	JMS I PHNDLER /"PHNDLER" CONTAINS THE ENTRY POINT THAT	29
30	430		/APPLIES; EITHER THE READ OR WRITE	30
31	431		/ENTRY POINT AS APPROPRIATE.	31
32	432	/	CORE /USER'S MEMORY BUFFER STARTING ADDRESS	32
33	433		/IN THE RANGE OF 00000-07400. [THE LAST	33
34	434		/ADDRESS TRANSFERRED MUST BE LESS THAN 07600.]	34
35	435	/	-BLOCKS /TWO'S COMPLEMENT OF THE DESIRED NUMBER OF	35
36	436		/BLOCKS TO BE TRANSFERRED AS A CONTIGUOUS	36
37	437		/GROUP IN A SINGLE SEARCH THEN READ [OR	37
38	438		/WRITTEN] IN ONE SMOOTH MOTION OF THE DECTAPE.	38
39	439	/	BLKNUM /THE STARTING BLOCK NUMBER FOR THE TRANSFER;	39
40	440	/	/ANY VALUE FROM 0000-2701 IS ALLOWED; THE LAST	40
41	441		/BLOCK TRANSFERRED MUST BE 2701 OR LESS.	41
42	442			42
43	443	/	THE SYSTEM DEVICE HANDLER IS RESIDENT IN MEMORY WHEN THE PRIMARY SYSTEM	43
44	444	/	COMPONENT, THE KEYBOARD MONITOR AND INTEGRATED EDITOR, IS MEMORY RESIDENT.	44
45	445	/	THE SYSTEM CAN BE RESTARTED BY ANY MEANS OF EXIT TO 07600. CALLS FOR THE	45
46	446	/	PURPOSE OF READING ARE LIMITED TO A MAXIMUM OF 31 BLOCKS; ATTEMPTING TO	46
47	447	/	TRANSFER OVER LOCATIONS 07600-07777 WILL CAUSE HANDLER SELF-DESTRUCTION.	47
48	448	/	CALLS FOR THE PURPOSE OF WRITING CAN BE PERFORMED UP TO 32 BLOCKS AT A TIME;	48
49	449	/	HOWEVER, THE IMAGE OF THE SYSTEM HANDLER ITSELF WOULD BE PART OF THE WRITTEN	49
50	450	/	DATA. THUS, WITHOUT ANY SUPPORT FOR EXTENDED MEMORY, PRACTICAL READ AND WRITE	50
51	451	/	OPERATIONS ARE LIMITED TO AT MOST 31 CONTIGUOUS BLOCKS.	51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	453	/	THERE IS NO ERROR RETURN; ALL CALLS ALWAYS RETURN TO THE CALLER PAST THE THREE	1
2	454	/	IN-LINE ARGUMENTS.	2
3	455			3
4	456	/	IF AN ACTUAL DEVICE ERROR OCCURS, THE HANDLER HALTS THE CPU; THE AC CONTAINS	4
5	457	/	ERROR-DEPENDENT CONTENTS WHICH CORRESPONDS TO THE TC01/TC08 CONTROLLER	5
6	458	/	STATUS REGISTER "B" CONTENTS [WHICH CONTAINS THE CURRENT ERROR STATUS AND THE	6
7	459	/	MEMORY TRANSFER FIELD, THE LATTER ALWAYS BEING 000]. THE OPERATOR CAN ALWAYS	7
8	460	/	SAFELY PRESS "CONTINUE" ON THE FRONT PANEL TO RETRY THE OPERATION. AS LONG AS	8
9	461	/	THE CALL TO THE HANDLER IS COMPLETELY LEGITIMATE [MEANING NO ATTEMPT TO ACCESS	9
10	462	/	NONEXISTENT BLOCKS], THE ERROR WILL LIKELY CLEAR AS LONG AS THE HARDWARE IS IN	10
11	463	/	GOOD WORKING ORDER. TRANSIENT READ ERRORS WILL GENERALLY SUCCEED WITHIN ONE	11
12	464	/	OR TWO RETRY ATTEMPTS.	12
13	465			13
14	466	/	MOST DETECTED ERRORS ARE CAUSED BY FAILURE TO MAKE THE DRIVE READY. IN THE	14
15	467	/	CASE OF WRITING, THE DRIVE MUST NOT BE WRITE-LOCKED. MANY USERS PREFERRED TO	15
16	468	/	DELIBERATELY WRITE-LOCK THE TAPE DRIVE; WHEN A WRITE WAS INITIATED, THE HALT	16
17	469	/	WITHIN THE SYSTEM DEVICE HANDLER WAS EXPECTED; THE WRITE-LOCK CONDITION WAS	17
18	470	/	CLEARED THEN "CONTINUE" WAS PRESSED AND THE WRITE OPERATION SUCCEEDED AS	18
19	471	/	INTENDED. ADEPT USERS NEVER CAUSED ANY ACCIDENTAL HARM TO THEIR PERSONAL	19
20	472	/	DECTAPE SYSTEM BY ALWAYS USING THIS PRUDENT METHOD.	20
21	473			21
22	474	/	THE R-L SYSTEM DEVICE HANDLER ONLY SUPPORTS DEVICE UNIT 0. [NOTE: THE PDP-8	22
23	475	/	AT BROOKLYN POLY ONLY HAD ONE DECTAPE DRIVE.] BY A GRADUAL PROCESS INVOLVING	23
24	476	/	MANY TAPE CHANGES, IT IS POSSIBLE TO EVENTUALLY TRANSFER AN ENTIRE DECTAPE IF	24
25	477	/	NEED BE, 31 BLOCKS AT A TIME; SINCE THE DRIVE WAS ONLY WRITE-ENABLED WHEN THE	25
26	478	/	TARGET TAPE WAS MOUNTED, USERS COULD PERIODICALLY CREATE BACKUPS OF THE SYSTEM	26
27	479	/	AND THEIR DATA [TIME PERMITTING; IT WAS A VERY BUSY SHARED ENVIRONMENT].	27
28	480			28
29	481	/	THE R-L SYSTEM DEVICE HANDLER INCLUDES THE ABILITY TO RESTART THE SYSTEM AT	29
30	482	/	07600. THE BOOTSTRAP CODE RELOADS THE SYSTEM DEVICE HANDLER OVER ITSELF BY	30
31	483	/	THREE-CYCLE DATA-BREAK [DMA]. WHILE THE TRANSFER IS OCCURRING, THE PDP-8	31
32	484	/	STANDARD WAIT LOOP FOR TC01/TC08 DECTAPE [07616-07617] IS IN CONTROL WAITING	32
33	485	/	FOR THE DECTAPE DONE FLAG TO SET. WHEN THE FLAG RAISES, THE NEW CONTENTS OF	33
34	486	/	MEMORY TAKE CONTROL BECAUSE KEY WORDS IN MEMORY ARE OVERLAID BY THE	34
35	487	/	CORRESPONDING CONTENTS OF BLOCK 0000, REPLACING THE WORDS IN MEMORY THAT WERE	35
36	488	/	SETUP FOR THIS OPERATION. THIS INCLUDES REPLACEMENT VALUES FOR OVERLAY OF THE	36
37	489	/	TC01/TC08 WORD COUNT [07754] AND CURRENT ADDRESS [07755] REGISTERS FOR PROPER	37
38	490	/	TRANSFER. [NOTE: WHEN THE CONTENTS OF BLOCK 0000 OVERLAY SIMILAR CODE IN	38
39	491	/	MEMORY, THE WORD COUNT IS EVENTUALLY OVERLAID BY THE COPY ON THE DECTAPE; THIS	39
40	492	/	CAN CHANGE THE OUTCOME OF THE TRANSFER TO A NEW VALUE DETERMINED BY THE	40
41	493	/	INTERNAL BOOTSTRAP OPERATION ITSELF. THE CURRENT ADDRESS MUST ALSO HAVE THE	41
42	494	/	PROPER VALUE SO AS TO CONTINUE THE DMA OPERATION AS INTENDED BY THE BOOTSTRAP	42
43	495	/	CODE ON THE DECTAPE.]	43
44	496			44
45	497	/	AT THIS POINT, THE FRESH COPY OF THE BOOTSTRAP BLOCK CODE IS IN CONTROL.	45
46	498	/	ADDITIONAL TRANSFERS ARE SETUP TO READ IN THE KEYBOARD MONITOR AND THE TINY	46
47	499	/	FILE SYSTEM [TFS] DIRECTORY BLOCKS BY SEVERAL INDEPENDENT CALLS TO THE SYSTEM	47
48	500	/	DEVICE HANDLER THAT ARE LOCATED IN THE LAST WORDS OF BLOCK 0000. [NOTE: THESE	48
49	501	/	LOCATIONS ARE LATER USED TO PASS A LIST OF FILE POINTERS; AT THE TIME OF	49
50	502	/	BOOTING UP, THEY ARE AVAILABLE FOR THE INSTRUCTIONS THAT CARRY OUT THE BOOTUP	50
51	503	/	PROCESS.]	51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	505	/	AFTER ALL OF THESE OPERATIONS HAVE COMPLETED, THE SYSTEM IS RUNNING AND	1
2	506	/	CAPABLE OF PROCESSING COMMANDS. THE AVAILABLE COMMANDS CAN BE BROKEN DOWN BY	2
3	507	/	CATEGORY:	3
4	508			4
5	509	/	1) FILE MANAGEMENT COMMANDS. FILES CAN BE ADDED TO THE DIRECTORY OR DELETED	5
6	510	/	FROM THE DIRECTORY; THE DIRECTORY CAN BE LISTED OUT ON THE SYSTEM CONSOLE.	6
7	511	/	SINCE THE ENTIRE TFS DIRECTORY IS CACHED IN MEMORY, ONLY OPERATIONS THAT	7
8	512	/	CHANGE THE DIRECTORY CONTENTS CAUSE IT TO BE WRITTEN OUT; IF A FILE IS	8
9	513	/	LOADED INTO THE EDITOR BUFFER AND UPDATED, A FILE SAVE COMMAND WILL WRITE	9
10	514	/	TO THE SAME DIRECTORY SLOT; NO DIRECTORY OPERATIONS ARE PERFORMED IN SUCH	10
11	515	/	CIRCUMSTANCES. [NOTE: THE PERFORMANCE IMPROVEMENT OF THIS DIRECTORY ACCESS	11
12	516	/	METHOD OVER MORE CONVENTIONAL SYSTEMS IS QUITE PROFOUND.]	12
13	517			13
14	518	/	2) EDITING COMMANDS. WHEN A FILE IS LOADED INTO THE EDITING BUFFER, VARIOUS	14
15	519	/	COMMANDS EXIST TO EITHER ADD OR REMOVE RANGES OF LINES. THE USER CAN ADD	15
16	520	/	NEW LINES BY SPECIFYING THE LINE NUMBER APPROPRIATE TO POSITIONING THE	16
17	521	/	LINE IN THE FILE. AN AUTO-SEQUENCING MODE EXISTS TO ALLOW MORE FREE-FORM	17
18	522	/	TEXT ENTRY. [NOTE: ALL LINE NUMBERS ARE KEPT IN THE FILE. COMMANDS EXIST	18
19	523	/	TO RENUMBER LINES BY AN ORDERED RESEQUENCE. AS NECESSARY THE REPLACEMENT	19
20	524	/	LINE NUMBERS WILL SUPERSEDE THE PREVIOUS LINE NUMBERS; THE TEXTUAL CONTENTS	20
21	525	/	OF ALL AFFECTED LINES IS NOT CHANGED.] [NOTE: THE LINE NUMBERS ARE PLACED	21
22	526	/	INTO A DIFFERENT AREA WITHIN THE FILE; THIS ALLOWS SYSTEM PROGRAMS TO	22
23	527	/	ACCESS THE TEXT WITHOUT PROCESSING THE LINE NUMBERS.]	23
24	528			24
25	529	/	EACH TFS FILE IS EXACTLY 2048 WORDS LONG. COMMANDS EXIST TO INDICATE HOW	25
26	530	/	FULL A FILE IS; IF A FILE IS GETTING CLOSE TO FULL, IT MAY BE SPLIT INTO	26
27	531	/	TWO FILES AS NECESSARY BY USING APPROPRIATE COMMANDS TO DELETE CERTAIN	27
28	532	/	LINE RANGES WHILE RETAINING OTHERS. THIS PROCESS IS NOT AS CLUMSY AS IT	28
29	533	/	MIGHT SOUND; MANY USERS HAVE ADAPTED TO THIS FORM OF "RANDOM ACCESS" FILE	29
30	534	/	EDITING OVER THE YEARS. [NOTE: INDEFINITE LENGTH FILE SYSTEMS GENERALLY	30
31	535	/	REQUIRE THAT THE ENTIRETY OF ANY ONE FILE BE AVAILABLE DURING EDITING PLUS	31
32	536	/	ADDITIONAL SPACE EQUAL TO OR GREATER THAN THE CURRENT LENGTH; THE EXISTING	32
33	537	/	FILE MUST BECOME A BACKUP COPY WHILE A NEW COPY IS BEING CREATED. ON A	33
34	538	/	LOW-PERFORMANCE DEVICE SUCH AS DECTAPE, THIS CAN BE EXTREMELY FRUSTRATING	34
35	539	/	TO ANY USER DUE TO THE EXTREME OVERHEAD INVOLVED. BY ALLOWING THE ABILITY	35
36	540	/	TO WORK ON ANY PROJECT SPANNING MULTIPLE FILES, ONLY A SINGLE 2048 WORD	36
37	541	/	SECTION IS UPDATED AT ANY GIVEN TIME. ANYONE WHO ROUTINELY EDITS TEXT FILES	37
38	542	/	CAN EASILY GET ACCUSTOMED TO LOCATING WHICH OF SEVERAL RELATED FILES	38
39	543	/	CONTAINS THE SECTION OF INTEREST AT ANY GIVEN MOMENT; THE REST OF THE TEXT	39
40	544	/	FILES ARE NOT EVEN READ, MUCH LESS WRITTEN.]	40
41	545			41
42	546	/	3) EXECUTION [RUN] COMMANDS. SYSTEM PROGRAMS [SUCH AS THE PAL III ASSEMBLER	42
43	547	/	MODIFIED TO TAKE INPUT FROM TFS TEXT FILES AND WRITE OUTPUT TO DESIGNATED	43
44	548	/	AREAS ON THE DECTAPE] CAN BE BROUGHT INTO MEMORY AND PASSED A LIST OF TEXT	44
45	549	/	FILES TO PROCESS. THE RESULTANT BINARY FILE[S] CAN SUBSEQUENTLY BE LOADED	45
46	550	/	INTO MEMORY FOR EXECUTION. [NOTE: THE R-L "SLURP" BINARY LOADER LOADS	46
47	551	/	BINARY FILES WITHOUT WRITING TO THE SYSTEM DEVICE. PROTECTING THE SYSTEM	47
48	552	/	DEVICE IS HIGHLY RECOMMENDED ESPECIALLY DURING PROGRAM DEVELOPMENT.]	48
49	553			49
50	554	/	SEVERAL OTHER SYSTEM PROGRAMS ARE AVAILABLE FOR SYSTEM MAINTENANCE.	50
51	555	/	ADDITIONAL LANGUAGES ARE SUPPORTED INCLUDING AN ADAPTATION OF FOCAL, 1969	51
52	556	/	MODIFIED TO ACCEPT INPUT FROM THE CONTENTS OF MULTIPLE TFS FILES. [NOTE:	52
53	557	/	THE MAXIMUM NUMBER OF FILES THAT CAN BE PASSED TO MOST PROGRAMS IS 17. A	53
54	558	/	12-BIT COMMAND-LINE PARAMETER EXPRESSED AS A FOUR-DIGIT OCTAL VALUE CAN	54
55	559	/	OPTIONALLY BE LOADED INTO 07756 WHEN THE SYSTEM PROGRAM IS STARTED.]	55
56				56
57				57
58				58

1	561	/	R-L MONITOR LIMITATIONS.	1
2	562			2
3	563	/	WHILE THE R-L MONITOR WAS QUITE USEFUL FOR PROGRAM DEVELOPMENT AND EXECUTION,	3
4	564	/	THERE ARE MANY SEVERE LIMITATIONS THAT WERE EVENTUALLY OVERCOME. P?S/8 AS	4
5	565	/	DESCRIBED IN THIS DOCUMENT IS THE CULMINATION OF ALL OF THE IMPROVEMENTS MADE	5
6	566	/	OVER THE YEARS; CERTAIN CONCEPTS REMAIN UNCHANGED, AND IN SOME CASES THE	6
7	567	/	FUNCTIONS ARE FULLY BACKWARD-COMPATIBLE. MANY PERFORMANCE IMPROVEMENTS WERE	7
8	568	/	MADE INCREMENTALLY OVER THE COURSE OF DECADES.	8
9	569			9
10	570	/	DURING THE LONG PROCESS OF GRADUAL DEVELOPMENT, CERTAIN LEVELS OF ABSOLUTE	10
11	571	/	COMPATIBILITY WERE LOST, GENERALLY REPLACED BY EQUIVALENT OR BETTER SYSTEM	11
12	572	/	FUNCTIONS OR COMPLETE REPLACEMENT PROGRAMS.	12
13	573			13
14	574	/	WITHIN THIS DOCUMENT, TOPICS OF DISCUSSION WILL BE LIMITED TO THOSE ASSOCIATED	14
15	575	/	WITH THE TC01/TC08-ORIENTED COMPONENTS OF THE R-L MONITOR AND P?S/8. THE	15
16	576	/	LARGER ISSUES OF OVERALL DESIGN CHANGES, ADD-ONS AND IMPROVEMENTS IS BEYOND	16
17	577	/	THE SCOPE OF THIS DOCUMENT; HOWEVER, SEVERAL ISSUES WILL BE DISCUSSED IN	17
18	578	/	DETAIL.	18
19	579			19
20	580	/	R-L MONITOR PROGRAM EXECUTION.	20
21	581			21
22	582	/	EXECUTION OF CORE-IMAGE PROGRAMS UNDER THE R-L MONITOR IS NOT A	22
23	583	/	STRAIGHTFORWARD PROCESS. THE KEYBOARD MONITOR PARSES A RUN COMMAND FOR THE	23
24	584	/	DESIRED SYSTEM PROGRAM. [NOTE: ONLY THE FIRST TWO CHARACTERS OF THE PROGRAM	24
25	585	/	NAME ARE SIGNIFICANT; ANY ADDITIONAL CHARACTERS ARE IGNORED.] THE	25
26	586	/	CORRESPONDING PROGRAM IS LOCATED IN A SPECIAL-PURPOSE SYSTEM PROGRAM DIRECTORY	26
27	587	/	TOTALLY DISTINCT FROM THE TFS FILE DIRECTORY. APPROPRIATE ENTRIES ARE LOCATED	27
28	588	/	AND USED FOR THE LOADING PROCESS OF THE PROGRAM SUCH AS STARTING BLOCK, LOWEST	28
29	589	/	LOADING ADDRESS, NUMBER OF BLOCKS TO LOAD, AND THE PROGRAM STARTING ADDRESS.	29
30	590			30
31	591	/	THESE PARAMETERS CANNOT BE APPLIED TO THE SYSTEM DEVICE HANDLER, WHICH CAN	31
32	592	/	ONLY PERFORM THE THREE FUNCTIONS STATED ABOVE [READ, WRITE, AND RESTART AT	32
33	593	/	07600]. THE KEYBOARD MONITOR LOADS IN THE CONTENTS OF A DESIGNATED BLOCK	33
34	594	/	WHICH IS USED AS A DEDICATED LOADER OF THE SYSTEM PROGRAM IMAGE.	34
35	595			35
36	596	/	THE R-L SYSTEM LOADER IS NOT A GENERAL-PURPOSE DEVICE HANDLER. ITS SOLE	36
37	597	/	FUNCTION IS TO PERFORM A SINGLE READ OPERATION TO LOAD THE PROGRAM IMAGE INTO	37
38	598	/	MEMORY AND THEN START AT THE ADDRESS PASSED BY THE KEYBOARD MONITOR. ALL OF	38
39	599	/	THE PARAMETERS FROM THE SYSTEM PROGRAM DIRECTORY ARE PLUGGED INTO DESIGNATED	39
40	600	/	LOCATIONS WITHIN THE SYSTEM LOADER FOR EVENTUAL EXECUTION.	40
41				41
42				42
43				43
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1	602	/	THE KEYBOARD MONITOR PARSES FILE NAMES USED IN THE COMMAND-LINE; THESE ARE	1
2	603	/	TRANSLATED INTO A SERIES OF STARTING BLOCKS FOR EACH FILE, ALL OF WHICH ARE	2
3	604	/	PASSED IN A LIST STARTING AT 07757. IF THERE ARE NO FILES, 07757 IS LOADED	3
4	605	/	WITH A SENTINAL 0000. AS MANY AS 17 FILE HANDLES CAN BE PASSED IN THE	4
5	606	/	ADDRESSES 07757-07777. WITH THE EXCEPTION OF THE MAXIMUM SIZE LIST, THE END	5
6	607	/	OF THE LIST ALWAYS CONTAINS A WORD OF 0000 TO SIGNIFY THE END OF ALL INPUT	6
7	608	/	FILES. [NOTE: IT IS NOT NECESSARY TO HAVE A SENTINAL 0000 ON A FULL-LENGTH	7
8	609	/	LIST; THIS IS BECAUSE THE PROGRAMMING USED TO PROCESS THE LIST AUTOMATICALLY	8
9	610	/	ENDS SHOULD LOCATION 07777 BE REFERENCED. AN EXAMPLE OF FILE LIST HANDLING IS	9
10	611	/	AS FOLLOWS:	10
11	612			11
12	613	/	ISZ FILPTR /BUMP TO NEXT FILE	12
13	614	/	TAD I FILPTR /GET THE NEXT ENTRY	13
14	615	/	SNA CLA /SKIP IF ANOTHER FILE	14
15	616	/	JMP FILDONE /JUMP IF ALL FILES PROCESSED.	15
16	617			16
17	618	/	FOR ALL LESSER ENTRIES, THE SENTINEL TEST ENCOUNTERS 0000 AS THE END OF THE	17
18	619	/	LIST; FOR A MAXIMUM LENGTH LIST, THE "ISZ FILPTR" INSTRUCTION SKIPS THE NEXT	18
19	620	/	INSTRUCTION. THUS, THE LIST IS SELF-TERMINATING WHEN THE MAXIMUM FILE LIST IS	19
20	621	/	PASSED.	20
21	622			21
22	623	/	THE LIST IS NOT INITIALLY PLACED IN THE LOCATIONS INDICATED ABOVE. THE LIST	22
23	624	/	IS CREATED AT THE END OF THE SYSTEM LOADER IMAGE READ IN BY THE KEYBOARD	23
24	625	/	MONITOR. WHEN ALL PARAMETERS ARE PROCESSED, THE SYSTEM LOADER IMAGE IS MOVED	24
25	626	/	TO 07600-07777 AND STARTED AT THE INTERNAL ADDRESS OF THE SYSTEM LOADER TO	25
26	627	/	READ IN THE SYSTEM PROGRAM INTO AT MOST LOCATIONS 00000-07577.	26
27	628			27
28	629	/	ONCE THE SYSTEM LOADER HAS FINISHED ALL LOADING OPERATIONS, THE SYSTEM PROGRAM	28
29	630	/	IS STARTED WITH THE =XXXX VALUE [IF ANY] PASSED IN LOCATION 07756, THE LIST OF	29
30	631	/	FILES PASSED IN 07757-07777. [NOTE: THE SYSTEM LOADER ALSO INCLUDES A	30
31	632	/	TC01/TC08 BOOTSTRAP ROUTINE STARTING AT 07600.]	31
32	633			32
33	634	/	THERE IS A SEVERE PROBLEM AT THIS POINT AS THE SYSTEM DEVICE HANDLER IS NO	33
34	635	/	LONGER IN MEMORY. NO MEANS IS PROVIDED BY THE R-L MONITOR SYSTEM TO ALLOW THE	34
35	636	/	JUST-LOADED SYSTEM PROGRAM TO PERFORM ANY READ OR WRITE OPERATIONS. [THE	35
36	637	/	INPUT FILE HANDLES (JUST BARE BLOCK NUMBERS IN THE R-L MONITOR SYSTEM) ARE	36
37	638	/	AVAILABLE TO SETUP A READ OPERATION; HOWEVER, THERE IS NO MEANS PROVIDED TO	37
38	639	/	ACTUALLY IMPLEMENT THIS.	38
39				39
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1	641	/	TO WORK AROUND THIS, EVERY R-L SYSTEM PROGRAM MUST PROVIDE ITS OWN MEANS OF	1
2	642	/	ACCESSING THE SYSTEM DEVICE. THIS GENERALLY CONSISTS OF A GENERIC READ AND	2
3	643	/	WRITE SUBROUTINE PATTERNED AFTER THE SYSTEM DEVICE HANDLER; HOWEVER, THIS	3
4	644	/	IS AN INTERNAL SUBROUTINE WITHIN EACH SYSTEM PROGRAM. THERE ARE SEVERAL	4
5	645	/	ISSUES WITH THIS APPROACH:	5
6	646			6
7	647	/	1) ALL R-L SYSTEM PROGRAMS THAT REQUIRE ACCESS TO THE SYSTEM DEVICE	7
8	648	/	[PRESUMABLY TO READ TFS FILES AND/OR WRITE TO DESIGNATED BLOCKS ON THE	8
9	649	/	SYSTEM DEVICE SET ASIDE FOR WRITING OUT BINARY] MUST INCLUDE THIS CODE;	9
10	650	/	THIS CAUSES ALL SUCH PROGRAMS TO BE HARDWARE-DEPENDENT. IF THE R-L SYSTEM	10
11	651	/	DEVICE IS RECONFIGURED FOR AN ALTERNATE STORAGE DEVICE, ALL AFFECTED	11
12	652	/	PROGRAMS MUST PROVIDE AN APPROPRIATE SET OF ALTERNATE READ/WRITE	12
13	653	/	SUBROUTINES; SUFFICIENT CODE SPACE MUST BE SET ASIDE; CERTAIN STORAGE	13
14	654	/	DEVICES REQUIRE MORE CODE SPACE TO SUPPORT COMPARABLE READ/WRITE ROUTINES.	14
15	655			15
16	656	/	2) SINCE THE TYPICAL READ/WRITE SUBROUTINES USED WERE NEARLY AN ENTIRE PAGE OF	16
17	657	/	PDP-8 BINARY CODE, THE CONSTRUCTIVE MAXIMUM SIZE OF SYSTEM PROGRAMS IS NOT	17
18	658	/	THE ENTIRE RANGE OF 00000-07577; CERTAIN PROGRAMS WERE MODIFIED WHEN	18
19	659	/	ADAPTED TO THE R-L MONITOR SYSTEM TO LIMIT THE FORMER MEMORY RANGE	19
20	660	/	SUFFICIENTLY TO ALLOW THE INCLUSION OF THE INTERNAL SUBROUTINES DESPITE THE	20
21	661	/	ORIGINAL DESIGN EXPECTATIONS OF FULL ACCESS TO THE ENTIRE MEMORY RANGE OF	21
22	662	/	00000-07577; ONLY PROGRAMS HAVING NO EXPECTATION OF ACCESS TO THE SYSTEM	22
23	663	/	DEVICE CAN BE ACCOMMODATED WITHOUT COMPROMISE [SUCH AS THE DEC STANDARD	23
24	664	/	DECTAPE FORMATTING PROGRAM WHICH WAS SAVED AS AN IMAGE PROGRAM].	24
25	665			25
26	666	/	WHILE THERE ARE MANY OTHER SPECIFIC LIMITATIONS WITHIN THE R-L MONITOR SYSTEM,	26
27	667	/	MOST OF WHICH WERE EVENTUALLY OVERCOME DURING THE DEVELOPMENT OF P?S/8, THE	27
28	668	/	MOST IMPORTANT WAS ELIMINATION OF THE PROBLEM DESCRIBED ABOVE. AFTER MANY	28
29	669	/	YEARS OF SPORADIC DEVELOPMENT, P?S/8 EMERGED AS A DEVICE-INDEPENDENT SYSTEM;	29
30	670	/	ALL PROGRAMS WILL WORK ON ANY P?S/8 CONFIGURATION, AND NO MEMORY NEED BE	30
31	671	/	RESERVED FOR INTERNAL HARDWARE-SPECIFIC READ/WRITE SUBROUTINES.	31
32	672			32
33	673	/	A DISCUSSION OF SPECIFIC PERFORMANCE IMPROVEMENTS WILL BE FOUND LATER IN THIS	33
34	674	/	DOCUMENT REGARDING DETAILS OF THE P?S/8 SYSTEM DEVICE HANDLER. [NOTE: THE	34
35	675	/	READER IS REFERRED TO EXTERNAL DOCUMENTATION FOR THE MANY OTHER IMPROVEMENTS	35
36	676	/	FOUND IN P?S/8 THAT DO NOT PERTAIN TO THE SYSTEM DEVICE HANDLER OR RELATED	36
37	677	/	HARDWARE-SPECIFIC COMPONENTS.]	37
38	678			38
39	679	/	THE OTHER COMPONENTS OF P?S/8 SPECIFIC TO THE TC01/TC08 DECTAPE HARDWARE WILL	39
40	680	/	BE DISCUSSED BELOW. IN GENERAL, THEY ARE IMPROVEMENTS OVER THEIR COUNTERPARTS	40
41	681	/	IN THE R-L MONITOR; ADEQUATE DESCRIPTIONS WILL BE PROVIDED IN EACH SECTION TO	41
42	682	/	GIVE SOME INSIGHT AS TO HOW THE R-L MONITOR WAS GRADUALLY REPLACED WITH NEW	42
43	683	/	P?S/8 EQUIVALENTS.	43
44				44
45				45
46				46
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1	685	/	SECTION 1: THE P?S/8 SYSTEM DEVICE HANDLER.	1
2	686			2
3	687	/	P?S/8 SYSTEM DEVICE HANDLER CALLING SEQUENCE.	3
4	688			4
5	689	/	CDF MYFLD /SET THE DATA FIELD TO THE CALLING FIELD.	5
6	690	/	CIF 00 /THE HANDLER IS IN FIELD 0.	6
7	691	/	JMS I (SYSIO)/(7640) /CALL THE HANDLER IN 07640.	7
8	692	/	BUFFER /12-BIT READ/WRITE BUFFER ADDRESS.	8
9	693	/	FUNCTION /FUNCTION WORD; SEE BELOW.	9
10	694	/	BLOCK /STARTING LOGICAL BLOCK NUMBER. [NOTE: ALL	10
11	695		/LOGICAL BLOCKS ARE 128 WORDS EACH.]	11
12	696			12
13	697	/	THE CALL ALWAYS RETURNS HERE WITH THE DATA FIELD SET TO THE CALLING FIELD, THE	13
14	698	/	AC CLEAR AND THE LINK UNDEFINED. [NOTE: THE TC01/TC08 SYSTEM DEVICE HANDLER	14
15	699	/	RETURNS WITH THE LINK CLEAR AS RECOMMENDED.]	15
16	700			16
17	701	/	FUNCTION WORD BREAKDOWN.	17
18	702			18
19	703	/	BIT[0]: READ=0, WRITE=1.	19
20	704			20
21	705	/	BITS[1-5]: BLOCK/PAGE COUNT 1 THROUGH 31, 00000=32 PAGES [AN ENTIRE MEMORY	21
22	706	/	FIELD]. NOTE: IF A TRANSFER CROSSES 7777 IN ANY MEMORY FIELD,	22
23	707	/	THE TRANSFER GENERALLY CONTINUES IN 0000 OF THE SAME MEMORY	23
24	708	/	FIELD, NOT THE NEXT HIGHER FIELD. HOWEVER, THIS BEHAVIOR IS	24
25	709	/	HARDWARE-DEPENDENT AND NOT TO BE ATTEMPTED OUTSIDE OF SPECIFIC	25
26	710	/	WELL-DOCUMENTED CIRCUMSTANCES. [NOTE: THE TC01/TC08 SYSTEM	26
27	711	/	DEVICE HANDLER PERFORMS THIS SPECIFIC TRANSFER INTERNALLY	27
28	712	/	DURING THE NORMAL BOOTUP PROCESS TO OPTIMIZE PERFORMANCE.]	28
29	713			29
30	714	/	BITS[6-8]: READ/WRITE BUFFER MEMORY FIELD EXTENSION.	30
31	715			31
32	716	/	BITS[9-11]: THE LOGICAL UNIT FOR THE CALL IN THE RANGE OF 0-7 [SEE BELOW].	32
33	717			33
34	718	/	NOTE: ALL CALLS TO THE SYSTEM DEVICE HANDLER HAVE NO SPECIAL CONSIDERATIONS	34
35	719	/	REGARDING THE AC, LINK OR INTERRUPT STATE. CERTAIN DEVICE HANDLERS FOR	35
36	720	/	OMNIBUS DEVICES, SUCH AS THE TD8E, WILL INHIBIT INTERRUPTS WITHIN THE HANDLER;	36
37	721	/	THE PREVIOUS INTERRUPT STATE WILL BE RESTORED UPON EXIT. THE SYSTEM DEVICE	37
38	722	/	MUST NOT GENERATE INTERRUPTS, GENERALLY ACHIEVED BY CLEARING INTERRUPT-ENABLE	38
39	723	/	WITHIN THE DEVICE; THIS ALLOWS DEVICE FLAGS TO REMAIN RAISED IF NECESSARY	39
40	724	/	[AS IS IMPLEMENTED FOR THE RX DISKETTE CONTROLLERS]. INTERRUPT-DRIVEN	40
41	725	/	PROGRAMS SHOULD INCLUDE CUSTOM PATCHING SECTIONS TO HANDLE STRAY INTERRUPTS.	41
42	726			42
43	727	/	THE TC01/TC08 CONTROLLER CAN BE CONNECTED TO THE EARLIEST COMPATIBLE SYSTEMS	43
44	728	/	SUCH AS THE CLASSIC PDP-8 COMPUTER; NO MECHANISM TO LIMIT INTERRUPTS EXISTS	44
45	729	/	WITHIN THIS DEVICE HANDLER. AS LONG AS WORST-CASE INTERRUPT-HANDLING LATENCY	45
46	730	/	IS WITHIN REASON, DECTAPE PERFORMANCE SHOULD NOT BE COMPROMISED WHILE ALSO	46
47	731	/	MAINTAINING GOOD OVERLAP WITH OTHER DEVICES [SUCH AS THE SYSTEM CONSOLE].	47
48	732			48
49	733	/	DECTAPES ARE BASED ON THE EARLIER LINCTAPES AS USED IN VARIOUS SYSTEMS.	49
50	734	/	APPENDIX "B" CONTAINS ADDITIONAL DETAILS ABOUT VARIOUS LINCTAPE AND DECTAPE	50
51	735	/	ISSUES INCLUDING IMPLEMENTATION DETAILS.	51
52				52
53				53
54				54
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56				56
57				57
58				58

1	737	/	LOGICAL UNIT CONSIDERATIONS.	1
2	738			2
3	739	/	THE ASSIGNMENT OF LOGICAL UNITS VERSUS PHYSICAL MEDIA IS HARDWARE-DEPENDENT.	3
4	740	/	ON THE TC01/TC08 DECTAPE CONTROLLER THIS REFERS TO THE PHYSICAL TU55/TU56	4
5	741	/	DRIVE UNITS ATTACHED TO THE CONTROLLER. THE ACTUAL UNIT ASSIGNMENT IS	5
6	742	/	USER-CONTROLLED BY SETTING DRIVE-RESIDENT ADJUSTABLE SELECT SWITCHES. SIMILAR	6
7	743	/	CONSIDERATIONS APPLY TO ALL DECTAPE AND LINCTAPE SYSTEMS AS WELL AS CERTAIN	7
8	744	/	DISKETTE SYSTEMS; THE ABILITY TO CHANGE DRIVE SELECTION VARIES WITH THE EXACT	8
9	745	/	CONTROLLER HARDWARE. FOR CERTAIN HARDWARE CONFIGURATIONS, LOGICAL DEVICE UNITS	9
10	746	/	MAY [PARTIALLY] BE IMPLEMENTED BY SOME FORM OF DRIVE MAPPING SCHEME.	10
11	747			11
12	748	/	APPENDIX "C" CONTAINS ADDITIONAL INFORMATION ABOUT THE P?S/8 IMPLEMENTATION OF	12
13	749	/	LOGICAL UNITS ON VARIOUS SUPPORTED DEVICES.	13
14	750			14
15	751	/	LOGICAL BLOCKSIZE CONSIDERATIONS.	15
16	752			16
17	753	/	REGARDLESS OF UNDERLYING HARDWARE, ALL P?S/8 SUPPORTED STORAGE DEVICES	17
18	754	/	ARE TREATED AS AN ORDERED SERIES OF LOGICAL BLOCKS, EACH 128 12-BIT	18
19	755	/	WORDS LONG. THIS SIZE WAS CHOSEN AS A GOOD COMPROMISE BETWEEN PHYSICAL	19
20	756	/	SIZE AND LOGICAL SIZE GIVEN THE MANY ISSUES THAT APPLY.	20
21	757			21
22	758	/	A LOGICAL BLOCKSIZE THAT MATCHES THE EXACT SIZE OF A SINGLE PDP-8 PAGE	22
23	759	/	OF MEMORY IS A CONVENIENT UNIT ALLOWING A SIMPLE CORRESPONDANCE OF	23
24	760	/	IMAGES OF MEMORY TO STORAGE BLOCKS WITHOUT REQUIRING ANY FORM OF MAPPING	24
25	761	/	BE PERFORMED.	25
26	762			26
27	763	/	ONE IMPORTANT CONSIDERATION REGARDING BLOCKSIZE IS THE REQUIREMENT FOR	27
28	764	/	DEDICATED BUFFERS WHEN ACCESSING THE BLOCKS. SHOULD THE BLOCKSIZE BE	28
29	765	/	LARGER, THE MINIMUM BUFFER WOULD ALSO NECESSARILY BECOME LARGER. FOR AN	29
30	766	/	OPERATING SYSTEM THAT REQUIRES AS LITTLE AS 32 PAGES OF MEMORY, IF THE	30
31	767	/	BLOCKSIZE IS MADE LARGER, CERTAIN PROGRAMS WOULD BECOME IMPOSSIBLE TO	31
32	768	/	IMPLEMENT. [NOTE: CERTAIN P?S/8 SYSTEM PROGRAMS CHECK FOR EXTENDED	32
33	769	/	MEMORY; IF AVAILABLE, LARGER BUFFERS WILL BE ALLOCATED. BY ALLOWING THE	33
34	770	/	MINIMUM BLOCKSIZE OF 128 WORDS, ANY MULTIPLE BUFFERS AVAILABLE CAN BE	34
35	771	/	ALLOCATED UP TO A MAXIMUM OF 32, THE UPPER LIMIT OF A PDP-8 MEMORY FIELD	35
36	772	/	WHICH IS GENERALLY A HARDWARE CONSTRAINT OF MOST PDP-8 DMA DEVICES.]	36
37	773			37
38	774	/	APPENDIX "D" CONTAINS ADDITIONAL INFORMATION REGARDING LOGICAL BLOCKSIZE	38
39	775	/	CONSIDERATIONS, ESPECIALLY AS IT PERTAINS TO VARIOUS SUPPORTED DEVICES.	39
40	776			40
41	777	/	ADDITIONAL DECTAPE PERFORMANCE CONSIDERATIONS.	41
42	778			42
43	779	/	WHILE P?S/8 DEVICE HANDLERS FOR DECTAPE ACHIEVE A REASONABLE RESULT CARRYING	43
44	780	/	OUT THE VARIOUS READ, WRITE AND BOOTUP OPERATIONS REQUIRED, THERE ARE	44
45	781	/	ADDITIONAL ISSUES THAT PERTAIN EXCLUSIVELY TO DECTAPES [AND LINCTAPES]. DUE	45
46	782	/	TO THE DYNAMICS OF THE DEVICE, THERE ARE OTHER POTENTIAL ACCESS SCHEMES THAT	46
47	783	/	MIGHT BE BETTER SUITED [BUT ONLY TO CERTAIN APPLICATIONS].	47
48	784			48
49	785	/	THE DESIGN OF P?S/8 SYSTEM DEVICE HANDLERS INCLUDED CONSIDERATION OF SOME OF	49
50	786	/	THESE ISSUES, EVEN IF NOT ACTUALLY USED. WITH THE ARRIVAL OF THE P?S/8 SHELL,	50
51	787	/	SOME OF THESE ISSUES MAY BE REVISITED FOR NON-SYSTEM HANDLERS.	51
52	788			52
53	789	/	APPENDIX "E" CONTAINS ADDITIONAL INFORMATION REGARDING MANY OF THE DESIGN	53
54	790	/	CONSIDERATIONS THAT GO INTO PRODUCING A PRACTICAL HANDLER [INCLUDING SOME OF	54
55	791	/	THE SPECIFIC ISSUES THAT COULD BE APPLIED IN THE FUTURE].	55
56				56
57				57
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1	793	/	P?S/8 DEVICE HANDLER ERROR ISSUES.	1
2	794			2
3	795	/	BY DESIGN, EVERY CALL TO THE P?S/8 SYSTEM DEVICE HANDLER IS MADE WITH AN	3
4	796	/	EXPECTATION OF SUCCESS; THERE IS NO MECHANISM FOR THE CALLING PROGRAM TO	4
5	797	/	INTERACT WITH POTENTIAL ERRORS WHICH ARE ALWAYS DEVICE-DEPENDENT. BY	5
6	798	/	ISOLATING INTERNAL ERROR HANDLING FROM THE CALLING PROGRAM, PROGRAMS CAN BE	6
7	799	/	WRITTEN IN A DEVICE-INDEPENDENT MANNER.	7
8	800			8
9	801	/	ACTUAL ERROR HANDLING IS DEPENDENT ON THE OVERALL SYSTEM CONFIGURATION; THE	9
10	802	/	LOGICAL CONSOLE OVERLAY INTERCEPTS ALL SYSTEM DEVICE ERRORS PRESENTING THE	10
11	803	/	USER WITH SEVERAL ALTERNATIVES REGARDING THE DETECTED ERROR. IN EVERY	11
12	804	/	CONFIGURATION THERE IS ALWAYS A MECHANISM AVAILABLE TO RETRY THE LATEST	12
13	805	/	OPERATION.	13
14	806			14
15	807	/	MOST REAL-WORLD PROBLEMS WITH STORAGE DEVICES ARE CAUSED BY FAILING TO ALLOW	15
16	808	/	THE DEVICE TO FUNCTION PROPERLY, SUCH AS HAVING THE DEVICE OFF-LINE, OR THE	16
17	809	/	DEVICE IS WRITE-PROTECTED [IF APPLICABLE]; CORRECTING THE CONDITION WILL ALLOW	17
18	810	/	THE LATEST OPERATION TO SUCCEED ON THE NEXT RETRY. FOR CERTAIN DEVICES, THERE	18
19	811	/	IS A SMALL POSSIBILITY OF A SOFT ERROR SITUATION, GENERALLY CLEARED BY	19
20	812	/	RETRYING THE OPERATION. IF ERRORS ARE OCCURRING OFTEN, THIS IS LIKELY A SIGN	20
21	813	/	OF HARDWARE FAILURE REQUIRING MAINTENANCE BEYOND THE CONTROL OF HANDLER	21
22	814	/	SOFTWARE. THE TC01/TC08 DECTAPE CONTROLLER AND DECTAPE DRIVES ARE SUBJECT TO	22
23	815	/	ALL OF THESE POTENTIAL ISSUES.	23
24	816			24
25	817	/	THE P?S/8 SYSTEM LOADER [AN IMBEDDED CALL TO THE SYSTEM DEVICE HANDLER LOCATED	25
26	818	/	WITHIN THE RESIDENT SYSTEM KERNEL] AVOIDS ERROR HANDLING COMPLICATIONS BY THIS	26
27	819	/	DESIGN PHILOSOPHY. APPENDIX "F" DISCUSSES P?S/8 SYSTEM DEVICE HANDLER ERROR	27
28	820	/	HANDLING IN MORE DETAIL, INCLUDING COMPARISON WITH OTHER OPERATING SYSTEMS	28
29	821	/	THAT HAVE CONCEPTUAL FLAWS IN CORRESPONDING INTERNAL SITUATIONS.	29
30	822			30
31	823	/	P?S/8 CODE OPTIMIZATION TECHNIQUES.	31
32	824			32
33	825	/	THE P?S/8 SYSTEM DEVICE HANDLER DESCRIBED IN THIS DOCUMENT IS THE PRODUCT OF	33
34	826	/	TREMENDOUS EFFORT ON THE PART OF A RELATIVELY SMALL GROUP OF INDIVIDUALS.	34
35	827	/	DESPITE THE OBJECTIVE OF CREATING A FUNCTIONAL DEVICE HANDLER THAT FITS IN	35
36	828	/	AT MOST 128 WORDS, THE EFFORT INVOLVED EXCEEDS THAT OF MOST OTHER PDP-8	36
37	829	/	SOFTWARE PROJECTS OVER 100 TIMES AS LARGE. MOST OF THIS EFFORT CONSISTS OF	37
38	830	/	APPLYING VARIOUS OPTIMIZATION TECHNIQUES; ONLY A FEW PDP-8 EXPERTS ARE	38
39	831	/	FAMILIAR WITH ALL OF THE "TRICKS" ALLOWED [AS WELL AS CERTAIN OTHER TECHNIQUES	39
40	832	/	NOT ALLOWED]. HAD THIS EFFORT NOT BEEN APPLIED, P?S/8 WOULD STILL LARGELY	40
41	833	/	RESEMBLE THE ORIGINAL R-L MONITOR SYSTEM INTERNALLY [WITH ALL THE ATTENDANT	41
42	834	/	DRAWBACKS]; MOST OF THE OVERALL IMPROVEMENTS OF P?S/8 OVER THE R-L SYSTEM	42
43	835	/	DEPEND ON THE PROGRESS MADE WITHIN THE SYSTEM DEVICE HANDLER.	43
44	836			44
45	837	/	A COMPLETE DISCUSSION OF ALL RELEVANT TECHNIQUES BOTH ACTUAL AND THEORETICAL	45
46	838	/	IS AVAILABLE WITHIN APPENDIX "G". WHERE RELEVANT, CERTAIN DISCUSSIONS	46
47	839	/	REGARDING TENTATIVE CONTENTS OF SPECIFIC MEMORY LOCATIONS WITHIN THE SYSTEM	47
48	840	/	DEVICE HANDLER ARE RETAINED FOR FUTURE REFERENCE. [NOTE: ANY PRACTICAL	48
49	841	/	HANDLER IS ALWAYS THE PRODUCT OF VARIOUS TRADE-OFFS OF POTENTIAL OPTIMIZATION	49
50	842	/	TECHNIQUES AS CERTAIN METHODS ARE BY NATURE IN DIRECT CONFLICT WITH EACH	50
51	843	/	OTHER.]	51
52				52
53				53
54				54
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1	845	/	P?S/8 SYSTEM GENERATION CONSIDERATIONS.	1
2	846			2
3	847	/	SINCE P?S/8 IS A COMPLETE AND INDEPENDENT OPERATING SYSTEM, MEANS MUST BE	3
4	848	/	AVAILABLE TO CREATE AN ENTIRE SYSTEM FROM SYSTEM COMPONENTS LOADED INTO MEMORY	4
5	849	/	BY ANOTHER OPERATING SYSTEM.	5
6	850			6
7	851	/	APPENDIX "H" DISCUSSES DETAILS ABOUT THE SYSTEM GENERATION PROCESS AND VARIOUS	7
8	852	/	METHODS USED TO ACCOMPLISH THIS. SINCE THERE ARE SEVERAL POTENTIAL WAYS TO	8
9	853	/	GO ABOUT THIS [NOT ALL OF WHICH HAVE BEEN IMPLEMENTED TO DATE], A COMPLETE	9
10	854	/	DISCUSSION IS AVAILABLE TO GIVE SOME INSIGHT ON THIS IMPORTANT PROCESS.	10
11	855	/	HISTORIC METHODS ARE DETAILED AS WELL AS SOME TENTATIVE FUTURE METHODS TO	11
12	856	/	ALLOW SYSTEM GENERATION BY USERS INSTEAD OF SYSTEM GURUS. CERTAIN AUXILIARY	12
13	857	/	ISSUES ARE DISCUSSED THAT HAVE AFFECTED THE PERFORMANCE OF OTHER OPERATING	13
14	858	/	SYSTEMS [IN SOME CASES CAUSING NEEDLESS OVERHEAD DUE TO EITHER INADEQUATE	14
15	859	/	PLANNING OR IMPLEMENTATION SHORTCOMINGS]. IN ONE PARTICULAR CASE, REPURPOSING	15
16	860	/	AN OBSCURE FEATURE OF THE PAL LANGUAGE MAY LEAD TO AN ELEGANT GENERATION	16
17	861	/	METHOD IN FUTURE SYSTEM BUILDING UTILITIES.	17
18	862			18
19	863	/	ADDITIONAL SYSTEM COMPONENTS.	19
20	864			20
21	865	/	APART FROM SYSTEM PROGRAMS AND OTHER UTILITIES, CERTAIN SYSTEM COMPONENTS ARE	21
22	866	/	ESSENTIAL TO P?S/8 DEPENDING ON THE CONFIGURATION. WHILE NOT DISCUSSED IN	22
23	867	/	DETAIL IN THIS DOCUMENT, CERTAIN REFERENCES TO THE FOLLOWING WILL BE MADE:	23
24	868			24
25	869	/	1) SYSTEM HANDLER DEVICE EXTENSION. WHILE NOT RELEVANT TO THE TC01/TC08	25
26	870	/	CONFIGURATION, CERTAIN P?S/8 SYSTEM DEVICE HANDLERS REQUIRE ANOTHER SECTION	26
27	871	/	OF CODE TO BE LOADED INTO THE LAST QUARTER OF THE HIGHEST MEMORY FIELD.	27
28	872			28
29	873	/	2) LOGICAL CONSOLE OVERLAY. ALL P?S/8 SYSTEMS NEWER THAN VERY EARLY RELEASES	29
30	874	/	SUPPORT AN ALTERNATE CONSOLE AND LINEPRINTER INTERFACE TO ALLOW LOGICAL	30
31	875	/	CONFIGURATIONS INSTEAD OF THE HARD-WIRED DEVICE 03/04 CONSOLE AND DEVICE 66	31
32	876	/	LINEPRINTER DEVICES. ALL SYSTEM DEVICES, INCLUDING THE TC01/TC08	32
33	877	/	CONFIGURATION, INTERFACE WITH THE LOGICAL CONSOLE OVERLAY FOR SPECIFIC	33
34	878	/	PURPOSES [HANDLER PERFORMANCE ENHANCEMENT, ENHANCED ERROR RECOVERY].	34
35	879			35
36	880	/	3) P?S/8 SHELL OVERLAY FILE AND COMMUNICATIONS AREA. THIS IS PART OF THE	36
37	881	/	OPTIONAL SHELL OVERLAY CURRENTLY UNDER DEVELOPMENT. ALL SYSTEM DEVICE	37
38	882	/	HANDLERS MUST INCLUDE CERTAIN STATUS INFORMATION REGARDING THE DYNAMICS	38
39	883	/	OF THIS FACILITY.	39
40	884			40
41	885	/	IMAGE OF BLOCK 0000 FOR THE P?S/8 TC01/TC08 SYSTEM.	41
42	886			42
43	887	/	A COMPLETE IMAGE OF THE SYSTEM DEVICE HANDLER IS DEFINED BELOW; IT IS INTENDED	43
44	888	/	TO BE WRITTEN OUT AS THE CONTENTS OF BLOCK 0000 OF THE SYSTEM DEVICE.	44
45	889			45
46	890	/	THE P?S/8 SYSTEM DEVICE HANDLER PROVIDES AN EXIT ROUTINE AT 07600 TO RELOAD	46
47	891	/	THE OPERATING SYSTEM. ANY PROGRAM CAN TAKE THIS STANDARD EXIT EVEN IF NOT	47
48	892	/	EXPRESSLY WRITTEN FOR P?S/8; MANUAL RESTART AT 07600 IS ALSO SUPPORTED.	48
49	893			49
50	894	/	A FULLY DEVICE-INDEPENDENT LOGICAL READ/WRITE INTERFACE IS PROVIDED TO GIVE	50
51	895	/	ACCESS TO THE SYSTEM DEVICE; ALL P?S/8 SYSTEMS USE THE IDENTICAL CALLING	51
52	896	/	CONVENTION REGARDLESS OF UNDERLYING SYSTEM DEVICE HARDWARE. SUBROUTINE CALLS	52
53	897	/	CAN BE MADE BY ANY P?S/8 SYSTEM UTILITY OR FULLY COMPLIANT USER PROGRAM.	53
54				54
55				55
56				56
57				57
58				58

1	899	/			A LOADING TEMPLATE IS PROVIDED TO FINALIZE ALL EXECUTION-CLASS COMMANDS THAT	1
2	900	/			LOAD OVER THE OPERATING SYSTEM [OR SHELL PROGRAM]. THIS INTERNAL CALL	2
3	901	/			MECHANISM ALLOWS ALL USER MEMORY TO BE AVAILABLE TO ANY EXECUTABLE PROGRAM	3
4	902	/			WITHOUT RESTRICTION; IT IS ALSO USED TO IMPLEMENT OVERLAY STRUCTURES WITHIN	4
5	903	/			P?S/8 SYSTEM PROGRAMS AS REQUIRED.	5
6	904					6
7	905	/			[NOTE: FOR HISTORICAL PURPOSES, THE INTERNAL CALL IS REFERRED TO AS THE	7
8	906	/			"SYSTEM LOADER". THE ORIGINAL R-L MONITOR SYSTEM CONTAINS A VAGUELY ANALOGOUS	8
9	907	/			ROUTINE THAT WAS IMPLEMENTED IN A VERY AWKWARD MANNER WITH MANY SERIOUS	9
10	908	/			LIMITATIONS. CERTAIN READERS OF THIS DOCUMENT ARE MORE FAMILIAR WITH THE R-L	10
11	909	/			MONITOR SYSTEM AND ITS ORIGINAL EXECUTABLE PROGRAM LOADING MECHANISM; AS SUCH,	11
12	910	/			THE SIGNIFICANT DIFFERENCES WILL BE DETAILED ELSEWHERE IN THIS DOCUMENT.]	12
13	911					13
14	912	/			NUMEROUS RESERVED WORDS EXIST FOR A VARIETY OF SYSTEM PARAMETERS USED BY	14
15	913	/			VARIOUS PORTIONS OF THE OPERATING SYSTEM AND ITS UTILITIES. MOST NOTABLY,	15
16	914	/			THIS INCLUDES THE ABILITY TO PASS INPUT AND OUTPUT FILES TO EXECUTABLE	16
17	915	/			PROGRAMS, OPTION SWITCHES, AN =XXXX PARAMETER [WHERE XXXX IS ANY FOUR-DIGIT	17
18	916	/			OCTAL VALUE], ACCESS TO THE CURRENT DATE AND VARIOUS STATUS WORDS AFFECTING	18
19	917	/			THE PRESENCE OR STATUS OF CERTAIN OPTIONAL SYSTEM COMPONENTS. [NOTE: SOME OF	19
20	918	/			THESE VALUES ARE DEPENDENT ON OPTIONAL COMPONENTS OF THE SYSTEM NOT YET	20
21	919	/			[RE-]WRITTEN, BUT VIRTUALLY ALL OF THE RESERVED WORDS HAVE BEEN DEFINED FOR	21
22	920	/			OVER THREE DECADES WHILE VARIOUS COMPONENTS WERE IN THE PROCESS OF	22
23	921	/			DEVELOPMENT].	23
24	922					24
25	923	/			LOADING CONSIDERATIONS FOR SYSTEM GENERATION.	25
26	924					26
27	925	/			THE IMAGE OF BLOCK ZERO MUST BE WRITTEN OUT BY MEANS DEFINED ELSEWHERE. THIS	27
28	926	/			CANNOT BE ACCOMPLISHED DIRECTLY FROM THE CODE DEFINED BELOW; THE EXACT	28
29	927	/			MECHANISM USED FOR SYSTEM GENERATION MUST BE LOADED BY SOME MEANS EXTERNAL TO	29
30	928	/			THIS IMAGE; IN ALL LIKELIHOOD THIS WILL BE SOME FORM OF DEVICE HANDLER THAT	30
31	929	/			COULD BE IN CONFLICT WITH THIS CODE.	31
32	930					32
33	931	/			AS SUCH, THIS IMAGE WILL BE LOADED INTO AN ALTERNATE MEMORY BUFFER MADE	33
34	932	/			AVAILABLE DURING THE SYSTEM GENERATION PROCESS; HOWEVER, THE CODE WILL BE	34
35	933	/			DEFINED ACCORDING TO ITS NORMAL ADDRESS [AND FIELD] CONSIDERATIONS.	35
36	934					36
37	935	0001	FIELD	SDHFLD%10	/BINARY LOADING FIELD FOR SYSTEM IMAGE.	37
38		*0200				38
39	936					39
40	937		NOPUNCH		/FOOL THE ASSEMBLER.	40
41	938					41
42	939	0000	FIELD	0	/WHERE THIS LOADS WHEN EXECUTED.	42
43		*0200				43
44	940					44
45	941		ENPUNCH		/UN-FOOL THE ASSEMBLER.	45
46	942					46
47	943	*7400	*SDHADR		/BINARY LOADING ADDRESS FOR SYSTEM IMAGE.	47
48	944					48
49	945	*7600	RELOC	SBOOT	/WHERE THIS EVENTUALLY EXECUTES AFTER BOOTUP.	49
50						50
51						51
52						52
53						53
54						54
55						55
56						56
57						57
58						58

1	947	/	THIS IS THE SYSTEM REBOOT ADDRESS. WHEN ANY PROGRAM EXITS HERE THE DATA FIELD			1
2	948	/	MUST BE 0; THE CONTENTS OF THE AC AND LINK ARE IGNORED. NOTE: INTERRUPTS			2
3	949	/	SHOULD BE TURNED OFF BECAUSE THE CONTENTS OF 00000-07577 WILL BE LOADED WITH			3
4	950	/	P?S/8 KEYBOARD MONITOR/EDITOR CODE [AND DESIGNATED DATA BUFFERS] WHICH RUNS			4
5	951	/	WITHOUT INTERRUPTS [THUS OVERLAYING ANY POTENTIAL INTERRUPT HANDLER].			5
6	952					6
7	953	07600* 4240	SBOOT,	JMS	SYSIO	7
8	954	07601* 7600	S7600,	SB00T		8
9	955		/BOOT BY CALLING THE HANDLER TO SELF-OVERLAY.			9
10	956	07602* 0200	S200,	2^100		10
11	957		/CONSTANT 0200; ALSO USEFUL FUNCTION WORD			11
12	958	07603* 0000		SYSBLK		12
13	959		/VALUE TO CAUSE SELF-OVERLAY.			13
14	960		/CONSTANT 0000; PROPER BLOCK NUMBER FOR			14
15	961	/	/SELF-OVERLAY; MUST BE 0000 WHEN BOOTING UP.			15
16	962	/	ALTHOUGH SEEMINGLY A SUBROUTINE CALL, THERE IS NO ACTUAL RETURN TO THE CALLER.			16
17	963	/	ONCE INITIATED, DATA BREAK TRANSFERS OVERLAY MEMORY; KEY CONTROLLING ADDRESSES			17
18	964	/	AND OTHER PARAMETERS READ IN FROM BLOCK 0000 TAKE CONTROL OF THE OVERALL			18
19	965	/	PROCESS IN MEMORY. THE ACTUAL TRANSFER IS 07600-03577 [USING ADDRESS			19
20	966	/	WRAP-AROUND KNOWN TO CONTINUE IN FIELD 0]. DESPITE THE READ ARGUMENT 0200 [A			20
21	967	/	CONSTANT NEEDED ELSEWHERE], THE EFFECTIVE FUNCTION ARGUMENT IS TO READ 20			21
22	968	/	[16 DECIMAL] BLOCKS INTO MEMORY. THE NET EFFECT IS TO READ BLOCKS 0000-0017			22
23	969	/	INTO MEMORY WHICH IS THE INITIAL STEP TOWARDS LOADING THE KEYBOARD MONITOR AND			23
24	970	/	BUILT-IN LINE EDITOR. [NOTE: FINALIZATION IS PERFORMED WITHIN THE KEYBOARD			24
25	971	/	MONITOR ITSELF.]			25
26	972	/	ONCE-ONLY CODE WITHIN THE KEYBOARD MONITOR [USUALLY RESIDING IN THE MEMORY			26
27	973	/	LATER USED AS THE COMMAND-LINE BUFFER FOR THE KEYBOARD MONITOR] MOVES THE			27
28	974	/	CONTENTS OF BLOCKS 0015-0017 FROM 03000-03577 TO THE INTENDED ADDRESS RANGE OF			28
29	975	/	07000-07577. ONCE INITIALIZED, THE MEMORY LAYOUT IS AS FOLLOWS:			29
30	976					30
31	977	/	BLOCK[S]	MEMORY	FUNCTION	31
32	978	/	0000	07600-07777	THE SYSTEM DEVICE HANDLER	32
33	979	/	0001-0014	00000-02777	THE RESIDENT KEYBOARD MONITOR AND LINE EDITOR	33
34	980	/	N/A	03000-06777	INTERNAL FILE BUFFER FOR THE LINE EDITOR	34
35	981	/	0015-0016	07000-07377	CACHED COPY OF THE CURRENT UNIT TFS DIRECTORY	35
36	982	/	0017	07400-07577	THE TRANSIENT AREA OF THE KEYBOARD MONITOR	36
37	983	/	TRANSIENT AREA DESCRIPTION.			37
38	984	/	THE TRANSIENT AREA IS USED FOR A VARIETY OF FUNCTIONS USUALLY AVAILABLE WHILE			38
39	985	/	P?S/8 BATCH IS NOT RUNNING. DURING BATCH OPERATIONS, THE TRANSIENT AREA IS			39
40	986	/	USED AS AN INPUT FILE BUFFER FOR ALL KEYBOARD MONITOR AND LINE EDITOR			40
41	987	/	COMMANDS; AS SUCH, SEVERAL COMMANDS ARE TEMPORARILY INHIBITED DURING BATCH			41
42	988	/	OPERATION.			42
43	989	/	SINCE BATCH CANNOT PASS CONTROL CHARACTERS BACK TO THE KEYBOARD MONITOR, A FEW			43
44	990	/	INTERNAL FEATURES ARE NOT AVAILABLE DURING BATCH OPERATIONS. AS SUCH, BATCH			44
45	991	/	OVERLAYS THE CODE SPACE NORMALLY OCCUPIED BY THESE ROUTINES. A SMALL CHANGE			45
46	992	/	IS MADE TO THE KEYBOARD MONITOR TO ACCEPT COMMANDS DERIVED FROM THE INPUT FILE			46
47	993	/	BUFFER CONTENTS [UNPACKED BY BATCH] TO ALLOW MOST NORMAL FUNCTIONALITY.			47
48	994	/				48
49	995	/				49
50	996	/				50
51	997	/				51
52	998	/				52
53	999	/				53
54	1000	/				54
55	1001	/				55
56						56
57						57
58						58

1	1003	/	A NOTABLE TEMPORARILY SUPPRESSED COMMAND REQUIRING CONTROL CHARACTER INPUT IS	1
2	1004	/	A SINGLE-LINE MINI-EDITOR FULLY COMPATIBLE WITH FOCAL, 1969 [WITH SEVERAL	2
3	1005	/	ADDED EXTENSIONS UNIQUE TO THE P?S/8 EDITING ENVIRONMENT]. [NOTE: DEPENDING	3
4	1006	/	ON P?S/8 REVISION, THIS COMMAND IS KNOWN AS EITHER THE "EDIT" OR "MODIFY"	4
5	1007	/	COMMAND.] AUTOMATED FILE EDITING OF COMPLETE LINES OR LINE GROUPS CAN BE	5
6	1008	/	PERFORMED FROM BATCH USING THE ADDITIONAL EDITOR COMMANDS AVAILABLE WITHOUT	6
7	1009	/	RESTRICTION. BY CLEVER DEPLOYMENT OF LINE EDITOR AND KEYBOARD MONITOR	7
8	1010	/	COMMANDS, A BATCH FILE STREAM CAN SELF-MODIFY TO PRODUCE A VARIETY OF	8
9	1011	/	INTERESTING RESULTS.	9
10	1012			10
11	1013	/	CERTAIN SYSTEM UTILITIES [FOR EXAMPLE SEARCH AND REPLACE OPERATIONS] ARE	11
12	1014	/	WRITTEN AS AUXILIARY FUNCTIONS OF THE KEYBOARD MONITOR AND WILL LOAD INTO THE	12
13	1015	/	TRANSIENT AREA AS REQUIRED VIA EXECUTION-CLASS COMMANDS. THE KEYBOARD MONITOR	13
14	1016	/	PROVIDES A SPECIAL EXIT FOR TRANSIENT OVERLAYS USUALLY TAKEN TO RELOAD THE	14
15	1017	/	NORMALLY RESIDENT CODE. HOWEVER, THIS DOES NOT DETER BATCH OPERATIONS AS ALL	15
16	1018	/	EXECUTION-CLASS COMMANDS ARE SUBJECT TO CHECKPOINT BY BATCH; NORMAL BATCH	16
17	1019	/	PROCESSING CONTINUES AFTER RELOADING THE PROPER BLOCK INTO THE INPUT BUFFER.	17
18	1020			18
19	1021	/	SINCE THE TFS DIRECTORY AND THE TRANSIENT AREA ARE LOADED ALONG WITH THE	19
20	1022	/	KEYBOARD MONITOR [ALBEIT TO ADDRESSES CONTIGUOUS TO THE KEYBOARD MONITOR AND	20
21	1023	/	NOT THE FINAL MEMORY LOCATIONS], THE SYSTEM LOADS WITH A SINGLE TAPE MOVEMENT	21
22	1024	/	FORWARD. THE SYSTEM PROMPT WILL BE ISSUED AFTER THE THREE PAGES OF CODE AND	22
23	1025	/	DATA ARE MOVED THROUGH MEMORY TO THEIR INTENDED OPERATING ADDRESSES AS	23
24	1026	/	DESCRIBED ABOVE. AS SUCH, THE P?S/8 KEYBOARD MONITOR WITH BUILT-IN LINE	24
25	1027	/	EDITOR LOADS IN FAR FASTER THAN MOST OTHER OPERATING SYSTEMS; THIS IS	25
26	1028	/	ESPECIALLY NOTICEABLE ON SYSTEMS BASED ON DECTAPES AND LINCTAPES.	26
27	1029			27
28	1030	/	RESERVED INTERNAL LOCATIONS.	28
29	1031			29
30	1032	/	THE FOLLOWING LOCATIONS ARE RESERVED FOR PASSING PARAMETERS TO EXECUTABLE	30
31	1033	/	PROGRAMS AS REQUIRED. THE EXACT INTERPRETATION OF CERTAIN OF THESE WORDS HAS	31
32	1034	/	EVOLVED OVER TIME; REFER TO PARTICULAR P?S/8 RELEASES FOR EXACT MEANING WHERE	32
33	1035	/	APPLICABLE.	33
34	1036			34
35	1037	/	OPTION SWITCH WORDS.	35
36	1038			36
37	1039	/	ALL SYSTEM PROGRAMS CAN REFERENCE THE 36 OPTION SWITCHES "/A" THROUGH "/Z" AND	37
38	1040	/	"/0" THROUGH "/9". NOTE: COMMAND SPECIFICS MAY OFFER ALTERNATE SYNTAX TO	38
39	1041	/	EXPRESS THESE OPTIONS SUCH AS "(ABC)" WHERE THE RIGHT PARENTHESIS IS OPTIONAL.	39
40	1042			40
41	1043	/	NOTE: OPTION BITS ARE ORDERED LEFT TO RIGHT; TWELVE OPTIONS ARE STORED IN EACH	41
42	1044	/	WORD.	42
43	1045			43
44	1046	07604* 0000 SWAL, ..	/THESE ARE THE PASSED OPTION SWITCHES A-L	44
45	1047	07605* 0000 SWMX, ..	/THESE ARE THE PASSED OPTION SWITCHES M-X	45
46	1048	07606* 0000 SWY9, ..	/THESE ARE THE PASSED OPTION SWITCHES Y-9	46
47				47
48				48
49				49
50				50
51				51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	1050	/	OUTPUT FILE COUNT AND STATUS WORD.	1
2	1051			2
3	1052	/	THE ORIGINAL R-L MONITOR CAN PASS A MAXIMUM OF 21 [17 DECIMAL] INPUT FILES AND	3
4	1053	/	DOES NOT SUPPORT THE CONCEPT OF DEFINED OUTPUT FILES. P?S/8 ADDS THE CONCEPT	4
5	1054	/	OF A COUNT OF HOW MANY OF THE PASSED FILES ARE OUTPUT FILES [WHICH ARE ALWAYS	5
6	1055	/	PASSED FIRST IN THE LIST, IF ANY, FOLLOWED BY THE INPUT FILES, IF ANY].	6
7	1056			7
8	1057	/	ALTHOUGH THE FILE COUNT IS INHERENTLY A QUANTITY THAT CAN BE EXPRESSED IN FIVE	8
9	1058	/	BITS, MOST PROGRAMS DEPEND ON THE HIGH-ORDER SEVEN BITS OF THIS WORD TO BE ALL	9
10	1059	/	ZEROS; THIS WAS THE INITIAL IMPLEMENTATION OF SUPPORT FOR OUTPUT FILES.	10
11	1060			11
12	1061	/	LATER RELEASES OF P?S/8 REQUIRE PROGRAMS TO MASK OFF THE HIGH-ORDER BITS TO	12
13	1062	/	PRESERVE THE PROPER OUTPUT FILE COUNT. THE NEWER COMPOSITE MEANING OF THE	13
14	1063	/	WORD IS AS FOLLOWS:	14
15	1064			15
16	1065	/	BIT[S] FUNCTION	16
17	1066			17
18	1067	/	0 IF SET THE P?S/8 SHELL FILE AND COMMUNICATIONS AREA IS VIRTUALIZED	18
19	1068	/	WHEN THE SHELL IS ENABLED. SEE P?S/8 SHELL DOCUMENTATION FOR MORE	19
20	1069	/	INFORMATION.	20
21	1070			21
22	1071	/	1-6 SYSTEM DATE GROUP OF EIGHT YEARS; 000000 INDICATES 1900-1907, 001010	22
23	1072	/	INDICATES 1980-1987, 001110 INDICATES 2012-2019; THE LARGEST VALUE OF	23
24	1073	/	111111 INDICATES 2404-2411. THE LOW-ORDER 12 BITS OF THE DATE ARE	24
25	1074	/	STORED IN "SDATE" [07610].	25
26	1075			26
27	1076	/	7-11 OUTPUT FILE COUNT IN THE RANGE OF 0000 THROUGH 0021 [17 DECIMAL].	27
28	1077			28
29	1078	07607* 0000	SOUTCT, .. /PASSED FILE OUTPUT COUNT [AND OTHER BITS]	29
30	1079		/STORED HERE.	30
31	1080			31
32	1081	/	NOTE: FILE HANDLES FOR ALL PASSED FILES ARE STORED STARTING IN "SFILES"	32
33	1082	/	[07757]; ALL OUTPUT FILES PRECEDE ALL INPUT FILES. SHORT FILE LISTS ARE	33
34	1083	/	TERMINATED WITH 0000 AS A SENTINEL INDICATOR.	34
35	1084			35
36	1085	/	SYSTEM DATE WORD.	36
37	1086			37
38	1087	/	THE LOW-ORDER 12 BITS OF THE CURRENT SYSTEM DATE [WHICH CAN BE SET BY THE	38
39	1088	/	P?S/8 "DATE" PROGRAM] ARE STORED HERE.	39
40	1089			40
41	1090	07610* 6524	SDATE, 6524/.. /LOW-ORDER 12 BITS OF THE CURRENT DATE.	41
42	1091			42
43	1092	/	APPENDIX "I" DISCUSSES DETAILS OF THE SYSTEM DATE WORD; THE EXACT MEANING OF	43
44	1093	/	THE VALUE STORED HERE HAS CHANGED OVER TIME. AS SUCH, THIS IS EXTREMELY	44
45	1094	/	RELEASE-SPECIFIC.	45
46	1095			46
47	1096	/	[NOTE: THE SPECIFIC VALUE SHOWN REPRESENTS THE DATE STORED WHEN THE BINARY	47
48	1097	/	CONTENTS OF BLOCK 0000 WERE USED TO CREATE THIS SOURCE FILE; THIS WAS DONE TO	48
49	1098	/	VALIDATE THE PROCESS. ANY LEGITIMATE DATE VALUE MAY BE PRESENT; UNLIKE OTHER	49
50	1099	/	SYSTEMS, P?S/8 RETAINS THE LAST DATE ENTERED VIA THE DATE PROGRAM.]	50
51				51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	1101	/		P?S/8 CORE SIZE AND STATUS WORD.	1
2	1102				2
3	1103	/		THE P?S/8 CORE SIZE AND STATUS WORD MANAGES MEMORY AVAILABLE TO ALL P?S/8	3
4	1104	/		UTILITIES AND CONFORMING USER PROGRAMS. ADDITIONALLY, THE STATUS OF THE	4
5	1105	/		CONSOLE OVERLAY AND SOFTWARE ABORT BITS ARE STORED HERE.	5
6	1106				6
7	1107	/		BITS MEANING	7
8	1108				8
9	1109	/	0-2	THE PHYSICAL HIGHEST MEMORY FIELD AVAILABLE. THIS IS BASED ON DYNAMIC	9
10	1110	/		DISCOVERY AT BOOTUP TIME. VALUES RANGE FROM 000 FOR A 4K MACHINE TO	10
11	1111	/		111 FOR A 32K MACHINE. [NOTE: APPENDIX "J" CONTAINS ADDITIONAL	11
12	1112	/		INFORMATION REGARDING ALTERNATE EXTENDED MEMORY SUPPORT FOR CERTAIN	12
13	1113	/		HARDWARE CONFIGURATIONS.]	13
14	1114				14
15	1115	/	3-5	THE LOGICAL HIGHEST MEMORY FIELD POSSIBLE. ON MOST CONFIGURATIONS	15
16	1116	/		THIS VALUE IS IDENTICAL TO THE PHYSICAL HIGHEST MEMORY FIELD CONTAINED	16
17	1117	/		IN BITS[0-2] AS DESCRIBED ABOVE; ON CERTAIN SYSTEMS THIS VALUE IS ONE	17
18	1118	/		MEMORY FIELD LOWER DUE TO ONE OR MORE CONFIGURATION ISSUES. [NOTE:	18
19	1119	/		APPENDIX "J" CONTAINS ADDITIONAL INFORMATION REGARDING CONFIGURATION	19
20	1120	/		SPECIFICS THAT AFFECT THIS VALUE.]	20
21	1121				21
22	1122	/	6-8	THE LOGICAL HIGHEST FIELD CURRENTLY ALLOWED. POSSIBLE SETTINGS RANGE	22
23	1123	/		FROM 000 THROUGH THE LOGICAL HIGHEST MEMORY FIELD CONTAINED IN	23
24	1124	/		BITS[3-5] AS DESCRIBED ABOVE. THE P?S/8 "CORE" PROGRAM CAN ADJUST	24
25	1125	/		THE LOGICAL CORE SIZE TO ANY ALLOWED VALUE.	25
26	1126				26
27	1127	/	9	LOGICAL CONSOLE OVERLAY STATUS. IF THIS BIT IS SET, ALL	27
28	1128	/		OVERLAY-COMPLIANT PROGRAMS DIRECT INPUT AND OUTPUT FOR THE CONSOLE	28
29	1129	/		TERMINAL AND LINEPRINTER TO DEDICATED CONSOLE OVERLAY ROUTINES LOCATED	29
30	1130	/		IN THE PHYSICAL HIGHEST MEMORY FIELD CONTAINED IN BITS[0-2] AS	30
31	1131	/		DESCRIBED ABOVE. [NOTE: APPENDIX "J" CONTAINS ADDITIONAL INFORMATION	31
32	1132	/		ON THE LOGICAL CONSOLE OVERLAY WITH REGARD TO INTERACTION WITH THE	32
33	1133	/		CORE SIZE AND STATUS WORD.]	33
34	1134				34
35	1135	/	10	LOGICAL CONTROL-B ABORT STATUS.	35
36	1136				36
37	1137	/	11	LOGICAL CONTROL-C ABORT STATUS.	37
38	1138				38
39	1139	07611* 0770	SCORE,	0770/.-. /SYSTEM CORE SIZE AND STATUS WORD.	39
40	1140				40
41	1141	/		APPENDIX "J" DISCUSSES LOGICAL ABORT IMPLEMENTATION AS WELL AS DETAILS ON ALL	41
42	1142	/		OTHER INTERACTION WITH THE SYSTEM CORE SIZE AND STATUS WORD. [NOTE: CERTAIN	42
43	1143	/		P?S/8 RELEASES DO NOT FULLY SUPPORT ALL STATUS BITS OF THE SYSTEM CORE SIZE	43
44	1144	/		AND STATUS WORD.]	44
45	1145				45
46	1146	/		[NOTE: THE SPECIFIC VALUE SHOWN REPRESENTS THE STATUS STORED WHEN THE BINARY	46
47	1147	/		CONTENTS OF BLOCK 0000 WERE USED TO CREATE THIS SOURCE FILE; THIS WAS DONE TO	47
48	1148	/		VALIDATE THE PROCESS. IN CERTAIN RELEASES, SOME OF THE MEMORY CONTROL BITS	48
49	1149	/		ARE WRITTEN OUT TO BLOCK 0000; THIS CAN ONLY AFFECT DISMOUNTABLE SYSTEMS WHEN	49
50	1150	/		BOOTED TO A DIFFERENT HARDWARE CONFIGURATION. RUNNING THE P?S/8 CORE UTILITY	50
51	1151	/		["CORE ALL"] WILL RESET THE MEMORY CONTROL BITS TO THE CURRENT LARGEST VALUES.	51
52	1152	/		THE USER CAN OPTIONALLY RESTRICT THE LOGICAL CORE SIZE AT ANY TIME. APPENDIX	52
53	1153	/		"J" DISCUSSES ALL OPTIONS OF THE P?S/8 CORE PROGRAM.]	53
54					54
55					55
56					56
57					57
58					58

1	1155	/	MULTI-PURPOSE DECTAPE DONE FLAG WAIT ROUTINE ["SYSWAIT"].	1
2	1156			2
3	1157	/	ALL CALLERS ENTER HERE. WITH ONE EXCEPTION [SEE BELOW] THE ROUTINE ALWAYS	3
4	1158	/	RETURNS DIRECTLY TO THE CALLER.	4
5	1159			5
6	1160	/	THIS SUBROUTINE IS CALLED FROM SEVERAL DIFFERENT CODE SEGMENTS FOR A VARIETY	6
7	1161	/	OF PURPOSES:	7
8	1162			8
9	1163	/	1) INITIATING A SEARCH IN A DIRECTION DETERMINED BY THE CALLING PROGRAM.	9
10	1164	/	DETECTED BLOCK MARKS WILL BE USED BY THE CALLER TO CALCULATE THE CORRECT	10
11	1165	/	DIRECTION FOR SUBSEQUENT SEARCH OPERATIONS [AS REQUIRED; GENERALLY SEVERAL	11
12	1166	/	SEARCH OPERATIONS ARE NECESSARY FOR THE CURRENT OVERALL DEVICE HANDLER	12
13	1167	/	CALL UNTIL THE TAPE IS PROPERLY POSITIONED]. THE GOAL IS TO LOCATE THE	13
14	1168	/	DESIRED BLOCK FROM A TAPE POSITION JUST BEFORE THE BLOCK WHILE MOVING	14
15	1169	/	FORWARD. WHEN NECESSARY, REVERSE SEARCHES MUST BE PERFORMED UNTIL A BLOCK	15
16	1170	/	EXACTLY TWO HIGHER THAN THE TARGET BLOCK IS DETECTED BEFORE TURNING THE	16
17	1171	/	TAPE AROUND.	17
18	1172			18
19	1173	/	[NOTE: FOR LOW-NUMBERED BLOCKS THE REVERSE END ZONE WILL BE ENCOUNTERED	19
20	1174	/	WHILE SEARCHING IN REVERSE; THIS WILL NECESSITATE AN IMMEDIATE CHANGE TO	20
21	1175	/	FORWARD MOTION; BLOCK 0000 OR BLOCK 0001 MUST BE READ OR WRITTEN FORWARD	21
22	1176	/	COMING OUT OF THE REVERSE END ZONE.]	22
23	1177			23
24	1178	/	2) CHANGE A SUCCESSFUL SEARCH INTO A READ OR WRITE. [BY DESIGN, CHANGING THE	24
25	1179	/	CURRENT MODE FROM SEARCH TO READ OR WRITE JUST AFTER DETECTING THE REAR	25
26	1180	/	BLOCK MARK IS THE INTENDED METHOD.]	26
27	1181			27
28	1182	/	3) CONTINUING A MULTI-BLOCK READ OR WRITE TRANSFER IN PROGRESS. [NOTE: ONLY	28
29	1183	/	THE FIRST OF A SET OF CONTIGUOUS BLOCKS IS EXPLICITLY SEARCHED FOR. IT IS	29
30	1184	/	NEITHER NECESSARY NOR EVEN POSSIBLE TO SEARCH FOR EVERY BLOCK IN A	30
31	1185	/	MULTI-BLOCK TRANSFER AS THE SEARCH FUNCTION WOULD OF NECESSITY CAUSE THE	31
32	1186	/	NEED TO TURN THE TAPE AROUND TWICE. WHILE THE OPERATING MODE CAN BE	32
33	1187	/	CHANGED FROM SEARCH TO READ OR WRITE, IT IS NOT POSSIBLE TO CHANGE A READ	33
34	1188	/	OR WRITE TO A SEARCH AND IMMEDIATELY DETECT THE NEXT CONTIGUOUS BLOCK.]	34
35	1189			35
36	1190	/	IF AN END ZONE ERROR IS DETECTED DURING ANY CALL, THE SUBROUTINE DOES NOT	36
37	1191	/	RETURN TO THE CALLER. INSTEAD, A DEDICATED ROUTINE "STURNIT" [07673] IS	37
38	1192	/	ENTERED TO REVERSE THE CURRENT TAPE DIRECTION AND REWIND OUT OF THE END ZONE.	38
39	1193	/	THIS IS FOLLOWED BY A NEW SEARCH OPERATION FOR THE LATEST BLOCK.	39
40	1194			40
41	1195	/	ALL CALLS SHOULD HAVE AC[9-11] CLEAR. SETTING AC[9] INVERTS DECTAPE	41
42	1196	/	INTERRUPT ENABLE, GENERALLY ALREADY CLEAR; SETTING AC[10] WOULD PRESERVE THE	42
43	1197	/	ERROR FLAG FROM THE PREVIOUS OPERATION; SETTING AC[11] WOULD PRESERVE THE	43
44	1198	/	DECTAPE DONE FLAG FROM THE PREVIOUS OPERATION. ALL OF THESE CONDITIONS SHOULD	44
45	1199	/	BE AVOIDED IN THE TC01/TC08 SYSTEM DEVICE HANDLER FOR PROPER OPERATION.	45
46	1200			46
47	1201	/	ANY BITS SET IN AC[0-9] WILL INVERT THE CORRESPONDING BITS IN STATUS REGISTER	47
48	1202	/	"A". CALLS WITH THE AC CLEAR ARE MADE TO CONTINUE PREVAILING SEARCH, READ OR	48
49	1203	/	WRITE OPERATIONS. THE WORD COUNT REGISTER IS ALWAYS SET TO TRANSFER THE FIRST	49
50	1204	/	128 DATA WORDS OF THE CURRENT BLOCK IN CASE THE CALL IS FOR A READ OR WRITE	50
51	1205	/	OPERATION. [NOTE: WORD COUNT REGISTER SETTINGS ARE IRRELEVANT DURING NORMAL	51
52	1206	/	MODE SEARCH OPERATIONS.]	52
53				53
54				54
55				55
56				56
57				57
58				58

1	1208	/			DURING THE BOOTUP OVERLAY THE RETURN ADDRESS FROM THE READ NEXT BLOCK CALL	1
2	1209	/			"SBLKRET" [07727] IS STORED OVER THE SUBROUTINE HEADER FROM THE PERMANENT COPY	2
3	1210	/			ON BLOCK 0000; WHEN THE DECTAPE FLAG RAISES AT THE END OF THE INITIAL BOOTUP	3
4	1211	/			READ OF BLOCK 0000, CONTROL WILL BE PASSED TO "SBLKRET" [07727]. ADDITIONAL	4
5	1212	/			SEQUENTIAL READ OPERATIONS WILL BE IMMEDIATELY PERFORMED [WITHOUT SEARCHING]	5
6	1213	/			TO LOAD IN THE KEYBOARD MONITOR [AND INTEGRATED LINE EDITOR], THE TFS	6
7	1214	/			DIRECTORY AND THE KEYBOARD MONITOR TRANSIENT AREA FROM BLOCKS 0001-0017 IN ONE	7
8	1215	/			CONTINUOUS READ OPERATION. ALL RELEVANT PARAMETERS IN MEMORY ARE SET TO	8
9	1216	/			APPROPRIATE VALUES THAT ARE THE PERMANENT CONTENTS OF THE CORRESPONDING WORDS	9
10	1217	/			ON BLOCK 0000. THIS WILL CONTINUE THE OVERALL READ OPERATION ALREADY IN	10
11	1218	/			PROGRESS. THE NET EFFECT IS EQUIVALENT TO THE [ACTUALLY IMPOSSIBLE] DEVICE	11
12	1219	/			HANDLER CALL TO READ BLOCKS 0000-0017 INTO 07600-03577 EXITING TO "KBDMON"	12
13	1220	/			[02530]. ["KBDMON" [02530] IS THE ADDRESS WITHIN THE KEYBOARD MONITOR WHERE	13
14	1221	/			FINALIZATION IS PERFORMED TO ALLOW USER INPUT FROM THE CONSOLE OR FROM THE	14
15	1222	/			CONTENTS OF BATCH FILES IF BATCH IS IN EFFECT.]	15
16	1223					16
17	1224	07612*	7727	SYSWAIT, SBLKRET	/UPDATE AND WAIT ROUTINE; PRELOADED WITH PROPER	17
18	1225				/RETURN ADDRESS NEEDED FOR BOOTUP OPERATIONS.	18
19	1226					19
20	1227	/			SPECIFIC VALUES ARE PASSED IN THE AC TO MODIFY STATUS REGISTER "A" IN VARIOUS	20
21	1228	/			WAYS. AC[10-11] MUST BE CLEAR TO RESET THE DECTAPE AND ERROR FLAGS. THE REST	21
22	1229	/			OF THE AC IS SETUP FOR VARIOUS FUNCTION AND MOTION BIT UPDATES. THE PRIOR	22
23	1230	/			CONTENTS OF STATUS REGISTER "A" MUST BE KNOWN; ALL CHANGES TO STATUS REGISTER	23
24	1231	/			"A" [OTHER THAN BITS[10-11] ARE ACCOMPLISHED BY A BIT-WISE .XOR. BETWEEN	24
25	1232	/			STATUS REGISTER "A" AND THE AC [WHICH IS THEN CLEARED].	25
26	1233					26
27	1234	/			IF THE LATEST OPERATION IS SEARCH, STATUS REGISTER "A" WILL BE SET FOR NORMAL	27
28	1235	/			MODE OPERATION WHICH INHIBITS WORD COUNT OVERFLOW [WHILE INCREMENTING THE WORD	28
29	1236	/			COUNT REGISTER]; THE CURRENT ADDRESS REGISTER IS NOT INCREMENTED TO ALLOW A	29
30	1237	/			ONE WORD SEARCH BUFFER FOR DETECTED BLOCK MARKS.	30
31	1238					31
32	1239	/			IF THE LATEST OPERATION IS A REVERSAL OF TAPE DIRECTION [AFTER AN END ERROR],	32
33	1240	/			AC[4] MUST BE SET BECAUSE ALL ERRORS CLEAR TAPE MOTION. AC[3] WILL ALSO BE	33
34	1241	/			SET TO REVERSE THE CURRENT DIRECTION [WITHOUT KNOWING WHAT IT WAS PRIOR TO THE	34
35	1242	/			END ZONE ERROR]. THIS CONDITION CAN ONLY COME ABOUT DURING THE SEARCH PHASE	35
36	1243	/			OF DEVICE HANDLER OPERATION; THE SEARCH ROUTINE WILL LATER RESOLVE ALL MOTION	36
37	1244	/			AND BLOCK POSITION ISSUES BEFORE MOVING ON TO A READ OR WRITE OPERATION.	37
38	1245					38
39	1246	/			THE WORD COUNT WILL BE SET FOR A 128 WORD DATA TRANSFER. THIS IS REQUIRED	39
40	1247	/			SHOULD THE LATEST OPERATION BE A READ OR WRITE [AND OTHERWISE IGNORED]. [AS	40
41	1248	/			SUCH, THE 129TH WORD IS ALWAYS IGNORED IN MEMORY; IT WILL BE USED FOR CHECKSUM	41
42	1249	/			CALCULATION PURPOSES ONLY.]	42
43	1250					43
44	1251	/			THEORETICALLY, IT COULD BE NECESSARY TO INCLUDE THE NEXT INSTRUCTION [OR	44
45	1252	/			EQUIVALENT] TO PREVENT ACCIDENTALLY SETTING BITS THAT MUST REMAIN CLEAR.	45
46	1253					46
47	1254	/		AND (7770)	/MASK OFF POTENTIALLY PRESENT UNWANTED BITS	47
48	1255					48
49	1256	07613*	6764	DTXA	/INVERT STATUS REGISTER "A" BITS PER CONTENTS OF	49
50	1257				/AC[0-9]; IF AC-10-11] ARE CLEAR RESET ERROR	50
51	1258				/AND DONE FLAGS; CLEAR THE AC.	51
52	1259					52
53	1260	07614*	1201	TAD	S7600/(-200)	53
54	1261	07615*	3354	DCA	WC	54
55					/DATA TRANSFER IN CASE READING OR WRITING.	55
56						56
57						57
58						58

1	1263	/	BOOTSTRAP COMPATIBILITY ISSUES.	1
2	1264			2
3	1265	/	THIS CODE MUST BE CAREFULLY LOCATED FOR BOOTSTRAP COMPATIBILITY WITH THE	3
4	1266	/	DIRECT OVERLAY FORM OF BOOTSTRAP [AS OPPOSED TO OTHER METHODS THAT READ IN THE	4
5	1267	/	ENTIRE BLOCK THEN START THE CODE FROM AN EXTERNAL ADDRESS]. THE WAIT LOOP	5
6	1268	/	MUST BE PLACED AT EXACTLY 07616 AND 07617 BECAUSE MOST BOOTUP SEQUENCES DEPEND	6
7	1269	/	ON THE GRADUAL OVERLAY OF THE CODE WAITING FOR THE READ OPERATION TO COMPLETE.	7
8	1270	/	LOCATION 07616 MAY CONTAIN [OR BE OVERLAID] WITH A VARIANT INSTRUCTION SUCH AS	8
9	1271	/	"DTSF" [6771] AS OPPOSED TO "DTSF!DTRB" [6773] AS USED HERE. FULL BOOTSTRAP	9
10	1272	/	COMPATIBILITY EXISTS WHETHER P?S/8 IS ATTEMPTING TO OVERLAY THIS OR SIMILAR	10
11	1273	/	CODE [OR CONVERSELY, SIMILAR CODE IS ATTEMPTING TO OVERLAY P?S/8].	11
12	1274			12
13	1275	/	NOTE: AT THE POINT IN TIME 07616 IS OVERLAID, LOCATION 07617 IS STILL	13
14	1276	/	"JMP SWAITLP" [5216].	14
15	1277			15
16	1278	/	THE NEXT DATA BREAK TRANSFER WILL OVERLAY LOCATION 07617. THE REPLACEMENT	16
17	1279	/	CONTENTS MUST BE A "JMP" [OR "JMS"] INSTRUCTION TO AN APPROPRIATE ADDRESS IN	17
18	1280	/	THE RANGE OF 07600-07616 ["JMP 07600-07616" OR "JMS 07600-07615" DUE TO THE	18
19	1281	/	WAY THE "JMS" INSTRUCTION OPERATES]; AT THIS POINT IN TIME, ONLY THESE FEW	19
20	1282	/	LOCATIONS CONTAIN THE DATA TRANSFERRED FROM BLOCK 0000 OF THE BOOTING UP	20
21	1283	/	DECTAPE. [NOTE: MOST SYSTEMS USE THE "JMP SWAITLP" [5216] [OR EQUIVALENT] AS	21
22	1284	/	IMPLEMENTED WITHIN THIS DEVICE HANDLER.]	22
23	1285			23
24	1286	/	STANDARD TC01/TC08 DECTAPE WAIT LOOP.	24
25	1287			25
26	1288		IFNZRO 7616-. <ERROR .> /ENSURE COMPATIBILITY WITH ADDRESS STANDARDS.	26
27	1290			27
28	1291	07616* 6773	SWAITLP,DTSF!DTRB /SKIP ON DECTAPE FLAG OR ERROR FLAG SET,	28
29	1292		/.OR. TRANSFER STATUS REGISTER "B" INTO AC.	29
30	1293	07617* 5216	JMP SWAITLP /FLAG NOT UP YET, KEEP WAITING FOR IT.	30
31	1294			31
32	1295	/	THE WAIT LOOP EXITS WHEN EITHER THE DECTAPE DONE FLAG OR THE ERROR FLAG	32
33	1296	/	RAISES; THE AC NOW CONTAINS THE CONTENTS OF STATUS REGISTER "B".	33
34	1297			34
35	1298	/	DETERMINE IF AN END ZONE ERROR OCCURRED. [NOTE: IT IS IMPOSSIBLE FOR THE MARK	35
36	1299	/	TRACK, TIMING TRACK OR SELECT ERROR STATUS BITS TO BE SET IF THE END ZONE	36
37	1300	/	ERROR BIT IS SET.]	37
38	1301			38
39	1302	07620* 7006	RTL /MOVE END ZONE ERROR STATUS BIT TO AC[0]	39
40	1303			40
41	1304	/	NOTE: THE AC IS GUARANTEED TO BE NON-ZERO AS AC[9] NOW CONTAINS THE DONE FLAG	41
42	1305	/	STATUS BIT AND AC[11] NOW CONTAINS THE ERROR FLAG STATUS BIT. THIS ALLOWS THE	42
43	1306	/	TEST FOR THE END ZONE ERROR TO BE PERFORMED BY EITHER AN "SPA CLA" [7710] OR	43
44	1307	/	"SPA SNA CLA" [7750] INSTRUCTION.	44
45	1308			45
46	1309	07621* 7750	S7750, SPA SNA CLA /SKIP IF NOT END ZONE ERROR; ALSO USED	46
47	1310		/ELSEWHERE AS THE CONSTANT 7750.	47
48	1311	07622* 5273	JMP STURNIT /TURN THE TAPE AROUND IF WE ARE IN THE END ZONE.	48
49	1312	07623* 6772	DTRB /GET CONTENTS OF STATUS REGISTER "B" AGAIN.	49
50	1313	07624* 7500	SMA /SKIP IF ANY OTHER ERROR[S] SET.	50
51	1314	07625* 5612	JMP I SYSWAIT /RETURN WITH DIRTY AC [BUT NO ERROR BITS SET].	51
52	1315	07626* 7402	SHALT, HLT /HALT SHOWING STATUS REGISTER "B" IN THE AC.	52
53				53
54				54
55				55
56				56
57				57
58				58

1	1317	/	ERROR RECOVERY CONSIDERATIONS.	1
2	1318			2
3	1319	/	AT THIS POINT THE MACHINE HAS HALTED DUE TO SOME FORM OF ERROR [OTHER THAN END	3
4	1320	/	ZONE ERROR]. IT IS EXPECTED THE OPERATOR WILL PRESS "CONTINUE" ON THE CPU	4
5	1321	/	FRONT PANEL AFTER PERHAPS PERFORMING SOME CORRECTIVE STEPS [DEPENDING ON THE	5
6	1322	/	SPECIFICS OF THE ERROR CONDITION]. A COMPLETE RETRY OF THE LATEST DECTAPE	6
7	1323	/	OPERATION WILL THEN BE PERFORMED [IF POSSIBLE].	7
8	1324			8
9	1325	/	USERS FAMILIAR WITH STATUS REGISTER "B" BIT ASSIGNMENTS CAN OBSERVE THE EXACT	9
10	1326	/	ERROR CONDITION CAUSING THE CURRENT PROBLEM DISPLAYED IN THE AC INDICATOR	10
11	1327	/	LIGHTS [OR THE EQUIVALENT STATUS LAMPS ON THE TC01 OR TC08 STATUS PANEL].	11
12	1328	/	THERE ARE SEVERAL CATEGORIES OF ERRORS; CERTAIN ERROR CONDITIONS CAN BE CAUSED	12
13	1329	/	BY SEVERAL POSSIBLE FAILURE SITUATIONS.	13
14	1330			14
15	1331	/	SELECT ERROR CONSIDERATIONS.	15
16	1332			16
17	1333	/	SELECT ERROR CAN BE CAUSED BY ANY OF THE FOLLOWING:	17
18	1334			18
19	1335	/	1) DRIVE SELECT - NO DRIVES ARE SET TO THE CURRENT DRIVE UNIT. ALL DRIVE	19
20	1336	/	SELECT LIGHTS ARE OFF.	20
21	1337			21
22	1338	/	2) DRIVE SELECT - TOO MANY DRIVES ARE SET TO THE CURRENT DRIVE UNIT. ALL	22
23	1339	/	DRIVES SET TO THE CURRENT UNIT WILL HAVE LIT SELECT LIGHTS.	23
24	1340			24
25	1341	/	A UNIQUE TAPE DRIVE MUST BE SET TO THE CURRENT DRIVE UNIT TO AVOID A SELECT	25
26	1342	/	ERROR. FUNCTIONS THAT CAUSE TAPE MOTION ADDITIONALLY REQUIRE A MOUNTED	26
27	1343	/	DECTAPE; WITHOUT A TAPE, ONE OR MORE REEL MOTORS WILL SPIN. [NOTE: THIS	27
28	1344	/	ALSO DEPENDS ON THE SPECIFIC DRIVE MODEL; TU56 DRIVE MOTORS WILL ALWAYS	28
29	1345	/	SPIN WHEN THERE IS NO TAPE TENSION UNLESS THE DRIVE IS OFFLINE].	29
30	1346			30
31	1347	/	3) WRITE-PROTECT - A TAPE IS CORRECTLY SELECTED AND CAPABLE OF MOTION. THE	31
32	1348	/	DRIVE-MOUNTED WRITE-ENABLE SWITCH IS SET TO THE WRITE-LOCKED POSITION.	32
33	1349	/	WHERE RELEVANT, THE CORRESPONDING WRITE-ENABLE INDICATOR LAMP WILL BE OFF.	33
34	1350	/	ONCE INITIATED, TAPE MOTION OCCURS UNTIL THE DESIRED BLOCK IS LOCATED WHILE	34
35	1351	/	THE TAPE IS MOVING FORWARD. WHEN THE FUNCTION CHANGES FROM NORMAL SEARCH	35
36	1352	/	TO WRITE, THE TAPE STOPS DUE TO THE WRITE-PROTECT ERROR.	36
37	1353			37
38	1354	/	4) FORMAT SWITCH - THE WRITE TIMING AND MARK TRACK ENABLE SWITCH MUST BE OFF	38
39	1355	/	DURING NORMAL TAPE OPERATIONS. [NOTE: WITHIN THE TC01/TC08 DECTAPE	39
40	1356	/	FORMATTING PROGRAM, SELECT ERROR OCCURS IF THE SWITCH IS NOT THROWN WHEN	40
41	1357	/	DIRECTED. LATER IN THE PROGRAM THE OPERATOR IS INSTRUCTED TO TURN THE	41
42	1358	/	SWITCH OFF TO CONTINUE THE TAPE FORMATTING OPERATION.]	42
43	1359			43
44	1360	/	A FORMAT-RELATED SELECT ERROR HAPPENS WHEN THE SWITCH IS THROWN AND NORMAL	44
45	1361	/	OPERATIONS ARE ATTEMPTED, AS WELL AS WHEN FORMATTING OPERATIONS ARE	45
46	1362	/	ATTEMPTED WITH THE SWITCH OFF. [THE LATTER IS ONLY USED IN THE VARIOUS	46
47	1363	/	TC01/TC08 FORMATTING AND DIAGNOSTIC PROGRAMS.]	47
48	1364			48
49	1365	/	5) DIAGNOSTIC READ - THIS READ ALL FUNCTION REQUIRES THE FORMAT SWITCH TO BE	49
50	1366	/	THROWN TO THE READ MARK POSITION. ALL OTHER SETTINGS CAUSE A SELECT ERROR.	50
51	1367	/	[NOTE: THIS FUNCTION IS NOT USED WITHIN THE SYSTEM DEVICE HANDLER.]	51
52	1368			52
53	1369	/	6) OTHER - ISSUING COMMANDS WITH THE RESERVED FUNCTION CODE OF 111 CAUSES A	53
54	1370	/	SELECT ERROR. [NOTE: THIS FUNCTION IS NOT USED WITHIN THE SYSTEM DEVICE	54
55	1371	/	HANDLER.]	55
56				56
57				57
58				58

1	1373	/	TIMING AND MARK ERROR CONSIDERATIONS.	1
2	1374			2
3	1375	/	THEORETICALLY, TIMING ERRORS OCCUR ONLY WHEN DMA CYCLES CANNOT BE PERFORMED IN	3
4	1376	/	THE WINDOW OF TIME ALLOTTED PER WORD [3 CYCLES EVERY 133 MICROSECONDS]. THIS	4
5	1377	/	CAN HAPPEN IF THE CPU HALTS WHILE THE OPERATION IS IN EFFECT. THE ERROR HALT	5
6	1378	/	WITHIN THE SYSTEM DEVICE HANDLER SHOWS THE STATUS WHILE THE MACHINE WAS	6
7	1379	/	RUNNING; THUS, THIS NORMALLY CANNOT HAPPEN UNLESS THE BUSS IS BUSIED OUT BY	7
8	1380	/	EXTREME LOADING FROM ANOTHER DMA DEVICE ALSO IN USE AT THE TIME. THIS	8
9	1381	/	SITUATION, AS UNLIKELY AS IT IS, CAN BE PREVENTED BY SETTING THE DATA BREAK	9
10	1382	/	PRIORITY FOR THE DECTAPE CONTROLLER HIGHER THAN FOR OTHER DEVICES. [NOTE:	10
11	1383	/	TIMING ERRORS CAN ALSO OCCUR IF IMPROPERLY WRITTEN SOFTWARE IS USED: THE DONE	11
12	1384	/	FLAG MUST BE CLEAR BEFORE IT SETS TO AVOID A TIMING ERROR; THE SYSTEM DEVICE	12
13	1385	/	HANDLER PERFORMS ALL TAPE CONTROL FUNCTIONS IN THE PROPER ORDER STARTING WITH	13
14	1386	/	CLEARING THE DONE AND ERROR FLAGS. THUS, THAT SITUATION CANNOT OCCUR WITHIN	14
15	1387	/	THIS DEVICE HANDLER.]	15
16	1388			16
17	1389	/	HOWEVER, WHEN THERE ARE READ ERRORS ON THE TIMING AND MARK TRACKS, THE TIMING	17
18	1390	/	ERROR CAN HAPPEN FOR UNEXPECTED REASONS. TIMING PULSES MUST BE CONSISTENT AND	18
19	1391	/	REGULAR [TO WITHIN A DEFINED TOLERANCE FACTOR]; TIMING TRACK READ ERRORS CAN	19
20	1392	/	AFFECT THE INTERVAL SUFFICIENTLY TO CAUSE AN OTHERWISE UNACCOUNTABLE ERROR.	20
21	1393			21
22	1394	/	MARK ERRORS OCCUR WHEN THE SIX BIT MARK WINDOW DETECTS CERTAIN ILLEGAL CODES	22
23	1395	/	OR UNEXPECTED CODES DEPENDING ON THE CURRENT OPERATION AND DECODING LOGIC	23
24	1396	/	WITHIN THE CONTROLLER. SINCE THE MARK WINDOW IS DEPENDENT ON THE TIMING	24
25	1397	/	TRACKS FOR PROPER SYNCHRONIZATION, THE INDICATED MARK ERROR COULD BE THE	25
26	1398	/	INDIRECT RESULT OF TIMING TRACK READ PROBLEMS AS WELL AS MARK TRACK READ	26
27	1399	/	ERRORS.	27
28	1400			28
29	1401	/	GENERALLY, TIMING AND MARK ERRORS SHOULD NOT OCCUR ON PROPERLY OPERATING	29
30	1402	/	SYSTEMS. POTENTIAL ISSUES INCLUDE THE CONDITION AND SKEW [ALIGNMENT] OF THE	30
31	1403	/	TAPE HEAD, TAPE GUIDES, THE OVERALL CONDITION AND [PROPER] FORMATTING OF THE	31
32	1404	/	DECTAPE MEDIA AND THE OVERALL CLEANLINESS OF ALL EQUIPMENT. AS COMPONENTS AGE	32
33	1405	/	AND/OR WEAR, IT IS ESSENTIAL THAT PROPER MAINTENANCE BE PERFORMED.	33
34	1406			34
35	1407	/	OFTEN THE FIRST SIGN OF A PROBLEM IS THE DETECTION OF TIMING AND MARK ERRORS	35
36	1408	/	SOMEWHERE ON THE TAPE. AS SUCH, IT IS FORTUITOUS THE SYSTEM DEVICE HANDLER	36
37	1409	/	ERROR ROUTINE DETECTS THESE PROBLEMS AS THEY OCCUR. OFTEN THE LATEST TAPE	37
38	1410	/	OPERATION SUCCEEDS AFTER A SERIES OF RETRY ATTEMPTS.	38
39	1411			39
40	1412	/	THIS SHOULD BE TAKEN AS A WARNING THE TAPE HARDWARE REQUIRES MAINTENANCE	40
41	1413	/	AND/OR THE DECTAPE NEEDS TO BE EITHER REFORMATTED [PERHAPS AFTER CLEANING] OR	41
42	1414	/	[IN INSTANCES OF SEVERE WEAR OR DAMAGE] DISCARDED. DUE TO THE INHERENTLY SAFE	42
43	1415	/	DEVICE HANDLER RECOVERY TECHNIQUES EMPLOYED, THE CURRENT OPERATION WILL LIKELY	43
44	1416	/	[EVENTUALLY] SUCCEED AS DESCRIBED ABOVE.	44
45	1417			45
46	1418	/	THE DESIGN PRINCIPLES OF THIS DEVICE HANDLER SHOULD BE CONTRASTED WITH OTHER	46
47	1419	/	MORE EXPEDIENTLY DESIGNED SYSTEMS SUCH AS OS/8 WHERE ALL SUCH ERRORS ARE	47
48	1420	/	TOTALLY IGNORED. BY THE TIME A TAPE EXHIBITS ANY ERRORS USING OS/8, IT IS	48
49	1421	/	OFTEN TOO LATE TO RECOVER DATA. [NOTE: IT HAS BEEN SUGGESTED THAT THIS ASPECT	49
50	1422	/	OF THE OS/8 DECTAPE HANDLERS IS THE RESULT OF AN EDICT FROM A MISGUIDED	50
51	1423	/	PRODUCT MANAGER. CLEARLY THE HANDLER OPERATES AS IF WRITTEN WITH NO ACTUAL	51
52	1424	/	UNDERSTANDING OF THE PHYSICAL CHARACTERISTICS OF DECTAPE. A HANDLER THAT	52
53	1425	/	TAKES AN ERROR RETURN FOR THESE SERIOUS CONDITIONS CAN BE WRITTEN FOR OS/8;	53
54	1426	/	HOWEVER, THIS WAS NEVER PART OF THE DEC STANDARD RELEASE.]	54
55				55
56				56
57				57
58				58

1	1428	/	BY BEING MORE DIAGNOSTIC IN NATURE, THE P?S/8 SYSTEM DEVICE HANDLER GREATLY	1
2	1429	/	BENEFITS THE USER: DATA LOSS WILL BE EITHER ELIMINATED OR AT LEAST SERIOUSLY	2
3	1430	/	REDUCED. HARDWARE ERRORS ARE DETECTED WHEN THEY FIRST OCCUR BEFORE DATA LOSS	3
4	1431	/	IS LIKELY. SUBSEQUENT MAINTENANCE WILL ELIMINATE THE CAUSE OF THE ERRORS LONG	4
5	1432	/	BEFORE THE PROBLEM ESCALATES TO POTENTIALLY PERMANENT DATA LOSS.	5
6	1433			6
7	1434	/	THIS DESIGN PHILOSOPHY HAS SUCCESSFULLY BEEN USED FROM 1966 [STARTING WITH THE	7
8	1435	/	ORIGINAL R-L MONITOR SYSTEM] ONWARD WITH VIRTUALLY NO LOST DATA. AS	8
9	1436	/	NECESSARY, SPECIFIC DECTAPES ARE TAKEN OUT OF SERVICE AND REPLACED [AND	9
10	1437	/	POSSIBLY RECYCLED]; COPYING THE CONTENTS OF MARGINAL TAPES TO FRESHLY	10
11	1438	/	FORMATTED MEDIA ELIMINATES DATA LOSS. ALL RELEASES OF THE P?S/8 SYSTEM DEVICE	11
12	1439	/	HANDLER USE THIS PROVEN METHOD OF ERROR DETECTION.	12
13	1440			13
14	1441	/	READ PARITY ERROR CONSIDERATIONS.	14
15	1442			15
16	1443	/	PARITY ERRORS ONLY OCCUR WHEN ATTEMPTING TO READ THE CURRENT BLOCK[S]. RETRY	16
17	1444	/	OPERATIONS GENERALLY CLEAR THE PROBLEM [BUT NOT ALWAYS]. REASONABLE EFFORT	17
18	1445	/	SHOULD BE MADE TO ATTEMPT DATA RECOVERY IF POSSIBLE. [NOTE: THE ROOT CAUSE OF	18
19	1446	/	PARITY ERRORS IS GENERALLY THE SAME SET OF ISSUES THAT TEND TO CAUSE TIMING	19
20	1447	/	AND MARK TRACK ERRORS AS DISCUSSED ABOVE.]	20
21	1448			21
22	1449	07627* 5257	JMP SRETRY /"CONTINUE" WAS PRESSED; RETRY THE ENTIRE CALL.	22
23	1450			23
24	1451	/	LOGICAL CONSOLE OVERLAY MODIFICATIONS.	24
25	1452			25
26	1453	/	IF THE LOGICAL CONSOLE OVERLAY IS IN EFFECT, THE PREVIOUS TWO INSTRUCTIONS ARE	26
27	1454	/	REPLACED AS FOLLOWS:	27
28	1455			28
29	1456	/	*SHALT /OVER THE HLT INSTRUCTION	29
30	1457			30
31	1458	/	CIF XX /WILL BE "CIF" TO THE HIGHEST PHYSICAL FIELD;	31
32	1459	/	JMP ERRTRAP /NOTE: "ERRTRAP" IS ON PAGE ZERO OF THE	32
33	1460		/CONSOLE OVERLAY THUS DIRECTLY ADDRESSABLE.	33
34	1461			34
35	1462	/	THE ERROR TRAP HANDLES ALL ERROR PARTICULARS AS GUIDED BY A TABLE OF	35
36	1463	/	INFORMATION SPECIFIC TO THIS SYSTEM DEVICE HANDLER WRITTEN OUT TO A SPECIFIC	36
37	1464	/	BLOCK RESERVED FOR THE PURPOSE. [NOTE: THE EXACT BLOCK NUMBER IS DEPENDENT ON	37
38	1465	/	THE PARTICULAR RELEASE OF P?S/8 AND IS SUBJECT TO CHANGE.]	38
39	1466			39
40	1467	/	THE ERROR INTERCEPTION IS GENERALLY DISPLAYED ON THE CURRENT LOGICAL CONSOLE	40
41	1468	/	DISPLAY DEVICE IN THE FORM OF AN ERROR MESSAGE WITH COMPLETE PARTICULARS OF	41
42	1469	/	THE CALL TO THE SYSTEM DEVICE HANDLER AND EXPLICIT STRINGS OF CHARACTERS TO	42
43	1470	/	DESCRIBE THE PARTICULAR ERROR. THIS CAN INCLUDE SUCH INFORMATION AS THE	43
44	1471	/	CALLING ADDRESS, WHETHER THE CALL WAS A READ OR WRITE OPERATION AND THE	44
45	1472	/	LOGICAL DRIVE UNIT, ETC. [NOTE: THE EXACT NATURE OF THE ERROR HANDLING IS	45
46	1473	/	CUSTOMIZABLE WITHIN THE CONSOLE OVERLAY CONFIGURATION. SLOWER CONSOLE DEVICES	46
47	1474	/	CAN BE CONFIGURED TO USE SHORTER ERROR MESSAGES. DEVICES SUCH AS THE VT8E AND	47
48	1475	/	THE PDP-12 VR12/VR14 DISPLAY THAT ARE ESSENTIALLY TERMINAL EMULATORS GENERALLY	48
49	1476	/	CAN PROVIDE ENTIRE SCREENS DEVOTED TO THE PARTICULARS OF ERROR RECOVERY. WHEN	49
50	1477	/	THE ERROR CLEARS, THE NORMAL CONSOLE SCREEN WILL BE DISPLAYED.]	50
51	1478			51
52	1479	07630* 0010 S10, 10	/CONSTANT 0010	52
53	1480	07631* 6203 S6203, CIF CDF 00	/CONSTANT 6203	53
54				54
55				55
56				56
57				57
58				58

1	1482	/	NOTE: THE FOLLOWING SECTION WILL BE REMOVED IN THE NEXT RELEASE OF THE	1
2	1483	/	TC01/TC08 SYSTEM DEVICE HANDLER.	2
3	1484			3
4	1485	/	*****	4
5	1486	/	*	5
6	1487	/	* NOTE: AS STATED ELSEWHERE IN THIS DOCUMENT, THIS IS NOT THE FINAL VERSION	6
7	1488	/	* OF THE TC01/TC08 SYSTEM DEVICE HANDLER. THERE IS A KNOWN THEORETICAL BUG	7
8	1489	/	* IN THIS CODE THAT CAN ONLY SURFACE IN A HIGHLY UNLIKELY SCENARIO:	8
9	1490	/	*	9
10	1491	/	* ASSUME THAT SOMEHOW AN END ZONE ERROR OCCURS DURING A SUBSEQUENT READ OR	10
11	1492	/	* WRITE OPERATION INVOLVING A MULTIPLE BLOCK TRANSFER ALSO INVOLVING THE USE	11
12	1493	/	* OF EXTENDED MEMORY FOR THE CALLER'S DATA BUFFER. THIS WOULD MEAN AN END	12
13	1494	/	* ZONE ERROR WOULD HAVE UNEXPECTEDLY OCCURRED WHICH WOULD THEN LEAD TO	13
14	1495	/	* RECOVERY ATTEMPTS TO ALLOW THE LATEST BLOCK TO TRANSFER. THIS WOULD	14
15	1496	/	* INVOLVE A NEW SEARCH OPERATION TO LOCATE THE CURRENT BLOCK MOVING FORWARD.	15
16	1497	/	* HOWEVER, IN THIS ODD SCENARIO [WHICH IMPLIES THERE IS SOME FORM OF FAILURE	16
17	1498	/	* IN THE CONTROLLER, TAPE OR DRIVE], IT IS POSSIBLE FOR THE DEVICE HANDLER	17
18	1499	/	* TO FAIL TO PROPERLY SEARCH FOR THE NEXT BLOCK ON ALL RETRY OPERATIONS.	18
19	1500	/	*	19
20	1501	/	* AN ALTERNATE SCENARIO INVOLVES AN IMPROPER CALL FOR MULTIPLE BLOCKS WHERE	20
21	1502	/	* AT LEAST ONE [BUT NOT EVERY ONE] IS NONEXISTENT [SUCH AS AN ATTEMPT TO	21
22	1503	/	* TRANSFER BLOCK 2702 AFTER BLOCK 2701 ON A STANDARD LENGTH DECTAPE. [NOTE:	22
23	1504	/	* LONGER TAPES MERELY MOVE THE APPLICATION POINT OF THE PROBLEM.]	23
24	1505	/	*	24
25	1506	/	* THE ROOT CAUSE OF THE BUG IS THE IMPROPER PLACEMENT OF A "DTLB"	25
26	1507	/	* INSTRUCTION TO CAUSE DATA BREAK TRANSFER OF THE BLOCK MARKS TO FIELD 0.	26
27	1508	/	* DURING THE READ OR WRITE OPERATIONS THAT FOLLOW THE FIRST SUCCESSFUL	27
28	1509	/	* SEARCH, THE NEXT DATA BREAK FIELD SETTING OPERATION SETS DATA TRANSFERS TO	28
29	1510	/	* THE CALLER'S BUFFER FIELD. IF THE SEARCH IS RESTARTED AFTER AN END ZONE	29
30	1511	/	* ERROR, THE FIELD WILL NOT BE RESET TO 0.	30
31	1512	/	*	31
32	1513	/	* A TRIVIAL FIX IS AVAILABLE: MERELY MOVE THE "DTLB" INSTRUCTION [CURRENTLY	32
33	1514	/	* LOCATED AT 07666] TO "STURNIT" [WHICH WOULD CHANGE TO 07672 AS A RESULT].	33
34	1515	/	* THE AC IS KNOWN TO BE 0000 AT THAT POINT; THIS WILL CORRECTLY RESET THE	34
35	1516	/	* DATA BREAK FIELD FOR PROPER SEARCH OPERATIONS. THE CODE CAN BE CORRECTED	35
36	1517	/	* BY PATCHING TAPE BLOCK 0000 USING THE P?S/8 DUMP UTILITY. NOTE: ALL	36
37	1518	/	* AFFECTED LOCATIONS MUST BE PATCHED SIMULTANEOUSLY BEFORE THE NEXT BOOTUP.	37
38	1519	/	*	38
39	1520	/	* SUBSEQUENT RELEASES OF THIS DEVICE HANDLER WILL INCLUDE FIXING THIS HIGHLY	39
40	1521	/	* IMPROBABLE BUG; OTHER MORE IMPORTANT IMPROVEMENTS WILL BE INCLUDED THAT	40
41	1522	/	* WERE ORIGINALLY CREATED SEVERAL YEARS AFTER THIS VERSION WAS RELEASED.	41
42	1523	/	*	42
43	1524	/	* THE SOURCE CODE HAS BEEN MODIFIED TO ALLOW ELIMINATION OF THE BUG OR	43
44	1525	/	* MAINTAINING IT FOR HISTORICAL PURPOSES [AS PART OF THE CIRCA 1976 RELEASE	44
45	1526	/	* OF P?S/8 AVAILABLE AS AN IMAGE FILE]. FUTURE RELEASES OF THIS FILE WILL	45
46	1527	/	* ELIMINATE THIS SUPPORT AS PART OF AN EFFORT TO INCLUDE ADDITIONAL WORK	46
47	1528	/	* PERFORMED AS MUCH AS A DECADE AFTER THIS RELEASE; ADDITIONAL NEWER CHANGES	47
48	1529	/	* MAY BE INCLUDED AS PART OF A NEWER OVERALL RELEASE OF P?S/8 IN THE FUTURE.	48
49	1530	/	*	49
50	1531	/	* THE PROGRAM CAN BE ASSEMBLED FOR EITHER CASE BY APPROPRIATE DEFINITION OF	50
51	1532	/	* THE SYMBOL "ASMBUG" IN THE DEFINITIONS SECTION ELSEWHERE IN THIS DOCUMENT.	51
52	1533	/	*	52
53	1534	/	*****	53
54				54
55				55
56				56
57				57
58				58

1	1536	/	P?S/8 SYSTEM PROGRAM LOADER.	1
2	1537			2
3	1538	/	PERHAPS THE SINGLE MOST SIGNIFICANT IMPROVEMENT UPON THE ORIGINAL R-L MONITOR	3
4	1539	/	SYSTEM, THE P?S/8 SYSTEM PROGRAM LOADER [THE "SYSTEM LOADER" AS IT WAS FIRST	4
5	1540	/	IDENTIFIED WITHIN THE R-L MONITOR SYSTEM] IS THE MECHANISM BY WHICH IMAGE	5
6	1541	/	BINARY PROGRAMS ARE LOADED INTO USER MEMORY [00000-07577] WITHOUT RESTRICTION	6
7	1542	/	AND THEN STARTED. IN GENERAL, LOADING RULES ARE FULLY COMPATIBLE WITH THE	7
8	1543	/	BINARY PAPER TAPE LOADERS. THUS, OTHER THAN OPTIONALLY INTERFACING WITH	8
9	1544	/	ADDITIONAL FEATURES OF P?S/8, GENERAL PROGRAMMING RULES FOR THE PDP-8 ARE	9
10	1545	/	FULLY SUPPORTED WITHOUT RESTRICTION AS DOCUMENTED WITHIN BIN LOADER	10
11	1546	/	SPECIFICATIONS.	11
12	1547			12
13	1548	/	APPENDIX "K" DISCUSSES THE MANY ISSUES SURROUNDING THE SYSTEM LOADER, WHICH IS	13
14	1549	/	USED IN SEVERAL CONTEXTS THROUGHOUT P?S/8. ADDITIONALLY, THERE IS A DETAILED	14
15	1550	/	DISCUSSION OF HOW THE ORIGINAL CONVOLUTED R-L MONITOR SYSTEM EQUIVALENT	15
16	1551	/	FUNCTION WAS CARRIED OUT [WITH INVOLVES A SERIOUS AMOUNT OF OVERHEAD AND MANY	16
17	1552	/	LIMITATIONS] THAT, IF THIS HAD NOT COMPLETELY REDESIGNED, WOULD HAVE GREATLY	17
18	1553	/	CRIPPLED THE OVERALL P?S/8 SYSTEM.	18
19	1554			19
20	1555	/	IT IS ONLY BECAUSE OF THE TREMENDOUS EFFORT ON THE PART OF MANY INDIVIDUALS	20
21	1556	/	OPTIMIZING THE ENTIRE SYSTEM DEVICE HANDLER CODE IN VARIOUS WAYS OVER THE	21
22	1557	/	YEARS THAT THE SYSTEM LOADER AS NOW DEFINED HERE BECAME POSSIBLE. [NOTE: VERY	22
23	1558	/	EARLY RELEASES OF P?S/8 WERE FULLY COMPATIBLE WITH ALL OF THE R-L MONITOR	23
24	1559	/	CALLS REPLETE WITH SYSTEM OVERHEAD; THE SYSTEM DEVICE HANDLER DESCRIBED IN	24
25	1560	/	THIS DOCUMENT IS THE FIRST GENERATION THAT REMOVED ALL OF THIS OVERHEAD ALL AT	25
26	1561	/	ONCE; FUTURE RELEASES OF THE HANDLER ARE MERELY REFINEMENTS IN OTHER WAYS TO	26
27	1562	/	ADD ADDITIONAL FUNCTIONALITY WITHOUT ADDING ANY STRUCTURAL CHANGES.]	27
28	1563			28
29	1564	/	THIS ABSOLUTELY *MUST* BE AT LOCATION 07632 UNLESS MANY PROGRAMS ARE ALTERED.	29
30	1565			30
31	1566		IFNZRO 7632-. <ERROR .> /ENSURE COMPATIBILITY WITH DESIGN STANDARDS.	31
32	1568			32
33	1569	/	NOTE: THE PRELOADED CONTENTS OF THE SYSTEM LOADER ARE USED TO SAVE ALL OF	33
34	1570	/	FIELD 0 BEFORE RESTARTING THE SYSTEM AT 07600; THIS IS USED BY VARIOUS	34
35	1571	/	MEMORY-SAVING OPERATIONS THROUGHOUT P?S/8; OTHER SYSTEM COMPONENTS WILL LOAD	35
36	1572	/	ALTERNATE PARAMETERS INTO THE MEMORY-RESIDENT COPY AS REQUIRED.	36
37	1573			37
38	1574	07632* 4240 SLOAD, JMS SYSIO	/CALL THE SYSTEM DEVICE HANDLER ON ITS OWN PAGE.	38
39	1575	07633* 0000 SLDADR, 0000	/LOADING ADDRESS; PRELOADED FOR MEMORY SAVE.	39
40	1576	07634* 4000 SFUN, 00^100!WRITE	/FUNCTION WORD; PRELOADED FOR MEMORY SAVE.	40
41	1577	07635* 0020 SLDBLK, PERFILE	/STARTING BLOCK; PRELOADED FOR MEMORY SAVE.	41
42	1578	07636* 5637 JMP I SSTART	/START IT UP.	42
43	1579	07637* 7600 SSTART, SBOOT	/STARTUP ADDRESS; PRELOADED FOR MEMORY SAVE.	43
44				44
45				45
46				46
47				47
48				48
49				49
50				50
51				51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	1581	/	SYSTEM DEVICE HANDLER ENTRY POINT.	1
2	1582			2
3	1583	/	THIS ABSOLUTELY *MUST* BE AT LOCATION 07640.	3
4	1584			4
5	1585		IFNZRO 7640-. <ERROR .> /ENSURE ENTRY POINT MATCHES "SZA CLA" VALUE.	5
6	1587			6
7	1588	07640* 2527	SYSIO, KBDMON-1 /HANDLER ENTRY POINT; PRELOADED FOR BOOTUP.	7
8	1589	07641* 7240	NL7777 /SET AC TO 7777.	8
9	1590	07642* 1640	TAD I SYSIO /NOW HAVE BUFFER ADDRESS MINUS 1.	9
10	1591	07643* 3346	DCA SADDR /STASH IT FOR DATA TRANSFER LATER.	10
11	1592	07644* 6214	RDF /GET CALLING FIELD.	11
12	1593	07645* 1231	TAD S6203/(CIF CDF 00) /TURN IT INTO "CIF CDF <RETURN FIELD>".	12
13	1594	07646* 3342	DCA SRETFLD /STORE IN-LINE FOR EXIT LATER.	13
14	1595	07647* 2240	ISZ SYSIO /BUMP TO NEXT.	14
15	1596	07650* 1640	TAD I SYSIO /GET FUNCTION WORD.	15
16	1597	07651* 3352	DCA SFUNCT /SAVE FOR LATER.	16
17	1598	07652* 2240	ISZ SYSIO /BUMP TO NEXT.	17
18	1599			18
19	1600	/	THE SUBROUTINE HEADER NOW POINTS TO THE REQUESTED STARTING BLOCK. ALL DECTAPE	19
20	1601	/	OPERATIONS MUST SEARCH FOR THIS BLOCK IN THE FORWARD DIRECTION BEFORE READING	20
21	1602	/	OR WRITING [AS INDICATED IN THE FUNCTION WORD].	21
22	1603			22
23	1604	/	INITIAL SEARCH DIRECTION CONSIDERATIONS.	23
24	1605			24
25	1606	/	STATISTICALLY, MOST SEARCH OPERATIONS MUST BE INITIATED IN THE REVERSE	25
26	1607	/	DIRECTION FOR ONE OF THE FOLLOWING REASONS:	26
27	1608			27
28	1609	/	1) SEQUENTIAL TRANSFERS OFTEN INVOLVE PHYSICALLY ADJACENT BLOCKS. HOWEVER,	28
29	1610	/	SINCE THE PREVIOUS SUBROUTINE CALL STOPPED THE TAPE DRIVE, THE NEXT BLOCK	29
30	1611	/	CANNOT BE DIRECTLY ACCESSED FORWARD; THE TAPE ROLLS WELL PAST THE NEXT	30
31	1612	/	BLOCK DUE TO INERTIA. THE ONLY WAY TO TRANSFER THE ADJACENT BLOCK IS TO	31
32	1613	/	SEARCH IN REVERSE FOR THE BLOCK TWO HIGHER THAN REQUESTED, THEN TURN THE	32
33	1614	/	TAPE AROUND TO LOCATE THE REQUESTED BLOCK GOING FORWARD. [NOTE: THIS IS THE	33
34	1615	/	BEST POSSIBLE OUTCOME; THE DECTAPE CONTROLLER CAUSES A TWO BLOCK UNDERSHOOT	34
35	1616	/	BEFORE THE REQUESTED BLOCK CAN BE LOCATED WHILE SEARCHING FORWARD.]	35
36	1617			36
37	1618	/	2) MANY TAPE OPERATIONS NEED TO SEARCH FOR SYSTEM COMPONENTS LOCATED NEAR THE	37
38	1619	/	BEGINNING OF THE TAPE. THE LATEST REQUESTED BLOCK IS LIKELY MUCH LOWER	38
39	1620	/	THAN THE PRESENT TAPE POSITION.	39
40	1621			40
41	1622	/	MOST OPERATING SYSTEMS ALWAYS USE REVERSE SEARCH BECAUSE SIMPLISTIC SEARCH	41
42	1623	/	CALCULATION LIMITED TO ALWAYS USING ONE DIRECTION GREATLY FAVORS THE USE OF	42
43	1624	/	INITIAL REVERSE SEARCHES. HOWEVER, THERE IS ENOUGH CODE SPACE IN THE P?S/8	43
44	1625	/	TC01/TC08 SYSTEM DEVICE HANDLER TO CALCULATE FORWARD INITIAL SEARCHES WHEN	44
45	1626	/	BENEFICIAL. [NOTE: CONTRAST THIS WITH THE CLEARLY INFERIOR METHODOLOGY USED	45
46	1627	/	IN OS/8. THE CALLING PROGRAM CAN OPTIONALLY SPECIFY A FORWARD SEARCH BY	46
47	1628	/	SETTING BIT[11] OF THE FUNCTION WORD; HOWEVER, AS A PRACTICAL MATTER, THE	47
48	1629	/	CALLING PROGRAM GENERALLY HAS NO BASIS FOR DOING THIS. THE ENTIRE USAGE OF	48
49	1630	/	THIS FACILITY CONSISTS OF THREE HARD-WIRED CALLS SCATTERED THROUGHOUT OS/8	49
50	1631	/	WHERE THE CERTAINTY OF FORWARD INITIAL SEARCH BEING BENEFICIAL IS KNOWN ON A	50
51	1632	/	STATIC BASIS. THE ACTUAL OPPORTUNITIES FOR DYNAMIC SEARCH IMPROVEMENT IN ANY	51
52	1633	/	OPERATING SYSTEM ARE FAR GREATER IN NUMBER; THUS, OS/8'S PERFORMANCE IS	52
53	1634	/	NEEDLESSLY INEFFICIENT DUE TO THIS ILL-CONCEIVED IMPLEMENTATION.]	53
54				54
55				55
56				56
57				57
58				58

1	1636	/				CALCULATING THE INITIAL SEARCH DIRECTION.	1
2	1637						2
3	1638	/				EVERY OPERATION OF THE P?S/8 TC01/TC08 SYSTEM DEVICE HANDLER MAINTAINS THE	3
4	1639	/				LAST BLOCK READ OR WRITTEN [INCLUDING THE INITIAL BOOTUP OPERATION, A FAUX	4
5	1640	/				SELF-INITIATED SUBROUTINE CALL]. UPON THE NEXT CALL TO THE DEVICE HANDLER,	5
6	1641	/				THIS VALUE IS USED TO DETERMINE THE OPTIMUM INITIAL SEARCH DIRECTION.	6
7	1642						7
8	1643	/				AT THIS POINT, THE SUBROUTINE HEADER POINTS TO THE BLOCK NUMBER ARGUMENT.	8
9	1644	/				ONCE THE ENTIRE CALL HAS BEEN PROCESSED, THE HEADER WILL BE INCREMENTED TO	9
10	1645	/				POINT TO THE CALLER'S RETURN ADDRESS. [NOTE: THROUGHOUT ALL OPERATIONS OF THE	10
11	1646	/				TC01/TC08 DEVICE HANDLER, THE DATA FIELD IS SET TO THE CALLER'S DATA FIELD	11
12	1647	/				ACCORDING TO THE PREVAILING CALLING CONVENTIONS. LOCAL DATA WITHIN THE	12
13	1648	/				DEVICE HANDLER IS DIRECTLY ADDRESSED. OTHER DEVICE HANDLERS MAY HAVE MORE	13
14	1649	/				COMPLEX DATA FIELD LOGISTICS REQUIRING RESTORATION OF THE CALLER'S DATA FIELD	14
15	1650	/				ONCE THE DATA FIELD IS CHANGED TO EITHER THE CALLER'S BUFFER FIELD OR THE	15
16	1651	/				EXTENDED DEVICE HANDLER FIELD IF APPLICABLE.]	16
17	1652						17
18	1653	07653*	1640	TAD I	SYSIO	/GET THE DESIRED STARTING BLOCK NUMBER.	18
19	1654	07654*	7161	CLL CML	CMA IAC	/INVERT FOR COMPARISON; LINK SET UNLESS 0000.	19
20	1655	07655*	1345	SGETBLK,	TAD SBLK	/COMPARE TO LATEST BLOCK FROM PREVIOUS CALL.	20
21	1656	07656*	1344	TAD	SOVSHT	/ADD ON OVERSHOOT FACTOR.	21
22	1657						22
23	1658	/				THE LINK NOW CONTAINS THE INVERSE OF INTENDED INITIAL SEARCH DIRECTION.	23
24	1659						24
25	1660	/				LOGICAL CONSOLE OVERLAY MODIFICATIONS.	25
26	1661						26
27	1662	/				IF THE LOGICAL CONSOLE OVERLAY IS IN EFFECT, THE PREVIOUS TWO INSTRUCTIONS ARE	27
28	1663	/				REPLACED AS FOLLOWS:	28
29	1664						29
30	1665	/				*SGETBLK /OVER "TAD SBLK" INSTRUCTION AND FOLLOWING.	30
31	1666						31
32	1667	/		CIF	XX	/WILL BE "CIF" TO THE HIGHEST PHYSICAL FIELD.	32
33	1668	/		JMP	GETSBLK	/THIS IS ON PAGE ZERO OF THE CONSOLE OVERLAY.	33
34	1669						34
35	1670	/				THE CONSOLE OVERLAY CODE FOR THE TC01/TC08 DEVICE HANDLER CHANGES THE LOGICAL	35
36	1671	/				FLOW TO REFERENCE THE LATEST BLOCK VALUE ASSOCIATED WITH THE CURRENT LOGICAL	36
37	1672	/				TAPE DRIVE UNIT INSTEAD OF THE SHARED VALUE. A TABLE OF LATEST BLOCK VALUES	37
38	1673	/				IS MAINTAINED WITHIN THE OVERLAY FOR ALL POSSIBLE TAPE DRIVES. WHEN THE	38
39	1674	/				DEVICE HANDLER CODE IS REJOINED, THE APPROPRIATE LATEST BLOCK IS ADDED TO THE	39
40	1675	/				AC FOLLOWED BY CAREFULLY ADDING ON THE CUSTOM OVERSHOOT FACTOR. THIS	40
41	1676	/				ELIMINATES ANY POTENTIAL INITIAL CURRENT BLOCK CALCULATION MISHAPS ASSOCIATED	41
42	1677	/				WITH USING THE SHARED VALUE.	42
43	1678						43
44	1679	/				INITIAL TC01/TC08 DECTAPE CONTROLLER SETUP.	44
45	1680						45
46	1681	/				THE TC01/TC08 DECTAPE CONTROLLER TAPE DIRECTION BIT IS SET FOR REVERSE MOTION,	46
47	1682	/				CLEAR FOR FORWARD MOTION. THE LINK IS SET TO THE INVERSE OF INTENDED SEARCH	47
48	1683	/				DIRECTION. STATUS REGISTER "A" WILL BE LOADED WITH THE LOGICAL DRIVE UNIT	48
49	1684	/				BITS SET, FUNCTION 010 [NORMAL MODE SEARCH] AND THE INVERSE OF INTENDED	49
50	1685	/				INITIAL SEARCH DIRECTION [TAKEN FROM THE LINK]. THE MOTION AND	50
51	1686	/				INTERRUPT-ENABLE BITS WILL BE CLEAR. AC[10-11] WILL BE CLEAR TO RESET THE	51
52	1687	/				DECTAPE AND ERROR FLAGS. AT A LATER TIME, THE MOTION AND DIRECTION BITS WILL	52
53	1688	/				BE SET VIA AN .XOR. OPERATION WITHIN THE GENERAL FLAG WAIT SUBROUTINE. AS A	53
54	1689	/				RESULT, THE SELECTED TAPE DRIVE UNIT WILL BE SET IN MOTION IN THE INTENDED	54
55	1690	/				DIRECTION.	55
56							56
57							57
58							58

1	1692	/	THE LAST TRANSFERRED BLOCK [FROM THE PREVIOUS SUBROUTINE CALL] IS COMPARED TO			1
2	1693	/	THE LATEST REQUESTED BLOCK; AN OVERSHOOT FACTOR IS ADDED TO COMPENSATE FOR			2
3	1694	/	PRESUMED WORST-CASE INERTIAL ROLL. [NOTE: IT MUST BE ASSUMED THE TAPE DRIVE			3
4	1695	/	HAS COASTED TO A COMPLETE STOP AFTER THE PREVIOUS OPERATION WHICH CLEARED THE			4
5	1696	/	CONTROLLER MOTION BIT. CONTEMPLATING CASES WHERE THE TAPE IS POSSIBLY STILL			5
6	1697	/	IN MOTION CAN LEAD TO INSTANCES OF "SHOESHINE" EFFECT.]			6
7	1698					7
8	1699	/	SYSTEMS THAT USE TU56 DRIVES EXCLUSIVELY MIGHT ALLOW THE OVERSHOOT FACTOR TO			8
9	1700	/	BE LOWERED BY ONE BLOCK; HOWEVER, AS A PRACTICAL MATTER, LITTLE IS GAINED BY			9
10	1701	/	CHANGING THIS VALUE AS MOST SEQUENTIALLY-ISSUED CALLS ARE EITHER FOR THE NEXT			10
11	1702	/	PHYSICAL BLOCK OR FOR A BLOCK LOCATED SIGNIFICANTLY AWAY FROM THE PRESENT			11
12	1703	/	POSITION. THE PURPOSE OF THE OVERSHOOT FACTOR IS TO CREATE A DIVIDING LINE			12
13	1704	/	FOR ALL POTENTIAL CIRCUMSTANCES WHERE AN INITIAL FORWARD SEARCH WOULD APPLY.			13
14	1705	/	THIS IS ESTIMATED AT 20 PERCENT OF ALL ACTUAL CALLS FOR TYPICAL SYSTEM USAGE.			14
15	1706					15
16	1707	/	REPRESENTATIVE INITIAL SEARCH DIRECTION EXAMPLES; FORWARD SEARCH IF LINK SET.			16
17	1708					17
18	1709	/	PRV. BLOCK	REQ. BLOCK	L	18
19	1710					19
20	1711	/	0000	0001	0	20
21	1712					21
22	1713	/	0001	0000	0	22
23	1714					23
24	1715	/	0000	0000	0	24
25	1716					25
26	1717	/	0000	0010	0	26
27	1718					27
28	1719	/	0000	0011	1	28
29	1720					29
30	1721	/	0000	0012	1	30
31	1722					31
32	1723	/	2000	2001	0	32
33	1724					33
34	1725	/	2000	2010	0	34
35	1726					35
36	1727	/	2000	2011	1	36
37	1728					37
38	1729	/	2000	2012	1	38
39	1730					39
40	1731	/	2000	0000	0	40
41	1732					41
42	1733	/	2000	0001	0	42
43	1734					43
44	1735	/	ABSOLUTE BLOCKS NUMBERED LESS THAN OR EQUAL TO THE OVERSHOOT FACTOR WILL			44
45	1736	/	RESULT IN AN INITIAL REVERSE SEARCH [INCLUDING BLOCK 0000 WHICH IS PROPERLY			45
46	1737	/	HANDLED] WHEN THE TAPE IS LOCATED NEAR THE REVERSE END ZONE. INITIAL SEARCH			46
47	1738	/	DIRECTION FOR HIGHER BLOCKS WILL BE FORWARD IF THE REQUESTED BLOCK IS MORE			47
48	1739	/	THAN THE OVERSHOOT FACTOR FORWARD OF THE PRESENT TAPE POSITION.			48
49	1740					49
50	1741	/	ALL [LINTAPE OR] DECTAPE-BASED OPERATING SYSTEMS INHERENTLY HAVE SOME			50
51	1742	/	OPPORTUNITIES TO EXPLOIT INITIAL FORWARD SEARCH DIRECTION; DYNAMIC DETECTION			51
52	1743	/	WILL ALWAYS ACHIEVE THIS TO THE FULLEST EXTENT POSSIBLE.			52
53	1744					53
54	1745	/	APPENDIX "L" ILLUSTRATES A WAY TO TAKE ADVANTAGE OF FORWARD INITIAL SEARCH IN			54
55	1746	/	THE P?S/8 KEYBOARD MONITOR/EDITOR ENVIRONMENT FOR FASTER PROGRAM ASSEMBLY.			55
56						56
57						57
58						58

1	1748	/			THE INITIAL SEARCH CODE CONTINUES HERE WITH THE LINK SET ACCORDING TO THE	1
2	1749	/			CALCULATED INITIAL SEARCH DIRECTION [INVERTED].	2
3	1750					3
4	1751	/			ERROR RECOVERY CONSIDERATIONS.	4
5	1752					5
6	1753	/			GENERAL ERROR RECOVERY FOR ALL DETECTED ERRORS [EXCEPT END ZONE ERRORS] STARTS	6
7	1754	/			HERE. THE AC CONTAINS STATUS REGISTER "B" DATA [AND WILL BE CLEARED]; THE	7
8	1755	/			LINK CONTAINS THE MARK ERROR STATUS. REVERSE INITIAL SEARCH IS RECOMMENDED	8
9	1756	/			FOR MOST RECOVERY SITUATIONS. IF A FORWARD SEARCH WAS IN PROGRESS WHEN A MARK	9
10	1757	/			ERROR OCCURRED, THE INITIAL SEARCH WILL BE FORWARD. SINCE MARK ERRORS ARE	10
11	1758	/			[HOPEFULLY] UNLIKELY, YET MOSTLY SEEN DURING SEARCH OPERATIONS, THIS IS A GOOD	11
12	1759	/			OVERALL COMPROMISE. IN MOST CASES THE TAPE WILL CONTINUE TO MOVE TOWARDS THE	12
13	1760	/			TARGET BLOCK; WHEN THERE ARE MAINTENANCE ISSUES, THIS COULD LEAD TO A SERIES	13
14	1761	/			OF REPEATING MARK ERROR EVENTS. SITUATIONS LIKE THIS REQUIRE REMEDIATION	14
15	1762	/			BEFORE THE TAPE DEGRADES FURTHER AND THERE IS THE RISK OF PERMANENT DATA LOSS.	15
16	1763	/			IT IS HOPED THAT HAVING TO DEAL WITH THE OBVIOUS SEQUENCE OF REPEAT FAILURES	16
17	1764	/			WILL ALERT THE USER TO THE SERIOUS NATURE OF THE PROBLEM AT HAND BEFORE THE	17
18	1765	/			SITUATION ESCALATES TO MASSIVE RELIABILITY PROBLEMS.	18
19	1766					19
20	1767	/			NOTE: THE NEXT TWO INSTRUCTIONS INVERT THE LINK TWICE THUS PRESERVING THE	20
21	1768	/			PREVIOUS VALUE WHICH IS THE REVERSE OF INTENDED SEARCH DIRECTION.	21
22	1769					22
23	1770	07657* 7260	SRETRY, CLA CMA CML		/SET AC TO 7777 AND INVERT THE LINK.	23
24	1771	07660* 1230	TAD S10/(10)		/INVERT LINK AGAIN; AC NOW SET TO 0007.	24
25	1772	07661* 0352	AND SFUNCT		/ISOLATE UNIT SELECT BITS.	25
26	1773	07662* 7012	RTR;RTR		/MOVE UNIT SELECT BITS, OPPOSITE OF	26
27		07663* 7012				27
28	1775				/INTENDED DIRECTION BIT TO AC[0-3].	28
29	1776	07664* 1230	TAD S10/(10)		/ADD ON SEARCH FUNCTION BITS WITHOUT MOTION BIT.	29
30	1777					30
31	1778	/			AT THIS POINT THE SEARCH FUNCTION AND UNIT SELECT BITS ARE SET. THE REVERSE	31
32	1779	/			OF INTENDED INITIAL SEARCH DIRECTION BIT IS SET; THE MOTION BIT IS CLEAR. THE	32
33	1780	/			FUNCTION BITS ARE SET TO NORMAL SEARCH; INTERRUPT ENABLE IS CLEAR; THE ERROR	33
34	1781	/			AND DECTAPE FLAGS WILL BE CLEARED.	34
35	1782					35
36	1783	07665* 6766	DTLA		/LOAD INITIAL "A" REGISTER BITS, CLEAR AC.	36
37	1784					37
38	1786		IFNZRO ASMBUG <		/HISTORIC CODE ASSOCIATED WITH FIXABLE BUG.	38
39	1787					39
40	1788	07666* 6774	DTLB		/SET BLOCK MARK TRANSFER FIELD TO 0.	40
41	1789					41
42	1790		>		/END OF HISTORIC CODE.	42
43	1791					43
44	1792	07667* 1640	TAD I SYSIO		/GET THE DESIRED STARTING BLOCK NUMBER.	44
45	1793	07670* 3345	DCA SBLK		/SAVE AS LATEST BLOCK TO BE TRANSFERRED.	45
46	1794	07671* 1221	TAD S7750/(SRCHBLK)		/GET THE BLOCK SEARCH ADDRESS AND	46
47	1795	07672* 3355	DCA CA		/STORE IN CURRENT ADDRESS REGISTER.	47
48	1796					48
49	1797	/			AT THIS POINT THE STATUS REGISTER "A" FUNCTION BITS ARE SET TO NORMAL SEARCH.	49
50	1798	/			THE SELECTED UNIT BITS ARE SET. THE MOTION BIT IS CLEAR AND THE DIRECTION BIT	50
51	1799	/			IS SET TO THE OPPOSITE OF INTENDED INITIAL SEARCH DIRECTION. AS THIS CODE	51
52	1800	/			CONTINUES BELOW, AC[3-4] WILL BE SET TO ENABLE MOTION AND REVERSE THE	52
53	1801	/			DIRECTION TO THE DESIRED VALUE.	53
54						54
55						55
56						56
57						57
58						58

1	1803	7673	STURNIT=.	/COMES HERE TO REVERSE DIRECTION WHEN	1
2	1804			/AN END ZONE ERROR IS DETECTED.	2
3	1805				3
4	1806	/	THIS IS ALSO THE JUNCTION POINT WHEN AN END ZONE ERROR IS DETECTED IN THE		4
5	1807	/	GENERAL WAIT ROUTINE. THE SAME OVERALL CONSIDERATIONS APPLY BECAUSE ALL ERROR		5
6	1808	/	CONDITIONS CLEAR THE MOTION BIT. THE ONLY DIFFERENCE IS THE REASON FOR		6
7	1809	/	DIRECTION REVERSAL; IN THIS INSTANCE, THE GOAL IS TO MOVE THE TAPE OUT OF THE		7
8	1810	/	CURRENT END ZONE AND TOWARDS THE CENTER OF THE TAPE TO ALLOW BLOCK MARKS TO BE		8
9	1811	/	DETECTED. [NOTE: BY DESIGN, END ZONE MARKS ARE NOT DETECTED AS THE TAPE LEAVES		9
10	1812	/	THE END ZONE.] THE FIRST DETECTED BLOCK MARK WILL BE EITHER 0000 IF LEAVING		10
11	1813	/	THE REVERSE END ZONE, OR 2701 [OR OTHER HIGHEST BLOCK NUMBER] IF LEAVING THE		11
12	1814	/	FORWARD END ZONE.		12
13	1815				13
14	1816	/	NOTE: THIS ROUTINE IS ENTERED FROM THE GENERAL FLAG WAIT ROUTINE UPON		14
15	1817	/	DETECTION OF AN END ZONE ERROR. THE AC WILL BE CLEAR [IN CASE THIS MATTERS].		15
16	1818				16
17	1819		IFZERO ASMBUG <	/START OF BUG FIX CODE	17
18					18
19		/	BLOCK MARKS MUST BE TRANSFERRED INTO FIELD 0. THE TRANSFER FIELD COULD BE		19
20		/	SET TO THE VALUE ASSOCIATED WITH THE CALLER'S BUFFER WHICH COULD BE ANY FIELD		20
21		/	FROM 0 THROUGH 7.		21
22					22
23		/	THIS SITUATION IS HIGHLY UNLIKELY, BUT DESERVES BEING FIXED. THERE IS NO NET		23
24		/	CHANGE IN PROGRAM SIZE; A "DTLB" INSTRUCTION IS BEING MOVED HERE; NO NEW		24
25		/	INSTRUCTIONS ARE ADDED.		25
26					26
27		/	AC[6-8] MUST BE CLEAR TO SET THE TRANSFER FIELD TO 0 FOR DETECTED BLOCK MARKS		27
28		/	DURING ALL SEARCH OPERATIONS.		28
29					29
30			DTLB	/LOAD THE FIELD BITS IN STATUS REGISTER "B".	30
31					31
32	1820	>		/END OF BUG FIX CODE	32
33	1821				33
34	1822	/	THE MOTION BIT IS OFF AT THIS POINT FOR ONE OF THE FOLLOWING REASONS:		34
35	1823				35
36	1824	/	1) THE MAIN LINE CODE HAS SET THE REVERSE OF INTENDED SEARCH DIRECTION WITH		36
37	1825	/	THE MOTION BIT CLEAR.		37
38	1826				38
39	1827	/	2) AN END ZONE ERROR WAS DETECTED; THE DIRECTION MUST BE REVERSED TO MOVE OUT		39
40	1828	/	OF THE END ZONE; ALL ERRORS CLEAR THE MOTION BIT.		40
41	1829				41
42	1830	/	THE LINK WILL BE CLEARED AT THIS POINT; IT WILL BE COMPLEMENTED AND MOVED TO		42
43	1831	/	AC[3].		43
44	1832				44
45	1833	/	AC[0] WILL BE SET AT THIS POINT; IT WILL BE MOVED TO AC[4].		45
46	1834				46
47	1835	/	THE CORRESPONDING BITS IN STATUS REGISTER "A" WILL BE SET BY AN .XOR. TRANSFER		47
48	1836	/	IN THE FLAG WAIT ROUTINE. THE DIRECTION WILL BE REVERSED AND DRIVE MOTION		48
49	1837	/	INITIATED IN THE INTENDED DIRECTION.		49
50	1838				50
51	1839	07673* 7330	NL4000	/CLEAR THE LINK AND SET AC[0]; THE SEARCH	51
52	1840			/OPERATION WILL BE INITIATED BELOW.	52
53					53
54					54
55					55
56					56
57					57
58					58

1	1842	7674	SNXTBLK=.	/THE SEARCH ROUTINE LOOPS HERE.	1
2	1843				2
3	1844	/	THIS IS THE ENTRY POINT FOR CONTINUING AN ONGOING SEARCH OPERATION. THE LINK		3
4	1845	/	WILL BE SET TO MAINTAIN THE CURRENT DIRECTION OR CLEAR TO REVERSE DIRECTION.		4
5	1846				5
6	1847	/	THERE ARE FOUR POSSIBILITIES:		6
7	1848				7
8	1849	/	1) FORWARD SEARCH IN PROGRESS - THE TAPE IS MOVING FORWARD TOWARDS THE TARGET		8
9	1850	/	BLOCK. THE LINK IS SET TO MAINTAIN FORWARD MOTION.		9
10	1851				10
11	1852	/	2) REVERSE SEARCH IN PROGRESS - THE TAPE IS MOVING IN THE REVERSE DIRECTION		11
12	1853	/	TOWARDS THE BLOCK TWO HIGHER THAN THE TARGET BLOCK. THE LINK IS SET TO		12
13	1854	/	MAINTAIN REVERSE MOTION.		13
14	1855				14
15	1856	/	3) REVERSE SEARCH ENDED - THE TARGET BLOCK PLUS TWO HAS BEEN LOCATED WHILE		15
16	1857	/	MOVING IN THE REVERSE DIRECTION. THE LINK IS CLEAR; THE TAPE WILL BE		16
17	1858	/	TURNED AROUND.		17
18	1859				18
19	1860	/	4) FORWARD SEARCH TOO FAR - THE TAPE IS MOVING FORWARD OF THE TARGET BLOCK.		19
20	1861	/	THE LINK IS CLEAR; THE TAPE WILL BE TURNED AROUND. [NOTE: THIS CASE IS NOT		20
21	1862	/	LIKELY TO OCCUR UNLESS THE TAPE IS MANUALLY REPOSITIONED OR THERE WAS AN		21
22	1863	/	ANOMALY INVOLVING DRIVE UNIT CHANGE FROM THAT USED IN THE PREVIOUS CALL.]		22
23	1864				23
24	1865	/	AC[0] IS SET IF THE MOTION BIT MUST BE INVERTED [FROM CLEAR TO SET]. THE LINK		24
25	1866	/	WILL BE INVERTED [AND POSSIBLY USED TO REVERSE DIRECTION IF CONDITIONS		25
26	1867	/	INDICATE THE TAPE SHOULD BE TURNED AROUND].		26
27	1868				27
28	1869	07674* 7032	CML RTR;RTR	/INVERT THE DIRECTION BIT IN THE LINK AND	28
29		07675* 7012			29
30	1871			/MOVE DIRECTION AND MOTION BITS TO AC[3-4].	30
31	1872				31
32	1873	/	AC[3] IS SET IF THE CURRENT DIRECTION IS TO BE REVERSED. AC[4] IS SET IF TAPE		32
33	1874	/	MOTION SHOULD BE ENABLED FOR A NEW SEARCH OPERATION [OR AFTER AN END ZONE		33
34	1875	/	ERROR].		34
35	1876				35
36	1877	/	NOTE: IF THE TAPE DIRECTION IS BEING REVERSED, TWO BLOCKS WILL BE IGNORED		36
37	1878	/	BEFORE THE NEXT BLOCK MARK IS DETECTED. [THIS IS WHY THE REVERSE SEARCH MODE		37
38	1879	/	ENDS TWO BLOCKS HIGHER THAN THE TARGET BLOCK; WHEN THE TAPE TURNS AROUND, THE		38
39	1880	/	FIRST BLOCK MARK FOUND WILL BE FOR THE TARGET BLOCK WHILE MOVING FORWARD.]		39
40	1881				40
41	1882	07676* 4212	JMS SYSWAIT	/[POTENTIALLY] UPDATE DIRECTION AND MOTION	41
42	1883			/BITS, WAIT FOR THE LATEST BLOCK MARK.	42
43	1884				43
44	1885	/	THE CURRENT TAPE DIRECTION MUST BE READ FROM STATUS REGISTER "A" TO DETERMINE		44
45	1886	/	WHAT TO DO NEXT. [FORWARD MOTION IS HANDLED DIFFERENTLY FROM REVERSE MOTION.]		45
46	1887				46
47	1888	07677* 6761	DTRA	/GET STATUS REGISTER "A".	47
48	1889	07700* 7046	CMA RTL;RTL	/INVERT AND MOVE CURRENT TAPE DIRECTION TO LINK.	48
49		07701* 7006			49
50	1891	07702* 7620	SNL CLA	/SKIP IF MOVING FORWARD; THE LINK IS SET.	50
51	1892	07703* 7026	CML RTL	/SET AC TO 0002 SO THE REVERSE SEARCH ENDS AT	51
52	1893			/THE BLOCK TWO BLOCKS HIGHER THAN THE FINAL	52
53	1894			/TARGET BLOCK; THE LINK REMAINS CLEAR.	53
54					54
55					55
56					56
57					57
58					58

1	1896	07704* 1350	TAD	SRCHBLK	/NOW HAVE THE LATEST BLOCK [POSSIBLY ADJUSTED].	1
2	1897	07705* 7061	CML	CIA	/INVERT THE LINK AND NEGATE AC FOR TESTING.	2
3	1898					3
4	1899	/			NOTE: THE LINK IS NOW CLEAR IF MOVING FORWARD [UNLESS LOCATED AT BLOCK 0000].	4
5	1900					5
6	1901	07706* 1345	TAD	SBLK	/COMPARE TO TARGET BLOCK.	6
7	1902					7
8	1903	/			IF THE TAPE IS MOVING FORWARD AT A BLOCK LOWER THAN THE TARGET BLOCK, THE LINK	8
9	1904	/			IS NOW SET; FORWARD MOTION WILL BE MAINTAINED WHILE WAITING FOR ADDITIONAL	9
10	1905	/			BLOCK MARKS.	10
11	1906					11
12	1907	/			IF THE TAPE IS NOW MOVING IN THE REVERSE DIRECTION AT A BLOCK HIGHER THAN THE	12
13	1908	/			TARGET BLOCK [PLUS TWO], THE LINK IS NOW SET. REVERSE TAPE DIRECTION WILL BE	13
14	1909	/			MAINTAINED WHILE WAITING FOR ADDITIONAL BLOCK MARKS.	14
15	1910					15
16	1911	/			IF THE CURRENT BLOCK MATCHES THE TARGET BLOCK, THE LINK REVERSED FROM THE	16
17	1912	/			CALCULATION [OR WAS ALREADY SET IF BOTH VALUES ARE 0000]. THE LINK IS NOW SET	17
18	1913	/			IF THE TAPE IS LOCATED AT THE TARGET BLOCK MOVING FORWARD. [NOTE: BLOCKS LOWER	18
19	1914	/			THAN 0002 WILL BE LOCATED GOING FORWARD OUT OF THE END ZONE.]	19
20	1915					20
21	1916	/			CHECK TO SEE IF AT THE TARGET BLOCK.	21
22	1917					22
23	1918	07707* 7650	SNA	CLA	/SKIP DIRECTION CHECK IF NOT A MATCH.	23
24	1919	07710* 7620	SNL	CLA	/CHECK IF AT THE TARGET BLOCK GOING FORWARDS.	24
25	1920	07711* 5274	JMP	SNXTBLK	/GO WAIT FOR ANOTHER BLOCK; THE LINK	25
26	1921				/DETERMINES IF THE DIRECTION WILL BE CHANGED.	26
27	1922					27
28	1923	/			AT THIS POINT, THE TAPE IS MOVING FORWARD AND POSITIONED AT THE DESIRED TAPE	28
29	1924	/			BLOCK. THE CURRENT ADDRESS REGISTER WILL BE SET TO THE CALLER'S BUFFER	29
30	1925	/			ADDRESS LESS ONE. AS EACH BLOCK IS READ OR WRITTEN, THE CURRENT ADDRESS	30
31	1926	/			REGISTER WILL AUTO-INCREMENT FOR EACH WORD TRANSFERRED.	31
32	1927					32
33	1928	07712* 1346	TAD	SADDR	/GET TRANSFER ADDRESS AND STUFF IT	33
34	1929	07713* 3355	DCA	CA	/INTO THE CURRENT ADDRESS REGISTER.	34
35	1930					35
36	1931	/			THE TC01/TC08 DECTAPE HARDWARE REQUIRES SETTING THE TRANSFER FIELD IN	36
37	1932	/			BITS[6-8] OF STATUS REGISTER "B" WHICH IS THE SAME RELATIVE POSITION AS THE	37
38	1933	/			CORRESPONDING BITS USED IN THE "CIF" AND "CDF" INSTRUCTIONS. THIS TENDS TO	38
39	1934	/			SHORTEN THE CODE CONCERNED WITH SETTING STATUS REGISTER "B" AS THE FIELD BITS	39
40	1935	/			FOR THE CALL ARE ALREADY POSITIONED IN THE PROPER PORTION OF THE FUNCTION	40
41	1936	/			WORD. FOR DEVICE HANDLERS THAT USE PROGRAM TRANSFER INSTEAD OF DATA BREAK,	41
42	1937	/			THIS FACILITATES SETTING UP "CDF" INSTRUCTIONS TO THE CALLER'S BUFFER FIELD.	42
43	1938					43
44	1939	/			THE R-L MONITOR SYSTEM DOES NOT SUPPORT EXTENDED MEMORY TRANSFERS. THUS, THIS	44
45	1940	/			IS AN ENTIRELY NEW FEATURE OF P?S/8. ALL R-L MONITOR CODE SETS STATUS	45
46	1941	/			REGISTER "B" TO FIELD 0 FOR THE BENEFIT OF THE SEARCH ROUTINE; NO CHANGE IS	46
47	1942	/			MADE WHEN DATA TRANSFERS OCCUR.	47
48	1943					48
49	1944	/			NOTE: THE IDENTICAL METHOD IS IMPLEMENTED IN DEVICE HANDLERS FOR PS/8-OS/8	49
50	1945	/			[AND COS-300/310] WHICH USE A SOMEWHAT SIMILAR FUNCTION WORD ARGUMENT IN THE	50
51	1946	/			ANALOGOUS DEVICE HANDLER CALLS.	51
52	1947					52
53	1948	07714* 1352	TAD	SFUNCT	/GET THE FUNCTION WORD [ONLY BITS[6-8] USED].	53
54	1949	07715* 6774	DTLB		/LOAD THE TRANSFER FIELD. NOTE: CLEARS AC.	54
55						55
56						56
57						57
58						58

1	1951	/			IN THE ORIGINAL R-L MONITOR SYSTEM, THE PAGE COUNT IS PASSED AS -[NUMBER OF	1
2	1952	/			BLOCKS TO TRANSFER]. IN P?S/8 [AND OS/8 ET AL.] BITS[1-5] OF THE FUNCTION	2
3	1953	/			WORD SERVE THIS PURPOSE; ADDITIONALLY, AN ENTIRE MEMORY FIELD CAN BE SPECIFIED	3
4	1954	/			BY SETTING THESE BITS TO ZERO. [NOTE: THIS IS MOOT IN THE R-L MONITOR SYSTEM	4
5	1955	/			AS EXTENDED MEMORY IS NOT SUPPORTED.]	5
6	1956					6
7	1957	07716*	1352	TAD	SFUNCT /GET THE FUNCTION WORD AGAIN	7
8	1958	07717*	7004	RAL	/SHIFT LEFT; READ/WRITE BIT TO LINK	8
9	1959	07720*	0201	AND	S7600/(7600) /ISOLATE PAGE COUNT BITS	9
10	1960	07721*	3351	DCA	SBLKCT /STORE FOR COUNTDOWN LATER	10
11	1961					11
12	1962	/			THE PAGE COUNT SHIFTED LEFT ONE BIT WILL RANGE FROM 0200 FOR A SINGLE PAGE	12
13	1963	/			TRANSFER THROUGH 7600 FOR A 31 PAGE TRANSFER; 0000 INDICATES A 32 PAGE	13
14	1964	/			[ENTIRE MEMORY FIELD] TRANSFER. THIS VALUE WILL BE COUNTED DOWN BY 200 PER	14
15	1965	/			TRANSFERRED BLOCK UNTIL ALL BLOCKS ARE PROCESSED.	15
16	1966					16
17	1967	/			DEPENDING ON EXACT CODING REQUIREMENTS, THIS SECTION MIGHT BE REWORKED INTO A	17
18	1968	/			MORE CONVENTIONAL "ISZ" INSTRUCTION COUNTING LOOP SET TO -[NUMBER OF PAGES TO	18
19	1969	/			TRANSFER]. HOWEVER, THE ALL-ZEROS CASE MUST BE PROPERLY DEALT WITH. THE CODE	19
20	1970	/			ALSO HAS TO PRESERVE THE READ/WRITE BIT IN THE LINK OR PROCESS THAT IN	20
21	1971	/			SEPARATE [AND LONGER] CODE. IT WOULD APPEAR THE CODE USED ABOVE OCCUPIES THE	21
22	1972	/			LEAST OVERALL CODE SPACE INCLUDING THE EASILY OBTAINED CONSTANT 7600. [NOTE:	22
23	1973	/			ALTERNATE METHODS MIGHT REQUIRE ONE OR MORE UNIQUE CONSTANTS AS PART OF THE	23
24	1974	/			COMPLETE IMPLEMENTATION LEADING TO A NET LARGER RESULT.]	24
25	1975					25
26	1976	/			SETTING THE CONTROLLER FOR DATA TRANSFER.	26
27	1977					27
28	1978	/			AT THIS POINT THE CONTROLLER MUST BE PUT IN READ MODE OR WRITE MODE AS	28
29	1979	/			DETERMINED BY THE HIGH-ORDER BIT OF THE FUNCTION WORD [CURRENTLY IN THE LINK].	29
30	1980					30
31	1981	/			STATUS REGISTER "A" UPDATES ARE PERFORMED BY .XOR. TRANSFERS IN THE GENERAL	31
32	1982	/			FLAG WAIT ROUTINE. THE BITS OF THE AC MUST BE SET AS FOLLOWS:	32
33	1983					33
34	1984	/			BITS[0-2] MUST BE 000 TO MAINTAIN THE CURRENT UNIT SELECT BITS.	34
35	1985					35
36	1986	/			BITS[3-4] MUST BE 00 TO MAINTAIN FORWARD TAPE MOTION.	36
37	1987					37
38	1988	/			BIT[5] MUST BE 0 TO MAINTAIN NORMAL MODE.	38
39	1989					39
40	1990	/			BITS[6-8] MUST BE SET TO 011 [FOR READ] OR 101 [FOR WRITE].	40
41	1991					41
42	1992	/			BIT[9] MUST BE 0 TO MAINTAIN CONTROLLER INTERRUPTS OFF.	42
43	1993					43
44	1994	/			BIT[10] IS ARBITRARY AS THERE ARE NO ERRORS AFTER A SUCCESSFUL SEARCH.	44
45	1995					45
46	1996	/			BIT[11] MUST BE 0 TO CLEAR THE DECTAPE FLAG.	46
47	1997					47
48	1998	07722*	7004	RAL	/MOVE READ/WRITE BIT TO AC[11]	48
49	1999	07723*	7001	IAC	/0001 IF READING, 0002 IF WRITING	49
50	2000	07724*	7126	CLL CML RTL	/0006 IF READING, 0012 IF WRITING	50
51	2001	07725*	7006	RTL	/0030 IF READING, 0050 IF WRITING	51
52	2002					52
53	2003	/			THE AC IS NOW SET CORRECTLY TO CHANGE OPERATING MODES. [NOTE: SUBSEQUENT CALLS	53
54	2004	/			SHOULD BE MADE WITH THE AC CLEAR TO MAINTAIN THE CURRENT MODE.]	54
55						55
56						56
57						57
58						58

1	2006	07726*	4212	SBLKLP, JMS	SYSWAIT	/WAIT FOR A BLOCK TO BE READ OR WRITTEN; THE	1
2	2007					/WORD COUNT REGISTER IS SET TO -200 EACH TIME.	2
3	2008						3
4	2009			/	NOTE: COMES HERE DURING THE BOOTUP PROCESS AFTER THE DECTAPE FLAG RAISES.		4
5	2010						5
6	2011	07727*	2345	SBLKRET, ISZ	SBLK	/BUMP LATEST BLOCK.	6
7	2012	07730*	7320	CLA CLL CML		/CLEAR STATUS REGISTER "B" JUNK; SET LINK NOW.	7
8	2013	07731*	1351	TAD	SBLKCT	/GET REMAINING WORD COUNT.	8
9	2014	07732*	1201	TAD	S7600/(-200)	/SUBTRACT 200; WILL CLEAR LINK WHEN 0000.	9
10	2015	07733*	7450	SNA		/SKIP IF MORE BLOCKS TO PROCESS.	10
11	2016	07734*	5337	JMP	SDONE	/JUMP IF DONE.	11
12	2017	07735*	3351	DCA	SBLKCT	/STORE BACK UPDATED VALUE FOR NEXT TIME.	12
13	2018	07736*	5326	JMP	SBLKLP	/GO TRANSFER ANOTHER BLOCK.	13
14	2019						14
15	2020			/	COMES HERE TO STOP THE TAPE AFTER ALL BLOCKS ARE TRANSFERRED.		15
16	2021						16
17	2022			/	NOTE: OVER THE YEARS THIS SECTION HAS BEEN REWRITTEN SEVERAL TIMES. THE		17
18	2023			/	OVERHEAD ASSOCIATED WITH PRESERVING THE DRIVE UNIT BITS PRECLUDES THE USE OF		18
19	2024			/	THE DTLA OPERATION TO UPDATE STATUS REGISTER "A". IT IS NOT NECESSARY TO		19
20	2025			/	CHANGE THE CONTROLLER FUNCTION AS LONG AS THE DIRECTION BIT IS MAINTAINED AND		20
21	2026			/	THE MOTION BIT IS CLEARED. THIS WILL BE IMPLEMENTED USING THE DTXA		21
22	2027			/	INSTRUCTION TO UPDATE STATUS REGISTER "A" BY AN .XOR. OPERATION.		22
23	2028						23
24	2029			/	ISSUES ASSOCIATED WITH STATUS REGISTER "A" UPDATE ARE AS FOLLOWS:		24
25	2030						25
26	2031			/	1) AC[0-2] MUST BE ZERO TO PREVENT THE UNIT BITS FROM CHANGING WHEN TURNING		26
27	2032			/	OFF THE MOTION BIT. UNEXPECTED PROBLEMS OCCUR WHEN CHANGING TOO MANY		27
28	2033			/	BITS AT ONCE.		28
29	2034						29
30	2035			/	2) AC[3] MUST BE ZERO TO PREVENT TURNING THE TAPE AROUND WHICH CAN LEAD TO		30
31	2036			/	UNEXPECTED PROBLEMS, POSSIBLY AFFECTING THE NEXT HANDLER CALL.		31
32	2037						32
33	2038			/	3) AC[4] MUST BE SET TO TURN OFF THE MOTION BIT WHILE HOLDING MOST OF THE		33
34	2039			/	OTHER BITS CONSTANT.		34
35	2040						35
36	2041			/	4) AC[5] IS ARBITRARY AS CONTINUOUS MODE IS IGNORED DURING MOTION COMMANDS.		36
37	2042						37
38	2043			/	5) AC[6-8] MUST BE 000 TO MAINTAIN THE PREVIOUS [READ OR WRITE] COMMAND.		38
39	2044			/	ALTERNATIVELY, THE COMMAND CAN BE CHANGED TO MOVE [000] WHICH REQUIRES		39
40	2045			/	PRIOR KNOWLEDGE OF BITS[6-8] OF STATUS REGISTER "A".		40
41	2046						41
42	2047			/	6) AC[9] SHOULD BE ZERO TO AVOID INTERRUPT-RELATED PROBLEMS.		42
43	2048						43
44	2049			/	7) AC[10] IS ARBITRARY AS THE ERROR FLAG IS KNOWN TO BE CLEAR.		44
45	2050						45
46	2051			/	8) AC[11] SHOULD BE ZERO TO CLEAR THE DECTAPE FLAG.		46
47	2052						47
48	2053			/	ALL THINGS CONSIDERED, SETTING THE AC TO 0200 IS THE PREFERRED METHOD. IF		48
49	2054			/	NECESSARY, ALTERNATE VALUES CAN BE CONSTRUCTED HEEDING THE ABOVE.		49
50	2055						50
51	2056	07737*	1202	SDONE, TAD	S200/(200)	/SET THE MOTION BIT [TO CLEAR IT].	51
52	2057	07740*	6764	DTXA		/STOP THE TAPE DRIVE; THE AC IS NOW CLEAR.	52
53							53
54							54
55							55
56							56
57							57
58							58

1	2059	/	LOGICAL CONSOLE OVERLAY MODIFICATIONS.	1
2	2060			2
3	2061	/	IF THE LOGICAL CONSOLE OVERLAY IS IN EFFECT, THE PREVIOUS TWO INSTRUCTIONS ARE	3
4	2062	/	REPLACED AS FOLLOWS:	4
5	2063			5
6	2064	/	*SDONE /OVER THE "TAD S200" INSTRUCTION AND FOLLOWING.	6
7	2065			7
8	2066	/	CIF XX /WILL BE "CIF" TO THE HIGHEST PHYSICAL FIELD.	8
9	2067	/	JMP ENDTRAP /THIS IS ON PAGE ZERO OF THE CONSOLE OVERLAY.	9
10	2068			10
11	2069	/	THE LOGICAL CONSOLE OVERLAY MUST CAREFULLY STORE THE LATEST BLOCK TRANSFERRED	11
12	2070	/	IN THE APPROPRIATE ELEMENT OF A TABLE OF LATEST BLOCK TRANSFERS PARTICULAR TO	12
13	2071	/	THE CURRENT DRIVE UNIT. THIS WILL BE USED ONLY ON SUBSEQUENT CALLS FOR THE	13
14	2072	/	SAME DRIVE UNIT; THIS WILL AVOID INITIAL SEARCH DIRECTION MISHAPS WHEN	14
15	2073	/	SWITCHING DRIVE UNITS.	15
16	2074			16
17	2075	07741* 2240	ISZ SYSIO /BUMP PAST THE BLOCK ARGUMENT.	17
18	2076	07742* 6203 SRETFLD,CIF CDF 00+.-.	/WILL BE "CIF CDF" TO CALLING FIELD; MUST BE	18
19	2077		/PRELOADED WITH "CIF CDF 00" FOR BOOTUP.	19
20	2078	07743* 5640	JMP I SYSIO /RETURN TO CALLER AFTER ALL IN-LINE ARGUMENTS.	20
21	2079			21
22	2080	07744* 0010 SOVSHT, 10	/CUSTOM OVERSHOOT FACTOR FOR DYNAMIC SEARCH	22
23	2081		/INITIAL DIRECTION CALCULATION. THIS VALUE	23
24	2082		/MIGHT NEED TWEAKING; NEWER VERSIONS OF THIS	24
25	2083		/CODE MAY CHANGE THIS SLIGHTLY.	25
26	2084			26
27	2085	07745* 0000 SBLK, BOOTBLK	/LATEST BLOCK TRANSFERRED; PRELOADED FOR	27
28	2086		/BOOTUP AND SLURP LOADER OVERLAY.	28
29	2087			29
30	2088	/	NOTE: ALL LOCATIONS ON THE REST OF THE PAGE ARE NONESSENTIAL FOR DEVICE	30
31	2089	/	HANDLER OR SYSTEM RESTART OPERATIONS WHEN THE SLURP LOADER OVERLAY IS	31
32	2090	/	PERFORMED TO RELOAD THE CONTENTS OF BLOCK 0000. THESE FUNCTIONS WILL WORK	32
33	2091	/	PROPERLY AS LONG AS ALL LOCATIONS FROM 07600 THROUGH THIS POINT ARE CORRECTLY	33
34	2092	/	LOADED; THIS ALLOWS SLURP LOADER STARTUP INFORMATION TO BE STORED IN MEMORY	34
35	2093	/	LOCATIONS DEFINED BELOW.	35
36	2094			36
37	2095	07746* 0000 SADDR, .-.	/BUFFER ADDRESS [-1] STORED HERE.	37
38	2096			38
39	2097	07747* 0000	ZBLOCK 1 /EMPTY SPACE!	39
40				40
41				41
42				42
43				43
44				44
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46				46
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1	2099	/	THREE-CYCLE DATA BREAK OVERLAP CONSIDERATIONS.	1
2	2100			2
3	2101	/	ON MACHINES THAT SUPPORT THREE-CYCLE DATA BREAK, THERE ARE THEORETICAL ISSUES	3
4	2102	/	REGARDING OVERLAPPED DEVICE ACTIVITY:	4
5	2103			5
6	2104	/	1) IF AN RF08 OR DF32 DISK SYSTEM IS PRESENT, LOCATIONS 07750-07751 ARE	6
7	2105	/	RESERVED.	7
8	2106			8
9	2107	/	2) IF A TC58 SERIES OR EQUIVALENT MAGNETIC TAPE CONTROLLER IS PRESENT,	9
10	2108	/	LOCATIONS 07752-07753 ARE RESERVED.	10
11	2109			11
12	2110	/	TO PREVENT INTERACTION PROBLEMS, DO NOT USE 07750-07753 FOR ANY DATA THAT MUST	12
13	2111	/	HAVE CONSTANT VALUE. AS WRITTEN CURRENTLY, THIS HANDLER DOES NOT VIOLATE THIS	13
14	2112	/	RULE AS ALL LOCATIONS IN THIS RANGE ARE ESSENTIALLY TEMPORARY STORAGE. ANY	14
15	2113	/	NEW CALL TO THE SYSTEM DEVICE HANDLER WILL INITIALIZE THE REFERENCED LOCATIONS	15
16	2114	/	AS NECESSARY.	16
17	2115			17
18	2116	/	HOWEVER, THIS WILL NOT FUNCTION CORRECTLY IF ONE OF THE OTHER DEVICES IS	18
19	2117	/	ACTUALLY PERFORMING DATA BREAK OPERATIONS SIMULTANEOUSLY. THIS COULD ONLY BE	19
20	2118	/	AN ISSUE IF SOME INTERRUPT-DRIVEN REAL-TIME MULTITASKING APPLICATION IS	20
21	2119	/	CONTEMPLATED. MANY HANDLERS ARE NOT CAPABLE OF SUCH OPERATION DUE TO THE	21
22	2120	/	UNCERTAINTY OF THE CONTENTS OF ANY OF THESE LOCATIONS. AS SUCH, A	22
23	2121	/	LOW-PRIORITY CONSIDERATION OF HANDLER DESIGN IS TO MINIMIZE THE CONSEQUENCES	23
24	2122	/	OF THIS THEORETICAL EXPOSURE. FUTURE RELEASES OF THIS HANDLER MAY ADDRESS	24
25	2123	/	THIS ISSUE IN A MORE EFFECTIVE MANNER THAN THE PRESENT CODE AS DOCUMENTED	25
26	2124	/	HERE.	26
27	2125			27
28	2126	/	DATA BREAK RESERVED LOCATIONS FOLLOW; PLACE ONLY TEMPORARIES HERE TO AVOID	28
29	2127	/	DATA CORRUPTION CAUSED BY ANOTHER DEVICE THAT USES THESE LOCATIONS.	29
30	2128			30
31	2129		IFNZRO 7750-. <ERROR .>	31
32	2131			32
33	2132	07750* 0000	SRCHBLK, .-. /BLOCK MARKS TRANSFER HERE IN SEARCH MODE.	33
34	2133	07751* 4000	SBLKCT, 4000 /BLOCK COUNT STORED HERE; PRELOADED FOR BOOTUP.	34
35	2134	07752* 2000	SFUNCT, 20^100 /FUNCTION WORD; PRELOADED FOR BOOTUP.	35
36	2135			36
37	2136	07753* 0000	ZBLOCK WC-. /GET PAST EMPTY SPACE IN THE DATA BREAK AREA.	37
38	2137			38
39	2138	/	THE FOLLOWING TWO LOCATIONS ARE EXCLUSIVELY RESERVED FOR TC01/TC08 DECTAPE USE	39
40	2139	/	ONLY. [NOTE: OTHER DEVICES MUST NOT STORE DATA INTO THESE LOCATIONS AS NORMAL	40
41	2140	/	DECTAPE OPERATIONS CHANGE THE CONTENTS.]	41
42	2141			42
43	2142	07754* 7755	WC, CA /WORD COUNT REGISTER; PRELOADED FOR BOOTUP.	43
44	2143	07755* 7755	CA, CA /CURRENT ADDRESS REGISTER; PRELOADED FOR BOOTUP.	44
45	2144			45
46	2145	/	END OF LOCATIONS RESERVED FOR 3-CYCLE DATA BREAK.	46
47				47
48				48
49				49
50				50
51				51
52				52
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1	2147	/	ADDITIONAL SYSTEM PARAMETERS.	1
2	2148			2
3	2149	/	THE REST OF THIS PAGE [WHEN LOADED INTO 07600-07777] IS USED TO PASS VARIOUS	3
4	2150	/	ADDITIONAL OPTIONAL PARAMETERS. DURING THE BOOTUP PROCESS, THESE LOCATIONS	4
5	2151	/	MAY PLAY A ROLE IN PLACING PART OF THE SYSTEM DEVICE HANDLER INTO DESIGNATED	5
6	2152	/	MEMORY LOCATIONS; THE PERMANENT CONTENTS WILL BE SET ACCORDINGLY DEPENDING ON	6
7	2153	/	THE SPECIFIC SYSTEM CONFIGURATION.	7
8	2154			8
9	2155	/	SYSTEM EQUALS PARAMETER.	9
10	2156			10
11	2157	07756* 0000	SEQUALS, .-. /=-XXXX PARAMETER PASSED HERE.	11
12	2158			12
13	2159	/	APPENDIX "M" CONTAINS MORE INFORMATION ABOUT THE USAGE OF THIS LOCATION BOTH	13
14	2160	/	IN MEMORY AND THE PERMANENT CONTENTS OF BLOCK 0000 WHERE RELEVANT.	14
15	2161			15
16	2162	/	SYSTEM FILE LIST AREA.	16
17	2163			17
18	2164	07757* 0000	SFILES, ZBLOCK 7777-.+1 /BASIC P?S/8 SYSTEM FILE LIST AREA.	18
19	2165			19
20	2166	/	APPENDIX "M" CONTAINS MORE INFORMATION ABOUT THE USAGE OF THE FILE LIST AREA	20
21	2167	/	BOTH IN MEMORY AND THE PERMANENT CONTENTS OF BLOCK 0000 WHERE RELEVANT.	21
22	2168			22
23	2169	IFNZRO . <ERROR .>	/SHOULD NOW BE JUST PAST THE END OF THE PAGE.	23
24	2171			24
25	2172	*7600	RELOC /TURN OFF RELOCATION.	25
26	2173			26
27	2174	/	THE SYSTEM DEVICE HANDLER CODE FOR THE TC01/TC08 CONFIGURATION ENDS HERE.	27
28	2175			28
29	2176	/	SUPPORT PROGRAMMING TO MANAGE THE SYSTEM DEVICE HANDLER CODE [AND THE OTHER	29
30	2177	/	TC01/TC08-SPECIFIC MODULES DESCRIBED ELSEWHERE IN THIS DOCUMENT] IS NOT	30
31	2178	/	CONTAINED WITHIN THIS DOCUMENT. ADDITIONAL UTILITIES ARE NEEDED TO CREATE A	31
32	2179	/	PRACTICAL SYSTEM; THIS CODE HAS BEEN ISOLATED FROM OTHER WORK TO ILLUSTRATE	32
33	2180	/	HOW THE SYSTEM DEVICE HANDLER AND OTHER TC01/TC08-SPECIFIC MODULES OPERATE.	33
34	2181	/	THE READER IS REFERRED TO OTHER P?S/8 SOURCE FILES FOR MORE INFORMATION.	34
35	2182			35
36	2183	/	P?S/8 TC01/TC08 SYSTEM DEVICE HANDLER IMPLEMENTATION NOTES:	36
37	2184			37
38	2185	/	1) WHEN THE LOGICAL CONSOLE OVERLAY IS ENABLED,THE SECTIONS OF CODE OVERLAID	38
39	2186	/	BY INTERCEPT ROUTINES MUST BE CAREFULLY IMPLEMENTED. THE PREVAILING DATA	39
40	2187	/	FIELD WITHIN THE TC01/TC08 SYSTEM DEVICE HANDLER IS THE CALLER'S DATA FIELD	40
41	2188	/	TO ALLOW ACCESS TO THE IN-LINE ARGUMENTS; INTERCEPT ROUTINES MUST PRESERVE	41
42	2189	/	THIS WHEN RETURNING TO THE MAINLINE DEVICE HANDLER CODE. ACCESS TO THE	42
43	2190	/	CUSTOM OVERSHOOT FACTOR MUST BE MADE WHILE THE DATA FIELD IS TEMPORARILY	43
44	2191	/	SET TO 00; THIS WILL ENSURE ANY MODIFICATIONS TO THE CUSTOM OVERSHOOT	44
45	2192	/	FACTOR ARE ALWAYS APPLIED.	45
46	2193			46
47	2194	/	THE OVERLAY IS RESPONSIBLE FOR STORING THE LAST BLOCK TRANSFERRED VALUE	47
48	2195	/	INTO THE CORRESPONDING ELEMENT OF A TABLE ACCORDING TO THE CURRENT LOGICAL	48
49	2196	/	DEVICE UNIT, AND ALSO RETURNING IT TO THE MAINLINE DEVICE HANDLER CODE. AS	49
50	2197	/	NECESSARY, THE DATA FIELD MAY NEEDED TO BE SET TO THE INSTRUCTION FIELD OF	50
51	2198	/	THE OVERLAY BEFORE BEING RESET TO THE CALLER'S DATA FIELD.	51
52				52
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1	2200	/	2) THE BINARY OUTPUT DERIVED FROM ASSEMBLY OF THIS SOURCE FILE IS LOADED INTO	1
2	2201	/	USER MEMORY SPACE AS A SEPARATE IMAGE IN CONJUNCTION WITH VARIOUS UTILITIES	2
3	2202	/	INCLUDING CERTAIN ROUTINES RELATED TO THE SYSTEM GENERATION PROCESS; TO BE	3
4	2203	/	FUNCTIONAL AS INTENDED, THE TC01/TC08 BOOTUP PROCESS MUST BRING SELECTED	4
5	2204	/	PORTIONS OF BLOCK 0000 INTO 07600-07777 TO CREATE A WORKING SYSTEM KERNEL.	5
6	2205			6
7	2206	/	IN GENERAL, THIS IS EQUIVALENT TO MOVING THE ENTIRE PAGE FROM THE USER	7
8	2207	/	MEMORY SPACE [AS LOADED] TO 07600-07777 BEFORE USE; DEPENDING ON THE	8
9	2208	/	SPECIFIC SYSTEM CONFIGURATION, OTHER CONSIDERATIONS MIGHT APPLY, ESPECIALLY	9
10	2209	/	FOR SYSTEMS WITH UNORTHODOX BOOTUP PROCEDURES AND/OR EXTENDED MEMORY	10
11	2210	/	REQUIREMENTS FOR A SYSTEM DEVICE EXTENSION. APPENDIX "M" CONTAINS GENERIC	11
12	2211	/	INFORMATION REGARDING THESE POSSIBILITIES ESPECIALLY AS IT RELATES TO	12
13	2212	/	DEFINING THE PERMANENT CONTENTS OF BLOCK 0000 LATER USED IN THE BASIC P?S/8	13
14	2213	/	SYSTEM FOR PASSING ADDITIONAL PARAMETERS OR FOR PROGRAM DEBUGGING BOTH IN	14
15	2214	/	THE BASIC AND SHELL OPERATING MODES. AS NECESSARY, ADDITIONAL INFORMATION	15
16	2215	/	CAN BE FOUND IN OTHER DOCUMENTS REGARDING SPECIFICS OF OTHER P?S/8 SYSTEM	16
17	2216	/	DEVICE HANDLERS AND RELATED COMPONENTS THAT DISTINGUISH EACH HARDWARE	17
18	2217	/	CONFIGURATION.	18
19	2218			19
20	2219	/	3) COPIES OF THE TC01/TC08 P?S/8 SYSTEM CAN BE MADE ON TD8E-BASED SYSTEMS	20
21	2220	/	USING A VARIETY OF AVAILABLE PROGRAMS INCLUDING THE STANDARD TDCOPY	21
22	2221	/	PROGRAM. [ON TC01/TC08 SYSTEMS, THE DEC STANDARD DTCOPY PROGRAM CAN BE	22
23	2222	/	USED FOR STANDARD LENGTH DECTAPES.]	23
24	2223			24
25	2224	/	PREMIUM LENGTH TAPES MUST BE COPIED USING LENGTH-AWARE PROGRAMS SUCH AS	25
26	2225	/	P?S/8 BLKCPY, ALSO AVAILABLE ON BOOTABLE P?S/8 TD8E SYSTEMS.	26
27	2226			27
28	2227	/	P?S/8 DECTAPES CAN ALSO BE COPIED ON THE PDP-12 EQUIPPED WITH THE TC12F	28
29	2228	/	OPTION. THERE ARE MODIFICATIONS AVAILABLE FOR THE LINC-8 TO ACCOMPLISH	29
30	2229	/	DECTAPE TRANSFERS IN A SIMILAR MANNER. THERE MAY BE ONLY LIMITED SUPPORT	30
31	2230	/	FOR PREMIUM LENGTH TAPES ON THESE SYSTEMS; FUTURE P?S/8 RELEASES MAY	31
32	2231	/	INCLUDE SPECIAL NON-SYSTEM HANDLERS FOR DECTAPES ON THESE LINCTAPE-BASED	32
33	2232	/	SYSTEMS. THIS WOULD ALLOW P?S/8 BLKCPY TO BE USED ON THESE SYSTEMS.	33
34	2233			34
35	2234	/	IN GENERAL, P?S/8 BLKCPY CAN BE USED ON ANY P?S/8 SYSTEM AS LONG AS THE	35
36	2235	/	REQUIRED NON-SYSTEM HANDLERS ARE AVAILABLE. SYSTEM IMAGES MAY BE COPIED	36
37	2236	/	VIA A CIRCUITOUS ROUTE INVOLVING SEVERAL INDEPENDENT SYSTEMS BEFORE A	37
38	2237	/	SYSTEM IS EVENTUALLY AVAILABLE TO RUN ON THE INTENDED HARDWARE	38
39	2238	/	CONFIGURATION.	39
40				40
41				41
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1	2240	/	SECTION 2: STAND-ALONE BOOTSTRAP FOR THE TC01/TC08 SYSTEM.	1
2	2241			2
3	2242	/	THE BOOTSTRAP PROGRAMMING BELOW IS ACTUALLY AN EXTRACT FROM THE SLURP LOADER	3
4	2243	/	DESCRIBED ELSEWHERE IN THIS DOCUMENT. THIS ALLOWS THE RELATIONSHIP BETWEEN	4
5	2244	/	THE STAND-ALONE BOOTSTRAP AND THE BIN AND RIM LOADERS [AS DESCRIBED ELSEWHERE	5
6	2245	/	IN THIS DOCUMENT] TO BE ILLUSTRATED. THERE ARE CODE SPACE ISSUES THAT MUST BE	6
7	2246	/	RESOLVED BETWEEN THE TWO COMPONENTS AS EACH COULD HAVE REASONS TO INCREASE IN	7
8	2247	/	SIZE. ACCORDINGLY, AREAS OF POTENTIAL FUTURE MODIFICATIONS WILL BE INDICATED.	8
9	2248			9
10	2249	/	BY INVOKING THE P?S/8 BIN UTILITY WITH CERTAIN PARTICULAR OPTIONS, THE	10
11	2250	/	STAND-ALONE BOOTSTRAP WILL BE LOADED EITHER WITH THE REST OF THE SLURP LOADER	11
12	2251	/	OR THE BIN AND RIM LOADERS. THERE IS ONLY ONE COPY OF THE BOOTSTRAP CODE	12
13	2252	/	WITHIN THE P?S/8 BIN UTILITY; INTERNAL ONCE-ONLY ROUTINES COPY THE CODE	13
14	2253	/	SECTIONS TOGETHER, THEN LOAD OVER 07600-07777; THE P?S/8 SYSTEM IS ENTIRELY	14
15	2254	/	DISPLACED FROM MEMORY. THE APPROPRIATE LOADER WILL BE STARTED ACCORDING TO	15
16	2255	/	OPTION SWITCHES PASSED TO P?S/8 BIN. P?S/8 [AND CERTAIN OTHER OPERATING	16
17	2256	/	SYSTEMS COMPATIBLE WITH THE STAND-ALONE BOOTSTRAP] CAN BE LOADED BY STARTING	17
18	2257	/	THE COMPUTER AT 07600. [NOTE: BOOT FUNCTIONALITY IS ONLY VIABLE IF THE	18
19	2258	/	MEMORY LOCATIONS WITHIN THE STAND-ALONE BOOTSTRAP ARE NOT MODIFIED.]	19
20	2259			20
21	2260	/	NOTE: THE CODE DESCRIBED HERE IS REDUNDANT TO THE SLURP LOADER; IT IS	21
22	2261	/	PRESENTED SEPARATELY FOR ILLUSTRATION PURPOSES ONLY. [FOR MORE INFORMATION,	22
23	2262	/	SEE THE DESCRIPTION OF THE SLURP LOADER ELSEWHERE IN THIS DOCUMENT.]	23
24	2263			24
25	2264	0001	FIELD SLUFLD%10 /GENERATION FIELD FOR SLURP LOADER.	25
26		*0200		26
27	2265			27
28	2266		NOPUNCH /FOOL THE ASSEMBLER.	28
29	2267			29
30	2268	0000	FIELD 0 /WHERE THIS LOADS WHEN EXECUTED.	30
31		*0200		31
32	2269			32
33	2270		ENPUNCH /UN-FOOL THE ASSEMBLER.	33
34	2271			34
35	2272	*7000	*SLUADR /GENERATION ADDRESS FOR SLURP LOADER.	35
36	2273			36
37	2274	*7600	RELOC SBOOT /WHERE THIS IS MOVED TO FOR EXECUTION.	37
38	2275			38
39	2276		NOPUNCH /PREVENT BINARY OUTPUT.	39
40	2277			40
41	2278	/	THE SYSTEM RESTARTS HERE USING THE STAND-ALONE BOOTSTRAP PLACED WHERE THE	41
42	2279	/	P?S/8 SYSTEM KERNEL WILL BE RELOADED; ONCE LOADED, THE SYSTEM KERNEL CAN ALSO	42
43	2280	/	BE RESTARTED AT THE SAME ADDRESS [07600].	43
44	2281			44
45	2282	/	RESTART CONSIDERATIONS:	45
46	2283			46
47	2284	/	1) THE DATA FIELD MUST BE SET TO 00.	47
48	2285			48
49	2286	/	2) INTERRUPTS SHOULD BE DISABLED AS THE P?S/8 KEYBOARD MONITOR LOADS OVER THE	49
50	2287	/	INTERRUPT LOCATIONS.	50
51	2288			51
52	2289	/	3) THE AC AND LINK ARE IGNORED.	52
53	2290			53
54	2291	/	NOTE: ALL PROGRAMMING ON THIS PAGE IS SPECIFIC TO THE TC01/TC08 CONFIGURATION;	54
55	2292	/	BOOTUP ROUTINES FOR OTHER DEVICES MUST FIT INTO THE SAME [OR LESS] CODE SPACE.	55
56				56
57				57
58				58

1	2294	/	THE SYSTEM RESTART ENTERS HERE.	1
2	2295			2
3	2296	07600* 7770 SL7770, SPA SNA SZL CLA/7770	/CLEAR THE AC AND POSSIBLY SKIP; ALSO USED	3
4	2297		/AS CONSTANT 7770 BY THE SLURP LOADER.	4
5	2298			5
6	2299	/	THE FOLLOWING INSTRUCTION MIGHT BE SKIPPED OR HARMLESSLY EXECUTED AS "AND 30".	6
7	2300			7
8	2301	07601* 0030 SL30, 30	/CONSTANT 0030 USED BY THE SLURP LOADER.	8
9	2302			9
10	2303	/	THE AC IS DEFINITELY CLEAR AT THIS POINT.	10
11	2304			11
12	2305	07602* 6774 DTLB	/ALL DMA TRANSFERS TO FIELD 0.	12
13	2306	07603* 1207 TAD SL600/(600)	/SETUP FOR REWIND ON UNIT 0.	13
14	2307	07604* 4213 JMS SLWAIT	/WAIT FOR REVERSE END ZONE ERROR.	14
15	2308	07605* 1223 TAD SL7577/(SB00T-1)	/GET PRE-DECREMENTED POINTER TO RELOAD ADDRESS.	15
16	2309	07606* 3355 DCA CA	/STASH IN THE CURRENT ADDRESS REGISTER.	16
17	2310			17
18	2311	/	THE FOLLOWING INSTRUCTION EXECUTES HARMLESSLY AS "AND I SL7770" [0600].	18
19	2312			19
20	2313	07607* 0600 SL600, 600	/CONSTANT 0600.	20
21	2314			21
22	2315	/	THE NEXT INSTRUCTION EXECUTES HARMLESSLY AS "AND I 0" [0400]; ADDITIONALLY,	22
23	2316	/	IT MUST BE LOCATED AT 7610 TO FORM ANOTHER CONSTANT USED BY THE SLURP LOADER.	23
24	2317			24
25	2318	IFNZRO .&177-10 <ERROR .>		25
26	2320			26
27	2321	07610* 0400 SL400, 400	/CONSTANT 0400.	27
28	2322			28
29	2323	07611* 1237 TAD SL220/(0220)	/GET READ FORWARD FUNCTION VALUE.	29
30	2324	07612* 4213 JMS SLWAIT	/GO WAIT THERE FOR DMA OVERLAY TO TAKE	30
31	2325		/CONTROL AWAY FROM US AND GIVE IT TO THE	31
32	2326		/JUST-READ BOOT-BLOCK CONTENTS.	32
33	2327			33
34	2328	/	THE BOOTSTRAP DEFINED HERE IS TAKEN FROM A CIRCA 1976 P?S/8 RELEASE; NEWER	34
35	2329	/	VERSIONS MAY SUPPORT ADDITIONAL FEATURES SUCH AS AN EXTENDED BOOTSTRAP FOR	35
36	2330	/	ALL DECTAPE DRIVES. [NOTE: ONLY NEWER VERSIONS OF P?S/8 SUPPORT THIS FEATURE;	36
37	2331	/	FOR ALL OLDER RELEASES AND MOST OTHER BOOT-COMPATIBLE OPERATING SYSTEMS, THE	37
38	2332	/	BOOTSTRAP MUST BE LIMITED TO DRIVE 0.]	38
39	2333			39
40	2334	/	FOR MORE INFORMATION ON THE STAND-ALONE BOOTSTRAP DESIGN, SEE DESCRIPTION OF	40
41	2335	/	THE SLURP LOADER ELSEWHERE IN THIS DOCUMENT; THE RESTART ROUTINE DEFINED HERE	41
42	2336	/	IS A COMPATIBLE SUBSET OF THE SLURP LOADER. THE P?S/8 "BIN" UTILITY WILL	42
43	2337	/	LOAD THE BOOTSTRAP ALONG WITH THE BIN AND RIM LOADERS IF PROPER COMMAND-LINE	43
44	2338	/	OPTIONS ARE PASSED.	44
45				45
46				46
47				47
48				48
49				49
50				50
51				51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	2340	/	RESTART OPERATION CONSIDERATIONS.	1
2	2341			2
3	2342	/	WHEN THE BOOTSTRAP CODE IS INITIATED, THE DECTAPE IS REWOUND INTO THE REVERSE	3
4	2343	/	END ZONE. THE BOOTSTRAP THEN INITIATES A FORWARD READ OPERATION; WHEN	4
5	2344	/	PERFORMED IN THIS MANNER, BLOCK 0000 WILL BE TRANSFERRED INTO MEMORY AS THE	5
6	2345	/	DECTAPE COMES OUT OF THE REVERSE END ZONE.	6
7	2346			7
8	2347	/	DURING THE TAPE BLOCK DMA TRANSFER, THE CONTENTS OF TAPE BLOCK 0000 GRADUALLY	8
9	2348	/	REPLACES THE CODE DEFINED HERE; THE ONLY REQUIREMENT IS THE REPLACEMENT WAIT	9
10	2349	/	LOOP CODE MUST BE SUFFICIENTLY COMPATIBLE WITH THE BOOTSTRAP-PROVIDED WAIT	10
11	2350	/	LOOP TO ALLOW THE DMA TO COMPLETE PROPERLY [SUBJECT TO POSSIBLE DATA TRANSFER	11
12	2351	/	LENGTH ALTERATION ONCE THE WORD COUNT AND CURRENT ADDRESS LOCATIONS	12
13	2352	/	[07754-07755] ARE OVERLAID BY THE CORRESPONDING DATA ON THE DECTAPE BLOCK].	13
14	2353			14
15	2354	/	NOTE: CONFORMING WITH THE STANDARD TC01/TC08 BOOTSTRAP CONVENTION REQUIRES THE	15
16	2355	/	LOOP WAITING FOR THE DONE FLAG BE PLACED IN MEMORY LOCATIONS 07616-07617.	16
17	2356			17
18	2357	/	VERSIONS OF THIS BOOTSTRAP CODE WRITTEN FOR LATER P?S/8 RELEASES MIGHT SUPPORT	18
19	2358	/	EXTENDED BOOTSTRAP FUNCTIONALITY AS A SUPERSET OF THE STANDARD TC01/TC08	19
20	2359	/	BOOTSTRAP ROUTINE. THIS ALLOWS BOOTING TO DRIVES OTHER THAN UNIT 0 TO BE USED	20
21	2360	/	AS THE SYSTEM DEVICE.	21
22	2361			22
23	2362	/	EXTENDED UNIT SUPPORT REQUIRES SUPPLYING A UNIT SELECT VALUE IN THE RANGE OF	23
24	2363	/	1000 THROUGH 7000; THE TC01/TC08 HARDWARE REQUIRES UNIT INFORMATION PASSED IN	24
25	2364	/	AC[0-2] DURING THE "DTLA" OR "DTXA" OPERATION. THERE MAY BE ADDITIONAL	25
26	2365	/	CONSIDERATIONS AFFECTING OVERALL PROGRAM OPERATION AND CODE EFFICIENCY; THIS	26
27	2366	/	WILL BE DEALT WITH IN FUTURE RELEASES OF THIS DOCUMENT.	27
28	2367			28
29	2368	/	[NOTE: TRADE-OFFS EXIST WITH REGARD TO DECTAPE STATUS REGISTER "A" ACCESS.	29
30	2369	/	THE "DTLA" OPERATION REQUIRES NO PRIOR KNOWLEDGE OF THE STATUS "A" REGISTER.	30
31	2370	/	THE "DTXA" OPERATION UPDATES THE EXISTING STATUS REGISTER INFORMATION VIA AN	31
32	2371	/	.XOR. TRANSFER BETWEEN THE AC CONTENTS AND THE REGISTER CONTENTS; END-ZONE	32
33	2372	/	HANDLING IS GENERALLY EASIER TO IMPLEMENT USING "DTXA"; THE DIRECTION IS	33
34	2373	/	MERELY INVERTED USING THE CONSTANT 0400 WHEN THE END ERROR IS DETECTED.]	34
35	2374			35
36	2375	/	UNLIKE THE SYSTEM DEVICE HANDLER EQUIVALENT, THIS ROUTINE IS ONLY USED TO	36
37	2376	/	REBOOT P?S/8 [OR ANY OTHER OPERATING SYSTEM COMPATIBLE WITH THIS BOOTSTRAP	37
38	2377	/	CONVENTION]. AS SUCH, IT DOES NOT CHECK FOR ERRORS; WHEN USED TO REWIND THE	38
39	2378	/	SYSTEM DECTAPE, THE ERROR FLAG WILL RAISE DUE TO CONTROLLER DETECTION OF THE	39
40	2379	/	REVERSE END ZONE. WHEN USED TO READ BLOCK 0000 MOVING FORWARD OUT OF THE	40
41	2380	/	REVERSE END ZONE, THE WAIT LOOP WILL BE OVERLAID BY THE CODE CONTAINED ON	41
42	2381	/	BLOCK 0000 ACCORDING TO WELL-ESTABLISHED CONVENTIONS. ALL MEMORY LOCATIONS	42
43	2382	/	OTHER THAN THE WAIT LOOP [WHICH MIGHT LEGITIMATELY BE OVERLAID WITH SLIGHTLY	43
44	2383	/	DIFFERENT CODE] ARE EXPECTED TO BE COMPLETELY DIFFERENT AS A RESULT OF THE	44
45	2384	/	READ OPERATION; ALL SUBSEQUENT ACTIONS ARE DETERMINED BY THE OPERATING SYSTEM	45
46	2385	/	BOOT BLOCK CONTENTS [WHICH ARE LIKELY COMPLETELY DIFFERENT FROM THE REST OF	46
47	2386	/	THE CODE LOADED HERE TO IMPLEMENT THE STAND-ALONE BOOTSTRAP AND SLURP LOADER].	47
48	2387			48
49	2388	/	THE WAIT ROUTINE AS DEFINED HERE REQUIRES THE NEW CONTENTS OF THE DECTAPE	49
50	2389	/	STATUS REGISTER "A" TO BE PASSED IN THE AC BECAUSE THE "DTLA" INSTRUCTION IS	50
51	2390	/	USED. AS SUCH, THERE IS NO DEPENDENCY ON PRIOR STATUS REGISTER CONTENTS.	51
52	2391	/	[NOTE: THERE ARE CERTAIN RESTRICTIONS ON CHANGING PARTICULAR STATUS REGISTER	52
53	2392	/	BITS; IF CERTAIN COMBINATIONS OF CHANGES ARE DONE IN AN IMPROPER ORDER, THE	53
54	2393	/	RESULT COULD BE UNCONTROLLED TAPE MOTION.]	54
55				55
56				56
57				57
58				58

1	2395	7613	SLRPBLK=.	/ALSO USED FOR LATEST SLURP BLOCK VALUE STORAGE.	1
2	2396				2
3	2397	07613* 0000	SLWAIT, .-. .	/WAIT ROUTINE.	3
4	2398	07614* 6766	DTLA	/LOAD STATUS "A" REGISTER WITH	4
5	2399			/PASSED FUNCTION BITS, CLEAR AC.	5
6	2400	07615* 3354	DCA WC	/CLEAR WORD COUNT REGISTER NOW IN CASE READING.	6
7	2401				7
8	2402	/	STANDARD TC01/TC08 DECTAPE WAIT LOOP; COMPATIBLE WITH MANY OPERATING SYSTEMS.		8
9	2403				9
10	2404		IFNZRO 7616-. <ERROR .>	/ENSURE COMPATIBILITY WITH THE STANDARD ADDRESS.	10
11	2406				11
12	2407	07616* 6771	SWAITLP, DTSF	/DECTAPE [OR ERROR] FLAG UP?	12
13	2408	07617* 5216	JMP SWAITLP	/FLAG NOT UP YET, KEEP WAITING FOR IT.	13
14	2409				14
15	2410	/	IF READING, EXECUTION CONTINUES HERE WITH THE NEW CONTENTS OF MEMORY TAKEN		15
16	2411	/	FROM BLOCK 0000 OF THE NOW-BOOTED SYSTEM DECTAPE. [NOTE: OFTEN A SUBROUTINE		16
17	2412	/	WITH PRELOADED "RETURN" CONTENTS OVERLAYS THIS CODE. THE P?S/8 SYSTEM DEVICE		17
18	2413	/	HANDLER USES THIS METHOD TO START UP WHEN BOOTSTRAPPED BY THIS CODE OR		18
19	2414	/	EQUIVALENT.]		19
20	2415				20
21	2416	/	THE NEXT INSTRUCTION EXECUTES HARMLESSLY AS "NOP" [7000]; ADDITIONALLY,		21
22	2417	/	IT MUST BE LOCATED AT 7620 TO FORM ANOTHER CONSTANT USED WITHIN THE SLURP		22
23	2418	/	LOADER [AS WELL AS THE STAND-ALONE BOOTSTRAP].		23
24	2419				24
25	2420		IFNZRO .&177-20 <ERROR .>		25
26	2422				26
27	2423	07620* 7000	SL7000, 7000	/CONSTANT 7000.	27
28	2424	07621* 5613	JMP I SLWAIT	/RETURN TO CALLER [IF NOT OVERLAID].	28
29	2425				29
30	2426	07622* 4002	SL4002, 4002	/CONSTANT 4002; USED BY THE SLURP LOADER.	30
31	2427	07623* 7577	SL7577, SB00T-1	/CONSTANT 7577.	31
32	2428				32
33	2429	/	NOTE: THE BOOTSTRAP CODE IS COMPLETELY CONTAINED ABOVE WITH THE EXCEPTION OF		33
34	2430	/	THE CONSTANT 0220 ["SL220"] DEFINED BELOW. REWORKING THE CODE CAN SHORTEN THE		34
35	2431	/	REQUIREMENT IF NECESSARY; HOWEVER, THERE COULD BE CODE SPACE TRADE-OFFS.		35
36	2432				36
37	2433	/	THE FOLLOWING ROUTINE IS ACTUALLY PART OF THE SLURP LOADER; IT IS LOCATED		37
38	2434	/	WITHIN THE AREA DEFINED FOR THE STAND-ALONE BOOTSTRAP. FOR DEVICES OTHER THAN		38
39	2435	/	THE TC01/TC08 DECTAPE, THE AMOUNT OF SPACE ALLOCATED FOR STAND-ALONE		39
40	2436	/	BOOTSTRAP AND SLURP LOADER MAY DIFFER GREATLY FROM THE TC01/TC08 VERSION.		40
41	2437				41
42	2438	/	THE SLURP LOADER COMES HERE TO STOP THE TAPE AFTER THE LATEST FILE HAS BEEN		42
43	2439	/	COMPLETELY LOADED.		43
44	2440				44
45	2441	07624* 1237	SLPSTP, TAD SL220/(0220)	/GET READ FUNCTION VALUE.	45
46	2442	07625* 6764	DTXA	/TURN OFF ONLY MOTION AND READ BITS.	46
47	2443				47
48	2444	/	THE CODE FALLS THROUGH TO PROCESS THE NEXT FILE [IF ANY].		48
49					49
50					50
51					51
52					52
53					53
54					54
55					55
56					56
57					57
58					58

1	2446	/	THE SLURP LOADER STARTS HERE AFTER BEING MOVED TO 07600-07777.	1
2	2447			2
3	2448	07626* 2353	SLPSTR, ISZ SLFLPTR /BUMP TO THE NEXT INPUT FILE.	3
4	2449	07627* 1753	TAD I SLFLPTR /GET LATEST FILE HANDLE [SKIPPED IF AT 7777].	4
5	2450	07630* 0200	AND SL7770/(7770) /JUST BLOCK BITS.	5
6	2451	07631* 7450	SNA /SKIP IF MORE FILES TO LOAD.	6
7	2452	07632* 5346	JMP SLUSTRT /ELSE STARTUP THE USER PROGRAM THERE.	7
8	2453	07633* 3213	DCA SLRPBLK /SAVE FOR SEARCH ROUTINE.	8
9	2454	07634* 1753	TAD I SLFLPTR /GET LATEST FILE HANDLE AGAIN.	9
10	2455	07635* 7012	RTR;RTR /MOVE UNIT BITS TO AC[0-2].	10
11		07636* 7012		11
12	2457	07637* 0220	SL220, AND SL7000/(7000) /ISOLATE UNIT BITS; ALSO CONSTANT 0220.	12
13	2458			13
14	2459	/	THE ENTIRE STAND-ALONE BOOTSTRAP LOADS ABOVE THIS POINT. THE CONSTANT 0220	14
15	2460	/	["SL220"] IS USED WITHIN THE STAND-ALONE BOOTSTRAP CODE.]	15
16	2461			16
17	2462	7640	DECLIM=. /UPPER LIMIT FOR CURRENT TC01/TC08 BOOTSTRAP.	17
18	2463			18
19	2464	/	THE SLURP LOADER ROUTINE CONTINUES; THE CURRENT DEFINITION OF A MAXIMUM LENGTH	19
20	2465	/	STAND-ALONE BOOTSTRAP INCLUDES THESE WORDS.	20
21	2466			21
22	2467	07640* 1254	TAD SL210/(210) /ADD ON FORWARD MOTION AND SEARCH BITS.	22
23	2468	07641* 6766	DTLA /START THE TAPE IN SEARCH MODE, CLEAR AC.	23
24	2469			24
25	2470	/	COMES HERE DURING THE SEARCH AFTER FINDING A BLOCK OTHER THAN THE DESIRED ONE.	25
26	2471			26
27	2472	07642* 7710	SLOOP, SPA CLA /SKIP IF POSITIVE AC.	27
28	2473			28
29	2474	/	FOR THE BENEFIT OF THE BIN AND RIM LOADER USAGE, ALL BOOTSTRAP CODE MUST LOAD	29
30	2475	/	BEFORE THIS POINT. [NOTE: EASILY MET IN THE CASE OF TC01/TC08 DECTAPE.]	30
31	2476			31
32	2477	7643	BOOTLIM=. /UPPER LIMIT OF CODE DEFINED IN THE BOOTSTRAP.	32
33	2478			33
34	2479	/	WHEN INITIATED BY THE P?S/8 BIN PROGRAM [USING APPROPRIATE COMMAND-LINE SWITCH	34
35	2480	/	OPTIONS], THE BIN AND RIM LOADER WILL LOAD INTO THE REST OF THE PAGE. FOR	35
36	2481	/	MORE INFORMATION, SEE DESCRIPTION OF THE P?S/8 MODIFIED BIN AND RIM LOADERS	36
37	2482	/	ELSEWHERE IN THIS DOCUMENT.	37
38	2483			38
39	2484	*7043	RELOC /TURN OFF RELOCATION.	39
40	2485			40
41	2486		ENPUNCH /RE-ENABLE BINARY OUTPUT.	41
42				42
43				43
44				44
45				45
46				46
47				47
48				48
49				49
50				50
51				51
52				52
53				53
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56				56
57				57
58				58

1	2488	/	SECTION 3: P?S/8 MODIFIED VERSIONS OF THE DEC STANDARD BIN AND RIM LOADERS.	1
2	2489			2
3	2490	/	IMAGE OF P?S/8 MODIFIED BIN AND RIM [READ-IN-MODE] LOADERS [07643-07777 ONLY].	3
4	2491			4
5	2492	/	THIS IS THE P?S/8 MODIFIED VERSION OF THE BIN LOADER FOR STANDALONE USAGE	5
6	2493	/	[SUCH AS RUNNING DIAGNOSTIC PROGRAMS ON BINARY PAPER TAPES] AND THE RIM LOADER	6
7	2494	/	[WHICH IS ESSENTIALLY UNMODIFIED FROM THE DEC ORIGINAL CODE]. THE BIN LOADER	7
8	2495	/	HAS BEEN MADE TO OCCUPY THE SHORTEST OVERALL CODE SPACE BY THE USE OF VARIOUS	8
9	2496	/	OPTIMIZATION TECHNIQUES.	9
10	2497			10
11	2498	/	THE P?S/8 "BIN" UTILITY WILL LOAD THE STAND-ALONE BOOTSTRAP AND THE BIN AND	11
12	2499	/	RIM LOADERS INTO 07600-07777; AS A RESULT, THE P?S/8 SYSTEM KERNEL IS	12
13	2500	/	DISPLACED FROM MEMORY. THIS IS ACCOMPLISHED BY SPECIFYING CERTAIN	13
14	2501	/	COMMAND-LINE OPTION SWITCHES TO P?S/8 BIN. [NOTE: BY DEFAULT, P?S/8 BIN	14
15	2502	/	LOADS BINARY FILES INTO MEMORY; BIN CAN ALSO CONVERT BINARY PAPER-TAPES TO	15
16	2503	/	BINARY FILES AND PUNCH BINARY FILES TO PAPER-TAPE IN EITHER BIN OR RIM FORMAT.	16
17	2504	/	FOR A COMPLETE LIST OF OPTIONS FOR THE P?S/8 BIN UTILITY, CONSULT ADVANCED	17
18	2505	/	DOCUMENTATION OF P?S/8 BIN AVAILABLE SEPARATELY.]	18
19	2506			19
20	2507	/	THE CODING FOR THE P?S/8 VERSION OF THE BIN AND RIM LOADERS IS INCLUDED IN	20
21	2508	/	THIS DOCUMENT TO ILLUSTRATE THE MODIFICATIONS THAT WERE MADE; CERTAIN FEATURES	21
22	2509	/	HAVE BEEN ADDED TO IMPROVE RELIABILITY; OTHERS HAVE BEEN REMOVED FOR DESIGN	22
23	2510	/	REASONS AND/OR TO LIMIT THE AMOUNT OF CODE SPACE USED.	23
24	2511			24
25	2512	/	THERE ARE CERTAIN TRADEOFFS WITHIN THE DESIGN OF THE BIN AND RIM LOADERS WITH	25
26	2513	/	REGARD TO POTENTIAL INCREASE IN OVERALL PROGRAM SIZE. THE P?S/8 STAND-ALONE	26
27	2514	/	BOOTSTRAP DESIGN ALSO HAS TRADEOFFS CONCERNING OVERALL PROGRAM SIZE. SINCE	27
28	2515	/	BOTH PROGRAMS MUST COEXIST WITHIN 07600-07777, ALL RELATED FACTORS MUST BE	28
29	2516	/	CONSIDERED AT ONCE. CERTAIN OF THESE ISSUES WILL BE DISCUSSED BELOW.	29
30	2517			30
31	2518	/	NOTABLE FEATURES OF THE P?S/8 VERSION OF THE BIN LOADER.	31
32	2519			32
33	2520	/	BIN LOADER START AT 07777.	33
34	2521			34
35	2522	/	THIS FEATURE HAS BEEN RETAINED FOR FULL COMPATIBILITY. THERE ARE ACTUALLY	35
36	2523	/	SEVERAL DIFFERENT IMPLEMENTATIONS OF WHAT IS COMMONLY REFERRED TO AS THE BIN	36
37	2524	/	LOADER AS DEC [AND OTHERS] HAVE MADE CHANGES TO THIS UTILITY PROGRAM OVER THE	37
38	2525	/	YEARS. INSTEAD OF DOCUMENTING THE INTERNAL STARTING ADDRESS, LOCATION 07777	38
39	2526	/	CONTAINS A "JMP BEGIN" [5306] INSTRUCTION SPECIFIC TO THIS IMPLEMENTATION.	39
40	2527			40
41	2528	/	DYNAMIC CHOICE OF LOW-SPEED OR HIGH-SPEED READER.	41
42	2529			42
43	2530	/	IN MOST OF THE DEC VERSIONS OF THE BIN LOADER, THE CHOICE OF INPUT DEVICE IS	43
44	2531	/	DETERMINED ON THE FRONT PANEL OF THE COMPUTER USING THE HIGH-ORDER BIT OF THE	44
45	2532	/	CONSOLE SWITCH REGISTER. IN THE P?S/8 IMPLEMENTATION, THIS IS CHOSEN BY	45
46	2533	/	SETTING PROPER COMMAND LINE OPTIONS WHEN INVOKING THE P?S/8 BIN UTILITY	46
47	2534	/	[WHICH IS ALSO USED FOR SEVERAL OTHER MODES OF OPERATION SUCH AS THE LOADING	47
48	2535	/	OF P?S/8 BINARY FILES INTO MEMORY]. ASSUMING APPROPRIATE COMMAND-LINE	48
49	2536	/	PARTICULARS ARE USED TO INVOKE THE LOW-SPEED VERSION OF THE RIM AND BIN	49
50	2537	/	LOADERS, ADDING "/F" TO THE SAME COMMAND CAUSES THE PROGRAM TO USE THE	50
51	2538	/	HIGH-SPEED READER HARDWARE FOR ALL PAPER-TAPE READ FUNCTIONS. AS NECESSARY,	51
52	2539	/	COMMENTS ARE PROVIDED BELOW TO INDICATE THE ALTERED CONTENTS OF ALL MEMORY	52
53	2540	/	LOCATIONS AFFECTED BY THE USE OF THE HIGH-SPEED READER OPTION.	53
54				54
55				55
56				56
57				57
58				58

1	2542	/	[NOTE: THE P?S/8 VERSION OF THE RIM LOADER IS ESSENTIALLY UNMODIFIED FROM THE	1
2	2543	/	DEC ORIGINAL; HOWEVER, CERTAIN RELEASES OF P?S/8 MAY HAVE NON-STANDARD	2
3	2544	/	HIGH-SPEED READER MODIFICATIONS APPLIED TO RIM WHEN THE "/F" OPTION SWITCH IS	3
4	2545	/	USED ON THE COMMAND LINE INVOKING P?S/8 BIN. ON THE FEW SYSTEMS TESTED CIRCA	4
5	2546	/	1976, NO PROBLEMS WERE OBSERVED; HOWEVER, THERE MAY BE CERTAIN MODEL-DEPENDENT	5
6	2547	/	PROBLEMS. NEWER RELEASES WILL CORRECT THESE MINOR ISSUES BY PROVIDING THE	6
7	2548	/	STANDARD HIGH-SPEED READER MODIFICATIONS TO RIM. WHERE RELEVANT, THE PATCHED	7
8	2549	/	VALUES APPLIED IN THE 1976 RELEASE OF P?S/8 WILL BE DISCUSSED BELOW, ALONG	8
9	2550	/	WITH THE CORRECTED VALUES THAT WILL APPEAR IN A NEWER RELEASE OF P?S/8 BIN.]	9
10	2551			10
11	2552	/	RUBOUT PROTECTION SUPPORT.	11
12	2553			12
13	2554	/	AS OF THIS WRITING, THE P?S/8 VERSION OF THE BIN LOADER SUPPORTS THE OBSCURE	13
14	2555	/	FEATURE OF IGNORING IMBEDDED ERROR MESSAGES WITHIN BINARY PAPER TAPES WHEN	14
15	2556	/	PRECEDED AND FOLLOWED BY <R0> [CODE 377] CHARACTERS. VIRTUALLY ALL RELEASED	15
16	2557	/	VERSIONS OF BIN FORMAT BINARY PAPER TAPES DO NOT REQUIRE THIS FEATURE; IT IS	16
17	2558	/	ONLY POSSIBLE WHEN INCOMPLETELY DEVELOPED PROGRAMS [THAT STILL HAVE DETECTED	17
18	2559	/	ASSEMBLY ERRORS] ARE BEING LOADED [PRESUMABLY FOR DEBUGGING PURPOSES] AND ONLY	18
19	2560	/	IF PUNCHED ON THE CONSOLE TELETYPE PUNCH USING PROGRAMS SUCH AS THE PAPER TAPE	19
20	2561	/	VERSION OF PAL III. [NOTE: IF THE HIGH-SPEED PUNCH IS USED TO CREATE THE	20
21	2562	/	BINARY PAPER TAPES, ERROR MESSAGES ARE STILL ISSUED TO THE CONSOLE TERMINAL.	21
22	2563	/	AS SUCH, THE PROBLEMATIC CIRCUMSTANCES BEING ADDRESSED COULD NOT OCCUR.]	22
23	2564			23
24	2565	/	FOR THE VAST MAJORITY OF USERS, THIS FEATURE IS TRULY MEANINGLESS. AS SUCH,	24
25	2566	/	FUTURE RELEASES OF THE P?S/8 BIN PROGRAM MAY DROP SUPPORT OF THIS FEATURE IN	25
26	2567	/	FAVOR OF SUPPORT OF LONGER REBOOT CODE AND/OR GREATER PROGRAM RELIABILITY.	26
27	2568	/	SHOULD LOADING OF PROBLEMATIC PAPER TAPES BE REQUIRED, THE DEC STANDARD BIN	27
28	2569	/	LOADER CAN BE LOADED BY THE RIM LOADER; IN THE PROCESS, THE P?S/8 RESTART AT	28
29	2570	/	07600 WILL BE DESTROYED; MANUAL ENTRY OF HARDWARE-SPECIFIC BOOTSTRAP CODE FOR	29
30	2571	/	THE P?S/8 SYSTEM DEVICE WILL BE REQUIRED TO RESTART P?S/8. [NOTE: THE CURRENT	30
31	2572	/	RELEASE OF THE P?S/8 BIN UTILITY CAN CREATE P?S/8 BINARY FILES FROM BINARY	31
32	2573	/	PAPER-TAPE INPUT IN EITHER BIN OR RIM FORMAT USING APPROPRIATE COMMAND-LINE	32
33	2574	/	OPTIONS. P?S/8 BIN CAN ALSO BE USED WITH ALTERNATE COMMAND-LINE OPTIONS TO	33
34	2575	/	PUNCH BINARY PAPER TAPES FROM THE CONTENTS OF P?S/8 BINARY FILES ON THE	34
35	2576	/	SYSTEM DEVICE. AS SUCH, P?S/8 CAN BE USED TO NOT ONLY STORE ANY PROBLEMATIC	35
36	2577	/	BINARY DATA, BUT TO ALSO CREATE A DESCENDENT BINARY PAPER-TAPE WITHOUT THE	36
37	2578	/	PROBLEMATIC ERROR MESSAGES; THE RESULTANT PAPER-TAPE WOULD BE SOMEWHAT SHORTER	37
38	2579	/	WITH THE ERROR MESSAGE TEXT REMOVED.]	38
39	2580	/		39
40	2581	/	BY MINIMIZING THE CODE SPACE OCCUPIED BY THE BIN LOADER TO 07643-07755 [WHICH	40
41	2582	/	IS ADJACENT TO THE RIM LOADER THAT OCCUPIES 07756-07776], A P?S/8 BOOTSTRAP	41
42	2583	/	CAN BE LOADED INTO 07600-07642. P?S/8 IS NOT LOADED INTO MEMORY WHEN THE BIN	42
43	2584	/	AND RIM LOADERS ARE PRESENT, BUT RESTARTING AT 07600 WILL ALLOW P?S/8 TO BE	43
44	2585	/	REBOOTED WITHOUT MANAL ENTRY OF A TOGGLE-IN BOOTSTRAP. [NOTE: MOST DEVICES	44
45	2586	/	SPECIFY A SMALL BOOTSTRAP ROUTINE TO RELOAD THE OPERATING SYSTEM, BUT THERE IS	45
46	2587	/	NO CONSISTENCY AS TO THE PROGRAM ADDRESSES OR EXACT OPERATIONAL STEPS USED TO	46
47	2588	/	PERFORM THE BOOTUP OPERATION. BY PROVIDING A STANDARD RELOAD ROUTINE AT THE	47
48	2589	/	UNIVERSAL EXIT LOCATION OF 07600, P?S/8 CAN MANAGE THE SYSTEM IN THE OVERALL	48
49	2590	/	SENSE. THIS CONCEPT IS UNIQUE TO P?S/8 AND ITS PREDECESSOR, THE R-L MONITOR	49
50	2591	/	SYSTEM. HOWEVER, THE R-L MONITOR SYSTEM ONLY SUPPORTS SYSTEM RESTART AT 07600	50
51	2592	/	FOR USE WITH THE "SLURP" LOADER FOR BINARY FILES WHILE P?S/8 USES THE SAME	51
52	2593	/	BOOTSTRAP CODE WITH THE BIN AND RIM LOADERS AS WELL.]	52
53				53
54				54
55				55
56				56
57				57
58				58

1	2595	/	ANY DECISION TO DROP SUPPORT OF THE <R0>-ENCLOSED ERROR MESSAGES WILL BE BASED	1
2	2596	/	ON THE NEED TO SUPPORT P?S/8 RESTART CODE AT 07600 FOR FUTURE DEVICES. ALL	2
3	2597	/	CURRENTLY SUPPORTED CONFIGURATIONS CAN BE SUCCESSFULLY IMPLEMENTED WITHIN	3
4	2598	/	07600-07642. [NOTE: P?S/8 EXTENDED SYSTEM DEVICE HANDLERS REQUIRE ADDITIONAL	4
5	2599	/	MEMORY FOR BASIC FUNCTIONALITY; ADDITIONAL CODING MAY BE PRESENT TO SUPPORT	5
6	2600	/	THE SLURP LOADER AND/OR THE ABILITY TO RELOAD THE SYSTEM DEVICE HANDLER, OR	6
7	2601	/	A PORTION OF THE STAND-ALONE BOOTSTRAP CODE.]	7
8	2602			8
9	2603	0001	FIELD BRGFLD%10 /GENERATION FIELD FOR BIN AND RIM LOADERS.	9
10		*0200		10
11	2604			11
12	2605		NOPUNCH /FOOL THE ASSEMBLER.	12
13	2606			13
14	2607	0000	FIELD 0 /WHERE THIS CODE EXECUTES.	14
15		*0200		15
16	2608			16
17	2609		ENPUNCH /UN-FOOL THE ASSEMBLER.	17
18	2610			18
19	2611	*6600	*BRGADR /GENERATION ADDRESS FOR BIN AND RIM LOADERS.	19
20	2612			20
21	2613	*7600	RELOC SBOOT /PAGE WHERE THIS IS MOVED TO FOR EXECUTION.	21
22	2614			22
23	2615		NOPUNCH /TURN OFF BINARY OUTPUT.	23
24	2616			24
25	2617	07600* 0000	ZBLOCK BOOTLIM-. /THIS SPACE RESERVED FOR STAND-ALONE BOOTSTRAP.	25
26	2618			26
27	2619		ENPUNCH /RESTORE BINARY OUTPUT.	27
28	2620			28
29	2621	*7643	*. /ESTABLISH PROPER LOADING ORIGIN.	29
30	2622			30
31	2623		IFNZRO BINLOAD-. <ERROR .> /ENSURE PROPER LOADING ADDRESS.	31
32	2625			32
33	2626	/	"BEGG" SUBROUTINE; EXTRACT ERRORS, FIELD SETTINGS, LEADER/TRAILER.	33
34	2627			34
35	2628	/	"BEGG" OPERATES IN A DATA-DEPENDENT MANNER AFTER FIRST OBTAINING A DATA FRAME	35
36	2629	/	FROM THE PAPER-TAPE READER:	36
37	2630			37
38	2631	/	1) THE LATEST FRAME IS A LEADER/TRAILER CODE OF THE FORM "2ZZ" WHERE "ZZ" IS	38
39	2632	/	IGNORED [AND GENERALLY 00]. TAKE NON-SKIP RETURN TO CALLER; THE CALLER	39
40	2633	/	DETERMINES WHAT TO DO WHEN LEADER/TRAILER FRAMES ARE DETECTED. [NOTE:	40
41	2634	/	LEADER/TRAILER FRAMES DO NOT CONTRIBUTE TO THE OVERALL CHECKSUM	41
42	2635	/	CALCULATION.]	42
43	2636			43
44	2637	/	LEADER/TRAILER FRAME FORMAT.	44
45	2638			45
46	2639	/	OCTAL VALUE ! 2 ! Z ! Z !	46
47	2640	/		47
48	2641	/	BIT VALUE ! 1 ! 0 ! Z ! Z ! Z ! Z ! Z !	48
49	2642	/		49
50	2643	/	BIT POSITION ! 4 ! 5 ! 6 ! 7 ! 8 ! 9 ! 10! 11!	50
51				51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	2645	/	2) THE LATEST FRAME IS A FIELD SETTING TO FIELD X OF THE FORM "3XY" WHERE "Y"	1
2	2646	/	IS IGNORED. FORM A "CDF X0" INSTRUCTION [62X1] AND STORE FOR LATER USE.	2
3	2647	/	CONTINUE PROCESSING BY OBTAINING AN ADDITIONAL DATA FRAME. [NOTE: FIELD	3
4	2648	/	SETTINGS DO NOT CONTRIBUTE TO THE OVERALL CHECKSUM CALCULATION.]	4
5	2649			5
6	2650	/	FIELD SETTING FRAME FORMAT.	6
7	2651			7
8	2652	/	OCTAL VALUE ! 3 ! X ! Y !	8
9	2653	/	-----	9
10	2654	/	BIT VALUE ! 1 ! 1 ! X ! X ! X ! Y ! Y ! Y !	10
11	2655	/	-----	11
12	2656	/	BIT POSITION ! 4 ! 5 ! 6 ! 7 ! 8 ! 9 ! 10! 11!	12
13	2657			13
14	2658	/	CDF INSTRUCTION FORMAT.	14
15	2659			15
16	2660	/	OCTAL VALUE ! 6 ! 2 ! X ! 1 !	16
17	2661	/	-----	17
18	2662	/	BIT VALUE ! 1 ! 1 ! 0 ! 0 ! 1 ! 0 ! X ! X ! 0 ! 0 ! 1 !	18
19	2663	/	-----	19
20	2664	/	BIT POSITION ! 0 ! 1 ! 2 ! 3 ! 4 ! 5 ! 6 ! 7 ! 8 ! 9 ! 10! 11!	20
21	2665			21
22	2666	/	3) THE LATEST FRAME IS THE FIRST HALF OF EITHER AN ORIGIN SETTING OF THE FORM	22
23	2667	/	"01AB" WHERE THE COMPLETE ORIGIN WILL BE USED TO UPDATE THE STORAGE POINTER	23
24	2668	/	TO MEMORY LOCATION "ABCD", OR A DATA WORD OF THE FORM "00AB" WHERE THE	24
25	2669	/	COMPLETE DATA WILL BE "ABCD". TAKE SKIP RETURN TO CALLER. [NOTE: THE	25
26	2670	/	SECOND HALF OF THE LATEST ORIGIN OR DATA ["CD"] IS OBTAINED BY THE CALLER	26
27	2671	/	INDEPENDENTLY.] THE ENTIRE DATA FRAME CONTRIBUTES TO THE OVERALL CHECKSUM	27
28	2672	/	CALCULATION INCLUDING BIT[5] WHICH IS SET FOR THE FIRST HALF OF AN ORIGIN	28
29	2673	/	SETTING; SECOND HALF DATA FRAMES ARE ALWAYS OF THE FORM "00CD".	29
30	2674			30
31	2675	/	ORIGIN SETTING FIRST FRAME FORMAT.	31
32	2676			32
33	2677	/	OCTAL VALUE ! 1 ! A ! B !	33
34	2678	/	-----	34
35	2679	/	BIT VALUE ! 0 ! 1 ! A ! A ! A ! B ! B ! B !	35
36	2680	/	-----	36
37	2681	/	BIT POSITION ! 4 ! 5 ! 6 ! 7 ! 8 ! 9 ! 10! 11!	37
38	2682			38
39	2683	/	ORIGIN SETTING FIRST FRAME 12-BIT CHECKSUM WORD FORMAT.	39
40	2684			40
41	2685	/	OCTAL VALUE ! 0 ! 1 ! A ! B !	41
42	2686	/	-----	42
43	2687	/	BIT VALUE ! 0 ! 0 ! 0 ! 0 ! 0 ! 1 ! A ! A ! A ! B ! B ! B !	43
44	2688	/	-----	44
45	2689	/	BIT POSITION ! 0 ! 1 ! 2 ! 3 ! 4 ! 5 ! 6 ! 7 ! 8 ! 9 ! 10! 11!	45
46	2690			46
47	2691	/	DATA WORD FIRST FRAME FORMAT.	47
48	2692			48
49	2693	/	OCTAL VALUE ! 0 ! A ! B !	49
50	2694	/	-----	50
51	2695	/	BIT VALUE ! 0 ! 0 ! A ! A ! A ! B ! B ! B !	51
52	2696	/	-----	52
53	2697	/	BIT POSITION ! 4 ! 5 ! 6 ! 7 ! 8 ! 9 ! 10! 11!	53
54				54
55				55
56				56
57				57
58				58

1	2699	/	DATA WORD FIRST FRAME 12-BIT CHECKSUM WORD FORMAT.	1
2	2700			2
3	2701	/	OCTAL VALUE ! 0 ! 0 ! A ! B !	3
4	2702	/	-----	4
5	2703	/	BIT VALUE ! 0 ! 0 ! 0 ! 0 ! 0 ! 0 ! A ! A ! A ! B ! B ! B !	5
6	2704	/	-----	6
7	2705	/	BIT POSITION ! 0 ! 1 ! 2 ! 3 ! 4 ! 5 ! 6 ! 7 ! 8 ! 9 ! 10 ! 11 !	7
8	2706			8
9	2707	/	ORIGIN OR DATA SECOND FRAME FORMAT.	9
10	2708			10
11	2709	/	OCTAL VALUE ! 0 ! C ! D !	11
12	2710	/	-----	12
13	2711	/	BIT VALUE ! 0 ! 0 ! C ! C ! C ! D ! D ! D !	13
14	2712	/	-----	14
15	2713	/	BIT POSITION ! 4 ! 5 ! 6 ! 7 ! 8 ! 9 ! 10 ! 11 !	15
16	2714			16
17	2715	/	ORIGIN OR DATA SECOND FRAME 12-BIT CHECKSUM WORD FORMAT.	17
18	2716			18
19	2717	/	OCTAL VALUE ! 0 ! 0 ! C ! D !	19
20	2718	/	-----	20
21	2719	/	BIT VALUE ! 0 ! 0 ! 0 ! 0 ! 0 ! 0 ! C ! C ! C ! D ! D ! D !	21
22	2720	/	-----	22
23	2721	/	BIT POSITION ! 0 ! 1 ! 2 ! 3 ! 4 ! 5 ! 6 ! 7 ! 8 ! 9 ! 10 ! 11 !	23
24	2722			24
25	2723	/	ORIGIN OR DATA FRAME PAIR 12-BIT ASSEMBLED WORD FORMAT.	25
26	2724			26
27	2725	/	OCTAL VALUE ! A ! B ! C ! D !	27
28	2726	/	-----	28
29	2727	/	BIT VALUE ! A ! A ! A ! B ! B ! B ! C ! C ! C ! D ! D ! D !	29
30	2728	/	-----	30
31	2729	/	BIT POSITION ! 0 ! 1 ! 2 ! 3 ! 4 ! 5 ! 6 ! 7 ! 8 ! 9 ! 10 ! 11 !	31
32	2730			32
33	2731	07643* 0000	BEGG, .-. /GET A NEW FILTERED FRAME ROUTINE.	33
34	2732	07644* 4271	BEGG1, JMS READ /GET A CHARACTER.	34
35	2733	07645* 1305	TAD M376/(-376) /TEST FOR 377.	35
36	2734	07646* 7750	SPA SNA CLA /SKIP IF <R0> CODE.	36
37	2735	07647* 5255	SWINST, JMP NOTRUB /**** <R0> CODE NOT YET FOUND **** JMP NOTRUB	37
38	2736		**** <R0> CODE FOUND PREVIOUSLY **** JMP BEGG1	38
39	2737			39
40	2738	/	<R0> DETECTED. REVERSE THE STATE OF <R0> HANDLING.	40
41	2739			41
42	2740	07650* 1247	TAD SWINST /GET THE PRESENT INSTRUCTION VALUE.	42
43	2741	07651* 7041	CIA /INVERT IT.	43
44	2742	07652* 1353	TAD FLIPIT /ADD ON INSTRUCTION SUM YIELDING THE OTHER ONE.	44
45	2743	07653* 3247	DCA SWINST /STORE INLINE.	45
46	2744	07654* 5244	JMP BEGG1 /GO GET ANOTHER FRAME.	46
47				47
48				48
49				49
50				50
51				51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	2746	/	REVERSING <R0> HANDLING MODE.	1
2	2747			2
3	2748	/	IF A <R0> CODE HASN'T OCCURRED BEFORE, ALL CHARACTERS THAT FOLLOW MUST BE	3
4	2749	/	IGNORED. WHEN A SECOND <R0> CODE OCCURS, THE IGNORED SECTION IS COMPLETE;	4
5	2750	/	NORMAL CHARACTER DETECTION WILL BE RESTORED.	5
6	2751			6
7	2752	/	LOCATION "SWINST" CAN CONTAIN EITHER OF TWO VALUES ("JMP NOTRUB" [5255],	7
8	2753	/	"JMP BEGG1" [5244]) DEPENDING ON THE STATE OF <R0> CHARACTER DETECTION.	8
9	2754			9
10	2755	/	COMES HERE IF NO <R0> CONSIDERATION.	10
11	2756			11
12	2757	07655* 1350 NOTRUB, TAD CHAR	/GET LATEST CHARACTER.	12
13	2758	07656* 0265 AND MASK/(300)	/JUST CONTROLLING BITS.	13
14	2759	07657* 1307 TAD M200/(-200)	/COMPARE TO LEADER/TRAILER CODE.	14
15	2760	07660* 7510 SPA	/SKIP IF LEADER/TRAILER CODE OR FIELD SETTING.	15
16	2761	07661* 2243 ISZ BEGG	/TAKE SKIP RETURN ON DATA OR ORIGIN SETTING.	16
17	2762	07662* 7750 SPA SNA CLA	/SKIP IF FIELD SETTING.	17
18	2763	07663* 5643 JMP I BEGG	/TAKE IMMEDIATE RETURN ON LEADER/TRAILER CODE.	18
19	2764			19
20	2765	/	FIELD SETTING FOUND.	20
21	2766			21
22	2767	07664* 1350 TAD CHAR	/GET LATEST CHARACTER.	22
23	2768	07665* 0300 MASK, AND FMASK/(70)	/JUST FIELD BITS.	23
24	2769	07666* 1301 TAD CHANGE/(CDF 00)	/FORM "CDF" INSTRUCTION TO LATEST FIELD.	24
25	2770	07667* 3351 DCA MEMTEM	/SAVE FOR FURTHER LOADING.	25
26	2771	07670* 5244 JMP BEGG1	/GO GET ANOTHER FRAME.	26
27	2772			27
28	2773	07671* 0000 READ, .-. .	/READ A CHARACTER ROUTINE.	28
29	2774	07672* 6031 READ1, KSF	/FLAG UP? **** HIGH-SPEED READER **** RSF	29
30	2775	07673* 5272 JMP .-1	/NO, WAIT FOR IT.	30
31	2776	07674* 6036 READ3, KRB	/YES, READ IT IN.	31
32	2777		/**** HIGH-SPEED READER **** RFC RRB	32
33	2778	07675* 3350 DCA CHAR	/SAVE LATEST CHARACTER.	33
34	2779	07676* 1350 TAD CHAR	/GET IT BACK.	34
35	2780	07677* 5671 JMP I READ	/RETURN TO CALLER.	35
36	2781			36
37	2782	IFNZRO .&177-100 <ERROR .>		37
38	2784			38
39	2785	07700* 0070 FMASK, 70	/FIELD MASK CONSTANT 0070	39
40	2786	07701* 6201 CHANGE, CDF 00	/ "CDF 00" INSTRUCTION CONSTANT 6201	40
41	2787			41
42	2788	/	TRAILER CODE SEEN.	42
43	2789			43
44	2790	07702* 4341 BEND, JMS ASSEMB	/ASSEMBLE CHECKSUM FROM THE TAPE.	44
45	2791	07703* 7041 CIA	/INVERT.	45
46	2792	07704* 1352 TAD CHKSUM	/SUBTRACT ACCUMULATED CHECKSUM.	46
47	2793	07705* 7402 M376, HLT	/HALT AT END OF LOADING. IF AC IS CLEAR,	47
48	2794		/APPARENTLY THE TAPE LOADED WITHOUT ERRORS.	48
49	2795		/ALSO USED AS CONSTANT -376.	49
50	2796			50
51	2797	/	PRESS CONTINUE TO LOAD ANOTHER BIN PAPER TAPE.	51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	2799	/	THIS IS THE STARTING ADDRESS OF THE BIN LOADER. A "JMP BEGIN" [5306]			1
2	2800	/	INSTRUCTION IS PLACED INTO 07777 AS THE OFFICIAL DOCUMENTED STARTING ADDRESS.			2
3	2801					3
4	2802	07706* 6032	BEGIN,	KCC	/CLEAR THE KEYBOARD FLAG TO START THE READER.	4
5	2803				/**** HIGH-SPEED READER **** RFC	5
6	2804	07707* 7600	M200,	CLA!400	/CLEAN UP; ALSO FORMS CONSTANT 7600.	6
7	2805	07710* 6214		RDF	/GET CURRENT DATA FIELD.	7
8	2806	07711* 1301		TAD CHANGE/(CDF 00)	/CREATE "CDF" INSTRUCTION TO CURRENT DATA FIELD.	8
9	2807	07712* 3351		DCA MEMTEM	/SAVE FOR LOADING DATA.	9
10	2808	07713* 4243		JMS BEGG	/GET A NEW CHARACTER.	10
11	2809	07714* 5313		JMP .-1	/LEADER FOUND, IGNORE AND TRY AGAIN.	11
12	2810	07715* 3352	GO,	DCA CHKSUM	/STORE NEW CHECKSUM VALUE.	12
13	2811	07716* 1351		TAD MEMTEM	/GET "CDF" TO LATEST DATA FIELD.	13
14	2812	07717* 3327		DCA MEMFLD	/STORE IN-LINE.	14
15	2813	07720* 1350		TAD CHAR	/GET THE LATEST CHARACTER.	15
16	2814	07721* 3354		DCA WORD1	/SAVE IT.	16
17	2815	07722* 4271		JMS READ	/GET THE SECOND CHARACTER.	17
18	2816	07723* 3355		DCA WORD2	/SAVE IT.	18
19	2817	07724* 4243		JMS BEGG	/LOOK AHEAD FOR THE NEXT CHARACTER.	19
20	2818	07725* 5302		JMP BEND	/TRAILER CODE FOUND.	20
21	2819	07726* 4341		JMS ASSEMB	/ASSEMBLE LATEST WORD.	21
22	2820	07727* 7402	MEMFLD,	HLT	/WILL BE "CDF" TO LATEST DATA FIELD.	22
23	2821	07730* 7420		SNL	/SKIP IF ORIGIN SETTING.	23
24	2822	07731* 3776		DCA I TEMP	/STORE DATA WORD.	24
25	2823	07732* 2376		ISZ TEMP	/BUMP TO NEXT LOCATION.	25
26	2824	07733* 7430		SZL	/SKIP IF DATA AND NOT AT LOCATION 0000.	26
27	2825	07734* 3376		DCA TEMP	/STORE NEW ORIGIN SETTING [POSSIBLY 0000].	27
28	2826	07735* 1354	CHEX,	TAD WORD1	/GET LATEST FIRST CHARACTER.	28
29	2827	07736* 1355		TAD WORD2	/GET LATEST SECOND CHARACTER	29
30	2828	07737* 1352		TAD CHKSUM	/ADD ON RUNNING CHECKSUM	30
31	2829	07740* 5315		JMP GO	/KEEP GOING.	31
32	2830					32
33	2831	07741* 0000	ASSEMB,	...	/WORD ASSEMBLE ROUTINE.	33
34	2832	07742* 1354		TAD WORD1	/GET FIRST CHARACTER.	34
35	2833	07743* 7106		CLL RTL;RTL;RTL	/MOVE UP.	35
36		07744* 7006				36
37		07745* 7006				37
38	2836	07746* 1355		TAD WORD2	/ADD SECOND CHARACTER.	38
39	2837	07747* 5741		JMP I ASSEMB	/RETURN TO CALLER.	39
40	2838					40
41	2839	/	THE REST OF THE BIN LOADER IS STORED OVER THE THREE-CYCLE DATA BREAK LOCATIONS			41
42	2840	/	RESERVED FOR THE DF32/RF08, TC58 SERIES MAGNETIC TAPE CONTROLLER AND THE			42
43	2841	/	TC01/TC08 DECTAPE CONTROLLER.			43
44	2842					44
45	2843	/	IF AT ALL POSSIBLE, ONLY TEMPORARY DATA SHOULD BE STORED IN THESE SIX WORDS.			45
46	2844					46
47	2845	/	NOTE: A CURRENT IMPLEMENTATION RESTRICTION THAT VIOLATES THIS IS THE PLACEMENT			47
48	2846	/	OF THE <R0> STATE REVERSAL CONSTANT.			48
49	2847					49
50	2848	07750* 0000	CHAR,	...	/BUFFER FOR LATEST CHARACTER FRAME.	50
51	2849	07751* 0000	MEMTEM,	...	/WILL CONTAIN "CDF" TO LATEST DATA FIELD.	51
52	2850	07752* 0000	CHKSUM,	...	/CHECKSUM ACCUMULATION TEMPORARY.	52
53						53
54						54
55						55
56						56
57						57
58						58

1	2852		/	<R0> STATE REVERSAL CONSTANT.			1	
2	2853						2	
3	2854		/	LOCATION "FLIPIT" [07753] CONTAINS THE SUM OF THE TWO POSSIBLE VALUES			3	
4	2855		/	("JMP BEGG1" [5244] AND "JMP NOTRUB" [5255]) TO ENABLE FLIPPING BETWEEN THE			4	
5	2856		/	TWO INSTRUCTIONS STORED IN "SWINST" [07647] AS REQUIRED. THE CURRENT CONTENTS			5	
6	2857		/	OF "SWINST" [07647] WILL BE SUBTRACTED FROM THE CONTENTS OF "FLIPIT" [07753]			6	
7	2858		/	WHICH WILL THEN BE STORED IN "SWINST" [07647] REPLACING THE CURRENT CONTENTS			7	
8	2859		/	WITH THE OTHER INSTRUCTION.			8	
9	2860						9	
10	2861	5255		INST1=	JMP	NOTRUB	/FIRST POSSIBLE VALUE OF SWINST.	10
11	2862							11
12	2863	5244		INST2=	JMP	BEGG1	/SECOND POSSIBLE VALUE OF SWINST.	12
13	2864							13
14	2865	07753*	2521	FLIPIT,	INST1+INST2		/SUM OF BOTH POSSIBLE VALUES.	14
15	2866							15
16	2867	07754*	0000	WORD1,	.-.		/FIRST CHARACTER.	16
17	2868	07755*	0000	WORD2,	.-.		/SECOND CHARACTER.	17
18	2869							18
19	2870		/	STAND-ALONE PAPER-TAPE RIM LOADER STARTS HERE.				19
20	2871							20
21	2872	07756*	6032	BEG,	KCC		/CLEAR THE KEYBOARD FLAG, START THE READER.	21
22	2873						/**** HIGH-SPEED READER **** RFC	22
23	2874	07757*	6031	BEG1,	KSF		/FLAG UP? **** HIGH-SPEED READER **** RSF	23
24	2875	07760*	5357		JMP .-1		/NO, WAIT FOR IT.	24
25	2876	07761*	6036	BEG3,	KRB		/YES, READ IT LATEST AND START ANOTHER.	25
26	2877						/**** HIGH-SPEED READER **** RFC RRB	26
27	2878	07762*	7106		CLL RTL;RTL		/MOVE LEADER/TRAILER BIT TO AC[0]	27
28		07763*	7006					28
29	2880	07764*	7510		SPA		/SKIP IF NOT LEADER CODE.	29
30	2881	07765*	5374		JMP CLRAC		/FOUND LEADER, IGNORE IT.	30
31	2882	07766*	7006		RTL		/MOVE FIRST HALF TO HIGH-ORDER BITS.	31
32	2883	07767*	6031	BEG11,	KSF		/FLAG UP? **** HIGH-SPEED READER **** RSF	32
33	2884	07770*	5367		JMP .-1		/NO, WAIT FOR IT.	33
34	2885	07771*	6034	BEG13,	KRS		/YES, READ IT IN.	34
35	2886						/**** HIGH-SPEED READER **** RRB	35
36	2887	07772*	7420		SNL		/SKIP IF ORIGIN SETTING.	36
37	2888	07773*	3776		DCA I TEMP		/STORE LATEST DATA WORD.	37
38	2889	07774*	3376	CLRAC,	DCA TEMP		/STORE LATEST ORIGIN.	38
39	2890	07775*	5356		JMP BEG		/KEEP GOING.	39
40	2891							40
41	2892	07776*	0000	TEMP,	.-.		/STORAGE ORIGIN TEMPORARY.	41
42	2893							42
43	2894		/	OFFICIAL STARTING ADDRESS FOR ALL VERSIONS OF THE BIN LOADER.				43
44	2895							44
45	2896		/	WITH SO MANY IMPLEMENTATIONS OF THE BIN LOADER, DEC WISELY DOCUMENTED THIS				45
46	2897		/	OFFICIAL STARTING ADDRESS OF 07777, WHICH IS ACTUALLY A "JMP" INSTRUCTION TO				46
47	2898		/	THE INTERNAL STARTING ADDRESS REGARDLESS OF IMPLEMENTATION.				47
48	2899							48
49	2900	07777*	5306	BINSTRT,	JMP BEGIN		/START BIN LOADER AT OFFICIAL START ADDRESS.	49
50	2901							50
51	2902		/	END OF RIM AND BIN LOADER HERE.				51
52	2903							52
53	2904			IFNZRO	.	<ERROR .>	/SHOULD NOW BE JUST PAST THE END OF THE PAGE.	53
54	2906							54
55	2907	*7000		RELOC			/TURN OFF RELOCATION.	55
56								56
57								57
58								58

1	2909	/	BINARY LOADER IMPLEMENTATION CONSIDERATIONS:	1
2	2910			2
3	2911	/	1) APPLYING CODE OPTIMIZATION TECHNIQUES TO THE BINARY [BIN] LOADER.	3
4	2912			4
5	2913	/	THE DEC BINARY LOADER PROGRAM EVOLVED OVER SEVERAL GENERATIONS OF	5
6	2914	/	DEVELOPMENT. IT IS APPARENT THAT EXTENDED MEMORY LOADING AND OTHER	6
7	2915	/	CONSIDERATIONS CAUSED INCREMENTAL CHANGES TO THE CODING THAT, IN HINDSIGHT,	7
8	2916	/	ARE SHOWN TO NOT BE WELL ORGANIZED. [IT IS LIKELY THERE WAS LITTLE IMPETUS	8
9	2917	/	TO MODIFY THE CODE BEYOND DIRECT REPAIR OF SPECIFIC PROBLEMS AS THEY	9
10	2918	/	SURFACED.] THE P?S/8 IMPLEMENTATION ADDS THE ADDITIONAL REQUIREMENT OF	10
11	2919	/	USING MORE EFFICIENT CODING TECHNIQUES. THIS TENDS TO ACHIEVE A SMALLER	11
12	2920	/	OVERALL CODE SPACE TO MAXIMIZE THE SIZE OF THE SYSTEM RESTART AT 07600	12
13	2921	/	REQUIRED FOR CERTAIN DEVICES; THE TWO ROUTINES MUST COEXIST WITHIN	13
14	2922	/	07600-07777. [NOTE: THE TC01/TC08 SYSTEM RESTART CODE IS DESCRIBED	14
15	2923	/	ELSEWHERE IN THIS DOCUMENT; IT IS SOMEWHAT SMALLER THAN THE CURRENT	15
16	2924	/	DEFINITION OF MAXIMUM SIZE ALLOWED.]	16
17	2925			17
18	2926	/	TO MINIMIZE THE CODE SPACE REQUIRMENTS OF THE BINARY LOADER, CERTAIN SLOPPY	18
19	2927	/	"SPAGHETTI CODE" SEQUENCES HAVE BEEN REPLACED BY MORE COHERENT SECTIONS	19
20	2928	/	THAT ARE ACTUALLY MORE STRAIGHTFORWARD THAN THE ORIGINAL CODING IN THE DEC	20
21	2929	/	RELEASES. [NOTE: THE P?S/8 REPLACEMENT CODE DEPENDS ON QUIRKS OF THE PDP-8	21
22	2930	/	INSTRUCTION SET CLEARLY NOT FULLY UNDERSTOOD BY THE ORIGINAL PROGRAMMERS OF	22
23	2931	/	THE BINARY LOADER. THOSE INTERESTED IN SEEING HOW THE ORIGINAL CODING WAS	23
24	2932	/	REWORKED ARE REFERRED TO THE DOCUMENTATION OF THE ORIGINAL RELEASE OF THE	24
25	2933	/	BINARY LOADER DEC-08-LBAA-D, MAY 10, 1967.] [NOTE: THERE MAY HAVE BEEN AN	25
26	2934	/	EARLIER RELEASE OF THE BINARY LOADER KNOWN TO BE INCOMPATIBLE WITH THE	26
27	2935	/	TC01/TC08 DECTAPE CONTROLLER; THERE ARE HAND-WRITTEN CHANGES TO THE	27
28	2936	/	DOCUMENTATION INDICATING CORRECTIONS MADE TO BE COMPATIBLE WITH THE TC01	28
29	2937	/	DECTAPE CONTROLLER.]	29
30	2938			30
31	2939	/	THE ORIGINAL VERSION OF THE BIN LOADER USES SEVERAL VALID SELF-REFERENTIAL	31
32	2940	/	PROGRAMMING TECHNIQUES OF CONTRIVING INSTRUCTIONS TO FORM USEFUL CONSTANTS	32
33	2941	/	THAT TEND TO MAKE THE OVERALL PROGRAM SIZE SMALLER.	33
34	2942			34
35	2943	/	A CASE IN POINT IS THE NEED TO COMPARE THE LATEST PAPER-TAPE FRAME TO THE	35
36	2944	/	VALUE 377. AN INITIAL DESIGN PASS MIGHT START BY COMPARING THE LATEST	36
37	2945	/	CHARACTER TO 377 BY INVERTING THE LATEST CHARACTER AND ADDING THE CONSTANT	37
38	2946	/	0377, THEN CHECK FOR A ZERO ACCUMULATOR. AN OBVIOUS OPTIMIZATION IS TO	38
39	2947	/	INSTEAD COMPARE THE CHARACTER TO THE SIGNED CONSTANT -377 [7401] WHICH	39
40	2948	/	ELIMINATES THE INHERENT INVERTING INSTRUCTION ["CIA" 7041]. HOWEVER, THE	40
41	2949	/	BIN LOADER INCLUDES THE "HLT" [7402] INSTRUCTION WHICH IS ALSO THE SIGNED	41
42	2950	/	CONSTANT -376. BY CHANGING THE COMPARISON TO SKIP ON STRICTLY POSITIVE,	42
43	2951	/	ALL FRAME CODES LESS THAN 377 ARE SEPARATED FROM THE DETECTION OF THE 377	43
44	2952	/	FRAME CODE USING EQUIVALENT LOGIC; A MEMORY LOCATION IS SAVED BY	44
45	2953	/	REFERENCING THE "HLT" [7402] INSTRUCTION AS A CONSTANT.	45
46	2954			46
47	2955	/	IN ANOTHER PORTION OF THE PROGRAM, AN INSTRUCTION IS NEEDED TO CLEAR THE	47
48	2956	/	ACCUMULATOR; THE "CLA" [7200] GROUP I OPR INSTRUCTION IS GENERALLY USED TO	48
49	2957	/	ACCOMPLISH THIS FUNCTION. INSTEAD, THE GROUP II INSTRUCTION 7600 IS USED	49
50	2958	/	TO CLEAR THE ACCUMULATOR; THE CONSTANT 7600 IS USED ELSEWHERE IN THE	50
51	2959	/	PROGRAM THUS SAVING AN ADDITIONAL CODE SPACE WORD.	51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	2961	/	THE P?S/8 MODIFICATIONS TAKE THIS CONCEPT A FEW STEPS FURTHER. THE BINARY	1
2	2962	/	LOADER CODING REQUIRES A CONSTANT 0300 AND A CONSTANT 0070. BY REQUIRING	2
3	2963	/	THE "FMASK" CONSTANT [0070] TO BE LOCATED AT LOCATION 7700 WITHIN THE BIN	3
4	2964	/	LOADER PROGRAM, THE LOCATION THAT REFERENCES THE CONSTANT, THE INSTRUCTION	4
5	2965	/	"AND FMASK", CAN BE USED AS THE CONSTANT 0300 THUS SAVING AN ADDITIONAL	5
6	2966	/	CODE SPACE WORD.	6
7	2967			7
8	2968	/	THE CODING OF THE RUBOUT PROTECTION SUPPORT WAS SHORTENED BY USE OF AN	8
9	2969	/	ALTERNATE METHOD WHICH PERFORMS ARITHMETIC OPERATIONS ON AN INSTRUCTION	9
10	2970	/	WITHIN THE CODING ITSELF.	10
11	2971			11
12	2972	/	ALTHOUGH UNUSUAL, THESE TECHNIQUES ARE VALID WAYS TO MINIMIZE THE OVERALL	12
13	2973	/	CODE SPACE OF THE PROGRAM. IN THEORY, ADDITIONAL TECHNIQUES COULD BE USED	13
14	2974	/	TO IMPLEMENT FURTHER CODE SPACE OPTIMIZATION IN THE FUTURE.	14
15	2975			15
16	2976	/	2) EXTENDED MEMORY PROGRAMMING CONSIDERATIONS.	16
17	2977			17
18	2978	/	DUE TO THE WAY THE BIN LOADER DETECTS FIELD SETTINGS [IGNORING THEM FOR	18
19	2979	/	CHECKSUM PURPOSES], A "CDF <NEW DATA FIELD" INSTRUCTION IS CREATED AND	19
20	2980	/	BUFFERED FOR USE WITH LATER DATA TO BE LOADED INTO THE NEW FIELD. A	20
21	2981	/	DEFECTIVE VERSION OF DEC'S BIN LOADER WAS RELEASED KNOWN AS THE	21
22	2982	/	"SELF-STARTING BINARY LOADER". THIS RELEASE HAS A FLAWED DESIGN AS IT	22
23	2983	/	LACKS THE REQUIRED BUFFERING; IT WAS APPARENTLY WRITTEN BY SOMEONE IGNORANT	23
24	2984	/	OF THE NUANCES OF BINARY PAPER-TAPE FORMAT AS IT APPLIES TO LOADING DATA	24
25	2985	/	INTO EXTENDED MEMORY.	25
26	2986			26
27	2987	/	THE ONLY REASON THIS SERIOUS PROBLEM WAS NOT DETECTED UNTIL POST-RELEASE IS	27
28	2988	/	MOST FIELD SETTINGS IN PRACTICAL PDP-8 CODE ARE FOLLOWED BY AN ORIGIN	28
29	2989	/	SETTING; THE PREMATURE FIELD CHANGE IS UNABLE TO AFFECT AN ORIGIN SETTING	29
30	2990	/	IMMEDIATELY FOLLOWING THE MISHANDLED FIELD SETTING; LOADING AFTER THE	30
31	2991	/	ORIGIN SETTING WILL BE CORRECTLY STORED INTO THE NEW FIELD. HOWEVER, THERE	31
32	2992	/	IS NO RULE REGARDING MANDATORY ORIGIN SETTINGS IMMEDIATELY AFTER A FIELD	32
33	2993	/	SETTING. CERTAIN PAL ASSEMBLERS DO NOT AUTOMATICALLY CREATE A "PROTECTIVE"	33
34	2994	/	ORIGIN SETTING AFTER A FIELD SETTING. AS IS THE CASE IN ALL OTHER VERSIONS	34
35	2995	/	OF THE BINARY LOADER, THE P?S/8 RELEASE IMPLEMENTS EXTENDED MEMORY LOADING	35
36	2996	/	AS ORIGINALLY DEFINED.	36
37	2997			37
38	2998	/	3) THREE-CYCLE DATA BREAK ADDRESS CONSIDERATIONS.	38
39	2999			39
40	3000	/	STANDARD PDP-8 PERIPHERALS INCLUDE THREE DEVICES THAT RESERVE HARD-WIRED	40
41	3001	/	ADDRESS PAIRS USED TO CONTROL DATA TRANSFERS BY THE METHOD OF THREE CYCLE	41
42	3002	/	DATA BREAK [DMA]. ANY ACTIVITY ON THESE DEVICES WILL WRITE ON THE	42
43	3003	/	CORRESPONDING MEMORY LOCATIONS. THE FIRST IS USED AS A WORD COUNT REGISTER	43
44	3004	/	TO REGULATE THE SIZE OF THE TRANSFER; THE SECOND IS USED AS THE CURRENT	44
45	3005	/	ADDRESS REGISTER TO LOCATE THE DATA TRANSFER IN MEMORY. THE FOLLOWING	45
46	3006	/	ADDRESS RESERVATIONS ARE DEFINED:	46
47	3007			47
48	3008	/	A) 07750-07751 IS RESERVED FOR THE DF32 OR RF08. ONLY ONE OF THESE IS	48
49	3009	/	ALLOWED ON ANY SYSTEM.	49
50	3010			50
51	3011	/	B) 07752-07753 IS RESERVED FOR THE TC58 SERIES OF .5" MAGNETIC TAPE	51
52	3012	/	CONTROLLERS.	52
53	3013			53
54	3014	/	C) 07754-07755 IS RESERVED FOR THE TC01/TC08 DECTAPE CONTROLLER. THE USE	54
55	3015	/	OF THESE LOCATIONS IS DESCRIBED ELSEWHERE IN THIS DOCUMENT.	55
56				56
57				57
58				58

1	3017	/	WHILE THE RIM LOADER IS NOT AFFECTED BY RUNNING PROGRAMS FOR THESE DEVICES,	1
2	3018	/	THERE ARE CONSIDERATIONS AFFECTING PROGRAMS SUCH AS THE BIN LOADER THAT	2
3	3019	/	CAN BE DESTROYED IN MEMORY ONCE A PROGRAM ADDRESSING THESE LOCATIONS IS	3
4	3020	/	STARTED; THE BIN LOADER WILL BECOME NONFUNCTIONAL UNTIL MEMORY LOCATIONS	4
5	3021	/	AFFECTED BOTH DIRECTLY AND INDIRECTLY ARE RESTORED. THE GENERAL	5
6	3022	/	RECOMMENDATION IS TO RELOAD THE BIN LOADER USING THE RIM LOADER IF	6
7	3023	/	DIAGNOSTIC OR OTHER PROGRAMS FOR THESE DEVICES ARE EXECUTED. [NOTE: ANY	7
8	3024	/	DOCUMENTED REMEDY FOR THIS PROBLEM OUTSIDE OF P?S/8 DOES NOT CONSIDER THE	8
9	3025	/	PRESENCE OF A STORAGE-BASED OPERATING SYSTEM.]	9
10	3026			10
11	3027	/	ASSUMING THE P?S/8 STAND-ALONE BOOTSTRAP AT 07600 IS NOT CORRUPTED, THE	11
12	3028	/	OPERATING SYSTEM CAN BE RELOADED; THE BIN AND RIM LOADERS CAN THEN BE	12
13	3029	/	LOADED INTO MEMORY USING THE P?S/8 BIN UTILITY [PRESUMABLY BY REPEATING THE	13
14	3030	/	LAST COMMAND ENTERED INTO THE KEYBOARD MONITOR]. MORE ELABORATE BATCH	14
15	3031	/	FILES CAN BE CREATED TO HANDLE KNOWN TROUBLESOME CIRCUMSTANCES.	15
16	3032			16
17	3033	/	THE ORIGINAL RELEASE OF THE BIN LOADER WAS PRONE TO CORRUPTION BY THE USE	17
18	3034	/	OF ANY OF THESE THREE DEVICES; THE MAY 10, 1967 RELEASE [DEC-08-LBAA]	18
19	3035	/	ELIMINATES INTERACTION WITH THE TC01/TC08 THREE CYCLE DATA BREAK LOCATIONS	19
20	3036	/	[07754-07755] BUT CAN STILL BE DESTROYED BY INTERACTION WITH PROGRAMS FOR	20
21	3037	/	RF08/DF32 OR THE TC58 MAGNETIC TAPE CONTROLLER.	21
22	3038			22
23	3039	/	AS CURRENTLY DEFINED, THE P?S/8 VERSION OF THE BINARY LOADER CAN ONLY BE	23
24	3040	/	CORRUPTED BY THE USE OF PROGRAMS FOR THE TC58 MAGNETIC TAPE CONTROLLER.	24
25	3041	/	FUTURE RELEASES OF THE STAND-ALONE BOOTSTRAP AT 07600 OR THE BINARY LOADER	25
26	3042	/	MAY ELIMINATE THIS PROBLEM; ONLY ONE ADDITIONAL MEMORY LOCATION IS REQUIRED	26
27	3043	/	TO REMEDY THE SITUATION.	27
28	3044			28
29	3045	/	BY ALLOWING THE AVAILABLE CODE SPACE FOR THE BIN LOADER TO INCREASE BY ONE	29
30	3046	/	WORD, IT IS POSSIBLE TO ELIMINATE ALL CORRUPTION ASSOCIATED WITH THREE-CYCLE	30
31	3047	/	DATA BREAK DEVICES WITHOUT OTHERWISE CHANGING THE DESIGN AS PRESENTED IN THIS	31
32	3048	/	DOCUMENT. OVERALL BINARY LOADER ROBUSTNESS CAN BE IMPROVED STILL FURTHER IF	32
33	3049	/	SEVERAL MORE LOCATIONS ARE MADE AVAILABLE.	33
34	3050			34
35	3051	/	OVER THE LONG COURSE OF DECADES OF P?S/8 SYSTEM DESIGN, CERTAIN STAND-ALONE	35
36	3052	/	BOOTSTRAP ROUTINES HAVE BEEN WRITTEN ATTEMPTING TO COMPLETELY EXIST IN FIELD 0.	36
37	3053	/	IN THE CASE OF CERTAIN DEVICES, EXTENDED MEMORY IS ALWAYS REQUIRED TO RUN	37
38	3054	/	P?S/8. ASSUMING NO PORTION OF RESERVED MEMORY IS CORRUPTED BY IMPROPER	38
39	3055	/	OVERLAY BY USER PROGRAMS, THERE SHOULD NOT BE ANY MEANINGFUL DIFFERENCE	39
40	3056	/	BETWEEN THE FIELD 0 STAND-ALONE BOOTSTRAP AND AN EXTENDED MEMORY-BASED	40
41	3057	/	STAND-ALONE BOOTSTRAP. [NOTE: ARGUABLY, THE STAND-ALONE BOOTSTRAP LOCATED IN	41
42	3058	/	EXTENDED MEMORY IS LESS LIKELY TO BE CORRUPTED THAN THE LONGER FIELD 0-BASED	42
43	3059	/	BOOTSTRAP FOR CERTAIN OF THESE DEVICES; CORRUPTION CAN ONLY BE CAUSED BY	43
44	3060	/	EITHER IMPROPER USE OF THE FRONT PANEL OR ATTEMPTING TO LOAD KNOWN "ROGUE"	44
45	3061	/	DIAGNOSTIC PROGRAMS THAT VIOLATE STORAGE-BASED OPERATING SYSTEM LOADING	45
46	3062	/	CONVENTIONS (YET BARELY FIT IN THE SLIGHTLY MORE ACCOMMODATING ENVIRONMENT OF	46
47	3063	/	A PARTICULAR VERSION OF THE BINARY LOADER). ISSUES REGARDING ROGUE PROGRAMS	47
48	3064	/	ARE DISCUSSED ELSEWHERE IN THIS DOCUMENT.]	48
49	3065			49
50	3066	/	AS SUCH, THE SUBJECT OF LOWERING THE MAXIMUM STAND-ALONE BOOTSTRAP SIZE IS	50
51	3067	/	CURRENTLY BEING RESEARCHED. IT IS CONCEIVABLE THAT THE CURRENT DEFINITION CAN	51
52	3068	/	BE LOWERED IF IT TURNS OUT THE CURRENT MAXIMUM SIZE WAS DEFINED FOR A DEVICE	52
53	3069	/	THAT REQUIRES EXTENDED MEMORY FOR THE SYSTEM DEVICE HANDLER.	53
54				54
55				55
56				56
57				57
58				58

1	3071	/	THE OVERALL SIZE OF THE EXTENDED MEMORY EXTENSION FOR A SYSTEM DEVICE HANDLER	1
2	3072	/	WAS INCREASED AFTER ALL EXISTING HANDLERS WERE WRITTEN. THIS WAS DONE TO	2
3	3073	/	FORMALIZE THE DEFINITIONS OF THE MAXIMUM SIZE OF THE LOGICAL CONSOLE OVERLAY	3
4	3074	/	AND THE OPTIONAL SHELL OVERLAY FILE AND COMMUNICATIONS AREA; THIS WOULD ALSO	4
5	3075	/	FACILITATE WRITING SYSTEM DEVICE HANDLERS IN THE FUTURE FOR MORE PROBLEMATIC	5
6	3076	/	DEVICES SUCH AS 8-BIT MODE DISKETTE INTERFACES. AS SUCH, A SHORT STUB OF	6
7	3077	/	CODE IN FIELD 0 [WITH PERHAPS SEVERAL SHORT AUXILIARY ROUTINES] CAN REPLACE	7
8	3078	/	THE STAND-ALONE BOOTSTRAP FOR THE LARGEST DEVICE DEFINED PRIOR TO THE INCREASE	8
9	3079	/	IN THE EXTENDED MEMORY EXTENSION. [NOTE: FOR ALL SUCH SYSTEM DEVICES, THE	9
10	3080	/	FIELD 0 CODE WOULD NOT ONLY BE QUITE SHORT, IT WOULD LIKELY BE GENERIC TO ALL	10
11	3081	/	SUCH DEVICES; THE BULK OF THE CODE WOULD BE RESIDENT IN THE EXTENDED MEMORY	11
12	3082	/	SECTION ENTERED INDIRECTLY FROM THE FIELD 0 STUB DEFINED AT 07600 WHICH WOULD	12
13	3083	/	BE ONLY TWO INSTRUCTIONS LONG: A "CIF" INSTRUCTION TO THE HIGHEST FIELD OF	13
14	3084	/	EXTENDED MEMORY FOLLOWED BY A "JMP" INSTRUCTION TO THE REST OF THE RESTART	14
15	3085	/	CODE RESIDENT IN THE EXTENDED MEMORY SECTION. THE MAIN ISSUE IS SUCH SUPPORT	15
16	3086	/	CODE WOULD BE FAR SMALLER THAN THE TC01/TC08 STAND-ALONE BOOTSTRAP CODE OR ANY	16
17	3087	/	FUTURE VARIATION.]	17
18	3088			18
19	3089	/	ONCE AN EXHAUSTIVE STUDY IS CONCLUDED, THERE MAY BE A SHORTER DEFINITION OF	19
20	3090	/	STAND-ALONE BOOTSTRAP MAXIMUM SIZE; THIS WOULD ALLOW THE BIN LOADER TO BE	20
21	3091	/	DEFINED AS A LONGER PROGRAM TO ADD ADDITIONAL FEATURES.	21
22	3092			22
23	3093	/	RIM LOADER IMPLEMENTATION CONSIDERATIONS.	23
24	3094			24
25	3095	/	THE P?S/8 RIM LOADER IS NOT 100% IDENTICAL TO THE DEC VERSION DEFINED IN	25
26	3096	/	DEC-08-LRAA-D, FEBRUARY 23, 1965. MODIFICATIONS HAVE BEEN MADE TO SIMPLIFY	26
27	3097	/	SUPPORT FOR THE HIGH-SPEED READER.	27
28	3098			28
29	3099	/	1) THE OFFICIAL LOW-SPEED RIM LOADER USES THE "KRB" [6036] INSTRUCTION IN	29
30	3100	/	"BEG3" [07761] TO READ IN POTENTIAL LEADER FRAMES. AS SUCH, WHEN IGNORING	30
31	3101	/	THE INPUT, THE PROGRAM LOOPS BACK TO "BEG1" [07757] SINCE THE "KRB" [6036]	31
32	3102	/	INSTRUCTION HAS ALREADY STARTED THE READER FOR THE NEXT CHARACTER; FOR EACH	32
33	3103	/	DATA FRAME, THE PRIOR CONTENTS OF THE ACCUMULATOR CAN BE IGNORED BECAUSE	33
34	3104	/	THE "KRB" [6036] INSTRUCTION LOADS THE ACCUMULATOR DIRECTLY. THE "KCC"	34
35	3105	/	[6032] INSTRUCTION AT "BEG" [07756] IS ONLY USED ONCE TO CLEAR THE INPUT	35
36	3106	/	FLAG WHICH INITIATES A READ OPERATION FOR THE FIRST DATA FRAME. THE "RFC"	36
37	3107	/	[6014] INSTRUCTION REPLACES THE CONTENTS OF "BEG" [07756] TO INITIATE A	37
38	3108	/	READ ON THE HIGH-SPEED READER.	38
39	3109			39
40	3110	/	2) THE OFFICIAL HIGH-SPEED RIM LOADER USES THE "RFC RRB" [6016] INSTRUCTION	40
41	3111	/	IN "BEG3" [07761] TO READ IN POTENTIAL LEADER FRAMES; THE ACCUMULATOR MUST	41
42	3112	/	ALREADY BE CLEAR BECAUSE THE "RFC RRB" [6016] INSTRUCTION TRANSFERS THE	42
43	3113	/	LATEST FRAME DATA BY INCLUSIVE-OR INSTEAD OF LOADING THE ACCUMULATOR AS	43
44	3114	/	IN THE LOW-SPEED VERSION. WHILE IGNORING THE LEADER FRAMES, THE PROGRAM	44
45	3115	/	INSTRUCTION LOADED AT "BEG3+4" [07765] JUMPS TO "CLRAC" [07774] TO CLEAR	45
46	3116	/	THE ACCUMULATOR, AND THEN JUMPS TO "BEG1" [07757] TO EFFECT THE SAME	46
47	3117	/	OVERALL PROGRAM FLOW AS IN THE LOW-SPEED VERSION.	47
48	3118			48
49	3119	/	3) THE P?S/8 VERSION OF THE LOW-SPEED RIM LOADER USES THE SAME INSTRUCTION AS	49
50	3120	/	THE HIGH-SPEED VERSION IN "BEG3+4" [07765] THUS USES "CLRAC" [07774] TO	50
51	3121	/	ACCOMPLISH THE SAME RESULT; HOWEVER, THE INSTRUCTION AT "CLRAC+1" [07775]	51
52	3122	/	REMAINS AS "JMP BEG" [5356] INSTEAD OF THE CORRESPONDING VALUE IN THE	52
53	3123	/	OFFICIAL VERSION INSTRUCTION "JMP BEG1" [5357] IN THIS POSITION.	53
54				54
55				55
56				56
57				57
58				58

1	3125	/	DUE TO THE WAY THE LOW-SPEED TELETYPE READER MECHANISM FUNCTIONS, IT IS	1
2	3126	/	COMPLETELY SAFE TO START THE READER MULTIPLE TIMES WITHIN THE FIRST FEW	2
3	3127	/	MICROSECONDS ONCE INITIATED. SINCE THE RIM LOADER CAN BE STARTED MANUALLY,	3
4	3128	/	THE READER WILL BE STARTED BOTH BY THE CPU POWER CLEAR AND THE KCC	4
5	3129	/	INSTRUCTION THAT STARTS THE PROGRAM DURING ROUTINE USAGE. UNLESS THE	5
6	3130	/	PROGRAM IS EITHER MANUALLY STEPPED OR PERHAPS RUN ON THE PDP-12 USING THE	6
7	3131	/	VARIABLE CPU CLOCK OPTIONS, THIS SHOULD NOT BE OF ANY CONSEQUENCE.	7
8	3132			8
9	3133	/	4) ASSUMING THE INSTRUCTION AT "CLRAC+1" [07775] IS "JMP BEG" [5356] AND NOT	9
10	3134	/	"JMP BEG1" [5357], THE HIGH-SPEED RIM LOADER WILL REPEATEDLY START THE	10
11	3135	/	READER TWICE FOR EACH FRAME AFTER THE INITIAL FRAME. IT WOULD APPEAR THE	11
12	3136	/	SAME CONSIDERATION WOULD APPLY AS IN THE LOW-SPEED RIM LOADER AS DESCRIBED	12
13	3137	/	IN 3) ABOVE: STARTING THE READER MORE THAN ONCE WITHIN A FEW MICROSECONDS	13
14	3138	/	DOES NOT AFFECT THE READER OPERATION. [NOTE: AN EARLIER RELEASE OF THE	14
15	3139	/	P?S/8 HIGH-SPEED RIM LOADER IMPLEMENTED THIS SPECIFIC CODING.]	15
16	3140			16
17	3141	/	HOWEVER, THIS IS NOT THE ACTUAL CODE IN THIS RELEASE. THE GOAL WAS TO	17
18	3142	/	CREATE A COMPOSITE VERSION OF THE RIM LOADER MINIMIZING THE DIFFERENCES	18
19	3143	/	WHEN SWITCHING FROM THE LOW-SPEED RIM LOADER TO THE HIGH-SPEED RIM LOADER	19
20	3144	/	IN-PLACE JUST BEFORE RELOCATING THE CODE TO 07756-07776. THE INSTRUCTION	20
21	3145	/	PATCHED INTO "BEG13" [07771] IS ACTUALLY "RRB" [6012] AND NOT "RFC RRB"	21
22	3146	/	[6016] AS IN THE OFFICIAL VERSION; THE NEXT CHARACTER IS NOT INITIATED AT	22
23	3147	/	THIS PART OF THE PROGRAM AS IN THE OFFICIAL VERSION.	23
24	3148			24
25	3149	/	AS SUCH, THIS REQUIRES THE "JMP BEG" [5356] INSTRUCTION BE LOCATED AT	25
26	3150	/	"CLRAC+1" [07775] TO START THE READER FOR THE NEXT CHARACTER. HAD THIS	26
27	3151	/	MINOR CHANGE BEEN MADE TO THE OFFICIAL VERSION, THE LOW-SPEED AND	27
28	3152	/	HIGH-SPEED VERSIONS WOULD DIFFER BY ONE LESS WORD OVERALL. [NOTE: IT IS	28
29	3153	/	NOT LIKELY SUCH CONSIDERATIONS WERE PART OF THE ORIGINAL DESIGN; IT WAS	29
30	3154	/	ACCEPTABLE THAT BOTH VERSIONS MERELY HAVE THE SAME OVERALL LENGTH. THE	30
31	3155	/	DESIGN OF THE P?S/8 BIN IMPLEMENTATION WHICH DYNAMICALLY TRANSFORMS A	31
32	3156	/	LOW-SPEED VERSION INTO A HIGH-SPEED VERSION WAS INFLUENCED BY TECHNIQUES TO	32
33	3157	/	MINIMIZE ONCE-ONLY INITIALIZATION CODE.]	33
34	3158			34
35	3159	/	IT IS OUT OF BOUNDS FOR A PROGRAM TO EXPECT THE RIM LOADER'S PARTICULAR	35
36	3160	/	INSTRUCTIONS TO BE PRECISELY AS EXPECTED; THIS IS DOCUMENTED IN THE ORIGINAL	36
37	3161	/	RELEASE [DEC-08-LRAA-D, FEBRUARY 23, 1965] WHICH SUGGESTS THE EXISTENCE OF	37
38	3162	/	ALTERNATE RIM LOADERS FOR ADDITIONAL DEVICES [SUCH AS PT08-BASED ASR-33	38
39	3163	/	TELETYPES]. LOADED PROGRAMS MUST NOT REFERENCE THE ACTUAL CODE ACCOMPLISHING	39
40	3164	/	THE LOADING. [NOTE: THERE ARE KNOWN PROGRAMS THAT VIOLATE THIS PRINCIPLE WITH	40
41	3165	/	REGARD TO THE BIN LOADER WITH POTENTIAL TO CORRUPT THE CONTENTS OF MEMORY.]	41
42	3166			42
43	3167	/	THE UNDERLYING DESIGN ISSUE LEADING TO THIS DISCUSSION IS INHERENT IN THE	43
44	3168	/	NATURE OF THE RIM LOADER. UNLIKE THE BIN LOADER, RIM TAKES ADVANTAGE OF THE	44
45	3169	/	ABILITY TO TRANSFER THE LATEST DATA FRAME BY AN INCLUSIVE-OR TRANSFER TO JOIN	45
46	3170	/	THE LATEST FRAME DATA WITH THE PREVIOUS FRAME DATA TO FORM A 12-BIT WORD.	46
47	3171	/	THIS REQUIRES THE "KRS" [6034] INSTRUCTION IN THE LOW-SPEED VERSION WHICH DOES	47
48	3172	/	NOT CLEAR THE FLAG [OR START THE READER FOR THE NEXT FRAME]. AS SUCH, THE	48
49	3173	/	LOGICAL FLOW FOR DATA TRANSFER IN THE RIM LOADER DIFFERS FROM MOST OTHER	49
50	3174	/	CONSOLE DEVICE INPUT PROGRAMMING [INCLUDING THE BIN LOADER].	50
51	3175			51
52	3176	/	FOR FULL AUTHENTICITY AND TO ELIMINATE ANY POTENTIAL PROBLEMS WITH SYSTEMS	52
53	3177	/	THAT RUN EITHER FASTER OR SLOWER THAN EXPECTED, FUTURE RELEASES OF THE P?S/8	53
54	3178	/	"BIN" UTILITY WILL CHANGE THE PARTICULARS SUCH THAT THE AUTHENTIC VERSIONS OF	54
55	3179	/	BOTH THE LOW-SPEED AND HIGH-SPEED RIM LOADERS WILL BE SUPPORTED.	55
56				56
57				57
58				58

1	3181	/	SECTION 4: THE P?S/8 SLURP LOADER FOR TC01/TC08 SYSTEMS.	1
2	3182			2
3	3183	/	THIS IS AN IMAGE OF THE TC01/TC08 HARDWARE-SPECIFIC SLURP LOADER AS LOADED	3
4	3184	/	INTO 07600-07777 BY THE P?S/8 BIN UTILITY. THE CODE IS MOVED THROUGH MEMORY	4
5	3185	/	[FROM AN INTERNAL IMAGE WITHIN P?S/8 BIN] WHEN P?S/8 BIN IS INVOKED FOR THE	5
6	3186	/	PURPOSE OF LOADING BINARY FILES ON THE SYSTEM DEVICE OR RELATED LOGICAL DEVICE	6
7	3187	/	UNITS. [NOTE: BY DEFAULT, P?S/8 BIN LOADS BINARY FILES AS DESCRIBED HERE;	7
8	3188	/	HOWEVER, THERE ARE ADDITIONAL UTILITY FUNCTIONS AVAILABLE FOR BINARY	8
9	3189	/	PAPER-TAPE MANAGEMENT. USING APPROPRIATE OPTION SWITCHES, P?S/8 BIN CAN READ	9
10	3190	/	OR PUNCH BINARY PAPER-TAPES; THE BIN AND RIM LOADERS CAN BE PLACED INTO MEMORY	10
11	3191	/	ALONG WITH THE STAND-ALONE BOOTSTRAP AS DESCRIBED ELSEWHERE IN THIS DOCUMENT.]	11
12	3192			12
13	3193	/	THE CONTENTS OF BINARY FILES ARE LOADED INTO USER MEMORY WITHOUT USING A	13
14	3194	/	GENERAL-PURPOSE DEVICE HANDLER; AN INTERNAL ONE-WORD BUFFER IS USED TO LOAD	14
15	3195	/	THE FILE CONTENTS IN REAL-TIME BY DMA OPERATION UPDATED AFTER EACH INDIVIDUAL	15
16	3196	/	THREE-CYCLE DATA BREAK TRANSFER. DETECTION INVOLVES MONITORING THE WORD COUNT	16
17	3197	/	[OR CURRENT ADDRESS] REGISTER FOR THE INCREMENT THAT OCCURS EVERY 133	17
18	3198	/	MICROSECONDS [NOMINALLY]. THE LATEST WORD TRANSFERRED IS THEN USED AS THE	18
19	3199	/	NEXT DATA WORD TO BE PROCESSED IN SLURP FORMAT [WHICH IS DESCRIBED IN DETAIL	19
20	3200	/	ELSEWHERE IN THIS DOCUMENT] FROM THE LATEST BINARY INPUT FILE.	20
21	3201			21
22	3202	/	SLURP BINARY FORMAT IS DESIGNED FOR EFFICIENT HIGH-SPEED LOADING; ONCE	22
23	3203	/	INITIATED, THE SLURP LOADER PROGRAM EASILY MAINTAINS DATA SYNCHRONIZATION WITH	23
24	3204	/	THE DMA TRANSFER; THE FILES ARE TRANSFERRED WITHOUT ANY ADDITIONAL BUFFERING	24
25	3205	/	AND THE DECTAPE DRIVES MAY BE WRITE-LOCKED DURING THE ENTIRE LOADING PROCESS.	25
26	3206			26
27	3207	/	ALL FILES PASSED TO THE SLURP LOADER MUST BE IN SLURP BINARY FORMAT. SLURP	27
28	3208	/	BINARY FILES ARE GENERALLY CREATED DURING SOURCE FILE ASSEMBLY USING THE P?S/8	28
29	3209	/	PAL ASSEMBLY PROGRAM WITH APPROPRIATE OUTPUT OPTIONS SET. SLURP FILES CAN	29
30	3210	/	ALSO BE CREATED BY P?S/8 BIN WHEN CONVERTING BINARY PAPER-TAPES TO P?S/8	30
31	3211	/	BINARY OUTPUT FILES. [FOR FURTHER DETAILS, REFER TO DOCUMENTATION OF P?S/8	31
32	3212	/	PAL OR P?S/8 BIN AS NECESSARY.]	32
33	3213			33
34	3214	/	THE P?S/8 SLURP LOADER INCLUDES THE STAND-ALONE BOOTSTRAP AT 07600, WHICH	34
35	3215	/	MAINTAINS SEEMLESS COMPATIBILITY WITH THE CORRESPONDING P?S/8 SYSTEM DEVICE	35
36	3216	/	HANDLER. THE SYSTEM DEVICE HANDLER IS DISPLACED BY THE STAND-ALONE BOOTSTRAP	36
37	3217	/	AND THE SLURP LOADER AS DESCRIBED HERE; HOWEVER, THE REST OF THE PAGE PAST THE	37
38	3218	/	STAND-ALONE BOOTSTRAP MAY CONTAIN THE BIN AND RIM LOADERS IF APPROPRIATE	38
39	3219	/	COMMAND-LINE OPTIONS ARE USED. P?S/8 CAN BE RESTARTED AT 07600 AS LONG AS	39
40	3220	/	THE STAND-ALONE BOOTSTRAP IS NOT CORRUPTED.	40
41	3221			41
42	3222	/	NOTE: THE BOOTSTRAP FUNCTION AS AN INDEPENDENT ENTITY IS DESCRIBED ELSEWHERE	42
43	3223	/	IN THIS DOCUMENT; THIS ALLOWS FOCUS ON THE INTERACTION BETWEEN THE STAND-ALONE	43
44	3224	/	BOOTSTRAP AT 07600 AND THE PAPER-TAPE RIM AND BIN LOADERS WHEN ALL ARE LOADED	44
45	3225	/	INTO THE LAST PAGE OF FIELD 0. ALL INTERNAL ASPECTS OF THE STAND-ALONE	45
46	3226	/	BOOTSTRAP ARE DISCUSSED HERE IN GREATER DETAIL; THIS INCLUDES INTEGRAL	46
47	3227	/	SECTIONS THAT ARE SHARED WITH THE SLURP LOADER.	47
48	3228			48
49	3229	/	WHEN THE P?S/8 BIN UTILITY IS INVOKED FOR SLURP LOADING, A LIST OF FILE	49
50	3230	/	HANDLES IS PASSED IN LOCATIONS 07757-07777. A SENTINEL VALUE OF 0000 IS	50
51	3231	/	PLACED AFTER THE LAST PASSED FILE HANDLE. IF THE LIST ENDS AT 07777, THERE IS	51
52	3232	/	NO NEED FOR A SENTINEL ELEMENT; THE PROGRAM EXPLOITS A QUIRK OF THE PDP-8	52
53	3233	/	INSTRUCTION SET TO SELF-TERMINATE LISTS ENDING AT LOCATION 7777. [NOTE: THIS	53
54	3234	/	METHOD OF PROCESSING FILE HANDLES APPLIES TO ALL BASIC P?S/8 SYSTEM PROGRAMS;	54
55	3235	/	PROGRAMMING PARTICULARS ARE DISCUSSED ELSEWHERE IN THIS DOCUMENT.]	55
56				56
57				57
58				58

1	3237	/	FOR USE WITH THE SLURP LOADER, P?S/8 FILE HANDLE LISTS MAY START WITH	1
2	3238	/	SUPERFLUOUS PASSED OUTPUT FILES FOLLOWED BY INPUT FILES. WHEN THE SLURP	2
3	3239	/	LOADER AND ALL PASSED PARAMETERS ARE PLACED INTO 07600-07777, THE LIST POINTER	3
4	3240	/	"SLFLPTR" [07753] WILL BE ADJUSTED TO POINT TO THE FIRST INPUT FILE AS	4
5	3241	/	NECESSARY; ALL OUTPUT FILES ARE IGNORED. [NOTE: FUTURE RELEASES OF P?S/8 BIN	5
6	3242	/	MAY ISSUE ERROR MESSAGES IF OUTPUT FILES ARE PASSED WITH BINARY INPUT FILES.]	6
7	3243			7
8	3244	/	[NOTE: DUE TO THE WAY P?S/8 BIN FUNCTIONS, CERTAIN FILE HANDLE ELEMENTS MAY BE	8
9	3245	/	SUPPLIED BY P?S/8 BIN BASED ON COMMAND-LINE SWITCHES INSTEAD OF FILENAMES AS	9
10	3246	/	USUALLY PROCESSED BY THE KEYBOARD MONITOR EXECUTION-CLASS COMMANDS. FOR	10
11	3247	/	EXAMPLE, THE KEYBOARD MONITOR COMMAND "BIN", WHICH HAS NO ARGUMENTS, MAY	11
12	3248	/	RESULT IN AN INPUT FILE HANDLE ELEMENT FOR THE "%" FILE [BLOCK 0020-0037]	12
13	3249	/	BEING CREATED IN THE PASSED FILE HANDLE LIST. THE COMMANDS "BIN /B" OR	13
14	3250	/	"BIN %" WILL ALSO PASS THE SAME FILE HANDLE INFORMATION DUE TO THE INTERNAL	14
15	3251	/	OPERATIONAL FLOW OF P?S/8 BIN. IN CERTAIN CASES, THE FILE HANDLE LIST MAY BE	15
16	3252	/	EXPANDED INTO A COMPOSITE OF FILE HANDLES PASSED EXPLICITLY BY THE KEYBOARD	16
17	3253	/	MONITOR AND FILE HANDLE ENTRIES SUPPLIED BY P?S/8 BIN; WHEN USED TO INITIATE	17
18	3254	/	THE SLURP LOADER, THERE WILL ALWAYS BE AT LEAST ONE PASSED INPUT FILE. THE	18
19	3255	/	DEGREE OF INTEGRITY CHECK WITH REGARD TO POINTLESS CASES OF OPTION SWITCH	19
20	3256	/	USAGE AND DEFAULT ACTIONS VARIES WITH THE RELEASE OF P?S/8 BIN; FUTURE	20
21	3257	/	RELEASES ARE EXPECTED TO BE COMPLETELY CONSISTENT INCLUDING ISSUING ERROR	21
22	3258	/	MESSAGES WHEN IMPROPER OPTION "BLENDING" IS ATTEMPTED. ALL PURPOSEFUL	22
23	3259	/	COMBINATION OF OPTIONS ARE SUPPORTED BY THE 1976 RELEASE OF P?S/8 BIN.]	23
24	3260			24
25	3261	/	STARTING FIELD INFORMATION IS DERIVED FROM COMMAND-LINE OPTIONS; THE DEFAULT	25
26	3262	/	IS FIELD 0. [NOTE: CERTAIN RELEASES OF P?S/8 BIN MAY ALLOW ANOMALY OF	26
27	3263	/	CERTAIN SWITCH OPTIONS; NEWER RELEASES WILL ISSUE ERROR MESSAGES WHEN MORE	27
28	3264	/	THAN ONE FIELD OPTION SWITCH ["/0" THROUGH "/7"] IS ATTEMPTED.]	28
29	3265			29
30	3266	/	THE STARTING ADDRESS IS GIVEN BY THE "=XXXX" COMMAND-LINE OPTION [WHERE XXXX	30
31	3267	/	IS ANY VALID VALUE BETWEEN 0000 AND 7776]; THIS IS PASSED IN LOCATION	31
32	3268	/	"SEQUALS" [07756] CONSISTENT WITH ALL P?S/8 PROGRAM EXECUTION. WHEN THE	32
33	3269	/	"=XXXX" COMMAND-LINE OPTION IS NOT GIVEN, P?S/8 BIN WILL DETECT THE DEFAULT	33
34	3270	/	VALUE OF 7777. AS REQUIRED, THE ADDRESS OF A SAFE HALT OPERATION WILL BE	34
35	3271	/	PASSED AS THE STARTING ADDRESS BY INITIALIZATION CODE WITHIN P?S/8 BIN BEFORE	35
36	3272	/	THE SLURP LOADER IS MOVED TO 07600-07777.	36
37	3273			37
38	3274	/	WHEN P?S/8 BIN IS USED TO INVOKE ANY OF THE AVAILABLE LOADER FUNCTIONS [SLURP	38
39	3275	/	BINARY FILE LOADER, RIM AND BIN PAPER-TAPE LOADERS, VIRTUAL BINARY FILE	39
40	3276	/	LOADER], VARIOUS OPTION SWITCHES ARE AVAILABLE TO PREPARE USER MEMORY BEFORE	40
41	3277	/	THE LOADER STARTS. THE VALUE 0000 [THE LINC "HLT" INSTRUCTION] OR 7402 [THE	41
42	3278	/	PDP-8 "HLT" INSTRUCTION] MAY BE CHOSEN USING THE APPROPRIATE OPTION SWITCHES.	42
43	3279	/	[NOTE: THIS FEATURE IS ALWAYS RECOMMENDED TO PREVENT RUNAWAY PROGRAMS IN THE	43
44	3280	/	EARLY STAGES OF DEVELOPMENT.] [NOTE: FOR MORE INFORMATION ON HOW P?S/8 BIN	44
45	3281	/	OPERATES INTERNALLY WITH REGARD TO THE VARIOUS OPERATING MODES AND A COMPLETE	45
46	3282	/	LIST OF OPTION SWITCHES, REFER TO ADVANCED DOCUMENTATION OF THE P?S/8 BIN	46
47	3283	/	UTILITY.]	47
48	3284			48
49	3285	/	THE CODE USED HERE IS TAKEN FROM A 1976 RELEASE OF P?S/8 FOR TC01/TC08 SYSTEMS	49
50	3286	/	AND IS KNOWN TO NOT BE THE FINAL RELEASE; CHANGES WERE MADE IN BOTH THE SLURP	50
51	3287	/	LOADER CODE AND THE BOOTSTRAP CODE FOR VARIOUS REASONS. VARIOUS FUTURE	51
52	3288	/	DIRECTIONS FOR THE SLURP LOADER IN GENERAL AND THE TC01/TC08 VERSION ARE	52
53	3289	/	DISCUSSED BELOW.	53
54				54
55				55
56				56
57				57
58				58

1	3291	/	SLURP LOADER IMBEDDED BOOTSTRAP CODE SIZE CONSIDERATIONS.	1
2	3292			2
3	3293	/	AS IMPLEMENTED IN THE 1976 RELEASE OF P?S/8, THE DEFINED SIZE OF THE	3
4	3294	/	STAND-ALONE BOOTSTRAP IS 07600-07642, ALTHOUGH THE TC01/TC08 VERSION IS	4
5	3295	/	SOMEWHAT SHORTER. FUTURE RELEASES COULD CHANGE THE MAXIMUM SIZE TO ALLOW	5
6	3296	/	PERFORMANCE IMPROVEMENT IN EITHER THE SLURP LOADER FOR A SPECIFIC DEVICE, OR	6
7	3297	/	THE PAPER-TAPE BIN AND RIM LOADERS AS USED ON ANY P?S/8 CONFIGURATION.	7
8	3298			8
9	3299	/	WHEN THE P?S/8 BIN UTILITY LOADS THE PAPER-TAPE BIN AND RIM LOADERS INTO	9
10	3300	/	07643-07777, THE SAME BOOTSTRAP CODE THAT IS THE INITIAL PORTION OF THE SLURP	10
11	3301	/	LOADER WILL BE LOADED INTO 07600-07642; THE THREE COMPONENTS ARE DESIGNED TO	11
12	3302	/	COEXIST IN 07600-07777. [NOTE: SPECIFIC COMMAND-LINE OPTIONS TO P?S/8 BIN	12
13	3303	/	WILL LOAD AND START THE BIN OR RIM LOADERS INSTEAD OF THE SLURP LOADER.]	13
14	3304			14
15	3305	/	FOR CERTAIN CONFIGURATIONS, THE ABILITY TO REDEFINE ONE OF THESE COMPONENTS	15
16	3306	/	MAY BE AT ODDS WITH ANOTHER COMPONENT. LOWERING THE MAXIMUM SIZE OF THE	16
17	3307	/	STAND-ALONE BOOTSTRAP MAY BE FEASIBLE; THIS WOULD ALLOW ENHANCEMENT OF THE	17
18	3308	/	BIN LOADER AS DISCUSSED ELSEWHERE IN THIS DOCUMENT.	18
19	3309			19
20	3310	/	SPECIFIC MEMORY ALLOCATION FOR THE BOOTSTRAP-ONLY PORTION OF THE SLURP LOADER	20
21	3311	/	MAY ALSO VARY WITH OVERALL RELEASE OF P?S/8. CONTRADICTORY FACTORS INCLUDE:	21
22	3312			22
23	3313	/	1) THE LARGEST SIZE CONTIGUOUS CODE REQUIRED TO BOOTSTRAP THE MOST CUMBERSOME	23
24	3314	/	DEVICE. WHILE THE TC01/TC08 BOOTSTRAP SIZE IS NOT AT ISSUE, THE ANALOGOUS	24
25	3315	/	CODING FOR OTHER DEVICES MAY REQUIRE MORE MEMORY.	25
26	3316			26
27	3317	/	2) IMPLEMENTATION RESTRICTIONS WITHIN THE P?S/8 VERSION OF THE BIN LOADER ARE	27
28	3318	/	SUBJECT TO CHANGE; CERTAIN CONTEMPLATED MODIFICATIONS COULD LENGTHEN OR	28
29	3319	/	SHORTEN THE PROGRAM DEPENDING ON SPECIFIC GOALS. THE CURRENT P?S/8	29
30	3320	/	IMPLEMENTATION OF THE BINARY LOADER IS SOMEWHAT LESS PRONE TO INTERNAL	30
31	3321	/	CORRUPTION PROBLEMS CAUSED BY THREE-CYCLE DATA BREAK DEVICES THAN THE DEC	31
32	3322	/	STANDARD RELEASE [DEC-08-LBAA, 1967]. SHORTENING THE BOOTSTRAP LENGTH BY	32
33	3323	/	ONE WORD WOULD ELIMINATE THE ONLY REMAINING DMA-RELATED PROBLEM WHILE	33
34	3324	/	MAINTAINING THE CURRENT FEATURE SET. [NOTE: FOR MORE INFORMATION, REFER TO	34
35	3325	/	DOCUMENTATION OF THE P?S/8 IMPLEMENTATION OF THE BIN LOADER WITHIN THE	35
36	3326	/	P?S/8 BIN UTILITY.]	36
37	3327			37
38	3328	/	3) THE TC01/TC08 BOOTSTRAP DESCRIBED HERE IS A STAND-ALONE PROGRAM LOADED	38
39	3329	/	ENTIRELY IN FIELD 0. BY DESIGN, CERTAIN SYSTEM DEVICE HANDLERS REQUIRE	39
40	3330	/	EXTENDED MEMORY TO OPERATE. IN MANY SUCH CASES, MOST OF THE SYSTEM RESTART	40
41	3331	/	CODE CAN BE MADE RESIDENT IN EXTENDED MEMORY, REQUIRING ONLY A SHORT	41
42	3332	/	LINKAGE ROUTINE IN FIELD 0 TO INITIATE THE STANDARD RESTART. AS SUCH,	42
43	3333	/	FUTURE RELEASES OF P?S/8 BIN COULD SUPPORT A UNIVERSAL RESTART CONVENTION	43
44	3334	/	FOR ALL SYSTEMS REQUIRING EXTENDED MEMORY.	44
45	3335			45
46	3336	/	THE FIELD 0 SEGMENT WOULD BE EXACTLY TWO WORDS LONG: A) A "CIF" INSTRUCTION	46
47	3337	/	TO THE HIGHEST PHYSICAL FIELD OF MEMORY [62X3], B) A "JMP 7777" [5377] OR	47
48	3338	/	"JMP 7600" [5200] INSTRUCTION; THE HANDLER EXTENSION IS RESPONSIBLE FOR ALL	48
49	3339	/	ADDITIONAL RESTART CONSIDERATIONS.	49
50	3340			50
51	3341	/	WITH THIS SECONDARY CONVENTION IN PLACE, IT IS CONCEIVABLE THE LARGEST	51
52	3342	/	STAND-ALONE BOOTSTRAP WOULD BE THE TC01/TC08 VERSION AS DESCRIBED HERE	52
53	3343	/	[PERHAPS WITH CERTAIN MODIFICATIONS, DEPENDING ON SLURP LOADER INTERACTION]	53
54	3344	/	AND OTHER CONSIDERATIONS. SHOULD THIS BE THE CASE, ALL KNOWN PROBLEMS OF	54
55	3345	/	THE PAPER-TAPE BIN LOADER CAN BE ELIMINATED.	55
56				56
57				57
58				58

1	3347	0001	FIELD	SLUFLD%10	/GENERATION FIELD FOR SLURP LOADER.	1
2		*0200				2
3	3348					3
4	3349		NOPUNCH		/FOOL THE ASSEMBLER.	4
5	3350					5
6	3351	0000	FIELD	0	/WHERE THIS LOADS WHEN EXECUTED.	6
7		*0200				7
8	3352					8
9	3353		ENPUNCH		/UN-FOOL THE ASSEMBLER.	9
10	3354					10
11	3355	*7000	*SLUADR		/GENERATION ADDRESS FOR SLURP LOADER.	11
12	3356					12
13	3357	*7600	RELOC	SBOOT	/WHERE THIS IS MOVED TO FOR EXECUTION.	13
14	3358					14
15	3359	/			THE SYSTEM RESTARTS HERE USING THE STAND-ALONE BOOTSTRAP PLACED IN 07600; THIS	15
16	3360	/			ALLOWS THE UNIVERSAL EXIT CONVENTION TO APPLY IN NEARLY ALL SITUATIONS OF	16
17	3361	/			CONFORMING PROGRAMS AND CONFORMING OPERATING SYSTEMS. [NOTE: NEVER ASSUME ANY	17
18	3362	/			PARTICULAR OPERATING SYSTEM IS FULLY COMPATIBLE WITHOUT PROPER DOCUMENTATION;	18
19	3363	/			HOWEVER, ALL RELEASES OF P?S/8 AND THE R-L MONITOR SYSTEM ARE COMPATIBLE.]	19
20	3364					20
21	3365	/			RESTART CONSIDERATIONS:	21
22	3366					22
23	3367	/			1) THE DATA FIELD MUST BE SET TO 0.	23
24	3368					24
25	3369	/			2) INTERRUPTS SHOULD BE DISABLED AS THE P?S/8 KEYBOARD MONITOR LOADS OVER THE	25
26	3370	/			PRIMARY INTERRUPT LOCATIONS [00000-00001].	26
27	3371					27
28	3372	/			3) THE AC AND LINK CONTENTS ARE IGNORED.	28
29	3373					29
30	3374	/			NOTE: ALL PROGRAMMING ON THIS PAGE IS SPECIFIC TO THE TC01/TC08 CONFIGURATION;	30
31	3375	/			BOOTUP ROUTINES FOR OTHER DEVICES MUST FIT INTO THE SAME [OR GENERALLY LESS]	31
32	3376	/			CODE SPACE.	32
33	3377					33
34	3378	/			TC01/TC08 HARDWARE-SPECIFIC BOOTSTRAP DESIGN CONSIDERATIONS.	34
35	3379					35
36	3380	/			1) WHEN THE BOOTSTRAP CODE IS INITIATED, THE DECTAPE IS REWOUND INTO THE	36
37	3381	/			REVERSE END ZONE. THIS IS FOLLOWED BY A FORWARD READ OPERATION WHICH	37
38	3382	/			INITIATES DATA TRANSFER OF THE CONTENTS OF BLOCK 0000 INTO MEMORY USING	38
39	3383	/			THREE-CYCLE DATA BREAK [DMA].	39
40	3384					40
41	3385	/			2) DURING THE TAPE BLOCK DMA TRANSFER, THE CONTENTS OF TAPE BLOCK 0000	41
42	3386	/			GRADUALLY REPLACES THE CODE DEFINED BELOW; THE ONLY REQUIREMENT IS THE	42
43	3387	/			REPLACEMENT WAIT LOOP CODE MUST BE SUFFICIENTLY COMPATIBLE WITH THE	43
44	3388	/			BOOTSTRAP-PROVIDED WAIT LOOP TO ALLOW THE DMA TO COMPLETE PROPERLY (SUBJECT	44
45	3389	/			TO POSSIBLE DATA TRANSFER LENGTH ALTERATION ONCE THE WORD COUNT AND CURRENT	45
46	3390	/			ADDRESS LOCATIONS [07754-07755] ARE OVERLAID BY THE CORRESPONDING DATA ON	46
47	3391	/			THE DECTAPE BLOCK).	47
48	3392					48
49	3393	/			3) CONFORMING TO THE TC01/TC08 BOOTSTRAP CONVENTION REQUIRES THE LOOP WAITING	49
50	3394	/			FOR THE DONE FLAG BE PLACED IN MEMORY LOCATIONS 07616-07617.	50
51						51
52						52
53						53
54						54
55						55
56						56
57						57
58						58

1	3396	/	4) TRADE-OFFS EXIST WITH REGARD TO DECTAPE STATUS REGISTER "A" ACCESS. THE	1
2	3397	/	"DTLA" [6766] OPERATION REQUIRES NO PRIOR KNOWLEDGE OF STATUS REGISTER "A".	2
3	3398	/	THE "DTXA" [6764] OPERATION UPDATES THE EXISTING STATUS REGISTER DATA VIA	3
4	3399	/	.XOR. TRANSFER BETWEEN THE ACCUMULATOR AND STATUS REGISTER "A". END-ZONE	4
5	3400	/	HANDLING IS GENERALLY EASIER TO IMPLEMENT USING THE "DTXA" [6764]	5
6	3401	/	INSTRUCTION; THE DIRECTION IS MERELY INVERTED USING THE CONSTANT 0400 [OR	6
7	3402	/	EQUIVALENT] WHEN THE END-ZONE ERROR IS DETECTED.	7
8	3403			8
9	3404	/	5) UNLIKE THE SYSTEM DEVICE HANDLER, THE WAIT ROUTINE WITHIN THE STAND-ALONE	9
10	3405	/	BOOTSTRAP IS ONLY USED TO RESTART P?S/8 [OR ANY OTHER OPERATING SYSTEM	10
11	3406	/	SUFFICIENTLY COMPATIBLE WITH THIS BOOTSTRAP CONVENTION]. AS SUCH, THE	11
12	3407	/	WAIT ROUTINE USED HERE DOES NOT CHECK FOR ERRORS; WHEN USED TO REWIND THE	12
13	3408	/	OPERATING SYSTEM DECTAPE, THE ERROR FLAG IS RAISED DUE TO CONTROLLER-BASED	13
14	3409	/	DETECTION OF THE REVERSE END-ZONE. WHEN USED TO READ BLOCK 0000 WHILE	14
15	3410	/	MOVING FORWARD COMING OUT OF THE REVERSE END-ZONE, THE WAIT LOOP WILL BE	15
16	3411	/	OVERLAID BY THE CODE CONTAINED IN BLOCK 0000 AT THE DESIGNATED WAIT	16
17	3412	/	ROUTINE LOCATIONS.	17
18	3413			18
19	3414	/	6) AS A RESULT OF THE READ OPERATION, ALL MEMORY LOCATIONS FROM 07600 ONWARD	19
20	3415	/	[OTHER THAN THE WAIT LOOP] ARE EXPECTED TO BE REPLACED WITH VALUES OBTAINED	20
21	3416	/	FROM THE CONTENTS OF THE BOOT BLOCK [0000]; ALL SUBSEQUENT ACTIONS ARE	21
22	3417	/	DETERMINED BY THE CONTENTS OF THE BOOT-BLOCK. [NOTE: IT IS LIKELY THE	22
23	3418	/	BOOT-BLOCK CODE DIFFERS GREATLY FROM THE REST OF THE CODE LOADED PREVIOUSLY	23
24	3419	/	ALONG WITH THE BOOTSTRAP CODE; THE FORMER CODE IS PRIMARILY INTERNAL	24
25	3420	/	ROUTINES OF THE SLURP LOADER WHEN USED AS DEFINED HERE; ALTERNATIVELY THE	25
26	3421	/	REST OF THE CODE COULD BE PRIMARILY THE RIM AND BIN LOADERS. THE BOOTSTRAP	26
27	3422	/	ROUTINE IS MEANT TO BE LOADED WITH EITHER LOADING PROGRAM AS REQUIRED.]	27
28	3423			28
29	3424	/	7) NEWER RELEASES OF THIS BOOTSTAP SHOULD EXIST THAT HAVE BEEN WRITTEN FOR	29
30	3425	/	LATER RELEASES OF P?S/8. THE DISTINGUISHING FEATURE IS SUPPORT FOR DRIVES	30
31	3426	/	OTHER THAN UNIT 0. THIS IS ACCOMPLISHED BY EXTENDING THE STANDARD	31
32	3427	/	BOOTSTRAP TO INCLUDE DRIVE UNIT SUPPORT WITHIN THE STAND-ALONE BOOTSTRAP	32
33	3428	/	WHEN USED AS DEFINED HERE, AS WELL AS SUPPORT CODE ON THE DECTAPE	33
34	3429	/	BOOT-BLOCK USED DURING THE INITIAL BOOTUP PHASE THAT DETERMINES THE DRIVE	34
35	3430	/	UNIT IN EFFECT. AS SUCH, THIS FEATURE WILL LIKELY REMAIN A UNIQUE FEATURE	35
36	3431	/	OF P?S/8. [NOTE: MOST OF THE MANUAL BOOT PROCEDURES FOR TC01/TC08 CAN BE	36
37	3432	/	EXTENDED TO SUPPORT DRIVES OTHER THAN UNIT 0 BY SLIGHT MODIFICATION.]	37
38	3433			38
39	3434	/	IMPLEMENTING EXTENDED UNIT SUPPORT WITHIN THE STAND-ALONE BOOTSTRAP	39
40	3435	/	REQUIRES SUPPLYING A UNIT SELECT VALUE IN THE RANGE OF 0000-7000 DEPENDING	40
41	3436	/	ON THE DESIRED DECTAPE DRIVE UNIT; THE TC01/TC08 HARDWARE REQUIRES UNIT	41
42	3437	/	SELECT INFORMATION BE PASSED IN AC[0-2] DURING A "DTLA" [6766] OR "DTXA"	42
43	3438	/	[6764] OPERATION. THERE MAY BE ADDITIONAL CONSIDERATIONS AFFECTING OVERALL	43
44	3439	/	PROGRAM OPERATION AND/OR CODE EFFICIENCY; THESE ISSUES WILL BE DEALT WITH	44
45	3440	/	IN FUTURE RELEASES OF THIS DOCUMENT.	45
46				46
47				47
48				48
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1	3442	/	THE SYSTEM RESTART ENTERS HERE.	1
2	3443			2
3	3444	/	P?S/8 CONVENTIONS FOR RESTART ARE SIMILAR TO OTHER PDP-8 OPERATING SYSTEM	3
4	3445	/	CONVENTIONS. CONSIDERATIONS ARE AS FOLLOWS:	4
5	3446			5
6	3447	/	1) THE DATA FIELD MUST BE 0.	6
7	3448			7
8	3449	/	2) INTERRUPTS MUST BE DISABLED AS RESTARTING P?S/8 LOADS OVER MOST OF USER	8
9	3450	/	MEMORY INCLUDING THE INTERRUPT-HANDLING LOCATIONS [00000-00001].	9
10	3451			10
11	3452	/	3) THE AC AND LINK ARE IGNORED.	11
12	3453			12
13	3454		IFNZRO .-7600 <ERROR .> /ENSURE BOOTSTRAP COMPATIBILITY.	13
14	3456			14
15	3457	07600* 7770 SL7770, SPA SNA SZL CLA/7770	/CLEAR THE AC AND POSSIBLY	15
16	3458		/SKIP; ALSO CONSTANT 7770.	16
17	3459			17
18	3460	/	THE FOLLOWING INSTRUCTION MIGHT BE SKIPPED OR HARMLESSLY EXECUTED AS "AND 30".	18
19	3461			19
20	3462	07601* 0030 SL30, 30	/CONSTANT 0030.	20
21	3463			21
22	3464	/	THE AC IS DEFINITELY CLEAR AT THIS POINT.	22
23	3465			23
24	3466	07602* 6774 DTLB	/ALL DMA TRANSFERS TO FIELD 0.	24
25	3467	07603* 1207 TAD SL600/(600)	/SETUP FOR REWIND ON UNIT 0.	25
26	3468	07604* 4213 JMS SLWAIT	/WAIT FOR REVERSE END ZONE ERROR.	26
27	3469	07605* 1223 TAD SL7577/(SBOOT-1)	/GET PRE-DECREMENTED POINTER TO RELOAD ADDRESS.	27
28	3470	07606* 3355 DCA CA	/STASH IN THE CURRENT ADDRESS REGISTER.	28
29	3471			29
30	3472	/	THE FOLLOWING INSTRUCTION EXECUTES HARMLESSLY AS "AND I SL7770" [0600].	30
31	3473			31
32	3474	07607* 0600 SL600, 600	/CONSTANT 0600.	32
33	3475			33
34	3476	/	THE NEXT INSTRUCTION EXECUTES HARMLESSLY AS "AND I 0" [0400]; ADDITIONALLY,	34
35	3477	/	IT MUST BE LOCATED AT 7610 TO FORM ANOTHER CONSTANT USED ELSEWHERE.	35
36	3478			36
37	3479		IFNZRO .&177-10 <ERROR .>	37
38	3481			38
39	3482	07610* 0400 SL400, 400	/CONSTANT 0400.	39
40	3483			40
41	3484	07611* 1237 TAD SL220/(0220)	/GET READ FORWARD FUNCTION VALUE.	41
42	3485	07612* 4213 JMS SLWAIT	/GO WAIT THERE FOR DMA OVERLAY TO TAKE	42
43	3486		/CONTROL AWAY FROM US AND GIVE IT TO THE	43
44	3487		/JUST-READ BOOT-BLOCK CONTENTS.	44
45				45
46				46
47				47
48				48
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1	3489	/	BOOTSTRAP-ONLY WAIT LOOP ROUTINE.	1
2	3490			2
3	3491	/	THIS IMPLEMENTATION OF A TC01/TC08 WAIT ROUTINE REQUIRES THE NEW CONTENTS OF	3
4	3492	/	THE DECTAPE STATUS REGISTER "A" BE PASSED IN THE AC BECAUSE THE "DTLA" [6766]	4
5	3493	/	INSTRUCTION IS USED; THIS AVOIDS DEPENDENCY ON PRIOR STATUS REGISTER "A"	5
6	3494	/	CONTENTS. [NOTE: THERE ARE CERTAIN RESTRICTIONS ON UPDATING PARTICULAR	6
7	3495	/	STATUS REGISTER "A" BITS DEPENDING ON THE PRIOR CONTENTS; IF CERTAIN BIT	7
8	3496	/	CHANGES ARE ATTEMPTED IN AN IMPROPER ORDER, THE RESULT COULD BE UNCONTROLLED	8
9	3497	/	TAPE MOTION.]	9
10	3498			10
11	3499	7613	SLRPBLK=. /ALSO USED FOR LATEST SLURP BLOCK VALUE STORAGE.	11
12	3500			12
13	3501	/	NOTE: DEPENDING ON THE SPECIFIC CALLER, THE WAIT ROUTINE WILL RETURN DUE TO	13
14	3502	/	END-ZONE ERROR OR WORD-COUNT OVERFLOW WHEN THE DMA OPERATIONS COMPLETE.	14
15	3503			15
16	3504	07613* 0000	SLWAIT, ... /WAIT ROUTINE.	16
17	3505	07614* 6766	DTLA /LOAD STATUS "A" REGISTER WITH PASSED	17
18	3506		/FUNCTION BITS, CLEAR AC.	18
19	3507	07615* 3354	DCA WC /CLEAR WORD COUNT REGISTER NOW IN CASE READING.	19
20	3508			20
21	3509	/	ENSURE COMPATIBILITY WITH THE STANDARD TC01/TC08 WAIT LOOP.	21
22	3510			22
23	3511	IFNZRO 7616-. <ERROR .>	/MUST BE LOCATED AT 07616.	23
24	3513			24
25	3514	07616* 6771	SWAITLP, DTSP /DECTAPE [OR ERROR] FLAG UP?	25
26	3515	07617* 5216	JMP SWAITLP /FLAG NOT UP YET, KEEP WAITING FOR IT.	26
27	3516			27
28	3517	/	NOTE: IF ENTERED HERE WHILE A READ OPERATION IS IN PROGRESS, EXECUTION	28
29	3518	/	CONTINUES WITH THE NEW CONTENTS OF MEMORY TAKEN FROM BLOCK 0000 OF THE	29
30	3519	/	JUST-BOOTED SYSTEM DECTAPE; OFTEN A SUBROUTINE HEADER SIMILAR TO "SLWAIT"	30
31	3520	/	[07613] IS PRELOADED WITH A CONTRIVED "RETURN" ADDRESS. THE P?S/8 SYSTEM	31
32	3521	/	DEVICE HANDLER USES THIS METHOD TO BOOT UP WHEN STARTED BY THIS CODE OR	32
33	3522	/	EQUIVALENT. [NOTE: EXACT MEMORY LOCATIONS OF THE REST OF THE WAIT ROUTINE IN	33
34	3523	/	THE SYSTEM DEVICE HANDLER MAY VARY; VARIOUS OTHER OPERATING SYSTEMS CAN BE	34
35	3524	/	LOADED BY THIS BOOTSTRAP ROUTINE WITH LITTLE ELSE IN COMMON BESIDES THE WAIT	35
36	3525	/	LOOP REQUIREMENT.]	36
37	3526	/		37
38	3527	/	THE NEXT INSTRUCTION EXECUTES HARMLESSLY AS "NOP" [7000]; ADDITIONALLY,	38
39	3528	/	IT MUST BE LOCATED AT 07620 TO FORM ANOTHER CONSTANT USED ELSEWHERE.	39
40	3529			40
41	3530	IFNZRO .&177-20 <ERROR .>		41
42	3532			42
43	3533	07620* 7000	SL7000, 7000 /CONSTANT 7000.	43
44	3534	07621* 5613	JMP I SLWAIT /RETURN TO CALLER [IF NOT OVERLAID].	44
45	3535			45
46	3536	07622* 4002	SL4002, 4002 /CONSTANT 4002.	46
47	3537	07623* 7577	SL7577, 7577 /CONSTANT 7577.	47
48	3538			48
49	3539	/	NOTE: THE BOOTSTRAP CODE IS COMPLETELY CONTAINED ABOVE WITH THE EXCEPTION OF	49
50	3540	/	THE CONSTANT 0220 ("SL220" [7637]) DEFINED ELSEWHERE. REWORKING THE CODE CAN	50
51	3541	/	SHORTEN THE OVERALL CONTIGUOUS CODE SPACE SIZE IF NECESSARY; HOWEVER, THERE	51
52	3542	/	COULD BE OVERALL CODE SPACE TRADE-OFFS.	52
53				53
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1	3544	/	THE SLURP LOADER COMES HERE TO STOP THE TAPE AFTER THE LATEST FILE HAS BEEN	1
2	3545	/	COMPLETELY LOADED.	2
3	3546			3
4	3547	07624* 1237	SLPSTP, TAD SL220/(0220) /GET READ FUNCTION BITS.	4
5	3548	07625* 6764	DTXA /TURN OFF ONLY MOTION AND READ BITS.	5
6	3549			6
7	3550	/	NOTE: THE CONSTANT 0220 IS USED MERELY IT IS OTHERWISE AVAILABLE; IT IS USED	7
8	3551	/	ELSEWHERE WITHIN THE STAND-ALONE BOOTSTRAP AND SLURP LOADER ROUTINES.	8
9	3552	/	ALTERNATE VALUES CAN BE USED IF AVAILABLE. USING THE CONSTANT 0220 TURNS OFF	9
10	3553	/	THE TAPE MOTION BIT; THE READ FUNCTION IS CHANGED TO A TAPE MOVE OPERATION.	10
11	3554	/	THE CONSTANT 0200 CAN ALSO BE USED TO MERELY TURN OFF THE MOTION BIT. EITHER	11
12	3555	/	VALUE CAN BE INCREASED BY 0100 AS SETTING CONTINUOUS MODE IS IGNORED WHEN THE	12
13	3556	/	MOTION BIT IS CLEARED. (AN ALTERNATE VALUE IS ACTUALLY AVAILABLE, BUT	13
14	3557	/	UNREFERENCED; BY FORCING THE CONSTANT "SL7770" [7770] TO BE PLACED IN 07600,	14
15	3558	/	THE INSTRUCTION LOCATED AT "SLPSTRT+2" [7630] CAN BE USED AS THE CONSTANT 0200	15
16	3559	/	IF DESIRED.)	16
17	3560			17
18	3561	/	THE CODE FALLS THROUGH TO PROCESS THE NEXT FILE [IF ANY].	18
19	3562			19
20	3563	/	THE SLURP LOADER STARTS HERE AFTER BEING MOVED TO 07600-07777.	20
21	3564			21
22	3565	07626* 2353	SLPSTRT, ISZ SLFLPTR /BUMP TO THE NEXT INPUT FILE.	22
23	3566	07627* 1753	TAD I SLFLPTR /GET LATEST FILE HANDLE [SKIPPED IF AT 7777].	23
24	3567	07630* 0200	AND SL7770/(7770) /JUST BLOCK BITS.	24
25	3568	07631* 7450	SNA /SKIP IF MORE FILES TO LOAD.	25
26	3569	07632* 5346	JMP SLUSTRT /ELSE STARTUP THE USER PROGRAM THERE.	26
27	3570	07633* 3213	DCA SLRPBLK /SAVE FOR SEARCH ROUTINE.	27
28	3571	07634* 1753	TAD I SLFLPTR /GET LATEST FILE HANDLE AGAIN.	28
29	3572	07635* 7012	RTR;RTR /MOVE UNIT BITS TO AC[0-2].	29
30		07636* 7012		30
31	3574	07637* 0220	SL220, AND SL7000/(7000) /ISOLATE UNIT BITS; ALSO CONSTANT 0220.	31
32	3575			32
33	3576	/	THE ENTIRE STAND-ALONE BOOTSTRAP LOADS ABOVE THIS POINT. [THE CONSTANT 0220	33
34	3577	/	IS USED WITHIN THE BOOTUP CODE.]	34
35	3578			35
36	3579	7640	DECLIM=. /UPPER LIMIT FOR CURRENT TC01/TC08 BOOTSTRAP.	36
37	3580			37
38	3581	07640* 1254	TAD SL210/(210) /ADD ON FORWARD MOTION AND SEARCH BITS.	38
39	3582	07641* 6766	DTLA /START THE TAPE IN SEARCH MODE, CLEAR AC.	39
40	3583			40
41	3584	/	SINCE THE AC IS CLEAR, FORWARD TAPE MOTION WILL BE USED AS THE CURRENT TAPE	41
42	3585	/	DRIVE IS SEARCHED FOR THE INITIAL FILE BLOCK. IN THE GENERAL CASE, THIS WOULD	42
43	3586	/	NOT BE RECOMMENDED; HOWEVER, ALL TFS BINARY FILES OR EQUIVALENT STORAGE IN	43
44	3587	/	EXTENDED DIRECTORIES [OTHER THAN % AND \$] ARE LOCATED FORWARD OF THE CURRENT	44
45	3588	/	TAPE POSITION WHERE THE P?S/8 BIN UTILITY [WHICH LOADS THIS PROGRAM INTO	45
46	3589	/	MEMORY] IS LOCATED. APPENDIX "N" DISCUSSES A TECHNIQUE TO OPTIMIZE SYSTEM	46
47	3590	/	PERFORMANCE WHEN USING THE SLURP LOADER AS DEFINED IN THIS VERSION. [NOTE:	47
48	3591	/	FUTURE RELEASES MAY DIFFER ON THIS POINT RENDERING THE METHODS DESCRIBED	48
49	3592	/	EITHER OBSOLETE OR SUPERFLUOUS.]	49
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1	3594	/	ADDITIONAL SPECIFIC SEARCH ROUTINE CONSIDERATIONS.	1
2	3595			2
3	3596	/	1) DUE TO SEVERELY LIMITED CODE SPACE, THE CAPABILITIES OF THIS SEARCH ROUTINE	3
4	3597	/	ARE INFERIOR TO THE CODING USED IN THE SYSTEM DEVICE HANDLER:	4
5	3598			5
6	3599	/	A) THERE IS NO RECOVERY FROM ERRORS SINCE THERE SHOULD NEVER BE AN END-ZONE	6
7	3600	/	ERROR. WITHOUT THE ABILITY TO TURN THE TAPE AROUND AND OTHERWISE IGNORE	7
8	3601	/	THIS SITUATION, IT IS NOT POSSIBLE TO RECOVER FROM THIS CONDITION	8
9	3602	/	WITHOUT RESTARTING THE SYSTEM MANUALLY THUS ABORTING THE SLURP LOADER.	9
10	3603	/	HOWEVER, THIS ONLY AFFECTS DECTAPE BLOCKS HIGHER THAN ANY VALID FILE	10
11	3604	/	BLOCK OR BLOCKS LESS THAN 0002. NONE OF THESE ARE VALID FILE BLOCKS FOR	11
12	3605	/	TFS FILES OR EQUIVALENT; THIS CONSIDERATION IS THUS IRRELEVANT TO THE	12
13	3606	/	SLURP LOADER.	13
14	3607			14
15	3608	/	B) SINCE VERY LOW BLOCK NUMBERS ARE NEVER SEARCHED FOR, IT IS ACCEPTABLE TO	15
16	3609	/	SEARCH FOR DECTAPE BLOCKS USING SIGNED ARITHMETIC; THIS LIMITS THE	16
17	3610	/	LARGEST VALID SEARCHABLE BLOCK TO 3776. THE LONGEST KNOWN PHYSICAL	17
18	3611	/	DECTAPES ARE LESS THAN 3400 BLOCKS; NO PROBLEMS ARE EXPECTED UNLESS THE	18
19	3612	/	SLURP LOADER IS USED ON A COMPUTER SIMULATOR THAT ALLOWS FOR LARGER	19
20	3613	/	DECTAPES. [IF NEED BE, P?S/8 BIN CAN BE INVOKED WITH THE "/V" SWITCH	20
21	3614	/	SET TO FORCE THE USAGE OF THE VIRTUAL BINARY FILE LOADER; THIS WILL	21
22	3615	/	ALLOW LOADING OF BINARY FILES LOCATED UP TO BLOCK 7776, THE INHERENT	22
23	3616	/	LIMITATION OF THE SYSTEM DEVICE HANDLER SEARCH ROUTINE.]	23
24	3617			24
25	3618	/	C) THE USUAL UNDERSHOOT FACTOR IS USED TO ALLOW THE TAPE TO TURN AROUND	25
26	3619	/	WHEN LOCATED FORWARD OF THE DESIRED DECTAPE BLOCK; HOWEVER, ALL FILE	26
27	3620	/	SEARCHES ARE INITIATED IN THE FORWARD DIRECTION. SUBSEQUENT SEARCH	27
28	3621	/	OPERATIONS WILL CORRECTLY TURN THE TAPE AROUND TO LOCATE THE DESIRED	28
29	3622	/	DECTAPE BLOCK WHILE MOVING FORWARD; HOWEVER, THE SEARCH MAY PHYSICALLY	29
30	3623	/	START FORWARD THEN IMMEDIATELY TURN AROUND DUE TO THE LIMITATIONS OF	30
31	3624	/	ALWAYS STARTING THE SEARCH INITIALLY FORWARD. THIS SLIGHT POTENTIAL	31
32	3625	/	OVERHEAD IS STILL OUTWEIGHED BY THE MUCH LARGER OVERHEAD ASSOCIATED WITH	32
33	3626	/	ALL OTHER FORMS OF BINARY LOADING [SUCH AS USING THE VIRTUAL LOADER].	33
34	3627			34
35	3628	/	WHEN CONTIGUOUS FILES ARE LOADED IN THE USUAL MANNER, THE OVERHEAD DUE	35
36	3629	/	TO INITIAL FORWARD SEARCH DIRECTION FOR THE SECONDARY FILES CANNOT BE	36
37	3630	/	DISCERNED BY THE USER UNLESS AN ACCURATE STOPWATCH IS USED; MOST OF THE	37
38	3631	/	OVERHEAD IS UNAVOIDABLE AND ACTUALLY CAUSED BY TAPE COASTING. THE	38
39	3632	/	EXAMPLES DISCUSSED IN APPENDIX "N" OVERCOME BOTH THE ISSUE OF TAPE	39
40	3633	/	COASTING AND DELIBERATE FORWARD INITIAL SEARCH DIRECTION [WHICH IS	40
41	3634	/	CONVERTED FROM A SMALL LIABILITY INTO AN ASSET]. TAKING ADVANTAGE OF	41
42	3635	/	THE FIXED SIZE OF TFS FILES, PALPABLE PERFORMANCE IMPROVEMENT IS	42
43	3636	/	IMMEDIATELY NOTICED.	43
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1	3638	/	2)	ALL DETECTED ERRORS ARE FATAL. IF THERE ARE HARDWARE CONSIDERATIONS THAT	1
2	3639	/		PREVENT THE USE OF THE SLURP LOADER, THE USER IS DIRECTED TO USE THE	2
3	3640	/		VIRTUAL LOADER AS DESCRIBED IN 1B) ABOVE. USING THE VIRTUAL BINARY FILE	3
4	3641	/		LOADER REQUIRES WRITE-ENABLE ON THE DECTAPE DRIVE; OVERALL PERFORMANCE WILL	4
5	3642	/		BE DEGRADED COMPARED TO THE SLURP LOADER; HOWEVER, ALL TAPE ERRORS CAN BE	5
6	3643	/		RETRIED UNTIL SUCCESSFUL.	6
7	3644				7
8	3645	/		NOTE: USING A SYSTEM EXHIBITING HARDWARE PROBLEMS SHOULD BE CONSIDERED A	8
9	3646	/		TEMPORARY MEASURE AS THE OCCURRENCE OF EXCESSIVE ERRORS IS A CLEAR SIGN OF	9
10	3647	/		IMPENDING HARDWARE FAILURE OF THE TAPE GUIDES OR DECTAPE MEDIA. IN SOME	10
11	3648	/		CASES, THE ONLY REMEDY IS PROPER DRIVE MAINTENANCE, WHICH COULD INCLUDE	11
12	3649	/		THOROUGH TAPE AND TAPE GUIDE CLEANING AND PERHAPS HEAD SKEW ALIGNMENT OR	12
13	3650	/		REPLACEMENT OF TAPE GUIDE PARTS OR THE TAPE MEDIA ITSELF DEPENDING ON THE	13
14	3651	/		SEVERITY OF THE PROBLEM.	14
15	3652				15
16	3653	/		USE OF THE VIRTUAL LOADER SHOULD NOT BE CONSIDERED AS A SUBSTITUTE FOR	16
17	3654	/		PROPER PHYSICAL MAINTENANCE! OCCURRENCE OF FREQUENT ERRORS IS A SIGN OF A	17
18	3655	/		SERIOUS HARDWARE PROBLEM THAT WILL ONLY TEND TO GET WORSE WITH TIME AND	18
19	3656	/		USAGE AS THE UNDERLYING CONDITIONS WORSEN.	19
20	3657				20
21	3658	/	3)	FUTURE VERSIONS OF THE SLURP LOADER MAY INTERACT WITH THE LOGICAL CONSOLE	21
22	3659	/		OVERLAY TO PROVIDE ERROR RECOVERY; THIS WILL DEPEND ON SOME REDESIGN OF	22
23	3660	/		BOTH THE P?S/8 BIN UTILITY AS WELL AS THE LOGICAL CONSOLE OVERLAY.	23
24	3661				24
25	3662	/		COMES HERE DURING THE SEARCH AFTER FINDING A BLOCK OTHER THAN THE DESIRED ONE.	25
26	3663				26
27	3664	07642* 7710	SLOOP, SPA CLA	/SKIP IF POSITIVE AC.	27
28	3665				28
29	3666	/		FOR THE BENEFIT OF THE BIN AND RIM LOADER AS DEFINED IN THIS OVERALL RELEASE	29
30	3667	/		OF P?S/8, ALL BOOTSTRAP CODE MUST LOAD BEFORE THIS POINT. [NOTE: THIS IS	30
31	3668	/		EASILY MET IN THE CASE OF TC01/TC08 DECTAPE SUPPORT.]	31
32	3669				32
33	3670	7643	BOOTLIM=.	/UPPER LIMIT OF CODE DEFINED IN THE BOOTSTRAP.	33
34	3671				34
35	3672	07643* 1210	TAD SL400/(400)	/GET DIRECTION REVERSING BIT IF NEGATIVE AC.	35
36	3673	07644* 6764	NXTSRCH,DTXA	/POSSIBLY REVERSE DIRECTION BY .XOR. TRANSFER	36
37	3674			/INTO STATUS REGISTER "A"; CLEAR DONE FLAG.	37
38	3675	07645* 1334	TAD SL7750/(7750)	/GET SEARCH BUFFER ADDRESS VALUE.	38
39	3676	07646* 3355	DCA CA	/SETUP CURRENT ADDRESS REGISTER FOR SEARCH.	39
40	3677	07647* 6773	DTSF!DTRB	/WAIT FOR LATEST BLOCK; ALSO CHECK FOR ERRORS.	40
41	3678	07650* 5247	JMP .-1	/WAIT UNTIL THE DONE [OR ERROR] FLAG RAISES.	41
42	3679	07651* 7510	SPA	/SKIP IF NO ERRORS.	42
43	3680	07652* 7402	HLT	/HALT IF A TAPE ERROR. [NOTE: NOT RECOVERABLE!]	43
44					44
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1	3682	/	AT THIS POINT THE LATEST TAPE BLOCK HAS BEEN FOUND WITHOUT ERRORS. NOTE:			1
2	3683	/	AC[3] IS CLEAR BECAUSE A SELECT ERROR DID NOT OCCUR.			2
3	3684					3
4	3685	07653* 6761	DTRA		/GET STATUS REGISTER "A".	4
5	3686	07654* 0210	SL210, AND	SL400/(400)	/JUST CURRENT DIRECTION BIT; ALSO CONSTANT 0210.	5
6	3687	07655* 7640	SZA CLA		/SKIP IF GOING FORWARDS.	6
7	3688	07656* 1222	TAD	SL4002/(4002)	/ADD ON UNDERSHOOT FACTOR IF GOING IN REVERSE.	7
8	3689	07657* 1350	TAD	SRCHBLK	/GET LATEST BLOCK FOUND.	8
9	3690	07660* 7041	CIA		/INVERT AND	9
10	3691	07661* 1213	TAD	SLRPBLK	/COMPARE TO DESIRED BLOCK.	10
11	3692	07662* 7440	SZA		/SKIP IF AT THE DESIRED BLOCK.	11
12	3693	07663* 5242	JMP	SL00P	/SIGN BIT WILL DETERMINE TAPE DIRECTION THERE.	12
13	3694					13
14	3695	/	THE DESIRED BLOCK HAS BEEN FOUND WHILE MOVING FORWARD; SWITCH TO READ MODE.			14
15	3696					15
16	3697	07664* 1201	TAD	SL30/(30)	/GET NEW FUNCTION BITS.	16
17	3698	07665* 6764	DTXA		/.XOR. INTO THE FUNCTION REGISTER TO START READ.	17
18	3699					18
19	3700	/	EACH DATA WORD TRANSFER WILL BE MONITORED INDIVIDUALLY. SLURP FILE			19
20	3701	/	PARTICULARS WILL BE DEALT WITH IN REAL-TIME.			20
21	3702					21
22	3703	/	NOTE: APPENDIX "O" DISCUSSES THE INTERNAL DESIGN OF SLURP FORMAT BINARY FILES.			22
23	3704	/	ALL SLURP DATA WORDS ARE QUICKLY PROCESSED BY THE TIGHTLY CODED SECTION BELOW;			23
24	3705	/	DATA SYNCHRONIZATION IS EASILY ACHIEVED. SIMILAR IMPLEMENTATIONS HAVE BEEN			24
25	3706	/	WRITTEN FOR THE TD8E SIMPLE DECTAPE CONTROLLER AND OTHER NON-DMA DEVICES.			25
26	3707					26
27	3708	/	PROCESS THE LATEST BINARY BLOCK IN SLURP FORMAT.			27
28	3709					28
29	3710	07666* 1340	TAD	SL7756/(-22)	/SETUP THE	29
30	3711	07667* 3352	DCA	GRPCTR	/GROUP COUNTER.	30
31	3712	07670* 1345	NXTGRP, TAD	SL7772/(-6)	/SETUP THE LATEST	31
32	3713	07671* 3350	DCA	GWRDCTR	/GROUP WORD COUNTER.	32
33	3714	07672* 4327	JMS	GETWRD	/GET LATEST GROUP CONTROL WORD.	33
34	3715	07673* 7110	CLL RAR		/MOVE OVER INITIALLY.	34
35	3716	07674* 7410	SKP		/DON'T USE PREVIOUS GROUP CONTROL WORD.	35
36	3717	07675* 1342	NXTWRD, TAD	SLPCTRL	/GET LATEST CONTROL WORD.	36
37	3718	07676* 7006	RTL		/MOVE OVER TO PROCESS LAST DATA WORD.	37
38	3719	07677* 3342	DCA	SLPCTRL	/STORE BACK FOR NEXT TIME.	38
39	3720	07700* 1342	TAD	SLPCTRL	/GET THE UPDATED GROUP CONTROL WORD.	39
40	3721	07701* 7710	SPA CLA		/SKIP IF DATA OR ORIGIN SETTING.	40
41	3722	07702* 5322	JMP	FLDCHK	/JUMP IF FIELD SETTING OR END-OF-FILE.	41
42	3723	07703* 4327	JMS	GETWRD	/GET A FILE WORD	42
43	3724	07704* 6201	SLPFLD, CDF	00+.-.	/WILL BE "CDF" TO THE LATEST LOADING FIELD.	43
44	3725	07705* 7420	SNL		/SKIP IF AN ORIGIN SETTING	44
45	3726	07706* 3741	DCA I	SLRPORG	/STORE DATA WORD.	45
46	3727	07707* 6201	CDF	00	/BACK TO FIELD 0.	46
47	3728	07710* 2341	ISZ	SLRPORG	/BUMP TO NEXT LOADING LOCATION.	47
48	3729	07711* 7430	SZL		/SKIP IF DATA WORD	48
49	3730	07712* 3341	DCA	SLRPORG	/STORE NEW LOADING ORIGIN.	49
50	3731	07713* 2350	SLPNEXT, ISZ	GWRDCTR	/FINISHED THIS GROUP?	50
51	3732	07714* 5275	JMP	NXTWRD	/NO, GO PROCESS NEXT WORD.	51
52	3733	07715* 2352	ISZ	GRPCTR	/YES, FINISHED ALL GROUPS IN THIS BLOCK?	52
53	3734	07716* 5270	JMP	NXTGRP	/NO, GO BACK AND DO ANOTHER ONE.	53
54						54
55						55
56						56
57						57
58						58

1	3736	/	ALL RELEVANT DATA WORDS ON THIS BLOCK HAVE BEEN PROCESSED; THE REST OF THE			1
2	3737	/	BLOCK WILL BE IGNORED; NOW PROCESS THE NEXT BLOCK IN THE CURRENT FILE.			2
3	3738					3
4	3739	07717* 2213	ISZ	SLRPBLK	/BUMP TO NEXT BLOCK.	4
5	3740	07720* 1201	TAD	SL30/(30)	/GET FUNCTION BITS TO CHANGE MODE TO SEARCH.	5
6	3741	07721* 5244	JMP	NXTSRCH	/SEARCH FOR THE NEXT FILE BLOCK.	6
7	3742					7
8	3743	/	COMES HERE TO CHECK FOR FIELD SETTING OR END-OF-FILE.			8
9	3744					9
10	3745	07722* 7620	FLDCHK,	SNL CLA	/SKIP IF FIELD SETTING.	10
11	3746	07723* 5224	JMP	SLPSTP	/JUMP IF END-OF-FILE.	11
12	3747	07724* 4327	JMS	GETWRD	/GET THE "CDF" NEW FIELD INSTRUCTION.	12
13	3748	07725* 3304	DCA	SLPFLD	/STORE IN-LINE.	13
14	3749	07726* 5313	JMP	SLPNEXT	/GO PROCESS THE NEXT WORD.	14
15	3750					15
16	3751	/	THIS IS THE HEART OF THE SLURP LOADER. THREE-CYCLE DATA BREAKS ARE IN			16
17	3752	/	PROGRESS; A 12-BIT DATA WORD IS TRANSFERRED EVERY 133 MICROSECONDS. EACH			17
18	3753	/	TRANSFER BUMPS UP THE WORD COUNT; HOWEVER, THE WORD COUNT IS SET TO ZERO			18
19	3754	/	BEFORE EACH WORD TRANSFER; THE CURRENT ADDRESS REGISTER IS RESET TO ALWAYS			19
20	3755	/	POINT TO THE SAME MEMORY LOCATION USED AS THE DATA BUFFER. WAIT FOR THE WORD			20
21	3756	/	COUNT TO CHANGE FROM 0000-0001; THIS INDICATES THE LATEST DATA WORD IS			21
22	3757	/	AVAILABLE.			22
23	3758					23
24	3759	/	SINCE ALL DATA TRANSFERS ARE WITHIN THE SYSTEM RESERVED SPACE [07600-07777]			24
25	3760	/	THERE ARE NO RESTRICTIONS ON USER SPACE LOADING.			25
26	3761					26
27	3762	/	NOTE: THIS ROUTINE MUST BE CALLED WITH THE AC CLEAR.			27
28	3763					28
29	3764	07727* 0000	GETWRD,	...	/GET A WORD ROUTINE.	29
30	3765	07730* 3354	DCA	WC	/CLEAR WORD COUNT REGISTER.	30
31	3766	07731* 1334	TAD	SL7750/(ONEWORD-1)	/GET POINTER TO ONE-WORD BUFFER.	31
32	3767	07732* 3355	DCA	CA	/STASH IN CURRENT ADDRESS REGISTER.	32
33	3768	07733* 1354	WATLUP,	TAD WC	/GET CURRENT WORD COUNT REGISTER VALUE.	33
34	3769	07734* 7750	SL7750,	SPA SNA CLA	/SKIP IF IT CHANGED; ALSO CONSTANT 7750.	34
35	3770	07735* 5333	JMP	WATLUP	/JUMP BACK IF IT HASN'T CHANGED YET.	35
36	3771	07736* 1351	TAD	ONEWORD	/GET THE LATEST DATA WORD.	36
37	3772	07737* 5727	JMP I	GETWRD	/RETURN TO CALLER.	37
38	3773					38
39	3774	07740* 7756	SL7756,	7756	/CONSTANT 7756.	39
40	3775	07741* 0000	SLRPOrg,	...	/SLURP LOADER LOADING ORIGIN.	40
41	3776	07742* 0000	SLPCTRL,	...	/SLURP GROUP CONTROL WORD.	41
42	3777					42
43	3778	07743* 0000	ZBLOCK	2	/EMPTY SPACE!	43
44	3779					44
45	3780	/	NOTE: FUTURE RELEASES OF THE SLURP LOADER WILL LIKELY USE THESE FREE WORDS.			45
46						46
47						47
48						48
49						49
50						50
51						51
52						52
53						53
54						54
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56						56
57						57
58						58

1	3782	/	INVOKING THE SLURP LOADER FROM THE P?S/8 "BIN" UTILITY DOES NOT DEMAND AN	1
2	3783	/	EXPLICIT PROGRAM STARTING ADDRESS. IF GIVEN BY THE USER, "SLUSTRT" [07746]	2
3	3784	/	AND "SEQUALS" [07756] WILL BE SET TO THE STARTING FIELD AS A "CIF CDF" [62X3]	3
4	3785	/	INSTRUCTION AND THE 12-BIT STARTING ADDRESS RESPECTIVELY. THE CORRESPONDING	4
5	3786	/	DEFAULT VALUES ARE "CIF CDF 00" [6203] AND "DEFSTRT" [07745]. THIS IS THE	5
6	3787	/	ADDRESS OF A SAFE HALTING INSTRUCTION WITHIN THE SLURP LOADER. PRESSING	6
7	3788	/	"CONTINUE" WILL ALWAYS LOOP BACK TO THE SAME ADDRESS AS A SAFETY FEATURE.	7
8	3789	/	THIS IS IMPLEMENTED BY EITHER A "JMP" INSTRUCTION BACK TO "SLUSTRT" [07746] OR	8
9	3790	/	PLACING "DEFSTRT" [07745] IN THE PREVIOUS MEMORY LOCATION AS IS THE CASE IN	9
10	3791	/	THIS RELEASE; PRESSING "CONTINUE" WILL CAUSE THE CODE TO FALL THROUGH THUS	10
11	3792	/	REPEATING THE DEFAULT STARTING ACTION INDEFINITELY.	11
12	3793			12
13	3794	7745	DEFSTRT=. /DEFAULT STARTING ADDRESS IF NONE GIVEN.	13
14	3795			14
15	3796	07745* 7772	SL7772, SPA SNA SZL CLA HLT /HALTS HERE BY DEFAULT; ALSO CONSTANT 7772.	15
16	3797		/NOTE: THE AC IS ZERO; THIS WILL NOT SKIP.	16
17	3798			17
18	3799	/	ENSURE THE DEFAULT START IS HERE.	18
19	3800			19
20	3801		IFNZRO SL7772-DEFSTRT <ERROR .>	20
21	3803			21
22	3804	/	THE USER PROGRAM IS STARTED UP HERE AFTER ALL SLURP LOADING IS CONCLUDED.	22
23	3805			23
24	3806	07746* 6203	SLUSTRT,CIF CDF 00+.. /WILL BE "CIF CDF" TO STARTING FIELD.	24
25	3807	07747* 5756	JMP I SEQUALS /GO START IT UP THROUGH THERE.	25
26	3808			26
27	3809	/	ENSURE THE USER STARTUP IS JUST PAST THE DEFAULT START; PRESSING "CONTINUE"	27
28	3810	/	WILL JUST HARMLESSLY LOOP BACK INDEFINITELY.	28
29	3811			29
30	3812		IFNZRO DEFSTRT-SLUSTRT+1 <ERROR .>	30
31	3814			31
32	3815	/	THE POINTER TO THE SEARCH BUFFER [7750] IS ALSO AN "SPA SNA CLA" INSTRUCTION.	32
33	3816			33
34	3817		IFNZRO 7750-. <ERROR .>	34
35	3819			35
36	3820	7750	GWRDCTR=. /ALSO USED AS SLURP GROUP	36
37	3821		/WORD COUNTER IN DATA MODE.	37
38	3822			38
39	3823	07750* 0000	SRCHBLK, .. /BLOCK MARKS TRANSFER HERE IN SEARCH MODE.	39
40				40
41				41
42				42
43				43
44				44
45				45
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1	3825	/	THE SLURP LOADER CONSTANTLY RESETS THE CURRENT ADDRESS REGISTER TO FORCE ALL	1
2	3826	/	DATA TRANSFERS INTO THE SAME ONE-WORD BUFFER.	2
3	3827			3
4	3828		IFNZRO 7751-. <ERROR .> /DATA BREAK WORDS TRANSFER HERE IN READ MODE.	4
5	3830			5
6	3831	07751* 0000 ONEWORD,...	/ONE WORD DATA BUFFER.	6
7	3832			7
8	3833	/	BY PLACING THE SEARCH MODE BLOCK MARK BUFFER ONE BEFORE THE CONTINUOUSLY RESET	8
9	3834	/	DATA BREAK BUFFER, THE SAME ADDRESS VALUE CAN BE USED TO SETUP THE CURRENT	9
10	3835	/	ADDRESS REGISTER WITHIN THE RESPECTIVE ROUTINES. [NOTE: IN SEARCH MODE THE	10
11	3836	/	CURRENT ADDRESS REGISTER DOES NOT INCREMENT WHILE IN READ MODE THE CURRENT	11
12	3837	/	ADDRESS REGISTER INCREMENTS FOR EACH DATA WORD TRANSFERRED.]	12
13	3838			13
14	3839		IFNZRO SRCHBLK-ONEWORD+1 <ERROR .>	14
15	3841			15
16	3842	07752* 0000 GRPCTR, ...	/SLURP BLOCK GROUP COUNTER.	16
17	3843	07753* 7756 SLFLPTR,SFILES-1	/INPUT FILE LIST POINTER; WILL BE	17
18	3844		/PRE-INCREMENTED BEFORE USE. THE VALUE	18
19	3845		/WILL BE ADJUSTED FOR PASSED OUTPUT FILES.	19
20	3846			20
21	3847	/	TC01/TC08 DECTAPE 3-CYCLE DATA BREAK ADDRESSES.	21
22	3848			22
23	3849	07754* 0000 WC, ...	/WORD COUNT REGISTER.	23
24	3850	07755* 0000 CA, ...	/CURRENT ADDRESS REGISTER.	24
25	3851			25
26	3852	07756* 7745 SEQUALS,DEFSTRT	/WILL BE 12-BIT STARTING ADDRESS; DEFAULT VALUE	26
27	3853		/IS THE ADDRESS OF THE SAFE HALTING INSTRUCTION.	27
28	3854			28
29	3855	07757* 0000 SFILES, ZBLOCK -.	/EMPTY SPACE RESERVED FOR PASSED FILE HANDLES.	29
30	3856			30
31	3857		IFNZRO . <ERROR .> /SHOULD NOW BE JUST PAST THE END OF THE PAGE.	31
32	3859			32
33	3860	*7200 RELOC	/TURN OFF RELOCATION.	33
34	3861			34
35	3862	/	SYSTEM GENERATION AND OTHER CONSIDERATIONS.	35
36	3863			36
37	3864	/	THE BUFFER ADDRESS FOR LOADING THE SLURP LOADER MUST BE USED TO CREATE A	37
38	3865	/	PROPER WRITE ROUTINE FOR THE ENTIRE P?S/8 BIN PROGRAM; THE SLURP LOADER IS	38
39	3866	/	SELF-CONTAINED WITHIN P?S/8 BIN WHEN FIRST LOADED INTO MEMORY. P?S/8 BIN	39
40	3867	/	INTERNAL INITIALIZATION CODE MUST REFERENCE THE CODE WHERE LOADED WITHIN BIN	40
41	3868	/	WHICH IS AN ENTIRELY DIFFERENT ADDRESS SPACE THAN THE ONE DEFINED IN THIS	41
42	3869	/	DOCUMENT FOR THE PURPOSES OF SYSTEM GENERATION OR EXECUTION. IT IS NOT	42
43	3870	/	UNCOMMON FOR CERTAIN ROUTINES TO HAVE MULTIPLE LOADING DEFINITIONS DEPENDING	43
44	3871	/	ON COMPLEXITY.	44
45				45
46				46
47				47
48				48
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50				50
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1	3873	/	SECTION 5: THE P?S/8 HANDLER RELOADING "FILE" IMAGE FOR TC01/TC08 SYSTEMS.	1
2	3874			2
3	3875	/	WHEN THE USER PROGRAM BINARY FILES ARE LOADED BY THE SLURP LOADER, THE	3
4	3876	/	CONTENTS OF THE LAST PAGE OF MEMORY CONSISTS OF THE STAND-ALONE BOOTSTRAP AT	4
5	3877	/	07600 AND THE SLURP LOADER. AS SUCH, THE SYSTEM DEVICE HANDLER IS NOT	5
6	3878	/	AVAILABLE TO THE LOADED PROGRAM. BY INCLUDING AN ADDITIONAL "FILE" TO LOAD	6
7	3879	/	AFTER ALL USER PROGRAMS, THE CONTENTS OF THE LAST PAGE OF MEMORY IS	7
8	3880	/	TRANSFORMED INTO A FULLY FUNCTIONAL COPY OF THE SYSTEM DEVICE HANDLER; ALL	8
9	3881	/	READ/WRITE CALLS ARE FUNCTIONAL; THE SYSTEM CAN BE RESTARTED AT 07600 IN THE	9
10	3882	/	USUAL MANNER.	10
11	3883			11
12	3884	/	THE NET EFFECT OF USING THE SPECIAL RELOADING "FILE" IS THAT THE SYSTEM DEVICE	12
13	3885	/	HANDLER IS RESIDENT IN MEMORY WITH NO ADDITIONAL LOADING RESTRICTIONS ON THE	13
14	3886	/	USER PROGRAM SPACE. THIS IS EQUIVALENT TO HAVING USED THE VIRTUAL LOADER TO	14
15	3887	/	LOAD THE USER PROGRAM FILES; HOWEVER, THE SIGNIFICANT OVERHEAD OF THE VIRTUAL	15
16	3888	/	LOADER HAS BEEN ELIMINATED. AS SUCH, THE LOADING PROCESS IS NOTICEABLY FASTER	16
17	3889	/	AND THE SYSTEM DEVICE CAN REMAIN WRITE-LOCKED DURING THE ENTIRE LOADING	17
18	3890	/	PROCESS. [NOTE: THE SYSTEM DEVICE HANDLER CAN BE CALLED FOR THE PURPOSE OF A	18
19	3891	/	WRITE OPERATION, IN WHICH CASE A HARMLESS WRITE-PROTECT ERROR WOULD OCCUR; THE	19
20	3892	/	USUAL ERROR RECOVERY PROCESS WOULD APPLY: WRITE ENABLE THE DRIVE AND PRESS	20
21	3893	/	"CONTINUE" ON THE CPU FRONT PANEL. IN CERTAIN RELEASES OF P?S/8, IF THE	21
22	3894	/	LOGICAL CONSOLE OVERLAY IS IN EFFECT, ALL ERRORS ARE HANDLED IN THE OVERLAY.	22
23	3895	/	P?S/8 BIN WILL PASS THE INTENDED STARTING ADDRESS AND FIELD TO THE OVERLAY BY	23
24	3896	/	A SPECIAL-PURPOSE INTERFACE BETWEEN THE TWO PROGRAMS DURING THE ONCE-ONLY	24
25	3897	/	INITIALIZATION PHASE. THE STARTING FIELD AND ADDRESS INFORMATION WILL BE THAT	25
26	3898	/	OF AN ADDRESS WITHIN THE LOGICAL CONSOLE OVERLAY; THE USUAL SYSTEM DEVICE	26
27	3899	/	HANDLER HOOKS TO THE OVERLAY WILL THEN BE APPLIED. AFTER FULLY RESTORING THE	27
28	3900	/	SYSTEM DEVICE HANDLER ENVIRONMENT, THE USER PROGRAM WILL THEN BE STARTED AS	28
29	3901	/	INTENDED.]	29
30	3902			30
31	3903	/	THIS IS THE ACTUAL IMAGE OF BLOCK 0060 OF THE P?S/8 SYSTEM DECTAPE FOR	31
32	3904	/	TC01/TC08 SYSTEMS TAKEN FROM A 1976 P?S/8 RELEASE FOR THE TC01/TC08 DECTAPE.	32
33	3905			33
34	3906	/	[NOTE: NEWER RELEASES OF P?S/8 FOR THE TC01/TC08 DECTAPE ARE KNOWN TO EXIST;	34
35	3907	/	CERTAIN ASPECTS OF THE CODING MAY BE SLIGHTLY DIFFERENT OR ENHANCED AS	35
36	3908	/	NECESSARY.]	36
37	3909			37
38	3910	/	ACTUAL CONTENTS IS HARDWARE-SPECIFIC AND ALSO SLURP LOADER RELEASE-SPECIFIC.	38
39	3911	/	THE CODING PRESENTED HERE IS COMPATIBLE WITH THE TC01/TC08 SLURP LOADER FOR	39
40	3912	/	THE TC01/TC08 DECTAPE AS DESCRIBED ELSEWHERE IN THIS DOCUMENT.	40
41	3913			41
42	3914	/	DUE TO THE WAY FILE HANDLES ARE PASSED TO P?S/8 SYSTEM PROGRAMS [INCLUDING	42
43	3915	/	P?S/8 BIN], IT IS NECESSARY TO PLACE THIS CODE ON AN APPROPRIATE SYSTEM BLOCK	43
44	3916	/	THAT IS SUFFICIENTLY SIMILAR TO THE PLACEMENT OF A FILE WITHIN THE TFS	44
45	3917	/	DIRECTORY. SINCE TFS FILE STARTING BLOCKS MUST BE ALIGNED ON EIGHT-BLOCK	45
46	3918	/	BOUNDARIES, THE USE OF BLOCK 0060 MORE THAN MEETS THIS REQUIREMENT. [NOTE:	46
47	3919	/	ALL RELEASED P?S/8 SYSTEMS ARE ACTUALLY CONFIGURED SUCH THAT ALL TFS FILES ARE	47
48	3920	/	ACTUALLY ALIGNED ON 16-BLOCK BOUNDARIES; THIS WAS DONE FOR A NEVER-IMPLEMENTED	48
49	3921	/	FUTURE REQUIREMENT BEYOND THE BASIC EIGHT-BLOCK BOUNDARY; THE USE OF BLOCK	49
50	3922	/	0060 ALSO CONFORMS TO THIS ADDITIONAL REQUIREMENT. GIVEN THE DEVELOPMENT OF	50
51	3923	/	THE P?S/8 SHELL OVERLAY, IT IS LIKELY THIS EXTRA BIT OF UNUSED BOUNDARY	51
52	3924	/	ALIGNMENT WILL BE ELIMINATED IN FUTURE RELEASES.]	52
53				53
54				54
55				55
56				56
57				57
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1	3926	/	THE "FILE" AS DESCRIBED HERE MUST BE PLACED IN THE SENTINEL POSITION OF THE	1
2	3927	/	PASSED LIST OF FILE HANDLES AS THE LOADING OPERATION INITIATED CAUSES	2
3	3928	/	SELECTIVE DESTRUCTION OF THE SLURP LOADER. ONCE THIS LOADING COMPLETES, THE	3
4	3929	/	SYSTEM DEVICE HANDLER WILL BE IN MEMORY DISPLACING THE SLURP LOADER; THE	4
5	3930	/	ENTIRE OPERATION CAN BE PERFORMED WITH THE SYSTEM DECTAPE WRITE-LOCKED.	5
6	3931			6
7	3932	/	NOTE: THE SLURP LOADER NORMALLY ALLOWS UP TO 17 [OCTAL 21] FILES TO BE PASSED	7
8	3933	/	FOR A SINGLE SLURP LOADER EXECUTION. SINCE SLURP FORMAT IS QUITE EFFICIENT,	8
9	3934	/	THIS MAXIMUM NUMBER IS RARELY USED; MOST USER PROGRAMS LOAD IN THREE BINARY	9
10	3935	/	FILES OR LESS. FOR HUGE PROGRAMS THAT LOAD INTO MOST OF 32K [THE MAXIMUM SIZE	10
11	3936	/	OF MEMORY] THERE IS A SMALL POSSIBILITY THAT THE LOADING OPERATION MAY	11
12	3937	/	REQUIRE A TWO-STAGE OPERATION; FOR ANY PROJECT 24K OR SMALLER, THIS IS HIGHLY	12
13	3938	/	UNLIKELY; HOWEVER, WHEN THE "FILE" DESCRIBED HERE IS IN EFFECT, THE MAXIMUM	13
14	3939	/	NUMBER OF PASSED INPUT FILES IS LOWERED TO 16 FILES SINCE THE "FILE" IS	14
15	3940	/	INTERNALLY PASSED AS THE SENTINEL FILE IN THE LIST. CERTAIN VERSIONS OF P?S/8	15
16	3941	/	MAY IGNORE THE INHERENT PROBLEM AND TRUNCATE THE LIST WITHOUT ERROR MESSAGES.	16
17	3942	/	FUTURE RELEASES WILL ISSUE ERROR MESSAGES FOR THIS UNLIKELY CASE SUGGESTING	17
18	3943	/	THE FILES BE LOADED IN TWO INDEPENDENT LOADING OPERATIONS [PREFERABLY ALL	18
19	3944	/	EXTENDED MEMORY LOADING PERFORMED FIRST THEN ANOTHER COMMAND TO ACCOMPLISH	19
20	3945	/	FIELD 0 AND OTHER LOADING]. [NOTE: TO AVOID THE ENTIRE PROBLEM, THE VIRTUAL	20
21	3946	/	LOADER MAY BE USED INSTEAD OF THE SLURP LOADER AND THE "FILE" AS DISCUSSED	21
22	3947	/	HERE; HOWEVER, THE LOADING PROCESS WILL TAKE LONGER AND THE SYSTEM DEVICE MUST	22
23	3948	/	BE WRITE-ENABLED.]	23
24	3949			24
25	3950	/	METHOD OF IMPLEMENTATION FOR THE TC01/TC08 DECTAPE SYSTEM.	25
26	3951			26
27	3952	/	THE TC01/TC08 DECTAPE SYSTEM HANDLER IS FULLY OPERATIONAL IF ALL ADDRESSES	27
28	3953	/	FROM "SBOOT" [07600] THROUGH "SADDR" [07746] ARE LOADED CORRECTLY IN MEMORY.	28
29	3954	/	THE REST OF THE PAGE FROM 07747-07777 NEED NOT BE RELOADED FOR THE SYSTEM	29
30	3955	/	DEVICE HANDLER TO FUNCTION CORRECTLY INCLUDING THE SYSTEM RESTART AT 07600.	30
31	3956			31
32	3957	/	AS SUCH, THE "FILE" AS DEFINED HERE LOADS MOST OF ITS OPERATIONAL CODE INTO	32
33	3958	/	THE PASSED FILE LIST AREA, WHICH AT THAT POINT IS OTHERWISE MEANINGLESS	33
34	3959	/	[CONTAINING THE PASSED FILE HANDLES FOR THE NOW-LOADED INPUT FILES, INCLUDING	34
35	3960	/	THE SENTINEL VALUE OF BLOCK 0060 ON THE SYSTEM DEVICE BOOTUP UNIT REPRESENTING	35
36	3961	/	THE RELOADER "FILE"].	36
37	3962			37
38	3963	/	THE FINAL LOADING OPERATION IS TO PATCH THE SLURP LOADER CODE THAT ATTEMPTS TO	38
39	3964	/	LOCATE ADDITIONAL PASSED FILES [ALL FILES HAVE BEEN PROCESSED] TO INSTEAD	39
40	3965	/	START THE OPERATIONAL CODE DEFINED BELOW AS "SLIG0" [07757]. EXECUTION OF	40
41	3966	/	THIS CODE RELOADS THE SYSTEM DEVICE HANDLER PORTION NEEDED FOR COMPLETE	41
42	3967	/	FUNCTIONALITY AS DEFINED ABOVE. AFTER THE RELOAD IS COMPLETE, THE PROGRAM	42
43	3968	/	STARTS AS DEFINED IN THE SLURP LOADER AS DESCRIBED ELSEWHERE IN THIS DOCUMENT.	43
44	3969	/	[NOTE: THE SLURP LOADER STARTUP CODE AND DATA ARE DEFINED BEYOND THE RELOAD	44
45	3970	/	POINT; AS SUCH, THE STARTUP FUNCTION OF THE SLURP LOADER IS STILL INTACT.]	45
46	3971			46
47	3972	/	ONE OF THE OPERATIONS ASSOCIATED WITH THE RELOAD IS TO STOP THE DECTAPE DRIVE	47
48	3973	/	AFTER THE RELOAD IS COMPLETE. THE CONSTANT "S200" [07602] IS USED FOR THIS	48
49	3974	/	PURPOSE AS IT IS AVAILABLE AS A RESULT OF THE RELOAD OPERATION.	49
50	3975			50
51	3976	/	ONCE THE SYSTEM DEVICE HANDLER IS FULLY LOADED [OTHER THAN POSSIBLE	51
52	3977	/	INITIALIZATION BY THE LOGICAL CONSOLE OVERLAY], THE PROGRAM WILL BE STARTED AS	52
53	3978	/	SPECIFIED.	53
54				54
55				55
56				56
57				57
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1	3980	/	DEFAULT STARTING ADDRESS CONSIDERATIONS.	1
2	3981			2
3	3982	/	SINCE THE SLURP LOADER IS NO LONGER IN MEMORY, THE SAFE HALT ADDRESS USED AS	3
4	3983	/	THE DEFAULT STARTING ADDRESS WHILE THE SLURP LOADER WAS IN MEMORY IS NO	4
5	3984	/	LONGER APPLICABLE; HOWEVER, THE RELOAD ROUTINES INCLUDE A REPLACEMENT DEFAULT	5
6	3985	/	STARTING ADDRESS "DEFADR" [07776] WHICH IS FOLLOWED BY A "JMP DEFADR" [5376]	6
7	3986	/	INSTRUCTION TO PROVIDE A SAFE HALT ADDRESS EQUIVALENT TO WHAT WAS AVAILABLE	7
8	3987	/	WITHIN THE SLURP LOADER. [NOTE: P?S/8 BIN IMPLEMENTS THE "FILE" SYSTEM DEVICE	8
9	3988	/	HANDLER RELOAD BY THE COMMAND-LINE OPTION "/I". AS SUCH, INTERNAL ONCE-ONLY	9
10	3989	/	INITIALIZATION CODE WITHIN P?S/8 BIN WILL ADJUST THE DEFAULT STARTING ADDRESS	10
11	3990	/	AS NECESSARY; THERE ARE EQUIVALENT ISSUES WHEN THE VIRTUAL LOADER IS USED AND	11
12	3991	/	A DEFAULT SAFE HALT ADDRESS IS REQUIRED. IN CERTAIN IMPLEMENTATIONS, THE	12
13	3992	/	RELOADED SAFE HALT ADDRESS AND THE VIRTUAL LOADER SAFE HALT ADDRESS ARE	13
14	3993	/	IDENTICAL; HOWEVER, THERE ARE NO REQUIREMENTS FOR THIS AS SUCH. THIS ISSUE	14
15	3994	/	IS CONSIDERED AN INTERNAL DESIGN RECOMMENDATION ONLY. APPROPRIATE	15
16	3995	/	DOCUMENTATION WILL PROVIDE ALL THREE DEFAULT SAFE HALT ADDRESSES FOR ANY GIVEN	16
17	3996	/	HARDWARE CONFIGURATION REGARDLESS OF COINCIDENTAL SIMILARITY.]	17
18	3997			18
19	3998	/	SLURP LOADER AND RELOADER "FILE" SUPPORT ISSUES.	19
20	3999			20
21	4000	/	ONLY CERTAIN HARDWARE DEVICES SUPPORT THE SLURP LOADER. ON SYSTEMS THAT MUST	21
22	4001	/	INVOKE THE VIRTUAL LOADER, THE "/I" SWITCH IS IGNORED [WHICH OBVIATES THE	22
23	4002	/	ENTIRE PROBLEM REGARDING TRUNCATING THE PASSED FILE LIST AS DISCUSSED ABOVE].	23
24	4003	/	[NOTE: FUTURE RELEASES OF P?S/8 MAY ISSUE WARNING MESSAGES IF THE "/I" SWITCH	24
25	4004	/	IS NOT BEING HONORED; HOWEVER, SINCE BATCH FILES MAY BE WRITTEN TO PERFORM	25
26	4005	/	OPTIMAL LOADING ON ANY P?S/8 CONFIGURATION, THIS WOULD NOT BE CONSIDERED AN	26
27	4006	/	ERROR CAUSING P?S/8 BIN TO ABORT THE LOADING OPERATION.]	27
28	4007			28
29	4008	/	FUTURE RELEASES OF P?S/8 BIN MAY SUPPORT RELOADING THE SYSTEM DEVICE HANDLER	29
30	4009	/	FROM NON-ZERO LOGICAL DEVICE UNITS. AS SUCH, THE "FILE" AS DESCRIBED HERE	30
31	4010	/	MAY BE SOMEWHAT MORE COMPLEX. [IF NECESSARY, ADDITIONAL LOCATIONS WITHIN THE	31
32	4011	/	SLURP LOADER MAY BE TRANSIENTLY AVAILABLE FOR SOME OF THE PRELIMINARY	32
33	4012	/	RELOADING FUNCTIONS; AS SUCH, THE CODE DESCRIBED HERE MAY BE BROKEN UP INTO	33
34	4013	/	SEVERAL SECTIONS TO ALLOW A FEW EXTRA WORDS OF CODE TO HANDLE THE ADDITIONAL	34
35	4014	/	LOGICAL UNIT REQUIREMENT.]	35
36	4015			36
37	4016	/	THE RELOADER "FILE" IS IN SLURP BINARY FORMAT AS CREATED BY THE OUTPUT OF THE	37
38	4017	/	P?S/8 PAL ASSEMBLER AND OTHER UTILITIES. THE SOURCE CODE USES CERTAIN	38
39	4018	/	TECHNIQUES TO ALLOW THE IMAGE OF THE "FILE" TO BE WRITTEN OUT AS IF THIS WAS	39
40	4019	/	THE CASE.	40
41				41
42				42
43				43
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45				45
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1	4021	/	THEORETICAL CONSIDERATIONS REGARDING USING AN ACTUAL BINARY FILE INSTEAD OF	1
2	4022	/	THE "FILE" SUPPLIED BY P?S/8 BIN.	2
3	4023			3
4	4024	/	LIMITATIONS OF A USER-SUPPLIED FILE TO ACCOMPLISH THE IDENTICAL RESULTS AS	4
5	4025	/	THE INTERNAL "FILE" ARE:	5
6	4026			6
7	4027	/	1) THE USER MIGHT PLACE THE RELOAD FILE IN THE WRONG POSITION IN THE PASSED	7
8	4028	/	FILE LIST; THE NET EFFECT WOULD PREVENT CERTAIN PASSED BINARY FILES FROM	8
9	4029	/	LOADING; P?S/8 BIN FORCES THE "FILE" TO THE SENTINEL POSITION IN THE PASSED	9
10	4030	/	FILE LIST AVOIDING THIS ISSUE ENTIRELY.	10
11	4031			11
12	4032	/	2) THE USE OF THE RELOAD FEATURE IS GENERIC TO ALL P?S/8 SYSTEMS THAT SUPPORT	12
13	4033	/	THE SLURP LOADER AND THE RELOAD "FILE". AS SUCH, THE COMMAND LINE OPTION	13
14	4034	/	IS THE ONLY USER REQUIREMENT TO ACCOMPLISH THIS FUNCTION. FOR USERS OF	14
15	4035	/	MULTIPLE P?S/8 SYSTEMS, RELOAD "FILES" WOULD BE NEEDED, EACH FOR EVERY	15
16	4036	/	HARDWARE CONFIGURATION ACCESSED WHICH CAN LEAD TO NEEDLESS CONFUSION.	16
17	4037			17
18	4038	/	3) THE ENTIRETY OF THE "FILE" MUST BE CONTAINED IN A SINGLE 128-WORD BLOCK.	18
19	4039	/	THIS IS A LIMITATION OF P?S/8 AS CURRENTLY IMPLEMENTED; IF LONGER RELOAD	19
20	4040	/	"FILES" ARE NEEDED, THE SYSTEM WOULD HAVE TO BE SLIGHTLY REDESIGNED TO	20
21	4041	/	ALLOW SOMEWHAT LONGER RELOAD "FILE" SPACE.	21
22	4042			22
23	4043	/	TO DATE, ALL RELOAD "FILES" HAVE BEEN DEFINED IN 128 WORDS OF SLURP FORMAT	23
24	4044	/	BINARY OR LESS.	24
25	4045			25
26	4046	/	IN THE CASE OF THE RX8E DISKETTE CONTROLLER AND SIMILAR DEVICES, THE	26
27	4047	/	CONTENTS OF THE FILE MAY BE OF MIXED FORMAT: SLURP FORMAT UP TO SOME	27
28	4048	/	DEFINED POINT FOLLOWED BY CORE-IMAGE FOR THE ADDITIONAL WORDS. FOR DEVICES	28
29	4049	/	SUCH AS THE RX8E, IT MAY BE MORE EFFICIENT TO LOAD IN ROUTINES TO READ IN	29
30	4050	/	12-BIT WORDS; THE CODE IS REQUIRED TO RELOAD THE SYSTEM DEVICE HANDLER; AS	30
31	4051	/	SUCH, THE SAME ROUTINE CAN BE USED TO READ THE REST OF THE RELOAD "FILE"	31
32	4052	/	AS WELL.	32
33	4053			33
34	4054	/	FOR CERTAIN SYSTEM HARDWARE CONFIGURATIONS, EXTENDED MEMORY IS REQUIRED FOR	34
35	4055	/	THE SYSTEM DEVICE HANDLER EXTENSION. AS NECESSARY, PORTIONS OF THE SLURP	35
36	4056	/	LOADER MAY BE RESIDENT IN THE HANDLER EXTENSION AREA. [NOTE: MOST EXISTING	36
37	4057	/	SYSTEMS REQUIRING EXTENDED MEMORY WERE WRITTEN BEFORE THE AVAILABLE MEMORY	37
38	4058	/	WAS GENERALIZED TO 6000-7777 IN THE HIGHEST FIELD OF MEMORY. FOR NEW WORK,	38
39	4059	/	THIS REPRESENTS THE UPPER LIMIT OF AVAILABLE MEMORY FOR THE SYSTEM DEVICE	39
40	4060	/	HANDLER WHICH IS ESSENTIAL TO RUN P?S/8 ON PROBLEMATIC DEVICES; HOWEVER,	40
41	4061	/	FOR MOST OF THE EARLIER HANDLER CONFIGURATIONS, THERE IS OFTEN SEVERAL FREE	41
42	4062	/	PAGES OF CODE SPACE AVAILABLE TO ENHANCE THE SLURP LOADER OR OTHER SYSTEM	42
43	4063	/	COMPONENT. AS SUCH, PORTIONS OF WHAT MIGHT OTHERWISE BE IN THE RELOADER	43
44	4064	/	"FILE" COULD ACTUALLY BE RESIDENT IN THE DEVICE HANDLER EXTENSION AREA IN	44
45	4065	/	EXTENDED MEMORY. IF THIS IS THE CASE FOR ANY PARTICULAR HARDWARE	45
46	4066	/	CONFIGURATION, IT WILL TEND TO SIMPLIFY IMPLEMENTATION OF THE ENTIRE "FILE"	46
47	4067	/	DEFINITION WITHIN AT MOST 128 WORDS OF SLURP FORMAT BINARY. [NOTE: THE	47
48	4068	/	ACTUAL EXTENDED MEMORY FIELD USED WOULD GENERALLY HAVE BEEN DETERMINED	48
49	4069	/	WITHIN THE SYSTEM RESTART CODE AT 07600 AS LOADED WITH THE SLURP LOADER.	49
50	4070	/	SUFFICIENT SPACE WOULD GENERALLY BE AVAILABLE TO PROVIDE ANY NECESSARY	50
51	4071	/	LINKAGE ROUTINES REQUIRED FOR THE SYSTEM DEVICE HANDLER RELOAD FUNCTION.]	51
52				52
53				53
54				54
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56				56
57				57
58				58

1	4073	/	THE P?S/8 "GET" COMMAND CONSIDERATIONS.	1
2	4074			2
3	4075	/	THE GET COMMAND LOADS SLURP BINARY FILES INTO MEMORY AND WRITES OUT AN IMAGE	3
4	4076	/	OF 00000-07777 ON THE SYSTEM SCRATCH AREA [BLOCKS 0020-0057] THEN REBOOTS THE	4
5	4077	/	SYSTEM. THIS ALLOWS PROGRAM DEBUGGING WITH P?S/8 ODT AND OTHER FUNCTIONS.	5
6	4078			6
7	4079	/	THE WRITE FUNCTION IS ACTUALLY RESIDENT ON THE SYSTEM DEVICE HANDLER IMAGE AS	7
8	4080	/	THE DEFAULT CONTENTS OF THE SYSTEM LOADER [07632-07637]. AS SUCH, IF THE	8
9	4081	/	SYSTEM DEVICE HANDLER IS RELOADED, SETTING THE DEFAULT STARTING ADDRESS TO	9
10	4082	/	07632 WILL ACCOMPLISH THE MEMORY SAVE AND REBOOT. [NOTE: WHEN THE VIRTUAL	10
11	4083	/	LOADER IS IN EFFECT, THE SYSTEM LOADER WILL BE UPDATED TO INCLUDE THE SAME	11
12	4084	/	CALL AS IF FRESHLY RELOADED. ONCE THE PROGRAM IS STARTED AT 07632, THE NET	12
13	4085	/	RESULT IS THE SAVING OF MEMORY AND SYSTEM RESTART AS DESCRIBED ABOVE	13
14	4086	/	REGARDLESS OF WHICH LOADER WAS USED.]	14
15	4087			15
16	4088	/	NOTE: THE GET COMMAND IS ACTUALLY AN ALIAS ENTRY FOR THE P?S/8 BIN PROGRAM.	16
17	4089	/	THE ONLY OPERATIONAL DIFFERENCE IS FORCING THE STARTING FIELD TO 0 AND THE	17
18	4090	/	STARTING ADDRESS TO 7632. THE "/V" SWITCH IS ALLOWED TO FORCE THE USE OF THE	18
19	4091	/	VIRTUAL LOADER; THE DEFAULT IS TO USE THE SLURP LOADER WITH THE "/I" SWITCH	19
20	4092	/	FORCED ON TO RELOAD THE SYSTEM DEVICE HANDLER. ALL DISCUSSIONS REGARDING THE	20
21	4093	/	ISSUES OF THE "/I" SWITCH AND PASSED FILE LIST LENGTH ARE IDENTICAL TO THAT	21
22	4094	/	DESCRIBED ABOVE. FUTURE RELEASES OF P?S/8 BIN AND GET MAY HAVE MORE STRINGENT	22
23	4095	/	CONSISTENCY CHECKS TO PREVENT THE USE OF CONFLICTING OPTIONS [SUCH AS	23
24	4096	/	SPECIFYING "/1" THROUGH "/7" FROM THE GET COMMAND. WARNING MESSAGES MAY BE	24
25	4097	/	ISSUED IN FUTURE RELEASES IF THE "/I" SWITCH IS EXPRESSLY USED ON SYSTEMS	25
26	4098	/	WHERE THE VIRTUAL LOADER MUST BE USED; THE OPTIONAL USE OF "/V" IN SUCH	26
27	4099	/	SITUATIONS WOULD BE REDUNDANT BUT OTHERWISE IGNORED; THIS ALLOWS USER COMMANDS	27
28	4100	/	TO FORCE THE USE OF THE VIRTUAL LOADER ON ALL SYSTEMS.	28
29	4101			29
30	4102	/	THE INTERNAL LOGIC OF BIN AND GET IS SUCH THAT THE REQUIREMENT OF USING THE	30
31	4103	/	VIRTUAL LOADER IS KNOWN BY INTERNAL INSPECTION OF THE AREA RESERVED FOR AN	31
32	4104	/	IMAGE OF THE SLURP LOADER. FUTURE VERSIONS OF P?S/8 MAY SUPPORT A	32
33	4105	/	CONFIGURATION PROGRAM THAT WILL DEPEND ON THIS GENERIC IMPLEMENTATION TO	33
34	4106	/	ALLOW SYSTEM PREPARATION ON A DEVICE OTHER THAN THE INTENDED CONFIGURATION.	34
35	4107	/	THE RESULTING SYSTEM CAN THEN BE IMAGED TO BE ULTIMATELY MADE AVAILABLE ON THE	35
36	4108	/	INTENDED HARDWARE.	36
37	4109			37
38	4110	/	NOTE: IF THE VIRTUAL LOADER IS NOT IN EFFECT, OR THE "/I" OPTION SWITCH IS	38
39	4111	/	NOT USED TO RELOAD THE SYSTEM DEVICE HANDLER, THE USE OF =7632 IS NOT A VIABLE	39
40	4112	/	OPTION. STARTING THE SLURP LOADER AT INTERNAL LOCATIONS CAN CAUSE SYSTEM	40
41	4113	/	CORRUPTION. ALL BINARY LOADER OPERATIONS MUST BE CAREFULLY STARTED CONSISTENT	41
42	4114	/	WITH PROGRAM REQUIREMENTS. IN GENERAL, ALL STARTING INFORMATION IS THE	42
43	4115	/	USER'S RESPONSIBILITY. [NOTE: THIS IS ONE OF THE MOST IMPORTANT FEATURES OF	43
44	4116	/	THE SLURP LOADER; PROGRAM BEHAVIOR CAN BE UNPREDICTABLE, ESPECIALLY IN THE	44
45	4117	/	EARLY STAGES OF DEVELOPMENT; IF THE SYSTEM DEVICE IS WRITE-LOCKED, ONLY THE	45
46	4118	/	IN-MEMORY COPY OF ANY SYSTEM COMPONENT CAN BE CORRUPTED.]	46
47	4119			47
48	4120	/	FUTURE RELEASES OF P?S/8 BIN MAY IMPOSE RESTRICTIONS ON COMMAND-SUPPLIED	48
49	4121	/	STARTING ADDRESS AND FIELD VALUES; THIS WOULD PREVENT IMPROPER PROGRAM START	49
50	4122	/	WITHIN ANY SYSTEM RESERVED ADDRESS SPACE. PERMITTED ADDRESSES WOULD INCLUDE	50
51	4123	/	ANY ADDRESS FROM 00000-07577, 07600 [WHICH HARMLESSLY RESTARTS THE SYSTEM	51
52	4124	/	AFTER ALL LOADING] AND 07632 IF THE SYSTEM DEVICE HANDLER IS AVAILABLE AS THE	52
53	4125	/	PROGRAM STARTS. ANY EXTENDED MEMORY ADDRESS WOULD BE PERMITTED UNLESS THE	53
54	4126	/	SYSTEM DEVICE HANDLER REQUIRES EXTENDED MEMORY EXTENSION OR THE LOGICAL	54
55	4127	/	CONSOLE OVERLAY IS IN EFFECT.	55
56				56
57				57
58				58

1	4129	0001	FIELD	SLIFLD%10	/GENERATION FIELD FOR SLURP RELOAD "FILE".	1
2		*0200				2
3	4130					3
4	4131		NOPUNCH		/FOOL THE ASSEMBLER.	4
5	4132					5
6	4133	0000	FIELD	0	/WHERE THIS LOADS WHEN EXECUTED.	6
7		*0200				7
8	4134					8
9	4135		ENPUNCH		/UN-FOOL THE ASSEMBLER.	9
10	4136					10
11	4137	*7200	*SLIADR		/GENERATION ADDRESS FOR SLURP RELOAD "FILE".	11
12	4138					12
13	4139	*7600	RELOC	SBOOT	/PAGE WHERE EVERYTHING HAPPENS.	13
14	4140					14
15	4141	/			NOTE: THIS IS AN IMAGE OF A VALID [SHORT] SLURP FILE; HOWEVER, IT DOES NOT	15
16	4142	/			INCLUDE AN ORIGIN TO 0200 AS IS USUALLY THE CASE. [AS SUCH, WERE IT AN ACTUAL	16
17	4143	/			TFS BINARY FILE, CERTAIN UTILITIES COULD REJECT IT AS INVALID. SINCE IT IS	17
18	4144	/			ALWAYS THE CONTENTS OF BLOCK 0060, NO FILE SYSTEM ACCESS EXISTS.]	18
19	4145					19
20	4146	07600* 7000	FL^4+OR^4+DA^4+DA^4+DA^4+DA		/CONTROL WORD.	20
21	4147					21
22	4148	07601* 6201	CDF	00	/FIELD SETTING TO FIELD 0.	22
23	4149					23
24	4150	07602* 7757	SFILES		/*SFILES	24
25	4151					25
26	4152	*7757	RELOC	SFILES	/WHERE THIS ASSEMBLES.	26
27	4153					27
28	4154	/			WHEN CONTROL IS PASSED HERE, START THE BLOCK 0000 RELOADING PROCESS.	28
29	4155					29
30	4156	07757* 1207	SLIGO, TAD	SL600/(600)	/GET REWIND BITS.	30
31	4157	07760* 4213	JMS	SLWAIT	/WAIT FOR THE TAPE TO REWIND.	31
32	4158	07761* 1223	TAD	SL7577/(SBOOT-1)	/SETUP THE	32
33	4159	07762* 3355	DCA	CA	/CURRENT ADDRESS REGISTER.	33
34	4160					34
35	4161	07763* 0000	DA^4+DA^4+DA^4+DA^4+DA^4+DA		/CONTROL WORD.	35
36	4162					36
37	4163	*7763	RELOC	. -1	/RESET ASSEMBLY ADDRESS.	37
38	4164					38
39	4165	07763* 1374	TAD	TRNLEN/(SBOOT-SADDR)	/GET WORD COUNT VALUE TO TRANSFER JUST	39
40	4166				/ENOUGH OF THE SYSTEM HANDLER NEEDED.	40
41	4167	07764* 3354	DCA	WC	/SETUP THE WORD COUNT REGISTER.	41
42	4168	07765* 1237	TAD	SL220/(220)	/GET READ FORWARD COMMAND BITS.	42
43	4169	07766* 6766	DTLA		/START READING BLOCK 0000.	43
44	4170	07767* 6771	DTSF		/FLAG UP?	44
45	4171	07770* 5367	JMP	. -1	/NO, WAIT FOR IT.	45
46						46
47						47
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52						52
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54						54
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1	4173	07771* 0000	DA^4+DA^4+DA^4+DA^4+DA^4+DA	/CONTROL WORD.	1
2	4174				2
3	4175	*7771	RELOC .-1	/RESET ASSEMBLY ADDRESS.	3
4	4176				4
5	4177	/	AT THIS POINT, [MOST OF] BLOCK 0000 IS READ IN. WE CAN ACCESS CONSTANTS		5
6	4178	/	LOCATED WITHIN THE SYSTEM DEVICE HANDLER.		6
7	4179				7
8	4180	07771* 1202	TAD S200/(200)	/GET MOTION BIT.	8
9	4181	07772* 6764	DTXA	/STOP THE TAPE.	9
10	4182	07773* 5346	JMP SLUSTR	/GO START THE USER PROGRAM.	10
11	4183				11
12	4184	07774* 7632 TRNLEN, SBOOT-SADDR		/DESIRED TRANSFER LENGTH	12
13	4185				13
14	4186	07775* 0000	ZBLOCK 1	/EMPTY SPACE!	14
15	4187				15
16	4188	/	THIS IS THE NEW DEFAULT STARTING ADDRESS OF A SAFE HALT.		16
17	4189				17
18	4190	07776* 7402 DEFADR, HLT		/DEFAULT STARTING ADDRESS.	18
19	4191				19
20	4192	07777* 1025	DA^4+OR^4+DA^4+EF^4+EF^4+EF	/CONTROL WORD.	20
21	4193				21
22	4194	*7777	RELOC .-1	/RESET ASSEMBLY ADDRESS.	22
23	4195				23
24	4196	07777* 5376	JMP DEFADR	/IN CASE THE USER RESTARTS THE MACHINE.	24
25	4197				25
26	4198	00000* 7626	SLPSTR	/*SLPSTR	26
27	4199				27
28	4200	*7626	RELOC SLPSTR	/WHERE THIS ASSEMBLES.	28
29	4201				29
30	4202	07626* 5357	JMP SLIGO	/START OUR CODE AFTER WE'RE LOADED.	30
31	4203				31
32	4204	/	THE SLURP FORMAT "FILE" ENDS HERE; THE CONTROL WORD INDICATES ALL ADDITIONAL		32
33	4205	/	WORDS ARE TO BE IGNORED.		33
34	4206				34
35	4207	*7231	RELOC	/TURN OFF RELOCATION.	35
36	4208				36
37	4209	07231 0000	ZBLOCK .&7600+200-.	/EMPTY SPACE.	37
38					38
39					39
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1	4211	/	SECTION 6: P?S/8 KEYBOARD MONITOR REWIND AND UNLOAD ROUTINES FOR TC01/TC08.	1
2	4212			2
3	4213	/	THESE ARE THE ACTUAL REWIND AND UNLOAD ROUTINES TAKEN FROM A 1976 P?S/8 SYSTEM	3
4	4214	/	DECTAPE IMAGE [KNOWN TO NOT BE A FINAL RELEASE].	4
5	4215			5
6	4216	/	THIS SMALL ROUTINE IS THE P?S/8 KEYBOARD MONITOR AND IMBEDDED EDITOR CODING	6
7	4217	/	TO HANDLE CONTROL-R AND CONTROL-T CHARACTERS WHICH CAN BE TYPED DURING THE	7
8	4218	/	ENTRY OF ALL FORMS OF TEXT ENTRY INCLUDING EXECUTION-CLASS COMMANDS NOT YET	8
9	4219	/	COMPLETED BY PRESSING THE RETURN KEY.	9
10	4220			10
11	4221	/	CONTROL-R IS MEANT TO BE USED TO REWIND THE CURRENT SYSTEM DEVICE HANDLER	11
12	4222	/	DRIVE UNIT [ASSUMING THE HARDWARE IS SOME FORM OF TAPE DEVICE]. SIMILARLY,	12
13	4223	/	CONTROL-T IS MEANT TO BE USED TO UNLOAD THE SAME DEVICE. NOT ALL SYSTEMS	13
14	4224	/	SUPPORT THESE COMMANDS EQUALLY; ON SOME SYSTEMS THE COMMANDS ARE VIRTUALLY	14
15	4225	/	INTERCHANGEABLE; ON OTHER SYSTEMS BOTH COMMANDS ARE IGNORED.	15
16	4226			16
17	4227	/	EACH SPECIFIC HARDWARE-DEPENDENT RELEASE OF P?S/8 SUPPORTS A CUSTOM VERSION OF	17
18	4228	/	THIS CODE TAILORED FOR THE SYSTEM DEVICE HARDWARE OF THAT PARTICULAR SYSTEM	18
19	4229	/	[IF THE HARDWARE SUPPORTS THESE COMMANDS AT ALL]. THE ACTIONS TAKEN ARE	19
20	4230	/	PARTICULAR TO THE HARDWARE CONFIGURATION.	20
21	4231			21
22	4232	/	THE CODE DECRIBED HERE IS FOR THE TC01/TC08 DECTAPE CONFIGURATION. ANY MIX OF	22
23	4233	/	TU55 AND TU56 DRIVES IS SUPPORTED; HOWEVER, THE EXACT ACTIONS TAKEN ARE	23
24	4234	/	DEPENDENT ON THE DRIVE TYPE:	24
25	4235			25
26	4236	/	WHEN CONTROL-R IS TYPED, THE CURRENT TAPE DRIVE UNIT IS REWOUND TOWARDS THE	26
27	4237	/	REVERSE END ZONE. CONTROL IS IMMEDIATELY RETURNED TO THE KEYBOARD MONITOR TO	27
28	4238	/	ALLOW ADDITIONAL KEYBOARD INPUT WHILE THE TAPE IS REWINDING. WHEN THE TAPE	28
29	4239	/	ENTERS THE END ZONE, THE CONTROLLER DETECTS THIS AS AN END ERROR, AND STOPS	29
30	4240	/	THE TAPE AUTOMATICALLY.	30
31	4241			31
32	4242	/	THIS ALLOWS THE USER TO TYPE ADDITIONAL COMMANDS DURING THE TAPE REWIND	32
33	4243	/	OPERATION TAKING ADVANTAGE OF OVERLAPPED OPERATION. THIS COULD INCLUDE	33
34	4244	/	ENTERING AN EXECUTION-CLASS COMMAND KNOWN TO REQUIRE ACCESS TO BLOCKS NEAR THE	34
35	4245	/	PHYSICAL BEGINNING OF THE TAPE; BY INITIATING A REWIND DURING THE COMMAND	35
36	4246	/	ENTRY, SIGNIFICANT WINDING TIME CAN BE ACHIEVED BEFORE THE COMMAND IS	36
37	4247	/	COMPLETELY ENTERED.	37
38	4248			38
39	4249	/	SHOULD THE COMMAND REQUIRE READING IN PART OF THE SYSTEM DEVICE, THE REWIND	39
40	4250	/	OPERATION IS CANCELLED BY THE CALL TO THE SYSTEM DEVICE HANDLER. [NOTE: THIS	40
41	4251	/	IS TRUE FOR THE TC01/TC08 VERSION OF THE REWIND ROUTINE AS DESCRIBED HERE;	41
42	4252	/	OTHER IMPLEMENTATIONS MAY NOT DUPLICATE THE COMMAND ACTION SUFFICIENTLY. FOR	42
43	4253	/	EXAMPLE, THE TD8E VERSION OF THE REWIND ROUTINE MUST MONITOR THE TAPE POSITION	43
44	4254	/	UNTIL THE END ZONE IS LOCATED; AS SUCH, OVERLAP IS NOT POSSIBLE ON THAT	44
45	4255	/	CONFIGURATION.]	45
46				46
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1	4257	/	WHEN CONTROL-T IS TYPED, THE CURRENT TAPE DRIVE IS REWOUND; HOWEVER, THE	1
2	4258	/	ROUTINE WAITS FOR THE END ERROR; AS SUCH, OVERLAPPED OPERATIONS ARE NOT	2
3	4259	/	POSSIBLE.	3
4	4260			4
5	4261	/	ADDITIONAL OPERATIONS ARE PERFORMED AFTER THE END ZONE IS DETECTED FOR THE	5
6	4262	/	PURPOSE OF FORCING THE TAPE OFF THE TAKE-UP REEL [CAUSING A FULL TAPE UNLOAD].	6
7	4263	/	WHAT HAPPENS NEXT DEPENDS ON THE DRIVE TYPE:	7
8	4264			8
9	4265	/	1) IF THE SELECTED DRIVE IS A TU55, ADDITIONAL COMMANDS ARE ISSUED TO MODIFY	9
10	4266	/	TAPE DIRECTION AND MOTION BITS FORCING THE TAPE REELS TO LOCK.	10
11	4267			11
12	4268	/	2) IF THE SELECTED DRIVE IS A TU56 [WHICH LACKS MECHANICAL BRAKING], THE TAPE	12
13	4269	/	REELS ROTATE UNCONTROLLABLY UNTIL THE USER RESETS THE DRIVE; THE ADDITIONAL	13
14	4270	/	COMMANDS ARE IGNORED.	14
15	4271			15
16	4272	/	AFTER THE UNLOAD OPERATION HAS COMPLETED, CONTROL IS RETURNED TO THE KEYBOARD	16
17	4273	/	MONITOR TO ALLOW ADDITIONAL KEYBOARD INPUT.	17
18	4274			18
19	4275	/	IMPLEMENTATION OF SPECIFIC REWIND/UNLOAD ROUTINES IN THE KEYBOARD MONITOR.	19
20	4276			20
21	4277	/	HISTORICALLY, P?S/8 SYSTEM GENERATION CONSISTED OF PLACING ALL COMPONENTS	21
22	4278	/	AFFECTED BY CHOICE OF SYSTEM DEVICE INTO MEMORY AT THE SAME TIME. THE	22
23	4279	/	SPECIFIC CODE PARTICULAR TO THE CHOSEN DEVICE IS THEN OVERLAID OVER THE BASIC	23
24	4280	/	ROUTINES TO PRODUCE A CUSTOMIZED SYSTEM.	24
25	4281			25
26	4282	/	TO CUSTOMIZE P?S/8 IN THE FUTURE, NEW MECHANISMS NEED TO BE DEFINED TO	26
27	4283	/	COMPLETELY DEFINE THE SYSTEM SPECIFICS. SYSTEMS SUCH AS OS/8 MERELY NEED TO	27
28	4284	/	DEFINE A SYSTEM DEVICE HANDLER; OTHER SYSTEM COMPONENTS SUCH AS THE REWIND AND	28
29	4285	/	UNLOAD ROUTINES DO NOT EXIST ELSEWHERE.	29
30	4286			30
31	4287	/	FOR THE PRESENT, ACCESS TO THE SOURCE CODE IS NEEDED, INCLUDING ALL CODE	31
32	4288	/	SPECIFIC TO THE HARDWARE. IN SOME CASES, THIS WAS PARTIALLY CREATED BY HAND	32
33	4289	/	OR TRANSFERRED BY IMAGE COPYING FROM OTHER SYSTEMS THAT SUPPORTED P?S/8 BUT	33
34	4290	/	NOT THE CODE SPECIFICS [SUCH AS CREATING TD8E SYSTEMS ON THE TC01/TC08-BASED	34
35	4291	/	SYSTEM].	35
36	4292			36
37	4293	/	FUTURE SYSTEM GENERATION TECHNIQUES MAY SEPARATE THE SPECIFICS OF THE HARDWARE	37
38	4294	/	CONFIGURATION TARGET FROM THE CURRENT SYSTEM CONFIGURATION. THIS WILL ALLOW	38
39	4295	/	IMAGING OF A SYSTEM TO EVENTUALLY BE TRANSFERRED TO THE TARGET HARDWARE	39
40	4296	/	CONFIGURATION. APPENDIX "H" DISCUSSES CERTAIN SYSTEM GENERATION TECHNIQUES	40
41	4297	/	THAT APPLY TO THE REWIND AND UNLOAD ROUTINES AS WELL AS ALL OTHER	41
42	4298	/	HARDWARE-SPECIFIC CONSIDERATIONS.	42
43	4299			43
44	4300	/	NOTE: OS/8 SUPPORTS A COMMAND TO REWIND A TC01/TC08 DECTAPE WHICH CAN BE USED	44
45	4301	/	FOR ANALOGOUS PURPOSES, AND A SEPARATE UNLOAD COMMAND FOR THE TC01/TC08 THAT	45
46	4302	/	WILL ALSO WAIT UNTIL THE TAPE UNLOAD OPERATION COMPLETES. BOTH OF THESE	46
47	4303	/	SYSTEMS SHOULD NOT EXECUTE THE TC01/TC08-SPECIFIC ROUTINES ON CERTAIN	47
48	4304	/	CONFIGURATIONS OF THE PDP-12; THE COMPUTER WILL HANG IF THE "DTLA" [6766]	48
49	4305	/	OPERATION IS ATTEMPTED.	49
50				50
51				51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	4307	0000	FIELD	KMNFLD%10	/GENERATION FIELD FOR KEYBOARD MONITOR.	1
2		*0200				2
3	4308					3
4	4309		NOPUNCH		/FOOL THE ASSEMBLER.	4
5	4310					5
6	4311	0000	FIELD	0	/WHERE THIS LOADS WHEN EXECUTED.	6
7		*0200				7
8	4312					8
9	4313		ENPUNCH		/UN-FOOL THE ASSEMBLER.	9
10	4314					10
11	4315	*0000	*KMNADR		/GENERATION ADDRESS FOR KEYBOARD MONITOR.	11
12	4316					12
13	4317	*0000	RELOC	0000	/THE KEYBOARD MONITOR LOADS HERE.	13
14	4318					14
15	4319		NOPUNCH		/PREVENT BINARY OUTPUT.	15
16	4320					16
17	4321	00000 0000	ZBLOCK	REWUNLD-	/UNCHANGED SPACE IN THE KEYBOARD MONITOR.	17
18	4322					18
19	4323		ENPUNCH		/RESTORE BINARY OUTPUT.	19
20	4324					20
21	4325	*1342	*		/ESTABLISH PROPER LOADING ORIGIN.	21
22	4326					22
23	4327	/	THE TC01/TC08-SPECIFIC REWIND/UNLOAD ROUTINES LOAD HERE.			23
24	4328					24
25	4329	01342 1040	REWUNLD,TAD	L7770/[7770]	/SETUP THE MAIN	25
26	4330	01343 3305	DCA	RWTEMP	/TIMEOUT COUNTER.	26
27	4331	01344 1000	TAD	CURUNIT	/GET THE CURRENT LOGICAL DEVICE UNIT BITS.	27
28	4332	01345 7112	CLL	RTR;RTR	/MOVE TO AC[0-2].	28
29		01346 7012				29
30	4334	01347 1366	TAD	L700/(700)	/ADD REWIND FUNCTION BITS.	30
31	4335	01350 6766	DTLA		/START THE TAPE REWIND.	31
32	4336	01351 1100	TAD	KBDCHR	/GET THE COMMAND CHARACTER.	32
33	4337	01352 1064	TAD	L7756/(-22)	/COMPARE TO CONTROL-R.	33
34	4338	01353 7650	SNA CLA		/SKIP IF NOT CONTROL-R.	34
35	4339	01354 5216	JMP	KBDLUP	/GO GET MORE KEYBOARD INPUT.	35
36	4340	01355 6771	UNLWAIT,DTSF		/FLAG UP YET?	36
37	4341	01356 7410	SKP		/SKIP THE NEXT IF NOT.	37
38	4342	01357 5342	JMP	REWUNLD	/FLAG IS UP, GO FORCE THE TAPE TO MOVE.	38
39	4343					39
40	4344	/	THE FLAG ISN'T UP YET, SO WAIT FOR SOME TIME.			40
41	4345					41
42	4346	01360 2100	ISZ	KBDCHR	/INNER LOOP: ENOUGH TIMES?	42
43	4347	01361 5355	JMP	UNLWAIT	/NO, KEEP WAITING.	43
44	4348	01362 2305	ISZ	RWTEMP	/WAITED LONG ENOUGH?	44
45	4349	01363 5355	JMP	UNLWAIT	/NO, KEEP GOING.	45
46	4350					46
47	4351	/	AT THIS POINT, THE DONE FLAG WILL NO LONGER RAISE; THE TAPE MUST HAVE FALLEN			47
48	4352	/	OFF THE REEL AND IS SPINNING OUT OF CONTROL.			48
49	4353					49
50	4354	01364 1367	TAD	L400/(400)	/GET DIRECTION-CHANGE BIT.	50
51	4355	01365 6764	DTXA		/REVERSE DIRECTION.	51
52	4356					52
53	4357	/	THE CONTROLLER IS NOW SET TO FORWARD MOTION. THERE IS NO TAPE MARK DETECTION			53
54	4358	/	IN PROGRESS BECAUSE THE TAPE IS OFF THE REELS. WE NEED TO WAIT TO ALLOW THE			54
55	4359	/	COMMAND TO SETTLE BEFORE SENDING ANOTHER COMMAND.			55
56						56
57						57
58						58

1	4361	/	THE FOLLOWING EXECUTES AS "AND I 1300".	1
2	4362			2
3	4363	01366 0700 L700, 700	/CONSTANT 0700.	3
4	4364			4
5	4365	/	THE FOLLOWING EXECUTES AS "AND I CURUNIT".	5
6	4366			6
7	4367	01367 0400 L400, 400	/CONSTANT 0400.	7
8	4368			8
9	4369	/	THE FOLLOWING EXECUTES AS "AND CURUNIT".	9
10	4370			10
11	4371	ZBLOCK 0	/EMPTY SPACE RESERVED FOR ALTERNATE CODE.	11
12	4372			12
13	4373	01370 1104 TAD L200/[200]	/GET MOTION BIT.	13
14	4374	01371 6764 DTXA	/STOP THE TAPE, LOCKING THE BRAKES.	14
15	4375			15
16	4376	/	ALL OF THE FOLLOWING EXECUTE AS "AND CURUNIT".	16
17	4377			17
18	4378	01372 0000 ZBLOCK KBDLUP&7600+177-. .	/EMPTY SPACE RESERVED FOR ALTERNATE CODE.	18
19	4379			19
20	4380	01377 5216 JMP KBDLUP	/GO GET MORE KEYBOARD INPUT.	20
21	4381			21
22	4382	/	SHOULD NOW BE ON THE NEXT PAGE.	22
23	4383			23
24	4384	*1400 RELOC	/TURN OFF RELOCATION.	24
25	4385			25
26	4386	IFNZRO .&177 <ERROR .>		26
27	4388			27
28	4389	/	IMPLEMENTATION NOTES:	28
29	4390			29
30	4391	/	1) THE USE OF THE CONSTANT 0700 IS ARBITRARY AS BIT[5] [WHICH SETS CONTINUOUS	30
31	4392	/	MODE] IS IGNORED IN TC01/TC08 DECTAPE MOVE COMMANDS. AS SUCH, 0600 WILL	31
32	4393	/	ALSO WORK AND WOULD PERHAPS BE THE MORE LIKELY CHOICE; HOWEVER, OTHER	32
33	4394	/	CONSIDERATIONS MAY HAVE INFLUENCED THE DESIGN. THE SPECIFIC USE OF CERTAIN	33
34	4395	/	CONSTANTS AND TEMPORARIES IN THIS CODE COULD BE A CONSEQUENCE OF OTHER	34
35	4396	/	ROUTINES IN THE KEYBOARD MONITOR CODE NOT SHOWN HERE. [THE MAJORITY OF THE	35
36	4397	/	CODE IS KNOWN TO BE OPTIMIZED FOR MINIMAL CODE SPACE; CONSIDERATIONS DOWN	36
37	4398	/	TO THE BIT LEVEL HAVE BEEN MADE FOR MOST ROUTINES.]	37
38	4399			38
39	4400	/	2) THE SPECIFIC LENGTH OF THE RESERVED SPACE FOR THE REWIND/UNLOAD ROUTINES	39
40	4401	/	HAS LIKELY CHANGED SLIGHTLY DEPENDING UPON THE RELEASE. THE SOURCE CODE	40
41	4402	/	AUTOMATICALLY ADAPTS AS NECESSARY AS LONG AS THERE IS SUFFICIENT SPACE FOR	41
42	4403	/	THE REST OF THE PROGRAM. IT IS PRESUMED THAT "ZBLOCK 0" IS SUPPORTED AS	42
43	4404	/	GENERATING NOTHING; IF THIS IS NOT THE CASE, THE SOURCE CODE NEEDS TO ADD A	43
44	4405	/	CONDITIONAL LAYER TO PREVENT ANY UNEXPECTED PROBLEMS. THE ZBLOCK DIRECTIVE	44
45	4406	/	IS KNOWN TO BE DEFECTIVE IN THE STANDARD RELEASE OF PAL10 FOR THE TOPS-10	45
46	4407	/	OPERATING SYSTEM; P?S/8 PAL AND OS/8 PAL8 CORRECTLY HANDLE THIS SITUATION.	46
47	4408			47
48	4409	/	3) IN MOST RELEASES OF P?S/8, SYSTEM GENERATION LOADS THE KEYBOARD MONITOR IN	48
49	4410	/	ITS INTENDED FIELD AND ADDRESS [00000-02777]; AS SUCH, THE CODE IS NOT	49
50	4411	/	ACTUALLY RELOCATED PER SE. FUTURE RELEASES OF P?S/8 SOURCE MODULES COULD	50
51	4412	/	REQUIRE RELOCATION OR OTHER MODIFICATIONS TO IMPLEMENT ADDITIONAL SYSTEM	51
52	4413	/	GENERATION METHODS; AS SUCH, THE CODE INCLUDES PROPER RELOCATION STATEMENTS	52
53	4414	/	SHOULD THE VALUES BE CHANGED. [NOTE: THE CURRENT SYSTEM GENERATION METHOD	53
54	4415	/	USES RELOCATED ADDRESSES FOR ALL OTHER SYSTEM COMPONENTS WITH THE EXCEPTION	54
55	4416	/	OF THE IMAGE OF THE INITIAL TFS DIRECTORY.]	55
56				56
57				57
58				58

1	4418	/	APPENDICES.	1
2	4419			2
3	4420	/	APPENDIX A.	3
4	4421			4
5	4422	/	THE SHORTEST KNOWN MANUAL TC01/TC08 BOOTSTRAP.	5
6	4423			6
7	4424	/	THE FOLLOWING PROCEDURE IS CLAIMED TO BE THE SHORTEST METHOD TO BOOTUP ANY	7
8	4425	/	TC01/TC08 DECTAPE SYSTEM [WITH THE EXCEPTION OF THE DISK MONITOR SYSTEM].	8
9	4426			9
10	4427	/	IT IS ADVISABLE TO WRITE-PROTECT THE TAPE DRIVE WHEN PERFORMING MANUAL	10
11	4428	/	BOOTSTRAP PROCEDURES REGARDLESS OF OPERATING SYSTEM. P?S/8, THE R-L MONITOR	11
12	4429	/	SYSTEM, POLY BASIC, EDU-SYSTEM 30 AND PERHAPS SOME ADDITIONAL SYSTEMS DO NOT	12
13	4430	/	WRITE ON THE SYSTEM DRIVE DURING SUCCESSFUL BOOTUP OPERATIONS.	13
14	4431			14
15	4432	/	PRELIMINARY TAPE PLACEMENT.	15
16	4433			16
17	4434	/	MOUNT THE SYSTEM DECTAPE ON DRIVE 0 WITH SOME PORTION OF THE REVERSE END ZONE	17
18	4435	/	OVER THE TAPE HEAD. [SEE THE NOTE BELOW IF UNFAMILIAR WITH DECTAPE FORMAT	18
19	4436	/	SPECIFICS.] THE DRIVE SHOULD BE WRITE-PROTECTED TO PREVENT UNEXPECTED WRITING	19
20	4437	/	ISSUES; THIS OPERATION MUST BE PERFORMED PRIOR TO STEP 6) BELOW.	20
21	4438			21
22	4439	/	BOOTSTRAP PROCEDURE.	22
23	4440			23
24	4441	/	1) SET FRONT PANEL ADDRESS TO 07754.	24
25	4442			25
26	4443	/	2) DEPOSIT 7577 TWICE. [NOTE: CONTENTS OF 07754 MUST BE AN ACCEPTABLE VALUE	26
27	4444	/	FROM A WIDE RANGE THAT INCLUDES 7577; 07755 MUST BE SET TO 7577.]	27
28	4445			28
29	4446	/	3) SET FRONT PANEL ADDRESS TO 07615.	29
30	4447			30
31	4448	/	4) DEPOSIT 7604; 6766; 5217.	31
32	4449			32
33	4450	/	5) SET FRONT PANEL ADDRESS TO 07615.	33
34	4451			34
35	4452	/	6) SET SWITCHES TO 0220; PRESS START.	35
36	4453			36
37	4454	/	THE SYSTEM WILL START UP AUTOMATICALLY; ADDITIONAL TAPE MOTION WILL OCCUR AS	37
38	4455	/	REQUIRED, GENERALLY ACHIEVING A COMMAND PROMPT WITHIN THREE SECONDS [ASSUMING	38
39	4456	/	P?S/8 IS BEING BOOTSTRAPPED WITH THE OPTIONAL SHELL DISABLED; BOOTUP TIMES FOR	39
40	4457	/	OTHER SYSTEMS MAY VARY].	40
41	4458			41
42	4459	/	SHOULD THE BOOTSTRAP OPERATION FAIL, THE ENTIRE PROCEDURE NEEDS TO BE REPEATED	42
43	4460	/	AS MOST ERRORS DESTROY THE MANUALLY ENTERED VALUES.	43
44				44
45				45
46				46
47				47
48				48
49				49
50				50
51				51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	4462	/	*****	1
2	4463	/	* THE FOLLOWING STEPS ARE ONLY NEEDED IF THE FORMATTING AND POSITIONING OF	2
3	4464	/	* DECTAPES IS NOT PROPERLY UNDERSTOOD. THE TAPE MEDIA ARE INTENDED TO BE	3
4	4465	/	* FORMATTED BY *BARELY* WRAPPING ONE TURN OF TAPE ON THE TAKE-UP REEL. THE	4
5	4466	/	* FORMATTING SOFTWARE WRITES OUT A LONG REVERSE END ZONE FROM THAT POSITION	5
6	4467	/	* FORWARD. WINDING ADDITIONAL TAPE ONLY SERVES TO WASTE TAPE AND CAUSE	6
7	4468	/	* CONFUSION LOCATING THE REVERSE END ZONE. IN EXTREME CASES THE FORMATTING	7
8	4469	/	* PROGRAM COULD FAIL IF THE TAPE WINDS OFF OF THE SUPPLY REEL.	8
9	4470	/	*	9
10	4471	/	* THERE ARE ADDITIONAL P?S/8-SPECIFIC DETRIMENTAL EFFECTS CAUSED BY USING	10
11	4472	/	* MISFORMATTED TAPES. [NOTE: OTHER OPERATING SYSTEMS MAY HAVE ANALOGOUS	11
12	4473	/	* FEATURES CAUSING EQUIVALENT PROBLEMS.]	12
13	4474	/	*	13
14	4475	/	* THE P?S/8 KEYBOARD MONITOR AND BUILT-IN EDITOR SUPPORTS A PAIR OF DEVICE	14
15	4476	/	* CONTROL OPERATIONS WHICH TAKE IMMEDIATE ACTION AS DETERMINED BY THE SYSTEM	15
16	4477	/	* DEVICE HANDLER. ON TC01/TC08 SYSTEMS PRESSING CONTROL-R REWINDS THE	16
17	4478	/	* SYSTEM TAPE WHILE ALSO ALLOWING ADDITIONAL COMMAND TYPING. THIS ALLOWS	17
18	4479	/	* FOR REWINDING THE TAPE AHEAD OF COMPLETING A COMMAND THAT WOULD START THE	18
19	4480	/	* TAPE MOTION TYPICALLY TOWARDS THE BEGINNING OF THE TAPE. PROPERLY USED	19
20	4481	/	* THIS CAN SAVE NOTICEABLE TIME INITIATING EXECUTION-CLASS COMMANDS.	20
21	4482	/	*	21
22	4483	/	* ON TC01/TC08 SYSTEMS PRESSING CONTROL-T UNLOADS THE SYSTEM TAPE. THE	22
23	4484	/	* MACHINE HANGS WAITING FOR THE TAPE TO REWIND COMPLETELY AND FALL OFF THE	23
24	4485	/	* TAKE-UP REEL. IF THE DRIVE IS TYPE TU55, THE DRIVE BRAKES WILL LOCK THUS	24
25	4486	/	* STOPPING THE TAPE MOTION. KEYBOARD INPUT IS RE-ENABLED ONCE THE UNLOAD	25
26	4487	/	* OPERATION COMPLETES.	26
27	4488	/	*	27
28	4489	/	* IF THE DECTAPE IS MISFORMATTED THESE OPERATIONS COULD BECOME CONFUSED.	28
29	4490	/	* THIS COULD LEAD TO INCORRECT TAPE POSITIONING AND/OR A STUMBLING EFFECT	29
30	4491	/	* AS THE DRIVE ATTEMPTS TO NAVIGATE REGIONS OF THE TAPE NEVER INTENDED TO	30
31	4492	/	* EXIST. [MANUAL INTERVENTION COULD BE REQUIRED TO REMEDY THIS.]	31
32	4493	/	*	32
33	4494	/	* IF THE COMPUTER OPERATOR HAS NO IDEA ABOUT THE ACTUAL TAPE FORMAT OR JUST	33
34	4495	/	* WISHES TO EXERCISE EXTREME CAUTION, THE FOLLOWING STEPS CAN BE USED TO	34
35	4496	/	* ENSURE A SUCCESSFUL BOOTUP [IF AT ALL POSSIBLE]. AFTER STEP 4) ABOVE	35
36	4497	/	* INSERT THE FOLLOWING ADDITIONAL OPERATIONS BEFORE MOVING TO STEP 5) ABOVE.	36
37	4498	/	*	37
38	4499	/	* I) SET FRONT PANEL ADDRESS TO 07615.	38
39	4500	/	*	39
40	4501	/	* II) SET SWITCHES TO 0200; PRESS START. THE TAPE WILL WIND FORWARD THEN	40
41	4502	/	* STOP EITHER WHEN THE MACHINE IS HALTED MANUALLY OR THE FORWARD END ZONE IS	41
42	4503	/	* REACHED. IT IS NOT NECESSARY TO WAIT FOR THE TAPE TO STOP AT THE FORWARD	42
43	4504	/	* END ZONE. THE ONLY PURPOSE OF THIS STEP IS TO ENSURE THE TAPE IS	43
44	4505	/	* POSITIONED WELL FORWARD OF THE REVERSE END ZONE. THIS CAN ALSO BE	44
45	4506	/	* ACCOMPLISHED BY WINDING THE TAPE MANUALLY ABOUT 50 FEET. [IF THE REVERSE	45
46	4507	/	* END ZONE IS NOT LOCATED BEFORE THIS PORTION OF THE TAPE, THE ORIGINAL	46
47	4508	/	* FORMATTING OPERATION COULD NOT HAVE SUCCEEDED.]	47
48	4509	/	*	48
49	4510	/	* III) SET FRONT PANEL ADDRESS TO 07615.	49
50	4511	/	*	50
51	4512	/	* IV) SET SWITCHES TO 0600; PRESS START. THE TAPE WILL WIND BACKWARDS AND	51
52	4513	/	* STOP WITHIN THE REVERSE END ZONE; THE TAPE IS NOW PROPERLY POSITIONED.	52
53	4514	/	*****	53
54	4515			54
55	4516	/	END OF APPENDIX A.	55
56				56
57				57
58				58

1	4518	/	APPENDIX B.	1
2	4519			2
3	4520	/	DECTAPE AND LINCTAPE DETAILS.	3
4	4521			4
5	4522	/	DECTAPE FORMAT IS PARTIALLY BASED ON THE OLDER LINCTAPE FORMAT WITH SEVERAL	5
6	4523	/	DESIGN MODIFICATIONS:	6
7	4524			7
8	4525	/	1) DECTAPE FORMAT INCLUDES EXTRA INTER-BLOCK ZONES BEFORE BLOCK 0000 TO ALLOW	8
9	4526	/	STRAIGHTFORWARD SEARCH ALGORITHMS; SEARCHING IN REVERSE WILL CAUSE AN END	9
10	4527	/	ZONE ERROR AND THEN A SUBSEQUENT TAPE TURNAROUND. THERE WILL BE SUFFICIENT	10
11	4528	/	DECTAPE MEDIA PRESENT TO ALLOW BLOCK 0000 TO BE DETECTED WHILE COMING OUT	11
12	4529	/	OF THE REVERSE END ZONE DESPITE THE TWO-BLOCK DELAY ASSOCIATED WITH	12
13	4530	/	REVERSING THE TAPE DIRECTION.	13
14	4531			14
15	4532	/	LINCTAPE CONTROLLERS CAN DIRECTLY LOCATE BLOCK 0000 BECAUSE PHANTOM	15
16	4533	/	NEGATIVE BLOCKS ARE PLACED AFTER THE END ZONE; AUTOMATIC SEARCH OPERATIONS	16
17	4534	/	REVERSE THE TAPE AS NECESSARY. SUFFICIENT PHANTOM BLOCKS ARE PLACED TO	17
18	4535	/	ENSURE PROPER TURNAROUND ON THE ORIGINAL LOW-PERFORMANCE LINC [AND LINC-8]	18
19	4536	/	BELT-DRIVEN TAPE DRIVES.	19
20	4537			20
21	4538	/	2) LINCTAPE CAN ONLY BE TRANSFERRED IN THE FORWARD DIRECTION; DECTAPE CAN ALSO	21
22	4539	/	BE TRANSFERRED IN REVERSE; HOWEVER, THIS FEATURE IS NOT USED IN MOST PDP-8	22
23	4540	/	OPERATING SYSTEMS DUE TO THE OVERHEAD ASSOCIATED WITH THAT ACCESS SCHEME;	23
24	4541	/	MULTIPLE PASSES OF THE TAPE ARE REQUIRED TO TRAVERSE ALL BLOCKS. [NOTE:	24
25	4542	/	SUPPORT FOR DECTAPE IN MOST 16/18/36-BIT SYSTEMS IS ORGANIZED THIS WAY; THE	25
26	4543	/	DEVICE IS TREATED AS A NON-SYSTEM PERIPHERAL STORAGE UNIT; PERFORMANCE IS	26
27	4544	/	COMPROMISED TO GENERALLY AVOID ADDITIONAL SEARCHES CAUSED BY HEAVY SYSTEM	27
28	4545	/	LOADING; TAPE COASTING IS AN EXPECTED PART OF THE OVERALL DESIGN.] IN	28
29	4546	/	P?S/8, THE R-L MONITOR SYSTEM, OS/8 AND THE DISK MONITOR SYSTEM, THIS	29
30	4547	/	MECHANISM IS TOTALLY IGNORED OUTSIDE OF CUSTOM INTER-SYSTEM DATA CONVERSION	30
31	4548	/	PROGRAMS.	31
32	4549			32
33	4550	/	3) LINCTAPES USED ON THE CLASSIC LINC SYSTEM MUCH BE 256 WORDS PER BLOCK.	33
34	4551	/	DESPITE THE SMALL SYSTEM MEMORY SIZE [ORIGINALLY 1K OF 12-BIT WORDS LATER	34
35	4552	/	UPGRADED TO 2K], RELATIVELY UNWIELDY BUFFERS ARE REQUIRED TO ACCESS THE	35
36	4553	/	LINCTAPE BLOCKS. THIS WAS A DESIGN COMPROMISE BASED IN PART ON THE USE OF	36
37	4554	/	THE SHORTER 150 FOOT TAPES; DECTAPES ARE NOMINALLY 260 FEET LONG; CERTAIN	37
38	4555	/	SYSTEMS CAN TAKE ADVANTAGE OF SELECTED LONGER TAPES.	38
39	4556			39
40	4557	/	LINC SOFTWARE DOES NOT USE DEVICE HANDLERS AS THE LINCTAPE IS ACCESSED BY	40
41	4558	/	USING HARD-WIRED INSTRUCTIONS OF LINC PROCESSOR. ONLY NINE BITS OF BLOCK	41
42	4559	/	ADDRESSING CAN BE USED ON THE LINC. THE PDP-12 EXPANDS ON THE ORIGINAL	42
43	4560	/	LINC CAPABILITY ALLOWING OTHER FORMATS SUCH AS 128 12-BIT WORDS PER BLOCK;	43
44	4561	/	THIS FORMAT IS USED BY P?S/8, POLY BASIC, OS/8 AND OTHER SYSTEMS.	44
45	4562			45
46	4563	/	ON THE LINC THERE IS NO FORMAL DESCRIPTION OF AN INTERRUPT MECHANISM; ALL	46
47	4564	/	TAPE OPERATIONS MUST COMPLETE BEFORE ANY CUSTOM INTERRUPT HARDWARE ADDED ON	47
48	4565	/	TO THE BASIC MACHINE CAN BE RESPONDED TO. [FEW LINC SYSTEMS EVER HAD ANY	48
49	4566	/	INTERRUPT HARDWARE; NO STANDARDS EXIST ON EXACTLY HOW TO CREATE ANYTHING OF	49
50	4567	/	CONSEQUENCE.] [NOTE: THE LINC-8 IMPLEMENTATION OF LINCTAPE IS ACTUALLY A	50
51	4568	/	PDP-8 PERIPHERAL; LINC TAPE INSTRUCTIONS ARE TRAPPED AND EMULATED BY TRAP	51
52	4569	/	SOFTWARE WHEN THE MACHINE IS OPERATED FOR THAT PURPOSE; ALTERNATE SOFTWARE	52
53	4570	/	DOES NOT USE THIS MODE; THERE ARE VERSIONS OF P?S/8 AND OS/8 FOR THE LINC-8	53
54	4571	/	DEPENDING ON AVAILABLE MEMORY AND/OR HARDWARE MODIFICATIONS.]	54
55				55
56				56
57				57
58				58

1	4573	/	ALL DECTAPES MUST BE FORMATTED AS MULTIPLES OF 18 BITS PER WORD; THE	1
2	4574	/	PDP-8 STANDARD FORMAT IS 86 18-BIT [OR 129 12-BIT] WORDS. THE STANDARD	2
3	4575	/	FORMAT FOR MOST OTHER DEC COMPUTERS IS 256 18-BIT WORDS [OR 384 12-BIT	3
4	4576	/	WORDS.	4
5	4577			5
6	4578	/	ALL VIABLE PDP-8 OPERATING SYSTEMS, OTHER THAN THE DISK MONITOR SYSTEM,	6
7	4579	/	IGNORE THE 129TH WORD. SINCE THE DISK MONITOR SYSTEM REQUIRES THE 129TH	7
8	4580	/	WORD AS AN INTEGRAL PART OF ALL DATA STRUCTURES, IT IS IMPOSSIBLE TO	8
9	4581	/	IMPLEMENT THE DISK MONITOR SYSTEM ON MOST PDP-8 PERIPHERALS; IT WAS	9
10	4582	/	CONSIDERED OBSOLETE/WORTHLESS BEFORE THE R-L MONITOR SYSTEM WAS CREATED AND	10
11	4583	/	PLAYED NO ROLE IN THE DEVELOPMENT OF ANY OF THE OTHER PDP-8 SYSTEMS	11
12	4584	/	MENTIONED IN THIS DOCUMENT. WITHIN THE SOURCE CODE OF OS/8 IT IS	12
13	4585	/	EXPLICITLY THE OBJECT OF RIDICULE. AS THERE ARE NO EXPLETIVES IN THIS	13
14	4586	/	DOCUMENT, THE READER IS REFERRED TO OTHER FILES FOR FURTHER INFORMATION,	14
15	4587	/	SUCH AS THE SOURCE CODE FILE FOR THE OS/8 TC01/TC08 SYSTEM DEVICE HANDLER.	15
16	4588			16
17	4589	/	4) LINCTAPES DO NOT SUPPORT BUFFERED TRANSFER FORMAT; THERE IS MERELY A	17
18	4590	/	CONSIDERATION OF THE FINAL DATA WORD. DECTAPE FORMAT INCLUDES AN	18
19	4591	/	ADDITIONAL PRE-FINAL WORD. SINCE DECTAPE CAN BE WRITTEN IN THE REVERSE	19
20	4592	/	DIRECTION, THERE ARE PRE-FINAL, FINAL AND CHECKSUM WORDS PRESENT AT BOTH	20
21	4593	/	ENDS OF EVERY DATA BLOCK. SINCE THERE ARE LESS UNIQUE STATES A LINCTAPE	21
22	4594	/	CAN POSSIBLY BE IN AT ANY POINT IN TIME, THE MARK WINDOW IS ONLY FOUR BITS	22
23	4595	/	WIDE; DECTAPES USE A SIX BIT MARK WINDOW WHICH COMPORTS WELL WITH THE	23
24	4596	/	18-BIT ORIENTATION AS DEFINED. THE TC01/TC08 CONTROLLER IMPLEMENTS THE SIX	24
25	4597	/	BIT MARK WINDOW FOR COMPLETE COMPATIBILITY.	25
26	4598			26
27	4599	/	5) DECTAPES AND LINCTAPES SPIN IN OPPOSITE DIRECTIONS. AT THE BIT LEVEL, THE	27
28	4600	/	DATA FOR EACH FORMAT IS THE MIRROR IMAGE OF THE OTHER. TO READ OR WRITE	28
29	4601	/	DECTAPES ON LINCTAPE SYSTEMS REQUIRES THE ABILITY TO LOGICALLY INVERT THE	29
30	4602	/	TIMING TRACK PULSES TO TRIGGER THE DATA BITS AT THE PROPER TIME PHASE.	30
31	4603	/	HARDWARE TO ACCOMPLISH THIS IS A STANDARD OPTION FOR THE PDP-12 [TC12F]; ON	31
32	4604	/	THE LINC-8, A COMPREHENSIVE SET OF SIMPLE MODIFICATIONS MUST BE MADE TO THE	32
33	4605	/	COMPUTER BACKPLANE INVOLVING THE ADDITION OF A SMALL NUMBER OF MODULES.	33
34	4606	/	[NOTE: CONTACT CHARLES LASNER FOR MORE INFORMATION ON ALL SUCH HARDWARE	34
35	4607	/	MODIFICATIONS.] TAPE CONVERSION REQUIRES SPECIAL SOFTWARE TO ACCESS THE	35
36	4608	/	DECTAPES FOR READING OR WRITING; IT IS NOT POSSIBLE TO FORMAT DECTAPES ON	36
37	4609	/	THESE MACHINES.	37
38	4610			38
39	4611	/	6) AS A RESULT OF THE SIMPLER FORMAT OF 128 WORD/BLOCK LINCTAPES [AS COMPARED	39
40	4612	/	TO 129 WORD/BLOCK DECTAPES], THE OVERALL DATA STORAGE OF LINCTAPES IS	40
41	4613	/	SOMEWHAT LARGER. PREMIUM LENGTH LINCTAPES CAN OCCASIONALLY BE FOUND THAT	41
42	4614	/	ARE CAPABLE OF 4000 128-WORD BLOCKS [0000-3777]	42
43	4615			43
44	4616	/	7) THE SAME PREMIUM TAPES CAN BE FORMATTED AS 3300 129-WORD DECTAPE BLOCKS.	44
45	4617	/	VARIOUS P?S/8 RELEASES INCLUDE A MODIFIED VERSION OF DEC'S TOG-8 TC01/TC08	45
46	4618	/	DECTAPE FORMATTING PROGRAM THAT SUPPORT A COMMAND-LINE SWITCH TO ENABLE	46
47	4619	/	USING THE PASSED =XXXX OPTION TO SET THE LENGTH OF THE DECTAPE. THE USUAL	47
48	4620	/	PROGRAM USAGE FOR A PREMIUM LENGTH TAPE WOULD BE AS FOLLOWS:	48
49	4621			49
50	4622	/	.R DTFRMT /C=3300 [NOTE: THE DEFAULT IS 2702 BLOCKS.]	50
51	4623			51
52	4624	/	[NOTE: IN SOME P?S/8 RELEASES THE PROGRAM IS KNOWN AS "DT4MAT".]	52
53	4625			53
54	4626	/	END OF APPENDIX B.	54
55				55
56				56
57				57
58				58

1	4628	/	APPENDIX C.	1
2	4629			2
3	4630	/	P?S/8 LOGICAL UNIT EXTENSIONS FOR VARIOUS SUPPORTED SYSTEM DEVICES.	3
4	4631			4
5	4632	/	SINCE CERTAIN PDP-8 USERS ARE FAMILIAR WITH THE OS/8 OPERATING SYSTEM, IT IS	5
6	4633	/	OFTEN USEFUL TO DRAW COMPARISONS BETWEEN OS/8 AND P?S/8, ESPECIALLY FOR THE	6
7	4634	/	PURPOSE OF EXPLAINING LOGICAL UNIT EXTENSIONS.	7
8	4635			8
9	4636	/	OS/8 DEVICE HANDLERS DO NOT SUPPORT THE CONCEPT OF A LOGICAL UNIT EXTENSION;	9
10	4637	/	THE HANDLER ADDRESSES ONLY ONE UNIT OF STORAGE; IN QUITE A FEW CASES, A	10
11	4638	/	SIMILAR BUT DIFFERENT HANDLER IS NEEDED TO ADDRESS ANOTHER UNIT OF STORAGE,	11
12	4639	/	DESPITE THE FACT THAT BOTH LOGICAL UNITS ARE PART OF THE SAME HARDWARE DEVICE.	12
13	4640	/	[NOTE: IN SOME INSTANCES THERE CAN BE AS MANY AS EIGHT RELATED LOGICAL UNITS	13
14	4641	/	ON A PARTICULAR HARDWARE CONTROLLER SUCH AS THE TC12 LINCTAPE CONTROLLER OF	14
15	4642	/	THE PDP-12, THE TC01/TC08 DECTAPE CONTROLLER OR THE LINCTAPE CONTROLLER OF THE	15
16	4643	/	LINC-8.]	16
17	4644			17
18	4645	/	THE LOGICAL UNIT CONCEPT ALLOWS THE P?S/8 SYSTEM DEVICE HANDLER TO ADDRESS	18
19	4646	/	LARGER OVERALL STORAGE SPACE THAN THE OS/8 EQUIVALENT. THE OS/8 TC01/TC08	19
20	4647	/	SYSTEM DEVICE HANDLER CAN ONLY ADDRESS DRIVE 0; THE P?S/8 SYSTEM DEVICE	20
21	4648	/	HANDLER ADDRESSES ALL EIGHT PHYSICAL DRIVES AS LOGICAL UNITS 0 THROUGH 7. DUE	21
22	4649	/	TO LOGICAL BLOCK SIZE CONSIDERATIONS, ONE OS/8 HANDLER CAN ADDRESS AS MUCH AS	22
23	4650	/	TWICE THE STORAGE SPACE OF ONE P?S/8 LOGICAL UNIT; HOWEVER, THIS GENERALLY	23
24	4651	/	MEANS THE ENTIRE STORAGE SUPPORTED BY THE P?S/8 SYSTEM DEVICE HANDLER	24
25	4652	/	AGGREGATES TO FOUR TIMES AS MUCH STORAGE. [NOTE: THE SYSTEM DEVICE HANDLER IS	25
26	4653	/	USED FOR ALL STORAGE ACCESS IN THE BASIC P?S/8 SYSTEM; NO ADDITIONAL HANDLERS	26
27	4654	/	ARE EVER LOADED, AVOIDING THE OVERHEAD FOUND WITHIN OS/8 WHICH MUST PERFORM	27
28	4655	/	SEVERAL OPERATIONS MERELY TO PLACE A NON-SYSTEM HANDLER INTO MEMORY.]	28
29	4656			29
30	4657	/	IN THE P?S/8 SHELL ENVIRONMENT, NON-SYSTEM HANDLERS WILL BE USED TO ACCESS ALL	30
31	4658	/	STORAGE DEVICES [THE SYSTEM DEVICE HANDLER IS ONLY USED INTERNALLY FOR SUCH	31
32	4659	/	OPERATIONS AS FETCHING THE BINARY IMAGE OF THE NON-SYSTEM HANDLERS]; HOWEVER,	32
33	4660	/	EACH NON-SYSTEM HANDLER IS CAPABLE OF ADDRESSING AS MUCH AS 16384 TIMES AS	33
34	4661	/	MUCH STORAGE AS ONE OS/8 HANDLER.]	34
35	4662			35
36	4663	/	LOGICAL UNIT CONSIDERATIONS ON LARGER DISK DEVICES.	36
37	4664			37
38	4665	/	DEPENDING ON THE SIZE OF THE STORAGE DEVICE, LOGICAL UNITS MAY BE DETERMINED	38
39	4666	/	BY A MIX OF PHYSICAL AND MAPPING/PARTITIONING CONSIDERATIONS. THE LARGEST	39
40	4667	/	DISKS MAY BE MAPPED INTO A SERIES OF PERHAPS EIGHT PARTITIONS, EACH A LOGICAL	40
41	4668	/	UNIT IN THE RANGE OF 0 THROUGH 7, ALL RESIDENT ON THE SAME PHYSICAL DRIVE.	41
42	4669			42
43	4670	/	EXAMPLES OF LOGICAL UNIT CONFIGURATION ON THE RK8E/RK05.	43
44	4671			44
45	4672	/	THE RK8E/RK05 LOGICAL UNIT CONFIGURATION IS DEPENDENT ON BOTH DRIVE SELECT AND	45
46	4673	/	PARTITIONING CONSIDERATIONS. VARIOUS P?S/8 CONFIGURATIONS HAVE BEEN SUPPORTED	46
47	4674	/	ON THE RK8E/RK05 DISK WITH DIFFERENT OBJECTIVES.	47
48				48
49				49
50				50
51				51
52				52
53				53
54				54
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56				56
57				57
58				58

1	4676	/	THE EARLIEST CONFIGURATION PROVIDES SUPPORT FOR THE ENTIRETY OF UP TO TWO RK05	1
2	4677	/	DRIVES DEFINING LOGICAL UNITS 0 THROUGH 7. SINCE THE RK8E CAN BE BOOTSTRAPPED	2
3	4678	/	TO ONE OF FOUR PHYSICAL DRIVES, THE MAPPING OF LOGICAL UNITS IS DETERMINED	3
4	4679	/	DYNAMICALLY AS THE OPERATING SYSTEM LOADS. IF THE SYSTEM BOOTS FROM RK05	4
5	4680	/	DRIVE 0, EACH 1/4 OF THE DISK IS A LOGICAL UNIT IN THE RANGE OF 0 THROUGH 3;	5
6	4681	/	LOGICAL UNITS 4 THROUGH 7 RESIDE ON RK05 DRIVE 1 [IF PRESENT]. THE SAME	6
7	4682	/	CONFIGURATION APPLIES IF THE SYSTEM BOOTS FROM RK05 DRIVE 1. IN THIS	7
8	4683	/	INSTANCE, THE SYSTEM DEVICE CAN ACCESS THE ENTIRETY OF BOTH DRIVES AS LOGICAL	8
9	4684	/	UNITS 0 THROUGH 7 [ASSUMING BOTH DRIVES ARE PRESENT].	9
10	4685			10
11	4686	/	IF THE SYSTEM BOOTS FROM RK05 DRIVE 2 OR 3, ALL CONSIDERATIONS ARE THE SAME AS	11
12	4687	/	ABOVE EXCEPT THAT THE SYSTEM HANDLER CAN ACCESS THE ENTIRETY OF DRIVES 2 AND 3	12
13	4688	/	[ASSUMING BOTH DRIVES ARE AVAILABLE].	13
14	4689			14
15	4690	/	IT SHOULD BE NOTED THAT EACH LOGICAL UNIT IS EXACTLY HALF THE SIZE OF THE	15
16	4691	/	TYPICAL OS/8 SUPPORT FOR THE SAME DEVICE. THE AGGREGATE OF ALL EIGHT LOGICAL	16
17	4692	/	UNITS ADDRESSES FOUR TIMES AS MUCH AS THE OS/8 HANDLER.	17
18	4693			18
19	4694	/	IN THIS CONFIGURATION, IT IS POSSIBLE TO WRITE AN EXTENSION TO THE STANDARD	19
20	4695	/	RK8E BOOTSTRAP; THIS WOULD ALLOW P?S/8 TO BOOT TO ANY LOGICAL DEVICE UNIT 0	20
21	4696	/	THROUGH 7 [BEYOND THE STANDARD BOOTSTRAP DEFINITION WHICH ONLY SUPPORTS	21
22	4697	/	LOGICAL DEVICE UNITS 0 AND 4 SINCE THEY ARE THE ONLY LOGICAL UNITS AT THE	22
23	4698	/	BEGINNING OF THE PHYSICAL PACK]. EACH COPY OF THE SYSTEM IS CAPABLE OF	23
24	4699	/	DETERMINING THE RELATIVE LOGICAL UNIT [0 THROUGH 3] AS WELL AS WHETHER BOOTING	24
25	4700	/	TO THE EVEN [0 OR 2] OR ODD 1 OR 3] PACK AS APPROPRIATE TO DYNAMICALLY CREATE	25
26	4701	/	THE SYSTEM BOOTUP LOGICAL DRIVE UNIT. [NOTE: CERTAIN VERSIONS OF THE P?S/8	26
27	4702	/	TC01/TC08 SYSTEM DEVICE HANDLER SUPPORT AN EXTENSION TO THE TC01/TC08	27
28	4703	/	BOOTSTRAP PROGRAM ALLOWING STARTING FROM ANY DRIVE UNIT 0 THROUGH 7.]	28
29	4704			29
30	4705	/	THIS EARLIER CONFIGURATION WAS USED BEFORE THE IMPLEMENTATION OF THE P?S/8	30
31	4706	/	SHELL OVERLAY. IN THE SHELL VARIANT, THE SYSTEM DEVICE HANDLER CAN ACCESS UP	31
32	4707	/	TO THE FIRST 4096 BLOCKS OF THE DEVICE [AND OFTEN WOULD BE CONFIGURED FOR	32
33	4708	/	PERHAPS A SIGNIFICANTLY SMALLER FRACTION OF THE AVAILABLE STORAGE] WHILE	33
34	4709	/	THE NON-SYSTEM DEVICE HANDLERS ASSOCIATED WITH THE SHELL ADDRESS THE REST OF	34
35	4710	/	THE DEVICE.	35
36	4711			36
37	4712	/	AS SUCH, IN THE SHELL ENVIRONMENT, THE RK8E SYSTEM CONFIGURATION WILL ADDRESS	37
38	4713	/	ENTIRE DRIVES AS LOGICAL UNITS. P?S/8 ADDRESSES STORAGE IN TERMS OF 128-WORD	38
39	4714	/	LOGICAL BLOCKS; ON ANY RK05 PACK THERE ARE 31300 LOGICAL BLOCKS [00000-31277].	39
40	4715	/	PRACTICAL SYSTEMS MAY RESERVE SOMEWHAT LESS THAN THE FIRST 200 BLOCKS TO THE	40
41	4716	/	BASE OPERATING SYSTEM LEAVING THE REST OF THE DRIVE TO THE SHELL OVERLAY	41
42	4717	/	ENVIRONMENT. [NOTE: SYSTEMS THAT USE BOTH THE BASIC P?S/8 SYSTEM AND THE	42
43	4718	/	SHELL OVERLAY EXTENSIVELY WILL TEND TO BE CONFIGURED FOR ADDITIONAL STORAGE	43
44	4719	/	AVAILABLE TO THE BASIC SYSTEM, AS MUCH AS BLOCKS 00000-07777, THE MAXIMUM	44
45	4720	/	POSSIBLE FOR ANY SYSTEM DEVICE HANDLER.]	45
46	4721			46
47	4722	/	ASSUMING THE BASIC SYSTEM IS CONFIGURED TO THE MINIMUM VALUE SUITABLE TO	47
48	4723	/	MERELY SUPPORT THE SHELL OVERLAY, THE STORAGE SPACE AVAILABLE IN THE SHELL	48
49	4724	/	ENVIRONMENT IS SLIGHTLY LESS THAN TWICE THE SPACE AVAILABLE TO ANY ONE OS/8	49
50	4725	/	HANDLER. [NOTE: ON A NON-BOOTING P?S/8 DEVICE MADE EXCLUSIVELY AVAILABLE TO	50
51	4726	/	THE SHELL ENVIRONMENT, THE STORAGE SPACE IS EXACTLY TWICE THE SIZE OF ANY ONE	51
52	4727	/	OS/8 HANDLER FOR THE RK05 AS EACH OS/8 HANDLER CAN ONLY ADDRESS HALF OF THE	52
53	4728	/	STORAGE SPACE ON THE RK05 PACK.]	53
54				54
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56				56
57				57
58				58

1	4730	/	IN THIS NEWER CONFIGURATION, THE ENTIRE RK05 DRIVE 0 IS LOGICAL UNIT 0; THIS	1
2	4731	/	ADDRESSING SCHEME IS USED TO ACCESS THE OTHER RK05 DRIVES 1 THROUGH 3 AS THE	2
3	4732	/	SAME LOGICAL UNIT AS THE PACK NUMBER. [NOTE: THERE IS AN OBSCURE VARIANT OF	3
4	4733	/	THE RK8E CONTROLLER KNOWN AS THE RKS8E, WHICH SUPPORTS EIGHT RK05 DRIVES. IN	4
5	4734	/	THIS CONFIGURATION, ONE P?S/8 NON-SYSTEM HANDLER FOR RKS8E COULD ADDRESS ALL	5
6	4735	/	EIGHT DRIVES.] AS SUCH, THE SINGLE P?S/8 NON-SYSTEM HANDLER FOR RKS8E	6
7	4736	/	ADDRESSES THE EQUIVALENT OF 16 OS/8 HANDLERS. [NOTE: OS/8 CANNOT BE	7
8	4737	/	CONFIGURED TO HAVE AS MANY AS 16 HANDLERS IN ANY ONE CONFIGURATION. THE	8
9	4738	/	UNWIELDY PROCESS OF SYSTEM RECONFIGURATION WOULD HAVE TO BE APPLIED OFTEN TO	9
10	4739	/	WORK WITH THIS MUCH STORAGE IN OS/8 AS WELL AS HAVING TO USE TWICE AS MANY	10
11	4740	/	DIRECTORIES.]	11
12	4741			12
13	4742	/	[NOTE: THE RK05F FIXED DISK DRIVE IS ALSO SUPPORTED AS AN EVEN/ODD PAIR OF	13
14	4743	/	DRIVES. NO SPECIAL SOFTWARE EXISTS TO TREAT THIS DEVICE AS OTHER THAN THE	14
15	4744	/	EQUIVALENT OF TWO RK05/RK05J REMOVABLE DISKS. HOWEVER, IF A P?S/8	15
16	4745	/	CONFIGURATION CAN BE CREATED FOR A DRIVE PAIR SUCH AS THE RK05F, THE	16
17	4746	/	CONFIGURATION COULD BE MODIFIED TO TREAT THE ENTIRE RK05F DRIVE AS ONE DEVICE	17
18	4747	/	CONTAINING 62600 BLOCKS. BY COMPARISON, OS/8 WOULD REQUIRE FOUR DIFFERENT	18
19	4748	/	DEVICE HANDLERS, EACH ADDRESSING 1/4 OF THE RK05F PACK.]	19
20	4749			20
21	4750	/	THERE ARE STILL LARGER DEVICES FOR THE PDP-8/A SUCH AS THE MDC8 SCSI	21
22	4751	/	CONTROLLER FROM CESI. A STANDARD CONFIGURATION USES ONE OR TWO SEAGATE	22
23	4752	/	ST-4096 80 MB HARD DISKS EACH OF WHICH PROVIDE THE FORMATTED CAPACITY OF 64 MB	23
24	4753	/	[WHERE A BYTE IS SIX BITS]. THE ENTIRE DRIVE CAPACITY OF 32 MWORDS [WHERE A	24
25	4754	/	WORD IS 12-BITS] IS 32 TIMES THE MAXIMUM CAPACITY OF ANY ONE OS/8 HANDLER.	25
26	4755			26
27	4756	/	P?S/8 COULD BE CONFIGURED TO ACCESS THE ENTIRE DRIVE AS ONE LOGICAL UNIT.	27
28	4757	/	HOWEVER, IT WOULD BE PRUDENT TO CONSIDER SOMEWHAT SMALLER LOGICAL UNITS SUCH	28
29	4758	/	AS SUCH AS 4.0 MWORDS [WHERE EACH LOGICAL UNIT ADDRESSES FOUR TIMES THE	29
30	4759	/	MAXIMUM STORAGE ADDRESSED BY ANY ONE OS/8 HANDLER]. THIS WOULD ALLOW EACH	30
31	4760	/	OF EIGHT LOGICAL UNITS OF ONE P?S/8 NON-SYSTEM HANDLER TO ACCESS THE ENTIRE	31
32	4761	/	DISK. [NOTE: THIS WOULD REQUIRE A SLIGHT REWORKING OF THE CURRENT P?S/8	32
33	4762	/	SUPPORT FOR THE MDC8; THE CURRENT DEFINITION OF THE SYSTEM HANDLER MAPS THE	33
34	4763	/	FIRST 32768 BLOCKS OF THE DISK AS THE STORAGE SPACE ADDRESSED BY THE SYSTEM	34
35	4764	/	DEVICE HANDLER; THE MAPPING WOULD HAVE TO BE REWORKED TO ALLOW A PROPER OFFSET	35
36	4765	/	BETWEEN LOGICAL UNITS. THE MDC8 IS AN ELEGANT DEVICE, EASILY REPROGRAMMED TO	36
37	4766	/	ACCOMMODATE THE ALTERNATE MAPPING FOR SUCH A CONFIGURATION. SINCE THE	37
38	4767	/	ORIGINAL CONFIGURATION ALLOWS BOTH P?S/8 AND OS/8 TO SHARE PORTIONS OF THE	38
39	4768	/	DISK IN UNEQUAL PROPORTIONS, SOME HYBRID ARRANGEMENT CAN BE CONFIGURED TO	39
40	4769	/	ALLOW BOTH OPERATING SYSTEMS TO ACCESS A RATIONAL PORTION OF THE AVAILABLE	40
41	4770	/	STORAGE SPACE.	41
42	4771			42
43	4772	/	SINCE THE MAXIMUM ADDRESSIBILITY OF A P?S/8 NON-SYSTEM HANDLER IS 16777216	43
44	4773	/	LOGICAL BLOCKS, THE MDC8 ONLY ADDRESSES A SMALL FRACTION OF THEORETICALLY	44
45	4774	/	POSSIBLE STORAGE CONFIGURATIONS. NEWER DESIGNS COULD ADDRESS THIS MORE	45
46	4775	/	REALISTICALLY IN THE FUTURE.	46
47	4776			47
48	4777	/	END OF APPENDIX C.	48
49				49
50				50
51				51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	4779	/	APPENDIX D.	1
2	4780			2
3	4781	/	P?S/8 LOGICAL BLOCKSIZE CONSIDERATIONS.	3
4	4782			4
5	4783	/	EXACTLY HOW EACH P?S/8 DEVICE HANDLER ADDRESSES THE UNDERLYING STORAGE DEVICE	5
6	4784	/	IS A FUNCTION OF SEVERAL FACTORS DEPENDING ON HARDWARE PARTICULARS. ON THE	6
7	4785	/	TC01/TC08 CONTROLLER THE LOGICAL BLOCK DATA IS THE FIRST 128 WORDS OF EACH	7
8	4786	/	PHYSICAL DECTAPE BLOCK; ALL BLOCKS ARE READ AND WRITTEN AS 128 DATA WORDS.	8
9	4787	/	THE 129TH WORD IS IGNORED; ARBITRARY DATA IS WRITTEN INTO THE 129TH WORD AS A	9
10	4788	/	HARDWARE REQUIREMENT OF THE DECTAPE CONTROLLER. [NOTE: A FORMATTED DECTAPE	10
11	4789	/	MUST CONTAIN INTEGER MULTIPLES OF 18-BIT WORDS; STANDARD PDP-8 DECTAPES ARE	11
12	4790	/	FORMATTED AS 86 18-BIT WORDS OR 129 12-BIT WORDS.] IN P?S/8 [AND MOST OTHER	12
13	4791	/	OPERATING SYSTEMS FOR THE PDP-8] ALL BLOCKS ARE READ AND WRITTEN FORWARD.	13
14	4792			14
15	4793	/	LOGICAL BLOCK NUMBERS PASSED TO THE P?S/8 DEVICE HANDLER TRANSLATE DIRECTLY	15
16	4794	/	INTO PHYSICAL BLOCK NUMBERS ON THE DECTAPE. THE OVERALL DESIGN OF P?S/8	16
17	4795	/	DEPENDS ON THIS FOR EFFICIENT OPERATION, ESPECIALLY ON DEVICES SUCH AS	17
18	4796	/	DECTAPES AND LINCTAPES.	18
19	4797			19
20	4798	/	OS/8 IS A SYSTEM THAT USES A LOGICAL BLOCKSIZE OF 256 WORDS/LOGICAL BLOCK.	20
21	4799	/	[NOTE: WITHIN OS/8 THERE IS THE NOTION OF A LOGICAL RECORD SINCE AT TIMES	21
22	4800	/	MENTION IS MADE OF UNDERLYING BLOCKS, ESPECIALLY ON DECTAPE AND LINCTAPE	22
23	4801	/	HARDWARE. UNFORTUNATELY, SLOPPY HANDLING OF THIS ISSUE HAS LED TO MANY OS/8	23
24	4802	/	USERS CONFUSING THE TERMS AS INTERCHANGEABLE, DESPITE THE ORIGINAL INTENT	24
25	4803	/	CLEARLY BEING TO EQUATE THEM BY THE INHERENT 2:1 RATIO. AT THIS POINT, IT IS	25
26	4804	/	HARD TO EVEN NOTICE, AS VARIOUS OS/8 SYSTEM PROGRAMS COUNT THE LOGICAL RECORDS	26
27	4805	/	AND APPEND THE WORD "BLOCKS" INTO THE VARIOUS UTILITY LISTINGS.]	27
28	4806			28
29	4807	/	THE THEORETICAL ADVANTAGE OF COUNTING LOGICAL RECORDS WHERE THE LOGICAL	29
30	4808	/	BLOCKSIZE IS 256 WORDS COMES WHEN THE DEVICE IS LARGE ENOUGH THAT ADVANTAGE IS	30
31	4809	/	TAKEN ON A DEVICE WHILE ADDRESSING MORE STORAGE SPACE. [NOTE: THIS IS NOT THE	31
32	4810	/	CASE ON DECTAPES AS THE CAPACITY IS FAR LESS THAN 4096 BLOCKS.] FOR EXAMPLE,	32
33	4811	/	USING THE P?S/8 ACCESS METHOD OF LOGICAL BLOCKS OF 128 WORDS, ONLY THE FIRST	33
34	4812	/	4096 BLOCKS OF ONE HALF OF AN RK8E/RK05 PACK CAN BE ACESSED WHILE OS/8	34
35	4813	/	ADDRESSES THE ENTIRE HALF OF THE PACK AS 3248 RECORDS [EQUIVALENT TO 6496	35
36	4814	/	P?S/8 LOGICAL BLOCKS]. CERTAIN OS/8 HANDLERS CAN ADDRESS THE FULL RANGE OF	36
37	4815	/	4096 LOGICAL RECORDS; IN THESE CASES, OS/8'S HANDLERS CAN ADDRESS EXACTLY	37
38	4816	/	TWICE THE SPACE OF A P?S/8 HANDLER'S MAXIMUM CAPABILITY ON ANY INDIVIDUAL	38
39	4817	/	LOGICAL UNIT.	39
40				40
41				41
42				42
43				43
44				44
45				45
46				46
47				47
48				48
49				49
50				50
51				51
52				52
53				53
54				54
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56				56
57				57
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1	4819	/	THIS ADVANTAGE IS COUPLED WITH SOME DISADVANTAGES:	1
2	4820			2
3	4821	/	1) ALL OS/8 PROGRAMS MUST ALLOCATE MULTIPLES OF 256 WORDS FOR ANY FORM OF	3
4	4822	/	READ/WRITE BUFFER.	4
5	4823			5
6	4824	/	2) DUE TO THE ILL-DEFINED DEFINITION OF EXTENDED MEMORY, LOCATIONS 7600-7777	6
7	4825	/	ARE UNAVAILABLE IN ANY EXTENDED MEMORY FIELD. AS A RESULT, THE LARGEST	7
8	4826	/	POSSIBLE BUFFER SIZE IN OS/8 IS 30 PAGES.	8
9	4827			9
10	4828	/	BY COMPARISON, P?S/8 ALLOWS ANY NUMBER OF PAGES FROM ONE THROUGH 32. EVEN	10
11	4829	/	THOUGH OS/8 REQUIRES A MINIMUM OF 8K MEMORY [SOME SYSTEMS REQUIRE 12K	11
12	4830	/	MINIMUM DEPENDING ON HARDWARE REQUIREMENTS], PROGRAMS CAN CLEARLY BECOME	12
13	4831	/	COMPROMISED BY THESE AND OTHER CONSIDERATIONS THAT PUT CONSTRAINTS ON THE	13
14	4832	/	USAGE OF MEMORY ASSOCIATED WITH OS/8.	14
15	4833			15
16	4834	/	3) DESIGNING AN OS/8 HANDLER REQUIRES ADDRESSING EACH LOGICAL RECORD WITH THE	16
17	4835	/	ABILITY TO READ ONLY THE FIRST HALF OF THE RECORD; NOT ALL DEVICES ARE	17
18	4836	/	CAPABLE OF THIS. [NOTE: OMNI-8 IS A PDP-8-BASED TIME-SHARING SYSTEM THAT,	18
19	4837	/	AMONG OTHER THINGS, EMULATES OS/8 IN A VIRTUALIZED ENVIRONMENT WITH THE	19
20	4838	/	USE OF "ERSATZ" [TRAP EMULATED] HANDLERS; HOWEVER, THE USUAL STORAGE	20
21	4839	/	DEVICE FOR OMNI-8 IS THE CESI MDC8 SCSI CONTROLLER CONFIGURED FOR THE MOST	21
22	4840	/	COMMON HARD DISK BLOCKSIZE; THIS REQUIRES THE MINIMUM READ TO BE THE ENTIRE	22
23	4841	/	LOGICAL RECORD. AS SUCH, OS/8 CANNOT RUN ON THIS HARDWARE; OMNI-8 IS A	23
24	4842	/	STAND-ALONE BOOTABLE PROGRAM; IT DOES NOT SHARE OS/8'S REQUIREMENT OF READS	24
25	4843	/	LIMITED TO THE FIRST HALF OF A LOGICAL RECORD. P?S/8 CAN RUN ON THIS	25
26	4844	/	HARDWARE AS LONG AS 8K OF MEMORY [OR MORE] IS AVAILABLE TO ALLOW AN	26
27	4845	/	EXTENDED SYSTEM DEVICE HANDLER; SEVERAL P?S/8 IMPLEMENTATIONS CAN LOGICALLY	27
28	4846	/	ADDRESS 256 WORD RECORDS TO THE NEAREST 128 WORDS, INCLUDING RETAINING THE	28
29	4847	/	SECOND HALF OF THE PHYSICAL RECORD, A FUNCTION NOT REQUIRED WITHIN OS/8.]	29
30	4848			30
31	4849	/	CLEARLY, THAT OS/8 HAS TO CONFORM TO THE NOTION OF READING AND WRITING ANY	31
32	4850	/	NUMBER OF PDP-8 MEMORY PAGES TO THE STORAGE DEVICE IS THE ROOT CAUSE OF WHY	32
33	4851	/	THE NOTION OF LOGICAL RECORDS TWICE THAT SIZE CAN EASILY COME INTO CONFLICT	33
34	4852	/	WITH THE HARDWARE UNLESS ACCOMMODATIONS ARE MADE FOR OS/8. [NOTE: IT IS	34
35	4853	/	RUMORED THAT RICHARD LARY HAD TO "BEG" THE PDP-8 HARDWARE GROUP TO MODIFY THE	35
36	4854	/	RK8E CONTROLLER TO ALLOW THE IMPLEMENTATION OF A "FIRST HALF" READ/WRITE BIT	36
37	4855	/	IN THE CONTROLLER DESIGN; CLEARLY, WITHOUT THIS HARDWARE HELP-OUT, OS/8 CANNOT	37
38	4856	/	RUN ON THIS DEVICE.]	38
39				39
40				40
41				41
42				42
43				43
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45				45
46				46
47				47
48				48
49				49
50				50
51				51
52				52
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1	4858	/	SINCE OS/8 CANNOT RUN ON 4K MACHINES, P?S/8 IS THE ONLY VIABLE SYSTEM OF ITS	1
2	4859	/	KIND FOR THESE MACHINES; ONE OF THE REASONS P?S/8 IS SUCCESSFUL IS THE USE OF	2
3	4860	/	128 WORD BLOCKSIZE.	3
4	4861			4
5	4862	/	WHEREVER POSSIBLE, P?S/8 SYSTEM COMPONENTS ATTEMPT TO DYNAMICALLY ALLOCATE	5
6	4863	/	ANY AVAILABLE EXTENDED MEMORY. [NOTE: P?S/8 CONFIGURATIONS HAVE MORE COMPLEX	6
7	4864	/	CONVENTIONS REGARDING EXTENDED MEMORY; IN CERTAIN CIRCUMSTANCES, MORE MEMORY	7
8	4865	/	MAY BE AVAILABLE, PERHAPS AS A SUBSTANTIAL FRACTION OF AN ADDITIONAL MEMORY	8
9	4866	/	FIELD. ALL AVAILABLE MEMORY IS USED CONSTRUCTIVELY THROUGHOUT THE SYSTEM.]	9
10	4867			10
11	4868	/	IN MANY EXTENDED MEMORY SCENARIOS, THE AVAILABLE EXTENDED MEMORY IS USED TO	11
12	4869	/	ALLOCATE LARGER MEMORY BUFFERS FOR READ/WRITE PURPOSES ON ONE OR MORE DEVICES	12
13	4870	/	[OR LOGICAL UNITS]. SINCE THE BLOCKSIZE IS 128 WORDS/BLOCK, ANY WHOLE NUMBER	13
14	4871	/	OF MEMORY PAGES CAN BE USED FOR THIS PURPOSE [IN THE RANGE OF 1 THROUGH 32	14
15	4872	/	INCLUSIVE].	15
16	4873			16
17	4874	/	WHILE THE BASIC P?S/8 SYSTEM IS LIMITED TO AT MOST LOGICAL BLOCKS 0000-7777 ON	17
18	4875	/	ANY LOGICAL UNIT THE EXTENDED DEFINITION OF P?S/8 USING THE SHELL OVERLAY	18
19	4876	/	ALLOWS ACCESS TO AS MUCH AS 4096 TIMES THAT STORAGE; DOUBLE-PRECISION BLOCK	19
20	4877	/	ARGUMENTS ARE PASSED TO THE VARIOUS NON-SYSTEM HANDLERS. [NOTE: ALL P?S/8	20
21	4878	/	SYSTEMS SUPPORT SOME NUMBER OF LOGICAL DEVICE UNITS, UP TO EIGHT LOGICAL	21
22	4879	/	DEVICE UNITS PER HANDLER. FOR MOST HARDWARE CONFIGURATIONS, THIS MEANS THE	22
23	4880	/	SYSTEM DEVICE HANDLER ACCESSES FOUR TIMES THE TOTAL STORAGE OF THE OS/8 SYSTEM	23
24	4881	/	DEVICE HANDLER IN AGGREGATE.]	24
25	4882			25
26	4883	/	IN PRACTICAL CONFIGURATIONS, THE SYSTEM DEVICE HANDLER ADDRESSES THE FIRST	26
27	4884	/	4096 LOGICAL BLOCKS ON THE DEVICE WHILE THE ASSOCIATED NON-SYSTEM HANDLER CAN	27
28	4885	/	ADDRESS AS MUCH AS 4096 TIMES THAT SPACE. [NOTE: THIS IS PER LOGICAL DEVICE	28
29	4886	/	UNIT; MOST DEVICES ARE FAR SMALLER THAN THIS MAXIMUM. SINCE THERE COULD BE AS	29
30	4887	/	MANY AS EIGHT LOGICAL DEVICE UNITS, THE AMOUNT OF SPACE A SUITABLY CONFIGURED	30
31	4888	/	P?S/8 SYSTEM CAN ADDRESS IS AS MUCH AS 16384 TIMES THE STORAGE SPACE OF AN	31
32	4889	/	OS/8 HANDLER.]	32
33	4890			33
34	4891	/	REGARDLESS OF HANDLER CONFIGURATION, ALL ACCESS TO ALL DEVICES IS MADE USING	34
35	4892	/	LOGICAL BLOCKS OF 128 WORDS/BLOCK; ANY CALL TO A P?S/8 DEVICE HANDLER CAN	35
36	4893	/	TRANSFER ONE THROUGH 32 BLOCKS PER CALL AS REQUIRED.	36
37	4894			37
38	4895	/	END OF APPENDIX D.	38
39				39
40				40
41				41
42				42
43				43
44				44
45				45
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47				47
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1	4897	/	APPENDIX E.	1
2	4898			2
3	4899	/	ADDITIONAL DECTAPE-RELATED PERFORMANCE CONSIDERATIONS.	3
4	4900			4
5	4901	/	CONTIGUOUS BLOCK ARRANGEMENT CONSIDERATIONS.	5
6	4902			6
7	4903	/	A SIGNIFICANT PERFORMANCE ADVANTAGE IS OBTAINED WHEN READING OR WRITING AS	7
8	4904	/	MANY CONTIGUOUS BLOCKS AS POSSIBLE IN A SINGLE SYSTEM DEVICE HANDLER	8
9	4905	/	SUBROUTINE CALL. THIS WAS FIRST DEVELOPED IN THE EARLIER LIMITED-PURPOSE	9
10	4906	/	SYSTEM KNOWN AS THE DECTAPE LIBRARY SYSTEM [WHICH CAN ONLY SAVE CORE IMAGES	10
11	4907	/	AND LATER LOAD THEM INDIVIDUALLY INTO MEMORY].	11
12	4908			12
13	4909	/	THE R-L MONITOR SYSTEM USED THE SAME BASIC ACCESS TO DECTAPE PRESERVING THE	13
14	4910	/	DYNAMICS OF TAPE TRANSFER TO AVOID EXCESS SEARCHES AND THE ATTENDANT OVERHEAD.	14
15	4911	/	THIS DESIGN COMPLETELY CARRIES OVER INTO P?S/8; THE ONLY DIFFERENCES ARE:	15
16	4912			16
17	4913	/	1) MUCH TIGHTER CODE AS A RESULT OF LITERALLY YEARS OF PROGRAM DEVELOPMENT	17
18	4914	/	AIMED AT ACHIEVING THE SAME OUTCOME IN LESS TOTAL CODE SPACE.	18
19	4915			19
20	4916	/	2) THE RELEASED CODE SPACE WAS USED TO IMPLEMENT ADDITIONAL FEATURES SUCH AS	20
21	4917	/	LOGICAL UNIT SUPPORT AND ADDITIONAL INFORMATION PASSED TO SYSTEM PROGRAMS	21
22	4918	/	RUN FROM THE SYSTEM LOADER; THE SYSTEM LOADER IS A DEDICATED CALL AND	22
23	4919	/	STARTUP ROUTINE LOCATED WITHIN THE RESERVED MEMORY AREA [07632-07637] THAT	23
24	4920	/	COEXISTS WITH THE SYSTEM HANDLER AND SYSTEM RESTART AT 07600. THIS GREATLY	24
25	4921	/	IMPROVES PERFORMANCE OVER THE R-L METHOD OF AN INTERMEDIARY DEDICATED	25
26	4922	/	SYSTEM LOADING PROGRAM AND SYSTEM RESTART ROUTINE.	26
27	4923			27
28	4924	/	3) ADDITIONAL INFORMATION AVAILABLE TO ALL PROGRAMS SUCH AS OPTION SWITCHES,	28
29	4925	/	SYSTEM DATE AND MEMORY MANAGEMENT INFORMATION, SUPPORT FOR OUTPUT FILES	29
30	4926	/	AND AN OUTPUT FILE COUNT WORD.	30
31	4927			31
32	4928	/	4) THE SUBROUTINE CALL WAS GREATLY MODIFIED TO BETTER HANDLE THE ADDITIONAL	32
33	4929	/	FUNCTIONALITY. [NOTE: THE R-L MONITOR DOES NOT SUPPORT MULTIPLE LOGICAL	33
34	4930	/	DRIVE UNITS OR EXTENDED MEMORY CALLS OR EXTENDED MEMORY TRANSFERS; IT	34
35	4931	/	REQUIRES TWO ENTRY POINTS, ONE FOR READING, THE OTHER FOR WRITING.]	35
36	4932			36
37	4933	/	UNLIKE THE DISK MONITOR SYSTEM ON DECTAPE, ALL BLOCKS ARE ACCESSED IN THE SAME	37
38	4934	/	OPERATION FOLLOWING A SUCCESSFUL INITIAL FORWARD SEARCH TO THE FIRST PHYSICAL	38
39	4935	/	BLOCK OF A GROUP; ANY ADDITIONAL BLOCKS TRANSFER IMMEDIATELY WITHOUT FURTHER	39
40	4936	/	SEARCH OPERATIONS. [NOTE: THE DEVICE HANDLER CALL IS LIMITED TO A 32-BLOCK	40
41	4937	/	MAXIMUM TRANSFER NOT ONLY BY SOFTWARE DESIGN BUT ALSO BY HARDWARE CONSTRAINTS	41
42	4938	/	OF THE TC01/TC08 DECTAPE CONTROLLER AND THE PDP-8 3-CYCLE DATA BREAK [DMA]	42
43	4939	/	MECHANISM.] [NOTE: IN THE DISK MONITOR ENVIRONMENT, ARRANGING BLOCKS IN A	43
44	4940	/	CONTIGUOUS MANNER IS ACTUALLY A DRAWBACK; EACH CALL IS LIMITED TO A SINGLE	44
45	4941	/	BLOCK TRANSFER. IF THE NEXT BLOCK NEEDED IS ADJACENT, IT IS NECESSARY TO	45
46	4942	/	FIRST PERFORM A REVERSE SEARCH UNTIL THE DESIRED BLOCK IS HAS BEEN PASSED,	46
47	4943	/	THEN TURN AROUND AGAIN TO TRANSFER THE BLOCK FORWARD. MANY OF THE DISK	47
48	4944	/	MONITOR STRUCTURES ARE ARRANGED IN THIS PESSIMISTIC WAY; ADDITIONALLY, THE	48
49	4945	/	DEVICE HANDLER DOESN'T SUPPORT THE CONCEPT OF ATTEMPTING TO INITIALLY SEARCH	49
50	4946	/	FORWARD EVEN WHEN ACTUALLY BENEFICIAL. THE NET EFFECT IS THE FAMILIAR	50
51	4947	/	"SHOESHINE" PATTERN OF TAPE MOTION THAT SEVERELY DEGRADES PERFORMANCE OF THIS	51
52	4948	/	SYSTEM.]	52
53				53
54				54
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57				57
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1	4950	/	SIMILAR IDEAS WERE APPLIED TO OS/8 [WHICH SUPPORTS EXTENDED MEMORY IN A	1
2	4951	/	MANNER SIMILAR TO P?S/8]; HOWEVER, USING OS/8 DEVICE HANDLERS TO TRANSFER AN	2
3	4952	/	ENTIRE MEMORY FIELD IS FROWNED UPON; WHAT MAY WORK ON A SPECIFIC SYSTEM	3
4	4953	/	CONFIGURATION COULD CRASH ON A DIFFERENT CONFIGURATION. THUS, THE GENERAL	4
5	4954	/	RULE IS TO NOT USE THIS FEATURE; ALL GENERIC OS/8 SYSTEM PROGRAMS CANNOT TAKE	5
6	4955	/	ADVANTAGE OF THIS FORM OF SUBROUTINE CALL. [NOTE: THIS CANNOT EVEN BE	6
7	4956	/	ATTEMPTED ON MACHINES SMALLER THAN 12K MEMORY; NO MEMORY SIZE IS CONSIDERED	7
8	4957	/	COMPLETELY SAFE TO ALLOW THE OPERATION OTHER THAN IN SPECIFIC CIRCUMSTANCES;	8
9	4958	/	OS/8 PROGRAMS EXIST THAT WILL CRASH IF THIS IS ATTEMPTED.]	9
10	4959			10
11	4960	/	[NOTE: THIS ISSUE IS A SORE POINT OF OS/8 AND REPRESENTS POOR SYSTEM DESIGN.	11
12	4961	/	THE AVAILABILITY OF LOCATIONS X7600-X7777 WHERE X IS ANY PARTICULAR EXTENDED	12
13	4962	/	MEMORY FIELD WAS NEVER PLANNED OUT; THIS MERELY EVOLVED ON A PRAGMATIC	13
14	4963	/	[READ KLUDGE HERE] BASIS.]	14
15	4964			15
16	4965	/	VIRTUALLY ALL PDP-8 SYSTEMS RESTRICT ACCESS TO FIELD 0 7600-7777; THIS IS A	16
17	4966	/	CLEAR REQUIREMENT BORN OF THE ORIGINAL PAPER-TAPE LOADER SPECIFICATIONS AND	17
18	4967	/	THE HARDWARE RESTRICTIONS OF THREE-CYCLE DATA BREAK DEVICES SUCH AS THE DF32,	18
19	4968	/	RF08, TC58 MAGNETIC TAPE CONTROLLER AS WELL AS THE TC01/TC08 DECTAPE HARDWARE.	19
20	4969	/	UNDER P?S/8, ALL EXTENDED MEMORY FIELDS ARE COMPLETELY AVAILABLE TO ALL	20
21	4970	/	PROGRAMS. [NOTE: THERE ARE NOTABLE EXCEPTIONS WHICH ARE INDICATED BY SYSTEM	21
22	4971	/	STATUS BITS WITHIN THE P?S/8 SYSTEM DEVICE HANDLER. AS REQUIRED, VARIOUS	22
23	4972	/	PORTIONS OF THE HIGHEST IMPLEMENTED MEMORY FIELD MAY BE RESERVED IF REQUIRED	23
24	4973	/	BY THE SYSTEM DEVICE HANDLER. ADDITIONAL PORTIONS OF THE HIGHEST MEMORY	24
25	4974	/	FIELD MAY BE IN USE BY PORTIONS OF THE LOGICAL CONSOLE OVERLAY, AN IMPORTANT	25
26	4975	/	FEATURE OF P?S/8 THAT HAS NO CONCEPTUAL EQUIVALENT IN OTHER PDP-8 OPERATING	26
27	4976	/	SYSTEMS AND/OR PORTIONS OF THE OPTIONAL P?S/8 SHELL. IF NONE OF THESE APPLY,	27
28	4977	/	ALL EXTENDED MEMORY IS AVAILABLE TO ALL PROGRAMS. CONFORMING PROGRAMS CAN	28
29	4978	/	EASILY ADAPT TO THESE RESTRICTIONS BY INSPECTING THE SYSTEM STATUS WORD.]	29
30	4979			30
31	4980	/	[NOTE: THE OVERALL MEMORY ALLOCATION SCHEME IS FOR LESS INTRUSIVE INTO	31
32	4981	/	PROGRAM DESIGN THAN OS/8. ASSUMING ENOUGH MEMORY IS AVAILABLE, CERTAIN	32
33	4982	/	FEATURES OF THE PDP-8 INSTRUCTION SET CAN BE USED TO CREATE POWERFUL BUFFERING	33
34	4983	/	AND RELATED STRUCTURES; CODING TECHNIQUES THAT USE AUTO-INDEX REGISTERS TO	34
35	4984	/	FULL ADVANTAGE CAN BE USED WITHOUT RESORTING TO USING DEFENSIVE CODING	35
36	4985	/	MEASURES TO RESERVE PORTIONS OF MEMORY FIELDS, SUCH AS IS REQUIRED UNDER OS/8.	36
37	4986	/	THIS IS THE MAIN REASON WHY SUCH APPLICATIONS ARE NEVER WRITTEN FOR OS/8; THE	37
38	4987	/	USE OF THESE TECHNIQUES IS ENCOURAGED IN P?S/8; SEVERAL SETS OF USER-WRITTEN	38
39	4988	/	REAL-TIME DATA COLLECTION PROGRAMS HAVE BEEN WRITTEN FOR USE ONLY WITH P?S/8	39
40	4989	/	THAT CANNOT BE RUN FROM OS/8 BY ANY POSSIBLE MEANS.]	40
41	4990			41
42	4991	/	DECTAPE HANDLING ISSUES BEYOND THE SCOPE OF THE SYSTEM DEVICE HANDLER.	42
43	4992			43
44	4993	/	IT IS NOT POSSIBLE TO WRITE A GENERAL SUBROUTINE FOR TC01/TC08 DECTAPE THAT	44
45	4994	/	CAN TRANSFER ADJACENT BLOCKS ACROSS CALLS UNLESS THERE IS CONCEPTUAL SUPPORT	45
46	4995	/	FOR MAINTAINING TAPE MOTION AFTER THE EARLIER TRANSFER. THERE ARE REAL-TIME	46
47	4996	/	CONSTRAINTS ON MAINTAINING TAPE MOTION AFTER THE COMPLETION OF A CALL; TIMING	47
48	4997	/	ISSUES COULD PREVENT THE NEXT CALL FROM BEING SUCCESSFUL WITHOUT EXCESSIVE	48
49	4998	/	TAPE TURNAROUND. AN INTERRUPT-DRIVEN REAL-TIME SYSTEM COULD HAVE AN AUTOMATIC	49
50	4999	/	SHUTDOWN TIMER TASK TO STOP TAPE MOTION IF THE WINDOW OF TIME IS EXCEEDED.	50
51	5000	/	OTHERWISE, THE DESIGN SHOULD REDUCE TO THAT IMPLEMENTED IN P?S/8 WITHOUT THE	51
52	5001	/	REQUIREMENT OF INTERRUPTS [PERHAPS AS A TASK PERFORMED ONLY IN THE	52
53	5002	/	BACKGROUND]. ELABORATE SCHEMES FOR HANDLING DECTAPE GENERALLY WILL ADD MANY	53
54	5003	/	LAYERS OF COMPLICATION WITH DUBIOUS PERFORMANCE IMPROVEMENT.	54
55				55
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1	5005	/	HISTORICALLY, NO ONE HAS BOTHERED TO IMPLEMENT SUCH DECTAPE HANDLING AT THE	1
2	5006	/	OPERATING SYSTEM LEVEL BECAUSE IT IS TOO CRITICAL AND ERROR-PRONE. [NOTE:	2
3	5007	/	CHARLES LASNER HAS EXTENSIVE EXPERIENCE WRITING REAL-TIME DECTAPE PROGRAMMING	3
4	5008	/	AT THE CUSTOM APPLICATION LEVEL.] CERTAIN 18/36-BIT SYSTEMS USE A DECTAPE	4
5	5009	/	LAYOUT THAT DOES NOT DEMAND ADJACENT BLOCK TRANSFER; THE BLOCKS THAT PASS OVER	5
6	5010	/	THE HEADS DURING THE TIME THE DECTAPE IS SLOWING TO A COMPLETE STOP ARE NOT	6
7	5011	/	INTENDED TO BE ACCESSED AT THAT POINT. THE OVERALL LOGICAL DECTAPE STRUCTURE	7
8	5012	/	IS INTERLEAVED TO AVOID PROBLEMS ASSOCIATED WITH COASTING. HOWEVER, SUCH	8
9	5013	/	SCHEMES REQUIRE THE ABILITY TO SPECIFY THE DIRECTION OF TRANSFER ON THE	9
10	5014	/	DECTAPE TO BE OF MUCH USE. AS SUCH, CERTAIN BLOCKS MUST BE WRITTEN IN THE	10
11	5015	/	REVERSE DIRECTION. ADDITIONALLY, THE BEST POSSIBLE HANDLING REQUIRES THREE	11
12	5016	/	COMPLETE PASSES ACROSS THE DECTAPE TO ACCESS ALL BLOCKS IN A MAPPED ORDER.	12
13	5017			13
14	5018	/	THE DEC PDP-8 STANDARD DECTAPE COPY PROGRAM USES THIS METHOD TO COPY DECTAPES	14
15	5019	/	SUCCESSFULLY WITHOUT THE USE OF INTERRUPTS AND REQUIRES TWO DECTAPE DRIVES.	15
16	5020	/	[NOTE: IF THE DISK MONITOR SYSTEM ADAPTED PART OF THE TECHNIQUE USED IN THE	16
17	5021	/	STANDARD DECTAPE COPY PROGRAM, AND ALSO COULD IMPLEMENT SMART INITIAL SEARCH	17
18	5022	/	DIRECTION SUPPORT, IT COULD PERFORM SIGNIFICANTLY BETTER; HOWEVER, THIS WOULD	18
19	5023	/	REQUIRE A TOTAL REDESIGN OF WHAT STORAGE BLOCKS ARE USED FOR WHICH PARTICULAR	19
20	5024	/	SYSTEM RESOURCES. THE NET EFFECT WOULD STILL BE A MEDIOCRE SYSTEM, BUT WOULD	20
21	5025	/	BE MATERIALLY BETTER THAN THE ACTUAL RELEASED VERSION. IT WOULD APPEAR THE	21
22	5026	/	DECTAPE VERSION OF THE DISK MONITOR SYSTEM IS AN EXPEDIENT DESIGN CREATED	22
23	5027	/	AFTER THE DF32/RF08 VERSIONS WERE CREATED; ON THOSE CONFIGURATIONS, MOST OF	23
24	5028	/	THESE CONSIDERATIONS ARE MOOT. WHEN YOU OPTIMIZE A SYSTEM DESIGN FOR A	24
25	5029	/	HEAD PER TRACK DISK STORAGE UNIT, RUNNING IT ON OTHER HARDWARE CAN EXHIBIT	25
26	5030	/	DRAMATICALLY POORER PERFORMANCE.]	26
27	5031			27
28	5032	/	IN THEORY, AN INTERRUPT-DRIVEN DECTAPE SUBSYSTEM CAN BE DEVELOPED WITH	28
29	5033	/	PRESUMPTIVE BUFFERING TECHNIQUES. ADJACENT BLOCKS CAN BE READ IN REAL TIME	29
30	5034	/	WHILE MAINTAINING DECTAPE MOTION UP TO THE LIMITS OF DEFINED BUFFERS. WHEN	30
31	5035	/	THE PROJECTED BLOCKS BECOME OBSOLETE [SUCH AS WHEN AN UNRELATED BLOCK READ IS	31
32	5036	/	REQUESTED], THE BUFFERED BLOCKS' CONTENTS ARE ABANDONED; A NEW SET OF	32
33	5037	/	ANTICIPATORY READS MUST BE INITIATED. THE OVERHEAD TO CREATE SUCH A SYSTEM	33
34	5038	/	WOULD TIE UP FAR TOO MANY MEMORY BUFFERS TO BE PRACTICAL FOR ANY REASONABLE	34
35	5039	/	FORM OF OPERATING SYSTEM, BUT PERHAPS SERVES SOME THEORETICAL DEDICATED	35
36	5040	/	APPLICATION PROGRAM. [NOTE: OBSOLETE VERSIONS OF THE COS-300 SYSTEM HAVE	36
37	5041	/	LIMITED SUPPORT FOR TC01/TC08 DECTAPE USING SOME OF THESE TECHNIQUES;	37
38	5042	/	REQUIREMENTS INCLUDE THE PRESENCE OF 32K OF MEMORY AND LIMITING EXECUTION TO	38
39	5043	/	OTHERWISE NOT ALLOWING ACCESS TO UPPER EXTENDED MEMORY FOR ANY OTHER PURPOSE;	39
40	5044	/	THIS WAS EVENTUALLY SCRAPPED IN FAVOR OF MORE GENERAL FEATURES WITHIN THIS	40
41	5045	/	SPECIAL-PURPOSE OPERATING SYSTEM.]	41
42	5046			42
43	5047	/	SYSTEMS SUCH AS THE R-L MONITOR SYSTEM, OS/8 AND P?S/8 DO NOT USE AN ELABORATE	43
44	5048	/	DECTAPE LAYOUT SCHEME, AS THE TRANSFERS ARE HARDWARE-LIMITED TO AT MOST 32	44
45	5049	/	BLOCKS AT A TIME. AS SUCH, IT IS A FAR BETTER COMPROMISE TO MERELY TRANSFER	45
46	5050	/	ALL DATA FORWARD WITH LINEAR ACCESS TO THE DECTAPE BLOCKS. EVEN WITH	46
47	5051	/	OCCASIONAL DECTAPE SEARCHES STARTED IN REVERSE, THE TOTAL TIME TO TRANSFER AN	47
48	5052	/	ENTIRE DECTAPE IS ACTUALLY LESS USING AN ADJACENT BLOCK SCHEME SUCH AS THE	48
49	5053	/	METHOD DESCRIBED ABOVE. REGARDING THE DEC STANDARD DECTAPE COPY PROGRAM.	49
50	5054	/	[NOTE: THE DECTAPE COPY PROGRAM TRANSFERS BOTH 18-BIT AND 12-BIT ORIENTED	50
51	5055	/	DECTAPES AND RUNS ENTIRELY IN 4K. LARGER TRANSFERS TO REALIZE THE IMPROVEMENT	51
52	5056	/	DESCRIBED HERE REQUIRE EXTENDED MEMORY TO ALLOW SUFFICIENTLY LARGE BUFFERS TO	52
53	5057	/	MINIMIZE OVERALL TRANSFER TIME.]	53
54	5058			54
55	5059	/	END OF APPENDIX E.	55
56				56
57				57
58				58

1	5061	/	APPENDIX F.	1
2	5062			2
3	5063	/	P?S/8 ERROR HANDLING SPECIFICS.	3
4	5064			4
5	5065	/	WHILE THIS IS A GENERIC ISSUE FOR ALL P?S/8 SUPPORTED DEVICES, THE TC01/TC08	5
6	5066	/	DECTAPE CONTROLLER WILL BE USED AS AN EXAMPLE.	6
7	5067			7
8	5068	/	IN THE BASIC P?S/8 SYSTEM [THAT CAN RUN IN 4K OF MEMORY], DETECTED ERRORS ARE	8
9	5069	/	EITHER HANDLED AUTOMATICALLY OR CAUSE AN INTERNAL HALT WITHIN THE HANDLER.	9
10	5070	/	END ZONE ERRORS ARE GENERALLY HANDLED BY REVERSING THE DIRECTION OF TAPE	10
11	5071	/	MOTION; THIS ONLY HAPPENS OCCASIONALLY WHEN THE DECTAPE IS LIKELY INITIALLY	11
12	5072	/	MOUNTED BY THE USER. ON THE TC01/TC08 CONTROLLER, TRANSFERS FROM BLOCKS LOWER	12
13	5073	/	THAN BLOCK 0002 CAN ONLY BE ACCOMPLISHED BY DRIVING THE TAPE INTO THE REVERSE	13
14	5074	/	END ZONE, THEN TURNING THE TAPE AROUND TO LOCATE THE DESIRED BLOCK WHILE THE	14
15	5075	/	TAPE IS MOVING FORWARD.	15
16	5076			16
17	5077	/	ALL OTHER ERRORS WILL CAUSE THE INTERNAL HALT. IN THE CASE OF THE TC01/TC08	17
18	5078	/	SYSTEM DEVICE HANDLER, THE CONTENTS OF THE ACCUMULATOR WILL CORRESPOND TO THE	18
19	5079	/	DECTAPE CONTROLLER STATUS REGISTER "B" CONTENTS. PRESSING "CONTINUE" ON THE	19
20	5080	/	CPU FRONT PANEL WILL RETRY THE LATEST OPERATION. ASSUMING THE CONDITIONS	20
21	5081	/	CAUSING THE PROBLEM WERE CORRECTED BY THE OPERATOR, THE DEVICE HANDLER CALL	21
22	5082	/	WILL THEN GENERALLY SUCCEED. CONTINUED OCCURRENCE OF ERRORS INDICATES ONE OF	22
23	5083	/	SEVERAL POSSIBLE HARDWARE ERRORS INVOLVING EITHER THE DECTAPE MEDIA OR A	23
24	5084	/	DEGRADED CONDITION OF THE DRIVE TAPE GUIDES. THIS SITUATION MUST BE REMEDIED	24
25	5085	/	QUICKLY AS THE UNDERLYING CONDITION CAN EASILY BECOME CHRONIC POSSIBLY LEADING	25
26	5086	/	TO DATA LOSS AND PERHAPS DESTRUCTION OF THE DECTAPE MEDIA. [NOTE: IN THE	26
27	5087	/	EARLY STAGES OF MOST OF THESE PROBLEMS, THERE ARE PERIODIC OCCURRENCES OF	27
28	5088	/	EITHER MARK OR TIMING ERRORS; RETRY OPERATIONS EVENTUALLY SUCCEED AFTER ENOUGH	28
29	5089	/	RETRIES TO MOVE THE TAPE TO THE TARGET BLOCK. AS THE HARDWARE FAILURES GET	29
30	5090	/	PROGRESSIVELY WORSE, THE RATE OF ERRORS CAN INCREASE DRASTICALLY. BY APPYING	30
31	5091	/	MAINTENANCE PROCEDURES BEFORE THIS HAPPENS, THE PROBLEM CAN BE CORRECTED MORE	31
32	5092	/	EASILY.]	32
33	5093			33
34	5094	/	WARNING ABOUT USING OS/8 ON DAMAGED DECTAPE HARDWARE.	34
35	5095			35
36	5096	/	OS/8 DECTAPE HANDLERS USE A DEFECTIVE ALGORITHM TO HANDLE DECTAPES THAT IS	36
37	5097	/	QUITE DANGEROUS TO USE ON MARGINAL DRIVES. DURING THE SEARCH PHASE, ALL	37
38	5098	/	ERRORS [OTHER THAN END ERROR WHICH TURNS THE TAPE AROUND] ARE IGNORED; THERE	38
39	5099	/	IS ONLY A SMALL OPPORTUNITY FOR TIMING AND MARK ERRORS TO BE DETECTED, ONLY	39
40	5100	/	DURING THE TIME A DATA TRANSFER IS IN PROGRESS. AS SUCH, IT IS EASY FOR A	40
41	5101	/	MARGINAL SITUATION TO BECOME A CHRONIC ONE. TYPICALLY, IF OS/8 IS USED	41
42	5102	/	EXCLUSIVELY, THE SITUATION IS ONLY NOTICED WHEN IT IS LIKELY TOO LATE TO	42
43	5103	/	RECOVER DATA AND PERHAPS IMMINENT HARDWARE FAILURE IS POSSIBLE. USING P?S/8	43
44	5104	/	WILL REVEAL THE PROBLEM AT AN EARLIER STAGE OF FAILURE POTENTIALLY SAVING	44
45	5105	/	BOTH THE DATA AND PERHAPS THE DECTAPE MEDIA [AND DRIVE GUIDE HARDWARE] ITSELF.	45
46	5106			46
47	5107	/	LOGICAL CONSOLE OVERLAY CONSIDERATIONS.	47
48	5108			48
49	5109	/	WHEN THE LOGICAL CONSOLE OVERLAY IS ENABLED, ALL ERRORS ARE INTERCEPTED AND	49
50	5110	/	HANDLED WITHIN THE OVERLAY. INFORMATIVE ERROR MESSAGES ARE ISSUED ON THE	50
51	5111	/	CONSOLE DEVICE ALLOWING THE USER TO DECIDE WHAT TO DO NEXT FROM A LIST OF	51
52	5112	/	OPERATIONAL CHOICES [RETRY, ABORT AND EXIT TO 07600, ETC.]. THE SYSTEM WILL	52
53	5113	/	NOT HALT UNDER ANY CIRCUMSTANCES. [NOTE: THE LOGICAL CONSOLE OVERLAY IS ONLY	53
54	5114	/	AVAILABLE IN CERTAIN RELEASES OF P?S/8; EXTENDED MEMORY IS ALWAYS REQUIRED.]	54
55				55
56				56
57				57
58				58

1	5116	/	[NOTE: THE LOGICAL CONSOLE OVERLAY ALSO INTERACTS WITH CERTAIN PORTIONS OF THE	1
2	5117	/	HANDLER TO IMPROVE PERFORMANCE; THIS MECHANISM IS DEVICE-DEPENDENT AND MAY	2
3	5118	/	ONLY BE PRESENT FOR SPECIFIC BOOTABLE DEVICES. THE TC01/TC08 HANDLER CODE	3
4	5119	/	ELSEWHERE WITHIN THIS DOCUMENT CONTAINS SPECIFIC REFERENCES TO THIS REGARDING	4
5	5120	/	IMPROVEMENT TO THE OVERALL SEARCH PERFORMANCE OF ALL DECTAPE DRIVES. FUTURE	5
6	5121	/	RELEASES OF THIS DOCUMENT WILL INCLUDE AN EXPLICIT SECTION COVERING THE	6
7	5122	/	INTERCEPTING CODE DESIGNATED FOR COMMUNICATIONS BETWEEN THE SYSTEM DEVICE	7
8	5123	/	HANDLER AND THE LOGICAL CONSOLE OVERLAY; AS OF THIS WRITING, THE CODE	8
9	5124	/	DESCRIBED IN THIS DOCUMENT PREDATES VERSIONS OF THE LOGICAL CONSOLE OVERLAY	9
10	5125	/	SUPPORTING THIS FEATURE.]	10
11	5126			11
12	5127	/	OS/8 AND P?S/8 SYSTEM DEVICE HANDLER ERROR CONSIDERATIONS.	12
13	5128			13
14	5129	/	ALL OS/8 PROGRAMS CALLING THE OS/8 SYSTEM DEVICE HANDLER ARE REQUIRED TO	14
15	5130	/	PROVIDE A ROUTINE TO HANDLE ERROR RETURNS REGARDLESS OF THE NATURE OF THE	15
16	5131	/	CALLER; THIS INCLUDES CALLS MADE TO THE HANDLER FROM WITHIN OS/8 ITSELF.	16
17	5132	/	WHILE PROGRAMS CALLING THE HANDLER MAY HAVE VARYING DEGREES OF ERROR RECOVERY,	17
18	5133	/	IT IS ALWAYS NECESSARY TO HAVE A VIABLE STRATEGY; HOWEVER, THE QUALITY OF	18
19	5134	/	IMPLEMENTATION VARIES GREATLY; TYPICAL OS/8 PROGRAMS PROVIDE GENERIC MESSAGES	19
20	5135	/	FOR HANDLER AND OTHER PROBLEMS THAT GENERALLY INVOLVE FAILURE OF THE LATEST	20
21	5136	/	OPERATION WITHOUT RECOURSE; THE OVERALL OPERATION MUST BE RESTARTED.	21
22	5137			22
23	5138	/	THERE IS A SERIOUS DESIGN FLAW WITHIN OS/8 WHEREIN A CALL TO THE SYSTEM DEVICE	23
24	5139	/	HANDLER IS MADE TO LOAD THE LAST SEGMENT OF MEMORY WHEN STARTING ANY PROGRAM	24
25	5140	/	UP. SINCE THERE IS NO CODE SPACE AVAILABLE, THE ERROR RETURN CONSISTS MERELY	25
26	5141	/	OF A "HLT" INSTRUCTION; AS SUCH, THE SITUATION IS UNRECOVERABLE. WORSE STILL,	26
27	5142	/	PRESSING "CONTINUE" ON THE CPU FRONT PANEL COULD POTENTIALLY HARM THE SYSTEM	27
28	5143	/	AS THE INTENDED PROGRAM SEGMENT IS NOT LOADED; THE UNINTENDED PRIOR CONTENTS	28
29	5144	/	OF MEMORY WILL BE EXECUTED WITH UNKNOWN CONSEQUENCES. [NOTE: IN RECENT MEMORY	29
30	5145	/	ALONE, SYSTEM DESTRUCTION HAS BEEN REPORTED WHEN THIS SCENARIO WAS ATTEMPTED	30
31	5146	/	IN AT LEAST TWO SPECIFIC INSTANCES. FOR MORE INFORMATION ON THIS PROBLEM, SEE	31
32	5147	/	THE SECTION OF THIS DOCUMENT REGARDING IMAGE PROGRAM LOADING AND THE P?S/8	32
33	5148	/	SYSTEM LOADER.]	33
34	5149			34
35	5150	/	UNLIKE OS/8, IN THE EVENT OF AN ERROR, P?S/8 PROVIDES AN EXPECTATION OF	35
36	5151	/	SUCCESSFUL RECOVERY FROM THE PROBLEM. CALLING PROGRAMS DO NOT HAVE TO PROVIDE	36
37	5152	/	ERROR RECOVERY ROUTINES; ALL ERRORS ARE HANDLED INTERNALLY BASED ON THE	37
38	5153	/	PARTICULARS OF THE ERROR CONDITION; THE CALLER PLAYS NO DIRECT ROLE IN THE	38
39	5154	/	RECOVERY OPERATION.	39
40	5155			40
41	5156	/	IN CERTAIN INSTANCES, THIS ERROR RECOVERY METHOD HAS ALLOWED MARGINAL DATA	41
42	5157	/	TO BE SUCCESSFULLY READ BEFORE THE MEDIA BECAME TOTALLY UNREADABLE. IN ONE	42
43	5158	/	PARTICULAR INSTANCE, P?S/8, BY PROVIDING PROPER ERROR RECOVERY, SOLVED A DATA	43
44	5159	/	RECOVERY ISSUE WHILE OS/8 ONLY AGGRAVATED THE SITUATION. [NOTE: WHILE THE	44
45	5160	/	ORGANIZATION OF P?S/8 AND OS/8 IS QUITE DIFFERENT ON THE RK8E/RK05, THE	45
46	5161	/	SUPERIOR P?S/8 ERROR RECOVERY ALLOWED USER DATA TO BE RECOVERED WHERE OS/8	46
47	5162	/	COULD NOT EVEN AFTER EXTENSIVE RETRY ATTEMPTS. COMPATIBLE P?S/8 GENERIC	47
48	5163	/	BLOCK-ORIENTED UTILITIES WERE USED TO COPY ALL AFFECTED DISK SECTORS TO	48
49	5164	/	ALTERNATE MEDIA RESULTING IN A COMPLETE DATA RECOVERY.]	49
50	5165			50
51	5166	/	END OF APPENDIX F.	51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	5168	/	APPENDIX G.	1
2	5169			2
3	5170	/	P?S/8 CODE OPTIMIZATION TECHNIQUES.	3
4	5171			4
5	5172	/	THE ORIGINAL GODS OF PDP-8 SOFTWARE, THE WILLIAM CLAUDE FIELDS MEMORIAL	5
6	5173	/	PROGRAMMING GROUP [WCFMPG] ARE RESPONSIBLE FOR MANY ASPECTS OF PDP-8	6
7	5174	/	PROGRAMMING CONCEPTS [AS WELL AS OTHER INDUSTRY-WIDE LEGENDS]. ONE OF THEIR	7
8	5175	/	IMPORTANT LEGACIES CONCERNS THE "MATHEMATICS" BEHIND WRITING TIGHT PDP-8 CODE.	8
9	5176	/	[NOTE: THE P?S/8 TC01/TC08 DECTAPE SYSTEM DEVICE HANDLER IS THE QUINTESSENTIAL	9
10	5177	/	EXAMPLE ADVANCING THEIR PRINCIPLES; THIS CODE HAS BEEN WORKED ON FOR DECADES	10
11	5178	/	AS FIRST THE R-L MONITOR SYSTEM DEVICE HANDLER AND RELATED CODE, THEN AS THE	11
12	5179	/	P?S/8 DESCENDANT. IN THE PROCESS, MANY OPTIMIZATIONS HAVE BEEN PERFORMED ON	12
13	5180	/	THIS PORTION OF THE OPERATING SYSTEM. FEATURES HAVE BEEN ADDED SOLELY BASED	13
14	5181	/	ON THE ABILITY TO FIND SUFFICIENT CODE SPACE IN THE HANDLER TO IMPLEMENT THE	14
15	5182	/	RESIDENT PORTIONS. SOMETIMES THIS COMES DOWN TO CREATING FREE BITS IN A	15
16	5183	/	SINGLE WORD. [NOTE: IT IS BEYOND THE SCOPE OF THIS DOCUMENT TO EXPLAIN THE	16
17	5184	/	CONSEQUENCES OF THE OPTIMIZATION ALREADY INHERENT IN THE CODE AS PRESENTED	17
18	5185	/	HERE. FURTHER, THIS IS NOT LIKELY THE FINAL VERSION OF THE CODE AS AT LEAST	18
19	5186	/	ANOTHER DECADE OF SERIOUS DEVELOPMENT EFFORT WAS EXPENDED AFTER THIS VERSION	19
20	5187	/	OF THE CODE WAS WRITTEN. FUTURE VERSIONS OF THIS FILE WILL UPDATE THE CODE	20
21	5188	/	TO SHOW A PORTION OF THE PROGRESSION TOWARDS THE ELUSIVE "PERFECTION" NEEDED	21
22	5189	/	TO EXPAND P?S/8 WITH ADDITIONAL FEATURES. AS OF THIS WRITING, SEVERAL ARE IN	22
23	5190	/	THE PLANNING STAGES PRESUMING SUFFICIENT SPACE WILL BECOME AVAILABLE].	23
24	5191			24
25	5192	/	SOME OF THE SPECIFIC CONCEPTS DISCUSSED HERE CAN BE PROPERLY EXPLAINED BY	25
26	5193	/	APPLYING THE WORK OF JACK BURNES AND LENNY ELEKMAN. ELEKMAN'S THEOREM STATES	26
27	5194	/	THAT, IN THEORY, ANY PROGRAM CAN ALWAYS BE FURTHER OPTIMIZED; THE DIFFICULTY	27
28	5195	/	OF IMPLEMENTATION IS PROPORTIONAL TO THE AMOUNT OF EFFORT ALREADY EXPENDED TO	28
29	5196	/	GET TO THE CURRENT PROGRAM REVISION. BURNES' COROLLARY STATES THAT THE TIME	29
30	5197	/	TO INTRODUCE FURTHER CODE OPTIMIZATION IS PROPORTIONAL TO THE AMOUNT OF TIME	30
31	5198	/	AND EFFORT ALREADY EXPENDED TO GET TO THE CURRENT OPTIMIZATION STATE. IN THE	31
32	5199	/	CASE OF THIS DEVICE HANDLER, THESE PRINCIPLES APPEAR TO BE QUITE ACCURATE IN	32
33	5200	/	PREDICTING THE COURSE OF PROGRAM DEVELOPMENT; SOME OF THE OPTIMIZATIONS	33
34	5201	/	DESCRIBED BELOW DEVELOPED OVER A PERIOD OF MANY YEARS.	34
35	5202			35
36	5203	/	THE PDP-8 IS A NEARLY UNIQUE MACHINE IN THE SENSE THAT CODE CAN BE WRITTEN TO	36
37	5204	/	OCCUPY LESS SPACE BY APPLYING SOME KNOWN TECHNIQUES COUPLED WITH WHAT CAN ONLY	37
38	5205	/	BE REFERRED TO AS MACHINE-SPECIFIC BRAINSTORMING. WHAT IS TRULY REMARKABLE IS	38
39	5206	/	AS VARIOUS TALENTED INDIVIDUALS LOOKED AT THE CODE CONSIDERED BY OTHERS TO BE	39
40	5207	/	"IMPOSSIBLE TO MAKE SMALLER", SOME OF THESE PEOPLE CAME UP WITH ADDITIONAL	40
41	5208	/	TECHNIQUES TO ACTUALLY ACHIEVE AN OVERALL CODE SPACE SIZE IMPROVEMENT.	41
42	5209			42
43	5210	/	THE OPTIMIZATION PROCESS IS OFTEN REGENERATIVE; THE LATEST IMPROVEMENT SPAWNS	43
44	5211	/	A SERIES OF REACTIONS THAT IN TURN MIGHT FURTHER OPTIMIZE OTHER CODE SECTIONS.	44
45	5212	/	WHAT WAS ONCE CONSIDERED "IMPOSSIBLE" TURNS INTO THE THEORETICALLY POSSIBLE.	45
46	5213	/	AT LEAST 12 INDIVIDUALS HAVE STUDIED THIS CODE TO SOME EXTENT. AT SEEMINGLY	46
47	5214	/	RANDOM INTERVALS, IMPROVEMENTS IN THE CODE REDUCED THE OVERALL CODE SPACE TO	47
48	5215	/	ROUGHLY THE SIZE AS PRESENTED IN THIS DOCUMENT.	48
49				49
50				50
51				51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	5217	/	AT THE SAME TIME AS THE OPTIMIZATION OF CODE SPACE WAS OCCURRING, OTHERS WERE	1
2	5218	/	SUGGESTING NEW FEATURES THAT IN TURN MADE THE PROGRAM LARGER, ALWAYS RAISING	2
3	5219	/	THE BAR FOR ANY FORM OF OPTIMIZATION. THE SET OF FEATURES AVAILABLE AS OF	3
4	5220	/	THIS WRITING IS CLEARLY NEARING STABILITY; SOME MIGHT CLAIM WE CAN GO NO	4
5	5221	/	FURTHER IN TERMS OF PASSING ADDITIONAL PROGRAM PARAMETERS. HOWEVER, ALL OF	5
6	5222	/	THE INDIVIDUALS DESIGNING THIS CODE WERE WRONG ABOUT THE NOTION OF FINALITY AT	6
7	5223	/	SOME POINT. MOST OF THEM WERE INCORRECT -- AT LEAST SIX TIMES. WHAT THE	7
8	5224	/	FUTURE HOLDS FOR THIS CODE IN TERMS OF OPTIMIZATION, NO ONE REALLY KNOWS.	8
9	5225			9
10	5226	/	SPECIFIC OPTIMIZATION TECHNIQUES.	10
11	5227			11
12	5228	/	1) USE OF INSTRUCTION PLACEMENT TO GENERATE USEFUL CONSTANTS. BY CAREFULLY	12
13	5229	/	DETERMINING EXACT REQUIREMENTS, IT IS POSSIBLE TO CREATE CONSTANTS OUT OF	13
14	5230	/	INSTRUCTIONS. ALTHOUGH PERHAPS NOT QUITE "CLEAN" IN SOME INSTANCES, AN	14
15	5231	/	ADEQUATE SOLUTION CAN OFTEN BE CONSTRUCTED.	15
16	5232			16
17	5233	/	AS NECESSARY, CERTAIN BIT COMBINATIONS MAY BE DISCOVERED THAT FULFILL ALL	17
18	5234	/	REQUIREMENTS. THIS COULD INCLUDE DON'T-CARE BITS AS WELL AS OTHER BITS	18
19	5235	/	THAT MUST BE SET WHILE OTHERS MUST BE CLEARED. OFTEN IT IS USEFUL TO	19
20	5236	/	DEFINE THE REQUIREMENTS OF EACH BIT TO PERFORM A FIRST-LEVEL ANALYSIS OF	20
21	5237	/	THE FEASIBILITY OF THIS TECHNIQUE.	21
22	5238			22
23	5239	/	2) USE OF EXTRANEEOUS BITS IN OPERATE-CLASS INSTRUCTIONS TO GENERATE USEFUL	23
24	5240	/	CONSTANTS. ALL SKIP-CLASS OPERATE INSTRUCTIONS HAVE COMMON SPECIFIC	24
25	5241	/	FEATURES SUCH AS CLEAR THE ACCUMULATOR AFTER THE TEST. BY SETTING CERTAIN	25
26	5242	/	BITS, THERE MAY BE SOME INSTRUCTION FLOW ISSUES REGARDING THE INTENDED	26
27	5243	/	PROGRAM LOGIC THAT MUST BE PROPERLY HANDLED. PROGRAM ANALYSIS MAY BE	27
28	5244	/	REQUIRED TO DETERMINE THE CONSEQUENCES OF UNINTENDED SKIP CONDITIONS.	28
29	5245			29
30	5246	/	AN AVAILABLE CONSTANT THAT IS THE SAME CODE AS AN INNOCUOUS INSTRUCTION MAY	30
31	5247	/	EXIST [AN OPERATE INSTRUCTION THAT DOESN'T EFFECT THE LOGIC WERE IT TO BE	31
32	5248	/	EXECUTED] THAT CAN BE PLACED AFTER THE PROBLEMATIC SKIP-CLASS OPERATE	32
33	5249	/	INSTRUCTION AS PROTECTION IN THE CASE OF AN UNINTENDED SKIP RESULT. A	33
34	5250	/	SIMILAR SOLUTION MIGHT BE OBTAINED USING AN AVAILABLE CONSTANT THAT FORMS	34
35	5251	/	AN "AND XXXX" INSTRUCTION WHICH WOULD BE HARMLESS IF EXECUTED WHILE ALSO	35
36	5252	/	PROTECTING AGAINST AN UNINTENDED SKIP OPERATION.	36
37	5253			37
38	5254	/	OTHER NEEDED CONSTANTS PLACED IN THE INSTRUCTION STREAM MIGHT ALWAYS SKIP	38
39	5255	/	[PERHAPS PERFORMING A NECESSARY FUNCTION SUCH AS CLEARING THE ACCUMULATOR].	39
40	5256	/	AS SUCH, ANY AVAILABLE USAGE COULD BE ASSIGNED TO THE FOLLOWING LOCATION,	40
41	5257	/	CONSTANT OR TEMPORARY. OTHER VARIATIONS OF THIS TECHNIQUE MIGHT APPLY.	41
42	5258			42
43	5259	/	3) MISCELLANEOUS TECHNIQUES MIGHT APPLY. A CONSTANT MAY BE NEEDED AS A	43
44	5260	/	SUBROUTINE ARGUMENT WITH MUCH LATITUDE AS TO ITS EXACT VALUE. THIS MIGHT	44
45	5261	/	REQUIRE BELONGING TO A NARROW RANGE OF CONTIGUOUS ADDRESSES OR A MEMBER OF	45
46	5262	/	A SMALL CLASS OF ALLOWED SIMILAR VALUES, ETC. THE VALUE CHOSEN TO FORM THE	46
47	5263	/	NEEDED CONSTANT MAY ALSO BE SUBJECT TO ANALYSIS FOR DON'T CARE BITS AS	47
48	5264	/	DESCRIBED IN 1) ABOVE.	48
49	5265			49
50	5266	/	SOME SPECIFIC TECHNIQUES CANNOT BE CATEGORIZED AND MAY ONLY APPLY TO THE	50
51	5267	/	EXACT CIRCUMSTANCES THEY WERE USED IN. THE OVERALL GOAL IS TO ACHIEVE CODE	51
52	5268	/	THAT FUNCTIONS. IT MAY NOT BE POSSIBLE TO CREATE SIMPLE LANGUAGE TO	52
53	5269	/	EXPLAIN THE RESULTS OF BRAINSTORMING. ONLY BY STUDYING THE CODE	53
54	5270	/	SUCCESSFULLY EXPLOITING THE TECHNIQUE MAY SOME UNDERSTANDING BE OBTAINED.	54
55				55
56				56
57				57
58				58

1	5272	/	OTHER TECHNIQUES NOT ENCOURAGED IN P?S/8 PROGRAMMING.	1
2	5273			2
3	5274	/	CERTAIN OPPORTUNISTIC PROGRAMMERS HAVE BEEN KNOWN TO "BEND" THE SPECIFIC	3
4	5275	/	REQUIREMENTS OF A PROGRAM DUE TO CIRCUMSTANCES OF TIGHT CODING. NEEDED VALUES	4
5	5276	/	MAY NOT EASILY BE SQUEEZED INTO AVAILABLE CODE SPACE. AS SUCH, INAPPROPRIATE	5
6	5277	/	EXPEDIENCY MAY CAUSE BIZARRE AND ARBITRARY IMPLEMENTATION OF A PROGRAM ELEMENT	6
7	5278	/	OR FUNCTION; DOCUMENTATION OF EXACTLY HOW THE PROGRAM WORKS MAY APPEAR	7
8	5279	/	BAFFLING TO THE END USER OR MAY REASONABLY BE DEEMED QUITE ARBITRARY. OTHER	8
9	5280	/	EFFECTS ON THE PROGRAM DESIGN MAY NOT BE DETECTABLE BY THE END USER WITHOUT	9
10	5281	/	APPLYING UNREASONABLE EFFORT OF PROGRAMMING ANALYSIS; PERHAPS THERE IS NO	10
11	5282	/	SPECIFIC YARDSTICK TO JUDGE THE DEGRADATION IN PERFORMANCE BY. INSPECTION OF	11
12	5283	/	THE SOURCE CODE MAY BE THE ONLY WAY TO DETERMINE THAT A BETTER JOB COULD HAVE	12
13	5284	/	BEEN DONE TO PRODUCE A BETTER OUTCOME.	13
14	5285			14
15	5286	/	BY AND LARGE THIS IS AN INDICATION OF EXPEDIENT PROGRAMMING AND THE HIGH	15
16	5287	/	LIKELIHOOD OF NOT APPLYING THE TECHNIQUES DISCUSSED ABOVE. TO THE FULLEST	16
17	5288	/	EXTENT POSSIBLE TECHNIQUES OF THIS SORT ARE NOT USED THROUGHOUT P?S/8. PROPER	17
18	5289	/	APPLICATION OF THE PROVEN METHODS DISCUSSED ABOVE GENERALLY WILL ALLOW	18
19	5290	/	MAINTAINING DESIGN GOALS WITHOUT COMPROMISE. AS NECESSARY, IMPLEMENTATION	19
20	5291	/	SCHEDULES SHOULD BE FLEXIBLE ENOUGH TO ALLOW APPLICATION OF ACCEPTABLE	20
21	5292	/	OPTIMIZATION TECHNIQUES. THIS WILL AVOID EMBARRASMENTS SUCH AS DESCRIBED	21
22	5293	/	HERE. [NOTE: CERTAIN INDIVIDUALS ARE KNOWN FOR REPEATED USE OF THIS SORT OF	22
23	5294	/	PROBLEMATIC TECHNIQUE; WE WILL NOT MENTION THEM BY NAME.]	23
24	5295			24
25	5296	/	FURTHER DEVELOPMENT HISTORY OF THE TC01/TC08 SYSTEM DEVICE HANDLER.	25
26	5297			26
27	5298	/	SOME USEFUL FEATURES WERE ADDED AT A POINT IN TIME WHERE IT SEEMED IMPOSSIBLE	27
28	5299	/	TO DO SO. BY INTENSIVELY STUDYING THE PROGRAM, MORE CODE SPACE WAS	28
29	5300	/	OCCASIONALLY FOUND, OFTEN BY EXPLOITING SOME ADDITIONAL QUIRK OF THE	29
30	5301	/	INSTRUCTION SET. IN SOME CASES THE IMPROVEMENT WAS FORMULATED BY TWO	30
31	5302	/	INDIVIDUALS BOUNCING IDEAS OFF OF EACH OTHER. THE SINGLE LARGEST EXAMPLE OF	31
32	5303	/	AN ADDED FEATURE IS THE IMPLEMENTATION OF THE COMPLETELY FUNCTIONAL	32
33	5304	/	REPLACEMENT SYSTEM LOADER INTERNAL CALL MECHANISM. [NOTE: THE SYSTEM LOADER	33
34	5305	/	IS DESCRIBED ELSEWHERE IN THIS DOCUMENT.]	34
35	5306			35
36	5307	/	A CONCEPT VIRTUALLY UNIQUE TO DATA BREAK DEVICES WAS DEVELOPED SPECIFICALLY	36
37	5308	/	FOR THE TC01/TC08 HANDLER. PROGRAM EXIT TO 07600 CAN BE INITIATED BY WHAT	37
38	5309	/	APPEARS TO BE A SUBROUTINE CALL TO THE SYSTEM DEVICE HANDLER WITH THREE	38
39	5310	/	CUSTOMARY IN-LINE ARGUMENTS. AS DESCRIBED WITHIN THE SYSTEM DEVICE HANDLER	39
40	5311	/	CODING [ELSEWHERE IN THIS DOCUMENT], THE SYSTEM DOES IN FACT BOOT UP WHEN THE	40
41	5312	/	ENTIRE PROCESS IS FINISHED. THIS UNORTHODOX TECHNIQUE SAVES SEVERAL WORDS OF	41
42	5313	/	CODE SPACE OVER THE FORMER CODE. [NOTE: TWO OF THE THREE ARGUMENT WORDS ALSO	42
43	5314	/	SERVE AS CONSTANTS NEEDED ELSEWHERE.] CERTAIN ASPECTS OF THIS TECHNIQUE MAY	43
44	5315	/	BE IMPLEMENTED IN SYSTEM DEVICE HANDLERS FOR OTHER HARDWARE CONFIGURATIONS.	44
45	5316			45
46	5317	/	AS OF SOME POINT IN TIME [NOT WELL REMEMBERED], IT WAS KNOWN THAT "ALL"	46
47	5318	/	REQUIRED FEATURES WERE IMPLEMENTED WITH EXACTLY ONE 12-BIT WORD TO SPARE, AND	47
48	5319	/	AT THE SAME TIME NO KNOWN FEATURE PLANNED TO TAKE ADVANTAGE OF THE SPACE;	48
49	5320	/	HOWEVER, AS NEW FEATURES WERE ADDED, THE NOTION OF "EXTRA" AVAILABLE WORDS	49
50	5321	/	DISAPPEARED. EVEN WITH SO MUCH EFFORT ALREADY APPLIED, IT IS NOT CLEAR IF WE	50
51	5322	/	WILL EVER EXCEED THE PRACTICAL UPPER LIMIT OF BURNESSE' COROLLARY.	51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	5324	/	DEVICE HANDLER ENTRY POINT DESIGN CONSIDERATION.	1
2	5325			2
3	5326	/	NOTE: THIS POINT IS ACTUALLY EXPLOITED IN THE REST OF THE OPERATING SYSTEM AND	3
4	5327	/	CONFORMING PROGRAMS.	4
5	5328			5
6	5329	/	THE SYSTEM HANDLER IS ENTERED BY A CALL TO LOCATION 07640 WITH APPROPRIATE	6
7	5330	/	IN-LINE ARGUMENTS. THE REASONS FOR CHOOSING THIS SPECIFIC ADDRESS ARE AS	7
8	5331	/	FOLLOWS:	8
9	5332			9
10	5333	/	IT IS QUITE COMMON FOR MOST PDP-8 CODE TO USE AN "SZA CLA" INSTRUCTION EITHER	10
11	5334	/	"NATURALLY" OR BY EASILY IMPLEMENTED CONTRIVANCE USING ONE OF THE TECHNIQUES	11
12	5335	/	SUGGESTED ABOVE [OR PERHAPS SOME OTHER TRICK NOT DOCUMENTED HERE]. AS SUCH,	12
13	5336	/	IT IS GENERALLY FAIRLY EASY TO DESIGN CALLS TO THE SYSTEM DEVICE HANDLER THAT	13
14	5337	/	USE "SZA CLA" AS USED IN AN INSTRUCTION ON THE SAME PAGE AS THE CALL, THUS	14
15	5338	/	ELIMINATING THE NEED FOR ANY FORM OF CURRENT PAGE [OR PAGE ZERO] LITERAL	15
16	5339	/	CONSTANT VALUE TO ACCOMPLISH IT.	16
17	5340			17
18	5341	/	CODE THAT USES TRICKS SUCH AS THIS IS REQUIRED TO HAVE FAILING CONDITIONAL	18
19	5342	/	ASSEMBLY EXPRESSIONS TO ENFORCE THE PARTICULARS THAT MATTER FOR THE TRICK TO	19
20	5343	/	WORK. WITHOUT THIS LEVEL OF SAFEGUARD, SOURCE CODE CAN EASILY BECOME BROKEN	20
21	5344	/	BY PROGRAMMERS UNAWARE OF THE TECHNIQUES IN EFFECT.	21
22	5345			22
23	5346	/	THIS HANDLER IS NO EXCEPTION [SEE THE SYSTEM DEVICE HANDLER ENTRY POINT CODE	23
24	5347	/	FOR MORE INFORMATION]. NOTE: THE MAIN P?S/8 KEYBOARD MONITOR AND INTEGRATED	24
25	5348	/	EDITOR COMPONENT OF P?S/8 IS 2048 WORDS OF CODE SPACE LONG NOT COUNTING	25
26	5349	/	OVERLAYS, BUT INCLUDING THE SYSTEM DEVICE HANDLER AND A COMPLETE IMAGE OF THE	26
27	5350	/	TINY FILE SYSTEM [TFS] DIRECTORY.	27
28	5351			28
29	5352	/	IN LIEU OF THE ORDINARILY DESERVED PAGE ZERO LITERAL CONSTANT 7640, ALL CODE	29
30	5353	/	REQUIRING THE ADDRESS POINTER INSTEAD USES LOCAL "SZA CLA" INSTRUCTIONS ON THE	30
31	5354	/	SAME PAGE AS THE SUBROUTINE CALL POINTER. BY USING THIS AND OTHER RELATED	31
32	5355	/	TECHNIQUES, THE PAGE ZERO OF THAT MODULE FUNCTIONS AS IF THERE WERE ABOUT 150	32
33	5356	/	MEMORY LOCATIONS INSTEAD OF THE NOMINAL 128. AT THE SOURCE CODE LEVEL, ALL	33
34	5357	/	CASES ARE PROTECTED BY APPROPRIATE FAILING CONDITIONAL ASSEMBLY EXPRESSIONS	34
35	5358	/	THAT MAINTAIN THE VIABILITY OF THE SOURCE CODE.	35
36	5359			36
37	5360	/	IN GENERAL, PROTECTION AGAINST SOURCE CODE DAMAGE IS ACHIEVED WITH EXPRESSIONS	37
38	5361	/	SUCH AS THIS:	38
39	5362			39
40	5363	/	IFNZRO .-7640 <ERROR .>	40
41	5364			41
42	5365	/	NOTE: THE ERROR DIRECTIVE IS A FEATURE OF P?S/8 PAL AND FOLLOW-ON ASSEMBLERS.	42
43	5366	/	SHOULD THE CODE BECOMES UNACCEPTABLY MODIFIED, THE CONDITIONAL DOES NOT FAIL	43
44	5367	/	AND INSTEAD THE ERROR DIRECTIVE IS ASSEMBLED. DURING THE ASSEMBLY PROCESS,	44
45	5368	/	ERROR MESSAGES ARE ISSUED DURING ALL ASSEMBLER PASSES. AN ERROR IS LOGGED IN	45
46	5369	/	THE OVERALL ASSEMBLY ERROR COUNT TO PREVENT SUBSEQUENT BINARY LOADING DURING	46
47	5370	/	CHAINED ASSEMBLIES. WHEN USING OS/8 PAL8, THIS MERELY GENERATES AN ORDINARY	47
48	5371	/	UNDEFINED SYMBOL [US] ERROR; EITHER WAY IT IS NOT POSSIBLE TO GET AN	48
49	5372	/	ERROR-FREE ASSEMBLY WHEN THE CODE IS MANGLED BY A NITWIT. [OF COURSE THERE IS	49
50	5373	/	NO PROTECTION FROM A NITWIT WHO DISABLES THE PROTECTION MECHANISM!]	50
51	5374			51
52	5375	/	END OF APPENDIX G.	52
53				53
54				54
55				55
56				56
57				57
58				58

1	5377	/	APPENDIX H.	1
2	5378			2
3	5379	/	P?S/8 SYSTEM GENERATION CONSIDERATIONS.	3
4	5380			4
5	5381	/	STUDYING THE DOCUMENTATION OF THE SYSTEM DEVICE HANDLER SHOULD LEAD TO AN	5
6	5382	/	UNDERSTANDING OF HOW IT FUNCTIONS ONCE IT IS LOADED INTO THE INTENDED MEMORY	6
7	5383	/	LOCATIONS; HOWEVER, THE PROCESS OF CREATING THE ACTUAL IMAGE ON BLOCK 0000 OF	7
8	5384	/	A BOOTABLE SYSTEM REQUIRES RELOCATING THE BINARY CODE TO AN ALTERNATE LOCATION	8
9	5385	/	FOR THE PURPOSE OF WRITING THE IMAGE. IF THE CODE WERE SIMPLY WRITTEN FROM A	9
10	5386	/	WORKING VERSION OF THE DEVICE HANDLER RUNNING IN THE INTENDED MEMORY SPACE IT	10
11	5387	/	IS DESIGNED FOR, A DEFECTIVE [MODIFIED] IMAGE WOULD RESULT. THUS, A STATIC	11
12	5388	/	IMAGE COPY OF THE DEVICE HANDLER IS REQUIRED TO PRODUCE THE INTENDED RESULT.	12
13	5389			13
14	5390	/	A PRIOR REQUIREMENT IS THAT ANOTHER OPERATING SYSTEM HAS TO LOAD THE BINARY	14
15	5391	/	FILE[S] INTO MEMORY. AS SUCH, ALL LOADING OF OPERATING SYSTEM COMPONENTS	15
16	5392	/	MUST BE ACCOMPLISHED FROM AVAILABLE MEMORY OUTSIDE OF THE RESERVED AREA OF THE	16
17	5393	/	CURRENT OPERATING SYSTEM ACCOMPLISHING THE LOADING. AS SUCH, ONLY MEMORY	17
18	5394	/	FULLY CONSIDERED AS USER MEMORY IS USED TO LOAD ALL OF THE COMPONENTS OF P?S/8	18
19	5395	/	INCLUDING A SYSTEM GENERATION ROUTINE DESIGNED TO WRITE OUT COMPONENTS LOADED	19
20	5396	/	INTO MEMORY AT ONCE. [NOTE: CREATING AN OPERATING SYSTEM IS ACCOMPLISHED IN	20
21	5397	/	SEVERAL STAGES TO MINIMIZE THE MEMORY REQUIREMENTS OF SYSTEM GENERATION.]	21
22	5398			22
23	5399	/	ALL P?S/8 SYSTEM COMPONENTS ARE THUS LOADED ONLY INTO LOCATIONS X0000-X7577	23
24	5400	/	WHERE X IS ANY FIELD USED. THIS WILL SATISFY THE REQUIREMENTS OF CREATING	24
25	5401	/	P?S/8 UNDER ANOTHER RUNNING COPY OF P?S/8, OS/8 OR THE PAPER-TAPE BINARY	25
26	5402	/	LOADER [BIN]. [FOR A DISCUSSION OF POTENTIAL ALTERNATE METHODS, SEE BELOW	26
27	5403	/	WITH REGARD TO PROPOSED SYSTEM BUILDING UTILITIES.]	27
28	5404			28
29	5405	/	THE CURRENT METHOD OF P?S/8 SYSTEM GENERATION REQUIRES 16K OF MEMORY. OTHER	29
30	5406	/	THAN LOCATIONS 7600-7777 IN FIELDS 0-3, ALL OF THIS MEMORY IS LOADED WITH	30
31	5407	/	VARIOUS SYSTEM COMPONENTS, GENERALLY RELOCATED TO AVAILABLE BUFFERS FOR THE	31
32	5408	/	PURPOSE OF WRITING OUT IMAGES OF THE VARIOUS COMPONENTS. IN A FEW CASES	32
33	5409	/	[SUCH AS THE KEYBOARD MONITOR AND INTEGRATED TEXT EDITOR] THE LOADING ADDRESS	33
34	5410	/	CORRESPONDS TO THE OPERATING ADDRESS USED WHEN THE SYSTEM IS BOOTED UP.	34
35	5411	/	[NOTE: 16K OF MEMORY IS GENERALLY REQUIRED TO ASSEMBLE SEVERAL OF THE LARGER	35
36	5412	/	SYSTEM COMPONENTS; AS SUCH, THE CURRENT METHOD WAS CHOSEN FOR SYSTEM CREATION	36
37	5413	/	ON A SUFFICIENTLY LARGE SYSTEM CAPABLE OF ASSEMBLING THE SOURCE FILES INTO	37
38	5414	/	BINARY FILES SUITABLE FOR LOADING AND CREATING A COMPLETE SYSTEM.]	38
39	5415			39
40	5416	/	MOST COMPONENTS ARE LOADED INTO ARBITRARY BUFFER ADDRESSES SUITABLE FOR	40
41	5417	/	WRITING OUT CONTIGUOUS IMAGES OF EACH MODULE. THIS ALLOWS EACH TO BE WRITTEN	41
42	5418	/	BY A SINGLE CALL TO A DEVICE HANDLER FOR EACH INSTANCE. EACH MODULE IS	42
43	5419	/	LOADED INTO THE DESIGNATED BUFFER; HOWEVER, THE CODE IS ASSEMBLED FOR USE IN	43
44	5420	/	THE OPERATIONAL ADDRESS IT IS INTENDED FOR. [NOTE: IN A FEW INSTANCES, THERE	44
45	5421	/	IS MORE THAN ONE UNIQUE ADDRESS ASSOCIATED WITH SPECIFIC USAGE. THIS OCCURS	45
46	5422	/	WHEN THE BLOCK WHERE THE CODE RESIDES IS READ DURING THE BOOTUP PROCESS INTO	46
47	5423	/	AN ALTERNATE BUFFER ADDRESS TEMPORARILY BEFORE FINALLY BEING MOVED TO THE	47
48	5424	/	INTENDED EXECUTION ADDRESS. WHERE NECESSARY, THIS MULTIPLE RELOCATION IS	48
49	5425	/	INDICATED IN THE SOURCE CODE.]	49
50				50
51				51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	5427	/	IN ALL CASES, THE LOADING ADDRESS AND FIELD MUST BE CLEARLY STATED, FOLLOWED	1
2	5428	/	BY RELOCATION STATEMENTS TO ENSURE THE PROGRAM SEGMENT ASSEMBLES AS NEEDED	2
3	5429	/	ONCE THE CODE IS LOADED WHERE INTENDED. EVERY SECTION OF CODE WILL THUS	3
4	5430	/	CONTAIN SEVERAL STATEMENTS REGARDING BOTH LOADING ORIGIN AND EXECUTION ORIGIN	4
5	5431	/	AS REQUIRED. [NOTE: THE USE OF LITERALS IS STRICTLY AVOIDED WITHIN P?S/8	5
6	5432	/	DUE TO THE INHERENT COMPLICATIONS INVOLVING LITERAL DUMP OPERATIONS.]	6
7	5433			7
8	5434	/	A SYSTEM GENERATION PROGRAM IS REQUIRED TO CARRY OUT ALL OF THE VARIOUS WRITE	8
9	5435	/	OPERATIONS NEEDED AT ANY STAGE OF SYSTEM CREATION. THIS IS ESSENTIALLY A	9
10	5436	/	SERIES OF CALLS TO A DEVICE HANDLER TO WRITE OUT APPROPRIATE BLOCKS ON THE	10
11	5437	/	TARGET DEVICE. WRITING THE SYSTEM DEVICE HANDLER AND THESE FUNDAMENTAL	11
12	5438	/	SYSTEM SEGMENTS CREATES A BOOTABLE P?S/8 OPERATING SYSTEM CAPABLE OF LOADING	12
13	5439	/	PROGRAMS AND EXECUTION OF A FEW ADDITIONAL UTILITIES. [NOTE: THIS PROCESS	13
14	5440	/	MERELY CREATES A STARTER SYSTEM; MANY STANDARD UTILITIES ARE NOT PRESENT.	14
15	5441	/	HOWEVER, THIS MINIMAL SYSTEM IS CAPABLE OF RUNNING ADDITIONAL PROGRAMS TO	15
16	5442	/	CONTINUE THE PROCESS OF BUILDING UP A MORE COMPLETE SYSTEM.]	16
17	5443			17
18	5444	/	AS A PRACTICAL MATTER, THE PROCESS OF CREATING A COMPLETE OPERATING SYSTEM IS	18
19	5445	/	CARRIED OUT IN STAGES TO ACCOMMODATE A PARTICULAR SYSTEM GENERATION SCHEME.	19
20	5446	/	CERTAIN RELEVANT ISSUES WILL BE DISCUSSED BELOW.	20
21	5447			21
22	5448	/	HISTORIC P?S/8 SYSTEM GENERATION METHOD.	22
23	5449			23
24	5450	/	GENERATION OF P?S/8 HAS BEEN PERFORMED ON 16K OR LARGER MACHINES. THE ENTIRE	24
25	5451	/	USER SPACE IS UTILIZED TO LOAD IN ALL COMPONENTS OF A DESIGNATED STARTER	25
26	5452	/	SYSTEM. THIS CONSISTS OF MORE THAN THE MINIMUM COMPLEMENT OF COMPONENTS	26
27	5453	/	REQUIRED TO BOOTSTRAP THE SYSTEM; SOME OF THE ADDITIONAL COMPONENTS ARE NEEDED	27
28	5454	/	IN SUBSEQUENT STEPS TO CREATE THE ADDITIONAL COMPONENTS FROM THE BOOTED UP	28
29	5455	/	STARTER SYSTEM. [NOTE: CREATING CERTAIN COMPONENTS REQUIRES 16K TO ASSEMBLE	29
30	5456	/	THE SOURCE CODE; THERE IS NO INTENTION TO ATTEMPT TO REDUCE SYSTEM DEVELOPMENT	30
31	5457	/	MEMORY REQUIREMENTS WHEN USING THIS METHOD.]	31
32	5458			32
33	5459	/	EACH OF THE ADDITIONAL SYSTEM COMPONENTS HAS ITS OWN MEMORY REQUIREMENT; AS OF	33
34	5460	/	THIS WRITING, ALL ADDITIONAL SYSTEM PROGRAMS REQUIRE LESS THAN 16K TO BE ADDED	34
35	5461	/	TO AN EXISTENT STARTER SYSTEM; MOST REQUIRE 4K MEMORY TO LOAD THE COMPONENT	35
36	5462	/	[ALTHOUGH AS MUCH AS 16K MAY BE REQUIRED TO ASSEMBLE THE SOURCE CODE INTO	36
37	5463	/	LOADABLE BINARY FILES]. SEVERAL SPECIFIC COMPONENTS REQUIRE 8K OR 12K FOR	37
38	5464	/	SYSTEM GENERATION PURPOSES. [NOTE: ALL OF THE SYSTEM PROGRAMS CAN MINIMALLY	38
39	5465	/	RUN ON A 4K SYSTEM ONCE CREATED; CERTAIN RESTRICTIONS MIGHT APPLY ON USAGE ON	39
40	5466	/	CERTAIN MEMORY CONFIGURATIONS. MANY PROGRAMS ATTEMPT TO ADAPT TO AVAILABLE	40
41	5467	/	MEMORY WITH VARYING DEGREES OF PERFORMANCE. IN GENERAL, MOST PROGRAMS ARE	41
42	5468	/	AT LEAST NOMINALLY USEFUL ON A 4K SYSTEM; IN SPECIFIC CASES, PERFORMANCE MAY	42
43	5469	/	DRAMATICALLY INCREASE WITH MORE MEMORY AVAILABLE. CERTAIN PROGRAMS ARE	43
44	5470	/	IMPLEMENTED WITH INTERNAL OVERLAY STRUCTURES; CREATING THESE PROGRAMS REQUIRES	44
45	5471	/	ALL SUB-COMPONENTS BE LOADED INTO MEMORY AT THE SAME TIME.]	45
46	5472			46
47	5473	/	ALTERNATE SYSTEM GENERATION METHODS WITH LESSER MEMORY REQUIREMENTS.	47
48	5474			48
49	5475	/	AS OF THIS WRITING, NO OTHER SYSTEM GENERATION METHOD HAS BEEN IMPLEMENTED;	49
50	5476	/	HOWEVER, MANY OF THE ISSUES HAVE BEEN CAREFULLY STUDIED. THE FOLLOWING	50
51	5477	/	SECTIONS OUTLINE ALTERNATE MECHANISMS, INCLUDING A METHOD THAT WILL SUCCEED	51
52	5478	/	ON 4K MEMORY SYSTEMS.	52
53				53
54				54
55				55
56				56
57				57
58				58

1	5480	/	THE SECONDARY SYSTEM GENERATION METHOD AND TENTATIVE MODIFICATION TO RUN ON A	1
2	5481	/	4K MEMORY SYSTEM.	2
3	5482			3
4	5483	/	NOTE: CERTAIN P?S/8 SYSTEM DEVICE HANDLERS REQUIRE A MEMORY EXTENSION DEVOTED	4
5	5484	/	TO ADDITIONAL PORTIONS OF THE DEVICE HANDLER. THE MEMORY EXTENSION MUST BE	5
6	5485	/	LOADED INTO EXTENDED MEMORY; LOCATIONS X6000-X7777 OF ANY EXTENDED MEMORY	6
7	5486	/	FIELD MUST BE AVAILABLE TO RUN THESE CONFIGURATIONS. WHEN THESE SYSTEMS FIRST	7
8	5487	/	BOOT UP, THE HIGHEST MEMORY FIELD IS DETERMINED; THE SYSTEM DEVICE HANDLER	8
9	5488	/	EXTENSION IS THEN LOADED INTO 6000-7777 OF THE HIGHEST MEMORY FIELD AVAILABLE.	9
10	5489	/	ANY CONSIDERATIONS REGARDING SYSTEM GENERATION CANNOT CONSIDER THIS RESERVED	10
11	5490	/	MEMORY; AS SUCH, FOR SYSTEM GENERATION PURPOSES, THE MACHINE WILL BE	11
12	5491	/	CONSIDERED AS HAVING ONE LESS MEMORY FIELD AVAILABLE.	12
13	5492			13
14	5493	/	AS OF THIS WRITING, ONCE THE STARTER SYSTEM IS CREATED [ON A SYSTEM WITH 16K	14
15	5494	/	OR MORE], THE SECONDARY SYSTEM GENERATION IS ACHIEVED USING AN INTERCHANGE	15
16	5495	/	MEDIA CREATED UNDER OS/8. [NOTE: THE RELEVANT INTERCHANGE MEDIA FOR THE	16
17	5496	/	TC01/TC08 DECTAPE SYSTEM IS AN OS/8 NON-SYSTEM DECTAPE WHICH STARTS WITH AN	17
18	5497	/	EMPTY OS/8 DIRECTORY; OTHER INTERCHANGE MEDIA ARE SUPPORTED FOR OTHER P?S/8	18
19	5498	/	CONFIGURATIONS.]	19
20	5499			20
21	5500	/	THE STARTER SYSTEM INCLUDES THE P?S/8 "DUMP" PROGRAM, A UTILITY THAT IS A	21
22	5501	/	DESCENDANT OF A FORMER PROGRAM WRITTEN FOR THE R-L MONITOR SYSTEM. THE MAIN	22
23	5502	/	FUNCTIONALITY IS THE ABILITY TO TRANSFER SEVERAL STORAGE BLOCKS AT ANY ONE	23
24	5503	/	TIME FROM ONE DEVICE UNIT TO ANOTHER; INDIVIDUAL BLOCKS CAN BE UPDATED BY	24
25	5504	/	APPROPRIATE PATCH-ORIENTED COMMANDS.	25
26	5505			26
27	5506	/	THE DUMP PROGRAM IS DESIGNED TO RUN FROM P?S/8 ONLY; IT NOT ONLY DEPENDS ON	27
28	5507	/	THE AVAILABILITY OF THE P?S/8 SYSTEM DEVICE HANDLER, BUT ALSO CAN TAKE INPUT	28
29	5508	/	FROM TFS TEXT FILES CONTAINING DUMP COMMANDS.	29
30	5509			30
31	5510	/	ONCE THE STARTER SYSTEM IS CREATED, THE INTERCHANGE MEDIA IS MOUNTED ON AN	31
32	5511	/	ACCESSIBLE LOGICAL UNIT [SUCH AS DECTAPE UNIT 1]. DUMP IS RUN FROM THE BOOTED	32
33	5512	/	P?S/8 SYSTEM ON DRIVE 0; COMMAND FILES ARE PASSED THAT ARE DESIGNED TO MOVE	33
34	5513	/	CORE IMAGE SUBSECTIONS FROM KNOWN AREAS ON THE OS/8 INTERCHANGE MEDIA TO	34
35	5514	/	DESIGNATED BLOCKS ON THE STARTER SYSTEM. ONCE THIS PROCESS IS COMPLETED, A	35
36	5515	/	COMPLETE [OR NEARLY COMPLETE] P?S/8 SYSTEM HAS BEEN CREATED.	36
37	5516			37
38	5517	/	AS WRITTEN, THIS PROCESS REQUIRES 8K OF AVAILABLE MEMORY; DUMP CAN TRANSFER	38
39	5518	/	AN ENTIRE MEMORY FIELD [32 BLOCKS] AT ANY ONE TIME. MANY OF THE SPECIFIC	39
40	5519	/	COMMANDS WITHIN THE DUMP COMMAND FILE REQUIRE 31 PAGES BE TRANSFERRED IN A	40
41	5520	/	SINGLE OPERATIONS. WHEN DUMP IS RUN ON A 4K SYSTEM, THE MAXIMUM AMOUNT THAT	41
42	5521	/	CAN BE TRANSFERRED IS LOWERED TO ABOUT 20 PAGES DUE TO THE PRESENCE IN FIELD 0	42
43	5522	/	OF THE ENTIRE PROGRAM AND THE READ/WRITE BUFFERS.	43
44	5523			44
45	5524	/	BY SLIGHTLY REWRITING THE DUMP COMMAND FILE, THE DEPENDENCY ON REQUIRING 8K OF	45
46	5525	/	MEMORY CAN BE LIFTED; SEVERAL COMMANDS CURRENTLY PERFORMED IN A SINGLE	46
47	5526	/	OPERATION WILL BE REQUIRED TO BE ACCOMPLISHED IN TWO STEPS.	47
48	5527			48
49	5528	/	THUS, WHILE THE CREATION OF THE STARTER SYSTEM REQUIRES 16K, THE SECONDARY	49
50	5529	/	SYSTEM GENERATION PROCESS CAN BE ACCOMPLISHED IN 4K [PRESENTLY 8K].	50
51				51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	5531	/	CREATING THE STARTER SYSTEM ON 4K SYSTEMS.	1
2	5532			2
3	5533	/	IF THE 16K MEMORY LOAD USED TO CREATE THE STARTER SYSTEM IS SAVED TO ANOTHER	3
4	5534	/	OS/8 INTERCHANGE MEDIA [OR PERHAPS A DIFFERENT SECTION OF THE SAME MEDIA USED	4
5	5535	/	FOR THE SECONDARY SYSTEM CREATION], THE ENTIRE PROCESS OF CREATING A NEW P?S/8	5
6	5536	/	SYSTEM CAN BE ACCOMPLISHED ON A 4K SYSTEM. [NOTE: A FUNCTIONING P?S/8 SYSTEM	6
7	5537	/	MUST ALREADY BE AVAILABLE.] [NOTE: DUE TO THE OVERALL SIZE OF ALL BINARY	7
8	5538	/	FILES, CERTAIN OS/8 DEVICES MAY NOT BE ABLE TO STORE ALL FILES REQUIRED TO	8
9	5539	/	SUPPORT THIS METHOD. ON RX01 SYSTEMS, THIS MAY REQUIRE AS MANY AS THREE	9
10	5540	/	DISKETTES; THE DUMP COMMAND FILES CAN BE MODIFIED AS REQUIRED.]	10
11	5541			11
12	5542	/	TRUE 4K-ORIENTED SYSTEM BUILDER ISSUES.	12
13	5543			13
14	5544	/	A STAND-ALONE SYSTEM BUILDER UTILITY CAN BE DEFINED TO ALLOW CREATING P?S/8	14
15	5545	/	FROM INDIVIDUAL BINARY COMPONENTS. THERE ARE SEVERAL ISSUES SPECIFIC TO	15
16	5546	/	P?S/8 SYSTEM BUILDER PROGRAMS:	16
17	5547			17
18	5548	/	1) DEFINING THE DEVICE HANDLER USED TO CARRY OUT THE VARIOUS WRITE OPERATIONS.	18
19	5549			19
20	5550	/	A STAND-ALONE SYSTEM BUILDER PROGRAM REQUIRES THE AVAILABILITY OF A DEVICE	20
21	5551	/	HANDLER APPROPRIATE TO THE TARGET HARDWARE FOR THE BUILD OPERATION. IN	21
22	5552	/	CERTAIN INSTANCES, THIS WOULD NOT BE THE SAME HARDWARE AS THE SYSTEM IS	22
23	5553	/	BEING CREATED FOR. FOR EXAMPLE, A TD8E-ORIENTED P?S/8 SYSTEM COULD BE	23
24	5554	/	CREATED ON A TC01/TC08 SYSTEM; THE RESULTING DECTAPE WOULD ONLY BE BOOTABLE	24
25	5555	/	ON THE CORRECT TD8E CONFIGURATION. [NOTE: THERE ARE TWO DISTINCT TD8E	25
26	5556	/	CONFIGURATIONS: A) REQUIRES 8K OR MORE MEMORY, B) REQUIRES THE MR8E-C ROM	26
27	5557	/	BE INSTALLED.]	27
28	5558			28
29	5559	/	WITHIN THE 16K BINARY SYSTEM GENERATION METHOD, A PORTION OF MEMORY IS	29
30	5560	/	RESERVED FOR A P?S/8 SYSTEM DEVICE CREATED SPECIALLY FOR THIS PURPOSE; IT	30
31	5561	/	MAY BE ORIENTED TOWARDS DIFFERENT HARDWARE FROM THE TARGET SYSTEM BEING	31
32	5562	/	GENERATED. ADDITIONALLY, SINCE THIS CODE IS TAKEN FROM A DIFFERENT P?S/8	32
33	5563	/	SYSTEM GENERATION CONFIGURATION, THE IMAGE OF THIS HANDLER MAY NOT RUN IF	33
34	5564	/	MERELY MOVED TO 07600-07777 [AND THIS DOES NOT ADDRESS THE NOTION THAT THE	34
35	5565	/	HANDLER MAY ALSO HAVE AN EXTENDED DEVICE HANDLER SECTION]. EVEN IF THE	35
36	5566	/	HANDLER IS FOR THE IDENTICAL HARDWARE AS THE TARGET SYSTEM, THE IMAGE OF	36
37	5567	/	THE HANDLER OFTEN IS NOT USABLE MERELY BY MOVING THE CODE. AS SUCH, THERE	37
38	5568	/	IS ALSO AN ASSOCIATED FIXUP ROUTINE THAT MAKES THE HANDLER OPERATIONAL	38
39	5569	/	AFTER THE HANDLER IS MOVED.	39
40	5570			40
41	5571	/	THIS ISSUE IS BEST SOLVED BY USING A P?S/8 NON-SYSTEM HANDLER. WHILE THE	41
42	5572	/	LARGEST THEORETICAL NON-SYSTEM HANDLER CAN OCCUPY AN ENTIRE MEMORY FIELD	42
43	5573	/	[WHICH WOULD RAISE THE MEMORY REQUIREMENT FOR A SYSTEM BUILDER PROGRAM TO	43
44	5574	/	8K], MOST CAN BE ACCOMMODATED IN EIGHT PAGES OF MEMORY [OR LESS]. A	44
45	5575	/	BUILDER PROGRAM MUST BE ABLE TO LOAD SUCH A HANDLER BEFORE ATTEMPTING TO	45
46	5576	/	WRITE OUT THE SYSTEM COMPONENTS. APPROPRIATE PARAMETERS EXPLAINING THE	46
47	5577	/	PARTICULARS OF ANY NON-SYSTEM DEVICE HANDLER CAN BE PROVIDED BY USING A	47
48	5578	/	SERIES OF SPECIAL ORIGIN SETTINGS WITHIN THE BINARY FILE OF THE DEVICE	48
49	5579	/	HANDLER TO GUIDE THE SYSTEM BUILDER PROGRAM IN LOADING THE HANDLER WITHIN	49
50	5580	/	THE UTILITY. [THIS WOULD ALSO INCLUDE THE FEW DEVICE HANDLERS THAT WOULD	50
51	5581	/	REQUIRE 8K MEMORY.] [NOTE: THE TOPIC OF EMBEDDED SPECIAL-PURPOSE ORIGIN	51
52	5582	/	SETTINGS IS DISCUSSED BELOW.]	52
53				53
54				54
55				55
56				56
57				57
58				58

1	5584	/	2) DEFINING THE MECHANISM FOR LOADING BINARY FILES OF THE SYSTEM COMPONENTS.	1
2	5585			2
3	5586	/	THE ONLY UNIVERSAL MECHANISM AVAILABLE TO A STAND-ALONE BUILDER PROGRAM IS	3
4	5587	/	THE USE OF BINARY PAPER-TAPES IN BIN FORMAT. HOWEVER, IF THE BUILDER	4
5	5588	/	PROGRAM IS LOADED FROM A P?S/8 SYSTEM, TFS BINARY FILES CAN BE PASSED TO	5
6	5589	/	THE BUILDER PROGRAM. HAVING THIS ABILITY CAN TRANSFORM A POTENTIALLY	6
7	5590	/	TEDIOUS SYSTEM GENERATION INTO A SERIES OF SMALLER OPERATIONS WHERE ALL OF	7
8	5591	/	THE SECONDARY OPERATIONS CAN BE PERFORMED WITHOUT THE USE OF PAPER-TAPES.	8
9	5592			9
10	5593	/	3) ABSOLUTE SYSTEM BLOCK NUMBERS VERSUS RELATIVE BLOCK NUMBERS.	10
11	5594			11
12	5595	/	CERTAIN SYSTEM COMPONENTS MUST BE LOADED TO DEFINED ABSOLUTE BLOCK NUMBERS.	12
13	5596	/	THIS INCLUDES TRULY ABSOLUTE DEFINED BLOCKS AS WELL AS THE ARBITRARY BLOCKS	13
14	5597	/	DESIGNATED FOR IMPORTANT SYSTEM PROGRAMS CONSIDERED PART OF THE STARTER	14
15	5598	/	SYSTEM. [NOTE: THE ONLY ABSOLUTELY REQUIRED P?S/8 SYSTEM PROGRAM IS THE	15
16	5599	/	"BIN" UTILITY PROGRAM; WITHOUT THE AVAILABILITY OF P?S/8 BIN, NO BINARY	16
17	5600	/	FILES CAN BE LOADED. P?S/8 DUMP WILL LIKELY ALSO BE DESIGNATED AS AN	17
18	5601	/	ABSOLUTE REQUIREMENT DUE TO ITS ABILITY TO RECTIFY SMALL DISCREPANCIES AS	18
19	5602	/	THEY ARISE.]	19
20	5603			20
21	5604	/	FURTHER SYSTEM BUILDING OPERATIONS PERTAIN TO ADDITIONAL SYSTEM UTILITY	21
22	5605	/	PROGRAMS SUCH AS P?S/8 PAL, FOCAL, THE LOGICAL CONSOLE OVERLAY AND OTHER	22
23	5606	/	OPTIONAL COMPONENTS. SINCE ALL OF THESE PROGRAMS CAN BE ADDED ON AT ANY	23
24	5607	/	TIME AFTER A STARTER SYSTEM IS CREATED, THE DESIGNATED METHOD WOULD BE TO	24
25	5608	/	USE THE SYSTEM BUILDER UTILITY WITH PASSED BINARY FILES CONTAINING RELATIVE	25
26	5609	/	BLOCK NUMBERS TO DEFINE THE INTERNAL STRUCTURE OF THE PROGRAM [SUCH AS	26
27	5610	/	INTERNAL OVERLAY STRUCTURES].	27
28	5611			28
29	5612	/	4) INITIAL STARTER DIRECTORY CREATION.	29
30	5613			30
31	5614	/	AS THE SYSTEM IS BEING BUILT UP, THE LOWEST AVAILABLE BLOCK FOR THE TFS	31
32	5615	/	DIRECTORY INHERENTLY INCREASES. AS SYSTEM PROGRAMS ARE ADDED, THE TFS	32
33	5616	/	DIRECTORY MUST BE REWRITTEN WITH CONSISTENT INTERNAL PARAMETERS. AS LONG	33
34	5617	/	AS THE SYSTEM BUILDER UTILITY IS USED TO ADD ADDITIONAL PROGRAMS, THE	34
35	5618	/	TFS DIRECTORY EXTENTS CAN BE UPDATED PROPERLY. [NOTE: IF FILES ARE ADDED	35
36	5619	/	TO THE TFS DIRECTORY, THEY MUST BE REMOVED BEFORE RUNNING THE SYSTEM	36
37	5620	/	BUILDER UTILITY. EXTERNAL MEANS MUST BE AVAILABLE TO DELETE ANY EXISTENT	37
38	5621	/	FILES.]	38
39	5622			39
40	5623	/	THE TFS DIRECTORY RESIDES IN BLOCKS 0015-0016 OF THE SYSTEM DEVICE; THE	40
41	5624	/	CURRENT 16K METHOD OF SYSTEM GENERATION WRITES OUT A STARTER COPY OF THE	41
42	5625	/	TFS DIRECTORY CONSISTENT WITH THE SYSTEM DEVICE TARGETED.	42
43	5626			43
44	5627	/	HOWEVER, A SYSTEM BUILDER MUST CHECK FOR VALIDITY OF THE TFS DIRECTORY; IF	44
45	5628	/	THE STRUCTURE IS INVALID, IT CAN BE CREATED ACCORDINGLY. HOWEVER, IF THE	45
46	5629	/	TFS DIRECTORY IS ALREADY VALID, IT MUST BE CONFIRMED AS EMPTY TO ALLOW THE	46
47	5630	/	SYSTEM BUILDER PROGRAM TO PROCEED; ALTERNATIVELY, THE USER CAN BE ASKED TO	47
48	5631	/	HAVE ALL FILES DESTROYED FROM THE BUILDER PROGRAM.	48
49				49
50				50
51				51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	5633	/	NOTE: IT IS BEYOND THE SCOPE OF A SYSTEM BUILDER PROGRAM TO REMOVE EXISTING	1
2	5634	/	SYSTEM PROGRAMS. EITHER AN ALTERNATE PROGRAM CAPABLE OF THIS FUNCTION MUST	2
3	5635	/	BE WRITTEN, OR THE PROGRAM CAN BE PROVIDED AS A P?S/8 SYSTEM UTILITY	3
4	5636	/	PROGRAM [AS OPPOSED TO THE STAND-ALONE DEFINITION AS USED HERE]. ONCE THE	4
5	5637	/	PROGRAM IS PART OF AN EXISTING P?S/8 SYSTEM, ADDITIONAL FUNCTIONS CAN BE	5
6	5638	/	ADDED. [NOTE: P?S/8 SYSTEM PROGRAMS CAN DEFINE OVERLAY STRUCTURES; THE	6
7	5639	/	ABILITY TO REMOVE EXISTING PROGRAMS INCLUDING POTENTIALLY SQUEEZING OUT THE	7
8	5640	/	EMPTY BLOCKS FORMERLY OCCUPIED, COUPLED WITH THE POTENTIAL OF HAVING TO	8
9	5641	/	RECOVER THE SPACE FOR THE TFS DIRECTORY AND PERHAPS MOVING OF TFS FILES IS	9
10	5642	/	A FAR LARGER PROGRAM THAN A STAND-ALONE BUILDER PROGRAM. HOWEVER, THE	10
11	5643	/	ENVIRONMENT OF A P?S/8 SYSTEM PROGRAM INCLUDES ACCESS TO THE SYSTEM DEVICE	11
12	5644	/	HANDLER AS WELL AS PASSED TEXT AND BINARY FILES. AS SUCH, THERE ARE	12
13	5645	/	ACTUALLY TWO VARIANT FORMS OF SYSTEM BUILDER PROGRAMS.]	13
14	5646			14
15	5647	/	[NOTE: P?S/8 SYSTEMS ALSO INCLUDE PARTIAL SUPPORT FOR AN EXTENDED DIRECTORY	15
16	5648	/	STRUCTURE BEYOND THE TFS STRUCTURE. AS LONG AS THE OVERALL LENGTH OF THE	16
17	5649	/	DEVICE IS OBTAINABLE, THIS STRUCTURE CAN BE CREATED TO USE ALL AVAILABLE	17
18	5650	/	SPACE AFTER THE TFS DIRECTORY [IF ANY]. THIS FUNCTION WOULD ALSO BE PART	18
19	5651	/	OF A UTILITY CREATED AS A P?S/8 SYSTEM UTILITY.]	19
20	5652			20
21	5653	/	USING SPECIAL-PURPOSE ORIGIN SETTINGS.	21
22	5654			22
23	5655	/	CERTAIN READERS OF THIS DOCUMENT MAY BE FAMILIAR WITH THE OS/8 BUILD.SV	23
24	5656	/	PROGRAM, WHICH IS PARTIALLY ANALOGOUS TO THE P?S/8 SYSTEM BUILDER PROCESS.	24
25	5657			25
26	5658	/	A USEFUL TECHNIQUE FOR PASSING INFORMATION WITHIN A BINARY FILE INVOLVES THE	26
27	5659	/	USE OF DELIBERATE ORIGIN SETTINGS [OR GROUPS OF ORIGIN SETTINGS] FOR THE	27
28	5660	/	PURPOSE OF PASSING ADDITIONAL INFORMATION ABOUT THE FILE APART FROM THE BINARY	28
29	5661	/	FILE CONTENTS BEING LOADED.	29
30	5662			30
31	5663	/	THIS TECHNIQUES IS ACTUALLY AN OUTGROWTH OF AN EARLIER METHOD FIRST CREATED	31
32	5664	/	FOR THE TYPESET-8 GROUP BY THE IMPLEMENTORS OF THE PAL10 CROSS-ASSEMBLER FOR	32
33	5665	/	PDP-8 PROGRAMMING THAT RUNS UNDER THE TOPS-10 OPERATING SYSTEM ON THE PDP-10.	33
34	5666			34
35	5667	/	TYPESET-8 USES LOGICAL BLOCKS IDENTICALLY TO P?S/8 [AND TYPESET-8 ONLY RUNS ON	35
36	5668	/	THE TC01/TC08 DECTAPE]. EACH BLOCK IS 128 12-BIT WORDS. A MEANS WAS ADDED TO	36
37	5669	/	PAL10 TO ALLOW DEFINING THE ABSOLUTE BLOCK NUMBERS FOR DESIGNATED COMPONENTS	37
38	5670	/	OF TYPESET-8. [NOTE: TYPICAL TYPESET-8 DEVELOPMENT WAS TO ASSEMBLE SOURCE	38
39	5671	/	CODE USING PAL10, THEN PUNCHING OUT LENGTHY BINARY PAPER-TAPES TO BE WALKED TO	39
40	5672	/	REMOTE PDP-8 SYSTEMS WITH ONLY 4K OF MEMORY; THE ENTIRE TYPESET-8 OPERATION	40
41	5673	/	WAS DONE WITH NO KNOWLEDGE OF P?S/8. CLEARLY, P?S/8 COULD HAVE BEEN USEFUL TO	41
42	5674	/	THIS PROJECT ON SOME BASIS.]	42
43	5675			43
44	5676	/	THE DESIGNATED BLOCKS WHERE TO WRITE OUT PDP-8 PAGES OF LOADABLE BINARY DATA	44
45	5677	/	IS DEFINED USING THE "DTORG" DIRECTIVE AVAILABLE IN PAL10 [AND LATER IN PAL8].	45
46	5678			46
47	5679	/	HOWEVER, THE IMPLEMENTATION IS CONCEPTUALLY FLAWED: BINARY FILES CREATED BY	47
48	5680	/	PAL10 OR PAL8 CAN BE LOADED BY ALL CONVENTIONAL MEANS AS LONG AS THE DTORG	48
49	5681	/	DIRECTIVE IS NOT USED. DTORG STATEMENTS CREATE DELIBERATE CHECKSUM ERRORS	49
50	5682	/	BECAUSE THE TYPESET-8 IMPLEMENTORS WERE TOO SHORT-SIGHTED TO REALIZE THIS IS	50
51	5683	/	COMPLETELY UNNECESSARY. [NOTE: LENGTHY BINARY PAPER-TAPES WERE CREATED THAT	51
52	5684	/	WERE INCOMPATIBLE WITH THE BIN LOADER; ANY ATTEMPT TO LOAD THE TAPES WILL END	52
53	5685	/	WITH A CHECKSUM ERROR; INSTEAD, A SPECIAL TYPESET-8 PROGRAM WAS USED THAT HAD	53
54	5686	/	NO CHECKSUMMING CAPABILITY; THIS IS INHERENTLY UNRELIABLE.]	54
55				55
56				56
57				57
58				58

1	5688	/	BIN PAPER-TAPE FORMAT STRICTLY DEFINES ALL POSSIBILITIES. MANY PAPER-TAPE	1
2	5689	/	FRAMES CONTAIN SIX BITS OF A 12-BIT ORIGIN OR DATA WORD. THE HIGH-ORDER BITS	2
3	5690	/	OF THESE WORDS ARE USED TO DISTINGUISH BETWEEN THE TWO VARIANTS, EACH	3
4	5691	/	EXPRESSES AS A PAIR OF ADJACENT FRAMES, INCLUDING THE BIT THAT DISTINGUISHES	4
5	5692	/	BETWEEN LOADING ORIGIN AND DATA WORD; ALL BITS OF ALL DATA FRAMES EXCEPT FOR	5
6	5693	/	FIELD SETTINGS CONTRIBUTE TO THE OVERALL CHECMSUM.	6
7	5694			7
8	5695	/	SINCE THERE WERE NO MORE AVAILABLE COMBINATIONS [IF YOU SIMPLISTICALLY MAP	8
9	5696	/	OUT THE DATA STRUCTURES OF BIN FORMAT], THEY DECIDED TO CREATE AN INCOMPATIBLE	9
10	5697	/	EXTENSION WHERE THE BIT ASSOCIATED WITH THE ORIGIN SETTING DATA IS ALSO	10
11	5698	/	PRESENT IN THE SECOND FRAME OF THE PAIR DEFINING A 12-BIT VALUE. WHEN BOTH	11
12	5699	/	BITS ARE SET, THIS IS A DTORG DATA PAIR. [THE DATA IS USED WITHIN THE	12
13	5700	/	TYPESET-8 GENERATION PROGRAM TO SPECIFY THE ABSOLUTE DECTAPE BLOCK TO WRITE	13
14	5701	/	OUT THE [UP TO] 128 WORDS OF DATA THAT FOLLOW.] AS SUCH, BINARY PAPER-TAPES	14
15	5702	/	CONTAINING DTORG FRAMES ARE INCOMPATIBLE WITH THE BIN LOADER.	15
16	5703			16
17	5704	/	WHEN THIS FEATURE WAS ADDED TO PAL8, THE SAME EXACT FLAWED METHOD WAS USED;	17
18	5705	/	THE RESULTANT BINARY FILES OR PAPER-TAPES ARE INCOMPATIBLE WITH OS/8 ABSLDR.SV	18
19	5706	/	AND AS SUCH ARE USELESS FOR ALL OTHER PURPOSES.	19
20	5707			20
21	5708	/	BY THINKING THE PROBLEM THROUGH, THIS DILEMMA CAN EASILY BE AVOIDED. P?S/8	21
22	5709	/	PAL CURRENTLY IMPLEMENTS DTORG AS A NON-FUNCTIONAL DIRECTIVE TO MAINTAIN	22
23	5710	/	SOURCE CODE COMPATIBILITY; ANY TYPESET-8 SOURCE FILE CAN BE ASSEMBLED	23
24	5711	/	CORRECTLY WITH P?S/8 PAL [WHILE THE CREATED BINARY ISN'T USEFUL TO TYPESET-8	24
25	5712	/	AT THE PRESENT TIME].	25
26	5713			26
27	5714	/	BY REDEFINING THE BINARY OUTPT OF A DTORG STATEMENT AS A SERIES OF	27
28	5715	/	SPECIAL-PURPOSE ORIGIN SETTINGS, TOTAL COMPATIBILITY CAN BE ACHIEVED. [NOTE:	28
29	5716	/	TYPESET-8 COMPATIBILITY CAN BE ACHIEVED WITH A CONVERSION UTILITY; WHEN THE	29
30	5717	/	SEQUENCE OF SPECIAL-PURPOSE ORIGIN SETTINGS IS DISCOVERED WITHIN THE BINARY	30
31	5718	/	FILE, SUBSTITUTE THE DTORG FRAME PAIR AS TYPESET-8 REQUIRES ON THE BINARY	31
32	5719	/	PAPER-TAPE BEING CREATED FOR THE BENEFIT OF TYPESET-8 COMPATIBILITY. IN OS/8	32
33	5720	/	THIS COULD HAVE BEEN EASILY ACCOMPLISHED WITH A SWITCH ADDED FOR THE EXPRESS	33
34	5721	/	PURPOSE OF TYPESET-8 BINARY PAPER-TAPE PREPARATION; THE ENTIRE INCOMPATIBILITY	34
35	5722	/	PROGRAM WAS COMPLETELY AVOIDABLE.]	35
36	5723			36
37	5724	/	OS/8 BUILD DOES NOT USE DTORG STATEMENTS; HOWEVER, IT DOES MAKE GOOD USE OF	37
38	5725	/	SPECIFIC SPECIAL-PURPOSE ORIGIN SETTINGS TO PASS PARAMETERS ABOUT DEVICE	38
39	5726	/	HANDLERS WITHIN THE BINARY FILE OF THE HANDLER.	39
40	5727			40
41	5728	/	A P?S/8 SYSTEM BUILDER PROGRAM CAN ALSO USE THIS TECHNIQUE; HOWEVER, SINCE THE	41
42	5729	/	DTORG DIRECTIVE IS ALREADY DEFINED, FUTURE VERSIONS OF P?S/8 PAL WILL BE	42
43	5730	/	ENHANCED TO OUTPUT A DESIGNATED SPECIAL-PURPOSE ORIGIN SETTING SEQUENCE FOR	43
44	5731	/	EVERY USAGE OF THE DTORG DIRECTIVE ESSENTIALLY AS IT WAS INTENDED, EXCEPT FOR	44
45	5732	/	THE PURPOSE OF P?S/8 SYSTEM GENERATION [AND NOT FOR TYPESET-8 PURPOSES].	45
46	5733			46
47	5734	/	NOTE: PAL8 MAY CONTINUE TO PLAY A ROLE IN THE PROCESS OF P?S/8 SYSTEM	47
48	5735	/	GENERATION. AS SUCH, A BINARY CONVERSION UTILITY MAY BE WRITTEN [AS A PRELUDE	48
49	5736	/	TO "PIPOS8" A TENTATIVE PROGRAM PLANNED FOR THE P?S/8 SHELL ENVIRONMENT] TO	49
50	5737	/	ALLOW PAL8 OUTPUT TO BE CONVERTED TO TFS BINARY FILES. THIS WOULD INCLUDE	50
51	5738	/	CONVERSION OF DTORG STATEMENTS RECAST AS SPECIAL-PURPOSE ORIGIN SETTINGS.	51
52	5739	/	SUCH A PROGRAM CAN CALCULATE CHECKSUM OFFSETS CAUSED BY DTORG STATEMENTS TO	52
53	5740	/	PRODUCE A PROPER CHECKSUM DESPITE THE FACT THAT NO OS/8 UTILITY EXISTS TO	53
54	5741	/	VALIDATE THESE FILES.	54
55				55
56				56
57				57
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1	5743	/	NOTE: SINCE A COMPLETE P?S/8 SYSTEM BUILDER PROGRAM REQUIRES BOTH ABSOLUTE AND	1
2	5744	/	RELATIVE BLOCK NUMBERS FOR THE VARIOUS SYSTEM COMPONENTS, SOME MEANS MUST BE	2
3	5745	/	PROVIDED TO SWITCH OPERATING MODES WITHIN THE BUILDER PROGRAM. A SUGGESTED	3
4	5746	/	METHOD TO IMPLEMENT THIS WOULD BE THE USE OF SPECIFIC SYNTAX SUCH AS DTORG	4
5	5747	/	7777 AND DTORG 7776. BY NATURE, NO VALID SYSTEM COMPONENT WOULD NORMALLY	5
6	5748	/	SPECIFY SUCH A LARGE VALUE; THUS, THESE VALUES CAN BE USED TO SWITCH SYSTEM	6
7	5749	/	BUILDER DTORG BLOCK INTERPRETATION MODES. ADDITIONAL SIMILAR VALUES CAN BE	7
8	5750	/	DEFINED AS NECESSARY.	8
9	5751			9
10	5752	/	[NOTE: FOR FURTHER INFORMATION ON THE GENERAL TOPIC OF SPECIAL-PURPOSE ORIGIN	10
11	5753	/	GROUPS, THE SHORTCOMINGS OF TYPESET-8 USING THE DTORG DIRECTIVE AND ITS	11
12	5754	/	NEGATIVE IMPACT ON OS/8 PERFORMANCE, REFER TO A RELATED DOCUMENT ADDRESSING	12
13	5755	/	THESE SPECIFIC ISSUES. EXAMPLES ARE GIVEN TO SOLVE ANALOGOUS PROBLEMS SUCH AS	13
14	5756	/	AN ALTERNATIVE METHOD FOR THE EXPRESS BENEFIT OF TYPESET-8 SUPPORT AND A BRIEF	14
15	5757	/	DISCUSSION OF OS/8 BUILD WHICH DEALS WITH VAGUELY ANALOGOUS ISSUES. THE	15
16	5758	/	DOCUMENT IS WRITTEN FROM A UNIQUE VANTAGE POINT: NEEDLESS OVERHEAD WAS ADDED	16
17	5759	/	TO OS/8 MERELY TO PLACATE THE TYPESET-8 GROUP WITH NO COMPROMISE ALLOWED.	17
18	5760	/	THEY INSISTED ON NOT CHANGING ANYTHING WHATSOEVER DESPITE THE FACT THAT SOME	18
19	5761	/	OF THE IDEAS EXPRESSED WOULD HAVE AIDED THE DEVELOPMENT OF TYPESET-8 AS WELL	19
20	5762	/	AS ELIMINATING SOME OVERHEAD WITHIN OS/8. ADDITIONAL TOPICS INCLUDE AN	20
21	5763	/	UNRELATED ACCOMMODATION TO THE TYPESET-8 GROUP WITH REGARD TO SERIOUS OVERHEAD	21
22	5764	/	WITHIN THE OS/8 PAPER-TAPE PUNCH HANDLER THAT AFFECTS ALL USERS AT ALL TIMES;	22
23	5765	/	THE PURPOSE OF THE MODIFICATIONS WILL SURPRISE MANY READERS AS THE "PROBLEM"	23
24	5766	/	SOLVED FOR THE TYPESET-8 GROUP IS PARTICULARLY FANTASTIC AS WELL AS SOMEWHAT	24
25	5767	/	COMICAL.]	25
26	5768			26
27	5769	/	ADDITIONAL SYSTEM GENERATION CONSIDERATIONS.	27
28	5770			28
29	5771	/	A COMPLETE SYSTEM GENERATION METHOD MUST INCLUDE MEANS OF CREATING OR UPDATING	29
30	5772	/	THE "SLURP" LOADER AND THE RELATED SYSTEM DEVICE HANDLER RELOADER CODE "FILE".	30
31	5773	/	ADDITIONAL SECTIONS MUST INCLUDE DEVICE-SPECIFIC REWIND/UNLOAD ROUTINES	31
32	5774	/	FOR THE KEYBOARD MONITOR AND SYSTEM DEVICE HANDLER INTERACTION MODULES THAT	32
33	5775	/	ARE REQUIRED TO INTERFACE WITH THE LOGICAL CONSOLE OVERLAY.	33
34	5776			34
35	5777	/	SYSTEM DEVICE HANDLERS MAY INCLUDE EXTENDED MEMORY SECTIONS AND RELATED	35
36	5778	/	"HELPER" BLOCK IMAGES IF APPLICABLE.	36
37	5779			37
38	5780	/	A COMPREHENSIVE UTILITY SHOULD BE CAPABLE OF CREATING A SYSTEM FROM SCRATCH	38
39	5781	/	[WITH INITIALLY ONLY THE UTILITY IN MEMORY ON A 4K MACHINE] OR USABLE FROM A	39
40	5782	/	RUNNING SYSTEM TO UPDATE AN IMAGE OF A SYSTEM [POSSIBLY ORIENTED TOWARDS	40
41	5783	/	ALTERNATE HARDWARE]. [NOTE: APPROPRIATE NON-SYSTEM HANDLERS MAY BE REQUIRED	41
42	5784	/	BOTH WITHIN THE GENERATION PROGRAM AND ON THE P?S/8 SYSTEM THE UTILITY IS	42
43	5785	/	BEING RUN FROM TO HANDLE ALL CASES.]	43
44	5786			44
45	5787	/	END OF APPENDIX H.	45
46				46
47				47
48				48
49				49
50				50
51				51
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54				54
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1	5789	/	APPENDIX I.	1
2	5790			2
3	5791	/	THE P?S/8 SYSTEM DATE WORD HAS ALWAYS BEEN DEFINED AS "SDATE" [MEMORY LOCATION	3
4	5792	/	07610]; HOWEVER, THE EXACT IMPLEMENTATION HAS CHANGED SEVERAL TIMES.	4
5	5793			5
6	5794	/	WHEN THE CONCEPT OF A SYSTEM DATE WORD WAS FIRST DISCUSSED, THE ONLY EXAMPLE	6
7	5795	/	OF ANOTHER PDP-8 SYSTEM WITH A DATE WORD WAS PS/8 [LATER RENAMED TO OS/8].	7
8	5796	/	THAT SYSTEM DEFINED THE SYSTEM DATE AS ANY VALID DATE IN THE RANGE OF 1970	8
9	5797	/	THROUGH 1977. SINCE THE PDP-8 HAD ALREADY PROVEN TO BE A POPULAR SYSTEM	9
10	5798	/	STARTING IN 1965, IT WAS QUITE OBVIOUS THE DATE HAD TO BE DEFINED OVER A	10
11	5799	/	LONGER TIME FRAME. WHILE 1970 IS AN OBVIOUS STARTING DATE [ALSO USED BY THE	11
12	5800	/	MAJORITY OF UNIX SYSTEMS], IT IS ACTUALLY NOT A GOOD CHOICE FROM THE VANTAGE	12
13	5801	/	POINT OF 12-BIT COMPUTING.	13
14	5802			14
15	5803	/	1960 IS GENERALLY CONSIDERED THE DAWN OF 12-BIT COMPUTING WITH THE RELEASE OF	15
16	5804	/	THE CDC 160 AS DESIGNED BY SEYMOUR CRAY. THE DESIGN OF THE CDC 160 IS IN	16
17	5805	/	CERTAIN WAYS A CLEAR PRECURSOR TO THE PDP-8 ARCHITECTURE.	17
18	5806			18
19	5807	/	BY 1962, THE MIT LINC COMPUTER [LATER INCORPORATED INTO THE LINC-8 AND PDP-12	19
20	5808	/	DESIGN] ADDED SOME ADDITIONAL REFINEMENTS THAT INFLUENCED THE PDP-8; BY 1963	20
21	5809	/	THE FIRST [COMMERCIALY UNSUCCESSFUL] DEC 12-BIT MACHINE, THE PDP-5, WAS	21
22	5810	/	RELEASED.	22
23	5811			23
24	5812	/	PDP-6/PDP-10 SOFTWARE USES A DATE RANGE STARTING IN 1964. BY 1975 THE	24
25	5813	/	END-POINT OF THE ORIGINAL SUPPORTED RANGE WAS REACHED; FORTUNATELY, AN	25
26	5814	/	EXTENSION WAS CREATED [BY SCRAMBLING FOR UNUSED BITS WITHIN THE DIRECTORY	26
27	5815	/	STRUCTURE] SOME TIME BEFORE THE ACTUAL DATE RANGE EXPIRATION.	27
28	5816			28
29	5817	/	IT WAS CLEAR THAT OS/8'S SHORT DATE RANGE OF EIGHT YEARS WAS ENTIRELY	29
30	5818	/	INADEQUATE; THE EARLIEST IMPLEMENTATION OF THE P?S/8 DATE STROVE FOR BETTER	30
31	5819	/	UTILIZATION OF A 12-BIT VALUE.	31
32	5820			32
33	5821	/	OS/8'S INHERENT LIMITATION IS THE USE OF ONLY THREE BITS TO STORE THE YEAR	33
34	5822	/	OFFSET [FROM 1970]. THE REST OF THE OS/8 DATE WORD IS USED TO STORE THE	34
35	5823	/	MONTH OFFSET FROM THE BEGINNING OF THE YEAR AND THE DAY OFFSET FROM THE	35
36	5824	/	BEGINNING OF THE MONTH. TO CONVERT THE STORED VALUE INTO PRINTABLE FORM IS A	36
37	5825	/	FAIRLY STRAIGHTFORWARD PROCESS.	37
38	5826			38
39	5827	/	NOTE: OS/8 EXHIBITS A DESIGN BLUNDER: THE BIT-WISE ORDERING OF THESE ARBITRARY	39
40	5828	/	BIT GROUPINGS WAS NOT CHOSEN WELL. BY PROPER USE OF BIT-WEIGHTING, ANY DATE	40
41	5829	/	VALUE CAN BE DIRECTLY COMPARED TO ANOTHER DATE VALUE BY SIMPLE ARITHMETIC	41
42	5830	/	COMPARISON; THIS CAN BE USEFUL IN UTILITIES THAT SPECIFY A DESIRED DATE RANGE	42
43	5831	/	FOR SELECTION/FILTERING. HOWEVER, SINCE OS/8'S CHOICE OF BITS DEFIES	43
44	5832	/	ARITHMETIC ORDER, ANY SUCH CALCULATION MUST FIRST CONVERT THE VALUES INTO A	44
45	5833	/	VALUE WHERE THE RESULT IS WEIGHTED CORRECTLY; THIS IS NEEDLESS OVERHEAD THAT	45
46	5834	/	COULD HAVE BEEN AVOIDED [AND IS NOT PRESENT IN THE LATEST P?S/8 DATE	46
47	5835	/	IMPLEMENTATION WHICH IS OTHERWISE SOMEWHAT ANALOGOUS TO THE OS/8 METHOD].	47
48	5836			48
49	5837	/	THE DESIGNERS OF P?S/8 CHOSE AN ALTERNATIVE THAT IS SLIGHTLY MORE COMPLICATED	49
50	5838	/	TO PROCESS WITH THE ADVANTAGE OF HAVING A DATE RANGE OF JUST UNDER 12 YEARS.	50
51	5839	/	THE BASE YEAR CHOSEN AT THE TIME WAS 1976, TO COINCIDE WITH THE FIRST	51
52	5840	/	IMPLEMENTATION OF THE DATE FUNCTION WITHIN P?S/8.	52
53				53
54				54
55				55
56				56
57				57
58				58

1	5842	/	THE METHODOLOGY USED IS AS FOLLOWS:	1
2	5843			2
3	5844	/	ASSUMING ALL 12 MONTHS HAVE 31 DAYS, THERE WILL BE 372 DAYS IN A YEAR. THE	3
4	5845	/	VALUE FOR THE DATE PARTICULARS IS CALCULATED THUS:	4
5	5846			5
6	5847	/	YEAR OFFSET = (SDATE CONTENTS)/372.	6
7	5848			7
8	5849	/	THE RELATIVE DAY OF THE YEAR IS THE REMAINDER FROM THE ABOVE DIVISION.	8
9	5850			9
10	5851	/	MONTH OFFSET = (REMAINDER)/31	10
11	5852			11
12	5853	/	THE DAY OFFSET IS IN TURN THE REMAINDER OF THIS ADDITIONAL DIVISION.	12
13	5854			13
14	5855	/	THE YEAR OFFSET IS CALCULATED ORIGIN 0 TO MAXIMIZE THE DURATION OF THE DATE	14
15	5856	/	RANGE [APPROXIMATELY 11-3/4 YEARS]. THE BASE DATE YEAR OF 1976 IS ADDED ON TO	15
16	5857	/	CREATE THE ABSOLUTE YEAR. [NOTE: MODERN ERA YEARS ARE EASILY CALCULATED AS AN	16
17	5858	/	UNSIGNED 12-BIT VALUE.] THE MONTH OFFSET IS ALSO CALCULATED ORIGIN 0 TO BE	17
18	5859	/	USED AS AN OFFSET INTO A LOOKUP TABLE OF APPROPRIATE MONTH STRINGS. THE DAY	18
19	5860	/	OFFSET IS DELIBERATELY CALCULATED ORIGIN 1 TO ALLOW THE 12-BIT VALUE OF 0000	19
20	5861	/	TO INDICATE THE SYSTEM DATE HAS NOT BEEN SET.	20
21	5862			21
22	5863	/	THIS INITIAL RELEASE OF THE DATE FACILITY ALLOWED P?S/8 TO MAINTAIN A PROPER	22
23	5864	/	DATE THROUGH THE END OF 1986. [NOTE: THE PARTIAL 12TH YEAR WAS SUPPRESSED	23
24	5865	/	WITHIN THE P?S/8 "DATE" PROGRAM TO AVOID CONFUSION.]	24
25	5866			25
26	5867	/	SHORTLY AFTER THE INITIAL RELEASE OF OS/8 [WHEN THE NAME WAS CHANGED FROM	26
27	5868	/	PS/8 AT THE REQUEST OF A DEC SALESMAN], VARIOUS OS/8 USERS EXPRESSED DEEP	27
28	5869	/	CONCERN ABOUT OS/8'S DATE EXPIRATION. [MANY WERE WELL AWARE OF THE EFFORTS ON	28
29	5870	/	THE PDP-10 SYSTEMS THAT PREVENTED THE DATE RANGE EXPIRATION IN 1975.] DEC	29
30	5871	/	RELEASED AN "IMPROVED" OS/8 CLAIMING TO SUPPORT A DATE RANGE EXTENSION THROUGH	30
31	5872	/	THE YEAR 2001; HOWEVER, THE ACTUAL IMPLEMENTATION IS QUITE DEFICIENT:	31
32	5873			32
33	5874	/	1) OS/8 RECKONS FILE DATES IN ONLY ONE OF TWO POSSIBLE WAYS: A) THE DATE IS IN	33
34	5875	/	THE RANGE OF 1970-1977 EXPRESSED AS THE OFFSET PAST THE START OF 1970; THIS	34
35	5876	/	WAS EFFECTIVELY NOT CHANGED, B) THE DATE IS NOT STORED IN THE DIRECTORY.	35
36	5877			36
37	5878	/	A SURPRISING NUMBER OF OS/8 USERS DELIBERATELY CONFIGURE DIRECTORIES	37
38	5879	/	WITHOUT DATE SUPPORT. THE UNDERLYING REASON IS THAT OS/8 DIRECTORIES ARE	38
39	5880	/	QUITE INADEQUATE FOR MOST USERS; DUE TO THE OVERHEAD OF THE HALF-BAKED	39
40	5881	/	IMPLEMENTATION OF DIRECTORY SUPPORT, OS/8'S INTERNAL STRUCTURE WAS NEVER	40
41	5882	/	FINISHED. A LINKED-LIST STRUCTURE IS USED TO ACCESS A LINEAR LIST OF	41
42	5883	/	RECORDS THAT, AS A PRACTICAL MATTER, CANNOT BE EITHER EXPANDED, CONTRACTED	42
43	5884	/	OR REORDERED. IN HINDSIGHT, THIS ONLY MAKES THE ALREADY OVERLOADED	43
44	5885	/	DIRECTORY STRUCTURE MORE PRONE TO OVERFLOWING THE DIRECTORY CAPABILITY WELL	44
45	5886	/	BEFORE THE UNDERLYING STORAGE RUNS OUT. BY ELIMINATING DATE WORD STORAGE,	45
46	5887	/	AN APPRECIABLE QUANTITY OF ADDITIONAL FILES CAN BE STORED ON THE DEVICE.	46
47	5888			47
48	5889	/	THE IMPLEMENTATION OF FILE DATES IN AN OS/8 DIRECTORY IS CONTROLLED BY AN	48
49	5890	/	INTERNAL FACTOR KNOWN AS THE COUNT OF "ADDITIONAL INFORMATION WORDS". THIS	49
50	5891	/	WAS ONLY DEFINED FOR THE FIRST WORD AS THE FILE DATE VALUE FOR EACH FILE	50
51	5892	/	ENTRY. TO INCREASE THE DATE RANGE WOULD REQUIRE AN ADDITIONAL INFORMATION	51
52	5893	/	WORD BEYOND THE FIRST ONE; HOWEVER, THIS WOULD ONLY FURTHER ALIENATE USERS	52
53	5894	/	ALREADY REMOVING THE BASIC DATE INFORMATION TO INCREASE THE FILE CAPACITY	53
54	5895	/	OF THE DIRECTORY.	54
55				55
56				56
57				57
58				58

1	5897	/	2) OS/8 IMPLEMENTATION OF THE SYSTEM DATE IN MEMORY WAS EXPANDED BY TWO BITS.	1
2	5898	/	THIS THEORETICALLY MEANS THAT THE CURRENT YEAR OFFSET IS EXPANDED BY	2
3	5899	/	ADDING ON 0-3 GROUPS OF EIGHT YEARS EXTENDING THE DATE RANGE TO 1970	3
4	5900	/	THROUGH 2001. [NOTE: THE YEAR 2001 IS ALREADY WELL IN THE PAST AS OF THE	4
5	5901	/	WRITING OF THIS DOCUMENT; IT IS HIGHLY LIKELY THE RANGE CANNOT BE EXTENDED	5
6	5902	/	ANY FURTHER.] SINCE THE ACTUAL DATE IMPLEMENTATION DOES NOT STORE THE TWO	6
7	5903	/	BITS FROM SYSTEM MEMORY, A FICTION WAS CREATED TO DESCRIBE THE FILE DATE:	7
8	5904			8
9	5905	/	ALL FILE DATES ARE PRESUMED TO HAVE BEEN CREATED EITHER THIS YEAR OR THE	9
10	5906	/	PREVIOUS SEVEN YEARS. AS TIME PROGRESSES, THE ANOMALY OF THIS INCREASES.	10
11	5907	/	THIS IS ESPECIALLY TROUBLESOME FOR FILES ACTUALLY CREATED EIGHT YEARS	11
12	5908	/	EARLIER; THEY APPEAR TO BE IN THE FUTURE DAYS OF THE CURRENT YEAR.	12
13	5909			13
14	5910	/	DURING THE EARLY YEARS OF THIS SHAM MODIFICATION, FEW USERS ACTUALLY HAD FILES	14
15	5911	/	CREATED EIGHT YEARS EARLIER, AS WIDESPREAD ADOPTION OF OS/8 DIDN'T OCCUR UNTIL	15
16	5912	/	ROUGHLY 1973-1974; AS SUCH, OS/8'S TRULY DEFICIENT IMPLEMENTATION DIDN'T	16
17	5913	/	IMMEDIATELY AFFECT MANY USERS. MANY WERE FOOLED BECAUSE IT WOULD NOT BE UNTIL	17
18	5914	/	1980 BEFORE THE TRUE NATURE OF WHAT WAS IMPLEMENTED WOULD BE MEANINGFUL TO	18
19	5915	/	THEM PERSONALLY. THE DEVELOPERS OF P?S/8 REALIZED THE FOLLY OF THIS SCHEME	19
20	5916	/	IMMEDIATELY; ACTUAL SOURCE FILES WERE DEMONSTRATED THAT WERE CREATED IN 1970	20
21	5917	/	REVEALING THE ANOMALY OF THE DIRECTORY LISTINGS CLAIMING THE FILES WERE	21
22	5918	/	CREATED EIGHT YEARS LATER. [NOTE: SOME OF THE ACTUAL FILES WERE CREATED PRIOR	22
23	5919	/	TO 1970; THE CONTRIVED DATE OF 01-JAN-1970 WAS USED AS THE CLOSEST	23
24	5920	/	APPROXIMATION OS/8 COULD ALLOW TO THE PROPER FILE DATES.]	24
25	5921			25
26	5922	/	WHEN OS/8 WAS USED DURING THE WANING YEARS OF THE FICTIONAL IMPLEMENTATION,	26
27	5923	/	MANY USERS RESORTED TO MANUAL EDITING OF DIRECTORY LISTINGS TO CREATE AN	27
28	5924	/	ARTIFICIAL REALITY. [THE LISTINGS WERE ACCURATE, BUT NO OS/8 UTILITY COULD	28
29	5925	/	HAVE AUTOMATICALLY CREATED THE DIRECTORY PRINTOUT FILE.] CERTAIN USERS SET	29
30	5926	/	THE YEAR TO 1986 BECAUSE THE DAY OF THE WEEK CORRESPONDS TO THE SAME VALUES IN	30
31	5927	/	2014. [NOTE: CERTAIN OS/8 PROGRAMS SUFFER FROM THE INFAMOUS "Y2K" PROBLEM.	31
32	5928	/	JUNK PRINTOUT OCCURS IF THE DATE IS SET PAST 1999.]	32
33	5929			33
34	5930	/	AT THIS POINT, IT BECAME CLEAR SOMETHING HAD TO BE DONE TO EXTEND THE P?S/8	34
35	5931	/	DATE RANGE SIGNIFICANTLY. [THERE WAS A CLEAR NEED TO IMPLEMENT SOMETHING THAT	35
36	5932	/	ACTUALLY COVERED THE CURRENT DATE WITHOUT ANOMALY; A QUICK FIX WAS SOUGHT TO	36
37	5933	/	EXTEND THE DATE RANGE DRAMATICALLY.	37
38	5934			38
39	5935	/	REVISING THE THEN-CURRENT METHOD, AN EXTENSION WAS CREATED ON PAPER; THIS	39
40	5936	/	WOULD GENERALLY FOLLOW OS/8'S LEAD IN TERMS OF ADDING EXTENSION BITS TO	40
41	5937	/	EXPRESS A FAR GREATER RANGE THAN 11 YEARS. A DESIGN WAS ATTEMPTED TO EXTEND	41
42	5938	/	THE RANGE TO A TOTAL OF 512 YEARS. BECAUSE OF THE SIGNIFICANCE OF THE YEAR	42
43	5939	/	1960 TO 12-BIT COMPUTER HISTORY, THIS WAS CHOSEN AS THE NEW BASE YEAR.	43
44	5940			44
45	5941	/	FOR EACH INCREMENT BY ONE OF THE BASE GROUP FIELD OF SIX BITS, THE BASE YEAR	45
46	5942	/	INCREASES BY EIGHT YEARS. AS SUCH, 1976 STILL PLAYED A ROLE AS A BASE YEAR,	46
47	5943	/	BUT THE ABSOLUTE ORIGIN BECAME 1960; THE SIX-BIT DATE GROUP WOULD BE SET TO	47
48	5944	/	000010 TO ALLOW VALID DATES IN THE RANGE OF 1976-1983.	48
49				49
50				50
51				51
52				52
53				53
54				54
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57				57
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1	5946	/	UNFORTUNATELY, AT THE TIME OF THIS DESIGN MODIFICATION, THERE WERE NO	1
2	5947	/	ADDITIONAL RESERVED BITS AVAILABLE WITHIN THE P?S/8 SYSTEM DEVICE HANDLER; A	2
3	5948	/	MODIFIED SCHEME WAS IMPLEMENTED:	3
4	5949			4
5	5950	/	1) ALL PROGRAMS CONCERNED WITH THE PRESENT DATE RESERVED A PATCHABLE WORD	5
6	5951	/	WITHIN THE PROGRAM IMAGE TO ALLOW CUSTOM PATCHING OF THE CURRENT DATE GROUP	6
7	5952	/	VALUE. WHILE THERE WAS NO SPECIFIC DESIGNATED LOCATION, THE LOCATION	7
8	5953	/	WITHIN EACH IMAGE WAS CHOSEN TO BE THE INITIAL CONTENTS OF A PAGE ZERO	8
9	5954	/	LOCATION IF POSSIBLE; IN CERTAIN INSTANCES, EQUIVALENT LOCATIONS WERE	9
10	5955	/	DEFINED WHEN THE ONLY PURPOSE OF THE DATE CALCULATION WAS IN CONJUNCTION	10
11	5956	/	WITH OTHER ONCE-ONLY INITIALIZATION CODE. A STATIC DATE STRING OF	11
12	5957	/	CHARACTERS TO PRINT OUT THE CURRENT DATE ELSEWHERE IN THE PROGRAM WAS THE	12
13	5958	/	ONLY DESIRED RESULT; AFTER THE CORRESPONDING STRING WAS STORED ELSEWHERE,	13
14	5959	/	ALL OF THE ONCE-ONLY INITIALIZATION CODE WAS NO LONGER NEEDED; THE CODE	14
15	5960	/	SPACE WAS OFTEN RECLAIMED AS AN INPUT BUFFER FOR THE PURPOSE OF SUBSEQUENT	15
16	5961	/	FILE READING OR WRITING.	16
17	5962			17
18	5963	/	A VIABLE P?S/8 SYSTEM WAS ACHIEVED BY PROPERLY PATCHING ALL AFFECTED	18
19	5964	/	LOCATIONS WITHIN EACH IMAGE; THIS VALUE WOULD HAVE TO BE UPDATED NOMINALLY	19
20	5965	/	EVERY EIGHT YEARS.	20
21	5966			21
22	5967	/	2) RECOGNIZING THAT THE SYSTEM DATE WORD HAS AN INTRINSIC RANGE OF NEARLY	22
23	5968	/	TWELVE YEARS, IT BECAME CLEAR THAT NEARLY EVERY THEORETICALLY POSSIBLE	23
24	5969	/	CAN BE EXPRESSED IN TWO DIFFERENT WAYS: A) AN OFFSET OF LESS THAN EIGHT	24
25	5970	/	YEARS FROM ONE DATE BASE GROUP VALUE, B) AN OFFSET OF GREATER THAN EIGHT	25
26	5971	/	YEARS FROM THE PREVIOUS DATE BASE GROUP VALUE.	26
27	5972			27
28	5973	/	AS SUCH, THERE IS EFFECTIVELY A "GRACE PERIOD" ON PERFORMING THIS DATE	28
29	5974	/	GROUP MAINTENANCE OF NEARLY FOUR YEARS AFTER THE CHANGE OUGHT TO BE MADE.	29
30	5975	/	IT WAS ACCEPTABLE TO DEFER THE IMPLEMENTATION OF THE UPDATES. [NOTE: DUE	30
31	5976	/	TO THE RELATIVE FREQUENCY OF P?S/8 RELEASES TO A CONTROLLED DISTRIBUTION,	31
32	5977	/	OVER TIME MOST USERS RECEIVED A SYSTEM ALREADY SET TO MAXIMIZE THE VIABLE	32
33	5978	/	RANGE OF THE CURRENT DATE BASE GROUP; EXPECTATIONS WERE THAT AN EVENTUAL	33
34	5979	/	RELEASE MADE AFTER THE NEXT EIGHT YEAR GROUP VALUE UPDATED WOULD BE RESET	34
35	5980	/	UPON RELEASE; AT THE SOURCE CODE LEVEL, IT WAS ACCOMPLISHED BY MERELY	35
36	5981	/	CHANGING ONE LINE WITHIN EACH AFFECTED SOURCE FILE; THE ACTUAL TEXT CHANGE	36
37	5982	/	WAS ALWAYS THE SAME FOR ALL AFFECTED FILES; AS NEW PROGRAMS WERE RELEASED	37
38	5983	/	WITH SIMILAR DATE REQUIREMENTS, THE SAME SOURCE CODE WAS COPIED INTO THE	38
39	5984	/	SOURCE FILES OF THE NEW PROGRAMS.]	39
40				40
41				41
42				42
43				43
44				44
45				45
46				46
47				47
48				48
49				49
50				50
51				51
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1	5986	/	EVENTUALLY THE NEED TO ACCURATELY STORE FILE DATES WAS CONSIDERED. [NOTE: TFS	1
2	5987	/	FILES DO NOT SUPPORT FILE DATES.]	2
3	5988			3
4	5989	/	WHILE SEVERAL EXTENDED DIRECTORY ALTERNATIVES WERE BEING DISCUSSED [WHERE SOME	4
5	5990	/	DESIGNS MIGHT STORE THE CURRENT SYSTEM DATE IN COMPLETE FORM], THE	5
6	5991	/	IMPLEMENTATION OF ADDITIONAL FEATURES WITHIN THE EXTENDED DIRECTORY WAS	6
7	5992	/	TABLED. [NOTE: AS A PRACTICAL MATTER, THE EXTENDED DIRECTORY IS ONLY USEFUL	7
8	5993	/	IN APPLICATIONS RELATED TO TEMPORARILY CREATING INDEFINITE LENGTH SOURCE	8
9	5994	/	FILES, GENERALLY ACCOMPLISHED USING THE EXTENDED DIRECTORY FEATURES OF THE	9
10	5995	/	P?S/8 "CONVRT" PROGRAM, A BI-DIRECTIONAL TEXT FILE CONVERSION UTILITY BETWEEN	10
11	5996	/	OS/8 TEXT FORMAT AND AN EXTENDED VARIANT OF THE TEXT FORMAT CONTAINED WITHIN	11
12	5997	/	TFS FILES WITHOUT LINE NUMBERS. THIS FEATURE WAS AN OUTGROWTH OF AN EARLIER	12
13	5998	/	[NOW OBSOLETE] WORK KNOWN AS "BIGCON" WHICH PERFORMED THE SAME FUNCTION, BUT	13
14	5999	/	IS DANGEROUSLY PASSED THE BLOCK NUMBER OF A TFS FILE. ALL STORAGE ON THE	14
15	6000	/	DEVICE FROM THAT POINT ONWARD IS SUBJECT TO REPLACEMENT BY THE CONVERTED TEXT;	15
16	6001	/	THE CONSOLE SWITCH REGISTER IS USED TO SET THE UPPER BLOCK LIMIT OF THE	16
17	6002	/	CONVERSION. MULTIPLE FILES ARE SUPPORTED ON DIFFERENT LOGICAL DEVICE UNITS	17
18	6003	/	ADDRESSED BY THE SYSTEM DEVICE HANDLER. SINCE AT LEAST SIX LOGICAL UNITS ARE	18
19	6004	/	GENERALLY AVAILABLE ON A SYSTEM OF REASONABLE SIZE, THE EFFECTIVE SIZE OF THE	19
20	6005	/	CONVERSION EXCEEDS THE CAPABILITY OF MOST OS/8 SYSTEMS IN TERMS OF THE ABILITY	20
21	6006	/	TO EXPRESS A SINGLE OS/8 TEXT FILE. WHILE EIGHT LOGICAL UNITS ARE GENERALLY	21
22	6007	/	AVAILABLE, ONE MUST BE RESERVED FOR THE OS/8 MEDIA; ANOTHER IS GENERALLY	22
23	6008	/	RESERVED FOR OTHER FUNCTIONS SUCH AS SCRATCH STORAGE WHEN ASSEMBLING THE	23
24	6009	/	PROGRAM WITH P?S/8 PAL.]	24
25	6010			25
26	6011	/	EVENTUALLY, THE CONCEPT OF THE P?S/8 SHELL WAS CONSIDERED; THIS WOULD BE A	26
27	6012	/	COMPLETE OVERLAY TO THE BASIC P?S/8 SYSTEM THAT WOULD SUPPORT A MUCH LARGER	27
28	6013	/	FILE STRUCTURE ACCESSED THROUGH NON-SYSTEM HANDLERS WITH GREATLY IMPROVED	28
29	6014	/	ABILITY TO ACCESS HUGE AMOUNTS OF STORAGE [IF AVAILABLE] WITH A DIRECTORY	29
30	6015	/	STRUCTURE GEARED TOWARDS HANDLING A LARGE NUMBER OF FILES AS NECESSARY.	30
31	6016			31
32	6017	/	WHILE ADDITIONAL WORK WAS DONE ON REFINING THE SYSTEM HANDLER, IT WAS REALIZED	32
33	6018	/	THE SPACE FOR THE ADDITIONAL SIX BITS OF THE DATE BASE GROUP VALUE COULD BE	33
34	6019	/	PLACED INTO MEMORY, REQUIRING ONLY A SYSTEM WIDE MINOR MODIFICATION OF	34
35	6020	/	ONCE-ONLY INITIALIZATION CODE IN EVERY PROGRAM THAT PROCESSES THE POSSIBILITY	35
36	6021	/	OF PASSED OUTPUT FILES.	36
37	6022			37
38	6023	/	THE EARLIER DEFINITION OF THE OUTPUT FILE COUNT WORD "SOUTCT" [07607] WAS THE	38
39	6024	/	COUNT OF HOW MANY [IF ANY] OF THE PASSED FILE HANDLES STARTING AT "SFILES"	39
40	6025	/	[07757] [IF ANY] WERE OUTPUT FILES. [ANY ADDITIONAL ENTRIES IN THE LIST ARE	40
41	6026	/	INPUT FILES.]	41
42	6027			42
43	6028	/	SINCE THERE ARE AT MOST 17 FILES IN THE LIST, THIS COUNT CANNOT EXCEED FIVE	43
44	6029	/	SIGNIFICANT BITS. AS SUCH, ALL REFERENCES TO THE OUTPUT FILE COUNT WORD MADE	44
45	6030	/	IN ALL P?S/8 SYSTEM PROGRAMS THAT HANDLE THIS WORD SHOULD ADD ONE INSTRUCTION:	45
46	6031			46
47	6032	/	TAD I (SOUTCT) /GET THE OUTPUT FILE COUNT WORD	47
48	6033	/	AND (37) /STRIP OUT OTHER BITS	48
49	6034			49
50	6035	/	WHEN ALL P?S/8 SYSTEM PROGRAMS HAVE BEEN CONSISTENTLY UPDATED TO CONFORM TO	50
51	6036	/	THIS EASILY-IMPLEMENTED CHANGE, THE REST OF THE BITS IN "SOUTCT" CAN BE	51
52	6037	/	REPURPOSED. BITS [1-6] ARE NOW DEFINED AS THE IN-MEMORY COPY OF THE CURRENT	52
53	6038	/	DATE BASE GROUP VALUE; WHEN ADDED TO THE LOW-ORDER SYSTEM DATE WORD "SDATE"	53
54	6039	/	[07610], THE RESULT IS A UNIQUE SYSTEM DATE EXPRESSED IN 18 BITS.	54
55				55
56				56
57				57
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1	6041	/	SINCE THERE IS NO PARTICULAR NEED TO EXTEND THE LOW-ORDER DATE WORD'S RANGE			1
2	6042	/	BEYOND EIGHT YEARS [AND HAVING TO DEAL WITH THE NOW UNDESIRABLE ANOMALY], THE			2
3	6043	/	DATE HAS NOW BEEN REDEFINED THUS:			3
4	6044					4
5	6045	/	WORD	BITS	MEANING	5
6	6046					6
7	6047	/	SOUTCT	1-6	HIGH-ORDER SIX BITS OF THE FULL DATE.	7
8	6048					8
9	6049	/	SDATE	0-2	RELATIVE YEAR OFFSET IN THE RANGE 0-7.	9
10	6050					10
11	6051	/	SDATE	3-6	RELATIVE MONTH OFFSET IN THE RANGE OF 0-11.	11
12	6052					12
13	6053	/	SDATE	7-11	CURRENT DAY OF THE MONTH IN THE RANGE OF 1-31.	13
14	6054					14
15	6055	/	AS WAS FIRST USED WITH THE PREVIOUS DATE FORMAT, THE RELATIVE DAY OF THE			15
16	6056	/	MONTH IS CALCULATED ORIGIN 1; WHEN THE DAY BITS ARE SET TO 00000, THE SYSTEM			16
17	6057	/	IS CONSIDERED UNDATED. [NOTE: ALL OTHER BITS SHOULD BE SET TO ZERO WHEN THE			17
18	6058	/	SYSTEM IS UNDATED; VALIDATION ROUTINES CAN DECLARE THE COMBINATION OF ALL DAY			18
19	6059	/	BITS SET TO ZERO WHILE ANY OTHER BITS ARE NON-ZERO AS AN ILLEGAL COMBINATION.]			19
20	6060					20
21	6061	/	DATE RANGE SELECTION/FILTERING CAN STILL BE ACCOMPLISHED BY ARITHMETIC MEANS;			21
22	6062	/	HOWEVER, DOUBLE-PRECISION ROUTINES MUST BE USED TO MAINTAIN 18-BIT ACCURACY.			22
23	6063					23
24	6064	/	THE DATE RANGE STARTING YEAR WAS EVENTUALLY RECONSIDERED. ALL NEW WORK WILL			24
25	6065	/	ALLOW SYSTEM DATES [AND FILE DATES WITHIN THE P?S/8 SHELL DIRECTORIES] FROM			25
26	6066	/	1900 THROUGH 2411. WITH VERY FEW EXCEPTIONS, ANY FILE PERTINENT TO ANY			26
27	6067	/	COMPUTING SUBJECT SHOULD BE CORRECTLY EXPRESSED IN THE DATE RANGE STARTING			27
28	6068	/	WITH THE DAWN OF THE TWENTIETH CENTURY. [NOTE: MANY IBM SYSTEMS USE THE			28
29	6069	/	IDENTICAL STARTING DATE. MANY ACTUAL PDP-8 AND PDP-10 FILES CANNOT BE			29
30	6070	/	EXPRESSED PROPERLY WITHIN SO-CALLED "MODERN" OPERATING SYSTEMS WITH STARTING			30
31	6071	/	DATES OF 1970 OR EVEN 1980.]			31
32	6072					32
33	6073	/	END OF APPENDIX I.			33
34						34
35						35
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58						58

1	6075	/	APPENDIX J.	1
2	6076			2
3	6077	/	P?S/8 CORE SIZE AND STATUS WORD ISSUES.	3
4	6078			4
5	6079	/	THE P?S/8 CORE SIZE AND STATUS WORD MANAGES MEMORY AVAILABLE TO ALL P?S/8	5
6	6080	/	UTILITIES AND COMPLIANT USER PROGRAMS. ADDITIONALLY, THE STATUS OF THE	6
7	6081	/	CONSOLE OVERLAY AND SOFTWARE ABORT BITS ARE STORED HERE.	7
8	6082			8
9	6083	/	BITS MEANING	9
10	6084			10
11	6085	/	0-2 THE PHYSICAL HIGHEST MEMORY FIELD AVAILABLE. THIS IS BASED ON DYNAMIC	11
12	6086	/	DISCOVERY AT BOOTUP TIME. VALUES RANGE FROM 000 FOR A 4K MACHINE TO	12
13	6087	/	111 FOR A 32K MACHINE.	13
14	6088			14
15	6089	/	3-5 THE LOGICAL HIGHEST MEMORY FIELD POSSIBLE. ON MOST CONFIGURATIONS	15
16	6090	/	THIS VALUE IS IDENTICAL TO THE PHYSICAL HIGHEST MEMORY FIELD CONTAINED	16
17	6091	/	IN BITS[0-2] AS DESCRIBED ABOVE; ON CERTAIN SYSTEMS THIS VALUE IS ONE	17
18	6092	/	MEMORY FIELD LOWER DUE TO ONE OR MORE CONFIGURATION ISSUES.	18
19	6093			19
20	6094	/	6-8 THE LOGICAL HIGHEST FIELD CURRENTLY ALLOWED. POSSIBLE SETTINGS RANGE	20
21	6095	/	FROM 000 THROUGH THE LOGICAL HIGHEST MEMORY FIELD CONTAINED IN	21
22	6096	/	BITS[3-5] AS DESCRIBED ABOVE.	22
23	6097			23
24	6098	/	9 LOGICAL CONSOLE OVERLAY STATUS. IF THIS BIT IS SET, ALL	24
25	6099	/	OVERLAY-COMPLIANT PROGRAMS DIRECT INPUT AND OUTPUT FOR THE CONSOLE	25
26	6100	/	TERMINAL AND LINEPRINTER TO DEDICATED CONSOLE OVERLAY ROUTINES LOCATED	26
27	6101	/	IN THE PHYSICAL HIGHEST MEMORY FIELD CONTAINED IN BITS[0-2] AS	27
28	6102	/	DESCRIBED ABOVE.	28
29	6103			29
30	6104	/	10 LOGICAL CONTROL-B ABORT STATUS.	30
31	6105			31
32	6106	/	11 LOGICAL CONTROL-C ABORT STATUS.	32
33	6107			33
34	6108	/	THE PHYSICAL HIGHEST MEMORY FIELD IS DEFINED AS THE HIGHEST DETECTABLE MEMORY	34
35	6109	/	FIELD [UP TO FIELD 7] ASSUMING THE STANDARD ARCHITECTURE [KNOWN AS THE KM8A AS	35
36	6110	/	IMPLEMENTED ON THE PDP-8/A]. [NOTE: ALL PRIOR MODELS ARE COMPATIBLE WITH THIS	36
37	6111	/	EXTENDED MEMORY SPECIFICATION TO THE EXTENT EXTENDED MEMORY IS AVAILABLE;	37
38	6112	/	CERTAIN MACHINES CANNOT SUPPORT THE FULL MEMORY EXTENSION TO 32K.] ON THE	38
39	6113	/	PDP-8/A, MEMORY CAN BE EXTENDED BEYOND 32K IN SEVERAL WAYS:	39
40	6114			40
41	6115	/	1) THE DEC KT8A CAN EXTEND MEMORY TO EITHER 64K OR 128K. THE 64K EXTENSION IS	41
42	6116	/	ACCESSED USING EXTENDED IOT INSTRUCTIONS BASED ON THE STANDARD INSTRUCTIONS	42
43	6117	/	TO ACCESS THE FIRST 32K. EXPANSION IS ALSO POSSIBLE TO 128K BY ALSO USING	43
44	6118	/	ALL IOT CODES FROM DEVICE 30 THROUGH DEVICE 37; THIS CAUSES LOSS OF	44
45	6119	/	COMPATIBILITY WITH MANY EXISTANT PERIPHERAL DEVICES.	45
46	6120			46
47	6121	/	DMA CAPABILITY OF ALL DEVICES IS EXTENDED BY A SERIES OF 13 TWO-BIT	47
48	6122	/	REGISTERS; THERE ARE 13 DMA PRIORITIES ON THE OMNIBUS. SINCE THERE IS NO	48
49	6123	/	WAY TO DYNAMICALLY DETERMINE DMA PRIORITIES, EVERY SYSTEM ATTEMPTING TO	49
50	6124	/	PERFORM EXTENDED DMA TRANSFERS MUST BE CUSTOM-CONFIGURED AT THE DEVICE	50
51	6125	/	LEVEL. [NOTE: THE OMNIBUS DOES NOT DEMAND SETTING A DMA PRIORITY; AS SUCH,	51
52	6126	/	THERE IS PRIORITY 0 THOROUGH 11 AND NONE, WHERE 0 IS THE HIGHEST PRIORITY.]	52
53				53
54				54
55				55
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1	6128	/	P?S/8 CAN SUPPORT THE OPERATION OF THE KT8A, BUT AS IMPLEMENTED, P?S/8	1
2	6129	/	CANNOT LOAD ANY PROGRAMS OR DATA BEYOND THE FIRST 32K; IT IS THE PROGRAM'S	2
3	6130	/	RESPONSIBILITY FOR ALL ADDITIONAL MEMORY BANK ACCESS.	3
4	6131			4
5	6132	/	2) THE CESI MEC8 SUPPORTS EXTENDED MEMORY UP TO EITHER 128K OR 512K. AT THE	5
6	6133	/	BUSS LEVEL, THE 128K CONFIGURATION IS HARDWARE COMPATIBLE WITH THE KT8A.	6
7	6134	/	PROGRAMMING EXTENDED MEMORY IS NOT COMPATIBLE WITH THE KM8A. THE MEC8 CAN	7
8	6135	/	DISABLE EXTENDED SUPPORT; IN THIS MODE, THE MACHINE APPEARS TO HAVE 32K	8
9	6136	/	ONLY; KM8A COMPATIBILITY IS MAINTAINED IN THIS MODE. ALL PREVAILING PDP-8	9
10	6137	/	OPERATING SYSTEMS MUST USE THIS MODE TO OPERATE.	10
11	6138			11
12	6139	/	THE MEC8 NATIVE MODE IS EASILY EXTENDED TO 512K MAXIMUM. HARDWARE JUMPERS	12
13	6140	/	MUST BE MODIFIED TO CHANGE THE WAY THE MEMORY EXTENSION IS ADDRESSED; THE	13
14	6141	/	INSTRUCTIONS ARE EASILY EXTENDED FOR THE FULL 512K MEMORY SUPPORT; HOWEVER,	14
15	6142	/	NONE OF THE OPERATIONS ARE KM8A-COMPATIBLE. [NOTE: A MINOR ADVANTAGE OF	15
16	6143	/	THE MEC8 OVER THE 128K VARIANT OF THE KT8A IS DEVICE CODES 30 THROUGH 37	16
17	6144	/	ARE NOT RESERVED BY THE MEC8. HOWEVER, THE MEC8 DOES NOT MANAGE THE	17
18	6145	/	MEMORY BANK EXTENSION FOR DMA PERIPHERALS. THE ONLY PERIPHERAL THAT CAN	18
19	6146	/	PERFORM EXTENDED TRANSFERS IN THIS ENVIRONMENT IS THE MDC8 SCSI CONTROLLER.	19
20	6147	/	IN THIS MODE, TRANSFERS OF UP TO 32K ARE POSSIBLE FOR ANY MEMORY BANK. THE	20
21	6148	/	MDC8 CAN ONLY BE SET TO PRIORITY 0 OR 1. ANY OTHER PERIPHERALS MUST BE SET	21
22	6149	/	TO A LOWER PRIORITY SETTING [2 THROUGH 11 OR NONE]. HOWEVER, ALL OTHER	22
23	6150	/	PERIPHERAL TRANSFERS ARE LIMITED TO THE FIRST 32K OF MEMORY.	23
24	6151			24
25	6152	/	THE ONLY SOFTWARE THAT RUNS IN THIS MODE IS OMNI-8, WHICH WAS SPECIFICALLY	25
26	6153	/	TAILORED FOR THIS ENVIRONMENT. 32K DMA TRANSFERS ARE ADVANTAGEOUS TO ROLL	26
27	6154	/	IN AND OUT OF MEMORY ENTIRE 32K TASK IMAGES. [NOTE: OMNI-8 SIMULATES THE	27
28	6155	/	KM8A ENVIRONMENT IN EACH OF AS MANY AS 128 TASKS. AS MANY AS 15 CAN BE	28
29	6156	/	RESIDENT IN MEMORY AT ANY ONE TIME.]	29
30	6157			30
31	6158	/	OMNI-8 SUPPORT OF THE MDC8 REQUIRES 256 WORDS/BLOCK, A STANDARD VARIANT	31
32	6159	/	BLOCKSIZE FOR THIS DEVICE. HOWEVER, OS/8 CANNOT RUN ON THIS CONFIGURATION.	32
33	6160	/	P?S/8 CAN SUPPORT A VERSION FOR THIS SYSTEM REQUIRING EXTENDED MEMORY	33
34	6161	/	[HARDLY A PROBLEM SINCE ALL CONFIGURATIONS WILL HAVE 32K]. HOWEVER, THE	34
35	6162	/	MEC8 MUST BE SET BACK TO KM8A-COMPATIBILITY MODE BY THROWING A SWITCH ON	35
36	6163	/	THE MODULE [PHYSICALLY SIMILAR TO THE FORMAT SWITCH ON THE TD8E].	36
37	6164			37
38	6165	/	OMNI-8 DEVELOPMENT FROM OS/8 REQUIRES ANOTHER DEVICE SUCH AS THE RK8E/RK05	38
39	6166	/	WHEN THE MEC8 IS SET TO KM8A COMPATIBILITY MODE; THE MDC8 IS INCAPABLE OF	39
40	6167	/	OS/8 SUPPORT EVEN WITH NON-SYSTEM HANDLERS. GENERATION OF OMNI-8 IS	40
41	6168	/	ACHIEVED BY USING OS/8 TO LOAD UP TO 32K OF OMNI-8 SYSTEM GENERATION CODE,	41
42	6169	/	PAUSING THE MACHINE AND CHANGING THE HARDWARE OVER TO THE FULL MEC8 MEMORY	42
43	6170	/	MODE. THE OMNI-8 GENERATION PACKAGE THEN WRITES ITSELF OUT TO THE MDC8;	43
44	6171	/	OMNI-8 IS A BOOTABLE APPLICATION SYSTEM WITH ITS OWN INTERNAL RULES.	44
45	6172	/	WITHIN OMNI-8, THE ENTIRE OS/8 ENVIRONMENT IS EMULATED; A SIMILAR VERSION	45
46	6173	/	OF P?S/8 [OR COS-300/310] CAN RUN USING THE SAME EMULATED HARDWARE	46
47	6174	/	ENVIRONMENT AS OS/8. HOWEVER, A P?S/8 VARIANT FOR THE 256 WORD BLOCKS CAN	47
48	6175	/	ALSO BE BOOTED TO THE SAME HARDWARE [IN KM8A COMPATIBILITY MODE]. THUS,	48
49	6176	/	P?S/8 IS ARGUABLY A BETTER DEVELOPMENT ENVIRONMENT FOR OMNI-8 PROGRAM	49
50	6177	/	DEVELOPMENT. [ADDITIONAL HARDWARE WOULD NOT BE NECESSARY; CONSIDERING THE	50
51	6178	/	SCSI DISKS USED ON THE MDC8 ARE OVER 16 TIMES AS LARGE AS THE RK05, THIS	51
52	6179	/	WOULD ELIMINATE A LOT OF OVERHEAD.]	52
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1	6181	/	[NOTE: SINCE THE MDC8 CAN SUPPORT PRIORITY 0 OR 1, IT IS POSSIBLE TO HAVE	1
2	6182	/	TWO MDC8 SUBSYSTEMS ON THE SAME MACHINE; THIS IS THE RECOMMENDED WAY TO	2
3	6183	/	SUPPORT OMNI-8 TO AVOID USING DEC PERIPHERAL HARDWARE [ACCORDING TO CESI].	3
4	6184	/	BY USING P?S/8 TO SUPPORT OMNI-8 DEVELOPMENT, THIS BECOMES SUPERFLUOUS AS	4
5	6185	/	ALL FUNCTIONS CAN BE SUPPORTED ON ONE MDC8.]	5
6	6186			6
7	6187	/	CHARLES LASNER HAS WRITTEN OS/8 AND P?S/8 SYSTEM AND NON-SYSTEM HANDLERS	7
8	6188	/	FOR BOTH OS/8 AND P?S/8 WHEN USED WITH THE 128 WORD BLOCKSIZE MDC8 VARIANT.	8
9	6189	/	DUE TO THE EXCELLENT DESIGN OF THE MDC8, OS/8 CAN BE RUN IN 8K AND P?S/8	9
10	6190	/	CAN BE RUN IN 4K ON THIS VARIANT OF THE HARDWARE. [NOTE: MEMORY	10
11	6191	/	CONTROLLERS MUST BE KM8A-COMPLIANT UP TO 32K FOR THESE SYSTEMS TO OPERATE.]	11
12	6192			12
13	6193	/	[NOTE: CHARLES LASNER DEVELOPED THE MDC8 UNIVERSAL BOOTUP CONVENTIONS FOR	13
14	6194	/	ALL SOFTWARE CONFIGURATIONS INCLUDING OMNI-8; THE ACTUAL CODE IS ONLY	14
15	6195	/	SLIGHTLY LONGER THAN THE RK8E BOOTUP CONVENTIONS. MULTIPLE OPERATING	15
16	6196	/	SYSTEMS ARE SUPPORTABLE USING MENU-8, A MASTER BOOT MENU SYSTEM VAGUELY	16
17	6197	/	ANALOGOUS TO MASTER MENU FOR THE DECMATE II AND DECMATE III+ WITH THE RD51	17
18	6198	/	HARD DISK CONTROLLER OPTION. ALL DEVICE HANDLERS USE A DOUBLE-PRECISION	18
19	6199	/	OFFSET WORD STORED WITHIN THE SYSTEM DEVICE HANDLER RESIDENT IN 07600-07777	19
20	6200	/	TO DETERMINE EXACTLY WHERE THE CORRESPONDING DISK STORAGE IS. A TYPICAL	20
21	6201	/	CONFIGURATION HAS MENU-8 AS THE ABSOLUTE BOOTUP SYSTEM WHICH IN TURN CAN	21
22	6202	/	BOOT OTHER SYSTEMS BY SETTING THE APPROPRIATE OFFSET IN A LONGER BOOTUP	22
23	6203	/	ROUTINE. IF MENU-8 IS NOT USED, ANY ONE OPERATING SYSTEM CAN BE RESIDENT	23
24	6204	/	IN THE ABSOLUTE BOOTUP AREA; BY USING THE LONGER BOOTUP ROUTINE, MULTIPLE	24
25	6205	/	OPERATING SYSTEMS CAN EXIST ON THE MDC8 DISK. THE PRIMARY OPERATING SYSTEM	25
26	6206	/	OUGHT TO BE ABLE TO RUN THE LONGER BOOTUP PROGRAM TO ALLOW MULTIPLE	26
27	6207	/	SYSTEMS TO COEXIST ON THIS SYSTEM; BY USING MENU-8, THE SHORTEST BOOTSTRAP	27
28	6208	/	CAN BE USED TO ACCESS ALL OPERATING SYSTEMS. THE SHORTEST BOOTSTRAP CAN	28
29	6209	/	BE LOADED INTO ANY OF THE HARDWARE LOADERS FOR THE OMNIBUS INCLUDING THE	29
30	6210	/	M18E BOARD FOR THE PDP-8/E; THE MDC8 WOULD HAVE TO BE ADDED TO AN -8/A BOX	30
31	6211	/	IN A MANNER SIMILAR TO THE FPP-8/E.]	31
32	6212			32
33	6213	/	WITHIN P?S/8, THE PHYSICAL SIZE OF MEMORY CAN BE AS LARGE AS 32K; ON SYSTEMS	33
34	6214	/	THAT RUN IN 4K, FIELD 0 COULD BE THE LARGEST MEMORY FIELD AVAILABLE; THE	34
35	6215	/	LARGEST FIELD POSSIBLE IS FIELD 7. AS P?S/8 BOOTS UP, THIS VALUE IS	35
36	6216	/	DYNAMICALLY CALCULATED USING CODING KNOWN TO WORK PROPERLY ON ALL MODELS; THE	36
37	6217	/	VALUE IS STORED INTO BITS[0-2] OF THE CORE SIZE AND STATUS WORD.	37
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1	6219	/	THE LOGICAL HIGHEST MEMORY FIELD IS DETERMINED AFTER THE PHYSICAL HIGHEST	1
2	6220	/	MEMORY FIELD IS CALCULATED. CERTAIN P?S/8 COMPONENTS OCCUPY PART OF THE	2
3	6221	/	PHYSICAL HIGHEST FIELD; IF ANY OF THESE COMPONENTS IS PRESENT, THE LOGICAL	3
4	6222	/	HIGHEST MEMORY FIELD IS SET TO ONE MEMORY FIELD LOWER. ANY OF THE FOLLOWING	4
5	6223	/	COMPONENTS BEING PRESENT IN MEMORY WILL FORCE THE LOWERING OF THE MAXIMUM	5
6	6224	/	AVAILABLE MEMORY BY ONE MEMORY FIELD:	6
7	6225			7
8	6226	/	1) THE SYSTEM DEVICE HANDLER REQUIRES AN EXTENSION; THIS RESERVES LOCATIONS	8
9	6227	/	6000-7777 IN THE HIGHEST MEMORY FIELD.	9
10	6228			10
11	6229	/	2) THE LOGICAL CONSOLE OVERLAY IS ENABLED; THIS OCCUPIES 0000-5377 OR	11
12	6230	/	0000-5777 IN THE HIGHEST MEMORY FIELD.	12
13	6231			13
14	6232	/	3) THE P?S/8 SHELL OVERLAY IS ENABLED AND THE FILE AND COMMUNICATIONS AREA IS	14
15	6233	/	LOADED INTO 5400-5777 OF THE HIGHEST MEMORY FIELD. [NOTE: THE SHELL DESIGN	15
16	6234	/	IS NEWER THAN CERTAIN RELEASES OF THE LOGICAL CONSOLE OVERLAY. ALL CURRENT	16
17	6235	/	DEVICE SUPPORT WITHIN THE LOGICAL CONSOLE OVERLAY CAN BE ACHIEVED WITHOUT	17
18	6236	/	USING 5400-5777; HOWEVER, THE OVERLAY MUST BE SLIGHTLY MODIFIED TO CONFORM	18
19	6237	/	TO THE NEW REQUIREMENT. THE LARGEST KNOWN IMPLEMENTATION IS THE VT8E	19
20	6238	/	CONFIGURATION WITH INTERNAL DEVICE 66 SUPPORT. ALL AVAILABLE MEMORY	20
21	6239	/	THROUGH 5777 WAS USED AS ORIGINALLY RELEASED; HOWEVER, THE HIGHEST	21
22	6240	/	ADDRESSES ARE ONLY USED TO ALLOW THE INTERNAL PRINT BUFFER TO BE NEEDLESSLY	22
23	6241	/	LARGE. THE REST OF AVAILABLE MEMORY WAS USED TO DEFINE THE PRINT BUFFER;	23
24	6242	/	HOWEVER, THE FULL EFFECT OF PRINT BUFFERING CAN BE ACHIEVED WITH FAR LESS	24
25	6243	/	BUFFER SPACE. AS LONG AS JUST OVER TWO COMPLETE PRINTABLE LINES CAN BE	25
26	6244	/	STORED, PRINT BUFFERING WILL BE 100% EFFECTIVE; LARGER BUFFERS COULD	26
27	6245	/	ACTUALLY BE SLIGHTLY LESS EFFICIENT. THUS, LIMITING THE PRINT BUFFER TO	27
28	6246	/	APPROXIMATELY 270 CHARACTERS WILL FULLY SUPPORT WIDE-CARRIAGE PRINTERS; THE	28
29	6247	/	REST OF AVAILABLE MEMORY CAN BE RESERVED FOR THE POTENTIAL NEEDS OF THE	29
30	6248	/	SHELL-RELATED RESOURCE. THIS PROBLEM WILL BE REMEDIED IN THE NEXT FULL	30
31	6249	/	RELEASE OF P?S/8.]	31
32	6250			32
33	6251	/	NOTE: ALL THREE OF THE POTENTIAL SYSTEM COMPONENTS OUTLINED ABOVE ARE	33
34	6252	/	DESCRIBED BELOW IN GREATER DETAIL.	34
35	6253			35
36	6254	/	WITH THE EXCEPTION OF THE SYSTEM DEVICE HANDLER EXTENSION, ALL OF THE ABOVE	36
37	6255	/	SYSTEM COMPONENTS MAY BE DISABLED; SHOULD THE ENTIRE MEMORY FIELD BECOME	37
38	6256	/	AVAILABLE, THE LOGICAL HIGHEST MEMORY FIELD CAN BE RECALCULATED TO MATCH THE	38
39	6257	/	HIGHEST PHYSICAL MEMORY FIELD. [NOTE: THE P?S/8 "CORE" UTILITY CAN BE USED TO	39
40	6258	/	MAKE ALL ADJUSTMENTS TO MEMORY FIELD PARAMETERS; THE COMMAND "CORE ALL" WILL	40
41	6259	/	ALWAYS SET THE MAXIMUM AVAILABLE MEMORY. SEE BELOW FOR MORE DETAILS OF THE	41
42	6260	/	P?S/8 CORE COMMAND.]	42
43	6261			43
44	6262	/	THE LOGICAL HIGHEST MEMORY FIELD IS USED BY ALL SYSTEM COMPONENTS AND	44
45	6263	/	CONFORMING PROGRAMS TO DETERMINE POTENTIAL ADDITIONAL MEMORY ALLOCATIONS	45
46	6264	/	BEYOND THE STANDARD AVAILABILITY OF THE FIRST 4K OF MEMORY REQUIRED TO RUN	46
47	6265	/	P?S/8. THE LOGICAL HIGHEST MEMORY FIELD CAN BE SET BY THE P?S/8 CORE UTILITY	47
48	6266	/	TO ANY VALUE BETWEEN FIELD 0 AND THE FIELD INDICATED IN THE LOGICAL HIGHEST	48
49	6267	/	MEMORY FIELD. ONLY COMPLETELY AVAILABLE FIELDS ARE CONSIDERED. BY	49
50	6268	/	ARTIFICIALLY LOWERING THE QUANTITY OF EXTENDED MEMORY AVAILABLE, THE EXISTING	50
51	6269	/	CONTENTS OF HIGHER MEMORY FIELDS CAN BE PROTECTED; PERFORMANCE OF PROGRAMS	51
52	6270	/	THAT DYNAMICALLY ALLOCATE MEMORY BUFFERS CAN BE TESTED AS IF RUNNING ON ACTUAL	52
53	6271	/	SMALLER SYSTEMS.	53
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1	6273	/	BASIC P?S/8 PROGRAMS ARE ALLOWED TO INSPECT THE HIGHEST PHYSICAL MEMORY FIELD	1
2	6274	/	DATA TO DETERMINE IF A PARTIAL FIELD IS AVAILABLE. [NOTE: NOT EVERY PROGRAM	2
3	6275	/	IS CAPABLE OF USING A PARTIAL MEMORY FIELD; P?S/8 PAL IS A NOTABLE EXAMPLE OF	3
4	6276	/	A PROGRAM THAT CAN USE A PARTIAL MEMORY FIELD TO INCREASE THE MAXIMUM SIZE OF	4
5	6277	/	THE SYMBOL TABLE.] IF THE LOGICAL HIGHEST MEMORY FIELD IS ONE MEMORY FIELD	5
6	6278	/	LOWER THAN THE HIGHEST MEMORY FIELD, 3K OF MEMORY MIGHT BE AVAILABLE IF THE	6
7	6279	/	LOGICAL CONSOLE OVERLAY IS NOT PRESENT. [THIS CAN BE DETERMINED AS DESCRIBED	7
8	6280	/	BELOW.] THIS WOULD ONLY APPLY IF THE SYSTEM DEVICE HANDLER REQUIRES AN	8
9	6281	/	EXTENDED MEMORY SEGMENT OF 6000-7777 IN THE HIGHEST MEMORY FIELD.	9
10	6282			10
11	6283	/	NOTE: P?S/8 SYSTEMS RELEASED PRIOR TO THE ADDITION OF THE P?S/8 SHELL OVERLAY	11
12	6284	/	ARE NOT AWARE OF THE SHELL FILE AND COMMUNICATIONS AREA POTENTIAL RESERVATION	12
13	6285	/	OF LOCATIONS 5400-5777 IN THE HIGHEST MEMORY FIELD; HOWEVER, THIS IS MOOT	13
14	6286	/	WITHIN THE BASIC P?S/8 SYSTEM. LOGICAL CONSOLE OVERLAY DESIGN COMPATIBLE WITH	14
15	6287	/	THE POTENTIAL OF THE SHELL OVERLAY WILL AVOID CONFLICT IN THIS TWO-PAGE AREA.	15
16	6288	/	AS SUCH, PROGRAMS RUN UNDER THE SHELL OVERLAY MUST CHECK FURTHER TO DETERMINE	16
17	6289	/	IF THE FILE AND COMMUNICATIONS AREA IS RESIDENT IN THESE LOCATIONS WHICH IS A	17
18	6290	/	CONFIGURATION PARAMETER OF THE SHELL ITSELF. [NOTE: THE STATUS OF THE FILE	18
19	6291	/	AND COMMUNICATIONS AREA CAN BE DETERMINED BY INSPECTION OF BIT[0] OF THE	19
20	6292	/	"SOUTCT" [07607] STATUS WORD; HOWEVER, THIS IS ONLY MEANINGFUL FROM THE SHELL	20
21	6293	/	ENVIRONMENT. SHELL-BASED PROGRAMS ATTEMPTING TO ALLOCATE ADDITIONAL MEMORY	21
22	6294	/	IN A PARTIAL HIGHEST PHYSICAL FIELD MUST BE PREPARED TO LIMIT ALLOCATION TO	22
23	6295	/	0000-5377 OR 0000-5777 AS REQUIRED.]	23
24	6296			24
25	6297	/	P?S/8 CORE UTILITY.	25
26	6298			26
27	6299	/	BY INVOKING THE CORE UTILITY WITH APPROPRIATE ARGUMENTS, THE CURRENT LOGICAL	27
28	6300	/	CORE SIZE CAN BE SET TO ANY VALUE BETWEEN 0 AND THE MAXIMUM LOGICAL CORE SIZE	28
29	6301	/	ALLOWED ON THE CURRENT SYSTEM.	29
30	6302			30
31	6303	/	DELIBERATELY LOWERING THE LOGICAL CORE SIZE CAN BE USEFUL FOR DEBUGGING,	31
32	6304	/	TESTING OR PROTECTING DATA RESIDENT IN HIGH MEMORY [SUCH AS AN IMAGE OF THE	32
33	6305	/	MR8E-C ROM LOADED INTO WRITABLE MEMORY ON TD8E-BASED SYSTEMS UNDER TEST].	33
34	6306			34
35	6307	/	"CORE" [WITHOUT ARGUMENTS] WILL DISPLAY THE CURRENT SETTING OF THE THREE	35
36	6308	/	MEMORY SIZE PARAMETERS. AN OPTIONAL ARGUMENT CAN BE USED [SUCH AS "CORE 5"]	36
37	6309	/	TO LIMIT MEMORY TO ANY ALLOWED VALUE. [NOTE: ERROR MESSAGES WILL BE ISSUED	37
38	6310	/	WHEN ATTEMPTING TO USE INVALID ARGUMENTS FOR THE CURRENT SYSTEM OR ANY ATTEMPT	38
39	6311	/	TO USE INVALID SYNTAX; THE ONLY VALID ARGUMENTS ARE "0" THROUGH "7" AND "ALL";	39
40	6312	/	NUMERIC VALUES MUST BE VALID FOR THE CURRENT SYSTEM CONFIGURATION.] [NOTE:	40
41	6313	/	OS/8 SUPPORTS A VAGUE ANALOGY TO THE CORE COMMAND; HOWEVER, THE LOGICAL SIZE	41
42	6314	/	OF OS/8 MUST BE SET TO A MINIMUM OF 1 BECAUSE OS/8 REQUIRES AT LEAST 8K OF	42
43	6315	/	MEMORY TO OPERATE; IF THE OS/8 LOGICAL CORE SIZE IS NOT SET, EVERY PROGRAM	43
44	6316	/	SEEKING ADDITIONAL MEMORY MUST PROVIDE ITS OWN CORE SIZING ROUTINE OR ASSUME	44
45	6317	/	ONLY 8K IS AVAILABLE; NO OS/8 CONFIGURATION EXISTS WITH ONLY 4K AVAILABLE.	45
46	6318	/	ALL CORE SIZE MANAGEMENT IN OS/8 MUST BE ACHIEVED IN THREE BITS USED FOR ALL	46
47	6319	/	PURPOSES; THE ADDITIONAL MEMORY SIZE CONCEPTS SUPPORTED BY P?S/8 ARE	47
48	6320	/	CONCEPTUALLY IMPOSSIBLE WITHIN OS/8.]	48
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1	6322	/	IF THE SIZE OF PHYSICAL MEMORY IS CHANGED FOR ANY REASON [OR A REMOVABLE			1
2	6323	/	SYSTEM DEVICE IS BOOTED ON A DIFFERENT COMPUTER WITH A DIFFERENT MEMORY			2
3	6324	/	CONFIGURATION], CERTAIN OF THE MEMORY SIZE VALUES MAY BE CONFUSED; CERTAIN			3
4	6325	/	IMPLEMENTATIONS OF THE CORE COMMAND WRITE OUT THE MAXIMUM LOGICAL CORE SIZE AS			4
5	6326	/	WELL AS THE CURRENT LOGICAL CORE SIZE. THIS SITUATION CAN ALWAYS BE REMEDIED			5
6	6327	/	BY USING THE "CORE ALL" COMMAND TO RESET ALL VALUES TO THE MAXIMUM CURRENTLY			6
7	6328	/	AVAILABLE; AFTERWARDS, THE CURRENT LOGICAL CORE SIZE CAN OPTIONALLY BE LOWERED			7
8	6329	/	AS NECESSARY.			8
9	6330					9
10	6331	/	IF BIT[9] OF THE CORE SIZE AND STATUS WORD IS SET, THE LOGICAL CONSOLE OVERLAY			10
11	6332	/	IS IN EFFECT. ALL CONFORMING PROGRAMS WILL SUBSTITUTE LOGICAL CALLS TO THE			11
12	6333	/	OVERLAY INSTEAD OF THE NORMAL PHYSICAL SUBROUTINE CALLS MADE WITH RESPECT TO			12
13	6334	/	ALL INPUT AND OUTPUT TO THE DEVICE 03/04 CONSOLE TERMINAL AND THE DEVICE 66			13
14	6335	/	SYSTEM LINEPRINTER. CALLS ARE GENERALLY MADE IN A MANNER AS DESCRIBED BELOW:			14
15	6336					15
16	6337	/	CDF	MYFLD	/INDICATE CALLING FIELD WHICH WILL BE RETAINED	16
17	6338					17
18	6339	/	CIF	OVLYFLD	/THIS IS "CIF" TO THE HIGHEST MEMORY FIELD	18
19	6340	/	TAD	CHAR	/LATEST CHARACTER IF OUTPUT; CLEAR AC IF INPUT	19
20	6341	/	JMS	20	/CALL THE ROUTINE	20
21	6342	/	JMP	.-3	/IMMEDIATE RETURN; RESOURCE NOT AVAILABLE	21
22	6343	/	NOP		/SKIP RETURN, THE RESOURCE WAS PROCESSED; THE	22
23	6344	/			/AC IS UNCHANGED IF OUTPUT; THE LATEST	23
24	6345	/			/CHARACTER IS IN THE AC IF INPUT	24
25	6346					25
26	6347	/	ROUTINES EXIST TO SEPARATELY HANDLE CONSOLE INPUT AND CONSOLE OUTPUT. THE			26
27	6348	/	LOGICAL LINEPRINTER CALL RESEMBLES THE CONSOLE OUTPUT ROUTINE. FOR MOST			27
28	6349	/	LINEPRINTER OUTPUT, THERE IS AN INTERNAL BUFFER WITHIN THE CONSOLE OVERLAY			28
29	6350	/	THAT CAN DRAMATICALLY INCREASE PERFORMANCE WITHOUT USING INTERRUPTS DUE TO			29
30	6351	/	INTELLIGENT BUFFERING AND DEVICE FLAG HANDLING. THE CALL ALWAYS MAKES THE			30
31	6352	/	USUAL RETURN, BUT NOT NECESSARILY WHEN THE LATEST CHARACTER HAS BEEN PRINTED;			31
32	6353	/	THE RETURN MIGHT INDICATE THE CHARACTER HAS BEEN PLACED INTO THE INTERNAL			32
33	6354	/	BUFFER; ADDITIONALLY, ANY CHARACTERS WAITING IN THE BUFFER WILL BE PROCESSED			33
34	6355	/	AS LONG AS THERE IS LITTLE OR NO WAITING TIME; AS SOON AS ANY BUFFERED			34
35	6356	/	CHARACTER [OR THE LATEST CHARACTER IF THE BUFFER IS EMPTY] IS TAKING EXCESSIVE			35
36	6357	/	TIME TO CLEAR THE FLAG, NO FURTHER ATTEMPTS WILL BE MADE TO OUTPUT CHARACTERS			36
37	6358	/	AT THIS TIME. BY NATURE, LINEPRINTERS ARE CAPABLE OF PROCESSING MANY			37
38	6359	/	CHARACTERS IN 15 MICROSECONDS OR LESS EACH UNTIL INTERNALLY, THE PRINTER IS			38
39	6360	/	SUFFICIENTLY BACKED UP; AT THAT POINT, IT IS PREDICTABLE THE PRINTER FLAG WILL			39
40	6361	/	NOT RAISE AGAIN FOR MANY MILLISECONDS [OR PERHAPS LONGER].			40
41	6362					41
42	6363	/	TO GUARANTEE THE ENTIRE PRINT BUFFER IS PRINTED BEFORE ANY ORDERLY PROGRAM			42
43	6364	/	EXIT, RECEIPT OF A <FF> CHARACTER CAUSES THE CALL TO WAIT UNTIL THE ENTIRE			43
44	6365	/	BUFFER EMPTIES [INCLUDING THE <FF> CHARACTER]. [NOTE: LINEPRINTER FORMATTING			44
45	6366	/	RULES ARE SUCH THAT ALL PRINTING ENDS WITH A <FF> CHARACTER FOR ALL PRINTED			45
46	6367	/	PAGES; EVEN IF SLOPPILY ALLOWED TO OVERFLOW ONTO ADDITIONAL PAGES, THE PROGRAM			46
47	6368	/	IS OBLIGATED TO OUTPUT A <FF> CHARACTER TO FINALIZE THE LAST PRINTED PAGE			47
48	6369	/	JUST PRIOR TO EXIT.]			48
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1	6371	/	A FOURTH CALL IS ALSO IMPLEMENTED TO SUPPORT A POTENTIAL LINEPRINTER REVERSE	1
2	6372	/	CHANNEL INPUT. ON CERTAIN PRINTERS, A BI-DIRECTIONAL SERIAL INTERFACE IS IN	2
3	6373	/	EFFECT. IN THESE CONFIGURATIONS, THE LINEPRINTER OUTPUT CALL HANDLES THIS	3
4	6374	/	INTERNALLY; PROGRAMS THAT OPTIONALLY PARTICIPATE IN THE REVERSE CHANNEL INPUT	4
5	6375	/	PROTOCOL MAY SPEED LINEPRINTER THROUGHPUT; HOWEVER, FULL COMPATIBILITY IS	5
6	6376	/	AVAILABLE MERELY BY CALLING THE LOGICAL OUTPUT ROUTINE. [NOTE: SERIAL	6
7	6377	/	PRINTERS HAVE INTERNAL HARDWARE BUFFERS; THE LOGICAL CONSOLE OVERLAY	7
8	6378	/	CONFIGURATION FOR SERIAL PRINTERS DOES NOT PROVIDE ADDITIONAL SOFTWARE	8
9	6379	/	BUFFERS.]	9
10	6380			10
11	6381	/	AN OVERALL LOGICAL INTERRUPT ROUTINE IS ALSO IMPLEMENTED; IT IS CALLED BY	11
12	6382	/	EXECUTION OF A "JMS 0" INSTRUCTION REMINISCENT OF THE ACTION OF THE INTERRUPT	12
13	6383	/	HARDWARE. INTERRUPT-DRIVEN PROGRAMS SHOULD IMPLEMENT ALL FIVE SUBROUTINE	13
14	6384	/	CALLS TO PROPERLY HANDLE LOGICAL INTERRUPTS; ALL CALLS SHOULD BE MADE AHEAD OF	14
15	6385	/	ANY PHYSICAL DEVICES. IF IMPLEMENTED CORRECTLY, HARDWARE FLAG-HANDLING	15
16	6386	/	INTERRUPT SERVICE ROUTINES WILL NOT INTERFERE WITH THE LOGICAL ROUTINES FOR	16
17	6387	/	THE SAME DEVICES. THIS ALLOWS FULL CONSOLE SUPPORT ON OTHER DEVICES WHILE	17
18	6388	/	STILL ALLOWING FOR INPUT AND OUTPUT ON THE NORMAL DEVICES EVEN WHILE USING	18
19	6389	/	INTERRUPTS. [NOTE: SOME LOGICAL CONSOLE CONFIGURATIONS ADDRESS ALTERNATE	19
20	6390	/	CONSOLE TERMINALS ON A DIFFERENT INTERFACE SUCH AS DEVICE 10/11 FOR THE PT08	20
21	6391	/	OR DEVICE 40/41 FOR A SUITABLY CONFIGURED KL8-TYPE SERIAL INTERFACE. IT IS	21
22	6392	/	THE PROGRAM'S RESPONSIBILITY TO DEAL WITH THE NORMAL DEVICES WHEN THE INPUT	22
23	6393	/	AND OUTPUT IS LOGICALLY REDIRECTED; ONE RECOMMENDED METHOD IS TO LIMIT CONSOLE	23
24	6394	/	SUPPORT TO CHECKING ONLY FOR ABORT CHARACTERS SUCH AS CONTROL-C. THIS WOULD	24
25	6395	/	ALLOW PROGRAM EXIT INITIATED FROM EITHER THE PHYSICAL OR LOGICAL CONSOLE	25
26	6396	/	DEVICE WHEN BOTH SETS OF HARDWARE ARE IN USE.] [NOTE: RECENT RELEASES OF	26
27	6397	/	P?S/8 FOCAL IMPLEMENT CONSOLE HANDLING FULLY CONFORMING TO THIS DESIGN.]	27
28	6398			28
29	6399	/	LOGICAL ABORT ISSUES.	29
30	6400			30
31	6401	/	ALL DECMATE SYSTEMS ARE INCAPABLE OF PROPER CONSOLE INPUT DUE TO HARDWARE	31
32	6402	/	INCOMPATIBILITY WITH THE PDP-8 CONSOLE DEVICE. [NOTE: THE STANDARD CONSOLE	32
33	6403	/	HARDWARE IS COMPATIBLE IN ALL MODELS FROM THE PDP-5 THROUGH THE VT78. VARIOUS	33
34	6404	/	NON-DEC HARDWARE IS ALSO FULLY COMPATIBLE.]	34
35	6405			35
36	6406	/	WHILE CERTAIN OF THE DEVICE 03/04 INSTRUCTIONS ARE REMINISCENT OF THE ORIGINAL	36
37	6407	/	IMPLEMENTATION, ANY PROGRAM ATTEMPTING TO USE ANYTHING BUT THE MOST SIMPLISTIC	37
38	6408	/	DESIGN WILL SIMPLY NOT FUNCTION CORRECTLY ON DECMATES.	38
39	6409			39
40	6410	/	IN PARTICULAR, THE ABILITY TO PASS PROGRAM ABORT STATUS VIA THE KEYBOARD INPUT	40
41	6411	/	FLAG DOES NOT EXIST ON DECMATES; THE "KRS" [6034] INSTRUCTION WAS REDESIGNED	41
42	6412	/	IN A MANNER RENDERING IT USELESS FOR THE ORIGINAL PURPOSE. THE "KSF" [6031]	42
43	6413	/	INSTRUCTION WAS MODIFIED TO NOT ONLY SKIP IF THE INPUT FLAG IS SET, BUT TO	43
44	6414	/	ALSO CLEAR THE FLAG; THERE IS SIMPLY NO WAY TO PASS ABORT STATUS BACK TO THE	44
45	6415	/	SYSTEM KERNEL UPON PROGRAM EXIT AS HAS BEEN CLASSICALLY DEFINED IN ALL PDP-8	45
46	6416	/	OPERATING SYSTEMS [INCLUDING P?S/8].	46
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1	6418	/	TO SOLVE THIS PROBLEM, RECENT RELEASES OF P?S/8 SUPPORT LOGICAL ABORT BITS	1
2	6419	/	FOR BOTH SUPPORTED ABORT CHARACTERS.	2
3	6420			3
4	6421	/	BIT[11] OF THE CORE SIZE AND STATUS WORD IS SET IF A PROGRAM EXITS DUE TO	4
5	6422	/	DETECTION OF CONTROL-C. THIS IS THE UNIVERSAL EXIT METHOD USED IN MOST PDP-8	5
6	6423	/	OPERATING SYSTEMS. [NOTE: IF A P?S/8 SYSTEM PROGRAM DETECTS IT IS RUNNING ON	6
7	6424	/	A COMPATIBLE MACHINE, IT IS NOT NECESSARY TO HANDLE THE ABORT LOGICALLY; THE	7
8	6425	/	NORMAL CONSOLE HARDWARE IS ALSO TESTED; HOWEVER, THE HARDWARE ABORT CANNOT BE	8
9	6426	/	DETECTED ON DECMATES.]	9
10	6427			10
11	6428	/	BIT[10] OF THE CORE SIZE AND STATUS WORD IS SET IF A PROGRAM EXITS DUE TO	11
12	6429	/	DETECTION OF CONTROL-B. THIS IS A P?S/8-SPECIFIC EXTENSION FOR ABORT PURPOSES	12
13	6430	/	IMPORTANT TO P?S/8 BATCH OPERATIONS. WHEN BATCH IS RUNNING, CONTROL-C WILL	13
14	6431	/	ABORT THE CURRENT PROGRAM; HOWEVER, THE BATCH WILL ATTEMPT TO CONTINUE TO RUN	14
15	6432	/	TO PROCESS OTHER PROGRAMS. CONTROL-B WILL ABORT ALL BATCH OPERATIONS; WHEN	15
16	6433	/	RUN WITHOUT BATCH, BOTH CHARACTERS ABORT THE CURRENT PROGRAM.	16
17	6434			17
18	6435	/	ALL CONFORMING PROGRAMS CHECK TO SEE IF THE CPU IS THE HARRIS 6120; IF THIS IS	18
19	6436	/	THE CASE, THE PHYSICAL ABORT-CHECKING ROUTINES ARE MODIFIED INTO HANDLING	19
20	6437	/	LOGICAL ABORTS; THE APPROPRIATE ABORT BIT IS SET WHEN THE CORRESPONDING	20
21	6438	/	CHARACTER IS DETECTED.	21
22	6439			22
23	6440	/	AS SUCH, P?S/8 IS THE ONLY PDP-8 OPERATING SYSTEM THAT CAN ALSO RUN ON THE	23
24	6441	/	ENTIRE DECMATE SERIES WITHOUT BASTARDIZING THE ENTIRE SYSTEM. [NOTE: OS/8	24
25	6442	/	CANNOT RUN ON A DECMATE; THE VARIANT SYSTEM GENERALLY REFERRED TO AS "OS/278	25
26	6443	/	V2" IS A SERIOUSLY FLAWED SYSTEM. THERE ARE MANY QUIRKS STEMMING FROM THE	26
27	6444	/	HARDWARE-RELATED PROBLEM AS WELL AS FLAWS IN IMPLEMENTATION NOT TRACEABLE TO	27
28	6445	/	HARDWARE INCOMPATIBILITY. IT IS KNOWN THAT THE QUALITY OF PROGRAMMING	28
29	6446	/	EXPERTISE IN THE 12-BIT COMPUTER GROUP WITHIN DEC WENT DOWNHILL WHEN THE	29
30	6447	/	DECMATES WERE INTRODUCED; ONE SPECIFIC POINT IS WORTH NOTING: WITHIN CERTAIN	30
31	6448	/	OS/278 SOURCE FILES IS A SMALL CHART CLAIMING TO BE THE WAY TO PROGRAM THE	31
32	6449	/	CONSOLE INTERFACE COMPARING PDP-8 TO DECMATE TO POINT OUT THE DIFFERENCES.	32
33	6450	/	UNFORTUNATELY, WHILE MANY PROGRAMS SLAVISHLY FOLLOWED THIS CHART, IT HAS	33
34	6451	/	ERRONEOUS INFORMATION DESCRIBING SOMETHING QUITE DIFFERENT FROM ACTUAL DECMATE	34
35	6452	/	CONSOLE HARDWARE. THIS IS THE APPARENT ROOT CAUSE FOR VARIOUS BUGS IN	35
36	6453	/	PROGRAMS SUCH AS OS/278 "ODT" NOT ACTUALLY RELATED TO CONSOLE HARDWARE	36
37	6454	/	INCOMPATIBILITY PER SE; NONE OF THESE BUGS OUGHT TO EXIST; ONLY PROBLEMS	37
38	6455	/	RELATED TO PROGRAM EXIT SHOULD BE PRESENT ALBEIT UNFIXABLE; OS/8 CANNOT	38
39	6456	/	SUPPORT THE RIGOROUS CONSOLE ROUTINES UNIVERSALLY USED WITHIN ALL CONFORMING	39
40	6457	/	P?S/8 PROGRAMS; NOT ONLY IS THE DECMATE INCOMPATIBILITY PROGRAM SOLVED, ALL	40
41	6458	/	CONFORMING P?S/8 PROGRAMS CORRECTLY HANDLE ALL CONTROL-S/CONTROL-Q FLOW	41
42	6459	/	CONTROL ISSUES AS WELL, A KNOWN SORE POINT THROUGHOUT OS/8. PARADOXICALLY,	42
43	6460	/	SOME OF THE MODIFICATIONS MADE TO THE DECMATES WERE INITIATED TO PARTIALLY	43
44	6461	/	CORRECT THIS WEAKNESS WHILE ACTUALLY MAKING THE OVERALL SITUATION	44
45	6462	/	UNSUPPORTABLE. IT HAS BEEN REPORTED THAT DEC MANAGEMENT APPROVED THESE	45
46	6463	/	ARCHITECTURAL CHANGES WITHOUT ASKING COMPETENT PERSONELL WHO WERE OUTRAGED BY	46
47	6464	/	THE MODIFICATIONS SUGGESTED AND THEN IMPLEMENTED BY HARRIS SEMICONDUCTOR;	47
48	6465	/	APPARENTLY ALL RESPONSIBLE PARTIES SOON RESIGNED AND SOUGHT EMPLOYMENT	48
49	6466	/	ELSWHERE THUS EVADING RESPONSIBILITY.]	49
50				50
51				51
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1	6468	/	ADDITIONAL INFORMATION ON SYSTEM DEVICE HANDLER EXTENSIONS.	1
2	6469			2
3	6470	/	P?S/8 DEVELOPMENT METHODOLOGY STRIVES TO OBTAIN MAXIMUM PERFORMANCE AND	3
4	6471	/	EFFICIENCY FROM EVERY SUPPORTED HARDWARE CONFIGURATION. AS SUCH, EVERY EFFORT	4
5	6472	/	IS MADE TO SUPPORT THE SYSTEM STORAGE DEVICE IN 4K IF POSSIBLE; HOWEVER,	5
6	6473	/	SEVERAL SUPPORTED DEVICES CANNOT ACHIEVE THIS GOAL IN PRACTICE. IF THE DEVICE	6
7	6474	/	IS NOT SUPPORTABLE IN A 4K MEMORY CONFIGURATION, ADDRESSES 6000-7777 OF THE	7
8	6475	/	PHYSICAL HIGHEST FIELD OF EXTENDED MEMORY IS RESERVED FOR EXTENDED DEVICE	8
9	6476	/	SUPPORT.	9
10	6477			10
11	6478	/	TYPICAL SUPPORT CODE CONSISTS OF ADDITIONAL PROGRAMMING THAT EXTENDS THE	11
12	6479	/	SYSTEM DEVICE HANDLER CODING DIRECTLY. CERTAIN DEVICES REQUIRE AN INTERNAL	12
13	6480	/	MEMORY BUFFER TO DEAL WITH THE ISSUES OF LOGICAL SUPPORT OF DEVICES WITH	13
14	6481	/	PHYSICAL RECORDS LARGER THAN THE LOGICAL BLOCK SIZE [SUCH AS THE RK8E/RK05] OR	14
15	6482	/	DISJOINT STORAGE OF LOGICAL BLOCKS [SUCH AS VARIOUS DISKETTE FORMATS ON THE	15
16	6483	/	VARIOUS RX-SERIES DISKETTE CONTROLLERS WHEN OPERATING IN BYTE MODE]. WHERE	16
17	6484	/	APPLICABLE, ADDITIONAL EXTENDED SUPPORT CODE MAY BE REQUIRED TO IMPLEMENT THE	17
18	6485	/	SLURP LOADER OR THE RELATED DEVICE HANDLER RELOAD ROUTINES, OR THE SYSTEM	18
19	6486	/	BOOTSTRAP AT 07600. [NOTE: IN CERTAIN SYSTEM CONFIGURATIONS, THE LOGICAL	19
20	6487	/	CONSOLE OVERLAY INTERFACES WITH THE SYSTEM DEVICE HANDLER TO ENHANCE	20
21	6488	/	PERFORMANCE OR ERROR RECOVERY; IT IS CONCEIVABLE SOME OF THE AFFECTED ROUTINES	21
22	6489	/	COULD BE RESIDENT WITHIN THE SYSTEM DEVICE HANDLER EXTENSION. CERTAIN	22
23	6490	/	IMPLEMENTATIONS OF THE SLURP LOADER OR SYSTEM DEVICE HANDLER RELOAD ROUTINES	23
24	6491	/	MAY REQUIRE INTERACTION WITH THE LOGICAL CONSOLE OVERLAY; IN MANY CASES, IT	24
25	6492	/	MAY BE NECESSARY TO LOCATE THIS CODE WITHIN THE SYSTEM DEVICE HANDLER	25
26	6493	/	EXTENSION RESERVED AREA.]	26
27	6494			27
28	6495	/	BY PROVIDING AN ADDITIONAL EIGHT PAGES OF MEMORY FOR SYSTEM DEVICE SUPPORT,	28
29	6496	/	P?S/8 GENERALLY AVOIDS FRUSTRATING DESIGN DILEMMAS INVOLVING UNREASONABLE	29
30	6497	/	COMPROMISES OF FUNCTIONALITY, PERFORMANCE OR ERROR RECOVERY. [NOTE: IN	30
31	6498	/	VIRTUALLY EVERY CASE WHERE OS/8 REQUIRES 12K TO OPERATE, THE TWO-PAGE SYSTEM	31
32	6499	/	DEVICE HANDLERS ARE GREATLY COMPROMISED IN SEVERAL OF THESE AREAS. IN	32
33	6500	/	GENERAL, JUST UNDER TWO PAGES OF MEMORY CANNOT PROVIDED ADEQUATE SUPPORT TO	33
34	6501	/	DEAL WITH THE REAL-WORLD ISSUES OF VARIOUS COMMON STORAGE DEVICES; SPECIFICS	34
35	6502	/	REGARDING CERTAIN MARGINAL OS/8 DEVICE HANDLERS AND ATTENDANT LIMITATIONS ARE	35
36	6503	/	THE SUBJECT OF A NUMBER OF ARTICLES OUTSIDE OF THIS DOCUMENT.]	36
37	6504			37
38	6505	/	BY NATURE, EXTENDED SYSTEM DEVICE HANDLERS REQUIRE 8K OF MEMORY OR MORE. IF	38
39	6506	/	NONE OF THE OTHER OPTIONAL SYSTEM COMPONENTS ARE PRESENT, A P?S/8 SYSTEM	39
40	6507	/	PROGRAM CAN ALLOCATE AS MUCH AS 3K OF THE HIGHEST MEMORY FIELD AVAILABLE.	40
41	6508	/	[NOTE: SHELL-BASED PROGRAMS MUST ALSO CHECK FOR THE PRESENCE OF THE SHELL FILE	41
42	6509	/	AND COMMUNICATIONS AREA POTENTIALLY LOADED INTO 5400-5777 OF THE HIGHEST	42
43	6510	/	MEMORY FIELD. AS SUCH, A SHELL-BASED PROGRAM MIGHT ONLY BE ABLE TO ALLOCATE	43
44	6511	/	2.75K ADDITINAL MEMORY IN THE HIGHEST MEMORY FIELD DEPENDING ON THE SHELL	44
45	6512	/	CONFIGURATION.]	45
46	6513			46
47	6514	/	NOTE: THE TC01/TC08 SYSTEM DEVICE HANDLER DOES NOT USE ANY FORM OF EXTENDED	47
48	6515	/	MEMORY SUPPORT. IT IS CONSIDERED THE P?S/8 "STANDARD BEARER" CONFIGURATION	48
49	6516	/	BECAUSE THE R-L MONITOR [WHICH P?S/8 IS PARTIALLY BASED ON] ONLY SUPPORTS	49
50	6517	/	THE TC01/TC08 AND A SINGLE TU55/TU56 DRIVE. [NOTE: ENHANCEMENTS TO ANY P?S/8	50
51	6518	/	SYSTEM DEVICE HANDLER MAY BE MADE BY THE LOGICAL CONSOLE OVERLAY REGARDING	51
52	6519	/	EITHER SPECIFIC PERFORMANCE ISSUES OR ERROR RECOVERY INTERCEPTION.]	52
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1	6521	/	LOGICAL CONSOLE OVERLAY ISSUES.	1
2	6522			2
3	6523	/	P?S/8 SUPPORTS AN OPTIONAL SYSTEM PROGRAM EXTENSION KNOWN AS THE LOGICAL	3
4	6524	/	CONSOLE OVERLAY. WHEN ENABLED, ALL COMPLIANT P?S/8 SYSTEM UTILITIES WILL	4
5	6525	/	SELF-MODIFY TO REDIRECT CONSOLE INPUT AND OUTPUT TO LOGICAL SUBROUTINE CALLS	5
6	6526	/	WITHIN THE OVERLAY. ALL SUBROUTINE CALLS HAVE CONDITIONAL SKIP RETURN	6
7	6527	/	MECHANISMS TO INDICATE THE SUCCESS OF THE CALL; MOST PROGRAMS WILL RETRY THE	7
8	6528	/	CALL INDEFINITELY UNTIL THE COMMAND SUCCEEDS, LARGELY MIMICKING THE USUAL	8
9	6529	/	METHOD OF PROGRAMMING THE STANDARD CONSOLE HARDWARE.	9
10	6530			10
11	6531	/	LOGICAL CONSOLE REDIRECTED LINEPRINTER SUPPORT.	11
12	6532			12
13	6533	/	NORMALLY LINEPRINTER OUTPUT IS DIRECTED TO A DEVICE 66 LINEPRINTER; THERE IS	13
14	6534	/	NO SUPPORT FOR A REVERSE CHANNEL ISSUING CONTROL-S AND CONTROL-Q FLOW-CONTROL	14
15	6535	/	CHARACTERS [AS IS REQUIRED FOR PROPER SERIAL PRINTER SUPPORT] OR OTHER	15
16	6536	/	HARDWARE VARIANTS. WHEN THE LOGICAL CONSOLE OVERLAY IS ENABLED, ALL COMPLIANT	16
17	6537	/	SYSTEM UTILITIES SELF-MODIFY TO REDIRECT LINEPRINTER OUTPUT TO ADDITIONAL	17
18	6538	/	SUBROUTINE CALLS WITHIN THE LOGICAL CONSOLE OVERLAY. THE UNDERLYING PHYSICAL	18
19	6539	/	PRINTING DEVICE MAY BE TOTALLY DIFFERENT FROM THE STANDARD DEVICE 66 PRINTER.	19
20	6540	/	[THE LOGICAL CONSOLE OVERLAY IS EASILY MODIFIED TO SUPPORT ALTERNATE HARDWARE	20
21	6541	/	CONFIGURATIONS USING BINARY UPDATE FILES.]	21
22	6542			22
23	6543	/	LINEPRINTER FLOW CONTROL SUPPORT.	23
24	6544			24
25	6545	/	A LOGICAL REVERSE CHANNEL SUBROUTINE CALL IS PROVIDED FOR USE BY	25
26	6546	/	FULLY-COMPLIANT PROGRAMS TO MANAGE FLOW-CONTROL ON SERIAL PRINTERS. HOWEVER,	26
27	6547	/	USE OF THIS ROUTINE IS NOT ABSOLUTELY REQUIRED AS THE OVERLAY ITSELF WILL	27
28	6548	/	BUFFER AND SUPPORT THE REVERSE CHANNEL PROTOCOL WITHIN THE LOGICAL LINE	28
29	6549	/	PRINTER OUTPUT ROUTINES. PROGRAMS THAT ACTIVELY SUPPORT THE REVERSE CHANNEL	29
30	6550	/	CAN PERHAPS SLIGHTLY IMPROVE LINEPRINTER THROUGHPUT; HOWEVER, THE PRIMARY	30
31	6551	/	PURPOSE FOR THE SUPPORT IS TO ALLOW LOGICAL INTERRUPT HANDLING OF ALL RELATED	31
32	6552	/	RESOURCES.	32
33	6553			33
34	6554	/	LOGICAL INTERRUPT SUPPORT.	34
35	6555			35
36	6556	/	WHEN AN INTERRUPT-DRIVEN P?S/8-COMPLIANT PROGRAM IS EXECUTED WITH THE LOGICAL	36
37	6557	/	CONSOLE OVERLAY PRESENT, THE PROGRAM WILL CALL A LOGICAL INTERRUPT HANDLER	37
38	6558	/	AHEAD OF ALL CONSOLE [AND LINEPRINTER] INPUT AND OUTPUT ROUTINES AS PART OF	38
39	6559	/	A GENERAL INTERRUPT SERVICE ROUTINE. PROPERLY IMPLEMENTED, THIS PRODUCES	39
40	6560	/	COMPLETE FUNCTIONALITY FOR INTERRUPT-DRIVEN PROGRAMS SUCH AS THE P?S/8	40
41	6561	/	ENHANCED IMPLEMENTATION OF FOCAL, 1969.	41
42	6562			42
43	6563	/	SYSTEM DEVICE HANDLER ENHANCEMENTS.	43
44	6564			44
45	6565	/	CUSTOM SYSTEM DEVICE HANDLER ENHANCEMENTS CAN BE PROVIDED BY THE LOGICAL	45
46	6566	/	CONSOLE OVERLAY AS DEFINED WITHIN A SMALL AUXILIARY INFORMATION BLOCK	46
47	6567	/	ASSOCIATED WITH THE SYSTEM DEVICE HANDLER. ANY SUCH ENHANCEMENTS ARE SPECIFIC	47
48	6568	/	TO THE DEVICE HANDLER ITSELF. COMMONLY IMPLEMENTED COMPONENTS PROVIDE	48
49	6569	/	PERFORMANCE ENHANCEMENTS SUCH AS UNIT-SPECIFIC SEARCH TABLES OR THE ABILITY	49
50	6570	/	TO PERFORM FORWARD SEARCHES ON PDP-12 LINCTAPE SYSTEMS. [NOTE: PDP-12	50
51	6571	/	LINCTAPE HARDWARE ALWAYS FORCES INITIAL REVERSE SEARCH UNLESS SUBSTANTIAL	51
52	6572	/	CODING IN MAINTENANCE MODE IS IMPLEMENTED TO OVERRIDE THE HARDWARE DEFAULTS.]	52
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1	6574	/	INTERACTIVE ERROR RECOVERY SUPPORT.	1
2	6575			2
3	6576	/	WHEN THE LOGICAL CONSOLE OVERLAY IS IN EFFECT, THE SYSTEM DEVICE HANDLER DOES	3
4	6577	/	NOT HALT ON ERRORS. INSTEAD, ALL ERROR PARTICULARS ARE REDIRECTED TO THE	4
5	6578	/	LOGICAL CONSOLE OVERLAY WHICH INTERACTS WITH THE USER TO DETERMINE THE PROPER	5
6	6579	/	DISPOSITION OF DEVICE ERRORS [SUCH AS RETRY, ABORT, ETC.]. AS NECESSARY,	6
7	6580	/	ERROR MESSAGE STRINGS ARE PROVIDED TO EMBELLISH THE SUPPORT IF THE USER	7
8	6581	/	DESIRES VERBOSE ERROR MESSAGES DURING THE RECOVERY OPERATION.	8
9	6582			9
10	6583	/	LOGICAL CONSOLE OVERLAY MEMORY LAYOUT.	10
11	6584			11
12	6585	/	WHEN ENABLED, THE LOGICAL CONSOLE OVERLAY OCCUPIES AS MUCH AS 3K OF MEMORY	12
13	6586	/	OF THE HIGHEST PHYSICAL MEMORY FIELD. [THIS IS ONE OF THE REASONS THE	13
14	6587	/	LARGEST MEMORY FIELD ALLOWED MAY BE ONE LOWER THAN THE LARGEST PHYSICAL	14
15	6588	/	MEMORY FIELD.]	15
16	6589			16
17	6590	/	THE EARLIEST IMPLEMENTATIONS OF THE LOGICAL CONSOLE OVERLAY WERE ORIGINALLY	17
18	6591	/	SPECIFICALLY DESIGNED FOR THE VT8E DATA BREAK DISPLAY DEVICE IN TEXT MODE.	18
19	6592	/	EVENTUALLY THE PROGRAM WAS GENERALIZED TO ALLOW FOR A VARIETY OF ALTERNATE	19
20	6593	/	DEVICES IMPLEMENTED BY OVERLAYING THE STANDARD VERSION WITH CUSTOM BINARY	20
21	6594	/	FILES FOR A VARIETY OF ALTERNATE DEVICES. EACH CONFIGURATION CHANGE IS	21
22	6595	/	PERMANENT. HOWEVER, CUSTOM BINARY FILES ARE AVAILABLE FOR ALL SUPPORTED	22
23	6596	/	CONFIGURATIONS INCLUDING THE ORIGINAL VT8E CONFIGURATION AND SEVERAL	23
24	6597	/	VARIATIONS. THE NEW DEFAULT CONFIGURATION SUPPORTS THE DEVICE 03/04 CONSOLE	24
25	6598	/	TERMINAL. ALL OTHER LOGICAL OVERLAY ENHANCEMENTS ARE GENERIC TO THE SYSTEM	25
26	6599	/	DEVICE SUPPORTED, NOT THE PARTICULAR CONSOLE DEVICE SUPPORTED. A VARIETY OF	26
27	6600	/	ALTERNATE STANDARD CONSOLE DEVICES IS SUPPORTED [COMPATIBLE WITH THE STANDARD	27
28	6601	/	CONSOLE BUT IMPLEMENTED ON ALTERNATE DEVICE CODE PAIRS SUCH AS IS FOUND ON	28
29	6602	/	PT08 OR KL8 TYPE SERIAL INTERFACES].	29
30	6603			30
31	6604	/	THE STANDARD LINEPRINTER SUPPORT IS FOR THE USUAL DEVICE 66 LINEPRINTER. AS	31
32	6605	/	NECESSARY, ALTERNATE HARDWARE MAY BE SUPPORTED IN A CUSTOMIZED CONFIGURATION.	32
33	6606	/	FOR ANY SUPPORTED CONFIGURATION, ENHANCED BUFFERING IS RECOMMENDED; DEPENDING	33
34	6607	/	ON HARDWARE PARTICULARS, THIS CAN HAVE A PROFOUND EFFECT ON LINEPRINTER	34
35	6608	/	THROUGHPUT.	35
36	6609			36
37	6610	/	DUE TO THE NATURE OF THE PARALLEL INTERFACE, MOST DEVICE 66 LINEPRINTERS HAVE	37
38	6611	/	REAL-TIME QUIRKS THAT MUST BE DEALT WITH OR ELSE THROUGHPUT MAY SUFFER.	38
39	6612			39
40	6613	/	IN THE EARLIEST IMPLEMENTATION OF DEVICE 66 LINEPRINTERS, IT WAS PREDICTABLE	40
41	6614	/	THE INTERFACE WOULD HANG FOR LONGER PERIODS OF TIME WHEN PRINTING FORM-FEED,	41
42	6615	/	CARRIAGE RETURN OR LINE FEED CHARACTERS; ALL OTHER CHARACTERS WERE QUICKLY	42
43	6616	/	PLACED INTO AN INTERNAL BUFFER. [NOTE: SOME PRINTERS HAVE MORE COMPLEX TIMING	43
44	6617	/	ISSUES SUCH AS DRUM PRINTERS WHERE CHARACTER PRINTING TIME IS IN PART	44
45	6618	/	DETERMINED BY ROTATIONAL DELAY; HOWEVER, THE GENERAL RULE THAT NON-PRINTING	45
46	6619	/	PAPER MOTION TAKE MORE TIME FOR THE DONE FLAG TO RAISE THAN FOR PRINTABLE	46
47	6620	/	CHARACTERS STILL APPLIES.]	47
48	6621			48
49	6622	/	UNFORTUNATELY, NEWER DESIGNS ADD SOME INTERNAL BUFFERING BETWEEN THE PRINT	49
50	6623	/	MECHANISM AND THE INTERFACE; THIS ESPECIALLY APPLIES TO SERIAL PRINTERS WHICH	50
51	6624	/	ARE KNOWN TO BE HEAVILY BUFFERED TO ACHIEVE MAXIMUM THROUGHPUT ASSUMING THE	51
52	6625	/	HOST SUPPORT CODE PAYS CLOSE ATTENTION TO THE REVERSE INPUT CHANNEL.	52
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1	6627	/	MOST PDP-8 SOFTWARE DOES NOT NORMALLY ADDRESS THESE ISSUES; ACHIEVED	1
2	6628	/	THROUGHPUT MAY BE FAR LESS THAN DESIGN SPECIFICATIONS INDICATE. THE BUFFERED	2
3	6629	/	LINEPRINTER SUPPORT OF THE LOGICAL CONSOLE OVERLAY OVERCOMES THIS COMPLETELY:	3
4	6630			4
5	6631	/	EVERY TIME THE LINEPRINTER OUTPUT ROUTINES ARE ENTERED, CHARACTERS ARE	5
6	6632	/	POTENTIALLY BUFFERED TO AVAILABLE BUFFER SPACE WITHIN THE OVERLAY. AS THE	6
7	6633	/	PRINTER DONE FLAG RAISES, THE NEXT CHARACTER FROM THE BUFFER IS SENT TO THE	7
8	6634	/	PRINTER; FOR A VERY SHORT TIME INTERVAL, THE PRINTER DONE FLAG IS TESTED.	8
9	6635			9
10	6636	/	IF THE PRINTER DONE FLAG RAISES QUICKLY AND THERE ARE MORE CHARACTERS IN THE	10
11	6637	/	BUFFER TO PRINT, THE PROCESS LOOPS UNTIL EITHER ALL CHARACTERS ARE SENT	11
12	6638	/	[INCLUDING THE ONE ASSOCIATED WITH THE LATEST CALL, IN WHICH CASE THE PRINTER	12
13	6639	/	FULLY CONTAINS ALL PRINT OUTPUT THROUGH THAT MOMENT IN TIME], OR THE DONE FLAG	13
14	6640	/	FAILS TO RAISE QUICKLY. [NOTE: FOR MOST ACTUAL DEVICE 66 PRINTERS, THE TIME	14
15	6641	/	INTERVAL IS AROUND 25 MICROSECONDS; SHOULD THE FLAG NOT RAISE BY THAT TIME,	15
16	6642	/	IT WILL NOT RAISE FOR MANY MILLISECONDS OR PERHAPS LONGER.]	16
17	6643			17
18	6644	/	EACH LOGICAL LINEPRINTER OUTPUT OPERATION KEEPS THE PRINTER BUFFER IN PLAY	18
19	6645	/	TO THE FULLEST EXTENT POSSIBLE. SHOULD A CALL BE MADE AND THERE IS NO ROOM	19
20	6646	/	AVAILABLE FOR THE LATEST CHARACTER, THE OVERLAY WILL INDICATE [VIA A NON-SKIP	20
21	6647	/	RETURN TO THE CALLER] THE LATEST CHARACTER CANNOT BE PROCESSED. MOST CALLERS	21
22	6648	/	WILL LOOP BACK AND REPEAT THE CALL UNTIL A SKIP RETURN OCCURS AS NOTHING ELSE	22
23	6649	/	CAN BE DONE TO ALLOW OVERLAPPING PROCESSING WITHIN THE CALLING PROGRAM WHEN	23
24	6650	/	CALLING THE OVERLAY IN THIS STATE. WHEN AT LEAST ONE CHARACTER HAS BEEN	24
25	6651	/	PROCESSED, THE LATEST CHARACTER CAN BE BUFFERED [AND THE LOGICAL CALL TAKES	25
26	6652	/	THE SKIP RETURN INDICATING SUCCESSFUL TRANSMISSION OF THE LATEST CHARACTER].	26
27	6653			27
28	6654	/	ONCE ONE CHARACTER HAS BEEN PROCESSED, PERHAPS WELL OVER 100 ADDITIONAL	28
29	6655	/	CHARACTERS MAY ALSO BE PROCESSED AS LONG AS SUBSEQUENT DONE FLAG EVENTS RAISE	29
30	6656	/	THE FLAG QUICKLY. [GENERALLY, THE ONLY ACTUAL REAL-TIME EXCEPTION CASE WOULD	30
31	6657	/	BE JUST AFTER A CARRIAGE RETURN WAS PROCESSED ASSUMING THE NEXT CHARACTER SENT	31
32	6658	/	TO THE PRINTER IS A LINE FEED.] BY NOT BEING CONCERNED WITH THE ACTUAL	32
33	6659	/	CHARACTERS, MERELY CHECKING THE DONE FLAG TIMING WILL BUFFER THE PRINTED	33
34	6660	/	OUTPUT AT NEARLY THE SPEED OF A FULLY INTERRUPT-DRIVEN PROGRAM EVEN WITH	34
35	6661	/	INTERRUPTS DISABLED. [NOTE: OS/8 DOES NOT IMPLEMENT LINEPRINTER BUFFERING	35
36	6662	/	WHATSOEVER; AS SUCH, THE MAXIMUM LINEPRINTER SPEED IS LIMITED TO JUST UNDER	36
37	6663	/	300 LINES PER MINUTE REGARDLESS OF ANY ADDITIONAL PRINTER PERFORMANCE	37
38	6664	/	CAPABILITY. P?S/8 WITH THE LOGICAL CONSOLE OVERLAY ENABLED HAS SUCCESSFULLY	38
39	6665	/	DRIVEN 1200 LINE PER MINUTE PRINTERS TO VIRTUALLY FULL SPEED VIA THE METHOD	39
40	6666	/	DESCRIBED HERE.]	40
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1	6668	/	HISTORIC OVERLAY PRINTER BUFFERING IMPLEMENTATION.	1
2	6669			2
3	6670	/	TYPICALLY, LOGICAL CONSOLE OVERLAY CONFIGURATIONS OCCUPY A SIGNIFICANT PORTION	3
4	6671	/	OF THE AVAILABLE MEMORY SPACE [X0000-X5777 WHERE X IS THE LARGEST PHYSICAL	4
5	6672	/	FIELD AVAILABLE], BUT HARDLY THE ENTIRE MEMORY AREA. AS SUCH, THE SOURCE CODE	5
6	6673	/	DEFINED ALL ADDITIONAL SPACE PAST THE CODE [THROUGH X5777] AS THE GENERAL	6
7	6674	/	LINEPRINTER OUTPUT BUFFER. [NOTE: AT THE TIME OF IMPLEMENTATION, THIS WAS THE	7
8	6675	/	UPPER LIMIT OF MEMORY RESERVED FOR THE OVERLAY; THERE WAS NO REASON TO LEAVE	8
9	6676	/	THE MEMORY UNUSED.]	9
10	6677			10
11	6678	/	HOWEVER, IT IS GENERALLY TRUE THAT THE LARGEST BUFFER NEEDED TO ACCOMPLISH	11
12	6679	/	VIRTUALLY ALL PRINT BUFFERING THROUGHPUT ADVANTAGE IS JUST LARGER THAN AT MOST	12
13	6680	/	TWO COMPLETE PRINTING LINES; ASSUMING 132 COLUMNS/PAGE, THIS MEANS THERE IS NO	13
14	6681	/	PARTICULAR ADVANTAGE TO RESERVING MORE THAN 268 CHARACTERS OR SO [THIS	14
15	6682	/	INCLUDES CARRIAGE RETURN AND LINE FEED CHARACTERS FOR EACH LINE]. ANY	15
16	6683	/	ADDITIONAL BUFFER MEMORY USED CREATES NO PARTICULAR THROUGHPUT ADVANTAGE [OR	16
17	6684	/	DISADVANTAGE]. [NOTE: THE VT8E CONFIGURATION IS THE MOST COMPLEX OVERLAY	17
18	6685	/	IMPLEMENTED TO DATE. THE VT8E SUPPORT CODE IS NOT PARTICULARLY LONG,	18
19	6686	/	CONSISTING OF ONLY A BASIC TERMINAL EMULATOR WITH MINOR FRILLS. HOWEVER, AN	19
20	6687	/	INTERNAL BUFFER IS USED FOR THE 1280 TEXT POSITIONS ON THE VT8E SCREEN.	20
21	6688	/	THERE IS ALSO A MORE THAN ADEQUATE LINEPRINTER BUFFER PRESENT IN THE VT8E	21
22	6689	/	CONFIGURATION; THE DEVICE 66 LINEPRINTER IS A STANDARD FEATURE OF THE VT8E.]	22
23	6690			23
24	6691	/	IF FOR ANY REASON BIT[9] IS CLEARED ON THE WORD ON BLOCK 0000 CORRESPONDING TO	24
25	6692	/	THE CORE SIZE AND STATUS WORD [RELATIVE WORD 011], THE LOGICAL CONSOLE OVERLAY	25
26	6693	/	WILL BE ABANDONED IN FAVOR OF THE PHYSICAL DEVICE 03/04 CONSOLE DEVICE. THE	26
27	6694	/	"CONSOL" COMMAND MUST BE RUN WITH THE "/E" SWITCH ["CONSOLE /E"] TO ENABLE THE	27
28	6695	/	LOGICAL CONSOLE FUNCTION. IF NECESSARY, THE "CORE ALL" COMMAND CAN BE GIVEN	28
29	6696	/	TO RESET BITS[0-8] OF THE CORE SIZE AND STATUS WORD TO STANDARD MAXIMUM VALUES	29
30	6697	/	FOR THE CURRENT SYSTEM CONFIGURATION. [NOTE: FUTURE VERSIONS OF THE	30
31	6698	/	"CORE ALL" COMMAND WILL ALSO ENSURE THE PERMANENT COPY OF THE CORRESPONDING	31
32	6699	/	WORD ON BLOCK 0000 IS WRITTEN WITH THE LOGICAL ABORT BITS [BITS[10-11]] CLEAR	32
33	6700	/	TO PREVENT INDEFINITE LOGICAL ABORT EVENT LOOPS WITHOUT USER INPUT. THIS	33
34	6701	/	RARE EVENT CAN BE CLEARED USING THE P?S/8 "DUMP" FACILITY FOR THE PRESENT.]	34
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1	6703	/	P?S/8 SHELL IMPLEMENTATION.	1
2	6704			2
3	6705	/	THE P?S/8 SHELL IS CURRENTLY UNDER DEVELOPMENT. [NOTE: MOST OF THE PARAMETERS	3
4	6706	/	OF THE SHELL OVERLAY HAVE BEEN DEFINED. THE ISSUES ASSOCIATED WITH POTENTIAL	4
5	6707	/	INTERACTION WITH THE LOGICAL CONSOLE OVERLAY WILL BE ADDRESSED HERE.] OTHER	5
6	6708	/	THAN THE ISSUES RAISED HERE, IT IS NOT EXPECTED THE SHELL WILL HAVE ANY	6
7	6709	/	ADDITIONAL IMPACT ON THE SYSTEM DEVICE HANDLER OR OTHER SYSTEM COMPONENTS.	7
8	6710			8
9	6711	/	WHEN THE P?S/8 SHELL IS ENABLED, IT IS ENCOURAGED THE OVERALL SIZE OF THE	9
10	6712	/	LOGICAL CONSOLE OVERLAY BE SLIGHTLY REDUCED [FROM X0000-X5777 TO X0000-X5377	10
11	6713	/	WHERE X IS THE HIGHEST PHYSICAL FIELD OF MEMORY AVAILABLE]. THIS IS	11
12	6714	/	REQUIRED TO ALLOW THE 256 WORD SHELL FILE AND COMMUNICATIONS AREA TO BE	12
13	6715	/	RESIDENT IN EXTENDED MEMORY FOR MAXIMUM PERFORMANCE.	13
14	6716			14
15	6717	/	THE SHELL CAN OPERATE WITH THE FILE AND COMMUNICATIONS AREA NON-RESIDENT AND	15
16	6718	/	REDIRECT ALL PROGRAMS TO THE CONTENTS OF TWO LOW-ORDER ADJACENT SYSTEM BLOCKS	16
17	6719	/	FOR ALL RELEVANT PURPOSES. THIS IS GENERALLY NOT RECOMMENDED; IF AT ALL	17
18	6720	/	POSSIBLE, THE MEMORY RESIDENT VARIANT SHOULD BE USED TO AVOID NOTICEABLE	18
19	6721	/	DEGRADATION OF SYSTEM PERFORMANCE [ESPECIALLY ON DEVICES SUCH AS DECTAPES AND	19
20	6722	/	LINCTAPES].	20
21	6723			21
22	6724	/	AS SUCH, ALL NEW WORK ON THE LOGICAL CONSOLE OVERLAY WILL CONFORM TO THIS NEW	22
23	6725	/	MEMORY LAYOUT. IT IS NOT CONTEMPLATED THAT ANY EXISTENT CONFIGURATION WILL BE	23
24	6726	/	AFFECTED BEYOND THE MINOR BUFFER LIMIT DEFINITION CHANGES. FOR THE REMAINDER	24
25	6727	/	OF THIS DOCUMENT, ASSUME THE LOGICAL CONSOLE OVERLAY LOADS INTO X0000-X5377.	25
26	6728			26
27	6729	/	P?S/8 SHELL RESIDENT FILE AND COMMUNICATIONS AREA.	27
28	6730			28
29	6731	/	WHEN THE P?S/8 SHELL IS IN EFFECT, THERE IS A NEED FOR A 256 WORD FILE AND	29
30	6732	/	COMMUNICATIONS AREA TO COORDINATE ALL SHELL-RELATED FUNCTIONS. IF AT ALL	30
31	6733	/	POSSIBLE, THIS SHOULD BE LOADED INTO X5400-X5777 [X IS THE HIGHEST PHYSICAL	31
32	6734	/	MEMORY FIELD AS DESCRIBED ABOVE].	32
33	6735			33
34	6736	/	AS STATED ELSEWHERE, THE HIGH-ORDER BIT OF THE OUTPUT FILE COUNT WORD "SOUTCT"	34
35	6737	/	[07607] INDICATES IF THE FILE AND COMMUNICATIONS AREA IS RESIDENT OR	35
36	6738	/	VIRTUALIZED TO THE CONTENTS OF TWO ADJACENT LOW SYSTEM BLOCKS. [THIS PRESUMES	36
37	6739	/	THE SHELL IS IN EFFECT.] [NOTE: THE EXACT DEFINITION OF THE TWO SYSTEM BLOCKS	37
38	6740	/	HAS NOT BEEN COMPLETELY DETERMINED. AT THE TIME OF THIS WRITING, POSSIBLE	38
39	6741	/	IMPLEMENTATION SHOULD BE IN THE RANGE OF BLOCK 0061 THROUGH BLOCK 0130.	39
40	6742	/	ALTERNATIVELY, THE BLOCKS COULD BE LOCATED WITHIN THE SYSTEM PROGRAM [NAMED	40
41	6743	/	"SHELL"] USED TO ENABLE OR DISABLE THE SHELL AS REQUIRED. THIS COULD HELP	41
42	6744	/	FACILITATE LOCATING THESE BLOCKS ON AN ALTERNATE SYSTEM DEVICE LOGICAL UNIT TO	42
43	6745	/	IMPROVE THROUGHPUT WHEN THE FILE AND COMMUNICATIONS AREA IS VIRTUALIZED. THE	43
44	6746	/	CHALLENGE WILL BE TO DETERMINE WHAT CAPABILITIES CAN BE RECONFIGURED FROM	44
45	6747	/	WITHIN THE SHELL WHILE IT IS ENABLED; IN GENERAL THE SHELL HAS NO ACCESS TO	45
46	6748	/	THE BASIC SYSTEM PROGRAM DIRECTORIES TO LOCATE THIS INFORMATION UNLESS CERTAIN	46
47	6749	/	SYSTEM CODING IS ESSENTIALLY CREATED REDUNDANTLY [WHICH IS A RELATIVELY EASY	47
48	6750	/	TASK]. HOWEVER, THE SHELL IS CAPABLE OF DETERMINING ITS OWN LOADING BLOCK AND	48
49	6751	/	LOGICAL UNIT; OFFSETS CAN BE USED TO DERIVE THE NEEDED INFORMATION WITHOUT	49
50	6752	/	DIRECT LOOKUP CAPABILITY. THE LOGICAL UNIT FOR VIRTUALIZATION CAN BE WRITTEN	50
51	6753	/	INTO THE SHELL CODE FROM EITHER SYSTEM OPERATING MODE. IF PERMITTED, THE	51
52	6754	/	SHELL CONFIGURATION COULD ALSO MOVE THE FILE AND COMMUNICATIONS AREA TO	52
53	6755	/	EXTENDED MEMORY FURTHER REDUCING SYSTEM OVERHEAD.]	53
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1	6757	/	THE P?S/8 SHELL IS INTENDED FOR USE WITH THE FILE AND COMMUNICATIONS AREA	1
2	6758	/	RESIDENT IN MEMORY. VIRTUALIZATION IS CONSIDERED A PERFORMANCE TRADEOFF WHICH	2
3	6759	/	SHOULD ONLY BE USED IF PHYSICAL MEMORY IS DEEMED INADEQUATE FOR THE PURPOSE AT	3
4	6760	/	HAND. SHELL CONTROL UTILITIES CAN RECONFIGURE THE SHELL TO OPERATE IN EITHER	4
5	6761	/	MODE WITHOUT A MAJOR CONFIGURATIONAL CHANGE. THIS WILL FACILITATE OPERATING	5
6	6762	/	IN EITHER MODE AS APPROPRIATE. [NOTE: IT IS CONTEMPLATED THAT USERS OF THE	6
7	6763	/	SHELL ON SYSTEMS WITH LESS THAN 12K PHYSICAL MEMORY WILL EXPERIENCE PALPABLE	7
8	6764	/	LOSS OF EFFECTIVE FUNCTIONALITY; THIS ASSUMES THE SYSTEM DEVICE HANDLER	8
9	6765	/	REQUIRES EXTENDED MEMORY AND/OR ENABLING THE LOGICAL CONSOLE OVERLAY.]	9
10	6766			10
11	6767	/	CERTAIN FORMS OF PROGRAM DEBUGGING MAY EITHER BE COMPROMISED OR PERHAPS	11
12	6768	/	TOTALLY UNAVAILABLE DEPENDING ON PHYSICAL MEMORY SIZE. [NOTE: ALL EFFORTS	12
13	6769	/	WILL BE MADE TO MINIMIZE THE EFFECTS OF INADEQUATE MEMORY WHERE POSSIBLE;	13
14	6770	/	HOWEVER, THE USE OF THE SHELL IS NOT RECOMMENDED ON MINIMAL SYSTEMS. CURRENT	14
15	6771	/	SHELL DESIGN PLANS PREDICT THAT GENERAL PROGRAM DEBUGGING WILL NOT BE POSSIBLE	15
16	6772	/	ON 4K SYSTEMS WHICH FURTHER REQUIRE THE FILE AND COMMUNICATIONS AREA TO BE	16
17	6773	/	VIRTUALIZED. AS A COMPROMISE, A 4K-ONLY DEBUGGING MODE MAY BE OFFERED THAT	17
18	6774	/	CANNOT PROCESS EXTENDED MEMORY INFORMATION. IT IS NOT CLEAR IF THERE IS	18
19	6775	/	SUFFICIENT CODE SPACE ON A 4K SYSTEM; THIS SUBJECT WILL BE RESEARCHED AT SOME	19
20	6776	/	POINT FOR POTENTIAL IMPLEMENTATION LATER.]	20
21	6777			21
22	6778	/	THERE IS NO RESIDENT STATUS INDICATOR FOR THE SHELL ITSELF; THE KEYBOARD	22
23	6779	/	MONITOR IS RESPONSIBLE FOR HANDING OFF CONTROL TO THE SHELL WHEN IT IS ENABLED	23
24	6780	/	FROM THE BASIC SYSTEM. AS SUCH, SHELL-BASED PROGRAMS ARE IMPLICITLY AWARE OF	24
25	6781	/	THEIR OPERATING ENVIRONMENT [WHICH SHARES MOST OF THE SAME SYSTEM RESIDENT	25
26	6782	/	AREA PARAMETERS WITH THE BASIC SYSTEM].	26
27	6783			27
28	6784	/	P?S/8 SYSTEM PROGRAMS FOR THE BASIC SYSTEM DO NOT RUN WITHIN THE SHELL	28
29	6785	/	ENVIRONMENT. THUS, FOR ALL BASIC SYSTEM OPERATIONS, THE IMPLEMENTATION OF THE	29
30	6786	/	SHELL CAN BE IGNORED. IF DESIRED, A SYSTEM PROGRAM CAN ALLOCATE THE FIRST 3K	30
31	6787	/	OF THE HIGHEST FIELD OF MEMORY IF THE SYSTEM HANDLER EXTENSION PREVENTS ACCESS	31
32	6788	/	TO THE LAST 1K OF THE HIGHEST FIELD [ASSUMING THE LOGICAL CONSOLE OVERLAY IS	32
33	6789	/	DISABLED].	33
34	6790			34
35	6791	/	THE IMPLEMENTATION OF THE SHELL PLAYS NO ROLE IN ANY BASIC SYSTEM PROGRAM	35
36	6792	/	MEMORY ALLOCATION SCHEME. HOWEVER, IF THE SHELL IS ENABLED AND THE FILE AND	36
37	6793	/	COMMUNICATIONS AREA IS MEMORY-RESIDENT, SHELL-BASED PROGRAMS MUST RESPECT THIS	37
38	6794	/	AREA; IF THE SHELL IS ENABLED AND THE LOGICAL CONSOLE OVERLAY IS NOT ENABLED,	38
39	6795	/	ONLY 2.75K OF MEMORY IN THE HIGHEST FIELD CAN BE ALLOCATED. [SINCE ALL SHELL	39
40	6796	/	PROGRAMS ARE NEW WORK, IT IS RECOMMENDED THAT MOST PROGRAMS NOT CHECK FOR THE	40
41	6797	/	FILE AND COMMUNICATIONS AREA TO BE VIRTUALIZED FOR THE SOLE PURPOSE OF	41
42	6798	/	OBTAINING AN ADDITIONAL .25K OF MEMORY UNLESS THIS IS IMPORTANT TO THE	42
43	6799	/	SPECIFICS OF THE UTILITY.] [NOTE: ASSUMING THE SHELL IS ENABLED, SHOULD THE	43
44	6800	/	FILE AND COMMUNICATIONS AREA BE MEMORY RESIDENT, THE LARGEST AVAILABLE LOGICAL	44
45	6801	/	MEMORY SIZE WILL BE REDUCED BY ONE FIELD REGARDLESS OF OTHER POTENTIALLY	45
46	6802	/	MEMORY-RESIDENT COMPONENTS.]	46
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1	6804	/	OTHER THAN THE RELEVANT SOURCE CODE CHANGES FOR THE NEW BUFFER ALLOCATION	1
2	6805	/	PARAMETERS, NO PROBLEMS ARE FORESEEN WITHIN ANY LOGICAL CONSOLE OVERLAY	2
3	6806	/	IMPLEMENTATION; THE VT8E VERSION IS KNOWN TO FUNCTION WITH THE REDUCED MEMORY	3
4	6807	/	ALLOCATION. TO DATE, ALL OTHER LOGICAL CONSOLE OVERLAYS REQUIRE LESS OVERALL	4
5	6808	/	MEMORY TO IMPLEMENT ALL INTERNAL FUNCTIONS. THE LOSS OF .25K OF SPACE WILL	5
6	6809	/	NOT AFFECT EXISTING LINEPRINTER BUFFERING ROUTINES FOR DEVICES SUCH AS THE	6
7	6810	/	PARALLEL PRINTER INTERFACE. [NOTE: SERIAL PRINTERS GENERALLY HAVE EVEN LARGER	7
8	6811	/	BUFFERS WITHIN THE PRINTER. THE TYPICAL REVERSE CHANNEL IMPLEMENTATION SENDS	8
9	6812	/	CONTROL-S CHARACTERS WHEN THE BUFFER IS 3/4 FULL, THEN SENDS A CONTROL-Q	9
10	6813	/	CHARACTER WHEN THE BUFFER IS 3/4 EMPTY. AS SUCH, PRINTER DATA BUFFERING	10
11	6814	/	WITHIN THE LOGICAL CONSOLE OVERLAY CAN BE MINIMAL FOR SERIAL PRINTER SUPPORT.]	11
12	6815			12
13	6816	/	END OF APPENDIX J.	13
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1	6818	/	APPENDIX K.	1
2	6819			2
3	6820	/	IMAGE BINARY LOADING CONSIDERATIONS IN P?S/8.	3
4	6821			4
5	6822	/	THERE ARE TWO CATEGORIES OF BINARY LOADING IN MOST PDP-8 OPERATING SYSTEMS:	5
6	6823			6
7	6824	/	1) LOADING BINARY USER FILES INTO MEMORY.	7
8	6825			8
9	6826	/	2) LOADING BINARY IMAGES INTO MEMORY.	9
10	6827			10
11	6828	/	BINARY FILE LOADING IS A DYNAMIC PROCESS GENERALLY ASSOCIATED WITH USER	11
12	6829	/	PROGRAMS [OR PERHAPS CONVERTED PAPER-TAPE IMAGES]; LOADING IS NOT REGIMENTED	12
13	6830	/	INTO ANY PARTICULAR UNITS OF MEMORY OTHER THAN OVERALL LOADING LIMITS. SOME	13
14	6831	/	ASPECTS OF BINARY FILE LOADING ARE DISCUSSED ELSEWHERE IN THIS DOCUMENT.	14
15	6832			15
16	6833	/	LOADING OF BINARY IMAGES INVOLVES DIRECT LOADING OF AN IMAGE OF SOME PORTION	16
17	6834	/	OF MEMORY; THE SMALLEST UNIT OF LOADING IS THE 128 WORD BLOCK CORRESPONDING	17
18	6835	/	TO A PDP-8 PAGE. [NOTE: BINARY IMAGES ARE GENERALLY ALIGNED ON PAGE	18
19	6836	/	BOUNDARIES; HOWEVER, IT IS POSSIBLE TO LOAD AN IMAGE WHILE CROSSING A PAGE	19
20	6837	/	BOUNDARY AS LONG AS NO SYSTEM RESERVED AREA IS VIOLATED. GENERALLY, MOST	20
21	6838	/	OPERATING SYSTEMS DO NOT SUPPORT MECHANISMS TO CREATE NON-ALIGNED IMAGES;	21
22	6839	/	WITHOUT AN ALIGNMENT MECHANISM, THE LARGEST IMAGES COULD NOT BE LOADED DUE TO	22
23	6840	/	CONFLICT WITH THE SYSTEM RESERVED AREA. AS SUCH, MOST PDP-8 OPERATING SYSTEMS	23
24	6841	/	WILL RESTRICT IMAGES TO LOADING ON COMPLETE PDP-8 PAGE BOUNDARIES.]	24
25	6842			25
26	6843	/	P?S/8 SUPPORTS THE DIRECT LOADING OF ALL OF USER MEMORY AS DEFINED BELOW.	26
27	6844	/	[NOTE: THE DEFINITION OF USER MEMORY SPACE APPLIES TO BOTH METHODS OF BINARY	27
28	6845	/	LOADING; ONLY BINARY IMAGES ARE LAID OUT IN A REGIMENTED FORMAT.]	28
29	6846			29
30	6847	/	USER MEMORY ADDRESS SPACE CONSIDERATIONS.	30
31	6848			31
32	6849	/	OTHER THAN SOME NITPICKS REGARDING POOR PROGRAMMING TECHNIQUES GENERALLY	32
33	6850	/	FROWNED UPON [YET KNOWN TO OCCASIONALLY BE PRESENT IN SOME "ROGUE" DEC	33
34	6851	/	DIAGNOSTIC AND OTHER PROGRAMS], THE GENERAL CONVENTION, UNIVERSALLY ACCEPTED	34
35	6852	/	IN ALL PDP-8 OPERATING SYSTEMS, IS THAT ALL LOCATIONS IN THE FIRST 31 PAGES OF	35
36	6853	/	MEMORY [00000-07577] IS FULLY AVAILABLE TO ANY PROGRAM FOR ANY PURPOSE.	36
37	6854	/	[NOTE: ALL GENERAL-PURPOSE PDP-8 OPERATING SYSTEMS RUN WITHOUT INTERRUPTS.	37
38	6855	/	APPLICATION PROGRAMS CAN MAKE FULL USE OF THE INTERRUPT SYSTEM AS LONG AS	38
39	6856	/	APPROPRIATE INTERRUPT HANDLING IS PROVIDED WITHIN THE PROGRAM; ALL INTERRUPTS	39
40	6857	/	ARE HANDLED WITHIN THE USER MEMORY SPACE. INTERRUPTS SHOULD BE DISABLED WHEN	40
41	6858	/	EXITING TO THE OPERATING SYSTEM TO AVOID CONFLICTS AS THE OPERATING SYSTEM	41
42	6859	/	RESIDENT ROUTINES LOAD OVER THE INTERRUPT HANDLER LOCATIONS IN PAGE 0.]	42
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1	6861	/	THIS CONVENTION AROSE OUT OF A VARIETY OF HARDWARE AND SOFTWARE CONCERNS:	1
2	6862			2
3	6863	/	1) TO ALLOW THE USER MEMORY SPACE TO BE AS LARGE AS POSSIBLE.	3
4	6864			4
5	6865	/	2) TO HAVE A UNIVERSAL EXIT TO A STORAGE-BASED OPERATING SYSTEM; THIS IS	5
6	6866	/	DEFINED AS 07600, THE LOWEST LOCATION ON THE LAST PAGE OF MEMORY.	6
7	6867			7
8	6868	/	3) THE NEED TO HAVE THE THREE-CYCLE DATA BREAK LOCATIONS RESERVED TO AVOID	8
9	6869	/	CONFLICTS WITH USER-WRITTEN CODE; ONLY SPECIAL PROGRAMS SPECIFICALLY	9
10	6870	/	WRITTEN FOR THE SPECIFIED DEVICES [TC01/TC08 DECTAPE, DF32/RF08,	10
11	6871	/	TC58-SERIES MAGNETIC TAPE CONTROLLERS] CAN SAFELY ACCESS THESE LOCATIONS.	11
12	6872			12
13	6873	/	4) THE NEED TO SUPPORT PAPER-TAPE LOADERS THAT ARE TOTALLY RESIDENT IN THE	13
14	6874	/	RESERVED AREA. [NOTE: A STORAGE-BASED OPERATING SYSTEM OCCUPIES THIS AREA	14
15	6875	/	IF THE PAPER-TAPE LOADERS ARE NOT IN USE.]	15
16	6876			16
17	6877	/	5) THE NEED TO HAVE A SYSTEM RESIDENT KERNEL OF A STORAGE-BASED OPERATING	17
18	6878	/	SYSTEM, ESPECIALLY IF THE THREE-CYCLE DATA BREAK LOCATIONS ARE AN INTEGRAL	18
19	6879	/	PART OF THE KERNEL DEVICE [SUCH AS IS THE CASE WITH THE TC01/TC08 DECTAPE	19
20	6880	/	CONTROLLER].	20
21	6881			21
22	6882	/	NON-CONFORMING ["ROGUE"] PROGRAM CONSIDERATIONS.	22
23	6883			23
24	6884	/	CERTAIN PROGRAMS EXIST THAT ARE KNOWN TO VIOLATE THESE UNIVERSAL CONVENTIONS	24
25	6885	/	FOR NO APPARENT REASON OTHER THAN SHEER IGNORANCE ON THE PART OF CERTAIN	25
26	6886	/	PROGRAMMERS AND THEIR MANAGERS. THE UNDERLYING MECHANISM AT FAULT IS THE	26
27	6887	/	UNREASONABLE DEPENDENCY ON SEEMINGLY AVAILABLE MEMORY LOCATIONS PAST 07600.	27
28	6888			28
29	6889	/	IN EVERY CASE, THE PROGRAMMER ASSUMED THAT THE INCIDENTAL MEMORY SPACE BETWEEN	29
30	6890	/	07600 AND SOME SLIGHTLY HIGHER ADDRESS WAS SOMEHOW AVAILABLE MERELY BECAUSE	30
31	6891	/	THE PARTICULAR VERSION OF THE BINARY PAPER TAPE LOADER THEY WERE FAMILIAR WITH	31
32	6892	/	COULD ALLOW THIS.	32
33	6893			33
34	6894	/	UNFORTUNATELY, THE EXACT MEMORY MAP OF THE DEC BIN LOADER VARIES WITH SPECIFIC	34
35	6895	/	RELEASE; AS SUCH, THIS ILL-ADVISED PROGRAMMING TECHNIQUE COULD EASILY FAIL	35
36	6896	/	MERELY BY BEING LOADED BY A DIFFERENT VERSION OF THE BIN LOADER PROGRAM.	36
37	6897	/	NOTE: THERE HAVE BEEN AT LEAST THREE REVISIONS OF THE BINARY PAPERTAPE	37
38	6898	/	LOADING PROGRAM, EACH OCCUPYING A SLIGHTLY DIFFERENT PORTION OF THE MEMORY	38
39	6899	/	SPACE BETWEEN 07600-07755. [07756-07776 IS RESERVED FOR THE RIM LOADER; 07777	39
40	6900	/	IS RESERVED AS THE OFFICIAL START OF THE BIN LOADER; THE ACTUAL START IS	40
41	6901	/	ACHIEVED INDIRECTLY BY USING A "JMP BEGIN" INSTRUCTION AT 07777.]	41
42	6902			42
43	6903	/	AS SUCH, NOT ONLY ARE THESE OFFENDING PROGRAMS INCAPABLE OF BEING LOADED FROM	43
44	6904	/	ANY STORAGE-BASED OPERATING SYSTEM, THEY MAY BE EXCESSIVELY DEPENDENT ON A	44
45	6905	/	PARTICULAR RELEASE OF THE BIN LOADER NOT NECESSARILY AVAILABLE. [NOTE: OF THE	45
46	6906	/	THREE MOST COMMON RELEASES, ONE IS INCOMPATIBLE WITH THE TC01/TC08 CONTROLLER;	46
47	6907	/	ANOTHER IS KNOWN TO HAVE BUGS IN IT RELATED TO IMPROPERLY HANDLING FIELD	47
48	6908	/	SETTINGS; P?S/8 IMPLEMENTS A FOURTH VERSION THAT PARTIALLY CORRECTS SOME OF	48
49	6909	/	THE PROBLEMS ASSOCIATED WITH ALL OF THESE DEC RELEASES INCLUDING CERTAIN	49
50	6910	/	INCOMPATIBILITY PROBLEMS WITH DF32/RF08 PROGRAMS WHICH CAN CORRUPT THE BIN	50
51	6911	/	LOADER PROGRAM ONCE EXECUTED.]	51
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1	6913	/	WITH REGARD TO THE DEC-RELEASED ROGUE DIAGNOSTIC PROGRAMS:	1
2	6914			2
3	6915	/	THAT THESE PROGRAMS EVEN EXIST IS UNFORTUNATELY A LEGACY OF POOR MANAGEMENT OF	3
4	6916	/	ONE OR MORE PDP-8 SOFTWARE DEVELOPMENT GROUPS. WHILE IT IS PERFECTLY	4
5	6917	/	REASONABLE TO EXPECT OFFICIAL DIAGNOSTIC PROGRAMS TO CONFORM TO PREVAILING	5
6	6918	/	CONVENTIONS, UNFORTUNATELY THIS LEVEL OF SCRUTINY WAS NOT ALWAYS APPLIED.	6
7	6919	/	IRONICALLY, MOST OF THESE PROGRAMS WERE RELEASED YEARS AFTER THE SIMPLE SET	7
8	6920	/	OF PROGRAMMING RULES WERE FIRST DOCUMENTED. [NOTE: THIS IS CONSISTENT WITH	8
9	6921	/	THE WELL-KNOWN GRADUAL BREAKDOWN OF DEC INTERNAL COMPANY STANDARDS.]	9
10	6922			10
11	6923	/	THESE PROGRAMS ARE NOT LOADABLE BY ANY OF THE PREVAILING PDP-8 STORAGE-BASED	11
12	6924	/	OPERATING SYSTEMS. P?S/8 HAS A TENUOUS FACILITY [READ "KLUDGE" HERE] WHEREBY	12
13	6925	/	THESE PROGRAMS CAN BE LOADED, ALBEIT WITH DIFFICULTY. THE EXACT METHOD USED	13
14	6926	/	MUST BE TAILORED NOT ONLY TO THE PROGRAM'S SPECIFIC LOADING RANGES, BUT ALSO	14
15	6927	/	TO P?S/8 RELEASE SPECIFICS; LOADING ANY OF THESE PROGRAMS WILL CORRUPT THE	15
16	6928	/	MEMORY-RESIDENT OPERATING SYSTEM; MANUAL REBOOT [OR EQUIVALENT] WILL BE	16
17	6929	/	REQUIRED AFTER LOADING. [NOTE: FOR MORE INFORMATION, SEE DOCUMENTATION OF THE	17
18	6930	/	P?S/8 "BIN" PROGRAM REGARDING VIRTUAL LOADING OF IMPROPERLY WRITTEN PROGRAMS	18
19	6931	/	SUCH AS DEC DIAGNOSTICS.]	19
20	6932			20
21	6933	/	P?S/8 IMAGE BINARY PROGRAM LOADING CONSIDERATIONS.	21
22	6934			22
23	6935	/	SINCE P?S/8 SYSTEMS EXIST WITH OR WITHOUT THE OPTIONAL SHELL ENVIRONMENT,	23
24	6936	/	THERE ARE TWO DIFFERENT APPROACHES TO IMAGE BINARY PROGRAM LOADING:	24
25	6937			25
26	6938	/	1) EXECUTION-CLASS COMMANDS IN THE BASIC P?S/8 ENVIRONMENT.	26
27	6939			27
28	6940	/	2) EXECUTION-CLASS COMMANDS IN THE P?S/8 SHELL ENVIRONMENT.	28
29	6941			29
30	6942	/	THE GENERAL MECHANISM USED TO LOAD IMAGE BINARY PROGRAMS IN THESE TWO VERY	30
31	6943	/	DIFFERENT ENVIRONMENTS SHARES A COMMON UNDERLYING MECHANISM GENERALLY KNOWN AS	31
32	6944	/	THE "SYSTEM LOADER". [NOTE: THIS IS THE HISTORICAL NAME DERIVED FROM A	32
33	6945	/	VAGUELY ANALOGOUS COMPONENT OF THE R-L MONITOR SYSTEM; HOWEVER, THE ACTUAL	33
34	6946	/	IMPLEMENTATION IS TOTALLY DIFFERENT FROM THIS FORERUNNER OPERATING SYSTEM FROM	34
35	6947	/	WHICH P?S/8 IS PARTIALLY DERIVED; DETAILS OF THE R-L MONITOR SYSTEM COMPONENT	35
36	6948	/	ARE DESCRIBED BELOW ALONG WITH EQUIVALENT MECHANISMS IN SEVERAL OTHER	36
37	6949	/	OPERATING SYSTEMS.]	37
38	6950			38
39	6951	/	LOADING IMAGE BINARY PROGRAMS IN THE BASIC P?S/8 ENVIRONMENT.	39
40	6952			40
41	6953	/	IN THE BASIC P?S/8 ENVIRONMENT THE SYSTEM PROGRAM DIRECTORY CONTAINS ALL	41
42	6954	/	PARAMETERS TO LOAD ANY AVAILABLE SYSTEM PROGRAM. EXECUTION-CLASS COMMANDS	42
43	6955	/	CAN BE INVOKED BY THE USER TO LOAD ANY SYSTEM PROGRAM CONTAINED WITHIN THIS	43
44	6956	/	DIRECTORY; A VARIETY OF OPTIONAL PASSED PARAMETERS, OPTION SWITCHES, INPUT	44
45	6957	/	AND/OR OUTPUT FILES, ETC. CAN BE PASSED TO THE PROGRAM BY APPROPRIATE	45
46	6958	/	COMMAND-LINE PARAMETERS. [NOTE: THIS METHOD IS A DIRECT DESCENDANT OF THE	46
47	6959	/	ORIGINAL R-L MONITOR SYSTEM; SYNTACTICAL CHANGES, IMPROVEMENTS AND ADDITIONAL	47
48	6960	/	FEATURES HAVE BEEN ADDED AS P?S/8 HAS BEEN DEVELOPED; HOWEVER, THE CONCEPT OF	48
49	6961	/	A SEPARATE DIRECTORY EXCLUSIVELY FOR SYSTEM PROGRAM IMAGES HAS BEEN RETAINED.]	49
50				50
51				51
52				52
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54				54
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1	6963	/	THE SYSTEM PROGRAM DIRECTORY ALLOWS FOR A FLEXIBLE OVERLAY STRUCTURE; IT IS	1
2	6964	/	THE RESPONSIBILITY OF THE PROGRAMMER TO ORGANIZE OVERLAYS AS REQUIRED. SINCE	2
3	6965	/	THE SYSTEM PROGRAM DIRECTORY ONLY DEFINES LOADING INTO THE USER MEMORY AREA	3
4	6966	/	[00000-07577], IT IS UP TO THE PROGRAM ITSELF TO LOAD ANY ADDITIONAL SEGMENTS	4
5	6967	/	[OR DATA] FROM OVERLAYS THAT IMMEDIATELY FOLLOW THE PRIMARY SEGMENT. NOTE: AS	5
6	6968	/	AN IMPLEMENTATION RESTRICTION, SYSTEM PROGRAMS ARE REQUIRED TO START WITHIN	6
7	6969	/	FIELD 0 [GENERALLY 00000-07577]. [NOTE: STARTING AT 07600 IS ALLOWED; AFTER	7
8	6970	/	LOADING, THE PROGRAM HARMLESSLY EXITS. THIS IS THE RECOMMENDED METHOD FOR	8
9	6971	/	TEMPORARILY DISABLING A SYSTEM PROGRAM WHILE UNDER GENERAL DEVELOPMENT THAT IS	9
10	6972	/	PRESENTLY DORMANT.] [NOTE: SINCE THE SYSTEM PROGRAM DIRECTORY CONTAINS	10
11	6973	/	LOADING INFORMATION ABOUT MEMBER PROGRAMS, THIS IS NOT THE SAME AS LOADING	11
12	6974	/	FILES FROM THE TFS OR ANY EXTENDED VARIANT DIRECTORIES. TO DATE,	12
13	6975	/	ADDING/DELETING SYSTEM PROGRAMS HAS ONLY BEEN ACCOMPLISHED BY EXPERTS FAMILIAR	13
14	6976	/	WITH THIS DATA STRUCTURE. IT IS POSSIBLE TO WRITE A GENERAL UTILITY OR GROUP	14
15	6977	/	OF UTILITIES TO SIMPLIFY THE PROCESS; IT WILL BE NECESSARY TO MANAGE THE	15
16	6978	/	UNDERLYING STORAGE ALLOCATED FOR ALL SYSTEM PROGRAMS TO ENSURE SUFFICIENT	16
17	6979	/	SPACE EXISTS FOR A TENTATIVE SYSTEM PROGRAM. [NOTE: WHEN THE SYSTEM IS FIRST	17
18	6980	/	CONFIGURED, ADDITIONAL SPACE IS OFTEN PROVIDED FOR ADDING A SMALL NUMBER OF	18
19	6981	/	ADDITIONAL UTILITIES.] AS A RESULT, THE STARTING BLOCK OF TFS DIRECTORY	19
20	6982	/	FILE STORAGE MUST BE ADJUSTED ACCORDINGLY; ANY STRUCTURES FURTHER UP THE	20
21	6983	/	DEVICE MAY NEED ADDITIONAL ADJUSTMENT FOR CONSISTENCY. ONCE THIS IS	21
22	6984	/	ACCOMPLISHED, A SYSTEM PROGRAM BUILDER UTILITY CAN BE DEFINED; THIS WILL	22
23	6985	/	LIKELY BE IN CONJUNCTION WITH SIGNIFICANT CHANGES MADE TO THE OVERALL P?S/8	23
24	6986	/	SYSTEM GENERATION METHODOLOGY. ONCE COMPLETED, THIS UTILITY CAN BE USED TO	24
25	6987	/	CREATE OR MODIFY THE BASIC P?S/8 SYSTEM FAR MORE FLEXIBLY AND BE PERFORMED BY	25
26	6988	/	LESS QUALIFIED PERSONNEL. HISTORICALLY, ALL RELATED FUNCTIONS HAVE BEEN	26
27	6989	/	CARRIED OUT USING THE P?S/8 "DUMP" PROGRAM EITHER MANUALLY OR BY USING DUMP	27
28	6990	/	SCRIPT FILES. FOR MORE INFORMATION, SEE DOCUMENTATION OF THE P?S/8	28
29	6991	/	DUMP UTILITY.]	29
30	6992			30
31	6993	/	LOADING IMAGE BINARY PROGRAMS IN THE P?S/8 SHELL ENVIRONMENT.	31
32	6994			32
33	6995	/	WHEN THE P?S/8 SHELL IS IN EFFECT, IMAGE BINARY PROGRAMS ARE LOADED BY VARIOUS	33
34	6996	/	NON-SYSTEM DEVICE HANDLERS. DEPENDING UPON SPECIFICS, A FINAL SEGMENT MAY BE	34
35	6997	/	LOADED BY THE SYSTEM LOADER INTO FIELD 0. [NOTE: AS STATED ABOVE, THE SHELL	35
36	6998	/	CAN SUPPORT A LOADING PROCESS USING AVAILABLE EXTENDED MEMORY; MOVING DATA	36
37	6999	/	THROUGH MEMORY CAN BE ACCOMPLISHED WITHOUT WRITING ON THE SYSTEM SCRATCH	37
38	7000	/	AREA.]	38
39	7001			39
40	7002	/	THE SHELL ENVIRONMENT USES AN ALL-PURPOSE GENERAL DIRECTORY. EACH IMAGE	40
41	7003	/	PROGRAM INCLUDES AN INTERNAL LOAD MAP FULLY DESCRIBING UP TO 32K OF MEMORY	41
42	7004	/	USAGE. AS SUCH, IMAGE PROGRAMS OCCUPY MORE STORAGE SPACE THAN IN THE STANDARD	42
43	7005	/	P?S/8 ENVIRONMENT [WHICH DESCRIBES ONE CONTIGUOUS MEMORY SEGMENT IN FIELD 0].	43
44	7006	/	SINCE EACH IMAGE BINARY FILE INCLUDES ITS OWN LOADING PARAMETERS AND	44
45	7007	/	ADDITIONAL INFORMATION REGARDING INTERACTION WITH THE SHELL COMMAND	45
46	7008	/	INTERPRETER, THE AGGREGATE SIZE OF A SHELL-BASED IMAGE PROGRAM WILL EXCEED THE	46
47	7009	/	LOADABLE SIZE BY [A PROJECTED] TWO BLOCKS. [NOTE: THIS MAY EXPAND AS	47
48	7010	/	ADDITIONAL FEATURES ARE IMPLEMENTED SUCH AS AN EXTENDED INTERNAL OVERLAY	48
49	7011	/	STRUCTURE.]	49
50	7012			50
51	7013	/	ASSUMING A FIELD 0 SEGMENT IS INDICATED [NOT ABSOLUTELY REQUIRED BUT TYPICALLY	51
52	7014	/	PRESENT], THE SYSTEM LOADER IS USED TO PLACE THIS FINAL SEGMENT INTO MEMORY	52
53	7015	/	[EXCEPT IN CERTAIN CONFIGURATIONS AS INDICATED ABOVE]. UNLIKE THE STANDARD	53
54	7016	/	P?S/8 ENVIRONMENT, SHELL EXECUTABLE PROGRAMS CAN START ANYWHERE IN THE 32K	54
55	7017	/	MEMORY SPACE WITHOUT RESTRICTION.	55
56				56
57				57
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1	7019	/	REGARDLESS OF WHICH ENVIRONMENT IS IN EFFECT, THE SYSTEM LOADER IS GENERALLY	1
2	7020	/	USED TO FINALIZE PROGRAM LOADING AND INITIATE PROGRAM EXECUTION WITH ALL	2
3	7021	/	USER-SPECIFIED OPTIONS AVAILABLE TO THE PROGRAM AS REQUIRED. [SEE VARIOUS	3
4	7022	/	PORTIONS OF THIS DOCUMENT FOR A DESCRIPTION OF THE OPTIONAL PASSED PARAMETERS	4
5	7023	/	AND FILE HANDLE MECHANISMS IMPLEMENTED.]	5
6	7024			6
7	7025	/	COMPARING THE P?S/8 SYSTEM LOADER FUNCTION TO OTHER OPERATING SYSTEMS.	7
8	7026			8
9	7027	/	DESPITE THE OBVIOUS BASIC FUNCTIONALITY OF LOADING IMAGE BINARY PROGRAMS AS	9
10	7028	/	DESCRIBED, OTHER SMALL-SCALE PDP-8 OPERATING SYSTEMS FAIL TO PROPERLY PROVIDE	10
11	7029	/	THIS FACILITY [OR EQUIVALENT] AND/OR THE IMPLEMENTATION IS SEVERELY FLAWED.	11
12	7030	/	[NOTE: MANY READERS WILL SCOFF AT THIS STATEMENT BEFORE READING WHAT FOLLOWS.]	12
13	7031	/	THE RELEVANT ISSUES OF ALL OTHER PREVAILING SYSTEMS ARE DISCUSSED BELOW:	13
14	7032			14
15	7033	/	1) THE DISK MONITOR SYSTEM DOES NOT DEFINE ANY OPERATING SYSTEM SERVICES TO	15
16	7034	/	EXECUTABLE PROGRAMS. ALL INTERFACING WITH THE USER OR THE REST OF THE	16
17	7035	/	OPERATING ENVIRONMENT MUST BE LOADED BY THE PROGRAM. OTHER THAN THE	17
18	7036	/	AVAILABILITY OF THE CRUDE COMPONENTS TO INTERFACE FILES WITH SEVERAL SYSTEM	18
19	7037	/	PROGRAMS [THAT ABSOLUTELY REQUIRE DETAILED INFORMATION BE KNOWN REGARDING	19
20	7038	/	WHERE THESE COMPONENTS EXIST ON CERTAIN SYSTEM BLOCKS], THERE IS NO	20
21	7039	/	INFORMATION PROVIDED WITHOUT EXPENDING SIGNIFICANT PROGRAMMING EFFORT TO	21
22	7040	/	OBTAIN THE RELEVANT INFORMATION.	22
23	7041			23
24	7042	/	THE BEST THAT CAN BE SAID ABOUT THE DISK MONITOR SYSTEM KERNEL IS THAT BY	24
25	7043	/	ESSENTIALLY IMPLEMENTING NOTHING INTRINSIC IN THE SYSTEM KERNEL OTHER THAN	25
26	7044	/	THE ABILITY TO EXIT TO 07600 AND THE BASIC SYSTEM DEVICE HANDLER READ AND	26
27	7045	/	WRITE FUNCTIONS [ALBEIT EXHIBITING POOR PERFORMANCE, ESPECIALLY ON DECTAPE]	27
28	7046	/	IS THAT IT FAILS TO DO ANYTHING PARTICULARLY FAULTY. [CONTRAST THIS WITH	28
29	7047	/	THE OS/8 EQUIVALENT AS DESCRIBED BELOW.]	29
30	7048			30
31	7049	/	THE DISK MONITOR SYSTEM HAS THE DUBIOUS DISTINCTION OF BEING IMPOSSIBLE TO	31
32	7050	/	IMPLEMENT ON OTHER THAN A VERY SMALL FEW PDP-8 STORAGE DEVICES BECAUSE ALL	32
33	7051	/	DATA IS ORGANIZED IN 129 WORD LOGICAL BLOCKS. VIRTUALLY ALL PDP-8 STORAGE	33
34	7052	/	PERIPHERALS DEVELOPED AFTER THE TC01/TC08 AND DF32/RF08 ARE INCAPABLE OF	34
35	7053	/	THIS AT THE HARDWARE LEVEL.	35
36	7054			36
37	7055	/	2) OS/8 NEARLY ACCOMPLISHES THE IMAGE BINARY LOADING REQUIREMENT BUT ACTUALLY	37
38	7056	/	FAILS TO COMPLETELY IMPLEMENT A VIABLE SOLUTION DESPITE REQUIRING AT LEAST	38
39	7057	/	8K OF MEMORY. A FATAL FLAW IS ALWAYS A DEVICE ERROR AWAY FROM SURFACING.	39
40	7058			40
41	7059	/	THE ROOT CAUSE OF THE PROBLEM IS OS/8'S CRUDE IMPLEMENTATION OF AN ERROR	41
42	7060	/	RETURN AS PART OF THE CALLING REQUIREMENT OF EVERY INSTANCE OF A CALL TO	42
43	7061	/	ALL HANDLERS, EVEN INTERNAL USAGE BY THE OPERATING SYSTEM ITSELF. THIS IS	43
44	7062	/	CONCEPTUALLY IMPOSSIBLE TO IMPLEMENT AS THERE IS NO DEFINED CODE FOR	44
45	7063	/	HANDLING THE ERROR RETURN FOR THE PORTION OF OS/8 IMAGE LOADING ANALOGOUS	45
46	7064	/	TO THE P?S/8 SYSTEM LOADER. SHOULD AN ERROR OCCUR WHEN LOADING IN THE LAST	46
47	7065	/	SEGMENT OF ANY EXECUTABLE PROGRAM, OS/8 WILL HALT WITH A FATAL ERROR	47
48	7066	/	SOMEWHERE IN THE LAST PAGE OF FIELD 0. [NOTE: IT IS GENERALLY A TRIBUTE	48
49	7067	/	TO THE RELIABILITY OF MOST PDP-8 PERIPHERAL HARDWARE TO DATE THAT THE	49
50	7068	/	EXISTENCE OF THIS FATAL DESIGN BLUNDER IS NOT WIDELY KNOWN.]	50
51				51
52				52
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1	7070	/	THE CONSEQUENCES OF THIS PROBLEM CAN BE QUITE SERIOUS. SHOULD THE USER	1
2	7071	/	PRESS "CONTINUE" ON THE CPU FRONT PANEL PRESUMING THIS ERROR IS SOMEHOW	2
3	7072	/	INNOCUOUS, EXECUTION OF THE PRIOR CONTENTS OF MEMORY WILL OCCUR INSTEAD OF	3
4	7073	/	THE INTENDED FUNCTION. THE ONLY NOTICE GIVEN OF THIS UNEXPECTED PROBLEM IS	4
5	7074	/	THE CPU HALTING WITH NO POSSIBLE RECOURSE. [NOTE: ALL P?S/8 SYSTEM LOADER	5
6	7075	/	CALLS ARE HANDLED AS ANY OTHER CALL TO THE SYSTEM DEVICE HANDLER. EVEN	6
7	7076	/	THE BASIC 4K SYSTEM SUPPORTS RECOVERABLE HALTS. PRESSING "CONTINUE" ALWAYS	7
8	7077	/	RETRIES THE LATEST OPERATION. IF THE LOGICAL CONSOLE OVERLAY IS IN EFFECT	8
9	7078	/	ALL SUCH PROBLEMS ARE HANDLED ELEGANTLY BY THE OVERLAY; IN THIS	9
10	7079	/	CONFIGURATION THERE ARE NO DEFINED SYSTEM HALTS.]	10
11	7080			11
12	7081	/	3) THE R-L MONITOR SYSTEM USES AN INDEPENDENT SYSTEM LOADER THAT DISPLACES	12
13	7082	/	THE SYSTEM DEVICE HANDLER ENTIRELY. THE ONLY FUNCTIONALITY IS TO LOAD THE	13
14	7083	/	BINARY IMAGE PROGRAM AND START AT THE DESIGNATED STARTING ADDRESS. THE	14
15	7084	/	R-L SYSTEM CAN PASS A SUBSET OF THE INFORMATION THAT P?S/8 DOES. HOWEVER,	15
16	7085	/	LOADED PROGRAMS HAVE NO ABILITY TO ACCESS ANY PART OF THE OPERATING SYSTEM	16
17	7086	/	WITHOUT PROVIDING THEIR OWN DEVICE-DEPENDENT STORAGE DEVICE SUBROUTINES.	17
18	7087			18
19	7088	/	IF AN ERROR SHOULD OCCUR ON THE SYSTEM DECTAPE, THE SYSTEM LOADER WILL	19
20	7089	/	HALT THE MACHINE REGARDLESS OF ERROR [INCLUDING END ERROR]. THE OPERATION	20
21	7090	/	CAN BE RETRIED [IF POSSIBLE] OR EXIT CAN BE MADE TO 07600. THUS, AS	21
22	7091	/	NECESSARY, PRESSING "CONTINUE" IS RECOMMENDED TO RECOVER FROM AN ERROR.	22
23	7092			23
24	7093	/	THE R-L MONITOR FAILS TO PROVIDE EVEN THE MOST FUNDAMENTAL RESIDENT SERVICE	24
25	7094	/	OR THE ABILITY TO READ ANY PORTION OF THE OPERATING SYSTEM STRUCTURES. AS	25
26	7095	/	SUCH, IT ALSO FAILS TO PROVIDE THE NECESSARY SYSTEM SERVICES AS ARE	26
27	7096	/	AVAILABLE IN EITHER OF THE P?S/8 OPERATING ENVIRONMENTS.	27
28	7097			28
29	7098	/	4) CPS-4K IS AN OPERATING SYSTEM FOR THE 4K UNMODIFIED LINC-8 OR OTHER MACHINE	29
30	7099	/	EQUIPPED WITH THE EQUIVALENT HARDWARE TAKEN FROM THE LINC CPU REGISTERS	30
31	7100	/	ASSOCIATED WITH CONTROLLING THE LINTAPE HARDWARE. [NOTE: A POSITIVE BUSS	31
32	7101	/	PERIPHERAL THAT IS ADEQUATELY COMPATIBLE WITH THE LINC CPU REGISTERS WAS	32
33	7102	/	SOLD AS A THIRD-PARTY STORAGE DEVICE FOR THE PDP-8/L AND SIMILAR SYSTEMS	33
34	7103	/	WITH 4K AND A POSITIVE BUSS OR LEVEL CONVERTER. DUE TO THE LACK OF	34
35	7104	/	ADDITIONAL INSTRUCTIONS ASSOCIATED WITH CERTAIN LINC-8 MODIFICATIONS, P?S/8	35
36	7105	/	CAN ONLY RUN ON THIS DEVICE IF 8K OR MORE MEMORY IS AVAILABLE.] WHILE THE	36
37	7106	/	CPS-4K OPERATING SYSTEM CAN ACCESS ANY PARTICULAR COMPONENT OF THE DUAL	37
38	7107	/	LINTAPE DRIVE, EXECUTABLE PROGRAMS CAN ONLY BE PARTIALLY LOADED BY A	38
39	7108	/	SYSTEM LOADER SIMILAR TO THE R-L MONITOR ENVIRONMENT [ALBEIT SOMEWHAT MORE	39
40	7109	/	RESTRICTIVE].	40
41	7110			41
42	7111	/	UNDER CPS-4K, THERE IS A DISTINCTION MADE BETWEEN PROGRAMS THAT LOAD	42
43	7112	/	EXCLUSIVELY INTO 00000-07377 AND THOSE THAT LOAD INTO ANY PORTION OF	43
44	7113	/	07400-07577. THE CONTENTS OF 07600-07777 ARE CHANGED TO AN IMAGE LOADING	44
45	7114	/	PROGRAM THUS NOT PROVIDING FUNDAMENTAL RESIDENT SERVICES MUCH LIKE THE R-L	45
46	7115	/	MONITOR SYSTEM. ANALOGOUS TO THE R-L ENVIRONMENT, ALL IMAGE BINARY	46
47	7116	/	PROGRAMS MUST PROVIDE THEIR OWN COPY OF LINTAPE READ/WRITE ROUTINES.	47
48	7117	/	[NOTE: THE NOTION OF DEVICE DEPENDENCY IS MOOT IN CPS-4K AS IT ONLY RUNS ON	48
49	7118	/	LINC-8 LINTAPE HARDWARE. EXPERIMENTAL VERSIONS OF THE R-L MONITOR SYSTEM	49
50	7119	/	HAVE BEEN CREATED FOR OTHER DEVICES SUCH AS THE DF32 MINIDISK. AS IN THE	50
51	7120	/	CASE OF THE STANDARD R-L MONITOR SYSTEM, EACH SYSTEM PROGRAM MUST PROVIDE A	51
52	7121	/	COPY OF AN APPROPRIATE READ/WRITE SUBROUTINE FOR THE SYSTEM DEVICE. AS	52
53	7122	/	SUCH, CPS-4K AND THE R-L MONITOR SYSTEM ARE EQUALLY INCAPABLE OF ANY FORM	53
54	7123	/	OF DEVICE INDEPENDENCE.]	54
55				55
56				56
57				57
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1	7125	/	BEYOND THIS THERE IS AN ADDITIONAL CONSIDERATION: THE PROGRAM LOADER	1
2	7126	/	PLACED IN 07600-07777 FOR THE PURPOSES OF LOADING IN IMAGES THAT DO NOT	2
3	7127	/	INVOLVE LOADING 07400-07577 IS MOSTLY A SYSTEM BOOTSTRAP; IT IS ENTIRELY	3
4	7128	/	DEPENDENT UPON A COMPANION ROUTINE LOADED INTO 07400-07577 TO CARRY OUT	4
5	7129	/	MOST OF THE ACTUAL LOADING.	5
6	7130			6
7	7131	/	WHEN THE IMAGE PROGRAM INVOLVES LOADING INTO 07400-07577, THE LAST PAGE OF	7
8	7132	/	LOADING IS ACCOMPLISHED BY A SPECIAL DEDICATED LOADING ROUTINE LOCATED	8
9	7133	/	WITHIN 07600-07777 THAT LACKS THE ABILITY TO CHECKSUM THE CONTENTS OF THE	9
10	7134	/	LOADING BLOCK; THUS EACH AND EVERY ATTEMPT TO RUN PROGRAMS THAT INCLUDE	10
11	7135	/	07400-07577 AS A LOADABLE SECTION IS SUBJECT TO THE POSSIBILITY OF	11
12	7136	/	UNDETECTED LOAD FAILURE AND THE POSSIBILITY OF EXECUTION OF UNINTENDED	12
13	7137	/	CONTENTS OF MEMORY [SOMEWHAT ANALOGOUS TO OS/8'S INTERNAL LIMITATION WHEN A	13
14	7138	/	DEVICE FAILURE OCCURS DURING THE LOADING PROCESS].	14
15	7139			15
16	7140	/	AT THIS POINT IT SHOULD BE APPARENT THAT CREATING A TRULY DEVICE-INDEPENDENT	16
17	7141	/	SYSTEM PROGRAM LOADER WITHOUT LIMITATIONS IS NOT EASILY ACHIEVED. THE	17
18	7142	/	ADVANCED TECHNIQUES USED IN THE P?S/8 TC01/TC08 SYSTEM DEVICE HANDLER ALLOW	18
19	7143	/	FOR A PROPER KERNEL COMPONENT THAT COEXISTS WITH THE MAIN DEVICE HANDLER AND	19
20	7144	/	ASSOCIATED FEATURES. THAT P?S/8 ACCOMPLISHES THIS WHILE ALSO IMPLEMENTING A	20
21	7145	/	SUBSTANTIAL SUPERSET OF THE ORIGINAL R-L MONITOR PASSED PROGRAM PARAMETERS	21
22	7146	/	MAKES THIS ALL THE MORE REMARKABLE. [NOTE: THIS DOCUMENT ONLY DISCUSSES THE	22
23	7147	/	TC01/TC08 SYSTEM DEVICE HANDLER. THE SYSTEM LOADER IS DEVICE-INDEPENDENT AND	23
24	7148	/	PRESENT IN ALL P?S/8 SYSTEM DEVICE HANDLERS FOR ALL SUPPORTED DEVICES. ALL	24
25	7149	/	MAJOR COMPONENTS OF P?S/8 DEPEND ON THE PRESENCE OF THE SYSTEM LOADER AS	25
26	7150	/	DESCRIBED ELSEWHERE IN THIS DOCUMENT.]	26
27	7151			27
28	7152	/	HISTORY OF THE SYSTEM PROGRAM LOADING FUNCTION.	28
29	7153			29
30	7154	/	WHILE THE ENTIRE DEVELOPMENT STORY OF THE SYSTEM LOADER CANNOT EASILY BE	30
31	7155	/	RECOUNTED IN DETAIL, CERTAIN ASPECTS OF THIS EVOLUTIONARY PROCESS CAN BE	31
32	7156	/	HIGHLIGHTED, STARTING FROM THE ORIGINAL R-L MONITOR SYSTEM'S CRUDE EQUIVALENT.	32
33	7157			33
34	7158	/	THE R-L MONITOR SYSTEM DEVICE HANDLER IS INCAPABLE OF SUPPORTING AN EMBEDDED	34
35	7159	/	CALL TO THE SYSTEM DEVICE HANDLER. ALTERNATE MEANS WERE PRAGMATICALLY	35
36	7160	/	CREATED SINCE A WORKING SYSTEM WAS MORE IMPORTANT AT THE TIME; MOST OF THE	36
37	7161	/	ORIGINAL EFFORT DEALT WITH THE LARGER WORK OF A COMPLEX OF MANY COMPONENTS.	37
38	7162	/	LIKE MANY OTHER PROJECTS, THERE ARE INITIAL TRADEOFFS AT LEAST WHEN SOME	38
39	7163	/	SEMBLANCE OF OVERALL FUNCTIONALITY HAS BEEN ACHIEVED. [TAKEN AS A WHOLE, THE	39
40	7164	/	GRADUAL EMERGENCE OF P?S/8 REPRESENTS THE CONTINUING WORK STARTED BY THE	40
41	7165	/	PIONEERING EFFORTS OF THE EARLY IMPLEMENTORS STILL AS AN EVOLVING PROCESS EVEN	41
42	7166	/	NOW. BY ELIMINATING THE NOTION OF IMPOSED DEADLINES, INCREASED FUNCTIONALITY	42
43	7167	/	CAN ALWAYS BE ADDED.]	43
44	7168			44
45	7169	/	WHEN THE R-L MONITOR WAS CREATED USING THE MOST PRIMITIVE POSSIBLE ENVIRONMENT	45
46	7170	/	[CONSISTING OF MANUAL ASSEMBLY OF THE CODE AND MANUAL ENTRY OF THE BINARY	46
47	7171	/	VIA THE PDP-8 FRONT PANEL DIRECTLY], THE IMPETUS CLEARLY WAS TO CREATE SOME	47
48	7172	/	USEFUL FUNCTIONALITY. [UNFORTUNATELY, THE ORIGINAL IMPLEMENTORS WENT ON TO	48
49	7173	/	PURSUUE OTHER ACTIVITIES, PRIMARILY AS EMPLOYEES OF DEC WHERE MANAGERS DECIDED	49
50	7174	/	WHAT WAS THE BEST USE OF THEIR PRODUCTIVE TIME, ARGUABLY A QUITE UNFORTUNATE	50
51	7175	/	SET OF EVENTS; P?S/8 WAS CREATED BY OTHERS WHO "CUT THEIR PDP-8 TEETH" ON THE	51
52	7176	/	R-L MONITOR SYSTEM AND THUS HAD MUCH MORE OF A DEVELOPMENT ENVIRONMENT TO WORK	52
53	7177	/	FROM. HAVING MERELY THE ABILITY TO MAKE SMALL SOURCE CODE CHANGES THAT CAN	53
54	7178	/	HAVE LARGE IMPACT ON THE BINARY OUTCOME PROFOUNDLY CHANGES WHAT IS POSSIBLE.]	54
55				55
56				56
57				57
58				58

1	7180	/	THE MANY LIMITATIONS OF THE R-L MONITOR IMAGE PROGRAM LOADING PROCESS CAN BE	1
2	7181	/	"EXCUSED" GIVEN THE LACK OF SUFFICIENT SYSTEM DEVICE HANDLER PROGRAM	2
3	7182	/	DEVELOPMENT AT THE TIME. PRODUCING A MORE POWERFUL REPLACEMENT DEVICE HANDLER	3
4	7183	/	WAS ULTIMATELY THE RESULT OF TREMENDOUS EFFORT ON THE PART OF MANY INDIVIDUALS	4
5	7184	/	PRIMARILY CONCERNED WITH REDUCING TOTAL CODE SPACE WHILE ALSO ADDING	5
6	7185	/	IMPORTANT FEATURES; THIS PROCESS LITERALLY TOOK YEARS TO ACHIEVE A VIABLE	6
7	7186	/	RESULT. ARGUABLY THE HANDLER AS DESCRIBED IN THIS DOCUMENT [ALONG WITH THE	7
8	7187	/	ASSOCIATED INTERNAL FEATURES] IS THE MOST WORKED OVER PIECE OF PDP-8 CODE IN	8
9	7188	/	HISTORY. AS OF THE TIME OF THIS WRITING, EVEN THE CODE DESCRIBED IN THIS	9
10	7189	/	DOCUMENT IS NOT AT THE MOST ADVANCED STATE KNOWN. [SEE ELSEWHERE FOR A	10
11	7190	/	DISCUSSION OF ADDITIONAL CHANGES AND CRITICISMS OF THE CODE DESCRIBED IN THIS	11
12	7191	/	DOCUMENT. SOME OF THE ADDITIONAL CHANGES OCCURRED A FULL DECADE AFTER THIS	12
13	7192	/	CODE WAS WRITTEN. IT IS HOPED NEWER VERSIONS WILL BE MADE AVAILABLE;	13
14	7193	/	ADDITIONAL SPECIFIC CHANGES ARE PLANNED FOR THE FUTURE.]	14
15	7194			15
16	7195	/	THE R-L MONITOR SYSTEM LOADER AND RELATED ISSUES.	16
17	7196			17
18	7197	/	THE ORIGINAL R-L MONITOR SYSTEM [WITHOUT ANY OF THE LATTER DAY IMPROVEMENTS	18
19	7198	/	LEADING TO P?S/8] REQUIRES AN ASSORTMENT OF UGLY KLUDGES TO PRODUCE A VIABLE	19
20	7199	/	SYSTEM. EVERY PROGRAM REQUIRING ACCESS TO ANY ASPECT OF THE SYSTEM DEVICE	20
21	7200	/	[SUCH AS THE CONTENTS OF FILES] MUST INCLUDE AN IMBEDDED DEVICE-DEPENDENT	21
22	7201	/	READ [AND PERHAPS WRITE] SUBROUTINE. HAVING AN INTERNAL DEVICE HANDLER IN	22
23	7202	/	EVERY SYSTEM UTILITY WASTES OVERALL STORAGE SPACE AND INCREASES PROGRAM SIZE.	23
24	7203	/	[NOTE: ORIGINALLY ONLY THE TC01/TC08 DECTAPE DRIVE UNIT 0 WAS SUPPORTED. ON	24
25	7204	/	AN EXPERIMENTAL BASIS, ALTERNATE PROGRAMMING WAS CREATED TO ADDRESS OTHER	25
26	7205	/	DEVICES SUCH AS THE DF32. DUE TO ADDITIONAL LIMITATIONS OF THE OPERATING	26
27	7206	/	SYSTEM, ONLY DEVICES THAT CAN SUPPORT [RICHARD LARY'S] "SLURP" BINARY FILE	27
28	7207	/	LOADER CAN BE CONSIDERED. EVERY RELEVANT PROGRAM REQUIRES SIGNIFICANT	28
29	7208	/	PATCHING TO SUPPORT THE SYSTEM DEVICE. BY NATURE, MUCH OF THIS WORK IS HIGHLY	29
30	7209	/	REDUNDANT AT THE SOURCE CODE LEVEL BUT QUITE DIFFERENT AT THE BINARY LEVEL;	30
31	7210	/	THE R-L MONITOR NEVER HAD COMPLETE SOURCE CODE AVAILABLE COMPLICATING MATTERS	31
32	7211	/	STILL FURTHER.]	32
33	7212			33
34	7213	/	SUPPORTING R-L MONITOR TEXTUAL INPUT FILES REQUIRES ALL OF THE FOLLOWING:	34
35	7214			35
36	7215	/	1) AN INPUT UNPACKING ROUTINE THAT OCCUPIES ABOUT 0.5 PAGES OF CODE SPACE.	36
37	7216	/	[NOTE: THIS FIGURE MUST BE ADJUSTED DEPENDING ON ADDITIONAL PROGRAM	37
38	7217	/	REQUIREMENTS. CERTAIN PROGRAMS MERELY PROCESS THE CONTENTS OF THE FILES	38
39	7218	/	WHILE OTHERS REQUIRE KEYBOARD SUPPORT AFTER THE AGGREGATE FILE CONTENTS IS	39
40	7219	/	COMPLETELY READ. CERTAIN PROGRAMS MAY REQUIRE IMBEDDED COMMANDS TO	40
41	7220	/	TRANSLATE CHARACTER SEQUENCES INCOMPATIBLE WITH THE TEXT FILE CHARACTER	41
42	7221	/	SET. THE MORE COMPLEX THE REQUIREMENTS, THE MORE CODE SPACE REQUIRED.]	42
43	7222			43
44	7223	/	2) A ONE PAGE INPUT BUFFER TO READ THE RAW FILE CONTENTS. [NOTE: CERTAIN	44
45	7224	/	PROGRAMS HAVE LARGER READ BUFFERS TO INCREASE THROUGHPUT. HOWEVER, ANY	45
46	7225	/	PROGRAM DESIGNED TO REQUIRE MOST OF 4K MEMORY IS ALREADY QUITE TAXED BY THE	46
47	7226	/	MINIMUM REQUIREMENTS. A PROGRAM THAT MUST ALSO WRITE OUT DATA WOULD	47
48	7227	/	REQUIRE ADDITIONAL RESOURCES SUCH AS AN OUTPUT BUFFER OF ONE OR MORE PAGES	48
49	7228	/	AND AN OUTPUT DATA PACKING ROUTINE.]	49
50	7229			50
51	7230	/	3) AN ADDITIONAL PAGE FOR THE [TC01/TC08 DECTAPE] READ/WRITE ROUTINES. [NOTE:	51
52	7231	/	CHANGING THE SYSTEM DEVICE REQUIRES THIS CODE BE REPLACED WITH CODE FOR THE	52
53	7232	/	ALTERNATE DEVICE. MOREOVER, IT WOULD NOT BE FEASIBLE TO RESERVE MORE CODE	53
54	7233	/	SPACE TO SUPPORT MORE DIFFICULT DEVICES AS SPACE IS AT A PREMIUM IN	54
55	7234	/	VIRTUALLY ALL CASES.]	55
56				56
57				57
58				58

1	7236	/	ALL OF THESE REQUIREMENTS TAKEN TOGETHER APPROACHES THREE PAGES OF CODE SPACE,	1
2	7237	/	NEARLY TEN PERCENT OF THE TOTAL USER SPACE OF 00000-07577.	2
3	7238			3
4	7239	/	AS SUCH, DESIGNING SYSTEM PROGRAMS TO FUNCTION UNDER THE R-L MONITOR SYSTEM	4
5	7240	/	IS QUITE AN ARDUOUS TASK. FOR MANY PROGRAMS, THE REQUIREMENTS OF INTERFACE	5
6	7241	/	EXCEED THE FREE SPACE AVAILABLE IN 00000-07577. EARLIER PROGRAMS GENERALLY	6
7	7242	/	WERE DESIGNED TO BE LOADED UNDER THE PAPER-TAPE ENVIRONMENT WHERE ALL INPUT	7
8	7243	/	AND OUTPUT IS DIRECTED TO EITHER THE SYSTEM CONSOLE OR PERHAPS THE PAPER-TAPE	8
9	7244	/	READER/PUNCH; COMPARATIVELY LITTLE CODE SPACE IS NEEDED TO SUPPORT THESE	9
10	7245	/	DEVICES. WITHOUT EITHER COMPROMISE OR RADICAL REDESIGN, IT MAY PROVE	10
11	7246	/	IMPOSSIBLE TO INTERFACE CERTAIN PROGRAMS TO THIS OPERATING SYSTEM.	11
12	7247			12
13	7248	/	THE BASIC R-L MONITOR IMAGE PROGRAM LOADING PROCESS.	13
14	7249			14
15	7250	/	JUST TO LOAD A CONFORMING PROGRAM INTO MEMORY, THE R-L MONITOR SYSTEM USES A	15
16	7251	/	SEPARATE MODULE KNOWN AS THE "SYSTEM LOADER". WHILE THE MAIN OPERATING SYSTEM	16
17	7252	/	CONTAINS READ/WRITE ROUTINES LOCATED IN 07600-07777, THERE ISN'T SUFFICIENT	17
18	7253	/	CODE SPACE TO ACCOMPLISH ALL OF THE FOLLOWING REQUIREMENTS:	18
19	7254			19
20	7255	/	1) MAINTAIN THE VIABILITY OF THE SYSTEM DEVICE HANDLER FOR READ/WRITE	20
21	7256	/	OPERATIONS ONCE PROGRAM EXECUTION IS INITIATED.	21
22	7257			22
23	7258	/	2) PROVIDE THE MEANS TO LOAD IN THE DESIGNATED PROGRAM. THE PROGRAM COULD BE	23
24	7259	/	AS LARGE AS A SINGLE SEGMENT MEANT TO LOAD INTO 00000-07577. [ACCOMPLISHING	24
25	7260	/	THIS REQUIRES CODE RESIDENT IN 07600-07777.]	25
26	7261			26
27	7262	/	3) START THE PROGRAM UP FROM AN ARBITRARY STARTING ADDRESS OBTAINED FROM THE	27
28	7263	/	SYSTEM PROGRAM DIRECTORY.	28
29	7264			29
30	7265	/	4) PASS FILE HANDLES ASSOCIATED WITH FILE NAMES OPTIONALLY INCLUDED IN THE	30
31	7266	/	"RUN" COMMAND USED TO INITIATE THE PROGRAM. THE FILES WILL BE USED IN A	31
32	7267	/	MANNER DETERMINED SOLELY BY THE PROGRAM [OR PERHAPS IGNORED]. [NOTE: P?S/8	32
33	7268	/	IMPLEMENTS SEVERAL EXECUTION-CLASS COMMANDS; THE R-L MONITOR SYSTEM ONLY	33
34	7269	/	HAS A PRIMITIVE "RU" COMMAND WHERE ONLY THE FIRST TWO LETTERS OF THE	34
35	7270	/	PROGRAM ARE SIGNIFICANT.]	35
36	7271			36
37	7272	/	5) PASS USER-PROVIDED PARAMETER INFORMATION. [NOTE: IN THE R-L MONITOR SYSTEM	37
38	7273	/	THIS CONSISTS OF MERELY A SINGLE 12-BIT VALUE OBTAINED FROM AN "=XXXX"	38
39	7274	/	OPTION INCLUDED IN THE RUN COMMAND-LINE. PROGRAMS GENERALLY HAVE TO DEFINE	39
40	7275	/	BITWISE APPORTIONMENT OF THIS PARAMETER TO CHANGE THE DYNAMICS OF PROGRAM	40
41	7276	/	EXECUTION.]	41
42	7277			42
43	7278	/	6) MAINTAIN A PROPER EXIT AND OPERATING SYSTEM RESTART AT 07600.	43
44	7279			44
45	7280	/	AS SUCH, THE FIRST REQUIREMENT WAS COMPLETELY DROPPED. ALL PROGRAMS THAT	45
46	7281	/	REQUIRE ACCESS TO ANY ASPECT OF THE SYSTEM DEVICE HAVE TO INCLUDE ROUTINES TO	46
47	7282	/	READ OR WRITE TO THE SYSTEM DEVICE AS NECESSARY; THE OPERATING SYSTEM CANNOT	47
48	7283	/	PROVIDE EVEN BASIC ACCESS TO THE SYSTEM DEVICE. [NOTE: R-L MONITOR BATCH IS	48
49	7284	/	ABLE TO WORK WITH THE SYSTEM LOADER ON A SPECIALIZED BASIS; THIS IS NOT	49
50	7285	/	RECOMMENDED FOR GENERAL-PURPOSE USE. FUNCTIONALITY IS LIMITED TO SINGLE PAGE	50
51	7286	/	READ-ONLY OPERATIONS TO A DESIGNATED BUFFER WITHIN THE KEYBOARD MONITOR.]	51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	7288	/	THE R-L KEYBOARD MONITOR LOADS A ONE-PAGE SKELETAL SYSTEM LOADER PROGRAM INTO	1
2	7289	/	MEMORY AND EXPANDS ALL SPECIFICS AS REQUIRED BY THE SYSTEM PROGRAM SELECTED;	2
3	7290	/	LOADING ADDRESS, NUMBER OF BLOCKS TO LOAD AND STARTING ADDRESS ARE STORED INTO	3
4	7291	/	THE CODE AT DESIGNATED LOCATIONS. A TC01/08 COMPATIBLE DECTAPE BOOTSTRAP IS	4
5	7292	/	PRESENT AT RELATIVE 0000 OF THE PAGE DESIGNED TO FUNCTION ONLY WHEN MOVED TO	5
6	7293	/	07600-0777. THE "=XXXX" VALUE IS STORED INTO RELATIVE 0156 OF THE PAGE. WHEN	6
7	7294	/	THE PAGE IS LATER MOVED TO 07600-07777, THIS VALUE CAN BE READ BY THE LOADED	7
8	7295	/	SYSTEM PROGRAM BY ACCESSING 07756. [NOTE: THIS CONVENTION IS MAINTAINED IN	8
9	7296	/	P?S/8 FOR BACKWARD COMPATIBILITY.]	9
10	7297			10
11	7298	/	THE FILE HANDLE PARAMETERS REPRESENTING THE PASSED FILES INCLUDED IN THE RUN	11
12	7299	/	COMMAND ARE STORED INTO RELATIVE 0157 AND FOLLOWING UP TO RELATIVE 0177. IF	12
13	7300	/	THE LIST IS SHORTER THAN THE MAXIMUM ALLOWED [17 INPUT FILES], A SENTINEL 0000	13
14	7301	/	VALUE IS PLACED IN THE NEXT WORD AS A LIST TERMINATOR.	14
15	7302			15
16	7303	/	FULL LENGTH LISTS DO NOT REQUIRE A TERMINATOR DUE TO THE DESIGNATED WAY TO	16
17	7304	/	ACCESS THE FILES AS ILLUSTRATED IN THE EXAMPLE BELOW.	17
18	7305			18
19	7306	/	CODE FRAGMENT TO ACCESS PASSED R-L FILES.	19
20	7307			20
21	7308	/	TAD (SFILES-1) /SETUP THE	21
22	7309	/	DCA POINTER /LIST POINTER	22
23	7310			23
24	7311	/	AT THIS POINT THE "POINTER" ADDRESS NEEDS TO BE INCREMENTED TO POINT TO THE	24
25	7312	/	LATEST FILE HANDLE.	25
26	7313			26
27	7314	/UPLUP, ISZ POINTER /BUMP TO NEXT FILE HANDLE		27
28	7315	/ TAD I POINTER /GET THE NEXT FILE BLOCK		28
29	7316			29
30	7317	/	THE AC EITHER CONTAINS THE STARTING BLOCK OF A FILE OR 0000. THERE ARE TWO	30
31	7318	/	WAYS FOR THE 0000 CASE TO OCCUR:	31
32	7319			32
33	7320	/	1) THE LIST SENTINEL [0000] WAS LOADED SIGNIFYING THE END OF THE LIST.	33
34	7321			34
35	7322	/	2) THE "ISZ POINTER" INSTRUCTION SKIPPED SINCE THE LAST ELEMENT IN A	35
36	7323	/	FULL-LENGTH LIST WAS PROCESSED PREVIOUSLY. THE CONTENTS OF "POINTER"	36
37	7324	/	CHANGED FROM 7777 TO 0000.	37
38	7325			38
39	7326	/	SNA /SKIP IF ANOTHER PASSED FILE TO BE PROCESSED	39
40	7327	/	JMP ENDOFALLFILES /JUMP IF NO ADDITIONAL FILES TO PROCESS	40
41	7328	/	DCA BLOCK /STORE IN-LINE WHERE NEEDED	41
42	7329			42
43	7330	/	THE LATEST FILE IS PROCESSED HERE; EVENTUALLY LOOPING BACK TO THE START OF THE	43
44	7331	/	UNPACKING LOOP.	44
45	7332			45
46	7333	/	JMP UPLUP /GO PROCESS ANOTHER PASSED FILE [IF ANY MORE]	46
47	7334			47
48	7335	/	THE FINAL ACTION TAKEN BY THE R-L MONITOR SYSTEM IS TO MOVE THE ENTIRE PAGE	48
49	7336	/	NOW FULLY PRELOADED WITH ALL RELEVANT PARAMETERS TO 07600-07777; AS A RESULT	49
50	7337	/	THE SYSTEM DEVICE HANDLER IS DESTROYED. THE SYSTEM LOADER IS STARTED AT ITS	50
51	7338	/	DESIGNATED STARTING ADDRESS TO CARRY OUT THE REST OF THE ACTUAL PROGRAM LOAD.	51
52	7339	/	FINALLY, THE SYSTEM PROGRAM IS STARTED VIA A "JMP I" INSTRUCTION REFERENCING	52
53	7340	/	THE LOCATION WHERE THE PROGRAM STARTING ADDRESS WAS STORED EARLIER.	53
54				54
55				55
56				56
57				57
58				58

1	7342	/	AT THIS POINT, THE DESIGNATED PROGRAM IS IN MEMORY; NOTHING REMAINS OF THE R-L	1
2	7343	/	MONITOR SYSTEM OTHER THAN THE MEANS TO REBOOT THE SYSTEM AT 07600 AND	2
3	7344	/	ARTIFACTS OF THE SYSTEM LOADER CODE.	3
4	7345			4
5	7346	/	OTHER THAN PRESERVING THE ORIGINAL SYSTEM DEVICE HANDLER, ALL OTHER ASPECTS OF	5
6	7347	/	LOADING A SYSTEM PROGRAM HAVE BEEN ACHIEVED, ALBEIT BY HIGH-OVERHEAD	6
7	7348	/	TECHNIQUES INVOLVING READING IN THE SYSTEM LOADER AS WELL AS THE SYSTEM	7
8	7349	/	DIRECTORY. AS STATED ABOVE, IT IS THE RESPONSIBILITY OF THE SYSTEM PROGRAM TO	8
9	7350	/	PERFORM ANY FILE ACCESS OR OTHER OPERATIONS BY PROVIDING EMBEDDED READ/WRITE	9
10	7351	/	ROUTINES. THE ONLY FACILITIES PROVIDED ARE:	10
11	7352			11
12	7353	/	1) THE ABILITY TO RESTART AT 07600. [NOTE: THE AC MUST BE CLEAR.]	12
13	7354			13
14	7355	/	2) THE LIST OF PASSED FILE HANDLES IS PRESENT AT 07757 AND FOLLOWING.	14
15	7356			15
16	7357	/	3) THE "=XXXX" VALUE IS PRESENT IN 07756.	16
17	7358			17
18	7359	/	REPLACEMENT OF THE R-L MONITOR SYSTEM LOADER WITH THE P?S/8 SYSTEM LOADER.	18
19	7360			19
20	7361	/	REMOVAL OF THE OVERHEAD OF THE R-L MONITOR SYSTEM LOADING RESTRICTIONS AND	20
21	7362	/	KLUDGES WAS EVENTUALLY ACHIEVED WHILE RETAINING BACKWARD COMPATIBILITY. A	21
22	7363	/	RESIDENT DEVICE-INDEPENDENT SYSTEM LOADER ELIMINATES ALL ASSOCIATED OVERHEAD	22
23	7364	/	WHILE STILL SUPPORTING THE ENTIRE USER MEMORY SPACE [00000-07577] FOR PROGRAM	23
24	7365	/	LOADING. ALL EXISTING R-L MONITOR PASSED PARAMETERS ARE RETAINED WHILE NEW	24
25	7366	/	FEATURES ARE ADDED TO ENHANCE THE PROGRAM OPERATING ENVIRONMENT.	25
26	7367			26
27	7368	/	REPLACEMENT OF THE R-L MONITOR SYSTEM HANDLER WITH THE P?S/8 SYSTEM DEVICE	27
28	7369	/	HANDLER.	28
29	7370			29
30	7371	/	A SEVERE WEAKNESS OF THE R-L MONITOR SYSTEM IS THE LIMITED CAPABILITY OF THE	30
31	7372	/	SYSTEM DEVICE HANDLER [WHICH IS NOT PRESENT WHEN A PROGRAM IS RUNNING]. [WHEN	31
32	7373	/	AVAILABLE,] THE SYSTEM DEVICE HANDLER IS CALLED BY THE FOLLOWING METHOD:	32
33	7374			33
34	7375	/	JMS I (HANDLER) /CALL THE READ [OR WRITE] HANDLER.	34
35	7376	/	BUFFER /[READ OR WRITE] BUFFER ADDRESS.	35
36	7377	/	-BLKCNT /TWO'S COMPLEMENT OF NUMBER OF BLOCKS.	36
37	7378	/	BLOCK /STARTING BLOCK NUMBER	37
38	7379			38
39	7380	/	THERE IS NO SUPPORT FOR A LOGICAL UNIT NUMBER AS THE SYSTEM IS LIMITED TO	39
40	7381	/	SUPPORTING DECTAPE DRIVE UNIT 0 ONLY. NO SUPPORT EXISTS FOR EXTENDED MEMORY.	40
41	7382	/	ALL OF FIELD 0 CAN BE WRITTEN; HOWEVER, READING OVER 07600-07777 DESTROYS THE	41
42	7383	/	DEVICE HANDLER. [NOTE: IF EXTENDED MEMORY OPERATIONS WERE SUPPORTED, FULL	42
43	7384	/	FIELD READS COULD BE IMPLEMENTED. THIS COULD FACILITATE EFFICIENT COPY	43
44	7385	/	UTILITIES ASSUMING EXTENDED MEMORY IS AVAILABLE.]	44
45	7386			45
46	7387	/	ADDITIONALLY, TWO CALLING ADDRESSES ARE REQUIRED; ONE IS FOR READING WHILE THE	46
47	7388	/	OTHER IS FOR WRITING. NEITHER ADDRESS IS PARTICULARLY OPTIMAL IN ANY SENSE.	47
48	7389	/	ALL CALLING PROGRAMS REQUIRE ADDRESS CONSTANTS TO ACCESS EITHER ROUTINE.	48
49				49
50				50
51				51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	7391	/	THE P?S/8 DEVICE HANDLER IS FAR MORE CAPABLE:	1
2	7392			2
3	7393	/	JMS I (7640) /CALL THE HANDLER. [NOTE: "SZA CLA" IS 7640].	3
4	7394	/	BUFFER /BUFFER ADDRESS	4
5	7395	/	FUNCTION WORD /COMPOSITE FUNCTION WORD [SEE BELOW FOR DETAILS]	5
6	7396	/	BLOCK /STARTING BLOCK NUMBER	6
7	7397			7
8	7398	/	EXTENDED MEMORY CALLS ARE SUPPORTED BY THE CALLING PROGRAM SETTING ITS OWN	8
9	7399	/	DATA FIELD BEFORE CALLING THE SYSTEM DEVICE HANDLER IN FIELD 0.	9
10	7400			10
11	7401	/	THE FUNCTION WORD BREAKS DOWN AS FOLLOWS:	11
12	7402			12
13	7403	/	BIT[0] 1=WRITING, 0=READING.	13
14	7404			14
15	7405	/	BITS[1-5] NUMBER OF PAGES IN THE ENTIRE TRANSFER 01-37; 0 MEANS 40.	15
16	7406			16
17	7407	/	BITS[6-8] BUFFER FIELD FOR THE TRANSFER.	17
18	7408			18
19	7409	/	BITS[9-11] LOGICAL UNIT FOR THE TRANSFER.	19
20	7410			20
21	7411	/	LOGICAL UNITS ARE MEANINGFUL FOR ALL PROGRAMS AND FILES. FILE HANDLES ARE	21
22	7412	/	MADE MORE POWERFUL WITHOUT CHANGING OVERALL STORAGE SIZE. THIS IS ACHIEVED BY	22
23	7413	/	IMPOSING A BOUNDARY RESTRICTION ON THE STARTING BLOCK OF ALL FILES:	23
24	7414			24
25	7415	/	FILE SYSTEM CHANGES.	25
26	7416			26
27	7417	/	ONCE LOGICAL UNIT SUPPORT WAS ADDED, IT BECAME NECESSARY TO PASS LOGICAL UNIT	27
28	7418	/	AS WELL AS BLOCK INFORMATION WITHIN ALL FILE HANDLES. TO ACCOMMODATE THIS	28
29	7419	/	WITHOUT COMPROMISING THE PASSED FILE LIST, THE FILE SYSTEM IMPOSES A BOUNDARY	29
30	7420	/	RESTRICTION ON FILE PLACEMENT WITHIN EACH DEVICE. [NOTE: ALL R-L MONITOR AND	30
31	7421	/	EARLY P?S/8 RELEASES ALLOW FREE PLACEMENT OF THE FILE SYSTEM; NEWER RELEASES	31
32	7422	/	OF P?S/8 CAN DIRECTLY COPY INDIVIDUAL FILES TO A DEVICE CONFORMING TO THE NEW	32
33	7423	/	RESTRICTION VIA KEYBOARD MONITOR COMMANDS. AS SUCH, ALL USER FILES CAN ALWAYS	33
34	7424	/	BE COPIED TO A NEWER RELEASE DESPITE CHANGING SYSTEM REQUIREMENTS. EXISTING	34
35	7425	/	TEXT FILES CREATED ON THE R-L MONITOR SYSTEM WERE FIRST BE PASSED THROUGH AN	35
36	7426	/	IMPROMPTU IN-PLACE FILE CONVERSION PROGRAM THAT DESTROYS THE VIABILITY OF THE	36
37	7427	/	R-L SYSTEM IN PREPARATION FOR THE REST OF THE COPYING PROCESS. BINARY FILES	37
38	7428	/	ARE DIRECTLY COMPATIBLE; P?S/8 SUPPORTS A PROPER SUPERSET OF THE EARLIER R-L	38
39	7429	/	MONITOR BINARY FILE FORMAT WHICH INCLUDES EXTENDED MEMORY SUPPORT.] [NOTE: IF	39
40	7430	/	THERE IS SUFFICIENT INTEREST, A NON-DESTRUCTIVE CONVERSION PROGRAM CAN BE	40
41	7431	/	WRITTEN THAT PRESERVES THE ORIGINAL R-L MONITOR TEXT FILES; TWO DECTAPE DRIVES	41
42	7432	/	WILL BE REQUIRED ADDRESSABLE BY EXISTING P?S/8 DEVICE HANDLERS SUCH AS THE	42
43	7433	/	CURRENT TC01/TC08 SYSTEM DEVICE OR ONE OF THE TD8E-BASED P?S/8 SYSTEMS.]	43
44	7434			44
45	7435	/	THE NEWEST SYSTEMS REQUIRE ALL FILES BE ALIGNED ON BOUNDARIES OF EIGHT BLOCKS.	45
46	7436	/	[NOTE: SINCE ALL TFS FILES ARE SIXTEEN BLOCKS LONG, THIS RESTRICTION IS EASILY	46
47	7437	/	COMPLIED WITH. TO DATE, ALL ACTUAL FILE SYSTEMS HAVE BEEN CREATED WITH A	47
48	7438	/	SIXTEEN BLOCK BOUNDARY IN ANTICIPATION OF A FUTURE UNSPECIFIED REQUIREMENT.]	48
49	7439			49
50	7440	/	THE FILE HANDLES NOW CONSIST OF THE LOGICAL UNIT BITS ADDED TO THE FILE	50
51	7441	/	STARTING BLOCK; THIS IS A FUNCTIONAL SOLUTION AS THE LOW-ORDER THREE BITS OF	51
52	7442	/	THE STARTING BLOCK ARE ALWAYS ZERO [DUE TO THE BOUNDARY ALIGNMENT RESTRICTION	52
53	7443	/	IMPOSED]. THE FILE HANDLES MUST BE PROCESSED AS FOLLOWS:	53
54				54
55				55
56				56
57				57
58				58

1	7445	/	ACCESSING THE P?S/8 FILE HANDLES LIST.	1
2	7446			2
3	7447	/	PROCESSING INDIVIDUAL FILE HANDLES IS ACCOMPLISHED USING THE SAME OVERALL	3
4	7448	/	METHOD AS IN THE R-L MONITOR EXAMPLE ABOVE WITH THE FOLLOWING MODIFICATIONS:	4
5	7449			5
6	7450	/	TAD I POINTER /GET THE LATEST FILE HANDLE	6
7	7451	/	AND (7770) /JUST BLOCK BITS	7
8	7452	/	DCA BLOCK /STORE WHERE NEEDED [AS IN THE R-L ORIGINAL]	8
9	7453	/	TAD I POINTER /GET THE FILE HANDLE AGAIN	9
10	7454	/	AND (7) /ISOLATE UNIT BITS	10
11	7455			11
12	7456	/	GENERALLY THE LOGICAL UNIT IS COMBINED WITH ADDITIONAL FUNCTION WORD VALUES	12
13	7457	/	AND STORED AS A COMPOSITE IN-LINE ARGUMENT TO THE SYSTEM DEVICE HANDLER.	13
14	7458			14
15	7459	/	TAD (1^100) /READ ONE PAGE FROM CURRENT LOGICAL UNIT	15
16	7460	/	DCA FUNCTION /STORE IN-LINE	16
17	7461			17
18	7462	/	FOR CALLING PROGRAMS WITH LITTLE CODE SPACE TO SPARE, IT IS OFTEN HELPFUL THE	18
19	7463	/	SYSTEM DEVICE HANDLER ADDRESS IS THE SAME BINARY CODE AS THE "SZA CLA"	19
20	7464	/	INSTRUCTION; VIRTUALLY ALL PDP-8 PROGRAMMING WILL TEND TO CONTAIN THIS	20
21	7465	/	INSTRUCTION ON THE SAME PAGE AS A CALL TO THE SYSTEM DEVICE HANDLER. [NOTE: IF	21
22	7466	/	THIS IS NOT "NATURALLY" THE CASE, IT IS GENERALLY EASILY CONTRIVED MERELY BY	22
23	7467	/	SLIGHTLY REORGANIZING EXISTING PROGRAMMING AS DESCRIBED ELSEWHERE IN THIS	23
24	7468	/	DOCUMENT.]	24
25	7469			25
26	7470	/	P?S/8 SYSTEM LOADER USAGE.	26
27	7471			27
28	7472	/	P?S/8 PROVIDES A FAR RICHER SET OF OPERATING PARAMETERS PASSED TO THE LOADED	28
29	7473	/	IMAGE PROGRAM THAN THE R-L MONITOR SYSTEM IT REPLACES. THIS INCLUDES THE	29
30	7474	/	CURRENT SYSTEM DATE, ACCESS TO SPECIFIED OUTPUT FILES AND SINGLE-CHARACTER	30
31	7475	/	OPTION SWITCHES [/A THROUGH /9] AMONG OTHER FEATURES. [NOTE: THE FULL LIST OF	31
32	7476	/	PASSED PARAMETERS IS RELEASE-DEPENDENT AND SUBJECT TO CHANGE.] THIS WAS MADE	32
33	7477	/	POSSIBLE BECAUSE OF THE EXCEPTIONAL EFFORT EXPENDED TO OPTIMIZE THE TC01/TC08	33
34	7478	/	SYSTEM DEVICE HANDLER TO OCCUPY LESS CODE SPACE. AS THE INITIALLY SUPPORTED	34
35	7479	/	DEVICE, THIS HAS BECOME THE STANDARD TO WHICH ALL OTHER DEVICE HANDLERS MUST	35
36	7480	/	BE HELD; ALL SUPPORTED DEVICES MUST INCLUDE ALL DESIGNATED OPTIONS FOR TOTAL	36
37	7481	/	COMPATIBILITY. WHERE POSSIBLE, DEVICE SUPPORT IS IMPLEMENTED IN 4K FOR THE	37
38	7482	/	MINIMUM VIABLE SYSTEM; ALL SYSTEMS SUPPORT THE SAME PASSED PARAMETERS	38
39	7483	/	REGARDLESS OF ACTUAL MEMORY SIZE.	39
40	7484			40
41	7485	/	WITHIN THE P?S/8 SYSTEM DEVICE HANDLER SPACE IS A RESERVED AREA FOR THE	41
42	7486	/	SYSTEM LOADER, AN IMBEDDED DEDICATED CALL TO THE SYSTEM DEVICE HANDLER	42
43	7487	/	FOLLOWED BY A DEDICATED "JMP I" STARTING INSTRUCTION AND DEDICATED STARTING	43
44	7488	/	ADDRESS POINTER.	44
45				45
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1	7490	/	THE P?S/8 SYSTEM DEVICE HANDLER AND SYSTEM LOADER ARE ORDINARILY NEVER	1
2	7491	/	DISPLACED FROM 07600-07777. [NOTE: THERE IS ONE EXCEPTION: THE P?S/8 "BIN"	2
3	7492	/	PROGRAM CAN PROVIDE AN OPTIONAL DEVICE-SPECIFIC BINARY FILE "SLURP" LOADER OR	3
4	7493	/	AN EQUIVALENT TO DEC'S PAPER-TAPE BIN AND RIM LOADERS; EITHER OPTION ALSO	4
5	7494	/	INCLUDES A DEVICE-SPECIFIC BOOTSTRAP AT 07600 TO RESTART P?S/8. IN EITHER OF	5
6	7495	/	THESE OPERATING MODES, PROGRAMS ARE LOADED WITH NO EXPECTATION OF OPERATING	6
7	7496	/	SYSTEM SERVICES WITH THE EXCEPTION OF THE ABILITY TO REBOOT THE OPERATING	7
8	7497	/	SYSTEM BY EXITING TO 07600. AS AN IMPROVEMENT TO THE ANALOGOUS SITUATION IN	8
9	7498	/	THE R-L MONITOR, THE AC NEED NOT BE CLEAR. THE DATA FIELD MUST BE SET TO	9
10	7499	/	ZERO AND INTERRUPTS MUST BE TURNED OFF. REFER TO ADDITIONAL DOCUMENTATION OF	10
11	7500	/	THE BIN SYSTEM PROGRAM FOR MORE INFORMATION.] [NOTE: AS DESCRIBED ELSEWHERE	11
12	7501	/	IN THIS DOCUMENT, THE SLURP LOADER SUPPORTS A MEANS OF OVERLAY TO RESTORE THE	12
13	7502	/	SYSTEM DEVICE HANDLER AFTER LOADING ALL USER BINARY FILES; THE SYSTEM DEVICE	13
14	7503	/	CAN BE WRITE-PROTECTED AS THE ENTIRE RELOAD PROCESS TAKES PLACE IN MEMORY.]	14
15	7504			15
16	7505	/	AS NECESSARY, VARIOUS PORTIONS OF THE OPERATING SYSTEM INSERT VALUES INTO THE	16
17	7506	/	SYSTEM LOADER TO EFFECT DIFFERENT IMAGE-ORIENTED SERVICES. MOST IMPLEMENTED	17
18	7507	/	FUNCTIONS FALL INTO ONE OF THE FOLLOWING CATEGORIES:	18
19	7508			19
20	7509	/	1) IMAGE PROGRAM EXECUTION. VARIOUS EXECUTION-CLASS COMMANDS [REPLACING THE	20
21	7510	/	ORIGINAL R-L MONITOR RU[N] COMMAND AND PARTICULAR TO VARIOUS OPERATING	21
22	7511	/	MODES OF P?S/8 AND THE P?S/8 SHELL] WILL STORE LOADING AND STARTING	22
23	7512	/	PARAMETERS [LOADING ADDRESS, FUNCTION WORD, STARTING BLOCK AND STARTING	23
24	7513	/	ADDRESS] TO INITIATE PROGRAM EXECUTION THAT GENERALLY OVERLAYS THE SYSTEM	24
25	7514	/	SEGMENT THAT INITIATED THE COMMAND. UNLIKE THE R-L MONITOR SYSTEM, THIS	25
26	7515	/	IS ALL THAT IS REQUIRED TO INITIATE IMAGE PROGRAM LOADING. THE SYSTEM	26
27	7516	/	DEVICE HANDLER IS PRESENT AT ALL TIMES DURING PROGRAM EXECUTION. ALL	27
28	7517	/	PROGRAM REQUIREMENTS [OTHER THAN SHELL-BASED EXTENSIONS IF APPLICABLE] ARE	28
29	7518	/	MET BY THE RESIDENT SYSTEM WITHOUT REQUIRING DEVICE-DEPENDENT CODE. ANY	29
30	7519	/	FUTURE IMPLEMENTATION OF SYSTEM FEATURES WILL MAINTAIN COMPATIBILITY WITH	30
31	7520	/	THIS DESIGN FURTHER DISTINGUISHING P?S/8 FROM THE R-L MONITOR SYSTEM.	31
32	7521			32
33	7522	/	2) PROGRAM OVERLAY SEGMENTATION. PROGRAMS THAT REQUIRE OVERLAYS CAN CALCULATE	33
34	7523	/	ABSOLUTE BLOCKS FOR ADDITIONAL PROGRAM [OR DATA] SEGMENTS BY INSPECTING THE	34
35	7524	/	SYSTEM LOADER VALUES AND ADDING APPROPRIATE OFFSETS TO THE ORIGINAL LOADING	35
36	7525	/	PARAMETERS. [NOTE: THE ANALOGOUS FUNCTION IS AVAILABLE IN THE R-L MONITOR	36
37	7526	/	SYSTEM TO THE EXTENT THAT WHEN A SYSTEM PROGRAM IS LOADED, THE SYSTEM	37
38	7527	/	LOADER WAS PASSED THE STARTING BLOCK. ACTUAL ACCESS TO CALCULATED BLOCKS	38
39	7528	/	CAN ONLY BE ACCOMPLISHED USING PROGRAM-SUPPLIED DEVICE-DEPENDENT	39
40	7529	/	SUBROUTINES. THIS TECHNIQUE WAS USED TO IMPLEMENT THE R-L MONITOR VERSION	40
41	7530	/	OF FOCAL, 1969.] WHEN NECESSARY, SECONDARY PROGRAM SECTIONS CAN BE LINKED	41
42	7531	/	TO WITHOUT REQUIRING RESERVATION OF USER MEMORY. [THIS TECHNIQUE IS USED	42
43	7532	/	IN THE P?S/8 PAL ASSEMBLER TO IMPLEMENT END-OF-ASSEMBLY OVERLAYS SUCH AS	43
44	7533	/	THE SYMBOL TABLE SORT/PRINT ROUTINE AND THE OPTIONAL DYNAMIC LINK TO THE	44
45	7534	/	BITMAP PROGRAM AND/OR BINARY FILE LOADER IF THERE ARE NO ASSEMBLY ERRORS.]	45
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1	7536	/	3) SAVING A COMPLETE MEMORY IMAGE. THE PRELOADED CONTENTS OF THE SYSTEM	1
2	7537	/	LOADER IS A CALL TO THE SYSTEM DEVICE HANDLER TO SAVE ALL OF FIELD 0 TO THE	2
3	7538	/	DEFINED SAVE AREA: BLOCKS 0020-0057. [NOTE: THESE BLOCKS CORRESPOND TO THE	3
4	7539	/	"%" AND "\$" FILES IN THE BASIC SYSTEM; ALL IMAGE-ORIENTED UTILITIES OBEY	4
5	7540	/	THIS CONVENTION INCLUDING P?S/8 "ODT", "BSAVE", ETC. THE SHELL OVERLAY	5
6	7541	/	USES THE IDENTICAL CONVENTION ALLOWING THE COMMAND PROCESSOR TO LOAD INTO	6
7	7542	/	FIELD 0 WITH THE CONTENTS OF MEMORY AVAILABLE TO RELEVANT COMMANDS SUCH	7
8	7543	/	AS THE SYSTEM DEBUGGING FACILITY.] UNLIKE OTHER OPERATING SYSTEMS, THIS	8
9	7544	/	FACILITY IS NOT AUTOMATICALLY INVOKED; NECESSARY OPERATIONS ONLY OCCUR WHEN	9
10	7545	/	USER COMMANDS INITIATE SAVING MEMORY. [NOTE: THE ACTUAL REBOOT PROCESS IS	10
11	7546	/	READ-ONLY.]	11
12	7547			12
13	7548	/	APPROPRIATE SYSTEM PROGRAMS [SUCH AS "GET"] CAN UTILIZE THIS FACILITY;	13
14	7549	/	TYPICAL OPERATING SYSTEM USAGE CONSISTS OF RELOADING THE APPROPRIATE	14
15	7550	/	COMMAND PROCESSOR AFTER THE PROGRAM EXITS. BY AVOIDING UNNECESSARY WRITE	15
16	7551	/	OPERATIONS, TYPICAL SYSTEM RELOADS ARE FASTER WITH GREATER SAFETY. [NOTE:	16
17	7552	/	IF NOT WRITING OUT AN IMAGE OF MEMORY, THE TYPICAL RESTART OPERATION IS A	17
18	7553	/	SINGLE OPTIMIZED READ OPERATION. DEVICES THAT SUPPORT WRITE-PROTECT SUCH	18
19	7554	/	AS THE TC01/TC08 DECTAPE CAN BEST BE USED THIS WAY TO PREVENT SYSTEM	19
20	7555	/	DESTRUCTION CAUSED BY A PROGRAM LOGIC ERROR.] [NOTE: THE BIN PROGRAM MAY	20
21	7556	/	INCLUDE AN OPTIONAL BINARY FILE SLURP LOADER AS DISCUSSED ABOVE AND	21
22	7557	/	ELSEWHERE IN THIS DOCUMENT. AN ADDITIONAL OPTION CAN RELOAD THE SYSTEM	22
23	7558	/	DEVICE HANDLER AFTER LOADING ALL PASSED FILES. AS SUCH, SPECIFYING 07632	23
24	7559	/	AS THE STARTING ADDRESS WILL SAVE ALL OF FIELD 0 THEN RELOAD THE OPERATING	24
25	7560	/	SYSTEM/KEYBOARD MONITOR. THIS IS DISCUSSED IN GREATER DETAIL ELSEWHERE IN	25
26	7561	/	THIS DOCUMENT. A DEVICE-INDEPENDENT VIRTUAL LOADER CAN ALSO BE USED TO	26
27	7562	/	LOAD BINARY FILES ON ALL SUPPORTED SYSTEM CONFIGURATIONS; THE DEFAULT	27
28	7563	/	CONTENTS OF THE SYSTEM LOADER ARE RESTORED TO ALLOW THE IDENTICAL	28
29	7564	/	FUNCTIONALITY FOR EITHER BINARY LOADER. THE BIN LOADING PROGRAM SETS ALL	29
30	7565	/	PARAMETERS REGARDLESS OF OPERATING MODE. THE DEFAULT WRITE OPERATION	30
31	7566	/	DESCRIBED ABOVE WILL BE RESTORED AS NECESSARY. FOR MORE DETAILS ON THE	31
32	7567	/	VIRTUAL LOADING PROCESS, SEE DOCUMENTATION OF THE P?S/8 BIN PROGRAM.]	32
33	7568			33
34	7569	/	END OF APPENDIX K.	34
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1	7571	/	APPENDIX L.	1
2	7572			2
3	7573	/	A KEYBOARD MONITOR TIP TO ASSEMBLE PROGRAMS CONTAINED IN A GROUP OF TFS FILES	3
4	7574	/	FASTER TAKING ADVANTAGE OF INTELLIGENT INITIAL SEARCH SUPPORT ON TAPE SYSTEMS.	4
5	7575			5
6	7576	/	P?S/8 HAS SEVERAL STRUCTURES ORGANIZED TO MINIMIZE SEARCH TIME WHERE POSSIBLE,	6
7	7577	/	INCLUDING THE USE OF SEPARATE DIRECTORIES FOR SYSTEM PROGRAMS AND USER FILES.	7
8	7578	/	FORWARD INITIAL SEARCH OPPORTUNITIES WILL OCCUR TO AN APPRECIABLE PERCENT OF	8
9	7579	/	THE TIME ASSUMING TYPICAL SYSTEM USAGE.	9
10	7580			10
11	7581	/	A PRACTICAL EXAMPLE OF EXPLOITING FORWARD INITIAL SEARCH DIRECTION IN P?S/8	11
12	7582	/	INVOLVES THE USE OF SEVERAL ADJACENT TFS FILES. OVERALL THROUGHPUT CAN BE	12
13	7583	/	IMPROVED BY INTERLEAVING THE FILES ON THE TAPE. [P?S/8 USERS HAVE COMPLETE	13
14	7584	/	CONTROL OVER THE EXACT PLACEMENT OF FILES USED TOGETHER AS A CONCATENATED	14
15	7585	/	DATA STREAM.] EACH TFS FILE IS 16 BLOCKS LONG [WHICH IS MORE THAN THE	15
16	7586	/	WORST-CASE OVERSHOOT FACTOR].	16
17	7587			17
18	7588	/	BY ARRANGING THE FILES IN AN INTERLEAVED FASHION, THE OVERALL ACCESS TIME CAN	18
19	7589	/	BE LESSENED SINCE EACH PASS THROUGH THE FILES CONSISTS OF ONLY TWO SECTIONS	19
20	7590	/	EACH SEARCHED FORWARD WITHIN; THE DATA-WISE ADJACENT FILES ARE SEPARATED BY	20
21	7591	/	THE FILES ASSOCIATED WITH THE OTHER PASS; AS SUCH, ALL OTHER SEARCHES ARE	21
22	7592	/	INITIATED FORWARD. [NOTE: THIS EXAMPLE ASSUMES AT LEAST 8K MEMORY IS	22
23	7593	/	AVAILABLE; IN 4K SYSTEMS, EACH FILE MIGHT REQUIRE SEVERAL READ OPERATIONS EACH	23
24	7594	/	REQUIRING REVERSE INITIAL SEARCH FOR ANY SECONDARY READ CALLS; WHEN THE P?S/8	24
25	7595	/	PAL ASSEMBLER DISCOVERS MORE AVAILABLE MEMORY, THE INPUT BUFFER IS INCREASED	25
26	7596	/	SUBSTANTIALLY. IN GENERAL, THE TEXT PORTION OF A TYPICAL TFS FILE WILL BE	26
27	7597	/	READ IN COMPLETELY USING A SINGLE FORWARD SEARCH AS DESCRIBED IN THIS EXAMPLE.	27
28	7598	/	IT IS NOT NECESSARY TO READ THE ENTIRE TFS FILE AS THE REAR-MOST DATA CONSISTS	28
29	7599	/	ONLY OF FREE SPACE FOLLOWED BY LINE NUMBER AND TEXT POINTER INFORMATION. SEE	29
30	7600	/	DETAILS ON TFS INTERNAL FILE FORMAT ELSEWHERE IN THIS DOCUMENT FOR MORE	30
31	7601	/	INFORMATION.] FOR EXAMPLE, CONSIDER A PROGRAM THAT SPANS EIGHT TFS FILES:	31
32	7602			32
33	7603	/	FILE1 FILE2 FILE3 FILE4 FILE5 FILE6 FILE7 FILE8	33
34	7604			34
35	7605	/	A TYPICAL ASSEMBLY COMMAND MIGHT BE:	35
36	7606			36
37	7607	/	PAL FILE1,FILE2,FILE3,FILE4,FILE5,FILE6,FILE7,FILE8 /B	37
38	7608			38
39	7609	/	IN THIS SCENARIO, EVERY FILE MUST BE SEARCHED INITIALLY IN REVERSE BECAUSE THE	39
40	7610	/	NEXT BLOCK WHEN TRAVERSING FILES IS ADJACENT TO THE BLOCKS JUST READ.	40
41	7611			41
42	7612	/	BY REARRANGING THE FILES THUS:	42
43	7613			43
44	7614	/	FILE1 FILE5 FILE2 FILE6 FILE3 FILE7 FILE4 FILE8	44
45	7615			45
46	7616	/	THE SAME COMMAND WILL HAVE DRAMATICALLY BETTER THROUGHPUT AS REVERSE INITIAL	46
47	7617	/	SEARCH DIRECTION WILL ONLY APPLY WHEN ACCESSING FILES "FILE1" AND "FILE5".	47
48	7618	/	ALL OF THE TRANSFERS ACCESSING FILES WITH LOGICALLY ADJACENT CONTENTS WILL BE	48
49	7619	/	SUCCESSFULLY SEARCHED WHILE MOVING FORWARD. [NOTE: THE TIME SPENT WINDING	49
50	7620	/	FROM THE END OF FILE "FILE4" BACK TO JUST BELOW THE FIRST BLOCK OF FILE	50
51	7621	/	"FILE5" IS FAR LESS THAN THE OVERHEAD OF THE REVERSE INITIAL SEARCH THAT IS	51
52	7622	/	UNAVOIDABLE WHEN USING THE MORE CONVENTIONAL FILE LAYOUT.]	52
53	7623			53
54	7624	/	END OF APPENDIX L.	54
55				55
56				56
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1	7626	/	APPENDIX M.	1
2	7627			2
3	7628	/	USAGE OF THE P?S/8 OPTIONAL PARAMETER AREA.	3
4	7629			4
5	7630	/	DURING NORMAL EXECUTION, VARIOUS VALUES CAN BE PLACED ANYWHERE IN THE REST OF	5
6	7631	/	THE PAGE FROM 07756-07757. VARIOUS ASSIGNMENTS HAVE BEEN MADE SPECIFIC TO	6
7	7632	/	THE RELEVANT EXECUTION MODES.	7
8	7633			8
9	7634	/	SYSTEM EQUALS PARAMETER LOCATION.	9
10	7635			10
11	7636	/	IN MOST EXECUTABLE CIRCUMSTANCES, THE LATEST COMMAND LINE GENERALLY ALLOWS A	11
12	7637	/	NUMERIC PARAMETER SPECIFIED AS =XXXX [WHERE XXXX IS ANY ONE THROUGH FOUR	12
13	7638	/	DIGIT VALUE FROM 0000 TO 7776].	13
14	7639			14
15	7640	/	THIS PARAMETER IS STORED IN "SEQUALS" [07756] AND AVAILABLE TO ALL SYSTEM	15
16	7641	/	PROGRAMS IN THE BASIC P?S/8 SYSTEM AND ALSO THE SHELL OVERLAY CONFIGURATION.	16
17	7642			17
18	7643	/	THE R-L MONITOR SYSTEM ALLOWED THIS PARAMETER [LOADED INTO THE SAME LOCATION]	18
19	7644	/	TO HAVE ANY VALUE FROM 0000-7777; IN P?S/8, 7777 IS EXPLICITLY NOT ALLOWED.	19
20	7645	/	HOWEVER, THE VALUE 7777 WILL BE PASSED IF THERE IS NO PARAMETER USED ON THE	20
21	7646	/	COMMAND LINE. AS SUCH, THE EXECUTABLE PROGRAM CAN OPTIONALLY ASSIGN A	21
22	7647	/	DEFAULT VALUE [DETERMINED BY INTERNAL PROGRAM LOGIC].	22
23	7648			23
24	7649	/	FOR EXAMPLE, THE P?S/8 "BIN" UTILITY ALLOWS VARIOUS BINARY LOADER OPTIONS WITH	24
25	7650	/	OPTIONAL STARTING INFORMATION. IF THE =XXXX PARAMETER IS OMITTED, AFTER THE	25
26	7651	/	PROGRAM IS FULLY LOADED, THE ADDRESS OF A SAFE "HLT" [7402] INSTRUCTION WILL	26
27	7652	/	BE PASSED TO THE LOADER; PRESSING "CONTINUE" ON THE FRONT PANEL WILL JUST	27
28	7653	/	LOOP BACK TO THE SAFE HLT INSTRUCTION.	28
29	7654			29
30	7655	/	NOTE: IN CERTAIN P?S/8 RELEASES, THE KEYBOARD MONITOR CAN BE SET TO PASS	30
31	7656	/	VALUES OBTAINED FROM THE COMMAND LINE IN DECIMAL RADIX. HOWEVER, THE INTERNAL	31
32	7657	/	VALUE IS STILL LIMITED TO THE IDENTICAL RANGE; EXPRESSED IN DECIMAL, THIS IS	32
33	7658	/	0000 THROUGH 4094 WITH A DEFAULT OF 4095 IF THE COMMAND LINE PARAMETER IS NOT	33
34	7659	/	USED.	34
35	7660			35
36	7661	/	PASSED FILE PARAMETERS AREA.	36
37	7662			37
38	7663	/	THE P?S/8 "ODT" [OCTAL DEBUGGING TECHNIQUE] UTILITY PLACES VARIOUS SMALL	38
39	7664	/	ROUTINES IN THE OPTIONAL PARAMETER AREA STARTING AT "SFILES" [07757] AND	39
40	7665	/	EXTENDING TO 07777. UNDER P?S/8 ODT, NO FILE PARAMETERS ARE PRESERVED; IN	40
41	7666	/	CERTAIN RELEASES THESE ROUTINES MAY RESERVE LOCATION 07756 THUS PREVENTING	41
42	7667	/	PROGRAM ACCESS TO THE =XXXX PARAMETER NORMALLY STORED THERE AS WELL.	42
43	7668			43
44	7669	/	MORE RECENT IMPLEMENTATIONS OF P?S/8 SUPPORT MANY OTHER PARAMETERS WITHIN	44
45	7670	/	THE SYSTEM KERNEL; AS SUCH, 07756 IS SUPPORTED AS THE =XXXX PARAMETER AS IS	45
46	7671	/	THE CASE WITH ALL OTHER PROGRAMS. HOWEVER, THE FILE AREA IS ALWAYS DESTROYED	46
47	7672	/	BY ODT BREAKPOINT MECHANISMS. ODT INSERTS TRAP MECHANISMS INTO THE CURRENT	47
48	7673	/	PROGRAM BEING DEBUGGED; WHEN A BREAKPOINT TRAP IS TRIGGERED, THE CONTENTS OF	48
49	7674	/	FIELD 0 IS SAVED IN THE SYSTEM SCRATCH AREA [BLOCKS 0020-0057] AND ODT IS	49
50	7675	/	RELOADED IN BREAKPOINT DISPLAY MODE FOR FURTHER DEBUGGING.	50
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1	7677	/	IN ALL OTHER BASIC P?S/8 SYSTEM PROGRAMS, ALL PARAMETERS ARE SUPPORTED FULLY	1
2	7678	/	WITHOUT RESTRICTIONS; THE P?S/8 "START" COMMAND CAN BE USED TO CREATE A	2
3	7679	/	COMPATIBLE ENVIRONMENT WITHOUT ACTUALLY PLACING THE PROGRAM ENTRY IN THE	3
4	7680	/	SYSTEM PROGRAM DIRECTORY. HOWEVER, THE =XXXX PARAMETER MUST BE USED TO	4
5	7681	/	SPECIFY THE 12-BIT STARTING ADDRESS FOR THE PROGRAM; LEAVING THE EXPLICIT	5
6	7682	/	PARAMETER OUT OF THE COMMAND LINE WILL INTERPRET THE PASSED VALUE OF 7777 TO	6
7	7683	/	MEAN AN EXPLICIT START IN LOCATION 00200. ADDITIONALLY, THE OPTION SWITCHES	7
8	7684	/	"/0" THROUGH "/7" ARE RESERVED; THE LOWEST EXPLICITLY PASSED SWITCH BECOMES	8
9	7685	/	THE STARTING FIELD FOR THE PROGRAM. [NOTE: FUTURE RELEASES OF THE START	9
10	7686	/	COMMAND WILL DISALLOW MULTIPLE OPTION SWITCHES IN THE RANGE OF /0 THROUGH /7	10
11	7687	/	AS A STARTUP COMMAND ERROR; AS SUCH, IT WILL NOT BE POSSIBLE TO USE HIGHER	11
12	7688	/	NUMERICAL OPTION SWITCHES AS PROGRAM PARAMETERS WHEN USED WITH THE START	12
13	7689	/	COMMAND.]	13
14	7690			14
15	7691	/	IF NONE OF THE SWITCHES IS PASSED, FIELD 0 IS USED BY DEFAULT [AS WELL AS IF	15
16	7692	/	THE PASSED =XXXX VALUE IS 7777; FIELD 0 MUST BE FORCED FOR THIS SITUATION AS	16
17	7693	/	WELL]. OTHER THAN THESE FEW RESTRICTIONS, ALL OTHER OPTIONS ARE AVAILABLE TO	17
18	7694	/	THE PROGRAM BEING STARTED: A) ALL OTHER COMMAND-LINE OPTION SWITCHES "/A"	18
19	7695	/	THROUGH "/Z" AND "/8" THROUGH "/9". INPUT AND OUTPUT FILES MAY BE FREELY	19
20	7696	/	PASSED. OUTPUT FILES [IF ANY] MUST PRECEDE INPUT FILES [IF ANY]. THE COUNT OF	20
21	7697	/	PASSED OUTPUT FILES [WHICH RANGES FROM 0000-0017] IS AVAILABLE IN "SOUTCT"	21
22	7698	/	[07607] BITS[7-11]. [NOTE: ALL OUTPUT FILES WILL HAVE BEEN CREATED BY THE	22
23	7699	/	TIME THE PROGRAM IS LOADED; IN CERTAIN RELEASES OF P?S/8, THE KEYBOARD MONITOR	23
24	7700	/	WILL CREATE THE OUTPUT FILES IF NECESSARY; EARLIER RELEASES WILL ISSUE ERROR	24
25	7701	/	MESSAGES FOR NON-EXISTENT OUTPUT FILES.]	25
26	7702			26
27	7703	/	ALL PASSED FILES ARE CREATED IN A LIST STARTING AT "SFILES" [07757] EXTENDING	27
28	7704	/	AS FAR AS NECESSARY. THE MAXIMUM FILE COUNT IS 17. A SENTINEL VALUE OF 0000	28
29	7705	/	IS PLACED AFTER THE LAST PASSED FILE WITH ONE EXCEPTION:	29
30	7706			30
31	7707	/	MAXIMUM LENGTH LISTS [17 FILES PASSED] ARE SELF-TERMINATING BECAUSE THE	31
32	7708	/	DESIGNATED METHOD OF ACCESSING THE FILES ALWAYS USES A PRESCRIBED METHOD	32
33	7709	/	SIMILAR TO WHAT FOLLOWS:	33
34	7710			34
35	7711	/	ISZ FILPTR /BUMP TO NEXT FILE [WAS SET TO SFILES-1].	35
36	7712	/	TAD I FILPTR /GET LATEST FILE [OR 0000 IF AT END].	36
37	7713	/	SNA CLA /SKIP IF ACTUALLY A FILE HANDLE.	37
38	7714	/	JMP ENDFILES /JUMP IF NO MORE FILES.	38
39	7715			39
40	7716	/	IF A SENTINEL 0000 VALUE TERMINATES THE LIST, ALL OF THE ABOVE INSTRUCTIONS	40
41	7717	/	EXECUTE. IF THE LIST IS MAXIMUM LENGTH, THE "ISZ FILPTR" INSTRUCTION SKIPS	41
42	7718	/	THE NEXT INSTRUCTION THUS THE AC IS 0000 FOR THIS SPECIAL CASE LEADING TO THE	42
43	7719	/	END OF FILE PROCESSING. [NOTE: WE THANK RICHARD LARY FOR THIS PROGRAMMING	43
44	7720	/	GEM THAT ONLY APPLIES TO SITUATIONS SUCH AS R-L MONITOR AND P?S/8 FILE LIST	44
45	7721	/	PROCESSING; THERE ARE FEW APPLICATIONS FOR LISTS LOCATED AT THE TOP END OF	45
46	7722	/	MEMORY.]	46
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1	7724	/	ADDITIONAL USAGE OF THE PASSED FILE PARAMETER AREA [IN MEMORY].	1
2	7725			2
3	7726	/	WHILE ALL BASIC P?S/8 SYSTEM PROGRAMS [OTHER THAN ODT] POTENTIALLY PROCESS	3
4	7727	/	PASSED FILES BY READING THE CONTENTS OF THE PASSED FILE AREA, SOME PROGRAMS	4
5	7728	/	ARE KNOWN TO LOAD OVER THIS AREA IN SPECIFIC CIRCUMSTANCES. AS EXPLAINED	5
6	7729	/	ELSEWHERE IN THIS DOCUMENT, THE "SLURP" LOADER [WHICH PROCESSES BINARY INPUT	6
7	7730	/	FILE HANDLES PLACED IN THE PASSED FILE AREA] MAY BE SELECTIVELY "DESTROYED" BY	7
8	7731	/	THE LAST "FILE" PASSED TO THE SLURP LOADER; THIS IS ACTUALLY A RESERVED BLOCK	8
9	7732	/	ON THE SYSTEM DEVICE [BLOCK 0060] CONTAINING CODE DESIGNED TO REPLACE THE	9
10	7733	/	SLURP LOADER WITH THE SYSTEM DEVICE HANDLER.	10
11	7734			11
12	7735	/	DURING THIS PROCESS, THE CONTENTS OF THE MEMORY LOCATIONS WITHIN THE PASSED	12
13	7736	/	FILE PARAMETER AREA ARE REPLACED WITH CODE BEING LOADED INTO MEMORY FOR THE	13
14	7737	/	PURPOSE OF ALTERING THE NORMAL SLURP LOADING PROCESS. [NOTE: SINCE THE	14
15	7738	/	PASSED "FILE" CAUSING THIS OPERATION IS ALWAYS THE LAST INPUT FILE PROCESSED,	15
16	7739	/	THE ENTIRE PASSED FILE PARAMETER AREA IS NO LONGER NEEDED; THERE IS NO FURTHER	16
17	7740	/	USE FOR THE FILE INFORMATION PASSED BY THE OPERATING SYSTEM; AS SUCH, THESE	17
18	7741	/	WORDS ARE AVAILABLE FOR OVERLAY WITH ADDITIONAL INSTRUCTIONS AND DATA TO	18
19	7742	/	CARRY OUT THE OVERLAY PROCESS.]	19
20	7743			20
21	7744	/	AFTER THIS CODE IS LOADED, THE LAST OPERATION INITIATED BY THIS "FILE" CAUSES	21
22	7745	/	THE [SOMEWHAT MODIFIED] SLURP LOADER TO JUMP INTO THE JUST-LOADED CODE; THE	22
23	7746	/	CONTENTS OF NEARLY ALL LOCATIONS IN THE RANGE OF 07600-07777 IS CHANGED TO THE	23
24	7747	/	CORRESPONDING CONTENTS OF BLOCK 0000; THIS CAUSES ALL NECESSARY LOCATIONS	24
25	7748	/	WITHIN THE SYSTEM KERNEL TO BE RESTORED. THE SYSTEM DEVICE HANDLER AND SYSTEM	25
26	7749	/	RESTART AT 07600 ARE FUNCTIONAL AS THE USER BINARY PROGRAM IS STARTED. [NOTE:	26
27	7750	/	THE PROGRAM STARTUP INFORMATION IS STORED IN MEMORY THAT IS NOT OVERLAID BY	27
28	7751	/	RELOADING THE PARTIAL CONTENTS OF BLOCK 0000; THIS INCLUDES THE SAFE HALT	28
29	7752	/	LOCATION USED AS THE DEFAULT STARTING ADDRESS; THESE LOCATIONS ARE PLACED IN	29
30	7753	/	AN AREA OF MEMORY THAT SYSTEM DEVICE HANDLER OPERATIONS WILL EVENTUALLY STORE	30
31	7754	/	INTO; THE PRIOR CONTENTS DO NOT AFFECT EITHER THE SYSTEM HANDLER OR THE SYSTEM	31
32	7755	/	RESTART AT 07600. IT IS CRUCIAL THAT THERE ARE NO CONFLICTS BETWEEN THE	32
33	7756	/	SYSTEM DEVICE HANDLER AND THE SLURP LOADER IN THIS REGARD IN ORDER FOR THE	33
34	7757	/	RELOAD PROCESS TO SUCCEED; EACH CONFIGURATION SUPPORTING ALL THREE RELEVANT	34
35	7758	/	COMPONENTS MUST BE VERY CAREFULLY DESIGNED.] [NOTE: CERTAIN CONFIGURATIONS DO	35
36	7759	/	NOT SUPPORT A SLURP LOADER DUE TO HARDWARE RESTRICTIONS; BINARY FILE LOADING	36
37	7760	/	BY THE VIRTUAL LOADER IS FORCED; AS SUCH, THE MAXIMUM NUMBER OF PASSED INPUT	37
38	7761	/	FILES INCREASES BY ONE; NO RELOADER "FILE" IS DEFINED ON THESE SYSTEMS.]	38
39	7762			39
40	7763	/	ADDITIONAL USAGE OF THE PASSED FILE PARAMETER AREA [STORED IN BLOCK 0000].	40
41	7764			41
42	7765	/	THE PERMANENT CONTENTS OF THE CORRESPONDING WORDS OF BLOCK 0000 [RELATIVE	42
43	7766	/	156-177] MAY ALSO BE LOADED WITH CERTAIN VALUES NEEDED AS PART OF THE BOOTUP	43
44	7767	/	PROCESS; THIS ONLY APPLIES TO CERTAIN RELEASES OF PARTICULAR P?S/8 SYSTEM	44
45	7768	/	DEVICE HANDLERS, ESPECIALLY SYSTEM CONFIGURATIONS THAT REQUIRE A SYSTEM DEVICE	45
46	7769	/	HANDLER EXTENDED MEMORY EXTENSION SUCH AS THE RK8E, RX8E ETC.	46
47	7770			47
48	7771	/	CERTAIN BOOTSTRAP CONVENTIONS MAY REQUIRE THE CONTENTS OF BLOCK 0000 TO BE	48
49	7772	/	LOADED WITH AN UNUSUAL OFFSET [SUCH AS THE RX SERIES OF DISKETTE CONTROLLERS].	49
50	7773	/	AS SUCH, THE PERMANENT CONTENTS OF THESE WORDS MAY BE USED DURING THE	50
51	7774	/	EARLIEST STAGES OF A MULTI-TIERED BOOTSTRAP PROCEDURE DESIGNED TO BRING	51
52	7775	/	ANOTHER DEDICATED SYSTEM BLOCK INTO MEMORY TO FINALIZE THE OVERALL PROCESS.	52
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1	7777	/	IN THE CASE OF EXTENDED DEVICE HANDLERS, THE SPACE IS OFTEN USED TO LOAD IN	1
2	7778	/	THE DESIGNATED "HELPER" BLOCKS OF THE SYSTEM DEVICE HANDLER TO DETERMINE	2
3	7779	/	EXACTLY HOW MUCH EXTENDED MEMORY IS AVAILABLE; THE REST OF THE HANDLER IS THEN	3
4	7780	/	LOADED INTO THE HIGHEST MEMORY FIELD. AS NECESSARY, FIELD RELOCATION IS	4
5	7781	/	PERFORMED TO INITIATE OPERATION OF THE HANDLER AFTER THE CODE IS MOVED [NOTE:	5
6	7782	/	ONLY RELEVANT ON 12K OR LARGER MACHINES].	6
7	7783			7
8	7784	/	IN PERVERSE SITUATIONS, LARGER SECTIONS OF THE INTENDED FIELD 0 DEVICE HANDLER	8
9	7785	/	CODE MAY NOT BE LOADED WHEN STANDARD BOOSTRAP METHODS ARE USED; IN SUCH CASES,	9
10	7786	/	THE PERMANENT CONTENTS OF BLOCK 0000 MAY INCLUDE ADDITIONAL INTERMEDIARY	10
11	7787	/	BOOTUP CODE. THE ADDITIONAL CODE IS USED FOR THE PURPOSE OF LOADING IN	11
12	7788	/	DESIGNATED HELPER BLOCKS WHICH THEN ALSO PROVIDE THE PROPER CONTENTS OF THE	12
13	7789	/	DISPLACED WORDS. IN THE MOST COMPLEX CASES, IT MAY BE NECESSARY FOR THE	13
14	7790	/	EXTENDED DEVICE HANDLER CODE TO INCLUDE A PORTION OF THIS "FIX-UP" CODE TO	14
15	7791	/	AVOID THE OVERHEAD OF ACCESSING OTHER BLOCKS OF THE ON THE SYSTEM DEVICE	15
16	7792	/	EVERY TIME THE SYSTEM IS RESTARTED AT 07600.	16
17	7793			17
18	7794	/	IN SOME CASES, ATTEMPTS TO WRITE LOGICAL BLOCK ZERO FROM THE CONTENTS OF	18
19	7795	/	LOCATIONS 07600-07777 MAY HAVE TO BE INTERCEPTED TO ENSURE THE CONTENTS OF THE	19
20	7796	/	BLOCK ARE AS REQUIRED BY THE BOOSTRAP CONVENTION [WHICH WILL DIFFER FROM THE	20
21	7797	/	RUNNING CODE IN MEMORY.] THE ACTUAL UPDATED CONTENTS OF LOGICAL BLOCK 0000	21
22	7798	/	WILL BE A COMPOSITE OF THE DYNAMIC LOCATIONS WITHIN THE SYSTEM KERNEL THAT	22
23	7799	/	REQUIRE UPDATING [SUCH AS THE STORED COPY OF THE CURRENT DATE] WHILE OTHER	23
24	7800	/	LOCATIONS IN BLOCK 0000 ARE TAKEN FROM THE ACTUAL BLOCK 0000 CODE WHICH IS	24
25	7801	/	THEN AUGMENTED ONLY BY THE UPDATABLE MEMORY LOCATIONS WITHIN THE HANDLER.	25
26	7802			26
27	7803	/	A WORST-CASE SCENARIO MIGHT REQUIRE SPLITTING OFF SOME OF THESE WORDS INTO AN	27
28	7804	/	ALTERNATE BLOCK [PART OF ONE OF THE HELPER BLOCKS] TO STORE UPDATED WORDS THAT	28
29	7805	/	CANNOT BE STORED IN BLOCK 0000 DIRECTLY. [NOTE: TO DATE THIS ABSOLUTE	29
30	7806	/	WORST-CASE SCENARIO HASN'T BEEN REQUIRED; HOWEVER, ASSUMING THE REST OF THE	30
31	7807	/	CODE CAN FIT WITHIN THE DEVICE HANDLER EXTENSION, EVEN THIS CASE CAN BE	31
32	7808	/	HANDLED. AS OF THIS WRITING, THE SYSTEM DEVICE HANDLER FOR THE DECMATE II-III	32
33	7809	/	SERIES RX50 DISKETTES HAS NOT BEEN FULLY DESIGNED; DUE TO VARIOUS CONFLICTING	33
34	7810	/	REQUIREMENTS OF A BOOTABLE DECMATE RX50, SOME OF THESE PROBLEMS ARE LIKELY TO	34
35	7811	/	ARISE. IT SHOULD BE NOTED THAT OS/278 V2, THE ONLY OS/8 FAMILY OPERATING	35
36	7812	/	SYSTEM FOR THE DECMATES, IS INCAPABLE OF WRITING ON LOGICAL RECORD 0000; TO	36
37	7813	/	ATTEMPT TO DO SO CAUSES THE DISKETTE TO BECOME UNBOOTABLE. SPECIAL UTILITIES	37
38	7814	/	ARE REQUIRED TO CREATE THESE DISKETTES FROM DESIGNATED DIAGNOSTIC UTILITY	38
39	7815	/	DISKETTES; THE NORMAL METHODS OF SYSTEM COPY AND UPDATE USED IN OS/8 CANNOT BE	39
40	7816	/	USED IN OS/278.]	40
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1	7818	/	IN P?S/8, ANY OF THESE SCENARIOS CAN BE SUCCESSFULLY IMPLEMENTED IN MOST	1
2	7819	/	EXTENDED DEVICE HANDLERS DUE TO THE LARGE CODE SPACE [6000-7777] ALLOCATED.	2
3	7820	/	FOR MOST HANDLERS, THE CODE TO ACCOMPLISH THIS WOULD LIKELY BE LESS THAN ONE	3
4	7821	/	PAGE [128 WORDS], INCLUDING A SMALL ROUTINE USED TO MOVE THE CODE TO FIELD 0.	4
5	7822	/	[NOTE: THERE ARE TECHNIQUES TO IMPLEMENT SELF-MODIFYING CODE WITHOUT USING	5
6	7823	/	INDIRECT ADDRESSING; IF THIS IS FEASIBLE, IT WOULD NOT BE NECESSARY TO BE	6
7	7824	/	AWARE OF THE FIELD THE FIX-UP CODE RESIDES IN. SHOULD INDIRECT ADDRESSING BE	7
8	7825	/	REQUIRED, APPROPRIATE "CDF" INSTRUCTIONS CAN BE PLACED WITHIN THE CODE AS	8
9	7826	/	NEEDED; THE CORRECT FIELD FOR THE HANDLER EXTENSION CAN BE OBTAINED	9
10	7827	/	DYNAMICALLY. THIS FORM OF RELOCATION RESULTS IN SLIGHTLY LONGER MOVE	10
11	7828	/	ROUTINES. THE HANDLER EXTENSION HELPER BLOCKS ARE NOT MEANT TO BE PART OF THE	11
12	7829	/	HANDLER EXTENSION; RATHER, THEIR ONLY PURPOSE IS TO CAUSE THE EXTENSION TO BE	12
13	7830	/	FUNCTIONAL IN THE SMALLEST AMOUNT OF RESIDENT CODE. FOR MORE INFORMATION AND	13
14	7831	/	PRACTICAL EXAMPLES, REFER TO VARIOUS SPECIFIC RELOCATION ROUTINES IN THE	14
15	7832	/	VARIOUS EXTENDED SYSTEM DEVICE HANDLER FOR THE RK8E/RK05, RX8E/RX01, ETC.]	15
16	7833			16
17	7834	/	[NOTE: THE TC01/TC08 SYSTEM DEVICE HANDLER DESCRIBED ELSEWHERE IN THIS	17
18	7835	/	DOCUMENT DOES NOT USE THE PERMANENT CONTENTS OF THE PASSED FILE AREA AS	18
19	7836	/	DEFINED HERE; HOWEVER, FUTURE RELEASES OF THIS DEVICE HANDLER MAY ADD	19
20	7837	/	ADDITIONAL CODE ONLY USED DURING THE INITIAL STARTUP PHASE OF A P?S/8 BOOTUP	20
21	7838	/	PROCEDURE. THIS AREA COULD ALSO CONTAIN GENERIC CODE TO ASSIST IN OTHER	21
22	7839	/	ASPECTS OF THE BOOTUP PROCESS; AS SUCH, IT WOULD APPLY TO ALL HANDLER VARIANTS	22
23	7840	/	ASSOCIATED WITH A PARTICULAR P?S/8 RELEASE. THE PERMANENT CONTENTS OF THESE	23
24	7841	/	WORDS IS ONLY MEANINGFUL DURING THE EARLY STAGES OF THE BOOTUP PROCESS; ONCE	24
25	7842	/	LOADED, THE MEMORY LOCATIONS ARE USED AS DESCRIBED ELSEWHERE IN THIS	25
26	7843	/	DOCUMENT.]	26
27	7844			27
28	7845	/	ADDITIONAL USAGE OF THE PASSED FILE PARAMETER AREA BY THE P?S/8 SHELL OVERLAY.	28
29	7846			29
30	7847	/	WITHIN THE SHELL ENVIRONMENT, ALL RELEVANT EXTENDED MEMORY IS COMPLETELY	30
31	7848	/	PHYSICAL WHILE ALL OF FIELD 0 IS ENTIRELY VIRTUAL UNTIL AN EXECUTABLE PROGRAM	31
32	7849	/	IS STARTED. THIS ALLOWS THE SHELL TO OPERATE AS AN INDEPENDENT 4K SUBSYSTEM	32
33	7850	/	WITH ITS OWN OVERLAY STRUCTURES AS DEFINED IN THE BASIC P?S/8 SYSTEM. AS	33
34	7851	/	NECESSARY, INTERCOMMUNICATION ELEMENTS WILL BE PLACED IN THE SYSTEM FILE LIST	34
35	7852	/	AREA UNTIL AN EXECUTION-CLASS COMMAND IS INVOKED.	35
36	7853			36
37	7854	/	WHEN THE SHELL OVERLAY IS IN EFFECT, VARIOUS ROUTINES ARE PLACED INTO THE	37
38	7855	/	PASSED FILE PARAMETER AREA [WHICH IS NOT SUPPORTED FOR THE ORIGINAL PURPOSE AS	38
39	7856	/	IS THE CASE IN THE BASIC P?S/8 SYSTEM]. THIS ALLOWS THE SHELL TO EXECUTE	39
40	7857	/	PROGRAMS WITH POTENTIAL STARTING ADDRESSES ANYWHERE IN 32K, AS WELL AS	40
41	7858	/	SUPPORTING THE ABILITY FOR A PROGRAM TO SPECIFY A MEMORY SAVE OPERATION WHEN	41
42	7859	/	LOCATION 07600 IS BRANCHED TO; THE DEFAULT IS TO ONLY PERFORM THE READ-ONLY	42
43	7860	/	BOOTSTRAP ASSOCIATED WITH THE BASIC P?S/8 SYSTEM. [NOTE: THE BASIC P?S/8	43
44	7861	/	SYSTEM HAS SEVERAL EXPLICIT WAYS TO HAVE MEMORY SAVED BEFORE THE READ-ONLY	44
45	7862	/	RESTART OPERATION; HOWEVER, NONE OF THESE TECHNIQUES APPLY TO THE SHELL	45
46	7863	/	OVERLAY CONFIGURATION. INSTEAD, EVERY SHELL-BASED PROGRAM CAN POTENTIALLY	46
47	7864	/	REQUEST AN EXPLICIT MEMORY SAVE; HOWEVER, THE DEFAULT ACTION IS TO NOT SAVE	47
48	7865	/	MEMORY WHICH IS THE FAR MORE LIKELY SCENARIO. THIS LEADS TO LOWERING SYSTEM	48
49	7866	/	OVERHEAD WHILE INCREASING SYSTEM RELIABILITY.]	49
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1	7868	/	FOR THE SELDOM-USED MEMORY-SAVING OPERATIONS, THE BOOTSTRAP IS MODIFIED TO	1
2	7869	/	FIRST SAVE ALL OF FIELD 0 IN THE DEFINED SCRATCH AREA [BLOCKS 0020-0057], THEN	2
3	7870	/	PERFORM THE NORMAL READ-ONLY BOOTUP PROCEDURE. THIS IS ACCOMPLISHED BY	3
4	7871	/	MODIFYING LOCATION 07600 TO A "JMP" INSTRUCTION INTO THE SYSTEM FILE LIST AREA	4
5	7872	/	TO THE MEMORY SAVE ROUTINES FOLLOWED BY ANOTHER "JMP" INSTRUCTION BACK TO	5
6	7873	/	07600. [NOTE: THE INSTRUCTION NORMALLY EXECUTED AT 07600 WILL BE SAVED AND	6
7	7874	/	LATER RESTORED BEFORE THE SYSTEM RESTART; AS LONG AS THIS WORD IS RESTORED,	7
8	7875	/	THE NORMAL DEVICE HANDLER REBOOT FUNCTION WILL WORK PROPERLY.]	8
9	7876			9
10	7877	/	ALL P?S/8 SYSTEM DEVICE HANDLERS ARE REQUIRED TO FUNCTION FOR PURPOSES OTHER	10
11	7878	/	THAN SYSTEM RESTART WHILE THE NORMAL CONTENTS OF 07600 ARE DISPLACED. AS	11
12	7879	/	SUCH, LOCATION 07600 MUST NOT BE OTHERWISE REFERENCED BY THE DEVICE HANDLER TO	12
13	7880	/	ALLOW PROPER FUNCTIONALITY. AS A RESULT OF THIS MECHANISM, THERE IS NO NEED	13
14	7881	/	FOR AN ALTERNATE NON-STANDARD RESTART ADDRESS SUCH AS WAS UNSUCCESSFULLY	14
15	7882	/	ATTEMPTED WITHIN OS/8. THE UNIVERSAL PROGRAM EXIT/SYSTEM RESTART ADDRESS OF	15
16	7883	/	07600 ALWAYS APPLIES IN ALL OPERATING MODES.	16
17	7884			17
18	7885	/	THE SHELL DEBUGGING FACILITY ALSO OPERATES IN THE PASSED FILE PARAMETER AREA	18
19	7886	/	IN A MANNER SIMILAR TO THE P?S/8 BASIC ODT BREAKPOINT MECHANISM. HOWEVER, THE	19
20	7887	/	SHELL ENVIRONMENT PASSES FILES AND OTHER PARAMETERS IN AN INDEPENDENT AREA	20
21	7888	/	ONLY MEANINGFUL IN THE SHELL ENVIRONMENT [THE FILE AND COMMUNICATIONS AREA];	21
22	7889	/	AS SUCH, DEBUGGING SHELL PROGRAMS WITH THE USE OF BREAKPOINT TRAPS DOES NOT	22
23	7890	/	INTERFERE WITH SHELL PROGRAM COMPATIBILITY IN ANY WAY.	23
24	7891			24
25	7892	/	LOCATIONS 07776-07777 HAVE SPECIFIC MEANING TO SHELL PROGRAMS. LOCATION 07776	25
26	7893	/	IS ALWAYS A "CIF CDF" INSTRUCTION [62X3] TO THE PROGRAM STARTING FIELD;	26
27	7894	/	LOCATION 07777 IS A "JMP I" INSTRUCTION THAT WILL ALLOW INSPECTION OF THE	27
28	7895	/	STARTING ADDRESS [WHICH MAY BE IMPLEMENTED AS THE CONTENTS OF 07637 FOR ALL	28
29	7896	/	SHELL EXECUTION MODES] BY EXAMINING THE CORRESPONDING ADDRESS. AS SUCH, P?S/8	29
30	7897	/	SHELL PROGRAMS WILL HAVE THE ABILITY TO BE AWARE OF THEIR OWN STARTING ADDRESS	30
31	7898	/	AND FIELD; IT IS POSSIBLE TO WRITE A UTILITY THAT FUNCTIONS DIFFERENTLY BASED	31
32	7899	/	ON PARAMETERS SUCH AS THE STARTING ADDRESS OR EXECUTABLE FILE NAME. THIS CAN	32
33	7900	/	BE USEFUL WHEN PROGRAMS WITH ANALOGOUS OPERATING MODES AND THE USE OF PASSED	33
34	7901	/	PARAMETERS MAY ONLY DIFFER IN SOME INTERNAL OPERATING MANNER; IT WOULDN'T BE	34
35	7902	/	NECESSARY TO CREATE SEVERAL SIMILAR UTILITIES IF ONE MASTER UTILITY CAN	35
36	7903	/	INSPECT ITS OWN STARTUP INFORMATION AND SELF-MODIFY APPROPRIATELY.	36
37	7904			37
38	7905	/	END OF APPENDIX M.	38
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1	7907	/	APPENDIX N.	1
2	7908			2
3	7909	/	TAKING ADVANTAGE OF FORWARD INITIAL SEARCH WITHIN THE TC01/TC08 SLURP LOADER.	3
4	7910			4
5	7911	/	UNLIKE THE TC01/TC08 SYSTEM DEVICE HANDLER, THE CURRENT IMPLEMENTATION OF THE	5
6	7912	/	SLURP LOADER [TAKEN FROM A 1976 RELEASE OF P?S/8] ALWAYS INITIATES NEW FILE	6
7	7913	/	SEARCHES IN THE FORWARD DIRECTION. CONVENTIONAL REASONING MIGHT DICTATE THIS	7
8	7914	/	TO BE UNFORTUNATE IN ALL CASES IN TERMS OF OVERALL DECTAPE MOTION WHILE	8
9	7915	/	TRANSFERRING BINARY FILES; HOWEVER, IN THIS PARTICULAR INSTANCE:	9
10	7916			10
11	7917	/	1) WITH THE EXCEPTION OF THE % AND \$ FILES [LOCATED AT BLOCK 0020 AND 0040	11
12	7918	/	RESPECTIVELY], ALL TFS BINARY FILES ARE LOCATED FORWARD OF THE CURRENT TAPE	12
13	7919	/	POSITION WHEN THE P?S/8 "BIN" UTILITY HAS JUST BEEN READ. [THE SLURP	13
14	7920	/	LOADER IS AN INTERNAL COMPONENT OF P?S/8 BIN; ONCE INITIALIZED, THE SLURP	14
15	7921	/	LOADER AND STAND-ALONE BOOTSTRAP AT 07600 DISPLACE THE SYSTEM DEVICE	15
16	7922	/	HANDLER. THE FIRST OPERATION INITIATES A FORWARD SEARCH FOR THE FIRST	16
17	7923	/	PASSED INPUT FILE].	17
18	7924			18
19	7925	/	2) ALL ADDITIONAL BLOCKS WITHIN FILES ARE ALWAYS FORWARD OF THE PREVIOUS BLOCK	19
20	7926	/	JUST PROCESSED; THIS IS THE INTENDED FUNCTION.	20
21	7927			21
22	7928	/	3) RARELY ARE ADDITIONAL FILES ACTUALLY PASSED IN TYPICAL USAGE. [THE MAXIMUM	22
23	7929	/	FILE COUNT IS 17 ASSUMING THE SYSTEM HANDLER RELOADER IS NOT IN USE.]	23
24	7930			24
25	7931	/	HOWEVER, IF MULTIPLE PASSED BINARY FILES ARE LOCATED CONTIGUOUSLY ON THE	25
26	7932	/	DECTAPE, THIS COULD CAUSE UNDESIRABLE DOUBLE TAPE TURNAROUNDS FOR EACH FILE	26
27	7933	/	PROCESSED [DEPENDING ON RELATIVE TAPE POSITION BETWEEN SEQUENTIALLY-USED	27
28	7934	/	FILES].	28
29	7935			29
30	7936	/	THE PROBLEM DISCUSSED IN 3) ABOVE CAN BE ELIMINATED BY PLACING UNRELATED FILES	30
31	7937	/	BETWEEN FILES TO BE LOADED SEQUENTIALLY; ALL FORWARD SEARCHES WILL ALWAYS	31
32	7938	/	LOCATE THE STARTING BLOCK OF EVERY FILE WITHOUT TURNING AROUND. [NOTE: TFS	32
33	7939	/	FILES ARE PLACES FURTHER APART THAN THE BLOCK SPACING REQUIRED TO RESTART THE	33
34	7940	/	DECTAPE SEARCH ONCE THE MOTION BIT IS TURNED OFF.] AN EXAMPLE OF THIS FORM	34
35	7941	/	OF PERFORMANCE OPTIMIZATION IS AS FOLLOWS:	35
36	7942			36
37	7943	/	.CATALOG:0	37
38	7944			38
39	7945	/	BFILE1 TEXT07 BFILE2 TEXT08	39
40	7946	/	BFILE3 TEXT3A BFILE4 TEXT3R	40
41	7947	/	BFILE5 TXTHLP BFILE6 MYPROG	41
42	7948			42
43	7949	/	.BIN BFILE1,BFILE2,BFILE3,BFILE4,BFILE5,BFILE6=200	43
44				44
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1	7951	/	AFTER THE OPERATING SYSTEM LOADS THE P?S/8 BIN UTILITY [WHICH IN TURN LOADS	1
2	7952	/	THE SLURP LOADER], A FORWARD SEARCH WILL FIND "BFILE1" WITHOUT TURNING THE	2
3	7953	/	DECTAPE AROUND. SINCE EACH TFS FILE IS 16 BLOCKS LONG, THERE IS A 16 BLOCK	3
4	7954	/	SPACING BETWEEN THE BINARY FILES PROVIDED BY THE UNRELATED FILES. THIS	4
5	7955	/	EXCEEDS THE DECTAPE COASTING OFFSET REQUIREMENT WHEN SETTING TAPE MOTION.	5
6	7956	/	NOTE: TAPE MOTION IS CLEARED AFTER EACH FILE IS PROCESSED. THIS IS NORMAL	6
7	7957	/	PRACTICE AS EACH FILE IS AN INDEPENDENT SET OF BLOCKS TO PROCESS; THE NEXT	7
8	7958	/	INPUT FILE COULD BE LOCATED ON A DIFFERENT DECTAPE DRIVE UNIT; THE PREVIOUS	8
9	7959	/	TAPE UNIT MUST BE STOPPED BEFORE THE NEXT DRIVE UNIT IS STARTED.	9
10	7960			10
11	7961	/	ADJACENT FILES REQUIRE SUBSEQUENT FILES BE SEARCHED INITIALLY IN REVERSE DUE	11
12	7962	/	TO INERTIAL ROLL AFTER THE CONTROLLER CLEARS THE MOTION BIT; IN THE CURRENT	12
13	7963	/	IMPLEMENTATION OF THE SLURP LOADER THIS IS NOT AVAILABLE. BY UTILIZING FILE	13
14	7964	/	INTERLEAVING AS DESCRIBED ABOVE, THE ENTIRE OPERATION LOADS ALL FILES WITHOUT	14
15	7965	/	TAPE TURNAROUND. [NOTE: THE TAPE SEARCH ROUTINE WITHIN THE SLURP LOADER IS	15
16	7966	/	CAPABLE OF REVERSE SEARCH IF NECESSARY; THE SPECIFIC CONSIDERATION DISCUSSED	16
17	7967	/	HERE ONLY REFERS TO THE INITIAL SEARCH DIRECTION FOR EACH FILE.]	17
18	7968			18
19	7969	/	THE ONLY REQUIREMENT IS THE INTERLEAVED PLACEMENT OF FILES NOT PART OF THE	19
20	7970	/	BINARY FILE STREAM AT HAND; THE OTHER FILES CAN BE USED FOR ANY ALTERNATE	20
21	7971	/	PURPOSE, INCLUDING THE STREAM OF BINARY FILES FOR ANOTHER PROGRAM OR SOURCE	21
22	7972	/	FILES FOR ANY PROGRAM [INCLUDING THE SOURCE CODE OF THE CURRENT BINARY FILE	22
23	7973	/	STREAM].	23
24	7974			24
25	7975	/	P?S/8 TFS FILE NAMES CAN BE ENTERED DIRECTLY WITH KEYBOARD MONITOR COMMANDS;	25
26	7976	/	THE FILE LOCATIONS ARE PERMANENT UNLESS MODIFIED WITH DIRECTORY-ORIENTED	26
27	7977	/	MAINTENANCE COMMANDS. MOST USERS CAN QUICKLY ADAPT TO THIS METHOD USING ONLY	27
28	7978	/	THE DE[LETE] AND [EN]TER COMMANDS AS REQUIRED. ADVANCED P?S/8 USERS SETUP	28
29	7979	/	NEARLY ALL TFS DIRECTORY SLOTS IN ANTICIPATION OF PERFORMANCE IMPROVEMENT.	29
30	7980			30
31	7981	/	[NOTE: MOVING A TFS DIRECTORY ENTRY DESTROYS THE RELATIONSHIP BETWEEN THE	31
32	7982	/	FILE NAME AND THE FORMER CONTENTS; REVERTING FILE NAMES CAN RESTORE THE	32
33	7983	/	ABILITY TO ACCESS EXISTING FILES AS THE CONTENTS OF THE FILE BLOCKS ARE NOT	33
34	7984	/	MODIFIED WHEN THE TFS DIRECTORY IS UPDATED. AS SUCH, FILE PLACEMENT	34
35	7985	/	OPTIMIZATION TECHNIQUES SHOULD BE PERFORMED BEFORE THE FILES ARE ACTUALLY USED	35
36	7986	/	FOR NORMAL CONTENT PURPOSES.]	36
37	7987			37
38	7988	/	SINCE SOURCE FILE PLACEMENT CAN ALSO BE BENEFICIAL IN TERMS OF LOWERING	38
39	7989	/	ASSEMBLY TIMES USING A SIMILAR POSITIONING SCHEME, THE TWO METHODS CAN BE	39
40	7990	/	COMPLEMENTARY. [NOTE: THE SYSTEM DEVICE HANDLER WILL ALWAYS SET THE CORRECT	40
41	7991	/	INITIAL SEARCH DIRECTION. BY FILE PLACEMENT AS DESCRIBED, THE SEARCH WILL BE	41
42	7992	/	CORRECTLY SET TO INITIALLY FORWARD FOR BOTH PURPOSES.]	42
43	7993			43
44	7994	/	END OF APPENDIX N.	44
45				45
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1	7996	/	APPENDIX 0.	1
2	7997			2
3	7998	/	DETAILED DESCRIPTION OF SLURP FORMAT P?S/8 BINARY FILES.	3
4	7999			4
5	8000	/	EVERY BLOCK OF A SLURP FILE IS INDEPENDENT OF ALL OTHER BLOCKS OTHER THAN ANY	5
6	8001	/	CONSIDERATION OF CURRENT LOADING ORIGIN AND CURRENT LOADING FIELD INHERITED	6
7	8002	/	FROM THE PREVIOUS BLOCK PROCESSED. TO ENSURE PROPER LOADING, EVERY SLURP FILE	7
8	8003	/	CONTAINS AN INITIAL ORIGIN TO LOCATION 0200 FOLLOWED BY AN APPROPRIATE FIELD	8
9	8004	/	SETTING, GENERALLY TO FIELD 0. THIS ALLOWS SYSTEM PROGRAMS TO DETERMINE IF A	9
10	8005	/	MIXED STREAM OF PASSED INPUT FILES CONTAINS SLURP BINARY FILES FOR	10
11	8006	/	PROGRAM-SPECIFIC PURPOSE.	11
12	8007			12
13	8008	/	THERE ARE 18 GROUPS OF SEVEN WORDS WITHIN EACH SLURP BINARY FILE BLOCK. ANY	13
14	8009	/	ADDITIONAL WORDS WITHIN THE BLOCK ARE IGNORED.	14
15	8010			15
16	8011	/	ALL DECTAPES HAVE THREE ADDITIONAL WORDS SINCE PDP-8 DECTAPES ARE BASED ON	16
17	8012	/	129 WORDS PER BLOCK; MOST SUPPORTED DEVICES ARE BASED ON 128 WORDS PER BLOCK.	17
18	8013	/	BY IGNORING ALL ADDITIONAL WORDS AS NECESSARY, ALL IMAGED TFS BINARY FILES ARE	18
19	8014	/	COMPATIBLE WHEN COPIED TO ALTERNATE DEVICES. [NOTE: INDIVIDUAL SLURP LOADERS	19
20	8015	/	ARE HARDWARE-SPECIFIC WITH THE EXCEPTION OF THE P?S/8 VIRTUAL SLURP LOADER	20
21	8016	/	AVAILABLE ON ALL P?S/8 SYSTEM CONFIGURATIONS. SLURP FORMAT IS SURPRISINGLY	21
22	8017	/	UNIVERSAL; IT IS USED IN CERTAIN OTHER PDP-8 OPERATING SYSTEMS GENERALLY	22
23	8018	/	WITHOUT DOCUMENTATION.]	23
24	8019			24
25	8020	/	EACH OF THE 18 GROUPS IS PROCESSED IN SEQUENCE. THE CONTROL WORD FOR THE	25
26	8021	/	NEXT SIX DATA WORDS IS PROCESSED FIRST TO DETERMINE THE DISPOSITION OF EACH	26
27	8022	/	WORD.	27
28	8023			28
29	8024	/	EACH CONTROL WORD CONTAINS SIX PAIRS OF CONTROL BITS, EACH FOR THE	29
30	8025	/	CORRESPONDING DATA WORD. [BIT PAIRS ARE RECKONED LEFT TO RIGHT.]	30
31	8026			31
32	8027	/	WITHIN EACH BIT PAIR, THE FOLLOWING DEFINITIONS APPLY:	32
33	8028			33
34	8029	/	00 - THE CORRESPONDING WORD IS DATA TO BE LOADED INTO THE LOCATION POINTED TO	34
35	8030	/	BY THE LOADING ORIGIN; THE LOADING ORIGIN WILL BE INCREMENTED AFTERWARDS.	35
36	8031			36
37	8032	/	10 - THE CORRESPONDING WORD IS AN ORIGIN SETTING; THE LOADING ORIGIN WILL BE	37
38	8033	/	UPDATED TO THE DATA VALUE.	38
39	8034			39
40	8035	/	11 - THE CORRESPONDING WORD IS A FIELD SETTING EXPRESSED AS A "CDF X" [62X1]	40
41	8036	/	INSTRUCTION TO THE NEW LOADING FIELD X; THE VALUE WILL BE STORED IN-LINE	41
42	8037	/	WITHIN THE SLURP LOADER; NEW DATA [IF ANY] WILL BE LOADED INTO THIS FIELD	42
43	8038	/	AS LOADING CONTINUES. [NOTE: IN THEORY, BINARY FILES COULD END WITH A	43
44	8039	/	FIELD SETTING.]	44
45	8040			45
46	8041	/	01 - END OF FILE HAS BEEN DETECTED; IGNORE ALL ADDITIONAL DATA WORDS IN THIS	46
47	8042	/	DATA BLOCK AND ANY ADDITIONAL BLOCKS IN THE CURRENT FILE.	47
48	8043			48
49	8044	/	IT IS THE RESPONSIBILITY OF THE LOADER TO PROCESS MULTIPLE FILES IF REQUIRED.	49
50				50
51				51
52				52
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54				54
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56				56
57				57
58				58

1	8046	/	SLURP FORMAT BINARY FILE LENGTH CONSIDERATIONS.	1
2	8047			2
3	8048	/	P?S/8 TFS FILES CONSIST OF EXACTLY SIXTEEN CONTIGUOUS BLOCKS; THIS WOULD	3
4	8049	/	GENERALLY MEAN THAT P?S/8 BINARY FILES WOULD BE REQUIRED TO BE EXACTLY	4
5	8050	/	SIXTEEN BLOCKS. WHILE THIS IS GENERALLY THE CASE, VARIOUS EXPERIMENTAL	5
6	8051	/	DIRECTORIES HAVE BEEN DEVELOPED FOR USE WITH VARIOUS P?S/8 SYSTEM PROGRAMS	6
7	8052	/	THAT ALLOW FOR VARIABLE LENGTH FILES. [THE PRIMARY PURPOSE OF THESE EFFORTS	7
8	8053	/	HAS BEEN TO SUPPORT VARIABLE-LENGTH TEXT FILES WITHOUT LINE NUMBERS; HOWEVER,	8
9	8054	/	THERE IS NO INHERENT REASON THAT SUPPORT MUST BE LIMITED TO A SINGLE FILE	9
10	8055	/	FORMAT.]	10
11	8056			11
12	8057	/	SINCE SLURP FORMAT IS HANDLED ONE BLOCK AT A TIME, THERE IS NO INHERENT LENGTH	12
13	8058	/	LIMITATION; SPECIAL-PURPOSE UTILITIES COULD BE CREATED TO SUPPORT LONGER FILES	13
14	8059	/	AS REQUIRED. THIS IS THE INTENDED METHOD TO BE USED WITHIN THE P?S/8 SHELL	14
15	8060	/	OVERLAY.	15
16	8061			16
17	8062	/	COMPARING SLURP FORMAT TO OTHER BINARY FILE FORMATS.	17
18	8063			18
19	8064	/	IN CERTAIN OPERATING SYSTEMS, BINARY FILE FORMAT IS BASED ON THE 8-BIT FRAMES	19
20	8065	/	OF BINARY PAPER-TAPES. AT FIRST GLANCE, THIS WOULD APPEAR TO BE A REASONABLE	20
21	8066	/	DESIGN AS BINARY LOADING IS A DESCENDANT OF THE ORIGINAL PAPER-TAPE BIN FORMAT	21
22	8067	/	LOADER.	22
23	8068			23
24	8069	/	HOWEVER, THIS IS SIMPLY SHORT-SIGHTED. BINARY PAPER-TAPE FORMAT IS A VERY	24
25	8070	/	INEFFICIENT FORMAT; THERE IS INHERENT OVERHEAD IN "MASSAGING" THE DATA INTO A	25
26	8071	/	FORMAT GEARED TO LOADING 12-BIT WORDS DIRECTLY INTO MEMORY WHEN THE UNDERLYING	26
27	8072	/	STORAGE DEVICE IS 12-BIT ORIENTED; NO ADVANTAGE IS BEING TAKEN OF THIS	27
28	8073	/	IMPORTANT FACT AND WHY THIS WAS REJECTED IN THE DESIGN OF THE R-L MONITOR AND	28
29	8074	/	P?S/8 [WHICH IS PARTIALLY BASED ON THIS FORMER SYSTEM]. PAPER-TAPE BINARY	29
30	8075	/	FORMAT HAS CONSIDERATIONS BEYOND A STORAGE-BASED OPERATING SYSTEM SUCH AS THE	30
31	8076	/	NOTION OF SECTION HEADERS REQUIRING LEADER/TRAILER FRAMES. NOT ALL BIT	31
32	8077	/	COMBINATIONS ARE DEFINED IN PAPER-TAPE FRAMES AS IT WAS DEEMED SUFFICIENT TO	32
33	8078	/	ACHIEVE A PRACTICAL LOADER REGARDLESS OF PROGRAM SIZE; ONLY THE CASES THAT	33
34	8079	/	MATTER TO THE BIN LOADER ARE CONSIDERED, INCLUDING THE GENERATION OF A	34
35	8080	/	CHECKSUM DURING THE LOADING TO ENSURE DATA VALIDITY.	35
36	8081			36
37	8082	/	WHEN BINARY DATA IS AVAILABLE ON A STORAGE DEVICE, THE DATA VALIDITY IS	37
38	8083	/	CHECKED MORE STRINGENTLY BY THE CONTROLLER DEVICE [SUCH AS THE TC01/TC08	38
39	8084	/	DECTAPE CONTROLLER] AFFECTING ALL OPERATIONS, NOT MERELY THOSE RELATED TO	39
40	8085	/	PROCESSING AN IMAGE OF A PAPER-TAPE THROUGH A LOADER THAT IS STORAGE-BASED.	40
41	8086			41
42	8087	/	AS SUCH, IT IS PERFECTLY REASONABLE TO LIMIT THE FORMAT TO THE EFFICIENT	42
43	8088	/	PURPOSE OF RAPID AND RELIABLE 12-BIT BINARY LOADING. THE SLURP LOADER IS THE	43
44	8089	/	BEST POSSIBLE IMPLEMENTATION OF A LOADER FOR THIS PURPOSE. DESPITE THE	44
45	8090	/	ABILITY TO LOAD MEMORY TOTALLY AT RANDOM [UNLIKE CORE-IMAGE BINARY WHICH MUST	45
46	8091	/	BE EXPRESSED IN UNIFORM CONTIGUOUSLY LOADING BLOCKS], SLURP FORMAT IS JUST	46
47	8092	/	UNDER 6/7 THE EFFICIENCY OF CORE-IMAGE BINARY.	47
48	8093			48
49	8094	/	WHERE NEEDED, PAPER-TAPE COMPATIBILITY CAN ALSO BE ACHIEVED THROUGH THE USE OF	49
50	8095	/	APPROPRIATE CONVERSION PROGRAMS. ALL SUCH UTILITIES ARE AVAILABLE AS OPTIONS	50
51	8096	/	OF THE P?S/8 BIN PROGRAM BY INVOKING APPROPRIATE OPTION SWITCHES. [REFER TO	51
52	8097	/	THE FULL DOCUMENTATION OF THE P?S/8 BIN PROGRAM FOR MORE INFORMATION.]	52
53	8098			53
54	8099	/	END OF APPENDIX 0.	54
55				55
56				56
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1	8101	/	GLOSSARY OF TERMS USED IN THIS DOCUMENT.	1
2	8102			2
3	8103	/ 01) %		3
4	8104			4
5	8105	/	"%" IS THE SYMBOLIC SYNTAX USED IN THE BASIC P?S/8 SYSTEM FOR THE FILE	5
6	8106	/	LOCATED AT BLOCKS 0020-0037 ON ANY SYSTEM DEVICE LOGICAL UNIT. IT IS USED	6
7	8107	/	AS A SCRATCH AREA COMPONENT FOR A VARIETY OF TEMPORARY PURPOSES, SOME OF	7
8	8108	/	WHICH ARE INCOMPATIBLE WITH EACH OTHER. THIS IS ALSO THE FIRST HALF	8
9	8109	/	[00000-03777] OF THE DESIGNATED SCRATCH AREA FOR SAVING ALL OF FIELD 0.	9
10	8110	/	THIS NOTATION IS NOT AVAILABLE IN THE SHELL ENVIRONMENT; HOWEVER, THE	10
11	8111	/	SYSTEM SAVE AREA IS USED IDENTICALLY IN BOTH SYSTEM ENVIRONMENTS.	11
12	8112			12
13	8113	/ 02) \$		13
14	8114			14
15	8115	/	"\$" IS THE SYMBOLIC SYNTAX USED IN THE BASIC P?S/8 SYSTEM FOR THE FILE	15
16	8116	/	LOCATED AT BLOCKS 0040-0057 ON ANY SYSTEM DEVICE LOGICAL UNIT. IT IS USED	16
17	8117	/	AS A SCRATCH AREA COMPONENT FOR A VARIETY OF TEMPORARY PURPOSES, SOME OF	17
18	8118	/	WHICH ARE INCOMPATIBLE WITH EACH OTHER. THIS IS ALSO THE SECOND HALF	18
19	8119	/	[04000-07777] OF THE DESIGNATED SCRATCH AREA FOR SAVING ALL OF FIELD 0.	19
20	8120	/	THIS NOTATION IS NOT AVAILABLE IN THE SHELL ENVIRONMENT; HOWEVER, THE	20
21	8121	/	SYSTEM SAVE AREA IS USED IDENTICALLY IN BOTH SYSTEM ENVIRONMENTS.	21
22	8122			22
23	8123	/ 03) JACK BURNES		23
24	8124			24
25	8125	/	JACK IS A MEMBER OF THE WCFMPG AND CLOSE COLLEAGUE OF RICHARD LARY, FAMOUS	25
26	8126	/	FOR VARIOUS PDP-8 PROJECTS INCLUDING THE SINGLE-HANDLED CREATION OF	26
27	8127	/	LAP6-DIAL AND DIAL-MS FOR THE PDP-12. JACK IS THE AUTHOR OF THE 8K	27
28	8128	/	VARIANT OF LENNY ELEKMAN'S BASIC FOR THE PDP-8 [AKA LAB-8/E BASIC]. HE	28
29	8129	/	IS ALSO THE AUTHOR OF THE LEGENDARY MOONLANDER PROGRAM FOR THE GT-40 BASED	29
30	8130	/	ON AN OBSCURE PRIOR DEMO PROGRAM HE WROTE FOR SPECIFIC CONFIGURATIONS OF	30
31	8131	/	THE PDP-12. IT WOULD APPEAR THAT NONE OF HIS WORK IS ORIENTED TOWARDS	31
32	8132	/	OS/8. WITHOUT SOURCE CODE MODIFICATIONS, 8K BASIC CANNOT SAFELY BE USED	32
33	8133	/	FROM OS/8 DUE TO LOADING COMPLICATIONS AND LIMITATIONS OF OS/8 THAT DO	33
34	8134	/	NOT APPLY TO NEARLY ALL OTHER PDP-8 OPERATING SYSTEMS.	34
35	8135			35
36	8136	/ 04) COS 300/310 [AND DIBOL-8]		36
37	8137			37
38	8138	/	THE COMMERCIAL OPERATING SYSTEM [COS] FOR THE PDP-8 AND DECMATE SERIES IS	38
39	8139	/	A DEDICATED ENVIRONMENT TO RUN THE DIBOL LANGUAGE DIALECT AS IMPLEMENTED	39
40	8140	/	ON THE PDP-8. THE ORIGINAL VERSION WAS KNOWN AS DIBOL-8; INTERNALLY, IT	40
41	8141	/	IS A HACKED UP VERSION OF THE R-L MONITOR SYSTEM WITH CERTAIN KEYBOARD	41
42	8142	/	MONITOR COMMANDS CHANGED [A TRIVIAL CHANGE TO THE OPERATING SYSTEM]. THE	42
43	8143	/	COMPILER WAS WRITTEN ORIGINALLY BY HANK MAURER.	43
44	8144			44
45	8145	/	TWO MAJOR RELEASE FAMILIES EXIST KNOWN AS COS-300 AND COS-310. BOTH ARE	45
46	8146	/	BASED ON OS/8 WITH A CHANGED BASE DATE YEAR [AS A BAND-AID ON THE GENERAL	46
47	8147	/	DATE PROBLEM PLAGUING OS/8]. [NOTE: THIS ACTUALLY CREATES ADDITIONAL	47
48	8148	/	PROBLEMS WHILE SOLVING NONE OF THE UNDERLYING DATE PROBLEMS OF OS/8.]	48
49	8149	/	COS-310 RUNS ON THE DECMATES [AT LEAST THE DECMATE II AND III+ WITH THE	49
50	8150	/	HARD DISK OPTION]. THE OS/8 KEYBOARD MONITOR IS REPLACED BY AN INTERFACE	50
51	8151	/	LIFTED FROM THE R-L MONITOR SYSTEM; THE UNDERLYING FILE SYSTEM IS	51
52	8152	/	COMPATIBLE WITH OS/8 TO WITHIN THE DATE OFFSET. CERTAIN PROGRAMS CAN RUN	52
53	8153	/	UNDER EITHER SYSTEM, SOMETIMES BY IMPLEMENTING INTERNAL KLUDGES TO ATTEMPT	53
54	8154	/	TO DETERMINE WHICH SYSTEM IS HOSTING THE PROGRAM. [NOTE: THIS IS NOT	54
55	8155	/	ALWAYS SUCCESSFUL LEADING TO NEGATIVE CONSEQUENCES.]	55
56				56
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1	8157	/	A NOTABLE FEATURE OF COS IS THE USE OF COMMANDS THAT ARE EXPRESSED AS	1
2	8158	/	UNIQUE TWO CHARACTER NAMES OPTIONALLY FOLLOWED BY ADDITIONAL IGNORED	2
3	8159	/	CHARACTERS. THESE COMMANDS ARE CHOSEN TO AVOID THE NAMES OF OTHER	3
4	8160	/	PROGRAMS THAT MUST BE EXPRESSED AS LONGER COMMANDS UP TO SIX CHARACTERS IN	4
5	8161	/	LENGTH. [NOTE: FOR ANY SIX CHARACTER NAME, ADDITIONAL CHARACTERS WILL BE	5
6	8162	/	IGNORED; HOWEVER, SHORTER NAMES UP TO FIVE CHARACTERS MUST BE UNIQUELY	6
7	8163	/	GIVEN.] AS SUCH, THE ORIGINAL COMMANDS OF THE R-L MONITOR SYSTEM IN	7
8	8164	/	CONFLICT WITH THIS SCHEME WERE MODIFIED IN DIBOL-8. THE ENTIRE MECHANISM	8
9	8165	/	WAS TRANSFERRED TO THE COS INTERFACE THAT REPLACES THE OS/8 KEYBOARD	9
10	8166	/	MONITOR.	10
11	8167			11
12	8168	/	WHERE RELEVANT, IT IS NOT NECESSARY TO INVOKE THESE PROGRAMS VIA AN	12
13	8169	/	EXPLICIT EXECUTION-CLASS COMMAND. INSTEAD THE NAME ITSELF IMPLIES	13
14	8170	/	EXECUTION; THIS SUPERFICIALLY RESEMBLES OS/8 CCL. FOR EXAMPLE, THE	14
15	8171	/	SE[QUENCE] COMMAND [TO HAVE AUTOMATICALLY GENERATED LINE NUMBERS WITHIN	15
16	8172	/	THE BUILT-IN EDITOR] WAS CHANGED TO LN[UMBER]. THIS ALLOWS THE "SET"	16
17	8173	/	PROGRAM TO BE INVOKED DIRECTLY AS A COMMAND.	17
18	8174			18
19	8175	/	CLEARLY, THERE IS AN "INCESTUOUS" RELATIONSHIP BETWEEN THE R-L MONITOR AND	19
20	8176	/	THE VARIOUS COS IMPLEMENTATIONS. TO MAKE MATTERS SOMEWHAT MORE	20
21	8177	/	COMPLICATED, THE IDENTICAL EXECUTION SCHEME WAS FIRST IMPLEMENTED IN	21
22	8178	/	P?S/8, OFTEN WITH THE SAME NAMES OF UTILITY PROGRAMS FOR ANALOGOUS	22
23	8179	/	PURPOSES [SUCH AS "SET", FORMERLY KNOWN AS "VARY"].	23
24	8180			24
25	8181	/	EARLY VERSIONS OF P?S/8 ATTEMPT TO BE AS CLOSE TO THE R-L MONITOR AS	25
26	8182	/	POSSIBLE IN TERMS OF IMPLEMENTING NOT ONLY THE SAME FUNCTIONALITY BUT ALSO	26
27	8183	/	THE SAME COMMAND NAMES. MORE RECENT VERSIONS ARE TOTALLY COMPATIBLE WITH	27
28	8184	/	COS. THUS, USERS OF THE VARIOUS COS SYSTEMS ARE ALREADY SOMEWHAT FAMILIAR	28
29	8185	/	WITH P?S/8 EVEN IF THEY HAVE NEVER USED IT. [NOTE: THESE SIMILARITIES ARE	29
30	8186	/	NOT A CONINCIDENCE. THERE HAS ALWAYS BEEN A FLUID RELATIONSHIP BETWEEN	30
31	8187	/	MEMBERS OF THE WCFMPG AND P?S, THUS CERTAIN INDIVIDUALS CLAIMING DUAL	31
32	8188	/	MEMBERSHIP IN BOTH ORGANIZATIONS.]	32
33	8189			33
34	8190	/	DIBOL-RELATED COMMANDS IN COS 300/310 ARE IMPLEMENTED IN THE SAME MANNER	34
35	8191	/	AS THE ANALOGOUS COMMANDS IN P?S/8. HOWEVER, ALL OTHER OS/8 OPERATIONS	35
36	8192	/	RUNNING FROM COS ARE IMPLEMENTED WITH ALL OF THE OVERHEAD THAT APPLIES TO	36
37	8193	/	OS/8 IN GENERAL. IN P?S/8, THIS IS NOT ACTUALLY A CCL IMPLEMENTATION.	37
38	8194	/	AS SUCH, THE OVERHEAD OF CCL AS A LAYER OVER AN OPERATING SYSTEM DOES NOT	38
39	8195	/	APPLY. HOWEVER, MANY COMMANDS ARE ENTERED IN QUITE SIMILAR WAYS IN ALL	39
40	8196	/	THREE SYSTEMS WHERE APPLICABLE.	40
41	8197			41
42	8198	/	IF THERE IS SUFFICIENT INTEREST, A COS-LIKE OVERLAY TO THE P?S/8 SHELL MAY	42
43	8199	/	BE IMPLEMENTED TO ALLOW ONE UNIFIED SYSTEM TO REPLACE ALL OF THESE	43
44	8200	/	OPERATING SYSTEMS. IT IS ARGUABLE THAT COS SHOULD NEVER HAVE BEEN	44
45	8201	/	IMPLEMENTED AS A SEPARATE OPERATING SYSTEM THAT OVERWHELMED THE PDP-8	45
46	8202	/	DEVELOPMENT GROUPS WITHIN DEC. HOWEVER, THIS WAS A MANAGEMENT DECISION;	46
47	8203	/	MANY MANAGEMENT DECISIONS WERE EVENTUALLY PROVEN TO BE POOR JUDGEMENT.	47
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1	8205	/ 05) MARIO DENOBILI	1
2	8206		2
3	8207	/ MARIO IS THE EMPEROR OF THE P?S. MOST P?S MEMBERS AGREE THEIR PROJECTS GO	3
4	8208	/ SMOOTHER WITHOUT HIS PARTICIPATION. DESPITE THIS, MANY OF THEIR WORKS ARE	4
5	8209	/ DEDICATED TO HIM. HIS LAST KNOWN MAILING ADDRESS IS 333 JAY STREET, BOX D,	5
6	8210	/ BROOKLYN NY 11201. THIS IS LIKELY OBSOLETE AS HE GENERALLY AVOIDS CONTACT	6
7	8211	/ WITH ANYONE. MARIO CLAIMS TO BE THE MASTER OF HIS DOMAIN ON THE ISLAND	7
8	8212	/ REPUBLIC OF GUAYAQUIL, LOCATED SOMEWHERE OFF THE COAST OF ECUADOR; HE	8
9	8213	/ APPARENTLY INTENDS TO INVADE ECUADOR SOMEDAY. HOWEVER, HE MUST FIRST	9
10	8214	/ OVERCOME THE FACT THAT QUAYAQUIL IS ACTUALLY ECUADOR'S LARGEST CITY AND IS	10
11	8215	/ NOT AN ISLAND. SHOULD YOU EVER LOCATE MARIO, YOU MIGHT WANT TO PASS ON	11
12	8216	/ THIS GEOGRAPHIC REALITY TO HIM. THE P?S SHALL NOT BE HELD RESPONSIBLE FOR	12
13	8217	/ ANY BODILY HARM OR OTHER INJURY THAT MAY BEFALL YOU.	13
14	8218		14
15	8219	/ VARIOUS PORTIONS OF P?S/8 ARE DEDICATED TO MARIO DENOBILI, WITHOUT WHOSE	15
16	8220	/ PARTICIPATION, IT WOULD HAVE BEEN DONE A LOT SOONER.	16
17	8221		17
18	8222	/ 06) LENNY ELEKMAN	18
19	8223		19
20	8224	/ LEONARD W ELEKMAN IS A MEMBER OF THE WCFMPG AND CLOSE COLLEAGUE OF RICHARD	20
21	8225	/ LARY. TOGETHER, THEY CREATED THE R-L MONITOR SYSTEM AND SEVERAL NOTABLE	21
22	8226	/ IMPLEMENTATIONS OF THE BASIC LANGUAGE INCLUDING POLY BASIC [LATER SOLD BY	22
23	8227	/ DEC AS EDU-SYSTEM 30].	23
24	8228		24
25	8229	/ LENNY SINGLE HANDEDLY WROTE 4K BASIC, A STANDALONE IMPLEMENTATION THAT	25
26	8230	/ USES EVERY MEMORY LOCATION IN A 4K PDP-8 EXCEPT THOSE USED BY THE RIM	26
27	8231	/ LOADER. WHILE IT IS NOT POSSIBLE TO INTERFACE THE PROGRAM TO A	27
28	8232	/ STORAGE-BASED OPERATING SYSTEM [UNLESS IMPLEMENTED VIRTUALLY], 4K BASIC IS	28
29	8233	/ THE BASIS FOR 8K BASIC, A VARIANT CREATED BY JACK BURNES. AS NORMALLY	29
30	8234	/ DISTRIBUTED, THIS PROGRAM CAN BE LOADED AND STARTED FROM THE R-L MONITOR	30
31	8235	/ AND P?S/8 BUT NOT OS/8, WHICH IS DESTROYED IN MEMORY IN THE INITIALIZATION	31
32	8236	/ PHASE OF THE PROGRAM. [NOTE: OS/8 MUST BE COMPLETELY DISABLED AT PROGRAM	32
33	8237	/ STARTUP; SUFFICIENT OS/8 CODE IS STILL PRESENT IN MEMORY TO DAMAGE THE	33
34	8238	/ OS/8 SYSTEM DEVICE IF THE MACHINE IS RESTARTED AT 07600; FULL MANUAL	34
35	8239	/ BOOTSTRAP IS REQUIRED TO SAFELY RESTART THE MACHINE.]	35
36	8240		36
37	8241	/ A LONG-TERM P?S/8 PROJECT IS TO INTERFACE 8K BASIC TO P?S/8 INCLUDING FILE	37
38	8242	/ SUPPORT. AT LEAST 8K WILL BE REQUIRED WITH 12K RECOMMENDED FOR MAXIMUM	38
39	8243	/ PERFORMANCE. BASIC PROGRAMS OCCUPYING UP TO EIGHT TFS FILES WILL BE	39
40	8244	/ SUPPORTED. [NOTE: BASIC FILES CANNOT BE EDITED OUTSIDE OF BASIC; THE	40
41	8245	/ KEYBOARD MONITOR WILL BE CAPABLE OF FILE MAINTENANCE ONLY; TFS TEXT FILES	41
42	8246	/ REQUIRE LINE NUMBERS IMPLEMENTED WITHIN THE KEYBOARD MONITOR AND	42
43	8247	/ INTEGRATED EDITOR. BASIC FILES CONTAIN LINES WHERE THE LINE NUMBER IS	43
44	8248	/ PART OF THE FILE CONTEXT; THE DIGITS OF THE LINE NUMBERS ARE TEXT	44
45	8249	/ CHARACTERS. ADDITIONALLY, BASIC FILES CONTAIN INTEGRITY CHECK INFORMATION	45
46	8250	/ PARTICULAR TO THE P?S/8 IMPLEMENTATION.]	46
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57			57
58			58

1	8252	/ 07) EMPEROR	1
2	8253		2
3	8254	/ THIS IS AN HONORIFIC BESTOWED UPON MARIO DENOBILI [BY HIMSELF]. SOME OF	3
4	8255	/ HIS PSYCOPHANTS ARE KNOWN TO HAVE CONTRARIAN VIEWS REGARDING THE USE OF	4
5	8256	/ THE "0" CHARACTER WHERE THE MORE COMMON "O" CHARACTER MIGHT BE EXPECTED.	5
6	8257	/ IN CERTAIN INSTANCES, THEY HAVE BEEN KNOWN TO REVERSE ALL OCCURRENCES OF	6
7	8258	/ THESE CHARACTERS, ESPECIALLY IN DOCUMENTS PRINTED AS HARD COPY ON	7
8	8259	/ LINEPRINTER PAPER.	8
9	8260		9
10	8261	/ 08) JOE FISCHETTI	10
11	8262		11
12	8263	/ JOE IS A MEMBER OF THE WCFMPG AND P?S. JOE IS RESPONSIBLE FOR EXPLAINING	12
13	8264	/ PDP-8 CO-ROUTINES TO VARIOUS INDIVIDUALS SUCH AS CHARLES LASNER, CHIEF	13
14	8265	/ ARCHITECT OF P?S/8, WHO IN TURN HAS PASSED THIS EXTRAORDINARY KNOWLEDGE TO	14
15	8266	/ OTHERS; MOST OF THE PEOPLE INVOLVED ADMIT THAT THEIR ENTIRE APPROACH TO	15
16	8267	/ PROGRAMMING RADICALLY CHANGED WHEN THIS POWERFUL CONCEPT WAS ADDED TO	16
17	8268	/ THEIR PERSONAL ARSENAL OF PROGRAMMING TECHNIQUES. [NOTE: SOME CLAIM THEIR	17
18	8269	/ LIVES ARE PERMANENTLY RUINED AS A RESULT OF THEIR EXPOSURE TO THIS	18
19	8270	/ KNOWLEDGE.]	19
20	8271		20
21	8272	/ JOE WAS INSTRUMENTAL IN THE CREATION OF THE "RELOC" DIRECTIVE IN MODERN	21
22	8273	/ PDP-8 PAL ASSEMBLERS. HE ALSO PROVIDED THE COUNTER-INTUITIVE METHOD OF	22
23	8274	/ HOW TO TEST IF NUMERIC VALUES ARE/ARE NOT IN A SPECIFIED RANGE WHILE ONLY	23
24	8275	/ PERFORMING A SINGLE TEST. [NOTE: CONVENTIONAL WISDOM IS THAT THIS METHOD	24
25	8276	/ IS IMPOSSIBLE DESPITE EASY VERIFICATION OF RESULTS; THE TECHNIQUE IS USED	25
26	8277	/ THROUGHOUT P?S/8 TO SAVE CODE SPACE AND IMPROVE PERFORMANCE.]	26
27	8278		27
28	8279	/ 09) RICHARD LARY	28
29	8280		29
30	8281	/ RICHIE [AKA "MOUTH"] IS A FAMOUS MEMBER OF THE WCFMPG, P?S AND THE FRIENDS	30
31	8282	/ OF THE BOSTON MASSACRE. MOST OF THE WORK MENTIONED IN THIS DOCUMENT IS AN	31
32	8283	/ OUTGROWTH OF HIS ORIGINAL EFFORT KNOWN AS THE R-L MONITOR SYSTEM [AKA	32
33	8284	/ MS/8], WHICH WAS CREATED IN PARTNERSHIP WITH LENNY ELEKMAN [AND TO A	33
34	8285	/ LESSER EXTENT THE OTHER MEMBERS OF THE WCFMPG MENTIONED IN THIS DOCUMENT].	34
35	8286	/ THE TWO OF THEM ARE ALSO KNOWN FOR WRITING POLY BASIC [LATER SOLD BY DEC	35
36	8287	/ AS EDU-SYSTEM 30] AND THE ORIGINAL PRECURSOR TO POLY SPACEWAR. MOST OF	36
37	8288	/ THE MEMBERS OF THE WCFMPG ARE KNOWN FOR IMPLEMENTATIONS OF THE BASIC	37
38	8289	/ LANGUAGE.	38
39	8290		39
40	8291	/ THE INSPIRATION FOR THE R-L MONITOR IS THE SPECIFIC IMPLEMENTATION OF	40
41	8292	/ BASIC ON THE GE-235 TIMESHARING SYSTEM [FROM A USER'S PERSPECTIVE]. THE	41
42	8293	/ ACTUAL GE-235 SYSTEM WAS OFTEN HACKED BY THE WCFMPG; SOME OF ITS MEMBERS	42
43	8294	/ WERE EVEN PAID TO REVEAL SHORTCOMINGS OF THAT SYSTEM.	43
44	8295		44
45	8296	/ RICHIE ALSO WROTE THE ORIGINAL CRUDE HACK VERSION OF PS/8 OVER A WEEKEND.	45
46	8297	/ HE ALSO PARTICIPATED IN VARIOUS ASPECTS OF ITS EVENTUAL UPGRADE; HOWEVER,	46
47	8298	/ IT ALSO HAD MANY SERIOUS DOWNTURNS AS EVENTUALLY OS/8 WAS PERVERTED BY THE	47
48	8299	/ ACTIVE PARTICIPATION OF SEMI-COMPETENT PROGRAMMERS. DUE TO MANAGEMENT	48
49	8300	/ INTERFERENCE, THE TENTATIVE CODING WAS NEVER SCRUTINIZED AND THUS BECAME	49
50	8301	/ PERMANENT PREMATURELY FORCING VARIOUS SERIOUS DESIGN BUGS TO BECOME	50
51	8302	/ UNFIXABLE. ALL MEMBERS OF THE P?S ARE FAMILIAR WITH HOW TO USE THIS	51
52	8303	/ FAMILY OF PRODUCT; SOME EVEN GOT PAID TO CREATE CERTAIN PORTIONS OF IT.	52
53	8304	/ STILL OTHERS CRITICIZED IT AND INSTEAD PARTICIPATED IN THE CREATION OF	53
54	8305	/ P?S/8 WHICH HAS A DESIGN BASE THAT CAN OUTPERFORM AND TOTALLY OBSOLETE	54
55	8306	/ OS/8 WHEN SUFFICIENT CODE IS WRITTEN.	55
56			56
57			57
58			58

1	8308	/	RICHIE WAS ALSO LEAD PROGRAMMER FOR THE DEC STORAGE DIVISION IN COLORADO	1
2	8309	/	FOR MANY YEARS; HE PERSONALLY HOLDS MANY STORAGE TECHNOLOGY PATENTS	2
3	8310	/	RELATING TO SMART ADAPTATION OF HOST OPERATING SYSTEM REQUIREMENTS TO	3
4	8311	/	MAXIMIZE THE PERFORMANCE OF RELATIVELY MEDIOCRE DISK STORAGE WITHIN THE	4
5	8312	/	IMBEDDED CONTROLLER.	5
6	8313			6
7	8314	/	IT IS ARGUABLE THAT RICHIE'S WORK MADE SO MANY MILLIONS OF DOLLARS FOR DEC	7
8	8315	/	THAT WITHOUT HIS EFFORTS, THE COMPANY WOULD HAVE GONE OUT OF BUSINESS	8
9	8316	/	YEARS BEFORE BAD MANAGEMENT CAUSED THE EVENTUAL COMPANY DEMISE.	9
10	8317			10
11	8318	/ 10)	CHARLES LASNER	11
12	8319			12
13	8320	/	CHARLES IS A MEMBER OF THE P?S. HE IS THE CHIEF ARCHITECT OF P?S/8 AND	13
14	8321	/	AUTHOR OF KERMIT-12 FOR OS/8 INCLUDING ALL OF THE ASSOCIATED UTILITIES.	14
15	8322	/	CHARLES ALSO WROTE THE OPERATING SYSTEM INTERFACE TO FOCAL, 1969 FOR BOTH	15
16	8323	/	THE R-L MONITOR AND P?S/8.	16
17	8324			17
18	8325	/	CHARLES ALSO CREATED THE RELOADER CONCEPT FOR RICHARD LARY'S "SLURP"	18
19	8326	/	LOADER WHICH CAN BEST BE DESCRIBED AS A "BENEFICIAL COMPUTER VIRUS", AS	19
20	8327	/	WELL AS ALL OF THE AUXILIARY MODES OF THE P?S/8 BIN PROGRAM INCLUDING THE	20
21	8328	/	SQUEEZED-DOWN VERSION OF THE PAPER-TAPE BIN LOADER. MOST OF THE HANDLERS	21
22	8329	/	THROUGHOUT P?S/8 FOR ALL SUPPORTED DEVICES WERE WRITTEN BY HIM AS WELL AS	22
23	8330	/	HUNDREDS OF OPTIMIZATION TRICKS THROUGHOUT P?S/8 BASED ON PRINCIPLES	23
24	8331	/	DISCUSSED IN THIS DOCUMENT.	24
25	8332			25
26	8333	/	CHARLES HAS TAKEN TOTAL RESPONSIBILITY FOR ALL COMPONENTS OF P?S/8 SINCE	26
27	8334	/	THE CREATION OF HIS VARIOUS BUSINESS ENTITIES STATED ELSEWHERE IN THIS	27
28	8335	/	DOCUMENT. SOME COMPONENTS OF P?S/8 WERE WRITTEN BY VARIOUS INDIVIDUALS	28
29	8336	/	NOT STATED HERE DUE TO THEIR REQUEST FOR ANONYMITY. SOME OF THESE PEOPLE	29
30	8337	/	ARE KNOWN FOR THEIR EFFORTS NOT NECESSARILY MAKING ANY POSITIVE	30
31	8338	/	CONTRIBUTION TO THE OVERALL PROJECT. IN ALL RELEVANT CASES, ALL CODE HAS	31
32	8339	/	GENERALLY BEEN REWRITTEN AND NOW KNOWN TO BE FUNCTIONAL. FOR THOSE WHO	32
33	8340	/	CONSIDER THEIR WORK OF POSITIVE WORTH, THEY ARE ENCOURAGED TO CONTACT	33
34	8341	/	CHARLES LASNER IF THEY WISH TO BE IDENTIFIED; IT IS BELIEVED THE STATUTE	34
35	8342	/	OF LIMITATIONS NOW APPLIES IN ALL CASES. CHARLES WILL DENY ANY KNOWLEDGE	35
36	8343	/	OF WITNESS PROTECTION OR OTHER APPLICABLE PROGRAMS.	36
37	8344			37
38	8345	/	CHARLES HAS ALSO REPAIRED VARIOUS FORMS OF COMPUTER HARDWARE, MOST	38
39	8346	/	NOTABLY THE ORIGINAL PDP-8 FRONT PANEL THAT WAS TOTALLY WORN OUT BY THE	39
40	8347	/	MANUAL OPERATIONS ASSOCIATED WITH TOGGLING IN THE ENTIRE R-L MONITOR. BY	40
41	8348	/	BEING EXPOSED TO CO-ROUTINES, HIS ENTIRE APPROACH TO PROGRAMMING HAS BEEN	41
42	8349	/	CHANGED FOREVER; IN MANY WAYS HIS LIFE HAS BEEN TOTALLY RUINED FROM THAT	42
43	8350	/	MOMENT ONWARD [WHICH IS PROBABLY A GOOD THING IN THE END].	43
44	8351			44
45	8352	/	HE IS ALSO RESPONSIBLE FOR THIS ENTIRE DOCUMENT, EVERY LAST CHARACTER FROM	45
46	8353	/	THE BEGINNING TO THE END. [YOU MAY PERIODICALLY PONDER THIS AS YOU PORE	46
47	8354	/	THROUGH IT FOR A TIME.]	47
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52				52
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54				54
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1	8356	/ 11) LOGICAL CONSOLE OVERLAY	1
2	8357		2
3	8358	/ THE P?S/8 LOGICAL CONSOLE OVERLAY IS AN OPTIONAL COMPONENT OF P?S/8 THAT	3
4	8359	/ PROVIDES ALTERNATE CONSOLE AND LINEPRINTER SERVICES TO ALL COMPLIANT	4
5	8360	/ PROGRAMS THROUGHOUT P?S/8 [INCLUDING USER-WRITTEN PROGRAMS]. EXTENDED	5
6	8361	/ MEMORY IS REQUIRED TO ENABLE THE OVERLAY; AS A RESULT, THE LOGICAL MEMORY	6
7	8362	/ SIZE MUST BE REDUCED BY ONE ENTIRE MEMORY FIELD [UNLESS ALREADY REDUCED BY	7
8	8363	/ ANOTHER SYSTEM COMPONENT SUCH AS AN EXTENDED SYSTEM DEVICE HANDLER].	8
9	8364		9
10	8365	/ COMPLIANT PROGRAMS REPLACE THEIR INTERNAL CODE FOR THE DEVICE 03/04	10
11	8366	/ CONSOLE TERMINAL AND/OR DEVICE 66 LINEPRINTER WITH LOGICAL CALLS TO	11
12	8367	/ SPECIFIC LOCATIONS ON PAGE ZERO OF THE HIGHEST FIELD OF MEMORY. THE P?S/8	12
13	8368	/ MEMORY STATUS WORD PROVIDES ALL RELEVANT INFORMATION. CALLS ARE PROVIDED	13
14	8369	/ FOR CONSOLE INPUT, CONSOLE OUTPUT, LINEPRINTER OUTPUT AND LINEPRINTER	14
15	8370	/ INPUT [USED ONLY FOR ACTIVE PARTICIPATION IN POTENTIAL FLOW-CONTROL	15
16	8371	/ MECHANISMS]. A LOGICAL INTERRUPT HANDLER IS PROVIDED TO BE HOOKED INTO	16
17	8372	/ ANY INTERRUPT-DRIVEN PROGRAM [AHEAD OF ANY PHYSICAL CONSOLE DEVICE	17
18	8373	/ HANDLING]. THE P?S/8 IMPLEMENTATION OF FOCAL, 1969 USES THIS MECHANISM	18
19	8374	/ FOR ALL RELEVANT INTERRUPT HANDLING.	19
20	8375		20
21	8376	/ WHEN THE LOGICAL CONSOLE OVERLAY IS ENABLED, ALL SYSTEM DEVICE ERRORS ARE	21
22	8377	/ REDIRECTED TO THE OVERLAY; THE SYSTEM DEVICE HANDLER NO LONGER HALTS FOR	22
23	8378	/ ERROR RECOVERY PURPOSES; INSTEAD, AN ERROR-HANDLING PORTION OF THE OVERLAY	23
24	8379	/ IS USED TO ALERT THE USER TO ANY ERRORS WITH SEVERAL OPTIONS [RETRY, EXIT,	24
25	8380	/ ETC.]. DEPENDING ON THE CONFIGURATION, VERBOSE ERROR MESSAGES CAN BE USED	25
26	8381	/ SPECIFIC TO THE SYSTEM DEVICE HARDWARE. FOR EXAMPLE, WHEN USED WITH THE	26
27	8382	/ TC01/TC08 HANDLER, IT IS POSSIBLE TO GET EXPLICIT NOTICE OF SELECT,	27
28	8383	/ TIMING, MARK, PARITY AND END ERRORS.	28
29	8384		29
30	8385	/ CERTAIN SYSTEM DEVICE HANDLERS DEFINE ENHANCEMENTS TO BE CARRIED OUT BY	30
31	8386	/ THE LOGICAL CONSOLE OVERLAY. FOR EXAMPLE, THE TC01/TC08 HANDLER USES A	31
32	8387	/ COMMON VALUE FOR THE LAST BLOCK TRANSFERRED; THIS IS USED TO DETERMINE THE	32
33	8388	/ BEST INITIAL SEARCH DIRECTION BASED ON LAST KNOWN BLOCK TRANSFER.	33
34	8389	/ HOWEVER, IF THE SYSTEM DEVICE UNIT IS CHANGED, THIS CAN CAUSE AN	34
35	8390	/ OCCASIONAL WRONG GUESS [INSTANTLY CORRECTED ONCE REALIZED TO BE IN ERROR].	35
36	8391	/ THE LOGICAL CONSOLE OVERLAY WILL ENHANCE THE DEVICE HANDLER TO USE A TABLE	36
37	8392	/ OF LAST TRANSFERRED BLOCKS, ONE FOR EACH LOGICAL DRIVE UNIT TOTALLY	37
38	8393	/ PREVENTING THIS MINOR PROBLEM.	38
39	8394		39
40	8395	/ THE LOGICAL CONSOLE OVERLAY CAN BE CONFIGURED TO SUPPORT A VARIETY OF	40
41	8396	/ PHYSICAL ALTERNATE CONSOLE DEVICES SUCH AS THE VT8E AS AN EMULATED	41
42	8397	/ TERMINAL, OR THE VR12/VR14 DISPLAY ON THE PDP-12. DEVICES SIMILAR TO THE	42
43	8398	/ DEFAULT CONSOLE CAN ALSO BE SUPPORTED SUCH AS A TERMINAL CONNECTED TO A	43
44	8399	/ DEVICE 10/11 PT08 OR KL8JA OR OTHER INTERFACE. ALTERNATE PRINTING DEVICES	44
45	8400	/ ARE ALSO SUPPORTED SUCH AS THE LS-120 SERIAL PRINTER WHICH REQUIRES	45
46	8401	/ REVERSE CHANNEL FLOW CONTROL.	46
47	8402		47
48	8403	/ SINCE THE LOGICAL CONSOLE OVERLAY ONLY INTERFACES WITH COMPLIANT PROGRAMS	48
49	8404	/ AND THE SYSTEM DEVICE HANDLER, IT IS AVAILABLE FROM THE SHELL ENVIRONMENT	49
50	8405	/ AS WELL AS THE BASIC P?S/8 SYSTEM.	50
51			51
52			52
53			53
54			54
55			55
56			56
57			57
58			58

1	8407	/ 12) HANK MAURER	1
2	8408		2
3	8409	/ HANK IS A MEMBER OF THE WCFMPG AND AUTHOR OF VARIOUS HIGH-LEVEL LANGUAGE	3
4	8410	/ IMPLEMENTATIONS OF BASIC AND CERTAIN OTHER ODD LANGUAGES. SOME OF HIS	4
5	8411	/ WORK IS FOUND IN FULL RELEASES OF THE R-L MONITOR AND OTHER PROJECTS	5
6	8412	/ WITHIN DEC [SOMETIMES WITHOUT DEC'S KNOWLEDGE] SUCH AS DIBOL-8, THE	6
7	8413	/ PRECURSOR OF COS-300/310. [NOTE: THE ORIGINAL DIBOL-8 IS ACTUALLY A SYSTEM	7
8	8414	/ PROGRAM ON A HACKED-UP VERSION OF THE R-L MONITOR SYSTEM WITH THE COMMAND	8
9	8415	/ NAMES PATCHED; THERE IS AN INCESTUOUS RELATIONSHIP BETWEEN P?S/8 AND	9
10	8416	/ VARIOUS IMPLEMENTATIONS OF COS.]	10
11	8417		11
12	8418	/ 13) OS/8	12
13	8419		13
14	8420	/ OS/8 IS THE MOST WELL KNOWN OF LATER MARKETING NAMES OF THE ORIGINAL PS/8	14
15	8421	/ PROJECT STARTED AS A WEEKEND HACK BY RICHARD LARY. WHILE MANY WILL	15
16	8422	/ ATTRIBUTE THIS MEDIOCRE SYSTEM TO RICHIE, IT IS ACTUALLY THE WORK OF MANY	16
17	8423	/ PEOPLE [ARGUABLY TOO MANY PEOPLE] AND SUFFERS FROM A VARIETY OF WEAKNESSES	17
18	8424	/ TRACEABLE TO POOR DESIGN OR MANAGERIAL INTERFERENCE PREVENTING EVOLVEMENT	18
19	8425	/ OF THE INITIAL DESIGN TO SOMETHING BETTER. IT WAS NEVER PROPERLY VETTED	19
20	8426	/ BY THE PDP-8 PROGRAMMING COMMUNITY; MANAGEMENT INTERFERENCE EVENTUALLY	20
21	8427	/ PREVENTED CHANGES CAUSING IT TO BECOME ESSENTIALLY FIXATED WITH A LOT OF	21
22	8428	/ LIMITATIONS. FOR EXAMPLE, THE SYSTEM FILE DATE RECKONING REACHED	22
23	8429	/ EXPIRATION IN 1978, ALTHOUGH A SERIES OF KLUDGES MANAGED TO KEEP MOST	23
24	8430	/ USERS IGNORANT [YET VAGUELY SATISFIED] UNTIL 2001. BY THAT TIME DEC WAS	24
25	8431	/ NO LONGER MARKETING THE PDP-8 OR ITS DESCENDANTS.	25
26	8432		26
27	8433	/ OS/8 WAS ALSO ALLOWED TO DEVOLVE INTO SEVERAL INCOMPATIBLE VARIANTS AS	27
28	8434	/ QUICK-FIX ATTEMPTS TO SOLVE VARIOUS HARDWARE COMPATIBILITY PROBLEMS.	28
29	8435	/ WITHOUT MAJOR REDESIGN IT IS QUITE IMPOSSIBLE TO CREATE A UNIFIED SYSTEM	29
30	8436	/ THAT CAN HANDLE ALL OF THE NEEDLESS PROBLEMS THAT AROSE. SOME OF THESE	30
31	8437	/ VARIANTS ARE KNOWN AS OS/78, OS/278 ETC. AND ASSOCIATED WITH VARIOUS	31
32	8438	/ VERSION NUMBERS FOR THE RELEASE. IN SOME CASES, THESE DISTINCTIONS ARE	32
33	8439	/ MARKETING PLOYS WHILE IN OTHERS IT IS THE DIRECT RESULT OF PRAGMATIC	33
34	8440	/ ATTEMPTS TO RUN ON SPECIFIC DISK MODELS SUCH AS THE DECMATES. SOME VARIANTS	34
35	8441	/ ARE MORE DYSFUNCTIONAL THAN OTHERS DUE TO A VARIETY OF REASONS, OFTEN	35
36	8442	/ ASSOCIATED WITH BOTH MANAGERIAL AND PROGRAMMER INCOMPETENCE.	36
37	8443		37
38	8444	/ OS/8 COMMANDS AND INTERNAL STRUCTURE ARE BARELY ACCEPTABLE TO USERS	38
39	8445	/ SEEKING THE GENERAL SENSIBILITIES OF MOST DEC OPERATING SYSTEMS IF VARIOUS	39
40	8446	/ LIMITATIONS ARE ACCEPTED. TO MANY, THIS IS NOT THE CASE, ESPECIALLY DUE	40
41	8447	/ TO SORE POINT PROBLEMS SUCH AS MODERN TERMINAL CONTROL REQUIREMENTS AND	41
42	8448	/ PROPER HANDLING OF LARGER DISK STORAGE DEVICES, POOR FILE NAMING AND	42
43	8449	/ DATE ISSUES, ETC. [NOTE: THESE ARE NOT HINDSIGHT ISSUES AS VIRTUALLY ALL	43
44	8450	/ OF THESE PROBLEMS WERE DISCUSSED DURING THE PRODUCT'S ACTIVE DEVELOPMENT	44
45	8451	/ CYCLE.]	45
46			46
47			47
48			48
49			49
50			50
51			51
52			52
53			53
54			54
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1	8453	/	THE P?S/8 OPTIONAL SHELL WILL ADD POWERFUL CAPABILITY TO THE BASIC P?S/8	1
2	8454	/	SYSTEM, WHICH ALREADY OPERATES IN A MANNER QUITE SIMILAR TO MOST DEC	2
3	8455	/	OPERATING SYSTEMS. [NOTE: THERE ARE SOME DIFFERENCES, MOSTLY TRACEABLE TO	3
4	8456	/	THE PRIMITIVE TFS FILE SYSTEM. HOWEVER, OTHER ASPECTS OF P?S/8 ARE	4
5	8457	/	ACTUALLY MORE IN LINE WITH TYPICAL SYSTEMS DEVELOPED ON OTHER DEC	5
6	8458	/	MACHINES.] AS SUCH, THE IMPLEMENTATION OF THE P?S/8 SHELL WILL ESSENTIALLY	6
7	8459	/	OBSOLETE OS/8; IT WILL BE DIFFICULT TO FIND EVEN A NITPICK FEATURE OF OS/8	7
8	8460	/	THE SHELL ENVIRONMENT DOESN'T PROVIDE A PROPER EQUIVALENT TO [OR MORE	8
9	8461	/	LIKELY AN IMPROVEMENT]. IT IS INTENDED TO MIGRATE ALL RELEVANT OS/8-BASED	9
10	8462	/	SYSTEM PROGRAMS AND/OR PROVIDE P?S/8 SHELL EQUIVALENT UTILITIES. DUE TO	10
11	8463	/	THE NATURE OF MOST OS/8 SYSTEM PROGRAMS WITH VIRTUALLY IDENTICAL INTERNAL	11
12	8464	/	INTERFACE TO THE OPERATING SYSTEM, IT IS EXPECTED TO BE A RELATIVELY	12
13	8465	/	STRAIGHTFORWARD MATTER [AT THE SOURCE CODE LEVEL] TO MIGRATE ALL	13
14	8466	/	MEANINGFUL PROGRAMS. [NOTE: SOME HAVE NICKNAMED THE P?S/8 SHELL "THE OS/8	14
15	8467	/	KILLER".]	15
16	8468			16
17	8469	/ 14) PS/8		17
18	8470			18
19	8471	/	PS/8 IS THE ORIGINAL MARKETING NAME GIVEN TO A QUICK AND DIRTY OPERATING	19
20	8472	/	SYSTEM HACKED UP OVER A WEEKEND BY RICHARD LARY.	20
21	8473			21
22	8474	/	THE ORIGINAL NAME OF THIS WORK WAS THE FIRST UPWARD COMPATIBLE KEYBOARD	22
23	8475	/	EDITOR AND MONITOR [NOTE THE OBVIOUS ACRONYM]. [PLANNED BUT NEVER	23
24	8476	/	IMPLEMENTED WAS A SECOND UPWARD COMPATIBLE KEYBOARD EDITOR AND MONITOR.]	24
25	8477	/	THE INTERNALS OF THIS SYSTEM WERE NEVER FULLY VETTED LEADING TO MANY	25
26	8478	/	LIMITATIONS CRITICIZED BY MANY PEOPLE, MOST NOTABLY CHARLES LASNER, CHIEF	26
27	8479	/	ARCHITECT OF THE P?S/8 PROJECT. [NOTE: CHARLES WAS RECRUITED TO WORK ON	27
28	8480	/	THIS SYSTEM BUT DECLINED IN FAVOR OF IMPROVING THE R-L MONITOR. HE IS	28
29	8481	/	ALSO THE AUTHOR OF KERMIT-12 FOR OS/8 AND HAS HAD THE DUBIOUS DISTINCTION	29
30	8482	/	OF WRITING ADDITIONAL UTILITIES FOR OS/8 TO SUPPORT THE KERMIT PROJECT.]	30
31	8483			31
32	8484	/	BY THE TIME MANAGERIAL DECISION WAS MADE TO MARKET IT [AS PS/8] THE DESIGN	32
33	8485	/	BECAME STAGNATED WITH LITTLE ABILITY TO IMPROVE; HALF-FINISHED FEATURES	33
34	8486	/	THAT DON'T FUNCTION PROPERLY ABOUND AND WILL REMAIN SO INDEFINITELY.	34
35	8487	/	[NOTE: THE MOST NOTABLE LIMITATION IS A SEGMENTED DIRECTORY THAT WOULD	35
36	8488	/	SUGGEST THE ABILITY TO ADJUST THE OVERALL DIRECTORY SIZE; HOWEVER, IN NOW	36
37	8489	/	STAGNATED IMPLEMENTATION, THERE IS NO PRACTICAL ABILITY TO EITHER LENGTHEN	37
38	8490	/	OR SHORTEN THE DIRECTORY. AS SUCH, THE RESULT IS A WORST CASE OUTCOME; ON	38
39	8491	/	SMALL DEVICES THE DIRECTORY IS GENERALLY TOO LONG. ON LARGER DEVICES, IT	39
40	8492	/	IS ALL TOO EASY TO RUN OUT OF DIRECTORY ENTRY SPACE LONG BEFORE THE	40
41	8493	/	UNDERLYING STORAGE IS COMPLETELY USED. THE SEGMENTATION MERELY	41
42	8494	/	CONTRIBUTES TO THIS PROBLEM AS A POINTLESS SOURCE OF OVERHEAD THAT	42
43	8495	/	REQUIRES ADDITIONAL PROGRAMMING TO ACCESS.]	43
44	8496			44
45	8497	/	AT THE REQUEST OF A DEC SALESMAN, THE NAME WAS CHANGED TO OS/8 BECAUSE HE	45
46	8498	/	FELT IT WAS DIFFICULT TO SELL CUSTOMERS ON A "PROGRAMMING" SYSTEM AS	46
47	8499	/	OPPOSED TO AN "OPERATING" SYSTEM.	47
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49				49
50				50
51				51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	8501	/ 15) P?S	1
2	8502		2
3	8503	/ P?S IS THE SHORTHAND DESIGNATION FOR THE POLY QUESTION SOCIETY, AN	3
4	8504	/ AMORPHOUS GROUP OF QUESTIONERS ONCE LOCATED AT BROOKLYN POLYTECHNIC	4
5	8505	/ INSTITUTE, 333 JAY STREET, BROOKLYN, NY 11201. VARIOUS MEMBERS ARE KNOWN	5
6	8506	/ TO HAVE ACHIEVED EXTRAORDINARY FEATS OF COMPUTER PROGRAMMING, OFTEN	6
7	8507	/ PIONEERING EFFORTS. ONE OF ITS MOST FAMOUS MEMBERS IS RICHARD LARY [AKA	7
8	8508	/ "RICHIE" OR "MOUTH"], CREATOR OF THE R-L MONITOR SYSTEM AND MANY OTHER	8
9	8509	/ FAMOUS PDP-8 PROGRAMS AND OTHER EXTRAORDINARY THINGS INCLUDING SEVERAL	9
10	8510	/ VALUABLE SOFTWARE PATENTS.	10
11	8511		11
12	8512	/ 16) P?S/8	12
13	8513		13
14	8514	/ P?S/8 IS THE NAME OF AN OPERATING SYSTEM FOR THE PDP-8. IT IS BASED, IN	14
15	8515	/ PART, ON THE R-L MONITOR SYSTEM. ALL OF THE DEVELOPERS OF P?S/8 ARE	15
16	8516	/ EXPERIENCED PROGRAMMERS WHO STARTED THEIR PDP-8 PROGRAMMING CAREERS ON THE	16
17	8517	/ R-L MONITOR SYSTEM AND ARE THOROUGHLY FAMILIAR WITH IT INTERNALLY AS WELL	17
18	8518	/ AS EXPERT USERS OF THAT PIONEERING SYSTEM. BY KNOWING ITS WEAKNESSES AND	18
19	8519	/ OVERCOMING THEM IN NEW PROGRAMMING, A BETTER OVERALL SYSTEM WAS CREATED.	19
20	8520		20
21	8521	/ THE REGULAR USER INTERFACE TO P?S/8 IS BASED ON USER EXPERIENCE WITH	21
22	8522	/ ORIGINAL IMPLEMENTATIONS OF THE BASIC LANGUAGE AS PRACTICED ON VARIOUS	22
23	8523	/ GE-235 TIME-SHARING SYSTEMS. FILE COMMANDS ARE VERY MINIMAL AND GEARED	23
24	8524	/ TOWARDS BEING AS INOBTRUSIVE AS POSSIBLE. MOST USERS INTUITIVELY	24
25	8525	/ UNDERSTAND THIS INTERFACE WITHIN MOMENTS OF FIRST CONTACT WITH P?S/8.	25
26	8526		26
27	8527	/ WHILE MAINTAINING THE ORIGINAL STYLE AND SUBSTANCE OF THE R-L MONITOR, THE	27
28	8528	/ OUTGROWTH OF P?S/8 IS TOTALLY ALONG THE LINES OF OTHER OPERATING SYSTEMS	28
29	8529	/ FROM DEC FOR OTHER COMPUTERS. HOWEVER, UNLIKE OS/8, THE EMPHASIS HAS BEEN	29
30	8530	/ ON EFFICIENCY OF RESOURCES AND SPEED OF PERFORMANCE, ESPECIALLY ON SLOWER	30
31	8531	/ DEVICES SUCH AS DECTAPES AND LINCTAPES. MOST USERS, GIVEN AN INFORMED	31
32	8532	/ CHOICE, WILL PREFER P?S/8 TO OS/8 FOR A VARIETY OF REASONS.	32
33	8533		33
34	8534	/ P?S/8 HAS CERTAIN LIMITATIONS, NOT SO MUCH AS TO IMPLEMENTATION, BUT	34
35	8535	/ TO THE DELIBERATE ATTEMPT TO MAXIMIZE PERFORMANCE WHILE PROVIDING ADEQUATE	35
36	8536	/ PRODUCTIVITY FEATURES; THIS IS NOT TO EVERYONE'S LIKING. FOR THOSE WHO	36
37	8537	/ PREFER A LARGER FEATURE SET, P?S/8 IS BEING EXPANDED TO INCLUDE AN	37
38	8538	/ OPTIONAL SHELL OVERLAY. THIS WILL OPERATE UNDER AN ENTIRELY DIFFERENT	38
39	8539	/ SET OF TRADE-OFFS GENERALLY REQUIRING MORE MEMORY.	39
40	8540		40
41	8541	/ FOR MOST USERS, THE OPTIONAL SHELL WILL NOT BE NEEDED. HOWEVER, "POWER	41
42	8542	/ USERS" WILL PREFER THE P?S/8 SHELL OVER THE BASIC SYSTEM [AS WELL AS OS/8]	42
43	8543	/ FOR GENERAL USE WITHOUT RESTRICTIONS. WHILE THE SHELL ENVIRONMENT WILL	43
44	8544	/ REQUIRE MORE RESOURCES THAN THE BASIC ENVIRONMENT, PERFORMANCE WILL STILL	44
45	8545	/ EXCEED THAT OF OS/8; ALL ATTEMPTS WILL BE MADE TO OBTAIN MAXIMUM	45
46	8546	/ PERFORMANCE FROM THE RESOURCES AVAILABLE. HOWEVER, THE BASIC SYSTEM IS	46
47	8547	/ 4K-ORIENTED. [FOR REFERENCE, OS/8 IS 8K-ORIENTED.] THE SHELL ENVIRONMENT	47
48	8548	/ IS ORIENTED TOWARDS LARGER CONFIGURATIONS; CERTAIN FEATURES WILL WORK BEST	48
49	8549	/ IN AS MUCH AS 24K OR MORE; A FEW WILL REQUIRE 32K TO WORK AS BEST AS	49
50	8550	/ POSSIBLE; A SMALL FEW FEATURES WILL NOT BE AVAILABLE IN A 4K ENVIRONMENT,	50
51	8551	/ GENERALLY LEADING TO GRACEFUL ERROR MESSAGES.	51
52			52
53			53
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1	8553	/	BY ADJUSTING CERTAIN CONFIGURATION PARAMETERS, THE SCOPE OF FUNCTIONALITY	1
2	8554	/	CAN BE ROUGHLY THE SAME AS AN EQUIVALENT OS/8 SYSTEM GIVEN THE IDENTICAL	2
3	8555	/	HARDWARE. HOWEVER, PERFORMANCE WILL BE BETTER IN THE P?S/8 SHELL IN MANY	3
4	8556	/	WAYS. SINCE THE CONFIGURATION IS QUITE FLEXIBLE, THE USER CAN DETERMINE	4
5	8557	/	WHAT IS BEST FOR THE PRESENT USAGE. FOR THE MOST SOPHISTICATED USERS,	5
6	8558	/	SEVERAL VARIATIONS USED JUDICIOUSLY MAY BE THE BEST OVERALL SOLUTION	6
7	8559	/	UNLESS 32K IS AVAILABLE. [NOTE: CERTAIN SYSTEM CONFIGURATION PARAMETERS	7
8	8560	/	CAN BE ADJUSTED TO OBTAIN TRADEOFFS WITH DIFFERENT OUTCOMES.] DUE TO THE	8
9	8561	/	VARIOUS FEATURES OF P?S/8 THAT ARE IMPLEMENTED IN A FAR MORE COMPREHENSIVE	9
10	8562	/	WAY THAN OS/8, COUPLED WITH OPTIONAL FEATURES THAT HAVE NO OS/8	10
11	8563	/	EQUIVALENT, IT MAY BE DIFFICULT TO COMPARE PERFORMANCE OF THE TWO SYSTEMS	11
12	8564	/	DIRECTLY. [NOTE: OS/8 DOES NOT SUPPORT ALTERNATE CONSOLE DEVICES OR	12
13	8565	/	MODERN TERMINAL FLOW-CONTROL OTHER THAN WHEN USED WITH A SPECIFIC OUTPUT	13
14	8566	/	HANDLER [KNOWN AS THE KL8E HANDLER]. UNFORTUNATELY, MOST CONSOLE OUTPUT	14
15	8567	/	IS PERFORMED DIRECTLY BY EACH PROGRAM. THUS, COMPARING THE ERRATIC	15
16	8568	/	PERFORMANCE OF OS/8 PROGRAMS TO THE SYSTEM-WIDE APPROACH OF P?S/8 WHERE	16
17	8569	/	ALL CONFORMING PROGRAMS ARE FULLY COMPLIANT WITH RESPECT TO BOTH TERMINAL	17
18	8570	/	HANDLING AND LOGICAL CONSOLE OVERLAY INTERFACE IS NOT EASILY GAUGED. MANY	18
19	8571	/	USERS WITH TERMINALS SUCH AS THE VT100 FAMILY HAVE COMPROMISED THEIR	19
20	8572	/	SYSTEMS TO LIMIT THE AMOUNT OF FLAWED OUTPUT FROM OS/8; P?S/8 CAN RUN	20
21	8573	/	AT ANY CONNECTION RATE WITHOUT DROPPING A SINGLE CHARACTER.]	21
22	8574			22
23	8575	/	NOTE: P?S/8 WAS DEMONSTRATED TO DEC PERSONNEL WITH THE INTENTION OF GIVING	23
24	8576	/	THE SOFTWARE TO DEC IN EXCHANGE FOR HARDWARE DONATION TO THE BROOKLYN	24
25	8577	/	POLYTECHNIC INSTITUTE, WHERE ALL MEMBERS OF THE WCFMPG AND P?S ONCE MET	25
26	8578	/	[NOT NECESSARILY AT THE SAME TIME]. THIS WAS ACTUALLY ATEMPTED TWICE.	26
27	8579	/	HOWEVER, DEC MANAGEMENT REJECTED P?S/8 FOR REASONS THAT DEFY RATIONAL	27
28	8580	/	EXPLANATION DESPITE ACTIVE PARTICIPATION BY SOME OF THE INDIVIDUALS	28
29	8581	/	MENTIONED IN THIS DOCUMENT AS WELL AS OTHER MEMBERS OF THE DEC PDP-8	29
30	8582	/	PROGRAMMING STAFF LOBBYING FOR IT.	30
31	8583			31
32	8584	/	EVENTUALLY, A LIMITED EDITION OF P?S/8 [RENAMED IFDOS] WAS LICENSED TO	32
33	8585	/	INTERSIL CORPORATION FOR USE ON THEIR INTERCEPT I, A PDP-8-COMPATIBLE	33
34	8586	/	MACHINE BASED ON THE IM6100 AND SIMILAR TO AN INFERIOR OFFERING FROM DEC	34
35	8587	/	KNOWN AS THE VT78, THE SLOWEST SYSTEM CONFIGURATION SUPPORTED BY P?S/8.	35
36	8588	/	FOR SYSTEM DEVICE HANDLER CONSIDERATIONS ALONE, P?S/8 RUNS 13 TIMES AS	36
37	8589	/	FAST AS OS/8 ON THIS MACHINE. SINCE THE INTERCEPT I IS 20 PERCENT FASTER	37
38	8590	/	THAN THE VT78, OS/8 PERFORMANCE IS MODESTLY BETTER ON THIS MACHINE. ON	38
39	8591	/	EITHER MACHINE, PERFORMANCE COMPARISONS SHOW A DRAMATIC DIFFERENCE BETWEEN	39
40	8592	/	THE TWO OPERATING SYSTEMS.	40
41				41
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1	8594	/	[NOTE: P?S/8 ACTUALLY RUNS AT THE SAME SPEED ON THE RX01/RX02 DISKETTE	1
2	8595	/	DRIVES REGARDLESS OF THE SPEED OF THE HOST MACHINE. HOWEVER, OS/8 RUNS	2
3	8596	/	13 TIMES AS SLOW ON THE VT78 DUE TO HANDLER INEFFICIENCY THAT CANNOT BE	3
4	8597	/	OVERCOME DUE TO DESIGN LIMITATIONS OF OS/8. THE P?S/8 SYSTEM DEVICE	4
5	8598	/	HANDLER REQUIRES EXTENDED MEMORY AS DOES OS/8; HOWEVER, P?S/8 EXTENDED	5
6	8599	/	DEVICE HANDLER DESIGN ALLOWS MUCH MORE CODE SPACE ALLOWING	6
7	8600	/	HIGH-PERFORMANCE HANDLING OF RX01/RX02 DRIVES BY USING MORE SOPHISTICATED	7
8	8601	/	METHODS. EVEN ON A SYSTEM AS SLOW AS THE VT78 THE PERFORMANCE IS	8
9	8602	/	VIRTUALLY THE SAME AS ON MUCH FASTER HARDWARE. OS/8 ONLY ALLOWS ENOUGH	9
10	8603	/	CODE SPACE TO IMPLEMENT A SEVERELY COMPROMISED DEVICE HANDLER UNABLE TO	10
11	8604	/	KEEP UP WITH THE REQUIRED 2:1 INTERLEAVE. AS SUCH, THE HIGHEST	11
12	8605	/	TRACK/SECTOR CALCULATIONS FAIL TO MAINTAIN TRANSFER SPEED ON ALL MODELS;	12
13	8606	/	ON VERY SLOW MACHINES SUCH AS THE VT78, THIS LOSS OF PERFORMANCE APPLIES	13
14	8607	/	TO VIRTUALLY EVERY DISKETTE SECTOR. MOST USERS OF THE RX01/RX02 DISKETTE	14
15	8608	/	DRIVE, WHEN CONNECTED TO FASTER MACHINES SUCH AS THE PDP-8/E, ARE UNAWARE	15
16	8609	/	OF THIS PROBLEM SINCE ONLY THE HIGHEST SECTORS ARE AFFECTED; MOST	16
17	8610	/	OPERATIONS ADDRESS LOWER SECTORS WHERE THE PERFORMANCE LOSS DOESN'T SET	17
18	8611	/	IN; ON THE VT78 ALL SECTORS ARE AFFECTED WITH THE POSSIBLE EXCEPTION OF A	18
19	8612	/	PORTION OF TRACK 1. OS/8 USES A MINIMAL SIZE DIVISION ALGORITHM	19
20	8613	/	IMPLEMENTED BY THE USE OF REPETITIVE SUBTRACTION; ALL TRACK/SECTOR DATA	20
21	8614	/	MUST BE CALCULATED EACH AND EVERY TIME. P?S/8 RX01/RX02 HANDLERS USE A	21
22	8615	/	METHOD DERIVED BY STRIPPING DOWN AN EAE SIMULATOR TO A DIRECT	22
23	8616	/	IMPLEMENTATION OF THE REQUIRED DIVISION. HOWEVER, THIS IS ONLY USED ONCE	23
24	8617	/	PER CALL TO ESTABLISH THE INITIAL TRACK/SECTOR DATA. ALL ADDITIONAL	24
25	8618	/	SECTORS ARE INSTANTLY CALCULATED BY A PREDICTOR/CORRECTOR ROUTINE. THE	25
26	8619	/	LATEST TRACK AND SECTOR ARE DERIVED FROM THE PREVIOUS VALUES. THE AMOUNT	26
27	8620	/	OF CODE SPACE TO IMPLEMENT THIS TWO-STAGE METHOD IS FAR BEYOND THE UPPER	27
28	8621	/	LIMIT OF OS/8 HANDLER DESIGN.]	28
29	8622			29
30	8623	/	17) R-L MONITOR SYSTEM	30
31	8624			31
32	8625	/	THE R-L MONITOR SYSTEM IS THE PRECURSOR TO P?S/8 WRITTEN BY RICHARD LARY	32
33	8626	/	AND LENNY ELEKMAN. THE ENTIRE OPERATING SYSTEM WAS HAND ASSEMBLED AND	33
34	8627	/	ENTERED DIRECTLY INTO PDP-8 MEMORY MANUALLY ON THE FRONT PANEL OF AN	34
35	8628	/	ORIGINAL PDP-8 COMPUTER LOCATED IN THE BROOKLYN POLYTECHNIC INSTITUTE	35
36	8629	/	ELECTRICAL ENGINEERING COMPLEX. ALL OF THE PROGRAMMERS MENTIONED HERE	36
37	8630	/	EXTENSIVELY USED THE R-L MONITOR SYSTEM AS IT WAS A MAJOR ADVANCE OVER	37
38	8631	/	PREVIOUS SYSTEMS FOR THE SAME HARDWARE.	38
39	8632			39
40	8633	/	[NOTE: THE MANUAL ENTRY OF SO MANY THOUSANDS OF R-L MONITOR INSTRUCTIONS	40
41	8634	/	ON THIS COMPUTER CAUSED PREMATURE FAILURE OF THE FRONT PANEL. CHARLES	41
42	8635	/	LASNER HAD THE DUBIOUS DISTINCTION OF MANUALLY CHANGING EVERY ONE OF THE	42
43	8636	/	FRONT PANEL SWITCHES TO RESTORE FUNCTIONALITY. IT IS NOT ADVISABLE TO	43
44	8637	/	MENTION THE SWITCH MANUFACTURER'S NAME ["STACKPOLE"] IN HIS PRESENCE.]	44
45	8638			45
46	8639	/	THE R-L MONITOR SYSTEM ONLY SUPPORTS A SINGLE TU55/TU56 DECTAPE DRIVE AND	46
47	8640	/	REQUIRES EXTENSIVE MODIFICATION TO SUPPORT ANY ALTERNATE DEVICE. MEMBERS	47
48	8641	/	OF THE P?S INTENSELY STUDIED THE INTERNAL WORKINGS OF THE R-L MONITOR	48
49	8642	/	SYSTEM AND BODILY REPLACED VARIOUS SECTIONS OVER THE COURSE OF MANY YEARS,	49
50	8643	/	EVENTUALLY PRODUCING THE P?S/8 REPLACEMENT SYSTEM AS DESCRIBED ELSEWHERE	50
51	8644	/	IN THIS DOCUMENT.	51
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56				56
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1	8646	/	P?S MEMBERS EVENTUALLY SUBMITTED THE R-L MONITOR SYSTEM TO DECUS [DIGITAL	1
2	8647	/	EQUIPMENT CORPORATION USERS SOCIETY], COMPLETE WITH THE LAST ADDED	2
3	8648	/	COMPONENT: FOCAL, 1969 WITH R-L MONITOR MODIFICATIONS BY CHARLES LASNER.	3
4	8649	/	THE SUBMITTED NAME WAS MS/8 TO APPEAR MORE IN LINE WITH THE VARIOUS OTHER	4
5	8650	/	DEC OFFERINGS OF THE TIME PERIOD [PS/8, OS/8 ETC.].	5
6	8651			6
7	8652	/	A FEW FEATURES OF THE R-L MONITOR SYSTEM ARE RETAINED IN P?S/8 [WITH ALL	7
8	8653	/	NEW SOURCE CODE] INCLUDING FULL COMPATIBILITY WITH TFS BINARY FILES [WHICH	8
9	8654	/	ARE A PROPER SUBSET OF P?S/8 TFS BINARY FILES].	9
10	8655			10
11	8656	/ 18) JIM ROTH		11
12	8657			12
13	8658	/	THE LATE JIM ROTH WAS RESPONSIBLE FOR MANY PROGRAMMING GEMS THROUGHOUT	13
14	8659	/	P?S/8. HE WROTE THE ORIGINAL RAW IMPLEMENTATION OF THE P?S/8 "PAL"	14
15	8660	/	ASSEMBLER. THIS WORK IS A FAST, CLEAN IMPLEMENTATION OF A REPLACEMENT	15
16	8661	/	ASSEMBLER FOR DEC'S PAL III ASSEMBLER, WHICH WAS A VERY OLD AND POORLY	16
17	8662	/	CODED PROGRAM ALBEIT WITH A MODICUM OF GOOD DESIGN POINTS; HOWEVER, PAL	17
18	8663	/	III IS NOT ADAPTABLE TO ADDING ON ANY ADDITIONAL FEATURES. JIM'S CODE IS	18
19	8664	/	QUITE ELEGANT MAKING IT A RELATIVELY STRAIGHT-FORWARD [BUT LENGTHY] TASK	19
20	8665	/	TO ADD ON ALL THE ADDITIONAL FEATURES A FULL-FLEDGED PDP-8 ASSEMBLER	20
21	8666	/	REQUIRES. [NOTE: THE REST OF THE ASSEMBLER WAS WRITTEN BY CHARLES LASNER	21
22	8667	/	OVER A COURSE OF YEARS. ADDED FEATURES INCLUDE DUAL MODE LINC ASSEMBLY	22
23	8668	/	OPTIONS, LITERAL PROCESSING, QUIRKY SYNTAX SUPPORT [SUCH AS "JMP."],	23
24	8669	/	MISCELLANEOUS ASSEMBLER DIRECTIVES AND CROSS-REFERENCE DUMP.]	24
25	8670			25
26	8671	/	JIM WAS ALSO FASCINATED WITH RICHARD LARY'S "SLURP" LOADER AND WAS ABLE TO	26
27	8672	/	CREATE A WORKING VERSION FOR THE DF32 MINIDISK AS PART OF A COMPLETE	27
28	8673	/	ALTERNATIVE VERSION OF THE R-L MONITOR SYSTEM FOR THE DF32.	28
29	8674			29
30	8675	/	ALONG THE WAY HE INTENSELY STUDIED THE OPERATION OF THE TC01 CONTROLLER	30
31	8676	/	AND IS THE AUTHOR OF MOST OF THE SOMEWHAT CONFUSING SEARCH AND MODE-CHANGE	31
32	8677	/	CODE WITHIN THE TC01/TC08 SYSTEM DEVICE HANDLER, INCLUDING THE INITIAL	32
33	8678	/	SEARCH GUESS ROUTINE. AS A DIRECT RESULT OF HIS SIGNIFICANTLY LOWERING	33
34	8679	/	THE OVERALL CODE SPACE USED BY THE HANDLER, ENOUGH NEW SPACE WAS OBTAINED	34
35	8680	/	TO ENVISION THE IMBEDDED SYSTEM LOADER THAT IS ARGUABLY THE SINGLE MOST	35
36	8681	/	IMPORTANT CONTRIBUTION OF P?S/8 OVER THE R-L MONITOR SYSTEM AT THE KERNEL	36
37	8682	/	LEVEL.	37
38	8683			38
39	8684	/	WHILE AN EMPLOYEE AT DEC, JIM WROTE THE QUICKSORT ROUTINES FOR	39
40	8685	/	REORGANIZING THE SYMBOL TABLE FOR PRINTOUT WITHIN PAL8. HE IS ALSO	40
41	8686	/	RESPONSIBLE FOR EXTENSIVE REWRITE OF CERTAIN SPECIALIZED VERSIONS OF OS/8	41
42	8687	/	[INDUSTRIAL] BASIC. JIM ALSO REMOVED AN ENTIRE PAGE OF CODE SPACE WITHIN	42
43	8688	/	POLY SPACEWAR BY USING A MORE ADVANCED TABLE LOOKUP METHOD FOR CALCULATING	43
44	8689	/	VARIOUS TRIGONOMETRIC FUNCTIONS. AS A RESULT, MANY MORE SPECIAL EFFECTS	44
45	8690	/	AND OTHER FRILLS BECAME POSSIBLE WHILE STILL RUNNING AS A LEGITIMATE USER	45
46	8691	/	PROGRAM COMPATIBLE WITH MOST 4K ENVIRONMENTS.	46
47	8692			47
48	8693	/	JIM WAS THE ONLY MEMBER OF THE P?S WITH AN ACCOMPLISHED BACKGROUND IN	48
49	8694	/	ELECTRONICS BESIDES CHARLES LASNER. HE ALSO WORKED FOR DEC'S VLSI	49
50	8695	/	HARDWARE DIVISION FOR SOME TIME AS WELL AS VARIOUS PDP-8 PROGRAMMING	50
51	8696	/	GROUPS.	51
52				52
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54				54
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1	8698	/ 19) SLURP LOADER	1
2	8699		2
3	8700	/ RICHARD LARY IMPLEMENTED THE FIRST "SLURP" LOADER [THE TERM ATTEMPTS TO	3
4	8701	/ DESCRIBE SOME ASPECTS OF THE PROCESS] AS THE ONLY BINARY FILE LOADER ON	4
5	8702	/ THE R-L MONITOR SYSTEM. ALL P?S/8 SYSTEMS PROVIDE A VIRTUAL LOADER THAT	5
6	8703	/ PRODUCES SUBSTANTIALLY THE SAME RESULT; WHERE FEASIBLE, THE SLURP LOADER	6
7	8704	/ IS PROVIDED AS AN OPTION. MOST, BUT NOT ALL SUPPORTED DEVICES CAN	7
8	8705	/ IMPLEMENT A SLURP LOADER. EXCEPTIONS INCLUDE THREE-CYCLE DATA BREAK	8
9	8706	/ DEVICES THAT ARE TOO FAST [RF08] AND OTHER DMA DEVICES THAT CANNOT BE	9
10	8707	/ ACCESSED VIA SINGLE WORD TRANSFERS [RK08/RK01, RK8E/RK05, ETC.]. ALL	10
11	8708	/ ASYNCHRONOUS PROGRAM TRANSFER DEVICES SHOULD SUPPORT A SLURP LOADER.	11
12	8709		12
13	8710	/ SLURP LOADING IS BASED ON AN EFFICIENT LOADING FORMAT THAT IS NEARLY 6/7	13
14	8711	/ AS EFFICIENT AS CORE-IMAGE FORMAT [AND FAR MORE EFFICIENT THAN ANY	14
15	8712	/ PAPER-TAPE LOADING FORMAT]. THE FORMAT CONSISTS OF A 12-BIT CONTROL WORD	15
16	8713	/ CONTAINING SIX TWO-BIT CONTROL FIELDS FOLLOWED BY SIX 12-BIT DATA WORDS.	16
17	8714	/ EACH DATA WORD IS EITHER A) A LOADING ADDRESS FOR SUBSEQUENT LOADING, B) A	17
18	8715	/ WORD TO BE LOADED INTO THE LATEST ADDRESS; IMPLICITLY INCREMENT THE	18
19	8716	/ LOADING ADDRESS, C) A FIELD SETTING [FIELD 0 IS THE DEFAULT], D) END OF	19
20	8717	/ FILE. THE SEVEN-WORD SEQUENCE IS REPEATED 18 TIMES IN ANY SINGLE BLOCK;	20
21	8718	/ THE LAST TWO WORDS OF EACH 128-WORD BLOCK ARE IGNORED.	21
22	8719		22
23	8720	/ OTHER THAN THE PREVAILING LOADING ORIGIN AND CURRENT DATA FIELD SETTING,	23
24	8721	/ ALL DATA BLOCKS ARE INDEPENDENT. [NOTE: SLURP FORMAT FILES CAN	24
25	8722	/ THEORETICALLY BE CREATED PROVISIONALLY THAT ARE SIGNIFICANTLY LONGER THAN	25
26	8723	/ TFS FILES.]	26
27	8724		27
28	8725	/ A PRACTICAL SLURP LOADER WORKS WITHOUT A BLOCK-ORIENTED BUFFER, PROCESSING	28
29	8726	/ THE WORDS AS THEY ARE READ IN DIRECTLY. IN THE CASE OF THE TC01/TC08	29
30	8727	/ DECTAPE, THE WORD-COUNT REGISTER IS SET TO 0000; THE CURRENT ADDRESS IS	30
31	8728	/ SET TO ONE LESS THAN THE ADDRESS OF A ONE-WORD BUFFER; JIM ROTH PROVED	31
32	8729	/ THAT THE CURRENT ADDRESS REGISTER IS A PERFECTLY ACCEPTABLE BUFFER. THE	32
33	8730	/ PROGRAM WAITS FOR THE WORD-COUNT REGISTER TO CHANGE FROM 0000 -> 0001. AT	33
34	8731	/ THAT POINT, THE LATEST FILE WORD IS IN THE CURRENT ADDRESS REGISTER. EACH	34
35	8732	/ WORD IS PROCESSED AS DESCRIBED ABOVE.	35
36	8733		36
37	8734	/ WHEN ALL FILES HAVE BEEN LOADED, THE LOADED PROGRAM IS STARTED AT AN	37
38	8735	/ ADDRESS DETERMINED BY THE P?S/8 BIN PROGRAM THAT PLACES THE SLURP LOADER,	38
39	8736	/ IN CONJUNCTION WITH USER-SUPPLIED OPTION SWITCHES. ALL SLURP LOADERS	39
40	8737	/ INCLUDE AN INTERNAL ADDRESS OF A HLT INSTRUCTION [7402] TO BE USED AS THE	40
41	8738	/ DEFAULT ADDRESS IF THE USER DOESN'T SUPPLY ONE; STARTING ADDRESS AND	41
42	8739	/ INITIAL STARTING FIELD ARE PROCESSED AT THE END OF FILE LOADING.	42
43	8740		43
44	8741	/ A MAJOR ADVANTAGE OF THE SLURP LOADER IS THAT THE DEVICE CAN BE	44
45	8742	/ WRITE-PROTECTED AS ALL LOADING IS READ-ONLY. [NOTE: IF THE VIRTUAL LOADER	45
46	8743	/ IS CHOSEN, THE SYSTEM DEVICE MUST BE WRITE-ENABLED AS THE VIRTUAL AREA IS	46
47	8744	/ USED TO PROTECT LOADING OVER THE VIRTUAL LOADER INCLUDING ITS BUFFER USED	47
48	8745	/ TO READ IN PORTIONS OF BINARY FILES ETC.] THERE IS A NOTICABLE PERFORMANCE	48
49	8746	/ IMPROVEMENT BETWEEN THE TWO LOADER VARIANTS; USE OF THE SLURP LOADER IS	49
50	8747	/ ALWAYS RECOMMENDED FOR NORMAL USAGE [IF AVAILABLE].	50
51	8748		51
52	8749	/ THE SLURP LOADER IS INCLUDED IN THE R-L MONITOR SYSTEM, POLY BASIC	52
53	8750	/ [EDU-SYSTEM 30] AS WELL AS P?S/8. [NOTE: ONLY THE P?S/8 IMPLEMENTATION	53
54	8751	/ SUPPORTS EXTENDED MEMORY.]	54
55			55
56			56
57			57
58			58

1	8753	/ 20) SYSTEM SAVE AREA	1
2	8754		2
3	8755	/ THE SYSTEM SAVE AREA REFERS TO THE DEDICATED BLOCKS ON THE P?S/8 SYSTEM	3
4	8756	/ DEVICE [BLOCKS 0020-0057] USED TO PRESERVE THE ENTIRE USER AREA CONTENTS	4
5	8757	/ OF FIELD 0 [0000-7577]. THIS IS ACCESSED BY VARIOUS MEMORY-ORIENTED	5
6	8758	/ COMPONENTS OF BOTH THE BASIC P?S/8 SYSTEM AND THE OPTIONAL SHELL	6
7	8759	/ ENVIRONMENT.	7
8	8760		8
9	8761	/ IN THE BASIC SYSTEM, THE SYSTEM SAVE AREA CAN ALSO BE ADDRESSED BY TWO TFS	9
10	8762	/ FILE NAMES "%" AND "\$"; EACH TFS FILE IS 2K LONG. "%" STORES LOCATIONS	10
11	8763	/ 00000-03777]; "\$" STORES LOCATIONS 04000-07777]. [NOTE: THE LAST PAGE OF	11
12	8764	/ MEMORY WHERE THE SYSTEM DEVICE HANDLER IS RESIDENT IS NEVER OVERWRITTEN;	12
13	8765	/ HOWEVER, CERTAIN PROGRAMS MAY WRITE OUT AN IMAGE OF THE DEVICE HANDLER IN	13
14	8766	/ CONJUNCTION WITH SAVING THE REST OF FIELD 0.]	14
15	8767		15
16	8768	/ 21) P?S/8 SHELL	16
17	8769		17
18	8770	/ THE P?S/8 SHELL IS AN OPTIONAL SYSTEM OVERLAY THAT DISPLACES THE NORMAL	18
19	8771	/ KEYBOARD MONITOR IN FAVOR OF AN ALTERNATE INTERFACE AND FILE SYSTEM. THE	19
20	8772	/ SYSTEM DEVICE HANDLER IS USED ONLY AS AN INTERNAL MECHANISM TO BIND ALL OF	20
21	8773	/ THE SHELL COMPONENTS TOGETHER. THIS INCLUDES THE SYSTEM LOADER WHICH,	21
22	8774	/ UNLIKE THE OS/8 ANALOGOUS SEGMENT, IS SELF-SUFFICIENT AND CANNOT FAIL DUE	22
23	8775	/ TO INTERNAL ERROR [UNRECOVERABLE HALT]. IT IS INTENDED TO BE USED WITH	23
24	8776	/ THE OPTIONAL LOGICAL CONSOLE OVERLAY; AS SUCH, ALL ERRORS ON THE SYSTEM	24
25	8777	/ DEVICE WILL BE ELEGANTLY HANDLED BY THE OPERATING SYSTEM; PROGRAMS DO NOT	25
26	8778	/ PARTICIPATE IN THE ERROR RECOVERY PROCESS FOR ANY OPERATION ON THE SYSTEM	26
27	8779	/ DEVICE. [NOTE: THE P?S/8 SHELL CANNOT DIRECTLY ACCESS THE TFS DIRECTORY ON	27
28	8780	/ THE SYSTEM DEVICE; SPECIFIC UTILITIES CAN BE WRITTEN TO ACCOMPLISH THIS.	28
29	8781	/ THIS WILL FACILITATE FILE CONVERSION BETWEEN THE TWO P?S/8 OPERATIONAL	29
30	8782	/ VARIANTS AND/OR OS/8.]	30
31	8783		31
32	8784	/ NO OTHER BASIC P?S/8 FEATURE IS USED IN THE SHELL ENVIRONMENT. AS SUCH,	32
33	8785	/ IT IS FEASIBLE TO STRIP OUT MOST OF THE BASIC SYSTEM TO MINIMIZE THE	33
34	8786	/ OVERALL SYSTEM REQUIREMENT IF DESIRED. BY SUITABLE CONFIGURATION, THE	34
35	8787	/ FIRST [UP TO] 4096 BLOCKS CAN BE RESERVED FOR THE SYSTEM DEVICE, WHICH	35
36	8788	/ CORRESPONDS TO THE LARGEST POSSIBLE BASIC P?S/8 SYSTEM DEVICE STORAGE. IF	36
37	8789	/ BOTH SYSTEM VARIANTS ARE TO BE PRESENT ON ANY DEVICE, THE RESERVED SIZE	37
38	8790	/ CAN BE ADJUSTED ACCORDINGLY. HOWEVER, THE MINIMUM VIABLE SIZE FOR A	38
39	8791	/ SYSTEM DEVICE GEARED TOWARDS ROUTINELY RUNNING THE SHELL ON A DAILY BASIS	39
40	8792	/ CAN BE SET TO APPROXIMATELY 200 BLOCKS; THIS IS COMPARABLE TO THE	40
41	8793	/ DIFFERENCE BETWEEN AN OS/8 SYSTEM DEVICE AND AN OS/8 NON-SYSTEM DEVICE ON	41
42	8794	/ THE SAME MEDIA. AS NECESSARY, ANY ADDITIONAL STORAGE CAN BE ACCESSED AS A	42
43	8795	/ NON-SYSTEM DEVICE; P?S/8 INCLUDES A TABLE OF NON-SYSTEM HANDLERS THAT	43
44	8796	/ SPECIFIES PROTECTION OFFSETS TO DEAL WITH MIXED CONFIGURATIONS. [NOTE: THE	44
45	8797	/ BASIC P?S/8 SYSTEM IS ORIENTED TOWARDS THE SMALLER MAXIMUM STORAGE SPACE	45
46	8798	/ ADDRESSED BY THE SYSTEM HANDLER. NON-SYSTEM DEVICES CAN ADDRESS AS MUCH	46
47	8799	/ AS 4096 TIMES AS MUCH SPACE. ALL NON-SYSTEM DEVICE HANDLER CALLS ARE THUS	47
48	8800	/ POTENTIALLY SUBJECT TO AN ADJUSTMENT ON DEVICES WITH DUAL STRUCTURES	48
49	8801	/ PRESENT.] [NOTE: THE NOTION OF MIXED CONFIGURATIONS IS STILL BEING	49
50	8802	/ RESEARCHED AS OF THIS WRITING; ACTUAL IMPLEMENTATION MAY VARY GREATLY.]	50
51			51
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1	8804	/	THE SHELL PROCESSES COMMANDS THAT REFERENCE A FILE SYSTEM FAR MORE	1
2	8805	/	POWERFUL THAN THE OS/8 STRUCTURE. FILE NAMES ARE UP TO 12 CHARACTERS WITH	2
3	8806	/	UP TO FOUR CHARACTER FILE EXTENSIONS WITH CASE-RETENTION. [OS/8 FILE NAMES	3
4	8807	/	ARE UP TO SIX CHARACTERS WITH UP TO TWO CHARACTER FILE EXTENSIONS IN	4
5	8808	/	UPPER-CASE ONLY.] SINCE THE SYSTEM DEVICE HANDLER IS DESIGNED FOR EITHER	5
6	8809	/	THE BASIC P?S/8 SYSTEM OR SHELL INTER-COMPONENT COMMUNICATION, IT PLAYS NO	6
7	8810	/	DIRECT ROLE IN THE SHELL FILE ACCESS. POWERFUL NON-SYSTEM HANDLERS ARE	7
8	8811	/	USED FOR ALL OPERATIONS WHICH CAN ACCESS FAR MORE STORAGE THAN EITHER THE	8
9	8812	/	BASIC P?S/8 SYSTEM OR OS/8.	9
10	8813			10
11	8814	/	THE FOLLOWING IS A REALISTIC COMPARISON: THE BASIC P?S/8 SYSTEM CAN	11
12	8815	/	ADDRESS .5 MWORDS OF DATA PER LOGICAL UNIT. [NOTE: 1.0 MWORDS IS 1048576	12
13	8816	/	12-BIT WORDS OF STORAGE.] ASSUMING EIGHT LOGICAL UNITS PER SYSTEM DEVICE,	13
14	8817	/	THIS IS A TOTAL ADDRESSABLE SPACE OF 4.0 MWORDS PER SYSTEM DEVICE GROUP.	14
15	8818	/	OS/8 HANDLERS CAN ADDRESS AT MOST 1.0 MWORDS OF STORAGE PER HANDLER.	15
16	8819	/	THUS, OS/8 CAN ADDRESS WITH ANY ONE HANDLER TWICE AS MUCH STORAGE SPACE AS	16
17	8820	/	ANY ONE P?S/8 LOGICAL DEVICE UNIT WHILE THE ENTIRE P?S/8 DEVICE GROUP CAN	17
18	8821	/	ACCESS FOUR TIMES AS MUCH STORAGE AS OS/8'S SYSTEM HANDLER.	18
19	8822			19
20	8823	/	THE SHELL ENVIRONMENT USES NON-SYSTEM HANDLERS THAT ARE RADICALLY	20
21	8824	/	DIFFERENT FROM EITHER THE BASIC P?S/8 SYSTEM OR OS/8. THE MAXIMUM	21
22	8825	/	ADDRESSABLE SPACE IS 2048 MWORDS PER LOGICAL UNIT FOR A TOTAL ADDRESSABLE	22
23	8826	/	STORAGE OF 16384 MWORDS PER NON-SYSTEM DEVICE GROUP. THIS IS 2048 TIMES	23
24	8827	/	AS LARGE AS ANY OS/8 HANDLER CAN ACCESS AND 4096 TIMES AS LARGE AS ANY	24
25	8828	/	P?S/8 SYSTEM DEVICE HANDLER PER LOGICAL UNIT.	25
26	8829			26
27	8830	/	A FURTHER CONSIDERATION OF STORAGE SPACE IS THE ALLOWED NUMBER OF LOGICAL	27
28	8831	/	DEVICES. OS/8 ALLOWS AT MOST 15 LOGICAL DEVICES. [NOTE: INTERNAL DEVICE	28
29	8832	/	INFORMATION IS HELD IN 4-BIT FIELDS; THE CASE OF 0000 IS NEEDED TO	29
30	8833	/	INDICATE A LACK OF EXPLICIT DEVICE TO ALLOW INTERPRETATION AT ANOTHER	30
31	8834	/	LEVEL.] AS A PRACTICAL MATTER [FOR COMPARISON PURPOSES] THE NUMBER EIGHT	31
32	8835	/	WILL BE USED. [NOTE: ONE OF THE DEVICES IS DSK:, A LOGICAL STAND-IN FOR	32
33	8836	/	ONE OF THE OTHER DEVICES; SEVERAL ADDITIONAL DEVICES ARE USED FOR	33
34	8837	/	NON-STORAGE PURPOSES SUCH AS THE CONSOLE DEVICE, HIGH-SPEED READER/PUNCH	34
35	8838	/	IF APPLICABLE AND SYSTEM LINEPRINTER.] WHILE THE MAXIMUM NUMBER OF	35
36	8839	/	DEVICES AVAILABLE TO THE SHELL ENVIRONMENT HAS NOT BEEN DETERMINED, THE	36
37	8840	/	MAXIMUM NUMBER IS NOT LIMITED TO A SUBSET OF 16; BETTER OVERALL DESIGN	37
38	8841	/	ALLOWS FOR FAR MORE DEVICE HANDLER ENTRIES IN A PRACTICAL SYSTEM. FOR	38
39	8842	/	COMPARISON PURPOSES THE NUMBER 32 WILL BE USED [IT COULD EASILY BE TWICE	39
40	8843	/	AS MUCH].	40
41	8844			41
42	8845	/	MORE MODERN STORAGE DEVICES TEND TO BE LARGER HARD DISKS PARTITIONED INTO	42
43	8846	/	SMALLER DIVISIONS AS NECESSARY. THIS SUGGESTS THE LARGEST ADDRESSABLE	43
44	8847	/	SPACE ON A SUITABLY PARTITIONED DISK WOULD BE 512 MWORDS. MANY MORE	44
45	8848	/	RECENT DEVICES USE SIX-BIT BYTES IN THE UNDERLYING STORAGE. [NOTE:	45
46	8849	/	EXAMPLES INCLUDE RX-SERIES DISKETTES AS TYPICALLY OPERATED IN 12-BIT MODE,	46
47	8850	/	RD50 HARD DISK SUBSYSTEM ON CERTAIN DECMATES, CESI MDC8 ETC. ALL SHARE	47
48	8851	/	THE COMMON TRAIT OF USING 3/4 OF THE ACTUAL STORAGE OF 8-BIT BYTES.] THUS,	48
49	8852	/	ANOTHER WAY TO STATE CAPACITY IS 4 GB PER LOGICAL DEVICE UNIT, 32 GB PER	49
50	8853	/	LOGICAL DEVICE GROUP AND 1.0 TB PER SYSTEM WITHIN THE SHELL ENVIRONMENT.	50
51	8854	/	OS/8 ONLY SUPPORTS 2.0 MB MAXIMUM ON ANY DEVICE AND 32 MB ON THE LARGEST	51
52	8855	/	SUPPORTABLE SYSTEM.	52
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1	8857	/	THERE IS AN ADDITIONAL IMPORTANT TRADEOFF: OS/8 RECORD SIZE IS 256 WORDS	1
2	8858	/	WHILE ALL P?S/8 HANDLERS ADDRESS STORAGE IN 128 WORD BLOCKS. DEPENDING ON	2
3	8859	/	PROGRAM LOGISTICS, OS/8'S LARGER MINIMUM BUFFER REQUIREMENT CAN CAUSE	3
4	8860	/	PROGRAM SIZE TO BE NEEDLESSLY LARGE AND UNWIELDY. OFTEN A SMALL CONTROL	4
5	8861	/	FILE IS NEEDED AND THE SMALLEST FILE BUFFER WOULD BE APPROPRIATE. IN ANY	5
6	8862	/	P?S/8 SITUATION THIS WOULD BE A ONE PAGE BUFFER WHILE OS/8 WOULD DEMAND A	6
7	8863	/	TWO PAGE BUFFER; THE OVERALL REQUIREMENT QUICKLY BUILDS WHEN MULTIPLE	7
8	8864	/	FILES ARE OPEN SIMULTANEOUSLY, EACH NEEDING ITS OWN BUFFER.	8
9	8865			9
10	8866	/	EVEN WHEN BETTER OVERALL THROUGHPUT IS DESIRABLE, OS/8 STILL PROVIDES AN	10
11	8867	/	INFERIOR SOLUTION: DUE TO THE OS/8 PROHIBITION ON THE USE OF THE LAST PAGE	11
12	8868	/	OF EVERY MEMORY FIELD, THE MAXIMUM GENERALLY ALLOWED TRANSFER WOULD BE 15	12
13	8869	/	RECORDS [30 PAGES]. ONLY EVEN VALUES FROM TWO THROUGH 30 CAN BE SPECIFIED	13
14	8870	/	IN TERMS OF ALLOWABLE PAGE COUNT [DIVIDE BY TWO TO GET RECORD COUNT]. IN	14
15	8871	/	ALL P?S/8 HANDLER CALLS, ANY VALUE IS ALLOWED IN THE RANGE OF ONE THROUGH	15
16	8872	/	32. P?S/8 HAS BOTH THE LARGEST MAXIMUM BUFFER SIZE AND THE SMALLEST	16
17	8873	/	MINIMUM BUFFER SIZE AND ALL CASES IN-BETWEEN WHILE OS/8 CANNOT ALLOW ANY	17
18	8874	/	ODD SIZES OR THE MAXIMUM SIZE OF 32. [NOTE: CERTAIN EXISTENT HARDWARE	18
19	8875	/	DEVICES CANNOT RUN OS/8. OS/8 DEMANDS A DESIGN QUIRK: THE DEVICE MUST BE	19
20	8876	/	INHERENTLY PAGE-ORIENTED [AS IS IDEAL FOR ALL P?S/8 IMPLEMENTATIONS] OR,	20
21	8877	/	IF THE DEVICE IS DOUBLE-PAGE RECORD-ORIENTED, THERE MUST BE A WAY TO	21
22	8878	/	READ/WRITE THE FIRST 128 WORDS ONLY. MOREOVER, THIS HAS TO BE	22
23	8879	/	IMPLEMENTED WITHIN A DEVICE HANDLER THAT OCCUPIES AT MOST SOMEWHAT LESS	23
24	8880	/	THAN TWO PAGES OF CODE SPACE.	24
25	8881			25
26	8882	/	SINCE P?S/8 ALLOWS JUST UNDER NINE PAGES OF CODE SPACE, THIS CAN GENERALLY	26
27	8883	/	BE ACCOMPLISHED WITH EASE. FOR EXAMPLE: P?S/8 ON THE RK8E/RK05 ADDRESSES	27
28	8884	/	EACH 256-WORD RK05 RECORD TO THE NEAREST HALF SECTOR. WHEN NECESSARY, AN	28
29	8885	/	INTERNAL BUFFER WITHIN THE HANDLER IS USED TO IMPLEMENT FIXUPS FOR ALL	29
30	8886	/	SPECIAL CASE CONSIDERATIONS SUCH AS WRITING ONLY ON THE SECOND HALF OF A	30
31	8887	/	SECTOR; FOR ALL CASES ANALOGOUS TO WHAT OS/8 ALLOWS, THE BUFFERING SCHEME	31
32	8888	/	IS NOT USED.	32
33	8889			33
34	8890	/	THE CESI MDC8 WITH 256 WORD RECORDS IS THE DESIGNATED STORAGE HARDWARE FOR	34
35	8891	/	OMNI-8, A DEDICATED BOOTABLE APPLICATION SYSTEM THAT IS EXPLICITLY	35
36	8892	/	DESIGNED TO MAXIMIZE PERFORMANCE ON THAT DEVICE, OFTEN WITH A NET STORAGE	36
37	8893	/	SPACE OF 32 MWORDS PER HARD DISK. OS/8 IS ONE OF THE *EMULATED* OPERATING	37
38	8894	/	SYSTEMS SUPPORTED; HOWEVER, OS/8 CANNOT RUN ON THE MACHINE. P?S/8 CAN RUN	38
39	8895	/	ON THIS MACHINE IF 8K AVAILABLE; THE P?S/8 SHELL ENVIRONMENT CAN ADDRESS	39
40	8896	/	THE ENTIRE MDC8 ENVIRONMENT OF TWO COMPLETE HARD DISKS WITH EASE.]	40
41				41
42				42
43				43
44				44
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1	8898	/	THE FULL CAPABILITY OF THE P?S/8 SYSTEM DATE WILL BE AVAILABLE TO THE	1
2	8899	/	SHELL ENVIRONMENT. ALL FILES WILL BE CAPABLE OF A UNIQUE DATE IN THE	2
3	8900	/	RANGE OF 01-JAN-1900 THROUGH 31-DEC-2411. THIS IS THE ONLY OPERATING	3
4	8901	/	SYSTEM FOR A DEC-MADE PRODUCT WITH THAT WIDE A DATE RANGE. AS SUCH, THERE	4
5	8902	/	SHOULD NOT BE ANY FILE THAT REQUIRES ANY FORM OF DATE SPOOFING FROM THE	5
6	8903	/	DAWN OF THE TWENTIETH CENTURY FORWARD. THE LARGE STORAGE CAPABILITY WILL ALLOW	6
7	8904	/	THE ENTIRE HISTORICAL RECORD OF THE PDP-8 TO BE STORED ON A SINGLE DEVICE THAT	7
8	8905	/	CAN BE SHARED WITH ALL USERS FOR THE FORESEEABLE FUTURE. THIS INCLUDES TEXT	8
9	8906	/	FILES WRITTEN BY SEYMOUR CRAY IN 1960 DEFINING 12-BIT COMPUTING AND LATER	9
10	8907	/	ARTICLES FROM MIT LINCOLN LABORATORIES IN SUPPORT OF THE LINC PROJECT WITH	10
11	8908	/	ACCURATE ORIGINAL FILE DATES. THE PDP-8 SHARES A HERITAGE OF DESIGN WITH THESE	11
12	8909	/	OTHER TWO PREDECESSOR SYSTEMS; SUPPORTING DOCUMENTS SHOULD BE PART OF ANY PROPER	12
13	8910	/	MASTER FILE STORAGE COLLECTION.	13
14	8911			14
15	8912	/	SHELL COMMANDS WILL BE TOTALLY IN LINE WITH DEC STANDARDS FOR OPERATING	15
16	8913	/	SYSTEMS NEWER THAN OS/8. FROM THE STANDPOINT OF AN OS/8 USER, THIS	16
17	8914	/	ESSENTIALLY MEANS CCL-LIKE IMPLEMENTATION IN ALL SITUATIONS, WITHOUT THE	17
18	8915	/	OVERHEAD OF OS/8'S IMPLEMENTATION OF CCL. [NOTE: OS/8 CCL IS SO	18
19	8916	/	OVERHEAD-RIDDEN, DEC MANAGEMENT MADE OFFICIAL RECOMMENDATIONS NOT TO	19
20	8917	/	ENABLE CCL ON SYSTEMS BASED ON SLOWER DEVICES SUCH AS DECTAPES AND	20
21	8918	/	LINCTAPES.]	21
22	8919			22
23	8920	/	UNLIKE OS/8, PROGRAMS CAN DETERMINE THE EXACT MEANING OF DEFAULT DEVICES.	23
24	8921	/	[NOTE: OS/8 ONLY SUPPORTS ONE DEFAULT DEVICE AT A TIME; THE SHELL ALLOWS	24
25	8922	/	PROGRAMS TO SUPPLY UNIQUE POSITIONAL DEFAULT DEVICES WITHIN THE COMMAND.]	25
26	8923	/	BY FORCING DEFAULTS TO BE DETERMINED FROM A SMALL SYSTEM CACHE OF UNIQUE	26
27	8924	/	DEFAULT DEVICES, IT IS POSSIBLE TO OPEN MORE THAN ONE OUTPUT STREAM AT A	27
28	8925	/	TIME.	28
29	8926			29
30	8927	/	FOR EXAMPLE, THE SHELL PDP-8 ASSEMBLER [TENTATIVELY TO BE NAMED XPAL] WILL	30
31	8928	/	BE ABLE TO PERFORM ALL OPERATIONS IN TWO PASSES, NOT THREE AS IS DONE IN	31
32	8929	/	OS/8. [NOTE: THE P?S/8 "PAL" ASSEMBLER NEARLY DOES THIS CURRENTLY, WITH	32
33	8930	/	THE LONE EXCEPTION BEING THE LINEPRINTER OUTPUT MUST BE DIRECTED TO THE	33
34	8931	/	PHYSICAL DEVICE 66 LINEPRINTER OR THE LOGICAL LINEPRINTER IF THE LOGICAL	34
35	8932	/	CONSOLE OVERLAY IS ENABLED.] THE ONLY REQUIREMENT IS THAT THE TWO OUTPUT	35
36	8933	/	DEVICES ARE DIFFERENT FROM EACH OTHER. WHILE THE USER CAN CREATE A	36
37	8934	/	COMMAND ATTEMPTING TO VIOLATE THIS, THE SHELL WILL REJECT THIS AS	37
38	8935	/	IMPOSSIBLE. HOWEVER, BY DEFAULT, THE DEVICES WILL ALREADY BE UNIQUE.	38
39				39
40				40
41				41
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43				43
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47				47
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1	8937	/	THE ONLY CONFIGURATION THAT WOULD PREVENT SUCH OPERATION ENTIRELY WOULD BE	1
2	8938	/	IF THE SYSTEM IS CONFIGURED TO ONLY EXIST AS BLOCKS ON THE SYSTEM DEVICE	2
3	8939	/	WITH AN OFFSET AND THERE ARE NO OTHER LOGICAL DEVICES AVAILABLE. IN THIS	3
4	8940	/	CONTRIVED EXAMPLE, XPAL WOULD HAVE TO OPERATE IN A MANNER ANALOGOUS TO	4
5	8941	/	OS/8 PAL8 USING TWO ASSEMBLY OPERATIONS: ONE TO PRODUCE BINARY AND ANOTHER	5
6	8942	/	TO PRODUCE A LISTING FILE. EXECUTABLE PROGRAMS CAN DEMAND AS MANY AS	6
7	8943	/	THREE POSITIONAL DEFAULT DEVICES, ALL UNIQUE. [NOTE: THIS IS SUBJECT TO	7
8	8944	/	CHANGE; HOWEVER, IT IS HOPED THAT THIS SHOULD COVER ALL KNOWN CASES.]	8
9	8945	/	XPAL WILL REJECT ATTEMPTS TO USE THE SECOND DEVICE IN A DEFAULT SITUATION	9
10	8946	/	SHOULD THIS DEFAULT DEVICE NOT BE DEFINED. FOR MOST CONFIGURATIONS, TWO	10
11	8947	/	OR MORE DEVICE/LOGICAL UNITS WILL BE AVAILABLE TO ALLOW MAXIMUM	11
12	8948	/	FUNCTIONALITY. [CONTRAST THIS WITH OS/8 WHERE THIS FEATURE DOES NOT	12
13	8949	/	EXIST; OS/8 DOES NOT ALLOW MULTIPLE OUTPUT FILES TO BE OPEN FOR PAL8 EVEN	13
14	8950	/	IF LOCATED ON DIFFERENT DEVICES. THUS, ONLY DEFICIENT CONFIGURATIONS WILL	14
15	8951	/	BE REQUIRED TO OPERATE IN A LIKE MANNER TO OS/8 WITH REGARD TO ASSEMBLER	15
16	8952	/	PASS CONSIDERATIONS.]	16
17	8953			17
18	8954	/	FOR RETRO-COMPATIBILITY ONLY, A COMMAND DECODE OPTION MAY BE MADE	18
19	8955	/	AVAILABLE. ALL SYSTEM PROGRAMS WILL BE DESIGNED TO NOT REQUIRE IT,	19
20	8956	/	INSTEAD DEPENDING UPON THE SHELL TO PRE-PARSE THE ENTIRE COMMAND AND OPEN	20
21	8957	/	ALL EXISTING AND TENTATIVE FILES.	21
22	8958			22
23	8959	/	GIVEN THIS LEVEL OF OPERATING SYSTEM SUPPORT, THE INTERNAL INTERFACE	23
24	8960	/	PORTION OF MOST OS/8 PROGRAMS CAN EASILY BE REWRITTEN ALLOWING ALL USEFUL	24
25	8961	/	APPLICATIONS TO BE MIGRATED TO THE P?S/8 SHELL.	25
26	8962			26
27	8963	/	THE IMPLEMENTATION OF THE SHELL WILL TEND TO CHANGE CERTAIN SYSTEM	27
28	8964	/	CONFIGURATIONS SUCH AS THE RK8E. WHEN THE RK8E SYSTEM HANDLER WAS	28
29	8965	/	ORIGINALLY DESIGNED, IT WAS NOT CONTEMPLATED THERE WOULD BE A NEED FOR	29
30	8966	/	MORE LOGICAL BLOCKS PER DEVICE THAN THE BASIC P?S/8 UPPER LIMIT OF 4096	30
31	8967	/	LOGICAL BLOCKS; BLOCKS BEYOND THIS RANGE CAN ONLY BE ADDRESSED IF THE	31
32	8968	/	DEVICE MAPPING CONTINUES THE NEXT PHYSICAL BLOCK AS THE LOWEST BLOCK ON	32
33	8969	/	THE NEXT LOGICAL UNIT. THE P?S/8 SHELL CHANGES THE DESIGN AS THE	33
34	8970	/	NON-SYSTEM HANDLERS HAVE A LARGER SCOPE OF BLOCK ADDRESSIBILITY.	34
35	8971			35
36	8972	/	AS SUCH, INSTEAD OF SUPPORTING LOGICAL UNITS 0 THROUGH 3 ON A SINGLE DISK	36
37	8973	/	CARTRIDGE, LOGICAL UNIT 0 COULD ADDRESS AN ENTIRE RK05 PACK WHICH IS	37
38	8974	/	BEYOND THE CAPABILITY OF OS/8 [WHICH REQUIRES TWO HANDLERS PER PACK]. BY	38
39	8975	/	DEFINING THE SYSTEM CONFIGURATION TO INCLUDE TWO PHYSICAL PACKS, EACH	39
40	8976	/	LOGICAL UNIT OF AN RK8E NON-SYSTEM HANDLER COULD ADDRESS AN ENTIRE RK05F	40
41	8977	/	FIXED PACK OR A PAIR OF RK05J CARTRIDGE PACKS ARRANGED AS PHYSICAL	41
42	8978	/	DRIVE 0 AND 1, 2 AND 3, ETC. [NOTE: A CONTROLLER MODIFICATION EXISTS KNOWN	42
43	8979	/	AS THE RKS8E THAT ALLOWS UP TO 8 PACKS SIMILAR TO THE WAY RK05 PACKS ARE	43
44	8980	/	DEFINED ON PDP-11 SYSTEMS.]	44
45	8981			45
46	8982	/	IN OS/8 TERMS, EACH RK05 PACK CONTAINS TWO DEVICES OF 3248 256-WORD	46
47	8983	/	RECORDS WHERE THE MAXIMUM POSSIBLE IS 4096 RECORDS]. SINCE THE ENTIRE	47
48	8984	/	PACK IS MORE THAN ONE OS/8 DEVICE, TWO DEVICES ARE REQUIRED. BY	48
49	8985	/	COMPARISON, THE ORIGINAL P?S/8 SYSTEM SUPPORTS FOUR LOGICAL UNITS PER PACK	49
50	8986	/	EACH 3248 128-WORD SIZE BLOCKS. THE SHELL ENVIRONMENT CAN UTILIZE AN	50
51	8987	/	ENTIRE RK05 PACK THAT ADDRESSES 12992 128-WORD BLOCKS [25984 FOR RK05F	51
52	8988	/	DEVICES OR A PAIR OF RK05J PACKS] WHERE THE P?S/8 SYSTEM DEVICE ADDRESSES	52
53	8989	/	UP TO THE FIRST 4096 BLOCKS. [NOTE: THE SHELL DIRECTORY STRUCTURE PROVIDES	53
54	8990	/	ENTRIES TO PROTECT THE BASIC P?S/8 SYSTEM AREA; INTERNAL FILE ENTRIES WILL	54
55	8991	/	BE AVAILABLE AS HIDDEN UNMOVABLE DIRECTORY ENTRIES FOR THIS PURPOSE, ETC.]	55
56				56
57				57
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1	8993	/	ASSUMING MODEST USAGE OF THE BASIC P?S/8 SYSTEM AREA WITH ONLY A FEW TFS	1
2	8994	/	FILES ALLOWED, ONLY THE FIRST 400 BLOCKS SHOULD BE REQUIRED FOR FULL	2
3	8995	/	COMPATIBILITY WITH THE BASIC SYSTEM. AS SUCH, OVER 12000 BLOCKS WOULD BE	3
4	8996	/	AVAILABLE TO THE SHELL SYSTEM. [COMPARE THIS WITH OS/8 REQUIRING TWO	4
5	8997	/	DEVICE HANDLERS EACH ADDRESSING AROUND HALF OF THIS. AND THIS SYSTEM	5
6	8998	/	COULD BE CONFIGURED FOR LARGER DEVICE SIZE SUCH AS REQUIRING EITHER TWO OR	6
7	8999	/	FOUR RK05 PACKS.] BECAUSE OF THE SUPERIOR DIRECTORY DESIGN OF THE P?S/8	7
8	9000	/	SHELL, THIS WOULD CREATE A FAR MORE POWERFUL ENVIRONMENT FOR ANY USAGE	8
9	9001	/	RANGING FROM MANY SMALL FILES [WHERE OS/8 EASILY RUNS OUT OF DIRECTORY	9
10	9002	/	SPACE] TO HANDLING FAR LARGER FILES THAN OS/8 CAN ADDRESS IN ANY	10
11	9003	/	CONFIGURATION. YET DESPITE THESE CONFIGURATIONAL DIFFERENCES, THE P?S/8	11
12	9004	/	SHELL NON-SYSTEM DRIVERS CAN ADDRESS THE ENTIRE DISK USING SMALLER READ	12
13	9005	/	AND WRITE BUFFERS [FOR A MINIMUM CONFIGURATION], OR LARGER READ AND WRITE	13
14	9006	/	BUFFERS [DUE TO LIFTING RESTRICTIONS ON FILE BUFFERING THAT ONLY APPLY IN	14
15	9007	/	OS/8 DUE TO VARIOUS LIMITATIONS. OS/8 CAN ONLY ADDRESS FILES BY TWO-PAGE	15
16	9008	/	BUFFERS OR MULTIPLES THEREOF UP TO A MAXIMUM SIZE OF 30. P?S/8 PROGRAMS	16
17	9009	/	CAN ALWAYS ADDRESS ANY FILE USING A SINGLE-PAGE BUFFER OR MULTIPLES	17
18	9010	/	THEREOF UP TO A MAXIMUM OF 32.	18
19	9011			19
20	9012	/	AS SUCH, CHANGING THE SYSTEM CONFIGURATION ON THE RK8E WILL BETTER TAKE	20
21	9013	/	ADVANTAGE OF THE SHELL ENVIRONMENT. [NOTE: THE BASIC P?S/8 SYSTEM SUPPORTS	21
22	9014	/	THE CONVERT PROGRAM TO TRANSFER TEXT FILES BETWEEN OS/8 AND P?S/8 IN BOTH	22
23	9015	/	DIRECTIONS. THIS PROGRAM REQUIRES THAT OS/8 DEVICES CAN BE ADDRESSED IN	23
24	9016	/	THE SAME BOUNDARY CONDITIONS ON A DEFINABLE LOGICAL UNIT. THE ORIGINAL	24
25	9017	/	RK8E HANDLER WOULD THUS ALLOW CONVERT TO WORK ON LOGICAL UNITS 2, 4 AND 6	25
26	9018	/	ASSUMING TWO CARTRIDGES. IN THE NEW CONFIGURATION LOGICAL UNIT 1 WILL	26
27	9019	/	WORK IDENTICALLY TO THE ORIGINAL SYSTEM USING UNIT 4. THIS CONSIDERATION	27
28	9020	/	WOULD BE CONSIDERED AN INTERIM STEP AS THE SHELL ENVIRONMENT CAN SUPPORT	28
29	9021	/	AN INDEPENDENT UTILITY TO BE KNOWN AS PIPOS8 TO TRANSFER FILES BETWEEN THE	29
30	9022	/	SHELL ENVIRONMENT AND AN OS/8 DEVICE MOUNTED ON A P?S/8 ADDRESSABLE	30
31	9023	/	DEVICE. THIS WILL ALLOW FILE TRANSFERS ON AN EQUAL BASIS IN BOTH	31
32	9024	/	DIRECTIONS.]	32
33	9025			33
34	9026	/	22) TFS FILE SYSTEM	34
35	9027			35
36	9028	/	THE TINY FILE SYSTEM [TFS] IS THE PRIMARY FILE SYSTEM OF THE BASIC P?S/8	36
37	9029	/	OPERATING SYSTEM. IT IS A DIRECT DESCENDANT OF THE R-L MONITOR FILE	37
38	9030	/	SYSTEM WITH WHICH IT IS PARTIALLY COMPATIBLE.	38
39	9031			39
40	9032	/	TFS FILES ARE FIXED SIZE INDEPENDENT FILES OF 2048 12-BIT WORDS. ACTUAL	40
41	9033	/	FILE CONTENTS IS CONTEXT-DEPENDENT; THERE ARE NO RULES OTHER THAN THE	41
42	9034	/	FILE SIZE. ENTRIES CAN BE MADE IN THE TFS DIRECTORY OVER OTHERWISE	42
43	9035	/	UNDEFINED STORAGE SPACE IN ANTICIPATION OF FUTURE USE. DEPENDING ON	43
44	9036	/	SPECIFIC CIRCUMSTANCES, PERFORMANCE IMPROVEMENT CAN BE ACHIEVED BY CLEVER	44
45	9037	/	FILE NAME PLACEMENT WITHIN THE DIRECTORY, ESPECIALLY ON SLOWER DEVICES	45
46	9038	/	BASED ON DECTAPES OR LINCTAPES.	46
47	9039			47
48	9040	/	THE FILE STRUCTURE ONLY SUPPORTS AN UP TO SIX CHARACTER FILE NAME WITHOUT	48
49	9041	/	EXTENSION; THERE IS NO DATE SUPPORT FOR TFS FILES. THIS CAN BECOME EITHER	49
50	9042	/	ITS STRONGEST POINT OR ITS WEAKEST POINT DEPENDING ON ACTUAL USAGE. THE	50
51	9043	/	TFS FIXED FILE CONCEPT IS NOT TO THE LIKING OF ALL USERS; HOWEVER, ONCE	51
52	9044	/	ACCEPTED, IT GENERALLY PROVIDES GREATER PERFORMANCE UNLESS THE SIZE OF A	52
53	9045	/	PROJECT OVERWHELMS THE TFS DIRECTORY CAPABILITY IN TERMS OF UNWIELDINESS.	53
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1	9047	/	FOR SMALL TO MEDIUM SOURCE CODE PROJECTS, EDITING A STRING OF TFS FILES	1
2	9048	/	CAUSES FAR LESS OVERHEAD THAN MORE CONVENTIONAL FILE SYSTEMS [SUCH AS IS	2
3	9049	/	USED IN THE OPTIONAL P?S/8 SHELL OR OS/8]. USERS EMBRACING THE TFS	3
4	9050	/	DIRECTORY WILL QUICKLY DEVELOP A SENSE OF HOW TO ACCESS THE ACTIVE	4
5	9051	/	PORTION OF A SOURCE CODE STREAM CONSISTING OF THE CONCATENATION OF SEVERAL	5
6	9052	/	TFS FILES. IT IS NOT NECESSARY TO MODIFY UNRELATED TFS FILES TO APPLY	6
7	9053	/	EDITS TO ONE PORTION OF A PROGRAM; MERELY OPEN THE AFFECTED FILE, MAKE	7
8	9054	/	CHANGES AND WRITE THE FILE BACK TO EXACTLY THE SAME EXACT FILE BLOCKS. NO	8
9	9055	/	BACKUP FILES ARE WRITTEN [UNLESS THE USER CREATES THEM EXPLICITLY] AND NO	9
10	9056	/	OPERATIONS ARE PERFORMED TO UPDATE THE TFS DIRECTORY UNLESS ABSOLUTELY	10
11	9057	/	NECESSARY. [NOTE: THE BASIC P?S/8 KEYBOARD EDITOR AND MONITOR OPERATES	11
12	9058	/	WITH THE ENTIRE TFS DIRECTORY CACHED AT ALL TIMES. BACKUP FILES CAN BE	12
13	9059	/	CREATED AHEAD OF TIME AND FILE COPIES CREATED INSTANTLY WRITING ONLY TO	13
14	9060	/	THE AFFECTED FILE BLOCKS ON THE CURRENT SYSTEM DEVICE LOGICAL UNIT; THE	14
15	9061	/	TFS DIRECTORY WILL NEVER BE UPDATED NEEDLESSLY.]	15
16	9062			16
17	9063	/	A TYPICAL EXAMPLE OF CONVENTIONAL VARIABLE-LENGTH FILE EDITING VERSUS TFS	17
18	9064	/	FILE EDITING FOLLOWS:	18
19	9065			19
20	9066	/	ASSUME A SINGLE FILE IN A CONVENTIONAL VARIABLE SIZE ENVIRONMENT IS 128	20
21	9067	/	BLOCKS LONG AND IN A TFS ENVIRONMENT THE SAME SOURCE TEXT EXISTS IN EIGHT	21
22	9068	/	TFS FILES TO BE USED AS A CONCATENATED STREAM WHEN ASSEMBLED. EDITING IN	22
23	9069	/	THE CONVENTIONAL ENVIRONMENT REQUIRES OPENING A FREE SPACE ON THE DEVICE	23
24	9070	/	THAT IS AT LEAST 128 BLOCKS LONG; EACH SYSTEM MAY PROVIDE A WAY TO OBTAIN	24
25	9071	/	A SUITABLE SECTION, OFTEN THE LARGEST CONTIGUOUS FREE SPACE CURRENTLY	25
26	9072	/	AVAILABLE. [NOTE: THE EDIT CANNOT EVEN BE PERFORMED IF SUCH SPACE IS NOT	26
27	9073	/	PRESENTLY AVAILABLE; THE USER MAY BE FORCED TO PUT OFF THE EDIT AND DEAL	27
28	9074	/	WITH COMPRESSING EMPTY SPACE TOGETHER BEFORE THE EDIT CAN BE PERFORMED.]	28
29	9075	/	ONCE THE EDITING SESSION COMMENCES, THERE ARE NOW TWO AREAS THE SIZE OF	29
30	9076	/	THE ENTIRE FILE IN PLAY SIMULTANEOUSLY. CLOSING THE FILE LEADS TO THE	30
31	9077	/	DECOMMITMENT OF THE ORIGINAL FILE BLOCKS; THIS IN TURN TENDS TO FRAGMENT	31
32	9078	/	THE STORAGE DEVICE LEADING TO ADDITIONAL NEED TO COMPRESS OUT EMPTY SPACE	32
33	9079	/	THE NEXT TIME, ETC. THE NEW FILE ALSO TENDS TO LEAD TO FRAGMENTATION AS	33
34	9080	/	THE FINAL FILE SIZE WILL TEND TO BE LESS THAN THE TEMPORARY ALLOCATION OF	34
35	9081	/	STORAGE SPACE REQUESTED. UNLESS THE STORAGE SYSTEM IS ONLY LIGHTLY USED,	35
36	9082	/	DEFRAGMENTATION OF EMPTY SPACE WILL TEND TO BE NEEDED OFTEN. [NOTE: THE	36
37	9083	/	P?S/8 SHELL CANNOT HELP BUT SUFFER FROM THIS PROBLEM IN PRACTICAL USAGE;	37
38	9084	/	HOWEVER, THERE ARE SOME WAYS TO MITIGATE THE PROBLEM SOMEWHAT AS COMPARED	38
39	9085	/	TO OS/8'S RAW IMPLEMENTATION OF A VARIABLE SIZE FILE SYSTEM THAT WORKS	39
40	9086	/	EXACTLY AS DESCRIBED IN THIS EXAMPLE. THIS WILL INCLUDE SUCH AS FILE	40
41	9087	/	LOCALIZATION TECHNIQUES AND THE ABILITY TO DEFINE TIERS OF SPACE	41
42	9088	/	ALLOCATION; EDITING PROGRAMS CAN SPECIFY REQUESTING THE NEAREST EMPTY	42
43	9089	/	SPACE ON THE DEVICE AS WELL AS ACCEPTABLE LIMITS ON THE ACTUAL SIZE OF THE	43
44	9090	/	REQUESTED TENTATIVE FILE SPACE INSTEAD OF ALWAYS BEING PROVIDED THE	44
45	9091	/	LARGEST AVAILABLE EMPTY SPACE ANYWHERE ON THE DEVICE. SUCH TECHNIQUES	45
46	9092	/	WILL TEND TO SOMEWHAT LIMIT FRAGMENTATION.] [NOTE: DETAILS OF THE OVERHEAD	46
47	9093	/	OF CARRYING OUT THESE OPERATIONS HAS NOT BEEN EXAMINED; IT IS DEPENDENT ON	47
48	9094	/	SPECIFIC IMPLEMENTATION; NO VARIABLE-LENGTH FILE SYSTEM ACCOMPLISHES THESE	48
49	9095	/	OPERATIONS INSTANTANEOUSLY.]	49
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1	9097	/	ACCOMPLISHING THE EQUIVALENT EDIT IN THE TFS DIRECTORY INVOLVES MERELY	1
2	9098	/	LOCATING THE ONE FILE WHERE THE EDITS MUST BE PERFORMED. THE FILE IS	2
3	9099	/	LOADED INTO THE EDITING BUFFER, EDITED IN PLACE AND WRITTEN BACK TO THE	3
4	9100	/	SAME BLOCKS AS BEFORE THE EDIT. [NOTE: THE USER MAY OPTIONALLY LOAD THE	4
5	9101	/	FILE, SAVE TO A PRE-ALLOCATED DESIGNATED BACKUP FILE, EDIT THE FILE AND	5
6	9102	/	THEN WRITE IT BACK TO THE ORIGINAL STORAGE BLOCKS.] BECAUSE THE TFS	6
7	9103	/	DIRECTORY STRUCTURE IS 100 PERCENT CACHED, THERE IS *NO* OVERHEAD	7
8	9104	/	ASSOCIATED WITH THE EDIT!	8
9	9105			9
10	9106	/	THE LARGER THE FILE, THE INHERENTLY MORE DRAMATIC THE DIFFERENCE BETWEEN	10
11	9107	/	THE TWO METHODS. [NOTE: THERE ARE TRADE-OFFS RELATED TO SUCH AS WHEN A	11
12	9108	/	TFS FILE BECOMES TOO FULL INVOLVING SPLITTING THE FILE INTO TWO FILES; THIS	12
13	9109	/	MUST BE PERFORMED AS NEEDED. AS LONG AS THERE IS ENOUGH ADDITIONAL SPACE	13
14	9110	/	ON THE STORAGE DEVICE TO ALLOCATE ONE MORE TFS FILE, THIS CAN ALWAYS BE	14
15	9111	/	ACHIEVED; CONTRAST THIS WITH THE NEED FOR ADDITIONALLY AT LEAST THE	15
16	9112	/	STORAGE OF THE ENTIRETY OF THE CURRENT FILE SIZE IN A CONVENTIONAL	16
17	9113	/	EDITING ENVIRONMENT. THE BASIC P?S/8 SYSTEM PROVIDES ADEQUATE UTILITIES TO	17
18	9114	/	MANAGE EVERY ASPECT OF THIS PROCESS.]	18
19	9115			19
20	9116	/	ULTIMATELY, VERY LARGE PROJECTS CANNOT BE HANDLED BY THE TFS DIRECTORY	20
21	9117	/	BECAUSE OF RESOURCE LIMITATIONS OF P?S/8. [NOTE: THERE ARE PROVISIONAL	21
22	9118	/	WAYS AROUND MOST OF THE LIMITATIONS.] THERE IS A HARD UPPER LIMIT ON	22
23	9119	/	PASSING TFS FILE HANDLES TO PROGRAMS OF 17 MAXIMUM, INCLUDING ANY REQUIRED	23
24	9120	/	OUTPUT FILES. FOR EXAMPLE, A PROGRAM MAY BE CREATED IN 14 TFS FILES THAT	24
25	9121	/	CREATES THREE BINARY FILES. DUE TO P?S/8 BATCH SUPPORT BEING PRESENT IN	25
26	9122	/	THE OPERATING SYSTEM, THE USER CAN TYPE UP SHORT ["MAKE"-LIKE] COMMAND	26
27	9123	/	FILES TO BE USED REPEATEDLY TO AVOID THE TEDIUM OF MANUALLY ENTERING LONG	27
28	9124	/	COMMAND LINES. FOR TRULY LARGE SOURCE CODE PROJECTS, A LARGER SYSTEM WILL	28
29	9125	/	GENERALLY BE REQUIRED; THE P?S/8 SHELL WILL BE AN EXCELLENT WAY TO HANDLE	29
30	9126	/	LARGER PROJECTS ASSUMING SUFFICIENT RESOURCES ARE AVAILABLE [SUCH AS MORE	30
31	9127	/	MEMORY AND LARGER STORAGE DEVICES].	31
32	9128			32
33	9129	/	23) .XOR.	33
34	9130			34
35	9131	/	MANY OPERATIONS WITHIN THE TC01/TC08 SYSTEM DEVICE HANDLER ARE CARRIED OUT	35
36	9132	/	BY A BIT-WISE EXCLUSIVE-OR OPERATION BETWEEN THE DECTAPE CONTROLLER STATUS	36
37	9133	/	REGISTER "A" AND THE PDP-8 ACCUMULATOR [AC]. THE ".XOR." NOTATION IS USED	37
38	9134	/	AS SHORT-HAND WHEN DESCRIBING THESE OPERATIONS.	38
39	9135			39
40	9136	/	A SUBTLE ISSUE COMPLICATES DESCRIBING THESE OPERATIONS: ONLY THE TEN	40
41	9137	/	HIGHEST BITS ARE TRANSFERRED BY THE EXCLUSIVE-OR OPERATION; THE TWO LOWEST	41
42	9138	/	BITS OF STATUS REGISTER "A" ARE USED TO PRESERVE CORRESPONDING CONTROLLER	42
43	9139	/	FLAGS DIRECTLY IF SET. ALL BITS OF THE AC ARE APPLIED TO STATUS REGISTER	43
44	9140	/	"A" SIMULTANEOUSLY AS DESCRIBED ABOVE.	44
45	9141			45
46	9142	\$	/THAT'S ALL, FOLK!	46
47				47
48				48
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1	COMPLETE SYMBOL TABLE LISTING												1		
2													2		
3	ACL	7701P	CONSOL	6764D	I	1357D	LSW	0517P	READ3	7674	SLIFLD=0010	SWMX	=7605	3	
4	ADA	1100P	CUF	6264P	IAAC	6171P	LZE	0452P	RELOC	0754D	SLIGO	7757	SWP	7521P	4
5	ADD	2000M	CURUNI=0000		IAC	7001P	L200	=0104	REWUNL=1342		SLOAD	=7632	SWY9	=7606	5
6	ADM	1140P	DA	=0000	IACA	6167P	L400	1367	RFC	6014P	SLOOP	7642	SXL	0400P	6
7	AND	0000M	DAD	7443P	IACF	6175P	L700	1366	RIB	6234P	SLPCTR	7742	SYSBLK=0000		7
8	APO	0451P	DATE	6765D	IACS	6163P	L7756	=0064	RIF	6224P	SLPFLD	7704	SYSIO	=7640	8
9	ASC	7403P	DCA	3000M	IBAC	6143P	L7770	=0040	RMF	6244P	SLPNEX	7713	SYSWAI	7612	9
10	ASMBUG=0001		DCM	7575P	IBZ	0453P	MASK	7665	ROL	0240P	SLPSTP	7624	SZA	7440P	10
11	ASMIFM	3361D	DDZ	7665P	ICON	6141P	MEMFLD	7727	ROR	0300P	SLPSTR	7626	SZL	7430P	11
12	ASMIFN	3357D	DECIMA	0306D	ICS1	6151P	MEMTEM	7751	RPE	6010P	SLRPBL=7613		S10	7630	12
13	ASMIFZ	3360D	DECTLI=7640		ICS2	6153P	MQA	7501P	RRB	6012P	SLRPOR	7741	S200	7602	13
14	ASMSKP	3355D	DEFADR	7776	IFDEF	3601D	SQL	7421P	RSF	6011P	SLUADR=7000		S6203	7631	14
15	ASR	7415P	DEFSTR=7745		IFNDEF	3600D	MSC	0000P	RSW	0516P	SLUFLD=0010		S7600	7601	15
16	ASSEMB	7741	DEVICE	1150D	IFNZRO	3613D	MTB	0703P	RTA	0015P	SLUSTR	7746	S7750	7621	16
17	ATR	0014P	DIS	0140P	IFZERO	3614D	MUL	1240P	RTF	6005P	SLWAIT	7613	TAC	0003P	17
18	AXO	0001P	DJR	0006P	ILES	6145P	MUY	7405P	RTL	7006P	SL210	7654	TAD	1000M	18
19	AZE	0450P	DLD	7663P	IMBS	6155P	M200	7707	RTR	7012P	SL220	7637	TCF	6042P	19
20	BCL	1540P	DPIC	7573P	INC	2000M	M376	7705	RWTEMP=1305		SL30	7601	TEMP	7776	20
21	BCO	1640P	DPSZ	7451P	INST1	=5255	NL4000=7330		SADDR	7746	SL400	7610	TEXT	1101D	21
22	BEG	=7756	DSC	1740P	INST2	=5244	NL7777=7240		SAE	1440P	SL4002	7622	TFL	6040P	22
23	BEGG	7643	DST	7445P	INTS	6147P	NMI	7411P	SAM	7457P	SL600	7607	TITLE	3401D	23
24	BEGG1	7644	DTCA	=6762	IOB	0500P	NOBITS	1367D	SBLK	7745	SL7000	7620	TLS	6046P	24
25	BEGIN	7706	DTLA	=6766	IOF	6002P	NOP	7000P	SBLKCT	7751	SL7577	7623	TMA	0023P	25
26	BEG1	7757	DTLB	=6774	ION	6001P	NOPUNC	1176D	SBLKLP	7726	SL7750	7734	TPC	6044P	26
27	BEG11	7767	DTORG	2762D	ISSP	6165P	NOTRUB	7655	SBLKRE	7727	SL7756	7740	TRNLEN	7774	27
28	BEG13	7771	DTRA	=6761	ISZ	2000M	NXTGRP	7670	SBOOT	=7600	SL7770	7600	TSF	6041P	28
29	BEG3	7761	DTRB	=6772	ITAC	6157P	NXTSRC	7644	SCA	7441P	SL7772	7745	TSK	6045P	29
30	BEND	7702	DTSF	=6771	IZSA	6173P	NXTWRD	7675	SCL	7403P	SMA	7500P	TWC	0417P	30
31	BINLOA=7643		DTXA	=6764	JMP	5000M	OCTAL	0301D	SCORE	=7611	SNA	7450P	TYP	0514P	31
32	BINSTR=7777		DVI	7407P	JMS	4000M	ONEWOR	7751	SCR	0340P	SNL	7420P	U	0010P	32
33	BOOTBL=0000		EF	=0001	JSR	6000M	OPR	7000P	SDATE	=7610	SNS	0440P	UMB	0640P	33
34	BOOTLI=7643		EJECT	3400D	KBD	0515P	OR	=0002	SDHADR=7400		SNXTBL=7674		UNLWAI	1355	34
35	BRGADR=6600		ENBITS	1367D	KBDCHR=0100		OSR	7404P	SDHFLD=0010		SOUTCT=7607		WATLUP	7733	35
36	BRGFLD=0010		ENDBIN	2551D	KBDLUP=1216		PAGE	0565D	SDONE	7737	SOVSHT	7744	WC	=7754	36
37	BSE	1600P	ENPUNC	1175D	KBDMON=2530		PAUSE	0535D	SEGMNT	3364D	SPA	7510P	WCG	0705P	37
38	BSW	7002P	ERROR	0371D	KCC	6032P	PCE	6020P	SEQUAL=7756		SRCHBL	7750	WORD1	7754	38
39	CA	=7755	ESF	0004P	KCF	6030P	PCF	6022P	SET	0040P	SRETFL	7742	WORD2	7755	39
40	CAF	6007P	EXC	0740P	KIE	6035P	PDP	0002P	SFA	0024P	SRETRY	7657	WRC	0704P	40
41	CAM	7621P	EXPUNG	1367D	KMNADR=0000		PERFIL=0020		SFILES=7757		SRO	1500P	WRI	0706P	41
42	CDF	6201P	FIELD	0613D	KMNFLD=0000		PLS	6026P	SFUN	=7634	SRQ	6003P	WRITE	=4000	42
43	CHANGE	7701	FILENA	1146D	KRB	6036P	PMODE	2576D	SFUNCT	7752	SSTART=7637		XLIST	2144D	43
44	CHAR	7750	FIXMRI	2124D	KRS	6034P	PPC	6024P	SGETBL	7655	STA	7240P	XOA	0021P	44
45	CHEX	7735	FIXTAB	1367D	KSF	6031P	PQS	1367D	SGT	6006P	STB	0414P	XSK	0200P	45
46	CHK	0707P	FL	=0003	KST	0415P	PSF	6021P	SHALT	7626	STC	4000M	Z	1367D	46
47	CHKSUM	7752	FLDCHK	7722	LAM	1200P	QAC	0005P	SHD	1400P	STD	0416P	ZBLOCK	2154D	47
48	CIA	7041P	FLIPIT	7753	LAS	7604P	QLZ	0455P	SHL	7413P	STH	1340P	ZTA	0005P	48
49	CIF	6202P	FLO	0454P	LDA	1000P	RAL	7004P	SINT	6254P	STL	7120P	ZZZ	0455P	49
50	CINT	6204P	FMASK	7700	LDF	0640P	RAR	7010P	SIXBIT	1100D	STURNI=7673				50
51	CLA	7200P	GETWRD	7727	LDH	1300P	RCC	6016P	SKIP	2561D	SUF	6274P			51
52	CLL	7100P	GLK	7204P	LIF	0600P	RCG	0701P	SKON	6000P	SWAB	7431P			52
53	CLR	0011P	GO	7715	LINC	6141P	RDC	0700P	SKP	7410P	SWAITL	7616			53
54	CLRAC	7774	GRPCTR	7752	LMB	0600P	RDE	0702P	SLDADR=7633		SWAL	=7604			54
55	CMA	7040P	GTF	6004P	LMODE	2575D	RDF	6214P	SLDBLK=7635		SWBA	7447P			55
56	CML	7020P	GWRDCT=7750		LMR	6151P	READ	7671	SLFLPT	7753	SWD	0457P			56
57	COM	0017P	HLT	7402P	LSR	7417P	READ1	7672	SLIADR=7200		SWINST	7647			57
58															58

1		1
2		2
3	NO ERRORS DETECTED	3
4		4
5	NO LINKS GENERATED	5
6		6
7	6K MEMORY UTILIZED	7
8		8
9	1 FILE CREATED	9
10		10
11	377 SYMBOLS	11
12		12
13		13
14		14
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MAP OF FIELD 0
NO MAPPING RESTRICTIONS

P?S BITMAP V02F 18-AUG-14 PAGE M-1

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4		4
5	00000	5
6	00100	6
7		7
8	00200	8
9	00300	9
10		10
11	00400	11
12	00500	12
13		13
14	00600	14
15	00700	15
16		16
17	01000	17
18	01100	18
19		19
20	01200	20
21	01300 00000000 00000000 00000000 00000000 00111111 11111111 11111111 11111111	21
22		22
23	01400	23
24	01500	24
25		25
26	01600	26
27	01700	27
28		28
29	02000	29
30	02100	30
31		31
32	02200	32
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35	02400	35
36	02500	36
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38	02600	38
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42	03100	42
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44	03200	44
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47	03400	47
48	03500	48
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50	03600	50
51	03700	51
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5	04000	5
6	04100	6
7		7
8	04200	8
9	04300	9
10		10
11	04400	11
12	04500	12
13		13
14	04600	14
15	04700	15
16		16
17	05000	17
18	05100	18
19		19
20	05200	20
21	05300	21
22		22
23	05400	23
24	05500	24
25		25
26	05600	26
27	05700	27
28		28
29	06000	29
30	06100	30
31		31
32	06200	32
33	06300	33
34		34
35	06400	35
36	06500	36
37		37
38	06600	38
39	06700	39
40		40
41	07000	41
42	07100	42
43		43
44	07200	44
45	07300	45
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47	07400	47
48	07500	48
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50	07600	50
51	07700	51
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5	10000	5
6	10100	6
7		7
8	10200	8
9	10300	9
10		10
11	10400	11
12	10500	12
13		13
14	10600	14
15	10700	15
16		16
17	11000	17
18	11100	18
19		19
20	11200	20
21	11300	21
22		22
23	11400	23
24	11500	24
25		25
26	11600	26
27	11700	27
28		28
29	12000	29
30	12100	30
31		31
32	12200	32
33	12300	33
34		34
35	12400	35
36	12500	36
37		37
38	12600	38
39	12700	39
40		40
41	13000	41
42	13100	42
43		43
44	13200	44
45	13300	45
46		46
47	13400	47
48	13500	48
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50	13600	50
51	13700	51
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4		4
5	14000	5
6	14100	6
7		7
8	14200	8
9	14300	9
10		10
11	14400	11
12	14500	12
13		13
14	14600	14
15	14700	15
16		16
17	15000	17
18	15100	18
19		19
20	15200	20
21	15300	21
22		22
23	15400	23
24	15500	24
25		25
26	15600	26
27	15700	27
28		28
29	16000	29
30	16100	30
31		31
32	16200	32
33	16300	33
34		34
35	16400	35
36	16500	36
37		37
38	16600 00000000 00000000 00000000 00000000 00011111 11111111 11111111 11111111	38
39	16700 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111	39
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41	17000 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111	41
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43		43
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45	17300 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111	45
46		46
47	17400 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111	47
48	17500 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111	48
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50	17600	50
51	17700	51
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